KNEE & SPORTS — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations + TO lecture notes · 61 chapters (0 from TO notes) –

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Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer · "TO – verify" = TO figure that differs from standard teaching

– – Active space – –

ACL TEAR

– Knee & Sports · 1 / 61 –

1 Basics

Epidemiology

Incidence	common · **~400,000 ACL reconstructions / yr** · 100,000–200,000 ruptures / yr in US (OB) · **half of all knee injuries**
Most injured knee ligament (TO)	**MCL > ACL** · **ACL = most commonly operated** knee ligament (MCL mostly nonop) ✗ Wrong options: LCL · ALL
Female : male	**4.5 : 1** (Orthobullets, OB) · **2–9×** (OB) · **~8×** (TO) · "twice as likely" (OB) – read the options · gap **not seen until puberty**
Females	injured **younger**, on the **supporting leg** (males → kicking leg) · ACL = **24.5%** of ligamentous knee injuries in high-school athletes
Risk factors	female soccer, male basketball · **valgus moment at knee + adduction moment at hip on landing** · **previous concussion** (neuromuscular alteration) ✗ Wrong options: varus alignment · LLD · prior meniscal tear · ↑ hamstring strength (protective)
Associated injuries	**lateral meniscus tear 54% in acute tears** (most common intra-articular injury) · **medial meniscus in chronic** · PCL, LCL / PLC (high energy, knee dislocation) · chronic ACL-deficient knee → **chondral injury**, **complex unrepairable + bucket-handle MM tears**

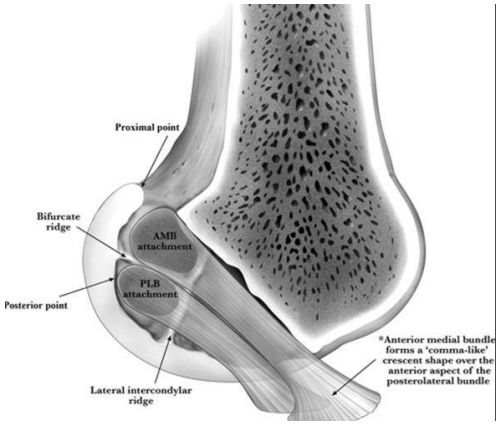
Anatomy – AM/PL bundles, footprints (resident's & bifurcate ridges), blood supply & ALL

	Anteromedial (AM)	**Posterolateral (PL)**
Named for	**Tibial** insertion	**Tibial** insertion
Tight in	**Flexion** (TO: taut in flexion + extension)	**Extension**; loose in flexion + mid-flexion
Isometry	**More isometric**	**Greatest length change**
Main job	**Anterior tibial translation** (esp. in flexion)	**Rotation** near extension; resists tibial IR + **pivot shift**
Exam	**Lachman / anterior drawer**	**Pivot shift**
Size (TO)	**Longer + bulkier** (~38 mm)	Shorter (~18 mm) (✗ Wrong options: "PL bulkier + longer")
Femoral footprint	**Proximal** to bifurcate ridge; high + deep; **10:30**	**Distal** to bifurcate ridge; low + shallow; **9:30**
Tibial footprint	Anterior	Posterior

Both bundles — **entire femoral insertion lies posterior to the lateral intercondylar ridge** (medial wall of LFC) · femoral insertion **nearly vertical in extension** (AM proximal to PL), **nearly horizontal at 90°** · parallel + taut in extension (TO) · **no fibre truly isometric** · **no histological separation** of bundles (TO)

Footprint areas (TO) — AM femoral 47 ± 13 mm², tibial 56 ± 21 mm² · PL femoral 49 ± 13 mm², tibial 53 ± 21 mm²

Femoral ridges — **lateral intercondylar ridge = resident's ridge → anterior edge** of ACL, runs proximo-distal · **bifurcate ridge →** separates **AM / PL**, runs antero-posterior (⊥ LIR) (Orthobullets, OB) · TO: "resident's ridge separates AM + PL" – read the options



ACL femoral footprint: lateral intercondylar and bifurcate ridges

Medial intercondylar ridge + bifurcate prominence — on **lateral wall of MFC → PCL** footprint
✗ Wrong options: ACL

Ideal femoral tunnel (OB) — **6.1 mm posterior to resident's ridge + 1.7 mm proximal to bifurcate ridge**
✗ Wrong options: distal to BR + anterior to resident's ridge

Tibial insertion (OB) — **11 mm anterior to anterior border of PCL** · ~6 mm anterior to median / medial tibial eminence · ~9 mm posterior to intermeniscal ligament · medial + anterior to posterior border of **anterior horn LM** · meniscal roots variable (unreliable alone)
✗ Wrong options: 15 mm anterior to PCL · 16 mm / 11 mm anterior to eminence · 17 mm behind IML

Dimensions — **32 mm long** (22–41 mm) × **7–12 mm wide** · 31–38 mm (TO) · **narrowest at mid-substance**; insertions **> 3.5×** larger (TO)

Dimensions (TO, primary-ACLR notes) — **31 mm** (29–35) long × **9 mm** (7–11) wide

Tibial insertion (TO) — **broader than femoral (~120%)**, fibres fan out · fibres pass under transverse intermeniscal ligament, blend with anterior horn LM · **stronger than femoral → avulsion on tibial side**, substance tear femoral side

Course (TO) — **intra-articular but extrasynovial** (intracapsular) · origin **posteromedial LFC (medial wall of LFC) →** runs inferomedially → tibia between spines

Composition — **90% type I** + 10% type III collagen
✗ Wrong options: type II (articular cartilage, nucleus pulposus) · type III (Dupuytren) · type IV (basement membrane)

– – Active space – –

Blood supply	**middle genicular artery** + **synovium** · MGA arises **directly from popliteal**, **pierces oblique popliteal ligament / posterior capsule**, runs along dorsal ACL · **inferior medial + lateral genicular** → distal ACL (minor) · **no significant supply from bony attachments** (Scapinelli) · proximal ACL better supplied ✗ Wrong options: superior genicular · inferior medial genicular as main supply
Avascular zone (TO)	**5–10 mm proximal to tibial attachment** (anterior fibrocartilage) → poor healing
MGA also supplies	**PCL** (+ fat pad) · ACL tear haemarthrosis = MGA branches · menisci → inferior / superior medial + lateral genicular (**LM = inferior lateral**, **MM = inferior medial**)
Innervation	**posterior articular nerve** (branch of tibial nerve)
Function	**85%** of restraint to **anterior tibial translation** · secondary restraint to **tibial IR** + varus / valgus (TO: resists varus at full extension) · strength **2200 N**
ACL reflex (TO)	mechanoreceptors (**Ruffini, Pacini, Golgi-like**) → proprioception via fusimotor system · **not a protective mechanism** · loss / inhibition → **quadriceps weakness** ✗ Wrong options: explains female tears · explains poor healing
Notch (TO)	narrower in women · **notch width index** (epicondylar : notch width) small → ↑ ACL tear · notch roof = **Blumensaat's line**; notch roof angle **37°** (23–60°)
Notch width index (Souryal & Freeman, TO)	normal **0.231** · narrow notch → ACL tear
PCL contrast (TO)	bundles named for **femoral** footprint · **AL 55%**, bulkier, tight in **flexion** · **PM 45%**, tight in **extension** · 38 mm long, 10–11 mm diameter · tibial insertion wider than femoral ✗ Wrong options: "ACL named for femoral, PCL for tibial"
ALL	origin **proximal + posterior to lateral epicondyle / LCL** · insertion **anterolateral tibia halfway between Gerdy's tubercle + fibular head** · anterolateral rotatory stabiliser · **Segond = ALL avulsion** · LET recreates it
Lateral knee distractors	tibial tubercle = patellar tendon · **anterior** fibular head = LCL (femur just posterior to lateral epicondyle) · **posterior** fibular head = biceps long head · lateral gastrocnemius → femur posterior + proximal to LCL · popliteus → femur anterior + distal to LCL · lateral epicondyle = ALL **origin**

Genes & lab markers

COL5A1 genotype	**↓ risk of ACL rupture in women** ✗ Wrong options: GH dislocation · cuff tear · patellar dislocation · discoid meniscus
Collagen	ACL **90% type I** / 10% type III
Post-ACLR septic arthritis	**ESR 105 + CRP 44** at 4 wk (TO) · **CRP 100 mg/L, ESR 40, aspirate 52,000 WBC/mm³** at 3 wk (OB) → infection

ACL-deficient kinematics & gait

Gait	**absent normal femoral IR in terminal swing** · ↓ anterior tibial translation in late swing (adaptive: ↓ quads / ↑ hamstrings) ✗ Wrong options: absent femoral ER · ↑ posterior femoral translation · ↑ midstance flexion · ↑ hip adduction
Normal gait	heel strike near full extension, femur IR on tibia → midstance flexion + femoral ER · late swing: anterior tibial translation (quads)
Kinematics	**↑ paradoxical anterior femoral translation with flexion** · **larger AP translation of MFC** · LFC **more posterior** in full extension · axial rotation same in first 30°, differs beyond ✗ Wrong options: no change
Medial tibial translation	at higher flexion in ACL-deficient knee → ↑ contact of tibial spine with inner MFC → degeneration
Dynamic imaging	**anterior shift + IR of tibia at low flexion** = **ACL** · posterior shift = PCL · ↑ IR + posterior translation = PCL + MCL · ↑ valgus + slight IR = MCL · ↑ ER = PLC (LCL, popliteus, arcuate)
Consequence	abnormal loading → **post-traumatic OA**, meniscal + osteochondral injury
ACL-deficient OA (TO)	up to **3×** higher frequency · **anteromedial compartment** most affected

2 Mechanism

Pathoanatomy

Non-contact pivot	tibia translates anteriorly with knee in **slight flexion + valgus** · or **blow to lateral knee** · soccer, basketball, skiing, football
Bone bruise mechanism	**valgus–ER of femur on fixed tibia** → LFC impacts **posterior lip of lateral tibial plateau** (= pivot-shift event)
Segond mechanism (TO)	**IR + varus in flexion** · capsular avulsion (mid-lateral capsule / ALL; meniscotibial ligament – OB)
Hyperextension	**kissing contusions of anterior tibia + femur** (PCL-type) ✗ Wrong options: ACL pattern
Female predisposition	landing biomechanics + **quadriceps-dominant** neuromuscular activation (biggest role)

Mechanism – PCL injury on flexed knee (TO)

Lands on flexed knee, foot plantarflexed (TO)	**dashboard-type** posterior force on tibia → **PCL** ✗ Wrong options: ACL · PLC · LCL

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Female athletes – risk factors & neuromuscular prevention

Female risk factor (TO)	Detail
Quadriceps dominance	More quad than hamstring activation; slower to peak hamstring torque · males decelerate with hamstrings
Erect landing	**Insufficient knee + hip flexion**; ↑ hip IR + adduction on landing
Pelvis / alignment	Wider pelvis, **↑ Q angle**, knee valgus
Notch / ligament	**Small intercondylar notch**, smaller ACL
Laxity	Greater ligament laxity
Hormones	Oestrogen, relaxin – indirect, via neuromuscular system
NOT a risk factor (TO)	**↑ external rotation at landing · hamstring dominance**

Other risk factors (TO) hard field surface, inappropriate footwear, **↑ BMI**, **high tibial slope**, **foot pronation**, female sex

Greatest contributor **neuromuscular factors** · notch far less important
✗ Wrong options: notch width · BMI · laxity · limb alignment

Highest-risk cohort **women with knee valgus moment on landing**
✗ Wrong options: hip adduction moment · men

Landing pattern (OB) **↓ hip + knee flexion, ↑ knee ER, dynamic valgus** · TO says ↑ ER at landing is NOT a risk factor – read the options

Hamstrings relative **hamstring weakness** / ↑ quad : hamstring ratio, electromechanical delay · after **menarche** quad strength ↑ more than hamstrings → hamstring-focused programmes

Menstrual cycle no consensus
✗ Wrong options: ovulatory · post-ovulatory phase

Prevention **neuromuscular training / plyometrics (jump training)** – core, abductor + hamstring synchrony, avoid dynamic valgus; land with more knee flexion, less valgus (Griffin; **FIFA 11+**) · skiers → teach how to fall

Bracing **no proven prevention** except **ACL-deficient skiers**
✗ Wrong options: brace to cut re-injury risk

3 Presentation

History **"pop"**, pain deep in knee, **immediate swelling (70%) / haemarthrosis**, instability, can't return to sport, hard to weight-bear

Non-contact + acute haemarthrosis **ACL tear > 60%** (literature > 70%; Maffulli 71 / 106) · acute effusion → ACL; slow effusion → meniscus
✗ Wrong options: MCL · MM tear · patellar dislocation · contusion

Inspection effusion · **quadriceps avoidance gait** · **varus → ↑ re-rupture risk**

Motion lacks extension → **meniscal block** (locked bucket handle) or arthrofibrosis · check McMurray, dial, varus / valgus

Lateral joint line tenderness likeliest extra finding with + pivot shift (LM tear)
✗ Wrong options: MJL tenderness · dial · varus at 30° · posterior drawer

Neurovascular peroneal function after high-energy / multiligament pattern

Diagnosis – Lachman grading, pivot shift mechanism, hemarthrosis

– – Active space – –

Lachman grade	Orthobullets	OB
1	3–5 mm	0–5 mm
2	5–10 mm	6–10 mm
3	> 10 mm	> 10 mm
A / B	A = firm endpoint · B = no endpoint	B = soft / no endpoint

Lachman — **20–30° flexion**, stabilise femur, translate tibia · compare sides (physiological laxity) · **2B = 9 mm, soft endpoint at 30°** · 4 mm firm = 1A · 12 mm firm = 3A · at 90° = anterior drawer

PCL tear — **"false" Lachman** (posterior subluxation)

Test	Sensitivity	Specificity	Note
Lachman	**87%**	93%	**Most sensitive**; mainly AM bundle
Pivot shift	61%	**97%**	**Most specific**; PL bundle / rotation; easier under anaesthesia
Anterior drawer (90°)	48%	–	Lower sensitivity
KT-1000	–	–	Quantifies anterior laxity; slight flexion, ER 10–30°

Pivot shift – mechanism — extension + **tibial IR + valgus → lateral tibial plateau subluxed anteriorly** (impinges LFC) → flex → **ITB becomes a reducing force → LTP reduces** with a clunk
✗ Wrong options: medial plateau · ER · reduced in extension

Reduction angle — **20–30°** (Orthobullets) · **30–40°** (OB) · 15–20° (Delhi active test, TO) – read the options

Pivot shift — mimics the giving-way event · patient fully relaxed · **better gauge of functional outcome** · stays abnormal with **vertical** graft
✗ Wrong options: anterior femoral tunnel

Pivot shift grade	Finding
0	Normal
I	Glide (may be physiological)
II	Clunk
III	Clunk with locking (gross / explosive)

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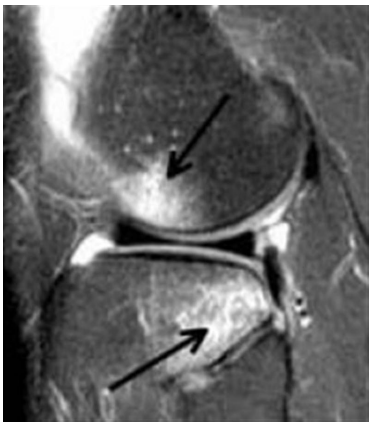
Delhi active test (TO) **standing**, active test for **complete ACL tear** · limb abducted, foot on table (tibial IR) · flex → at **15–20°** subluxed LTP relocates (shift / clunk) · extend → anterolateral subluxation (jerk)
✗ Wrong options: supine · PCL

Positive test	Points to
Pivot shift / Lachman	**ACL**
Quadriceps active test, reverse pivot shift, posterior drawer (12 mm at 90°)	**PCL**
Dial ↑ ER > 10° (~15°) at 30° only	**PLC**
Dial at 30° + 90°	**PLC + PCL**
Varus laxity at 30°	LCL / PLC (at 0° + 30° if ACL torn too)
McMurray, extending with **IR + varus**	**Lateral meniscus**
McMurray, extending with **ER + valgus**	**Medial meniscus**
Patellar apprehension	Patellar instability / MPFL

4 Diagnosis & Classification

Key radiology findings – MRI & bone bruises, Segond fracture

Study	Finding / use
X-ray – AP, lateral, sunrise / Merchant	Often **normal**; **Segond**; **deep sulcus sign**; tibial spine avulsion
Segond fracture	Avulsion of **proximal lateral tibia** by **ALL** (AP or **notch view**) · **pathognomonic** · ACL tear in **75–100%** · present in 9% of ACL tears
Deep sulcus (terminalis) / lateral femoral notch sign	**Lateral X-ray**: LFC depression at terminal sulcus (junction of tibial + patellar articular surfaces) · **> 2 mm** deep = diagnostic (uncommon, specific) · PD-FSE MRI: sulcus terminalis impaction + oedema
Post-ACLR X-ray	Tunnel position (vertical = both tunnels steep) · **tunnel osteolysis / widening** around screws · fluoroscopy: tibial centre at **40% along Amis–Jakob line**
MRI – confirm + concomitant injury	Sens **97%** / spec **100%** (Orthobullets) · accuracy **90–95%** (TO)
MRI direct signs	**Discontinuity on T2** (often proximal), ACL **too flat** vs Blumensaat, **non-visualisation** (subacute: only small distal stump), tibial spine avulsion
Femoral bony avulsion (adolescent)	AP X-ray fragment at femoral condyle · MRI: lax fibres + attached bony fragment → **repair**
MRI coronal	Fibres fail to reach femur · **empty notch / lateral wall sign** (fluid at femoral attachment → proximal avulsion)
Bone bruise	**Middle ⅓ LFC (sulcus terminalis) + posterior ⅓ lateral tibial plateau** · > ½ of acute tears (~50%) · **↑ T2 (fat-sat), ↓ T1** = trabecular haemorrhage
Other indirect (TO)	Anterior tibial translation, **PCL buckling** (non-specific), haemarthrosis (non-specific)
CT	**Revision – most sensitive + specific for bone loss / tunnel widening**



Classic ACL bone bruise: mid-LFC and posterior lateral tibial plateau

Bruise "not ACL" (TO)	**medial tibial plateau** bruise · **anterior proximal tibia oedema** (hyperextension) (✓ posterolateral tibia + LFC)
Chronic ACL on MRI	fibres scar to PCL → acute (flat) angle
Bone bruise significance	**delayed recovery of ROM** after ACLR · subchondral change persists years → long-term chondral damage · when present, pathognomonic ✗ Wrong options: re-rupture · laxity · infection
Double PCL sign	displaced **bucket-handle MM** tear (vs LM)



Segond fracture (arrows): ALL avulsion of the lateral tibia

Avulsion / pattern	Means
Segond (anterolateral proximal tibia)	**ACL** (ALL avulsion)
Arcuate sign (fibular head tip – LCL / biceps)	**PLC**
Intercondylar eminence fragment	ACL bony avulsion
Pellegrini–Stieda (medial femoral condyle ossification)	Chronic **MCL**
Medial patellar facet + LFC bruise	**Lateral patellar dislocation**
Widened lateral joint, squared condyle, cupped LTP	Discoid meniscus
Large lateral plateau fracture through weight-bearing surface	Plateau fracture (✗ Wrong options: Segond)

– – Active space – –

Diagnostic criteria (MRI signs, TO)

Sign	Threshold
ACL–Blumensaat angle	Normal **≤ 15°** · **> 15°** (apex anterior) → torn, collapsed ACL
Deep lateral femoral notch	**> 2 mm**
Anterior tibial translation	**≥ 5 mm** indicative · **> 7 mm** diagnostic
PCL index (B ÷ A: PCL chord ÷ max perpendicular to PCL)	**< 3** (buckled PCL) → ACL tear
PCL buckling after ACLR	**Graft laxity**

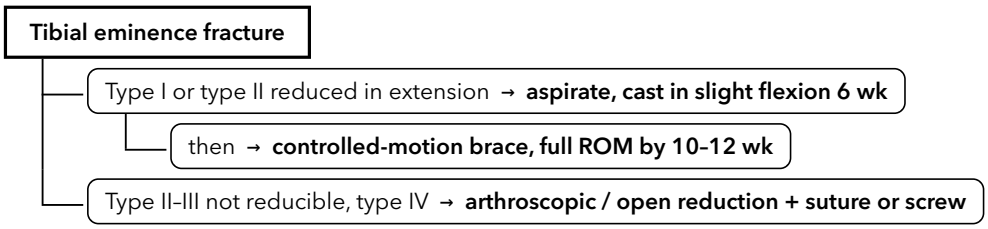
Tibial eminence avulsion – meniscal block (TO)

Meyers-McKeever	OB (3 types)	Modified (TO, 4 types)
I	Non- / minimally displaced anterior margin	Little or no displacement
II	Anterior aspect elevated, **posterior hinge intact**	Elevated anteriorly + proximally, **posterior cartilaginous hinge**
III	**Completely displaced**	Completely displaced, intact fragment
IV	–	**Comminuted**

Epidemiology (TO) **2–5%** of knee injuries with effusion, age **8–14** · unossified eminence (< 14 yr) fails through cancellous bone · ACL attaches to **medial spine** · same mechanism as ACL tear

Imaging X-ray → CT + MRI · subtle in adults → MRI (OB)

Block to closed reduction (TO) **anterior horn MM > intermeniscal ligament > anterior horn LM** · entrapment **26% type II (6/23)**, **65% type III (37/57)** (Kocher) (TO stem says "tibial tubercle" – means eminence)



Fixation (TO)	Pros	Cons
Screw	Less demanding, earlier mobilisation	Needs **large fragment**, irritation, not for comminuted, impingement, iatrogenic comminution, **physeal damage**, removal if malpositioned
Suture	**Minimal physeal damage**, comminuted OK (through ACL base)	Technically demanding, cinch loops constrict vascularity

Concomitant injury **meniscal** most common

Complications (TO) — **persistent ACL laxity ~40%** (types III–IV) · **malunion** (commoner; delayed diagnosis → extension block → debride fragment / notchplasty) · nonunion (more with nonop) · loss of extension · **arthrofibrosis 36% if ROM after 4 wk vs 0% early ROM** (Patel)

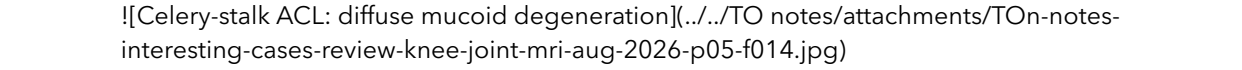
Posterior knee (ACL mucoid degeneration / ganglion) (TO)

Mucoid degeneration (TO) — **glycosaminoglycan** infiltration of ACL · **posterior knee pain** (impinges PCL + posterior capsule, bone erosion), pain in **flexion** (AM tightening) or terminal extension, heaviness on stairs

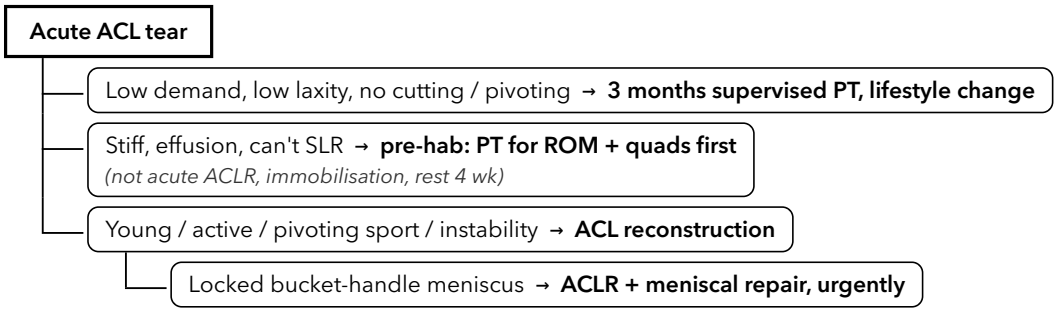
Treatment (TO) — **arthroscopic debulking / debridement**
✗ Wrong options: ACL reconstruction · CKC → OKC rehab · PCL reconstruction

Mucoid ACL type (TO, MRI)	Features
Cystic (ganglion)	Cysts in front of / behind ACL footprint; may track into **anterior root LM** (shared footprint)
Diffuse ("celery stalk")	Thick lax ACL, mucoid material between **intact** fibres → stuffs notch, **block to flexion** / locking
Mixed	Both

Mucoid ACL – who (TO) — usually **> 40 yr**; younger → dysplastic knee (trochlear dysplasia, high slope) · **US-guided fenestration** via medial gastrocnemius belly, 16–18 G needle, then steroid + LA



5 Management



– – Active space – –

Operative criteria

Decision	Indication / threshold
ACLR – patient	Young + active (↓ meniscal / chondral damage), **children** (activity limitation unrealistic), older active (**> 40 not a CI**), partial tear with instability, prior failure
ACLR – knee	**Full ROM restored first** (unless meniscal block)
Nonop	Low demand, ↓ laxity, recreational without cutting / pivoting · non-op ↑ meniscal / cartilage damage (buckling episodes, level I–II sport)
ACL repair	Femoral / tibial **avulsion** without mid-substance rupture, paediatric · (✗ Wrong options: mid-substance tear)
LET (OB)	Age **14–25** with **≥ 2**: return to contact pivoting sport, **pivot ≥ grade 2**, **Beighton > 4** · also Segond, chronic ACL, lateral femoral notch sign, lateral laxity, **recurvatum > 10°**
ALL reconstruction (TO)	**Grade 3 / explosive pivot**, pivoting athlete, **Segond**, **revision ACLR**, subjective rotational laxity, **chronic** ACL, high-level sport, **LFC depression**, **Telos > 10 mm** (✗ Wrong options: high-grade tear in young man alone)
Two-stage revision	Tunnel **> 15 mm**, malpositioned tunnels / hardware blocking, **infection**
Staged revision (TO)	Motion problem: **lacks 5° extension or 20° flexion** (stability fine)
Osteotomy with ACLR	**Varus** (> 3° between mechanical axes or WBA medial to medial tibial spine tip) · **slope > 12°**
HTO contraindications	Inflammatory arthritis, **BMI > 35**, **flexion contracture > 15°**, flexion < 90°, **> 20° correction**, **varus thrust**, PF arthritis, (lateral meniscectomy)
HTO indications	< 50 yr, active, non-obese, one compartment, good vascular status, compliant
Acute MCL repair	**Grade III distal avulsion** (pes entrapment) · grade III with valgus instability
Graft size	**≥ 8 mm** (< 8 mm → failure)

Timing – restoring motion before surgery; arthrofibrosis & cyclops lesion

Pre-hab — **regain full ROM + quad strength before ACLR**; wait for swelling / inflammatory phase to settle · acute ACLR in stiff, weak knee → **arthrofibrosis + weakness**

Timing (TO)	Risk
Early < 3 weeks	**Arthrofibrosis**
Late > 1 year	**Meniscal + chondral damage**

Stiff acute knee — e.g. ROM 5–70°, can't SLR → **PT for ROM** · Segond + ROM 10–75° + dial 5° → **ACL; PT to optimise ROM** (dial < 10–15° asymmetry = no PLC)

Adolescent with deep sulcus sign — **PT, then early reconstruction** (delay → instability, secondary meniscal damage, degeneration)
✗ Wrong options: immediate ACLR · ORIF · PT + RTS

Predictors of post-op motion loss — **female**, **acute surgery**, **limited preop ROM**, **classic bone bruise**

Prevention pre-hab, wait out inflammation, **correct tunnel position**, aggressive cryotherapy

Shelbourne arthrofibrosis (TO)	Extension loss	Flexion loss	Patella
1	< 10°	Normal	Normal
2	> 10°	Normal	Normal
3	> 10°	**> 25°**	↓ mobility (tight)
4	> 10°	**> 30°**	↓ mobility + **patella infera**

TO key > 10° extension + > 25° flexion loss, **patella normal → Type 2** (patellar mobility is the discriminator) – read the options · prevalence **4–38%** · Rx **arthroscopic debridement**

Loss of motion after ACLR (most common complication)

- < 12 weeks → **aggressive PT + serial splinting**
- > 12 weeks → **lysis of adhesions / MUA**

Cyclops lesion fibroproliferative nodule **blocks extension**, **"click" at terminal extension** · anterior tibial tunnel → cyclops (TO) · not graft-related (OB)

(TO)	Lesion	Key feature
Cyclops	**Hard** fibrovascular nodule at anterior tibial insertion	**Extension block**, **1–10%**
Cyclopoid	**Soft** nodule	**Asymptomatic**, no extension loss
Arthrofibrosis	Diffuse scarring	Loss of flexion + extension; after prolonged immobilisation

Shelbourne – TO revision notes III = flexion loss **< 25°** + ↓ patellar mobility · IV = flexion loss **> 25°** + **patella infera** · vs table above (III > 25°, IV > 30°) – read the options

Cyclops on MRI (TO) nodular heterogeneous tissue **anterior to an intact graft** · **3–9 months** post-op with loss of extension → **excision** · posterior cyclops possible after PCL revision (GRE no blooming → not PVNS) · ACL graft **ligamentises over ~1–1.5 yr** (some signal acceptable) · **LET → expected ITB defect** at harvest, not a tear

Arthroscopic arthrolysis sequence (TO) suprapatellar pouch → medial / lateral gutters → **anterior interval** (fat pad; restores extension) → retinacular release → notch clearance (cyclops) → **MUA only after arthrolysis** · **extension loss less tolerated → correct first**

Infrapatellar contracture syndrome uncommon · **↓ patellar mobility**, **patella infera (baja)** – poor prognostic sign · loss of extension + **active AND passive** flexion
✗ Wrong options: "active but not passive"

Reconstruction technique – tunnel targets, tensioning & fixation

Goal anatomic restoration of AP + rotational stability · arthroscopic · clear remnant to see bony landmarks (remnant kept vs removed – no difference)

Single vs double bundle no patient-reported difference · single most common · double may restore kinematics, ↓ laxity · no definitive clinical benefit

Double bundle (TO) separate AM + PL grafts → ↑ rotational stability, ↓ pivot shift · name: **Freddie Fu**

– – Active space – –

Karlsson principles of anatomical ACLR (TO) restore native insertion sites · recreate **two bundles** (AM + PL) · appropriate graft tension · correct graft diameter + type

Remnant preservation (TO) **SAMBBA** concept · keeps native stump → proprioception + biological healing · in revision preserve stumps → ↓ tunnel dilatation

Internal brace augmentation (TO) **FiberTape** protects graft in remodelling phase · ↓ dynamic elongation **50%**, ↑ peak load **64%**

PRP gel (TO) faster MRI graft incorporation (**~179 vs 369 days**) · **no proven clinical difference**

History of ACL surgery (TO)	Contribution
Hey Groves 1917	ACL repair + **ITB** reinforcement
Campbell 1936	Patellar tendon graft sutured to periosteum
Macey 1939	**Semitendinosus** graft, distal attachment preserved
Kenneth Jones 1963 → Clancy 1982	**BPTB** graft
MacIntosh, Lemaire, Lipscomb 1970s–80s	Extra- → intra-articular
2000s	Anatomical + double-bundle (trend: extra-articular → arthroscopic → anatomical)

Horizontal graft (TO) **prevents pivoting** · femoral tunnel **obliquity** sets rotational stability · isometric (vertical) graft → translation only · 10 o'clock also avoids PCL impingement in flexion
✗ Wrong options: vertical · roof-top · 11 o'clock

Femoral tunnel – coronal lateral wall **10 o'clock right / 2 o'clock left** (Orthobullets, OB) · **10:30 right / 1:30 left** (OB) · **9:30–10 o'clock right** (TO) – read the options

Femoral tunnel – sagittal **1–2 mm rim** to posterior cortex (Orthobullets) · **~2 mm**, knee at 90° (TO) · centre on **bifurcate ridge**, posterior to resident's ridge · OB: posterior to LIR, just proximal to BR, **~1 cm anterior to posterior articular margin of LFC** · drill in **> 70° (70–90°) flexion** → no posterior wall blowout

Pinczewski (radiographic) femoral tunnel **86% along Blumensaat's line** (0% anterior → 100% posterior) · tibial tunnel **48%** AP and **46%** medial → lateral across plateau · graft coronal inclination **19°**

Tibial tunnel – sagittal centre **10–11 mm anterior to PCL** (Orthobullets, OB) · **7 mm anterior to PCL = best guide** (TO) – read the options · 6 mm anterior to median eminence · 9 mm behind intermeniscal ligament · **15 mm behind anterior tibial edge** (TO) · **in line with posterior edge of anterior horn LM** (most used landmark) · **posterior + parallel to Blumensaat's line** in full extension · native insertion starts 10–14 mm behind anterior tibia (TO)

Tibial tunnel – coronal **< 75° from horizontal** (Orthobullets) · **60–75°**, well-placed ~75° (OB) · start halfway between tubercle and posteromedial tibial edge · lateral fluoroscopy: tibial centre at **40% of Amis–Jakob line**

Tensioning pre-conditioning ↓ stress relaxation **up to 50%** · 20 N vs 40 N → no difference · **fix in 20–30° flexion** · initial tension falls soon after fixation

Fixation **aperture** (interference screw) vs **suspensory** (cortical button, screw + washer post, staple) · alone or combined

Preferred site (TO) **femur → suspensory** (EndoButton) · **tibia → aperture** (interference screw; ↓ tunnel widening)

Tibial tunnel landmarks (TO) — **15 mm** behind anterior tibial edge · **7 mm** anterior to PCL · in line with **posterior horn LM** (TO – verify: usual landmark = posterior edge of **anterior** horn LM) · fix graft at **30°** flexion

No effect on outcome — graft type, one vs two incisions (OB) · tunnel position **does** (anterior to LIR = definitive adverse effect)

Femoral tunnel drilling – anteromedial portal vs transtibial

	Transtibial	Transportal (AM portal)
Femoral tunnel drilled	**Through the tibial tunnel**	**Through the AM portal** (AL portal = scope)
Clock position	≥ 11 o'clock / roof-top 12 o'clock	Anatomic **10 o'clock**
Graft	**Vertical**	**Oblique / horizontal – more anatomic**
Drawbacks	Poor access to lateral wall	**Shorter femoral tunnel**, **posterior wall breach**, risk to **posterior LFC cartilage + subchondral bone**; hyperflexion
Outcome	No clinical difference (Orthobullets) · not itself a re-rupture risk	

Too-anterior tibial tunnel + transtibial — **vertical (12:00) graft**
✗ Wrong options: oblique 10:30 / 1:30

Transtibial guide reaching 11:30 — **drill femur via AM portal (hyperflexed) or two-incision** · lower notch start point
✗ Wrong options: revise tibial tunnel (bone loss) · transtibial double bundle · hyperflex + same guide

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Graft choice – BTB vs hamstring vs allograft, biomechanics & graft size

	BPTB autograft	Quadrupled hamstring	Quadriceps tendon	Allograft
Load to failure	**2600 N** (Orthobullets) · 3000 N (TO)	**~4000 N – highest**; **~1.5× BPTB**	**2185 N**	–
Healing	**Bone-to-bone → fastest** incorporation	Slower (TO: delayed)	± bone block	**Longest**; more immune reaction
Fixation	**Rigid screws at joint line**	Suspensory → **windshield-wiper**, tunnel widening	As hamstring	–
Pros	**Gold standard**; low laxity, durable, ↑ RTS, low revision (TO)	Small incision, ↓ peri-op pain, **↓ anterior knee pain**, kneeling pain, extension loss	No kneeling-zone scar, spares physis, often free in revision	No donor morbidity; revisions
Cons	**Anterior knee pain 10–30%**, kneeling pain, extension loss, **patella fracture**, **patellar tendon rupture**	**Flexor torque deficit** (↓ flexion strength at 3 yr), numbness (saphenous), ↑ mid-term failure, female hamstring weakness	As hamstring (suspensory)	**↑ re-rupture in young**, cost, disease transmission
Re-rupture risk	**Age < 20, graft < 8 mm**	**Graft < 8 mm, age < 20**	–	**4.3× odds age 10–19**; revision > 2×

Native ACL tensile strength — **~2200 N** · 1725 N (Orthobullets graft section) – read the options · all grafts stronger than native ACL
✗ Wrong options: "approximate 4000 N" (the false statement, TO)

Hamstring lab facts — 2 equally tensioned ST strands = **220%** of one · single ST > single gracilis · **all strands equally tensioned** · quadrupled > 10 mm BPTB
✗ Wrong options: "lower than native ACL"

Not a BPTB advantage over hamstring (TO) — **"no loss of ROM"** (BPTB → extension loss) (✓ low revision, ✓ no flexor torque deficit, ✓ low laxity)

Most used allograft (TO) — **BPTB** · allograft integrates more slowly than autograft

Synthetic grafts (TO) — **LARS, Gore-Tex, Dacron → high failure + synovitis** · allograft (Achilles, fascia lata) → infection, delayed healing

Graft (TO table)	Ultimate load (N)	Stiffness (N/mm)
Quadrupled hamstring (STG)	**4090**	**776**
BPTB	2977	620
Quadriceps tendon	**2352** (TO) · 2185 (above) – read the options	463

Allograft vs autograft in young athletes — **↑ graft failure / re-rupture + revision**, mostly in first year · autograft better even in adolescents
✗ Wrong options: infection · growth disturbance · arthrofibrosis · OA · cyclops (no difference)

Hamstring failure risk — **graft < 8 mm**, **age < 20** · revision rates 5–20% (OB)
✗ Wrong options: BMI · transtibial drilling · staple fixation

Tunnel osteolysis more with **hamstring soft-tissue** grafts; no clinical effect at 2 yr

Hamstring in revision contralateral harvest if allograft undesirable

Allograft processing	Effect
Fresh-frozen	**Lower re-rupture** vs chemical / irradiated
Deep freezing	Kills cells, **strength unaffected**
4% chlorhexidine gluconate	Kills cells, **strength unaffected**
Supercritical CO_2	↓ structural + mechanical properties
> 3 Mrad (needed to kill HIV)	↓ structural + mechanical properties
2–2.8 Mrad	↓ stiffness 30%
1–1.2 Mrad	↓ stiffness 20%
Low-dose < 2 Mrad gamma	Failure = non-irradiated (OB)
HIV transmission	< 1 : 1.6 million; hepatitis higher

Graft-tunnel healing (TO)

Tendon–bone healing (TO) **fibrovascular interface at 12 weeks (3 months)** · continuous fibrous interface ~**6 months** · Sharpey-like fibres mature after **10 months**
✗ Wrong options: 3 wk · 6 wk · 6 months

Graft incorporation (TO)	Process	Time
Inflammatory	Partial graft **necrosis**	3–6 wk
Proliferative	**Neovascularisation**	6–8 wk
Matrix synthesis	Collagen formation	8–12 wk
Remodelling	Tendon–bone healing	6–9 months

Weakest graft (TO) **8–12 weeks** post-op · autografts heal faster

All-inside, BEAR & fixation devices (TO)

Common fixation combos (TO) femoral button + tibial interference screw · femoral + tibial interference screws · femoral cross-pin + tibial bio-screw + staple · **femoral button + tibial button = all-inside**

All-inside (TO) **single semitendinosus** (tripled / quadrupled), **gracilis spared** · **BPTB cannot be used** · **sockets** (not full tunnels) via **flip cutter** (retro-reamer) · graft ≤ **7.5 cm** (ideally < 6.5 cm) · **suspensory buttons both ends** (reverse tensioning loops) · **no interference screw**

All-inside advantages (TO) bone-preserving · **↓ pain** (little periosteal disruption) → faster rehab · smaller incision · spared gracilis shields graft · femoral socket drilled outside-in at **90°** (no hyperflexion, no muscle splitting)

All-inside "limitation" (TO) **false: "double bundle impossible"** – it is possible (4 buttons) · real limits: learning curve, larger AM portal / passport cannula · **ST allograft all-inside = most minimally invasive ACLR**

BEAR (TO) **bridge-enhanced ACL repair**: suture repair + **extracellular matrix scaffold** soaked in patient's blood · absorbed ~**8 weeks** → native tissue · BEAR trial: 2-yr results comparable
✗ Wrong options: syndesmosis · AVN · chondral defect

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ACL repair history — abandoned (high failure, poor cellularity) · renewed for **avulsions**, paediatrics

Tibial retro-screw (TO) — inserted **retrograde, proximal → distal** · **less tunnel widening** (head blocks tunnel mouth, stops synovial fluid)
✗ Wrong options: "more widening than antegrade"

Interference screw (TO)	Metal (titanium)	Bioabsorbable (PLA, PGA)
MRI	Artefact	MRI-friendly
Dissolves	No	3–5 yr
Strength	Stronger	Weaker
Breakage / failure	Less	**More**
Tunnel widening, effusion	Less	More
Revision surgery	Difficult	Easier
IKDC, Lysholm, Tegner, KT, ROM, pivot	Same	Same

Graft fixation – screw divergence

Graft–screw divergence — **> 30°** → loss of fixation, **early** failure (explanation range **15–30°**) · risk factor (e.g. 35°) · commoner on **femoral** side
✗ Wrong options: tibial · late failure · "eliminated by accessory AM portal"

Avoid — drill + screw through the **same portal** · hyperflex for screw if femur drilled transtibially · guide wire · suspect if screw hard to insert; check X-ray

Associated injuries – meniscal secondary restraint, MCL, LET for rotatory instability

Meniscus with ACL — **acute → lateral meniscus**, **chronic → medial meniscus**, **posterior horn MM** most common in chronic (TO) · posterior > anterior horn

PHMM — **secondary restraint to anterior tibial translation** in ACL-deficient knee (buttress wedging against posterior MFC)
✗ Wrong options: PCL · popliteus · anterior horn LM · quads

Partial PHMM meniscectomy — **↑ medial peak contact loads** · ↑ in situ ACL graft forces
✗ Wrong options: ↑ contact area · ↓ graft force

Meniscal tear timing — **repair / meniscectomy at time of ACLR** · ↑ healing when repaired with ACLR · LM repair doesn't raise ACL failure

Locked knee + ACL tear — lacks 40° extension, lateral McMurray → **ACLR + meniscal repair**
✗ Wrong options: ACL repair for mid-substance · MCL / PLC surgery when stable

ACL + MCL injury

- Low-grade / proximal (e.g. grade II) → brace 4–6 wk, MCL heals → **then ACLR**
- Grade III distal avulsion or high grade → **acute MCL repair with ACLR (within 7–10 days)**

Distal MCL — entrapped around pes ("**Stener lesion of the knee**") → won't heal · acute ACLR with untreated MCL → arthrofibrosis · unaddressed valgus instability → ↑ graft re-rupture

Not addressed acutely — LFC impaction fracture, minimally displaced fibular head fracture, grade II PCL, grade II proximal MCL

Chondral defects — treat at ACLR (or staged) · consistently ↓ long-term PROs

PCL / PLC reconstruct with ACLR or staged · missed → **varus instability, graft overload**

Malalignment correct coronal + sagittal before or with ACLR · **lateral closing wedge** corrects slope better than medial opening wedge

LET **↓ graft rupture + persistent rotatory laxity at 2 yr** (single-bundle hamstring + LET, young high-risk) · e.g. 16F, **Beighton 6, recurvatum 11°, pivot grade 2** → add LET
✗ Wrong options: "avoid vertical tunnel" (not an extra procedure) · suspensory / dual fixation · meniscal repair

ALL (TO) key stabiliser of **anterolateral tibial rotation** · affects pivot shift when ACL fails · aim: rotational control + ↓ retear · LET = ITB strip (attached at Gerdy's) under LCL → femur posterior + proximal to lateral epicondyle · Lumiere technique · no long-term data

Paediatric ACL – skeletal age & physeal-sparing techniques

Skeletal age **onset of menarche = best predictor** in girls · maturity ~**2 yr** after menarche (OB) · most growth in **18 months** after menarche (OB) · ~9 mm distal femur + 6 mm proximal tibia growth / yr · Tanner good; menarche delayed in elite athletes
✗ Wrong options: shoe size · doubled height at 2 · patellar ossification

Open physis < 14 yr · **60% of adolescent tears partial**

Nonop (peds) compliant, low demand, no other pathology, **partial tear with near-normal Lachman + pivot** · otherwise poor: instability, secondary meniscal + chondral damage, ↓ return to sport

Maturity	Technique
Pre-adolescent (M ≤ 12, F ≤ 11)	Combined intra- + extra-articular **ITB** (physeal-sparing)
Approaching maturity, open physes (12F premenarchal, Risser I)	**All-epiphyseal** or **partial transphyseal**, **soft-tissue (hamstring) autograft**, tunnel **< 8 mm**
Near / at maturity (M ≥ 16, F ≥ 14), Tanner IV, post-menarche	**Adult-type transphyseal** – BPTB, quad or hamstring autograft
Femoral-sided avulsion, near maturity	**Arthroscopic repair + transphyseal fixation** (femoral tunnels + cortical button)
Stiff knee (ROM 20–70°)	**PT for ROM first**, then decide

ITB technique ITB harvested free proximally, **left on Gerdy's tubercle → over-the-top of LFC** → through notch → **under anterior intermeniscal ligament** → sutured to LFC + proximal tibia (anteromedial tibial periosteum)
✗ Wrong options: under MM / LM / LCL / biceps

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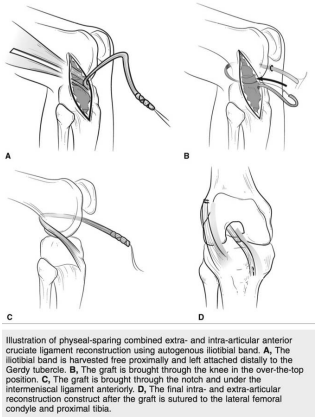


Illustration of physeal-sparing combined extra- and intra-articular anterior cruciate ligament reconstruction using autogenous iliotibial band. **A,** The iliotibial band is harvested free proximally and left attached distally to the Gerdy tubercle. **B,** The graft is brought through the knee in the over-the-top position. **C,** The graft is brought through the notch and under the intermeniscal ligament anteriorly. **D,** The final intra- and extra-articular reconstruction construct after the graft is sutured to the lateral femoral condyle and proximal tibia.

Physeal-sparing ITB reconstruction: over-the-top and under the intermeniscal ligament

Physeal injury — risk factors	**tunnel diameter** (most important) · **oblique / horizontal tunnel** · **interference screw** · **high-speed reaming** · LET · dissection near **perichondral ring of LaCroix** · sutures near tibial tubercle · BPTB > hamstring (bone block across physis)
Tunnel size	**> 12 mm most important**; 8 mm < 3% vs 12 mm > 7–9% of physis (Orthobullets) · **> 8 mm** ↑ injury; **7–9%** of physis enough for disturbance (OB) · **< 8 mm keeps < 5%** (OB) — read the options
Protective	vertical tunnel, slow reaming, small diameter, **soft-tissue grafts** (rarely disturb growth) · no difference in growth disturbance between techniques · physeal disruption without disturbance **10%**
BPTB in children	**physeal bridging → distal femoral valgus + proximal tibial recurvatum**; tibial tubercle apophysis injury · allograft → high failure (avoid in young)
Over-the-top femoral graft	transtibial over-the-top **partial transphyseal**: tibial tunnel transphyseal (anatomic), femoral graft passed over back of LFC → avoids femoral physis but **lax in flexion, tight in extension**

Rehabilitation – open-chain exercise, accelerated & eccentric protocols, return to sport

Immediate aggressive **cryotherapy** · **immediate WBAT** (↓ PF pain) · **early full passive extension** (esp. with MCL injury / patellar dislocation) · rehab starts immediately
✗ Wrong options: "start exercises ≥ 10 days after surgery" (TO false-statement stem, no key given)

Exercise	Early post-op	Why
Squat (low degree, wall, dumbbell)	✓ Closed chain	Foot fixed → compression + hamstring co-contraction ↓ anterior shear
Leg press	✓ Closed chain	As above
Stationary bike, stair climber	✓ Closed chain	No unopposed anterior translation
Quad sets, straight leg raise	Yes	Isometric about knee
Isometric hamstrings (any angle)	✓ **Lowest graft strain (60°)**	Pulls tibia posteriorly
Hamstring curls	Yes (avoid after **PCL** recon)	Open chain but no ACL load
Active motion **35–90°**	Yes	Orthobullets
Seated leg extension / resisted knee extension	**Avoid first 6 wk**	Open chain → quads pull tibia forward (mimics drawer / Lachman)
Open-chain extension **0–40°**	No	**↑ graft load 3–4×**, no hamstring co-contraction (OB)
Isokinetic quads **15–30°**	No	Orthobullets
Short-arc extension **0–30°**, unweighted	✓ acceptable (OB)	Conflicts with above – read the options

↑ strain isolated quads at 15° or 30°, co-contraction at 15°, active resisted extension 30° → 0°

Eccentric strengthening from week 3 **↑ quadriceps volume + strength at 1 yr**, no extra laxity
✗ Wrong options: concentric at wk 3 · eccentric from wk 12 · hinged brace 6 wk · CPM 2 wk

Accelerated vs nonaccelerated rehabilitation programs **equal outcomes** (laxity, ROM, strength, RTS) · accelerated = immediate WBAT with crutches · programme **19 wk** vs non-accelerated **32 wk** · ≠ earlier RTS · no ↑ re-injury

Blood flow restriction (BFR) tourniquet ↓ venous return → anaerobic milieu → **↑ quad strength + cross-sectional area** at low load · outcome benefit unproven
✗ Wrong options: Graston · TENS · ultrasound · acupuncture

Braces no added functional stability · no brace / duration better than none

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Return to sport no accepted criteria · historic **≥ 9 months** · pass hop / jump tests · **dynamic valgus → ↑ ipsilateral + contralateral rupture** · RTS before clearance → ↑ re-rupture · RTS at 11 months not a risk factor

Return-to-play criteria (TO)	Requirement
Time since surgery	**> 8–12 months** (TO) · historic ≥ 9 months (above) – read the options
Isokinetic quadriceps strength	**≥ 85%** of contralateral
One-leg hop test	**≥ 85%** of contralateral
Vertical jump / drop test	**No dynamic valgus**
Lachman + pivot shift	Negative / grade I
Effusion, pain	None

Revision ACLR rehab (TO) brace in **full extension 0–2 wk** · WBAT + isometrics 2–4 wk · full ROM (0–120°) by **8 wk** · weights + VMO 8–12 wk · jogging **3 months** · **RTS ~9 months** (later than primary)

RTS decision **psychological + lifestyle factors** dominate; **fear of re-injury** = commonest reason not to return · prior ACLR, poor hop symmetry → not returning
✗ Wrong options: surgeon advice · pain · sport duration · graft type

Revision & re-rupture risk factors (tibial slope, varus, PLC injury, tunnel widening, allograft)

Revision volume ~**13,000 / yr** in US · re-rupture ~**5%**

Re-rupture risk factors **young age**, **hyperextension (≥ 5°)**, high activity / contact sport, **slope > 12°**, **varus**, **missed PLC**, **allograft**, **graft < 8 mm**, female sex, **tunnel widening** (> 15 mm worst), open-chain / aggressive rehab

Hyperextension (MARS) 1145 revisions · passive HE **≥ 5° in 33%** · predictors: younger age, allograft, HE ≥ 5° (**> 2-fold** rupture) · e.g. recurvatum 5–10° → "preoperative exam findings"

Not risk factors concomitant MCL injury, Segond on X-ray, accelerated rehab, tibial tunnel 10 mm in front of PCL, interference screws, hamstring graft per se, 9 mm graft, slope 7–8°, new anatomic femoral tunnel, LET, two-stage revision, prior PLC reconstruction, LM repair, mild PF arthrosis

Female re-rupture female sex = biggest listed risk (OB) · re-rupture risk > never-injured

Missed PLC **varus laxity at 30°** (0° + 30° with ACL tear), **dial ↑ ER > 10° at 30° not 90°**, varus stress X-ray +5 mm → **reconstruct PLC with ACL** (or as stage 1) · PLC = **LCL, popliteus, PFL** · LCL resists varus 55% at 5°, **69% at 25°**

Intra-articular femoral bone plug 3 wk post-op, + Lachman → **revision ACLR**
✗ Wrong options: excise plug · modified rehab

Revision technique find cause of failure · **high-strength autograft** (quad, hamstring) · allograft **> 2×** re-rupture · **re-harvest BPTB contraindicated** · **dual fixation** (suspension + screw) · bone graft dilated tunnels (**15 mm**) · LET (STABILITY trial: ↓ rupture + rotatory laxity) · conservative rehab

Revision technique (TO) — **most common cause = tunnel malposition** (✗ Wrong options: graft selection) · tunnel **> 15 mm → stage 1 bone graft** (✗ Wrong options: 10 mm) · **remove hardware only if necessary** (✗ Wrong options: "femoral screw must come out") · bioabsorbable PLA screw still present ≥ 2 yr → leave or ream through · prosthetic ligament out en bloc · reuse incisions, **skin bridge ≥ 7 cm** · generous revision notchplasty

Failure category (TO)	Picture	Causes
Recurrent instability	Laxity, giving way, pivot shift	Malplaced tunnels (**vertical femoral**), missed **ALL / PLC**, re-rupture
Loss of motion	Nutcracker / captured knee	Anterior tunnels, cyclops, arthrofibrosis
Extensor dysfunction	Extensor lag	BPTB harvest, reflex quads inhibition, VMO weakness
Degenerative (OA)	Early arthritis	Abnormal kinematics – **Rosenberg view** (45° flexion WB PA) **most sensitive** for early change

Tunnel malposition (TO) — **> 50%** of failures · **ALL = most commonly missed** associated injury, then PLC · re-rupture with **slope > 12°** → consider HTO

Single- vs two-stage (TO) — tunnels **< 8 mm**, correct position → single-stage (reuse) · **> 8 mm**, malpositioned or merged → two-stage, graft then revise at **3–6 months** · threshold **> 15 mm** (above) vs **> 8 mm** (TO) – read the options

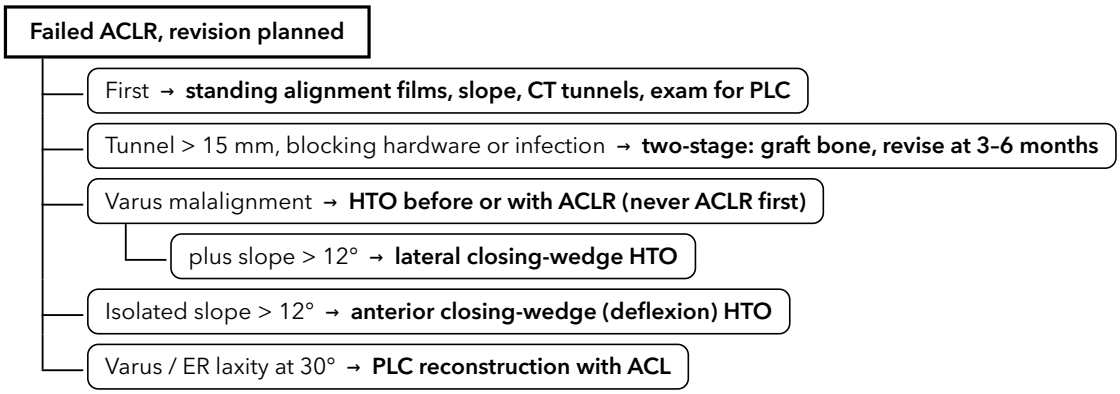
Revision tunnel planning (TO) — new tibial tunnel **more lateral + posterior** · **outside-in drilling** (avoids merging) · femoral tunnel on lateral X-ray: **posterior ¼ of condyle** · tibial tunnel posterior to Blumensaat line, exits between spines

Revision graft (TO)	Preference
Contralateral hamstrings	**1st choice**
Ipsilateral quadriceps tendon (bone plug)	**2nd** · multiligament / PLC cases
Peroneus longus	Alternative · strong but bulky (notch impingement)
BPTB	Rare now (if prior hamstring)
Allograft / synthetic	Selected cases

Revision fixation (TO) — **hybrid (cortical + aperture) preferred** · cortical (EndoButton / TightRope) femur → high pull-out, **low BMD** · aperture tibia near joint line → ↓ synovial seepage, ↓ widening

Tunnel widening (TO) — synovial seepage + graft micromotion · **bungee cord effect = longitudinal** motion · **windshield wiper effect = transverse** motion · prevent: screw near joint line, preserve tibial stump, snug tunnel

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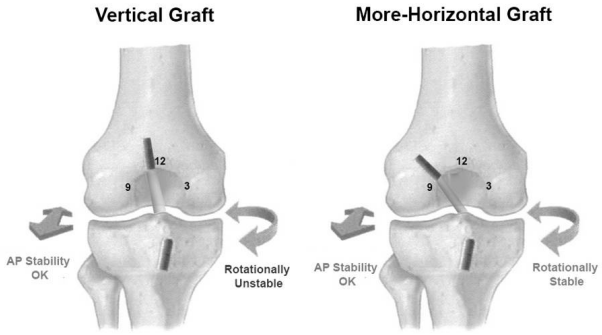
Case – 18 mm tibial tunnel, infantry	**standing alignment films first**, then two-stage revision
Case – varus 10°, slope 15°, mLDFA 87°	**single-stage lateral closing-wedge HTO + revision ACLR** (hamstring) ✗ Wrong options: DFO (femur normal) · isolated revision · revision before HTO · bracing
Case – varus 11°, slope 7°, mild medial OA, 35 yr	**medial opening-wedge HTO + ACLR** ✗ Wrong options: ACLR alone · biplanar (slope normal) · lateral DFO (for valgus) · UKA
Case – ROM 20–130°, varus	**flexion contracture > 15° = CI to HTO** ✗ Wrong options: age 32 · BMI 30 · partial medial meniscectomy · varus itself

6 Complications

Complication	Key fact
Arthrofibrosis / loss of motion	**Most common**; risk = **lack of preop motion**; ↓ patellar translation
Graft failure	~**5%** (Orthobullets, OB) · **5–15%** (OB)
Infection	**< 1%**
Patellar tendon rupture	**Patella alta**, lost extensor mechanism → **repair / reconstruct**
Patella fracture	BPTB / quad with bone block · **8–12 wk** post-op (rehab)
Infrapatellar contracture	↓ patellar mobility, patella infera
CRPS	–
Tunnel osteolysis	**Observe** unless laxity / instability
Late OA	Linked to **meniscal integrity**; ↑ if **> 50 yr** at ACLR
Saphenous nerve	Hamstring harvest (IBSN)
Cyclops lesion	Blocks extension, **click at terminal extension**

Tunnel malposition – effects & most common cause of failure

Malposition	Graft behaviour	Clinical picture
Femoral too anterior (anterior to resident's ridge)	**Tight in flexion, lax in extension** (cam effect → stretch)	**Loss of flexion** (**"captured knee"**), graft elongation · e.g. **ROM 0–75°, negative pivot**
Femoral too posterior / **over-the-top**	**Lax in flexion, tight in extension**	Limited extension, ↑ translation in flexion · TO: "positive error", only blowout risk
Femoral vertical (12 or 11 o'clock)	Controls AP translation only	**+ Pivot, – Lachman / drawer**, cutting instability, **PL bundle not restored**, ↓ Lysholm · TO: PCL impingement, extension block
Femoral anterior + vertical (TO)	–	**+ Pivot + ROM 0–70°**
Tibial too anterior	**Tight in flexion + roof impingement in extension**	**Loss of extension** (**"nutcracker knee"**), cyclops (✗ Wrong options: flexion deficit (TO))
Tibial too posterior	**Impinges PCL**	Loss of flexion, graft tear (TO) · gross + Lachman / pivot (OB)
Tibial too medial / lateral	Impinges MFC / LFC	–
Lax in flexion AND extension	**Under-tensioned graft** (not tunnel)	–
No knee cycling before tibial fixation	Residual laxity in flexion + extension	–



Vertical vs horizontal graft: AP stable but rotationally unstable

Most common cause of failure — **tunnel malposition** · surgical error in **65–75%**; tunnel malposition = **70%** of errors · **femoral > tibial** errors
✗ Wrong options: graft choice · fixation · obesity · smoking · early RTS · PT

Failure causes – conflict — tunnel malposition most common (Orthobullets, OB) · **trauma most common, technical error 2nd** (OB) – read the options

Most common technical error — **anterior femoral tunnel** (OB, poor visualisation, "resident's ridge" not cleared) · **MARS** (460 revisions, 87 surgeons): technical error **60%**; femoral **too vertical 35.9%**, too anterior 20.0%, both 26.5%; **82.1% re-drilled**

Early vs late failure — < 6 months → technical error, aggressive rehab, early RTS, failed incorporation · > 1 yr → **new injury**

– – Active space – –

Vertical graft	classic **transtibial** era · **1A Lachman, 2+ pivot, negative dial** → femoral tunnel missed PL bundle ✗ Wrong options: missed PLC (dial negative) · AM bundle · ALB / PMB = PCL bundles
Anterior femoral tunnel EXCEPT	**does not cause interference screw divergence** (✓ flexion loss, ✓ stretch-out, ✓ commonest error, ✓ poor visualisation)
Flexion loss → failure on X-ray	**anterior femoral tunnel** (forced flexion pulled tibial plug)

Intra-operative complications — graft-tunnel mismatch & posterior wall blowout

Graft–tunnel mismatch	BPTB longer than tunnels + intra-articular length → prominent tibial plug, poor fixation · risk: **BPTB allograft, patella alta, non-transtibial drilling** · Rx: measure precisely, **twist graft** to shorten
Posterior wall blowout	breach of posterolateral LFC cortex · risk: poor exposure, unchecked walls, **femoral drilling at < 70–90° flexion**

Posterior wall blowout

- Small defect at notch (3–5 mm) → **redrill deviating anteriorly, same fixation**
- Substantial defect → **keep tunnel, suspensory fixation ± interference screw**

Other graft failure causes	fixation / hardware failure (divergence > 30°) · **attritional (graft < 8 mm)** · femoral plug dislodgement (revision) · missed ligament injury / malalignment · open-chain rehab

Graft harvest complications – saphenous nerve & patellar tendon rupture

IBSN injury	**anteromedial pain + numbness over anterolateral infrapatellar knee / proximal leg → infrapatellar branch of saphenous nerve** ✗ Wrong options: suprapatellar branch · superficial peroneal · LFCN · CPN
Saphenous anatomy	femoral nerve branch → leaves adductor canal → pierces fascia between **gracilis + sartorius** → infrapatellar branch crosses below patella · at joint line **deep to sartorial fascia, superficial to gracilis**
Highest risk step	**hamstring harvest with knee extended** (tendon harvester) · also tibial tunnel incision, PT harvest · **hip ER + knee flexion** relaxes nerve · **oblique / horizontal incision** ↓ risk
Patellar tendon rupture after BPTB	early post-op, lost extension, **patella alta → patellar tendon repair / reconstruction** ✗ Wrong options: continue rehab · quad tendon repair · cast · revision ACLR

Postoperative septic arthritis

Incidence	**< 1%** · most superficial · **CoNS (S. epidermidis) commonest**, S. aureus 2nd · presents **2–14 days** post-op (Orthobullets) · first 10–14 days (OB) · TO case at 4 wk
Risk	graft contamination, **dropped graft** · vancomycin soak may ↓ risk · sequential antibiotic washes → no ↑ infection in dropped grafts
Diagnosis	↑ ESR / CRP → **aspirate (Gram stain + culture)**

Septic arthritis after ACLR

- Gächter 1–2 → **arthroscopic I&D, retain graft, ≥ 6 wk IV antibiotics (culture-based)**
- Gächter 3–4 → **medial / lateral arthrotomy**
- Persistent infection → **graft removal (no tunnel bone graft while infected)**

Gächter (TO)	Findings
1	Blurred effusion, hyperaemic synovium
2	**Purulent** effusion, fibrin, hypertrophic synovium
3	Synovial adhesions, necrotic synovium + cartilage
4	Diffuse cartilage necrosis, **bone erosion, osteolysis**

Graft retention first-line; often repeat I&Ds · works better with **S. epidermidis** than S. aureus
✗ Wrong options: single-stage revision · graft removal + staged revision first · open arthrotomy first

Outcomes – pivot shift & satisfaction; post-traumatic OA

Satisfaction **pivot shift** correlates best with functional instability / satisfaction
✗ Wrong options: KT-1000 · Lachman · drawer · Cybex

Natural history ACL-deficient knee → accelerated OA · after ACLR near-normal kinematics, high RTS

OA risk factors **meniscal tear** (painful pop on McMurray in non-op knee), osteochondral lesions, malalignment, **ACL + PCL** (✗ Wrong options: collateral injury · IIB Lachman · pivot shift · Ober) · **age > 50** + **female** after ACLR (✗ Wrong options: hardware failure · ligamentisation · instability · wound)

Gillquist meniscal tear / repair or isolated ACL tear → **10-fold** OA risk (**15–20%** vs 1–2%)

Return to play driven by **psychological** factors

Arthroscopy complications – hemarthrosis (TO)

Commonest arthroscopy complication (TO) **haemarthrosis** > infection > thromboembolism > anaesthetic > instrument failure > RSD > ligament injury > fracture / nerve injury
✗ Wrong options: cartilage injury · stiffness

X vs Y

	Acute ACL tear	Chronic ACL deficiency
Meniscus 241	**Lateral** (54%)	**Medial**, **posterior horn MM**
Other	Bone bruise, Segond	Chondral damage, unrepairable / bucket-handle MM

– – Active space – –

	Lachman	Pivot shift
Bundle	AM (translation)	PL (rotation)
Strength	**Most sensitive** (87%)	**Most specific** (97%)
After vertical graft	Negative	**Positive**

	Medial opening-wedge HTO	Lateral closing-wedge HTO
Cuts	One	Two
Slope	Little change	**Better for ↓ slope**
Pros	Simple	Immediate cortical contact → early WB
Cons	Relies on bone-graft healing	Near **peroneal nerve + PTFJ**; harder later TKA
Use	Varus, normal slope	Varus + ↑ slope

	Segond	Arcuate sign
Fragment	Anterolateral proximal tibia	Fibular head tip
Structure	ALL	LCL / biceps
Injury	**ACL**	**PLC**

	Closed chain	Open chain
Distal segment	Fixed (foot planted)	Free
Examples	Squat, leg press, bike, stair climber	Seated leg extension
Graft	Co-contraction, low shear	Unopposed anterior translation

How it's asked

"Graft tight in flexion, lax in extension?"	**→ femoral tunnel too anterior**
"Lacks 12° extension, stable Lachman, 3 months?"	**→ tibial tunnel too anterior**
"Over-the-top graft?"	**→ lax in flexion, tight in extension**
"Most common cause of failed primary ACLR?"	**→ tunnel malposition**
"Anterior femoral tunnel — all true EXCEPT?"	**→ causes interference screw divergence**
"Definitive adverse effect on outcome?"	**→ femoral tunnel anterior to lateral intercondylar ridge**
"12 o'clock femoral tunnel — exam?"	**→ positive pivot shift, negative Lachman**
"12 o'clock graft — what failed?"	**→ posterolateral bundle not reconstructed**
"Vertical graft associations?"	**→ positive pivot shift + lower Lysholm**
"Tunnel position — EXCEPT?"	**→ anterior tibial tunnel + transtibial gives oblique graft**
"Femur 9 mm anterior to ridge at 10 o'clock?"	**→ negative pivot, ROM 0–75°**

"Femur 10 mm anterior to ridge at 11 o'clock (TO)?"	→ positive pivot, ROM 0–70°
"To prevent pivoting, graft should be (TO)?"	→ horizontal
"NOT true of tunnel malposition (TO)?"	→ anterior tibial tunnel — flexion deficit
"Inability to fully extend matched to?"	→ anterior tibial tunnel
"MRI sign of leading failure cause?"	→ femoral tunnel centred on lateral intercondylar ridge
"Lost flexion then graft pulled out?"	→ anterior femoral tunnel
"ACL blood supply NOT from (TO)?"	→ superior genicular artery
"Main ACL supply — EXCEPT inferior medial genicular (TO)?"	→ middle genicular artery
"Middle genicular artery — true?"	→ from popliteal, pierces oblique popliteal ligament
"PL bundle bulkier + longer — true (TO)?"	→ false — AM longer + bulkier
"PL tight in flexion — true (TO)?"	→ false — PL tight in extension
"ACL named for femoral, PCL for tibial footprint (TO)?"	→ incorrect — reversed
"Fibres tight in extension?"	→ posterolateral
"Double bundle roles?"	→ AM limits translation, PL controls rotation
"Femoral ridges A and B?"	→ lateral intercondylar (A), bifurcate (B)
"PL bundle femoral insertion?"	→ posterior to LIR, distal to bifurcate ridge
"Ideal femoral tunnel?"	→ 6.1 mm posterior to resident's ridge, 1.7 mm proximal to bifurcate
"Tibial insertion landmark?"	→ 11 mm anterior to anterior border of PCL
"Segond — ALL insertion?"	→ halfway between Gerdy's tubercle and fibular head
"LET recreates which insertion?"	→ ALL: anterolateral tibia, between Gerdy's and fibular head
"Predominantly type I collagen?"	→ ACL
"COL5A1 genotype protects against?"	→ ACL rupture in women
"Transportal ACLR (TO)?"	→ femoral tunnel drilled through anteromedial portal
"Benefit of AM portal drilling?"	→ more anatomic graft placement
"Transtibial guide at 11:30?"	→ drill femur via AM portal or two-incision
"Native ACL ~4000 N — false (TO)?"	→ native ACL ~2200 N

– – Active space – –

"Highest load to failure?"	→ quadrupled semitendinosus + gracilis
"Hamstring vs BPTB load to failure?"	→ ~1.5×
"Hamstring vs patellar tendon graft?"	→ less anterior knee pain
"Not a BPTB advantage (TO)?"	→ no loss of ROM
"Allograft in young athlete?"	→ higher re-rupture / revision
"Hamstring graft failure risk?"	→ graft < 8 mm
"Adolescent, menarche at 12 — true?"	→ autograft beats allograft
"Graft–screw divergence?"	→ > 30° → loss of fixation
"Anterolateral infrapatellar numbness?"	→ infrapatellar branch of saphenous nerve
"Saphenous nerve most at risk?"	→ hamstring harvest with knee extended
"Lost extensor mechanism + patella alta after BPTB?"	→ patellar tendon repair / reconstruction
"Imaging NOT suggestive of ACL tear (TO)?"	→ medial tibial plateau bruise
"Doesn't indicate ruptured ACL (TO)?"	→ anterior proximal tibial oedema
"Delhi active test (TO)?"	→ standing manoeuvre for complete ACL tear
"Acute haemarthrosis, non-contact — chance of ACL?"	→ > 60%
"Classic ACL bone bruise?"	→ mid LFC + posterior lateral tibial plateau
"Segond on X-ray — exam?"	→ positive Lachman
"Segond represents?"	→ anterolateral ligament avulsion
"Pivot shift clunk mechanism?"	→ LTP subluxed in extension, reduces with flexion
"ITB role in pivot shift?"	→ reduces LTP in flexion
"Pivot shift — true?"	→ ITB reduces the plateau ✗ Wrong options: most sensitive
"2B Lachman?"	→ 9 mm, soft endpoint, 30° flexion
"Anterior shift + tibial IR at low flexion?"	→ ACL
"Pivot shift positive — extra finding?"	→ lateral joint line tenderness
"ACL tear — most likely associated structure?"	→ lateral meniscus (54%)
"Double PCL sign + empty lateral wall?"	→ ACL + bucket-handle MM tear
"Tibial ACL avulsion (Meyers–McKeever)?"	→ ACL avulsion from tibia
"Not a female risk factor (TO)?"	→ increased external rotation at landing
"NOT a risk factor (TO)?"	→ hamstring dominance

"Biggest female contributor?"	→ neuromuscular factors
"Female landing pattern?"	→ ↓ hip + knee flexion, ↑ knee ER, dynamic valgus
"Female muscle weakness?"	→ hamstrings
"Highest-risk cohort?"	→ women with knee valgus moment on landing
"Lowers future ACL risk?"	→ neuromuscular training programme
"Associated with ACL tear?"	→ previous concussion
"Revision ACL – true (TO)?"	→ tunnel malposition is the commonest cause
"Missed injury raising failure (exam figure)?"	→ dial test (PLC)
"Revision will fail – dial + at 30° only?"	→ concomitant PLC injury
"Re-rupture risk in primary ACLR?"	→ varus laxity (missed PLC)
"Revision – highest re-rupture risk?"	→ posterior tibial slope 13°
"Revision graft raising risk?"	→ semitendinosus allograft
"Varus 10°, BTB planned – re-tear risk?"	→ varus deformity
"Varus 11°, slope 7°?"	→ medial opening-wedge HTO + ACLR
"Varus + slope 15° re-rupture?"	→ lateral closing-wedge HTO + revision ACLR
"CI to HTO with ACLR?"	→ flexion contracture > 15°
"18 mm tibial tunnel – next step?"	→ standing alignment radiographs
"Femoral plug intra-articular at 3 wk?"	→ revision ACLR
"Biggest re-rupture risk, 23F BTB 9 mm?"	→ female gender
"Recurvatum 5–10°, re-tear?"	→ preoperative exam (hyperextension)
"Avoid early after ACLR (TO)?"	→ seated leg raising (open chain)
"Avoid early – options?"	→ seated / resisted knee extensions
"Lowest graft strain?"	→ isometric hamstrings at 60°
"ACL reflex (TO)?"	→ inhibition → quadriceps weakness
"Accelerated rehab?"	→ equal outcomes vs nonaccelerated programs
"Eccentric at week 3?"	→ ↑ quad volume + strength, no laxity
"Modality ↑ strength + hypertrophy?"	→ blood flow restriction
"Why not returning despite good function?"	→ lifestyle + psychological factors
"Arthrofibrosis type, patella normal (TO)?"	→ Type 2
"Acute ACL, ROM 5–70°, no SLR?"	→ physical therapy for ROM

– – Active space – –

"Segond, ROM 10–75°, dial 5°?"	→ ACL – PT to optimise ROM
"Deep sulcus sign, adolescent?"	→ PT then early reconstruction
"Bone bruise predicts?"	→ delayed recovery of ROM
"Infrapatellar contracture – EXCEPT?"	→ loss of active but not passive flexion
"Infection 4 wk post-ACLR (TO)?"	→ arthroscopic debridement, graft retention, IV antibiotics
"WBC 52,000 at 3 wk?"	→ arthroscopic I&D, retain graft, antibiotics
"Best predictor of growth remaining?"	→ onset of menarche
"Greatest tibial physeal injury?"	→ horizontal / oblique tunnel
"12 yo, ROM 20–70°?"	→ physical therapy first
"12 yo premenarchal, Risser I?"	→ all-epiphyseal with hamstring autograft
"14F, Tanner IV, femoral avulsion?"	→ arthroscopic repair, transphyseal fixation
"ITB passed under (figure)?"	→ anterior intermeniscal ligament
"Key step of ITB technique?"	→ over-the-top LFC, under anterior intermeniscal ligament
"Not an ALL indication (TO)?"	→ high-grade tear in young active man
"Chronic ACL – meniscal tear (TO)?"	→ posterior horn MM
"Most injured knee ligament (TO)?"	→ MCL
"Secondary restraint in ACL-deficient knee?"	→ posterior horn MM
"Partial PHMM meniscectomy effect?"	→ ↑ medial peak contact loads
"Non-op – predicts arthritis?"	→ painful pop on McMurray (meniscal tear)
"Lacks 40° extension, lateral McMurray?"	→ ACLR + meniscal repair
"Address during acute ACLR?"	→ grade III distal MCL tear
"Beighton 6, recurvatum, pivot 2?"	→ add LET
"ACL-deficient gait?"	→ no femoral IR in terminal swing
"ACL-deficient kinematics?"	→ ↑ paradoxical anterior femoral translation in flexion
"Best correlate of satisfaction?"	→ pivot shift
"Age 57 – increased risk of?"	→ post-traumatic OA
"Posterior knee pain, MRI ACL thickened (TO)?"	→ arthroscopic debulking (mucoid degeneration)
"Flexed knee, plantarflexed foot (TO)?"	→ PCL

"Tibial retro-screw EXCEPT (TO)?"	→ more tunnel widening than antegrade
"Metal vs bio-screw — true (TO)?"	→ bio-screws break / fail more
"Button both ends on X-ray (TO)?"	→ all-inside ACLR
"All-inside — not an advantage (TO)?"	→ "double bundle impossible"
"BEAR implant (TO)?"	→ ACL repair
"Eminence fracture — block to reduction (TO)?"	→ anterior horn MM
"Fibrovascular graft–bone interface (TO)?"	→ 12 weeks
"Commonest arthroscopy complication (TO)?"	→ haemarthrosis
"Weakest period of an ACL graft (TO)?"	→ 8–12 weeks post-op
"Preferred fixation femur / tibia (TO)?"	→ suspensory femur, aperture tibia
"Name behind double-bundle ACLR (TO)?"	→ Freddie Fu
"Cyclops vs cyclopoid (TO)?"	→ cyclops is hard and symptomatic
"Revision graft of first choice (TO)?"	→ contralateral hamstrings
"Most sensitive view for early OA after failed ACLR (TO)?"	→ Rosenberg view

– – Active space – –

ADDUCTOR STRAIN

– Knee & Sports · 2 / 61 –

1 Basics

Incidence	**10–30% of soccer and hockey players**
Adductor complex	**adductor longus (most commonly injured)**, adductor magnus, adductor brevis, gracilis, obturator externus, pectineus · all **obturator nerve** (Orthobullets)

2 Mechanism

"Pulled groin"	**forceful hip extension + external rotation of an abducted leg → strong eccentric contraction** of adductors
Avulsion	muscle **fails in tension as the leg is abducted**

Repetitive hip extension + abduction (hockey, soccer) → **athletic pubalgia** – adductor longus torn and retracted from the **common aponeurosis** with rectus abdominis (detail in Athletic Pubalgia & Adductor strain chapter)

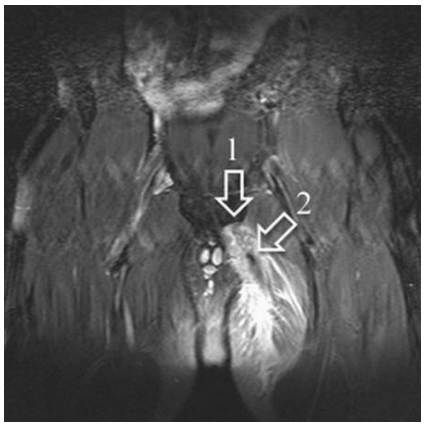
3 Presentation

Pain	**immediate, severe groin pain**
Tenderness	**subcutaneous border of the (inferior) pubic ramus** · medial pubic ramus
Strength	pain ± **weakness on resisted adduction** vs other leg

4 Diagnosis & Classification

Key radiology findings

Study	Finding
X-ray: **AP pelvis + lateral hip**	**Fleck of bone** if avulsion
MRI	**Avulsion from pubic ramus**, muscle oedema + haemorrhage · T2 ↑ fluid signal at tendon origin



T2 MRI: adductor origin on the pubic ramus (1) and avulsed adductor tendon (2)

Adductor strain classification

Degree	Findings
1st	Pain, **minimal loss** of strength + motion
2nd	**Compromised strength**
3rd	**Complete disruption**, loss of muscle function

5 Management

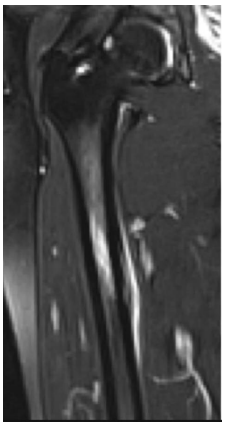
Groin injuries – adductor strain / avulsion management

Mainstay (incl. avulsion)	**rest, ice, protected WB as needed** → gentle stretching → resistance → **core strengthening** → gradual return to sport
Avoid immobilisation	promotes **muscle tightness + scarring** / contracture
Rugby player, MRI adductor avulsion	**immediate rehabilitation with increasing passive + active motion** ✗ Wrong options: strict immobilisation in flexion + adduction · tendon repair · tenodesis · repair + allograft
Open repair	**no data / no high-level evidence** it beats nonoperative care
Hockey groin syndrome	**external oblique aponeurosis tear + ilioinguinal nerve entrapment** → ilioinguinal nerve ablation + external oblique reinforcement

Adductor insertion avulsion syndrome ("thigh splints")

What	painful **proximal-to-mid femur at adductor insertions** · insidious thigh pain + limp, no trauma
Exam / labs	pain on **resisted adduction** only · afebrile, **normal WBC, ESR, CRP**
Imaging	**X-rays normal** · MRI (TSE inversion recovery) → **linear high signal along medial (posteromedial) periosteum** of proximal–mid femoral shaft, **4–9 cm** long

– – Active space – –



Inversion-recovery MRI: linear high signal along the medial proximal femur

Treatment	**activity modification + NSAIDs**, observe · resolves in **1–2 months** ✗ Wrong options: antibiotics · I&D · prophylactic fixation (not a stress fracture) · biopsy
Differential	osteomyelitis, neoplasm · biopsy only if pain **persists or increases**

X vs Y

	Adductor strain	**Thigh splints**
Onset	**Acute**, severe	**Insidious**
Site	**Pubic ramus** (origin)	**Femoral shaft** (insertion)
MRI	Avulsion / oedema at pubis	**Periosteal high signal**, posteromedial femur
Treatment	Rest, ice, protected WB, **early motion**	Activity modification, NSAIDs

How it's asked

"Adductor avulsion on MRI – treatment?"	**→ immediate rehabilitation with increasing passive and active motion**
"Insidious thigh pain, normal labs, periosteal signal medial femur – next step?"	**→ activity modification and NSAIDs**
"Repetitive motion causing adductor longus tear from common aponeurosis (sports hernia)?"	**→ hip extension and abduction**

– – Active space – –

ANTERIOR INFERIOR ILIAC SPINE AVULSION (AIIS)

– Knee & Sports · 3 / 61 –

1 Basics

Who adolescents **14–17 yr**, **males > females** · **kicking sports** (soccer)

Why adolescents **apophysis weaker than the musculotendinous unit** during forceful contraction

Anatomy

AIIS bony prominence **just above the acetabulum** → origin of the **direct head of rectus femoris** (**femoral n.**)

Reflected head **supra-acetabular groove + anterior hip capsule**

Insertion rectus femoris → **quadriceps tendon → superior pole of patella**

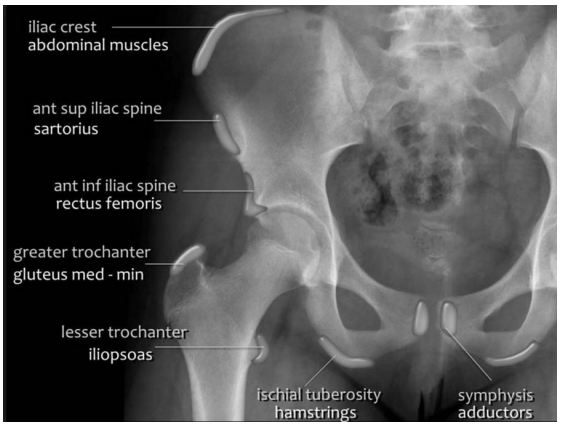
Pelvic apophyseal avulsions – sites and muscles

Site	Muscle	Frequency (Rossi, 203)
Ischial tuberosity	Hamstrings (BF long head, SM, ST) + adductor magnus	**~50% – most common**
AIIS	**Rectus femoris (direct head)**	**25%**
ASIS	**Sartorius** + TFL	20%
Superior corner of pubic symphysis	Adductors	4%
Iliac crest	Abdominals	**1% – least common**
Lesser trochanter	**Iliopsoas**	–
Greater trochanter	Gluteus medius + minimus	–

Most common site **ischial tuberosity ~50%** (Rossi, OB) · **AIIS most common** (Schuett, 228 avulsions, OB – see ASIS chapter) – read the options

Sports **soccer + gymnastics** most common

– – Active space – –



Common pelvic apophyseal avulsion sites and their muscles

2 Mechanism

Rectus femoris origin avulsion — mechanism, innervation & nonoperative care

Mechanism	**eccentric contraction of rectus femoris as hip extends + knee flexes** → avulses origin · **kicking** (missed ball, foot strikes ground) · hurdler's **trailing limb** striking hurdle
MRI label	rectus femoris = anterior central thigh muscle ✗ Wrong options: vastus lateralis · vastus intermedius · vastus medialis · sartorius
Nerve	**femoral** ✗ Wrong options: genitofemoral (cremaster) · pudendal (levator ani, bulbo- / ischiocavernosus) · obturator (obturator externus, adductor longus / brevis, gracilis) · superior gluteal (gluteus medius / minimus, TFL)
Distractor insertions	sartorius → **pes anserinus** · TFL → **Gerdy's tubercle** via ITB · iliopsoas → lesser trochanter · gluteus medius / minimus → greater trochanter

3 Presentation

History	**sudden "pop" in the pelvis**
Exam	antalgic gait · **anterior hip pain + hip flexion weakness** · pain on **resisted hip flexion**, local tenderness
Differential	SCFE, Perthes, congenital hip subluxation, malignancy, infection

4 Diagnosis & Classification

Key radiology findings

Radiographs	AP pelvis / **AP + frog lateral hip** show the AIIS avulsion · **X-rays sufficient** to diagnose + decide ✗ Wrong options: MRI before treatment



AIIS avulsion fragment above the acetabulum

5 Management

- **AIIS avulsion**
  - Minimally displaced (almost all) → **REST, ACTIVITY MODIFICATION, PT**
  - Selected cases → **SURGERY**
    - Displaced > 2 cm / painful nonunion → **ORIF**
    - Symptomatic heterotopic ossification → **ARTHROSCOPIC EXCISION**

Nonoperative	**almost all succeed** · bed rest, ice, **activity modification**, NSAIDs, supervised PT · **hip flexed 2 wk** (↓ stretch on muscle + apophysis) → **guarded WB 4 wk**
Pelvic avulsions (OB)	protected WB **2–4 wk** → stretching + strengthening → normal activity **~2 months**
16-yr soccer, minimally displaced AIIS	**rest, activity modification + physical therapy** ✗ Wrong options: corticosteroid injection (impairs healing) · early aggressive strengthening (displacement / delayed union) · MRI · ORIF

Operative criteria

Surgery for	Detail
Significant displacement	**Commonly > 2 cm**
Painful nonunion	ORIF
Symptomatic heterotopic ossification	Excision (arthroscopic)
High-level athlete	Persistent limitation despite adequate nonop

Skeletally mature elite athletes	complete / retracted proximal rectus femoris avulsion → surgery gives **faster RTP + lower recurrence** · individualise
Post-traumatic rectus femoris HO	**arthroscopic excision** → better pain, ROM, function at 2 yr

6 Complications

Complications	**loss of reduction · delayed union**

– – Active space – –

How it's asked

"Hurdler's AIIS avulsion — nerve to the muscle?"	**→ femoral nerve**
"AIIS avulsion — eccentric contraction of which labelled muscle?"	**→ rectus femoris (C)**
"Avulsion at origin of muscle inserting on superior pole of patella?"	**→ AIIS radiograph (Figure B)**
"Minimally displaced AIIS avulsion — next step?"	**→ rest, activity modification, physical therapy**

ANTERIOR SUPERIOR ILIAC SPINE (ASIS) AVULSION

– Knee & Sports · 4 / 61 –

1 Basics

Who adolescents **14–17 yr** approaching skeletal maturity, **males > females** · sprinting, running, baseball, soccer, hockey

Avulsion physiology sudden violent contraction or excess stretch across an **open apophysis** · often **no external trauma** · low-energy → **no associated life-threatening injuries**

Anatomy

Muscle from ASIS	Nerve	Insertion
Sartorius	**Femoral**	**Pes anserinus** → source of **pes anserine bursitis**
Tensor fascia lata	**Superior gluteal**	Gerdy's tubercle via ITB

Insertion distractors fibular head = **biceps femoris** · femoral insertion anterior + distal to LCL = **popliteus** · ACL autograft = **gracilis (inferior pubic ramus) + semitendinosus (ischial tuberosity)**, both at pes · extends hip + flexes knee = **hamstrings**

2 Mechanism

Sartorius origin avulsion – mechanism, innervation & protected weight bearing

Mechanism indirect · **sudden forceful (eccentric) contraction of sartorius ± TFL** · **hip extended** (+ knee flexed) → **sprinting** (out of blocks, 400 m) or **swinging a baseball bat**
✗ Wrong options: external oblique · rectus femoris · iliopsoas · gluteus medius

Avulsed muscle nerve sartorius → **femoral**
✗ Wrong options: sciatic · obturator · pudendal · inferior gluteal

3 Presentation

History **"pop" or snap** · may be **misdiagnosed as a muscle strain**

Exam tender over ASIS (superior to hip joint), groin ecchymosis · **pain with straight-leg raise** · weak **hip flexion + knee extension** · limp if severe

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Use
AP pelvis	Displaced fractures visible · small fragment **easily missed**
CT / MRI	Confirm **nondisplaced** fractures
CT (relative)	**Rarely needed** – measure displacement, or tell **ASIS vs AIIS** origin (✗ Wrong options: urgent CT · extensive work-up)



Left ASIS avulsion

Diagnostic criteria

Criterion	ASIS avulsion
History	Sudden contraction of sartorius + TFL
Exam	ASIS tenderness, pain on SLR
Age	Adolescent / young adult
Radiographs	Fracture on standard pelvis views

White classification

White	Muscle	Activity	Fragment
Type 1	**Sartorius**	**Sprinting**	**Small**, displaced **anteriorly + inferiorly**
Type 2	**TFL**	**Twisting trunk / start of bat swing**	**Larger**, displaced **laterally**

5 Management

ASIS avulsion

- Displacement < 3 cm (most) → **PROTECTED WB, CRUTCHES, EARLY ROM**
- Displacement > 3 cm or painful nonunion → **ORIF**

Nonoperative rest, ice, NSAIDs, **protected / progressive WB as tolerated with crutches**, early ROM + stretching, PT · position limb to ↓ stretch on muscle · **return when pain-free motion** · outcomes **equivalent to surgery**

14-yr sprinter, pop, ASIS fragment	**progressive weightbearing, return when pain-free** ✗ Wrong options: fix femoral neck stress fracture · iliopsoas recession (internal snapping hip has normal X-rays + can bear weight) · fix sartorius origin · fix rectus femoris direct head (that is AIIS)
1 wk after injury, improving, still on crutches	**WBAT + crutches for 3 weeks total** ✗ Wrong options: NWB 6 wk · closed reduction + pinning (operative choice is ORIF) · aspiration · urgent CT
12-yr bat swing, 2 cm displaced	**rest + protected WB with crutches** ✗ Wrong options: ORIF with rectus femoris / sartorius / iliopsoas · excision

Operative criteria

ORIF	**displacement > 3 cm** (Orthobullets, most OB) · **> 2 cm in elite athletes** or failed nonop (OB) – read the options · **painful nonunion**
High-energy paediatric acetabular fracture	ORIF if **> 2 mm** intra-articular or triradiate displacement (contrast with low-energy avulsions)

Iliac crest apophysis avulsion – abdominal wall innervation

Mechanism	**rapid lateral trunk flexion / twisting** → contraction of **external + internal oblique, transversus** · sprinter out of blocks · **rare**
Nerve	abdominal wall = **lower intercostal nerves** ✗ Wrong options: sciatic (hamstrings) · femoral (rectus femoris) · obturator (adductor longus) · superior gluteal (TFL)



Left iliac crest apophyseal avulsion

Treatment	**protected WB + activity as tolerated**, rest, analgesics, PT ✗ Wrong options: oncology work-up + biopsy · ORIF · abduction brace in 45° flexion · **120° knee flexion 24 h – quadriceps contusion**
Surgery	displaced **> 3 cm** or failed nonop

– – Active space – –

Apophyseal avulsion series

Series	Finding
Schuett (228, mean 14.4 yr, **76% male**)	**AIIS > ASIS > ischial tuberosity > iliac crest** · surgery **3%**
Schuett – nonunion	5 nonunions, **4 ischial tuberosity** · displacement **> 2 cm → 26× nonunion risk**
Rossi (203)	**Ischial tuberosity ~50%** > AIIS 25% > ASIS 20% > pubic symphysis 4% > iliac crest 1% · soccer + gymnastics
Schiller (lower-limb avulsions)	Surgery: displaced tibial tubercle, failed nonop, nerve impingement, **> 2 cm**

Most common site ischial tuberosity (Rossi) · AIIS (Schuett) · ASIS listed first (other OB) – read the options

6 Complications

Missed ASIS avulsion **exostosis** needing excision

Painful nonunion ORIF

How it's asked

"ASIS avulsion – eccentric contraction of?"	**→ sartorius**
"Sprinter, ASIS avulsion – treatment?"	**→ progressive weightbearing, return when pain-free motion achieved**
"True of the avulsed (ASIS) muscle's insertion?"	**→ can be source of pes anserinus bursitis**
"Sprinter, iliac crest avulsion – next step?"	**→ protected weight bearing and activity as tolerated**
"1 wk after ASIS avulsion, improving – management?"	**→ weight-bearing as tolerated and crutches for 3 weeks total**
"Bat swing, ASIS displaced 2 cm?"	**→ rest and protected weight bearing with crutches**
"ASIS avulsion – nerve to avulsed muscle?"	**→ femoral**
"Iliac crest avulsion – nerve to attached muscles?"	**→ lower intercostal nerves**

ARTICULAR CARTILAGE DEFECTS OF KNEE

– Knee & Sports · 5 / 61 –

1 Basics

Epidemiology & pathomechanics

Incidence	**5–10% of people > 40 yr** have high-grade chondral lesions · isolated defects in athletes est. **36%** (OB)
Chronic ACL tear	**anterior lateral femoral condyle + posterolateral tibial plateau**
OCD	**70%** on **posterolateral (lateral aspect of) medial femoral condyle**
Mechanism	acute trauma or chronic repetitive overload → softening, fissuring, flap tears, delamination
Threshold	impaction **> 24 MPa** disrupts normal cartilage
Healing	**limited spontaneous healing + propensity to worsen**; defects isolated from blood supply / marrow → **no repair response** (basis of microfracture)
Overloaded cartilage (runner)	chondropenia, chondrocyte apoptosis, subchondral bruising, sclerosis, fracture, necrosis; cartilage volume ↑ linearly with loading until a threshold
Focal lesion (TO)	localised defect **< 10 cm²**, often **< 5 cm²** · precursor of OA → treat while focal
Pathogenesis (TO)	insult → **MMPs** degrade collagen → exposed collagen acts as **antigen** → synovitis → **IL-1, IL-6, TNF-α** → **NF-κB + Ras** pathways → further breakdown

Genes & lab markers (collagen type of each repair tissue)

Tissue / procedure	Collagen	Note
Hyaline (articular) cartilage	**Type II**	Columnar organisation + **lamina splendens** (separates it from "hyaline-like")
Microfracture / drilling → fibrocartilage	**Type I > II** (mostly I + III, some II)	Stiffer, less durable → deteriorates over time
Microfracture (TO)	**Types I, II and III**	Fibrocartilage alone → mix of fibrocartilage + hyaline-like
ACI / MACI	**I + II, type II predominates**	"Hyaline-like"; ACI (TO) may give hyaline-like, hybrid or fibrocartilage (I + II)
OATS / mosaicplasty, OCA	**Type II**	Direct transfer of articular cartilage; chondrocytes viable
Particulated juvenile cartilage allograft	Type II	Live cells in native ECM
Fibrous tissue	No type II	Made by fibrocytes
Elastic cartilage	–	Ear, epiglottis, larynx

– – Active space – –

Labs	only to rule out inflammatory disease
Composition (TO)	**water 70–80%** · **type II collagen** (tensile strength) · **proteoglycans** (aggrecan, chondroitin sulphate, HA) resist compression · **aggrecan = main functional molecule** · cartilage type II vs meniscus type I

3 Presentation

History	precipitating trauma common · **some found incidentally** on MRI / arthroscopy
Symptoms	asymptomatic vs localised pain, effusion, motion loss, **mechanical symptoms** (catching, locking) – esp. unstable / loose lesions
Exam	look for background factors: **laxity, malalignment, compartment overload**; ROM, ligaments, gait

4 Diagnosis & Classification

Key radiology findings

Study	Use / finding
Standing AP, lateral, **Merchant**	Rule out arthritis, bony defects; check alignment
Semiflexed 45° PA	**Most sensitive for early joint-space narrowing**
Long-leg alignment	**Mechanical axis**
CT	Bone loss; **TT-TG** for patellofemoral joint
MRI	**Most sensitive for focal defects**; FS T2, PD, T2 FSE ↑ sens / spec
dGEMRIC, T2 mapping	Evolving – cartilage defects + repair
MRI after microfracture	**High fill rate** = good short-term outcome
MRI ↑ T2 signal under defect	**Subchondral bone involvement / deficiency** → favours **OCA** (✗ Wrong options: ACI)
MRI axial slices superimposed	**TT-TG > 20 mm** = patellar maltracking



Semiflexed 45° PA view – most sensitive for early joint-space narrowing

Outerbridge arthroscopic grading

Grade	Orthobullets	TO version
0	Normal	–
I	**Softening + swelling** (probe)	Softening + swelling
II	**Partial-thickness**, surface fissures, **not to subchondral bone, ≤ 1.5 cm diameter**	Fragmentation + fissuring, area **< 0.5 inch**
III	**Deep fissures to subchondral bone, > 1.5 cm diameter**	Fragmentation + fissuring, area **> 0.5 inch**
IV	**Exposed subchondral bone**	**Erosion to bone**

II / III cut-off **1.5 cm diameter** (Orthobullets) · **0.5 inch area** (TO) – read the options

Modified Outerbridge – TO lecture I softening · II fibrillation **< 50% depth** · III fissuring **> 50% depth** · IV exposed subchondral bone · **most practical / widely cited** system (ICRS detailed, less used)

Outerbridge grade IV **indication** for cartilage restoration, not a failure risk factor

ICRS grading

Grade	Finding
0	Normal
1	Nearly normal – superficial lesions
2	Abnormal – **< 50% cartilage depth**
3	Severely abnormal – **> 50% depth**
4	Severely abnormal – **through subchondral bone**

Chondromalacia patellae – diagnosis & grading

Gold standard (TO) **arthroscopy**
✗ Wrong options: X-ray · MRI · CT

Modified Outerbridge – MRI (TO)	FS PD finding
I	Focal hyperintensity, **normal contour**
II	**Fissure or blister-like swelling** reaching surface
III	**Focal ulceration + "crab meat" lesions**
IV	**Full-thickness loss** + reactive underlying bone

Tram-track sign (TO) **mucoid degeneration of PCL**
✗ Wrong options: crab meat

5 Management

Decision factors & size thresholds

Nonop first rest, NSAIDs, physio, weight loss when **mild**; viscosupplement, steroid, unloader brace – **controversial**, symptom relief, defect won't heal

– – Active space – –

Treatment spectrum (Alford) – **marrow stimulation (MST) → ACI → osteochondral grafting (OCG**, autograft / allograft)

Patient factors age, skeletal maturity, demand, rehab tolerance · **defect factors** – size, location, **contained vs uncontained**, **subchondral bone involvement**

First correct **malalignment, ligament instability, meniscal deficiency** (before / at time of cartilage surgery); patellofemoral → **maltracking + malalignment**

Femoral condyle defect (after correcting axis, ligaments, meniscus)

- < 4 cm² → **MICROFRACTURE or OATS (palliative if older / low demand)**
- > 4 cm² → **OSTEOCHONDRAL ALLOGRAFT or ACI**

Patellofemoral defect (after correcting maltracking)

- < 4 cm² → **MICROFRACTURE or OATS**
- > 4 cm² → **ACI (microfracture if older / low demand)**

Size / bone status (OB algorithm)	Procedure
< 1 cm², no osseous defect	**Microfracture**
< 1 cm², osseous defect	OATS
1–2 cm², no osseous defect	Microfracture or ACI
1–2 cm², osseous defect	OATS
> 2 cm², osteochondral defect or AVN	**OCA** or ACI + bone graft

Procedure	Size range quoted (source)
Microfracture	**< 2 cm²** / < 2 × 2 cm (Orthobullets, OB, TO) · < 1 cm² (OB patella, OB algorithm) · with OATS < 4 cm² (Orthobullets)
OATS	< 2 cm², some 2–4 cm² (OB) · up to 4–6 cm² (OB, OCD chapter) · grafting **up to 5 cm** (TO)
ACI	**1–10 cm²** (OB) · **2–10 cm²** large grade IV (TO) · > 4 cm² (Orthobullets)
OCA	**> 2 cm²** (OB) · > 4 cm² (Orthobullets, OB) · > 6 cm² (OB, OCD chapter)
Threshold debate	**4 cm²** usual; some use **2 cm²** (OB) – read the options

Depth / demand / size (TO, Mohindra)	Procedure
< 50% depth (grade I–II)	**Debridement + lavage + chondroplasty** (palliative)
> 50%, **low demand**	**Microfracture**
> 50%, high demand, **< 2.5 cm**	**OATS (autograft)**
> 50%, high demand, **2.5–5 cm**	**OATS (allograft) / mosaicplasty**
> 50%, high demand, **2.5–10 cm**	**ACI / BMAC**
> 10 cm	**UKA / HTO**

Kissing lesions (TO) both condyles → **contraindication** to cartilage restoration

Operative criteria

Indication	Detail
Failed nonop	Symptomatic defect
Acute osteochondral fracture	**Full-thickness cartilage loss**
Size (OB)	Defects **> 2 × 2 cm** failing nonop
Arthroplasty instead	**> 55 yr** (OB) · PF arthroplasty **40+ yr** with isolated PF arthritis

Debridement / chondroplasty

Debridement / chondroplasty — removes loose flaps → relieves mechanical symptoms · **short-term benefit 50–70%** · simple, fast rehab · leaves exposed bone; natural history unknown

Chondroplasty for large PF defect — only **older patients with diffuse degeneration**
✗ Wrong options: young

Microfracture & marrow stimulation — fibrocartilage, outcomes & overgrowth

Principle — perforate subchondral bone → **marrow MSCs** into defect → **fibrocartilage** · includes **microfracture, abrasion chondroplasty, drilling**
✗ Wrong options: true hyaline regeneration

Technique — **stable vertical walls**; **remove calcified cartilage layer** (✓ should be done); awl holes **3–4 mm apart**; entry **perpendicular (90°)** to avoid skipping / hole collapse

Osseous overgrowth — risk ↑ with **removal / violation of subchondral plate** (over-aggressive debridement)
✗ Wrong options: more holes per area · small awls · 90° entry · calcified layer removal

Awl size — smaller (**1.0 mm**) better than **1.2 mm** in some studies

Microfracture (TO) — awl holes **1 mm** wide, **3 mm deep, 3 mm apart** · any lesion size · fibrocartilage → elderly / low-demand

Drilling steps — debride **sclerotic bone** → trim unstable cartilage edges → drill **past subchondral plate (~4 mm)** → enough coverage between holes for a **mesenchymal clot**

Drilling — alters subchondral plate → micro-cysts + intralesional osteophytes (later fill with cancellous bone)

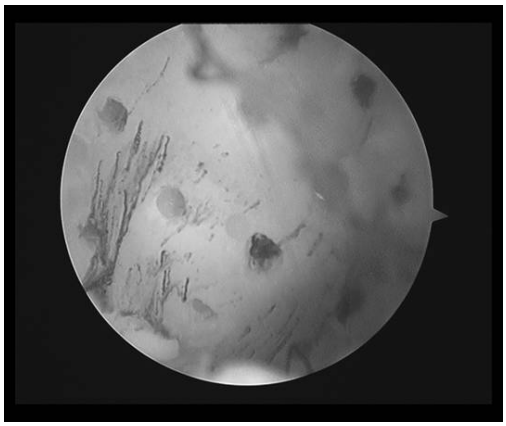
Benefits — cheap, **single-stage**, arthroscopic; **best for acute, contained lesions < 2 × 2 cm**

Limits — poor if **> 2 × 2 cm**; not for bone defects or **uncontained** lesions (no rim); **restricted WB 6–8 wk** (Orthobullets) · **~6 wk** (OB) · NWB 4–6 wk + CPM (Orthobullets OCD note)

Good short-term outcome — **high MRI fill rate**, **low BMI**, **short preop symptoms** · **patellar microfracture → poor**

Durability — gains maintained only **2–3 yr**; not maintained long term

– – Active space – –



Microfracture awl holes in a chondral defect

ACI / MACI — hyaline-like repair & patellofemoral indications

ACI	cell therapy → autologous **"hyaline-like"** cartilage · arthroscopic harvest from low-load area → culture → inject under **periosteal patch** at 2nd (open) surgery
Needs	**full-thickness cartilage margins**, healthy subchondral bone ✗ Wrong options: significant osseous defect: **< 8 mm deep** OK (OB) · add bone graft if bone loss **> 1 cm** (OB, OCD chapter)
Limits	open, **2-stage**, prolonged protection · better histology than marrow stimulation; long-term ≈ microfracture
Periosteal patch (TO)	harvested **proximal tibia / distal femur**, slightly oversized · **cambium layer faces the defect** (down to bone) · sewn **watertight** with interrupted **6-0 absorbable** sutures + **fibrin glue** at margins · patch is **chondrogenic** + paracrine ✗ Wrong options: cambium away · sewn into defect · mixed with matrix
1st-stage harvest (TO)	**200–300 mg** cartilage · cells expanded **2–3 wk** (TO) · implanted after **3–6 wk** (TO) — read the options
Collagen membrane (TO)	replaced periosteum → avoids **patch hypertrophy + harvest morbidity**
Generation limitations (TO, Mohindra)	1st: **periosteal hypertrophy** · 2nd: **uneven cell distribution** · 3rd (mechanically stimulated): cost · modified 3rd **CAIS**: minced cartilage, **no culture, single-stage**, up to **10 cm²** · culture time **3–6 weeks**
BMAC (TO)	**iliac crest** aspirate → centrifuge → **buffy coat** (stem cells + growth factors) → paste over defect · **single-stage**, cheap, low morbidity · hyaline-like aim · long-term data lacking

Chondroprotective agent (TO)	Mechanism / note
Diacerein	**IL-1 blocker** · **50 mg OD–BD × 3 months**, pause 2–3 months, repeat · a **drug** (others = supplements)
Glucosamine + chondroitin sulphate	Anabolic + anti-catabolic · mild relief
SAMe	Proteoglycan synthesis · limited evidence
Undenatured type II collagen	**Oral tolerance** → ↓ immune response to cartilage

ACI generation (TO)	Features
1st (Brittberg 1994)	**2-stage**: arthroscopic biopsy → open / mini-open implantation under **periosteal flap (PACI)** or collagen membrane
2nd = MACI	**Autologous chondrocytes on a scaffold for expansion**; 3D bioabsorbable scaffold press-fit (hyalograft C, bioseed-C) → **all-arthroscopic** implantation possible; NeoCart (bioreactor loading)
3rd	**Single stage**; allogeneic tissue or **stem cells**; CAIS (minced autologous chondrocytes + marrow MNCs on polycaprolactone + PGA + PDS scaffold); De Novo ET (minced juvenile allograft); CARTISTEM (umbilical cord MSCs, hyaluronate gel)

3rd vs 1st generation (TO)
all-arthroscopic, single-stage
✗ Wrong options: 2-stage · collagen cover · periosteal sheath

Generation conflicts
MACI = 2nd generation (TO) vs **3rd generation, first to use scaffold** (OB) · **CAIS = 2nd-gen single-stage** (one TO) vs **3rd-gen classic** (other TO) — read the options

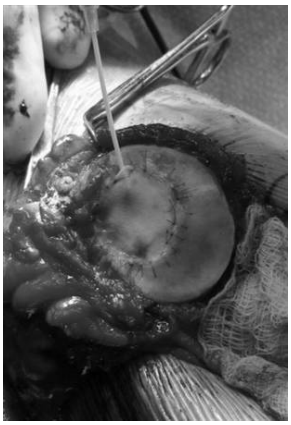
MACI
cells cultured on **porcine collagen membrane** · fixed with **fibrin glue or sutures** · **only FDA-approved cell therapy for cartilage (USA)** · 2-stage, expensive · **type II > I**

5 cm² MFC defect, 32 yr (TO)
ACI
✗ Wrong options: microfracture · UKA · mosaicplasty

ACI in PF joint
JSN on Merchant view = contraindication · acceptable: grade 4 MFC / trochlea lesion, varus axis (corrected), concomitant **Fulkerson AMZ** · **ACI not FDA-approved for patella** (off-label)

PF defect + recurrent dislocation, TT-TG 26 mm, 2.2 × 2.1 cm, exposed bone
MPFL reconstruction + autologous chondrocyte implantation + antero-medializing (Fulkerson) TTO
✗ Wrong options: lateralising TTO (↑ Q angle) · microfracture (< 1 cm² only) · PF arthroplasty (40+ yr)

PF lesions
inferior outcomes vs condyles (**high shear**) · **TT-TG > 20 mm** = maltracking → AMZ protects the repair · recurrent dislocation may avulse the **MPFL patellar insertion** (osseous fragment)



ACI for a patellar chondral defect

– – Active space – –

Osteochondral allograft – indications, contraindications & failure risk

Goal	**live chondrocytes in mature matrix + underlying bone** · **fresh, refrigerated** grafts · bulk (screws) or shell / dowel press-fit · donor = **cadaver hemicondyle** · size + **radius of curvature** matched ✗ Wrong options: freeze-dried
Indications	**large defects (> 2 cm² / > 4 cm²)**, **subchondral bone loss / oedema / AVN**, **uncontained** lesions, **revision** of failed techniques · **single-stage**
OCA over ACI	**depth of osseous involvement** ✗ Wrong options: footprint > 6 cm² · location · age · time since injury
Contraindications	**ligament instability, meniscal insufficiency, uncorrectable malalignment, multicompartment / tricompartmental arthrosis, inflammatory arthropathy** · ✓ defect > 2 cm² is an **indication**
Failure risk	**patellofemoral joint defect** (adverse biomechanics) · **bipolar** lesions ✗ Wrong options: unipolar · male sex · Outerbridge IV · concomitant AMZ
Outcomes	PF worse than condyles (contact pressure) · **> 80% return** to activity in young · limits: **availability, cost**, infection / immune risk ✗ Wrong options: more easily available
Immune response	only a concern with **allograft** · allografts **cleansed in antibiotics** before implantation ✗ Wrong options: OATS
Varus axis + large MFC lesion	correct alignment **before / at time** of OCA ✗ Wrong options: isolated OCA
MFC 25 × 25 mm, 32 yr	**osteochondral allograft** (ACI also correct if listed) ✗ Wrong options: 1–2 autograft plugs · microfracture · UKA · abrasion
MFC 1.7 × 1.3 cm (2.21 cm²) + loose bodies, neutral axis	**removal of loose osteochondral bodies + osteochondral allograft transplantation to the MFC** · loose bodies sit in **suprapatellar pouch**; hip-to-ankle film neutral, sunrise view normal ✗ Wrong options: osteotomy · microfracture (osseous defect) · OATS alone (loose bodies stay)

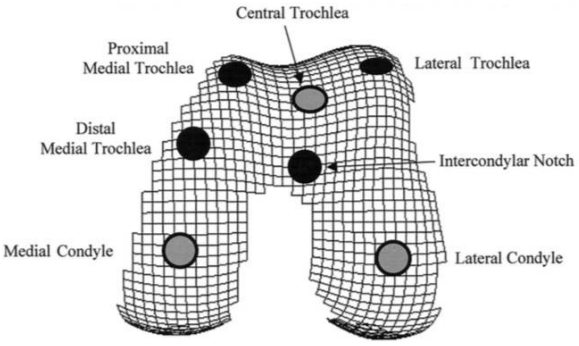


Osteochondral allograft dowel

Osteochondral autograft (OATS) – graft fixation & donor site

Principle	cylinders of normal cartilage + bone from **low-load areas** (periphery of trochlea, notch) press-fit into recipient socket · chondrocytes **viable**, cartilage **heals**, biopsy **type II** ✗ Wrong options: scaffold · replaced by fibrocartilage

Benefits	autologous, cheap, **single-stage**, arthroscopic possible · **limits** – size + **donor-site morbidity**, curvature matching
OATS technique (TO)	dowels **6, 8 or 10 mm** (Arthrex) · depth **15–25 mm**, donor **1–2 mm longer** (graft kept "proud") · up to **3 dowels (~2.5 cm)** · **NWB 6 wk** (TO) vs delay WB 3 months (below) – read the options · donor: medial / lateral **trochlear edge**, not central trochlea
OATS vs mosaicplasty (TO)	same principle · **mosaicplasty = multiple small 2–4 mm dowels** (Smith & Nephew) · OATS = fewer, thicker dowels · autograft up to 2.5 cm, allograft up to 5 cm
Fixation strength	**initially decreases in early healing, then increases with subchondral bone healing → delay WB 3 months**; early non-WB motion
Donor site	**distal medial trochlea** = **lowest PF contact pressure + no tibiofemoral contact** · also **peripheral / superior lateral trochlea** · grafts **< 10 mm wide** ✗ Wrong options: central trochlea · condyles (weight-bearing)
Ahmad map (figure)	1 proximal medial trochlea · 2 medial condyle · 3 central trochlea · **4 distal medial trochlea** ✓ · 5 lateral condyle



OATS donor sites – distal medial trochlea has the lowest contact pressure

Malalignment & unloading procedures (osteotomy, tibial tubercle transfer)

Procedure	Indication	Contraindication / risk
Maquet (anteriorisation)	**Distal pole** lesions only	**Superior patellar arthrosis**; elevate **≤ 1 cm** → else **skin necrosis**
Fulkerson AMZ (anterior + medial)	**Lateral + distal pole** lesions, ↑ Q angle, offload PF restoration, **bipolar PF lesions**	**Superior-medial patellar arthrosis**, **medial facet + MFC arthrosis**, skeletal immaturity
Distal femoral varus osteotomy	**Valgus** + lateral compartment chondral disease	–
HTO (medial opening wedge)	Varus-pattern disease	Not if axis neutral

Scope before tubercle transfer – need **intact proximal + medial cartilage**

Anteriorised 2.5 cm	**skin necrosis** / wound breakdown ✗ Wrong options: instability · nonunion · progression · fracture
Lateral FC + plateau chondromalacia, 10° valgus, 31 yr	**distal femoral varus osteotomy** · **5° valgus → 5-fold risk of progressing ≥ 1 grade in 18 months** (Sharma) ✗ Wrong options: OAT · UKA · HTO · Fulkerson
PF joint load	up to **6.5 × body weight** in flexion

– – Active space – –

Fixation of unstable fragments & instability-related lesions

Fixation	needs fragment with **adequate subchondral bone** · debride base, ± drill / local graft · absorbable or nonabsorbable screws · **best: unstable OCD, open physes** · lower healing when mature · **headless nonabsorbable screws removed at 3–6 months**
Acute patellar dislocation + osteochondral fragment + MPFL tear	**fragment fixation (ORIF) + MPFL reconstruction** (best: **14-year-old female athlete, first-time dislocation, imaging shown in Figure D**) · sites: **patella + LFC** · chronic, no fragment → OATS ✗ Wrong options: fixation without OC defect · partial-thickness chronic defect · bucket-handle tear · Outerbridge IV OA
Unfixable unstable MFC flap 1 × 1.5 cm, single procedure → type I + II	**microfracture**

Osteochondritis dissecans & instability-related fragments

Commonest site (TO)	**knee: lateral surface of medial femoral condyle** (classic, posterolateral MFC) · others: **talus anteromedial corner**, **femoral head superomedial**, **capitellum**, **1st MT head** ✗ Wrong options: LFC
Who (TO)	**males**, adolescents · **commonest cause of loose body** in knee in adolescents / young adults
Catchers (OB)	**posterior femoral condyle** (hyperflexion loading) ✗ Wrong options: anterior LFC / posterolateral plateau (ACL injury pattern) · patella · trochlea
Wilson test (TO)	seated, legs hanging · **pain on internal rotation**, relieved by external rotation
Imaging (TO)	X-ray: loose body / loosely attached fragment · **MRI = investigation of choice**, earliest diagnosis, guides management · **MRI = best to follow healing** ✗ Wrong options: CT (insensitive) · bone scan (can't separate healed vs unhealed; high Tc uptake → repair potential) · SPECT
"Healing" MRI criteria (TO17118770)	(1) round homogeneous high signal **≥ 5 mm** beneath lesion · (2) articular focal defect **≥ 5 mm** wide · (3) high-signal line through cartilage + subchondral plate into lesion — another TO lists the same ≥ 5 mm signs as **De Smet instability criteria** — read the options

JOCD (TO) — **activity modification** first · most heal · small lesions → **multiple drilling** · large → **mosaicplasty / OATS**
✗ Wrong options: in situ fixation · microfracture · excision

ROCK / De Smet (TO)	Detail
Nonop protocol	Stop impact sport, protected WB, unloader brace or cast · **≥ 3–6 months**; healing on 6-month MRI → extend to **12–18 months**
Strongest success predictor	**Younger age / open physes** (boys **< 14**, girls **< 12**)
Other success predictors	Small **normalised lesion width**; **no cysts** (< 1.3 mm may still heal at 12 months); extreme medial / lateral (coronal) or far posterior (sagittal) site
ROCK nomogram	Age + normalised width + cyst size → **> 48% healing probability at 6 months** → continue nonop
Strongest failure predictor	**Subchondral cysts > 5 mm on MRI** (fluid intrusion = instability) (✗ Wrong options: sclerotic rim (common, chronicity))
De Smet instability	**T2 fluid line ≥ 5 mm** behind fragment · **cysts ≥ 5 mm** · **focal cartilage defect ≥ 5 mm** · fluid through cartilage into bone
Other failure signs	Closing physes, **mechanical symptoms**, **patella / LFC** site, no progress after 3–6 months
Surgery options	Trans- / retroarticular **drilling** (intact cartilage), **internal fixation** (unstable), MACI / OCA (unsalvageable)

Stable lesion (TO) — no fluid behind fragment, **firm on probing** → failed conservative → **drilling alone**

Unstable but salvageable, immature (TO) — circumferential T2 fluid, **ballotable**, minimally displaced → **in situ fixation of the fragment using bioabsorbable screws, darts or compression screws** ± graft
✗ Wrong options: drilling alone · excision + microfracture · OATS

6 Complications

Complication	Fact
Osseous overgrowth (microfracture)	**Subchondral plate removal** → ↓ repair cartilage volume
Microfracture deterioration	Type I fibrocartilage; gains last **2–3 yr**
OCA dropped on floor	**40% contamination**, **Staphylococcus** commonest → **5-min povidone-iodine bath + saline rinse** (max cell viability, ↓ bacteria)
OCA	Infection / immune reaction risk
Maquet > 1 cm	**Skin necrosis**
OATS	**Donor-site morbidity**
ACI periosteal patch	**Hypertrophy** + harvest morbidity
Retained headless screws	Remove at **3–6 months**

– – Active space – –

X vs Y

	Microfracture	OATS	OCA	ACI / MACI
Stages	**1**	**1**	**1**	**2** (3rd-gen ACI: 1)
Tissue	**Fibrocartilage (type I)**	Hyaline (II)	Hyaline (II), live cells	**Hyaline-like (II > I)**
Healing source	**Marrow MSCs**	Transplanted autologous chondrocytes	Donor chondrocytes; bone by **creeping substitution**	Cultured **autologous** (✗ Wrong options: allogeneic) chondrocytes
Size	Small, **contained**	Small	**Large**, uncontained OK	Large (1–10 cm²)
Bone loss	No	✓ small	✓ **best**	No (needs bone graft)
Key issue	Overgrowth, durability	Donor site	Availability, cost, immune	Cost, open, rehab

How it's asked

"Reparative tissue after microfracture (figure)?"	→ **fibrocartilage** ✗ Wrong options: hyaline · fibrous tissue · elastofibroma
"Tissue after drilling an 8 × 7 mm defect?"	→ **fibrocartilage**
"Microfracture (B) vs mosaicplasty (C) – true?"	→ **biopsy of repair site at 3 months will reveal more type I collagen in technique B** ✗ Wrong options: both 2-stage (only ACI)
"Short-term microfracture success?"	→ **high fill rate of the defect on MRI** ✗ Wrong options: high BMI · long symptoms · patella · ACL deficiency
"Risk of osseous overgrowth after microfracture?"	→ **removal of the subchondral plate**
"Single procedure stimulating type I + II collagen?"	→ **microfracture**
"Awl through subchondral bone – true?"	→ **content of type I cartilage generated is greater than type II** ✗ Wrong options: lasts 5 yr · uncontained · no WB limit · > 4 cm²
"Predominantly type I collagen in defect?"	→ **microfracture**
"Type II > type I collagen on healed surface?"	→ **matrix-associated autologous chondrocyte implantation** ✗ Wrong options: single-core OATS (type II only)
"OATS – graft at 1 year?"	→ **transplanted chondrocytes viable, articular cartilage heals, biopsy shows type II collagen**
"5 cm² MFC defect, 32 yr (TO)?"	→ **ACI**

"ACI periosteal patch (TO)?"	→ **cambium layer faces towards chondral defect**
"Second-generation ACI (TO)?"	→ **autologous chondrocytes scaffold used for chondrocyte expansion**
"Third vs first generation ACI (TO)?"	→ **all-arthroscopic procedure**
"ACI in PF joint — EXCEPT?"	→ **joint space narrowing on Merchant view**
"Recurrent dislocation, TT-TG 26 mm, 2.2 × 2.1 cm medial facet defect?"	→ **MPFL reconstruction + ACI + antero-medializing tibial tubercle osteotomy**
"MFC 25 × 25 mm, 32 yr?"	→ **osteochondral allograft**
"OCA contraindications — EXCEPT?"	→ **osteochondral defect measuring > 2 cm²**
"Runner, 2.21 cm² MFC osteochondral defect + loose bodies?"	→ **removal of loose osteochondral bodies + osteochondral allograft transplantation to the MFC**
"Best candidate for isolated OCA (radiograph depicted)?"	→ **30-year-old man, chronic atraumatic knee pain, 4.2 cm² medial femoral condyle osteochondral lesion, neutral axis (Figure A)** ✗ Wrong options: 67 yr / 55 yr (arthroplasty) · varus axis · 1.5 cm² (nonop)
"OCA — true?"	→ **replaces cartilage defect with live chondrocytes in mature matrix with underlying bone** ✗ Wrong options: freeze-dried
"OCA rather than ACI — deciding factor?"	→ **depth of osseous involvement**
"Highest risk of OCA failure?"	→ **patellofemoral joint cartilage defect** ✗ Wrong options: male · unipolar · Outerbridge IV · AMZ
"OCA — true?"	→ **can be useful in addressing subchondral bone deficiency** ✗ Wrong options: 2-stage · PF better · uncontained excluded · easy availability
"Mosaicplasty graft fixation strength?"	→ **initially decreases during early healing, then increases with subchondral bone healing**
"Best knee donor site for capitellar OATS (figure)?"	→ **distal medial trochlea (location 4)**
"Medial facet + MFC arthrosis — contraindicated?"	→ **anteromedial (Fulkerson) tibial tubercle osteotomy**
"Lateral compartment chondromalacia, 10° valgus?"	→ **distal femoral varus osteotomy**
"Tubercle anteriorised 2.5 cm — complication?"	→ **skin necrosis**

– – Active space – –

"Most common site of OCD (TO)?"	**→ medial femoral condyle of femur** (lateral surface)
"Juvenile OCD femoral condyle — treatment (TO)?"	**→ activity modification**
"Best investigation to follow OCD healing (TO)?"	**→ MRI**
"Immature, failed 6 months, circumscribing T2 fluid line, ballotable (TO)?"	**→ in situ fixation of the fragment with bioabsorbable screws, darts or compression screws**
"Strongest predictor of nonop failure in JOCD — ROCK (TO)?"	**→ presence of subchondral cystic changes deep to the lesion measuring greater than 5 mm on MRI**
"OCD site in a baseball catcher?"	**→ posterior femoral condyle**
"Candidate for fragment fixation + MPFL reconstruction?"	**→ 14-year-old female athlete, first-time dislocation, imaging shown in Figure D**
"Gold standard for chondromalacia patellae (TO)?"	**→ arthroscopy**
"Crab meat lesions on MRI (TO)?"	**→ grade III chondromalacia patellae** ✗ Wrong options: tram track (PCL mucoid degeneration)
"Diacerein mechanism (TO)?"	**→ IL-1 blocker**
"Treatment of high-demand > 50%-depth defect < 2.5 cm (TO)?"	**→ OATS (autograft)**

ATHLETIC PUBALGIA & ADDUCTOR STRAIN

– Knee & Sports · 6 / 61 –

1 Basics

Definition (TO)	**chronic lower abdominal + groin pain without a true hernia** · aka **sports hernia, Gilmore's groin, core muscle injury (CMI)**
Who	**males > females** (TO: **narrower male pelvis** → greater force shifts, less stability) · **hockey + soccer** · high-level athletes (forces ↑ as strength ↑) (TO) ✗ Wrong options: more common in females
Sports hip triad	**adductor strain + osteitis pubis + athletic pubalgia** (adductor strain itself: see Adductor Strain chapter)

Anatomy – the pubic fulcrum (TO)

Fulcrum (TO)	**pubic symphysis** = common attachment of **rectus abdominis + adductor longus** ✗ Wrong options: main muscle transversus abdominis
Pubic aponeurosis (TO)	**rectus abdominis + conjoint tendon (internal oblique + transversus abdominis) + external oblique** → confluent with **adductor longus + gracilis** at the symphysis
Opposing pull (TO)	rectus abdominis vs adductors act in **opposite directions** on the symphysis → **imbalance = origin of pubalgia**
Pain source	injury to the **common aponeurosis between rectus abdominis and adductor longus** (Orthobullets)



Sports hernia: injured tissue at the rectus–adductor aponeurosis

– – Active space – –

2 Mechanism

Mechanism & surgical referral

Movement	**repetitive hip extension + abduction** (± external rotation) with **eccentric adductor contraction** · abdominal hyperextension + thigh abduction (Orthobullets) ✗ Wrong options: extension + adduction · flexion + abduction / adduction · extension + IR · flexion + IR · lumbar hyperextension
Imbalance	**strong adductors vs weak rectus** (lower-limb-focused training) (OB) · **strong adductors vs weak hip abductors** (OB)
Type	overuse microtrauma or acute trauma
Activities (TO)	running, kicking, cutting, twisting, explosive turns, change of direction
What tears	shear on rectus → **transversalis fascia, rectus, adductor magnus origin** (OB) · **adductor longus** (Orthobullets, coronal MRI) · inguinal hernia + **internal oblique** micro-tears at surgery (OB) – read the options

3 Presentation

Pain	lower abdominal + **inguinal pain at extremes of exertion** · **unilateral**, exercise-related, radiates to **perineum, inner thigh, scrotum** (TO) · settles with rest, **returns with play** (TO) ✗ Wrong options: bilateral
Exam	**tender adductor longus** · tender **pubic tubercle / inguinal region**, pain over pubic rami · **pain with Valsalva + sit-ups** (rectus aponeurosis) · pain on **resisted hip adduction** · **not** worse with hip rotation
No palpable hernia	on Valsalva · differentiate from **subtle true inguinal hernia**
Check hip	**concomitant FAI** may coexist

4 Diagnosis & Classification

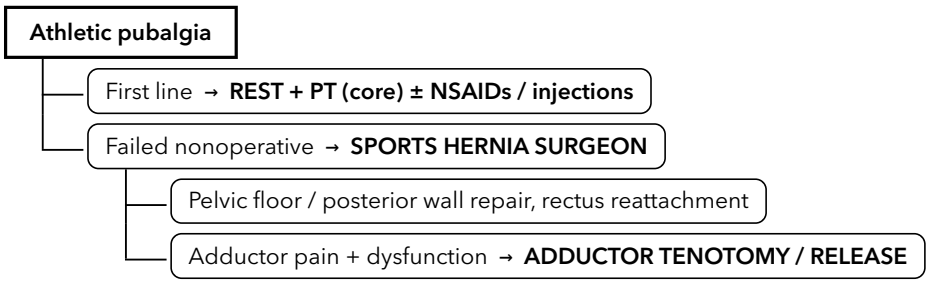
Key radiology findings

Study	Finding / use
X-ray, MRI, bone scan	**Rule out other causes** (Orthobullets)
MRI	**Oedema adjacent to the aponeurosis** · coronal T2: adductor longus tear **retracted from common aponeurosis** · "helpful for definitive diagnosis" (OB) vs "**not very helpful**" / diagnosis **largely clinical** (other OB) – read the options
CT pelvis	Rules out **bony injury (osteitis pubis)**
Scrotal US	**No added value**



Coronal T2: adductor longus tear retracted from the common aponeurosis

5 Management



Nonoperative

Duration	**6–8 weeks** (Orthobullets, OB) · core stability **≥ 2 months** before surgery (OB) · surgery if no improvement in **3 months** (TO) – read the options
Physiotherapy (TO)	**core strengthening + stabilisation**, restore **pelvic tilt**, postural training
Drugs (TO)	NSAIDs, **oral steroid taper**
Injections (TO)	selective **corticosteroid or PRP** into rectus abdominis and / or adductor longus origin

Operative criteria

Indication	Procedure
Failed extensive nonoperative care	**Pelvic floor repair (hernia operation)** · mesh hernia repair · **adductor / rectus recession**
Failed nonop (TO)	**Reinforce posterior wall + fix rectus abdominis or conjoint tendon**
Adductor pain + dysfunction (TO)	**Adductor tenotomy** / release
Failed nonop (Orthobullets)	**Decompress genital branch of genitofemoral nerve**

Pro soccer player, failed nonop, no hernia, malignancy excluded	**refer to a sports hernia surgeon** (strengthen anterior pelvic floor) ✗ Wrong options: more nonop · scrotal US · CT pelvis · **cortisone – doesn't strengthen pelvic floor**
Groin pain, normal imaging	tends **not to improve without hernia-type surgery** (OB)
Outcome	rectus / adductor **aponeurotic plate repair → 96% return to sport**

– – Active space – –

X vs Y

Sports hip triad	Athletic pubalgia	Adductor strain	Osteitis pubis
Lesion	Rectus–adductor **aponeurosis**	**Adductor longus** (pubic ramus)	**Pubic symphysis** inflammation
Onset	Chronic, exertional	**Acute**, immediate	Gradual, vague
Tender	**Inguinal region / lower abdomen**, adductor longus	Subcutaneous border of pubic ramus	**Directly over symphysis**
Provoked by	**Sit-ups, Valsalva**	**Resisted adduction**	Resisted rectus testing, adduction / abduction
Imaging	Oedema at aponeurosis	Avulsion / oedema at pubis	**Erosions, symphysis widening**, marrow oedema
Failed nonop	**Surgery**	Rehab (repair unproven)	Rest (self-limiting)

How it's asked

"True about athletic pubalgia (TO)?"	→ **fulcrum is the pubic symphysis** ✗ Wrong options: more common in females · bilateral radiating pain · transversus abdominis main muscle
"Movements contributing to athletic pubalgia?"	→ **hip extension and abduction**
"Failed nonop, no hernia, malignancy excluded – next?"	→ **referral to a sports hernia surgeon**

BURNERS & STINGERS

– Knee & Sports · 7 / 61 –

1 Basics

Definition & epidemiology

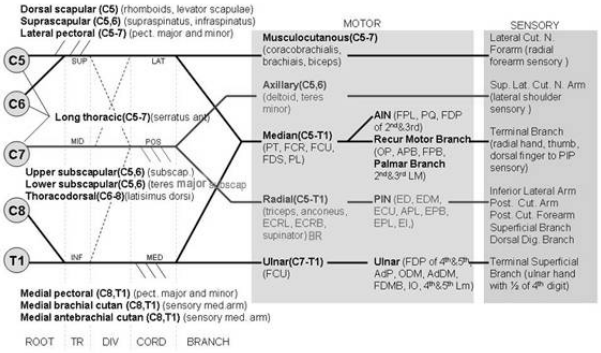
Definition	**transient brachial plexus (or cervical root) neurapraxia** = "**dead arm syndrome**" · **mild traction injury of the brachial plexus (TO)** · reversible peripheral nerve injury
Commonest	**most common nerve injury in sportspersons (TO)** · most common UE neurologic injury in athletes (OB)
Sports	collision: **football**, rugby, ice hockey, wrestling (TO) · under-reported
Recurrence	**1 stinger → 3× risk of another** · **stinger in previous season = strongest risk factor** for injury + lasting symptoms (rugby: 33.9% had a history)
Stenosis	**Torg ratio < 0.8 → 3× stinger risk** (47.5% of stinger group vs 25.1%) · cervical canal + **foraminal stenosis** ↑ risk → work up recurrent burners

Anatomy – C5-6 upper trunk

Commonest level	**C5-6 (upper trunk)** – greater midcervical mobility in young athletes ✗ Wrong options: C3-4, C4-5, C6-7, C7-T1

Root	Muscles tested
C5	**Deltoid, biceps**, rotator cuff (external rotators)
C6	Biceps (**elbow flexion + supination**), wrist extensors
✗ Wrong option: C7	Triceps, wrist flexors
✗ Wrong option: C8 / C8-T1	Finger flexors / interossei

Dorsal root ganglion	fills most of the neural foramen → takes the brunt → **purely sensory** stinger possible

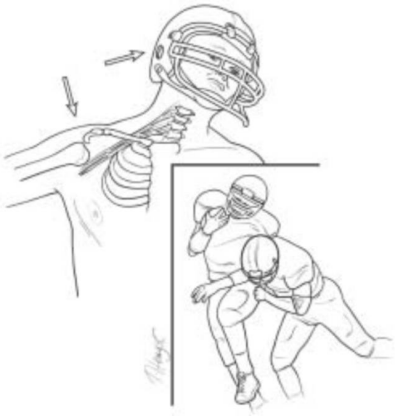


Brachial plexus – motor and sensory innervation

– – Active space – –

2 Mechanism

Mechanism	How	Typical
Traction (commonest)	**Arm / shoulder depressed + neck bent AWAY** from injured side · TO: shoulder + head distracted opposite ways	**Younger** athletes – **plexus**
Compression	**Head turned / extended TOWARD affected side** → root pinched in neural foramen	**Older** athletes – **foraminal stenosis**
Direct blow	**Erb's point**, just superior to clavicle	Plexus



Traction mechanism – shoulder depression with neck flexed away

3 Presentation

Symptoms **unilateral** burning / stinging pain + paresthesia, shoulder → hand · **not dermatomal / not isolated to one area** ✓

Weakness **C5-6: deltoid, biceps**, external rotators · often > 1 shoulder muscle group (ER, flexors, abductors) (TO)

Never bilateral (TO) **bilateral or lower-limb symptoms → spinal cord injury (cervical cord neurapraxia)** until proven otherwise

Neck exam **full, painless cervical ROM**, no tenderness · neck ROM **doesn't** reproduce symptoms · **positive Spurling's possible** ✓ · ROM often ↓ in chronic cases (TO)
✗ Wrong options: neck pain on extension

Course resolves quickly, **doesn't progress** · range: paresthesia → **complete arm monoplegia** · seconds → weeks if severe
✗ Wrong options: progresses 20–25 min

Duration of symptoms	Source
1–2 min	Orthobullets, OB
Seconds to several minutes	OB
15–20 min	OB
Several minutes; **5–10% last up to several weeks**	TO
"Resolution in **6–8 weeks**" accepted as **true**	TO (EXCEPT stem) – read the options

Subtle deltoid weakness mistaken for shoulder injury · **4/5 deltoid → withhold**
✗ Wrong options: greater tuberosity tenderness = cuff contusion · AC pain · O'Brien's · labral shear (play on if strength full)

Differentials

Condition	Distinguishing fact
Stinger	Unilateral UE, C5-6, resolves in minutes, no residual
Acute transient spinal cord injury	**Temporary complete paralysis UE + LE**, resolves **< 24 h**
Central cord syndrome	**UE > LE**, mostly **elderly**
Nerve root avulsion	**Permanent** deficit, poor prognosis · **neurotmesis = most severe "stinger"** (arm-tackle traction)
Guillain-Barré	**Ascending** paralysis, legs → arms → face, **loss of DTRs**

4 Diagnosis & Classification

Key radiology findings

Test	Indication / finding
C-spine radiographs	Usually normal · **recurrent stingers** → rule out **fracture + cervical stenosis**
MRI C-spine	**Bilateral symptoms** (not a stinger) → HNP, stenosis
EMG	**Persistent > 3 wk** (Orthobullets, OB) · **delay 3–4 wk** after injury (OB) · abnormal roots, cords, trunks, peripheral nerves

Formal work-up neurologic findings present, **persistent symptoms**, prior brachial plexus injury · = EMG + radiographs + MRI

Not MRI triggers unilateral symptoms > 15 min · two prior unilateral episodes (→ **radiographs**) · transient deltoid + biceps weakness · **brief LOC → CT head**, not C-spine MRI

5 Management

Arm numbness / tingling after contact

- Bilateral or lower-limb signs → cord injury → **MRI before any RTP**
- Unilateral, fully resolved, full strength + painless neck ROM → **same-day RTP**
- Recurrent stingers → out for game → **C-spine radiographs / imaging**
- Weakness persists → withhold → **EMG / NCS at 3–4 wk**

– – Active space – –

Return to play & red flags (bilateral or recurrent symptoms)

RTP criteria	Contraindications to RTP
Complete resolution of symptoms	**Bilateral symptoms** (until MRI)
Normal (5/5) strength, normal sensation	**Multiple stingers** (until off-field work-up)
Full, painless ROM of neck + shoulder	**Recurring symptoms** (until C-spine X-rays)
Normal cervical exam – compression, **Spurling's, Adson's**, resisted head pressure	**Persistent** symptoms / weakness

Not RTP criteria fitting a brace · MRI excluding avulsion · normal EMG · partial sensation + arm overhead

Number of stingers	RTP recommendation
First	RTP after **pain-free full ROM + normal strength**
Second	**Withhold rest of game** → re-examine → next game if no deficit
Third (any duration)	**Withhold rest of game + advanced C-spine imaging** (foraminal / uncovertebral stenosis, disc herniation)
Any, **persistent pain / weakness**	**Withhold** + **electrodiagnostic study**

Same-contest return **allowed after complete resolution** (Orthobullets, OB) · TO counts "**precludes participation in present contest**" as true – read the options

3rd stinger (2 today + 1 last week), resolved in 5 min **withhold from the remainder of the game + pursue further off-field workup**
✗ Wrong options: RTP after halftime · RTP once asymptomatic · 2 wk + EMG (only for persistent weakness)

Biceps weakness > 10 min + 2 stingers last season **hold until off-field work-up incl. C-spine radiographs**
✗ Wrong options: agility test · "benign" immediate RTP · graduated RTP protocol (not a concussion)

Prevention **prevent recurrence** = key · neck collars / rolls (football) · avoid spear tackling · fitted equipment · recurrent stingers → possible **chronic proximal arm weakness + pain**

X vs Y

	Stinger / burner	Cervical cord neurapraxia (transient quadriparesis)
Lesion	**Root or brachial plexus** (peripheral)	**Spinal cord**
Limbs	**Unilateral upper limb**	**Bilateral**, > 1 limb ± legs
Mechanism	Traction, compression, Erb's point blow	Hyperextension, hyperflexion, **axial load**
Resolution	Minutes	**10 min – 48 h** (OB) · minutes – hours (Orthobullets)
Imaging	Radiographs if recurrent	**MRI** before RTP
Same-day RTP	**Yes** if fully resolved	**No**

	Younger athlete	Older athlete
Cause	**Traction** of plexus (midcervical mobility)	**Foraminal stenosis + extension → root compression**

How it's asked

"All true about stingers EXCEPT (TO)?"	**→ bilateral arm weakness + paresthesia**
"Acute transient finding consistent with a stinger?"	**→ unilateral upper-extremity pain + weakness**
"Muscles most likely affected?"	**→ deltoid + biceps**
"Shoulder + finger numbness gone in 15 min, no residual?"	**→ stinger / burner** ✗ Wrong options: transient cord injury · central cord · root avulsion · GBS
"Most commonly involved level?"	**→ C5-6**
"Finding consistent with a stinger?"	**→ physical examination may show a positive Spurling's sign**
"Where were the paresthesias (figure)?"	**→ not isolated to any single area**
"Indication for C-spine MRI before RTP?"	**→ symptoms + signs in both upper extremities**
"Condition before RTP after a stinger?"	**→ no pain, normal sensation, full ROM, 5/5 strength**
"Exam finding that keeps him out?"	**→ full painless ROM but 4/5 deltoid strength**
"Third stinger, resolved in minutes — next step?"	**→ withhold from remainder of game, pursue further off-field workup**
"Persistent biceps weakness + 2 prior stingers?"	**→ hold until off-field work-up incl. C-spine radiographs**
"Risk of another stinger after one?"	**→ 3×**

– – Active space – –

CARDIAC CONDITIONS

– Knee & Sports · 8 / 61 –

1 Basics

Overview

Leading cause — cardiac conditions = **most common cause of sudden death in young athletes → HCM (most common)**, coronary artery disease, commotio cordis

Conditions covered — syncope, HCM, CAD, long QT syndrome, commotio cordis, sudden cardiac death

Sudden cardiac death – epidemiology & causes

Definition — death from a cardiovascular cause **within 1 hour of symptom onset**

Incidence — **0.76–13 per 100,000** (Orthobullets) · up to **1 in 40,000** young athletes (OB) · **2.8×** commoner in adolescent / teenage athletes than non-athletes

Mechanism — arrhythmia → ventricular ectopy → sustained **VT / VF**; usually a **pre-existing condition**

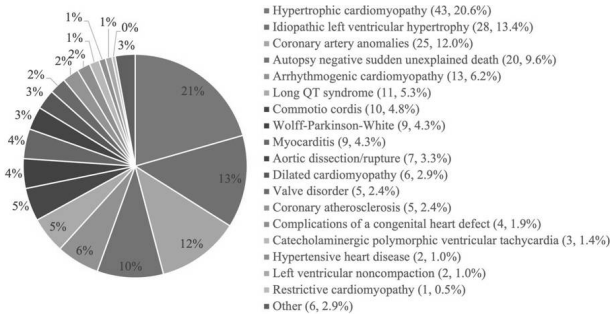
Age group	Usual cause
Birth – 13 yr	**Congenital abnormality**
14 – 24 yr	**HCM**, ARVC, congenital coronary anomalies, WPW, Marfan
Young athletes (TO)	**HCM + ARVC > 50%**; congenital coronary anomalies + premature coronary atherosclerosis **20%**
Structurally normal heart (TO)	**Ion channelopathies** – congenital long QT, Brugada, CPVT
> 35 yr	**Atherosclerotic CAD** – **> 80%** of SCD (TO)

Most common cause, < 35 yr — **hypertrophic cardiomyopathy** ("cardiomyopathy" in TO)
✗ Wrong options: CAD · Brugada · channelopathy · commotio cordis · aortic rupture · myocarditis · long QT · WPW

Cause at autopsy (OB, Garrick)	Share
HCM	**36%**
Coronary anomalies	19%
Increased cardiac mass	10%
Ruptured aorta (Marfan)	5%
Dilated cardiomyopathy	3%
Mitral valve prolapse	2%

HCM share — **36%** (OB, Garrick) · **20.6%**, then idiopathic LVH 13.4%, coronary anomalies 12.0% (OB illustration) – read the options

– – Active space – –



Causes of sudden death in athletes

2nd commonest cause	**coronary artery disease** (Orthobullets; mostly older) · coronary anomalies 19% (OB, Garrick) · **anomalous origin of left main coronary** → compression → ischaemia
Risk factors	**male**, **African-American (black)** race, **cardiomyopathy**, **contact sports**, **basketball** (OB) · false: "commoner in white American males" = the incorrect statement (TO)
High-risk sports (TO)	**basketball (most deaths)**, American football, soccer
Age trend	SCD **commoner in adolescent than adult athletes** (true statement, TO) · risk **increases with age** (Orthobullets) – read the options
High-risk markers (TO)	**unheralded syncope**, **LVH > 30 mm**, **non-sustained VT**, **family history of SCD**, **↓ BP in response to exercise**
Prognosis	**> 90% mortality**; early recognition + **AED** ↑ survival
Survival (TO)	**time from arrest to defibrillation** = single most important factor · **one shock → immediate CPR**
Sequence (TO)	**C-A-B** = chest compressions, airway, breathing · AHA **30:2** compression : ventilation (OB)
AED	primarily for **ventricular fibrillation**; may ↑ survival by **90%** (OB) ✗ Wrong options: AF · bradycardia · tachycardia · asystole

Genes & lab markers

Condition	Inheritance / gene / lab
HCM	**Autosomal dominant**, variable penetrance; mostly **missense** (less often frameshift)
HCM genes	**TNNT2** (cardiac troponin T), TNNI3 (troponin I), **MYL2** (regulatory light chain), **MYL3** (essential light chain)
Long QT	Mostly **AD** (Orthobullets) · AD or AR (TO) · genetic testing for all, **yield 70%** (TO)
Long QT labs	**Basic metabolic panel + magnesium** (electrolytes worsen QT)
SCD work-up	**Cardiac enzymes**; genetic testing for **Brugada, LQTS, ARVC**

Athlete's heart – physiology

Well-conditioned heart (OB)	**↑ wall thickness → ↑ stroke volume** · **↑ vagal tone → ↓ resting HR** ✓ · CO = SV × HR, ↑ in exercise ✗ Wrong options: ↓ SV · ↓ CO · ↓ wall thickness · ↓ vagal tone
Normal ECG in endurance athletes	**ventricular hypertrophy**, **1st-degree (primary) AV block**, **nonspecific ST-T changes in lateral leads**, **resting sinus bradycardia 40 bpm**

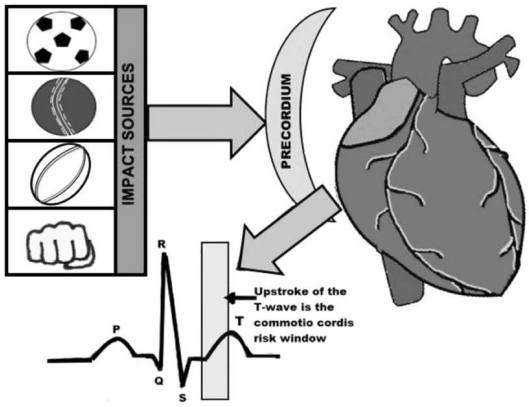
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Athletic heart syndrome	physiological **LVH may mimic HCM** ✓ (mild phenotypes overlap) · asymptomatic, normal exam · **no extra SCD risk**

2 Mechanism

Commotio cordis – mechanism

Mechanism	**blunt blow to anterior chest (precordium) during ventricular repolarization → upstroke of T wave** → aberrant depolarization → **ventricular fibrillation** ✗ Wrong options: during depolarization
Vulnerable window	T-wave upstroke = **1%** of cardiac cycle (Orthobullets) · **< 1%** (OB)
Physics	energy **> 50 J** → myocardial membrane stretch → ion channels via **mechanical-electrical coupling** · **smaller balls ↑ risk** (smaller impact area) · higher energy → structural damage rather than isolated VF
Heart structure	**no structural damage at autopsy** ever found · = sudden death **without structural cardiac damage** (OB) ✗ Wrong options: "often structural damage"



Commotio cordis: T-wave upstroke risk window

Epidemiology	**< 30 cases / yr**; **average age 15**; **M:F 20:1**; contact sports, hard-ball sports (baseball, lacrosse), karate punch, Little League batter hit by pitch
Screening	**not detectable on pre-season physical** ✗ Wrong options: ; unlike HCM
Prevention	**chest protectors / pads not shown to prevent** (baseball, hockey) · soft-core baseballs ≈ standard balls for fatal risk (OB, Janda) · educate coaches + **AED access**

3 Presentation

Red flags & syncope

Best screening tool	**history + physical exam** (high-school athletes)
Symptoms	**chest pain**, **palpitations**, **syncope** with exercise
Murmurs	**diastolic murmur → work-up**; **II/VI systolic murmur**
Syncope	**red flag for serious cardiac disease** in a young athlete · annual incidence **7%**, F > M · risk: cardiac disease, dehydration, female

Vasovagal (neurocardiogenic)	**50%** of syncope; vasodilation + **bradycardia** (↓ sympathetic, ↑ parasympathetic); prolonged standing, heat
Syncope RTP	**medical evaluation before return** to sport

Hypertrophic cardiomyopathy – clinical features

Pathology	abnormal thick / thin myofilament proteins → **asymmetric LV wall + septal hypertrophy → LVOT obstruction**, diastolic dysfunction, ↑ SCD · **interventricular septum in 66%**
Epidemiology	**1 in 500**; present from birth, diagnosed **20–30 yr**; **M:F 1:1** (females more often symptomatic); risk: hypertension, genetics
Symptoms	dyspnoea on exertion, chest pain, dizziness, presyncope / syncope · **> 50% minimal or no symptoms** (OB)
Murmur	**II/VI systolic crescendo-decrescendo** · obstruction at rest **25%**, with provocation **70%**
Murmur ↑	**↓ preload: Valsalva, standing, diuretics** → ↓ venous return → asymmetric septum obstructs LVOT
Squatting / handgrip	↑ preload / afterload; Orthobullets lists them with the provocative manoeuvres (wording inconsistent) – the tested fact is **murmur ↑ with standing / Valsalva**
Other signs	**double carotid pulse**, **double apical pulse**, arrhythmia, family history



Hypertrophic cardiomyopathy specimen

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4 Diagnosis & Classification

Key radiology findings

Test	Finding / role
Echocardiogram	**Nondilated, thickened LV wall** vs age-matched · **80%** accuracy · outflow obstruction, valves · **confirms HCM**; best test with **FHx of SCD** (OB)
Cardiac MRI	**Gold standard for HCM** – when echo inconclusive
ECG	**Most sensitive diagnostic test** (Orthobullets) · **LVH + right atrial enlargement** highly suggestive · Q waves inferior + lateral, **left axis deviation**, inverted T
Ambulatory ECG (Holter)	**All HCM**, 24–48 h → ventricular arrhythmias, SCD risk
Exercise stress test	Preferred over pharmacological · **LVOT gradient → risk stratification** · does not diagnose HCM (OB)
CAD	**Coronary angiography + MR angiography** = gold standard
Commotio cordis	ECG → **ventricular fibrillation**
Long QT	**ECG gold standard** (Orthobullets) · long QT on resting ECG in **~60%** (TO)

SCD work-up cardiac MRI, coronary angiography, echo, exercise stress test, **electrophysiology testing**, ECG

Best HCM test **ECG most sensitive** (Orthobullets) · **echo most effective screen / confirms** (OB) · **cardiac MRI gold standard** (Orthobullets) – match the stem's wording

Pre-participation screening (history, ECG, echocardiogram)

AHA careful **personal + family history + physical exam** = best, most practical screening at any age

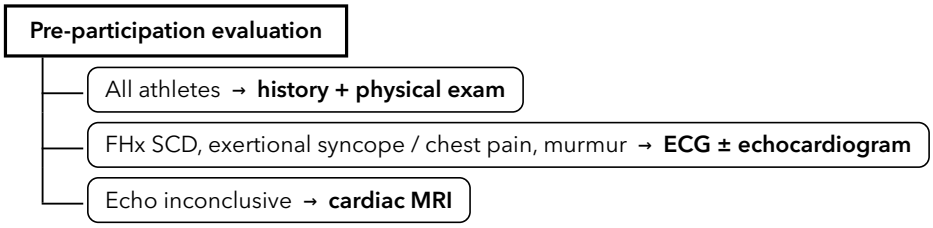
ECG not routine in US **cost**; **98.8%** of mass-ECG cost spent on **false positives** (O'Connor) · H&P more cost-effective · ECG **can** detect SCD causes
✗ Wrong options: "cannot detect" = the EXCEPT answer

Most cost-effective HCM screen **history of chest pain, syncope, FHx of early cardiac death**
✗ Wrong options: echo · ECG · auscultation (may be normal at rest) · stress test

Screening echo cost **> $500,000** to diagnose one HCM case

Black male basketball player, no history **routine history + physical** · ECG only if personal / family cardiac history
✗ Wrong options: ECG · stress test · Hb electrophoresis · lung diffusion

2 relatives died playing sport (AD trait) **echocardiogram**
✗ Wrong options: ECG · renal US · PFT · cervical MRI



Diagnostic criteria

Condition	Criteria
SCD	Cardiovascular death **< 1 h** from symptom onset
HCM	**Echo + ECG + genetic testing + family history**
Commotio cordis	**Witnessed chest blow → collapse + ECG VF**
Long QT, symptomatic (TO)	**QTc > 440 ms (male)**, **> 460 ms (female)**
Long QT, no symptoms / FHx (TO)	**QTc > 500 ms** · or shorter QT + exercise paradoxical QT prolongation, polymorphic VT, or long QT in first-degree relative
Long QT extras (TO)	FHx positive in **40%** · **Holter** for polymorphic VT

Long QT syndrome

Definition prolonged ventricular action potential (QT) → arrhythmia · congenital (ion-channel genes) or acquired (electrolytes, K-channel blockers → ↑ intracellular positive charge)

Prevalence **1 in 2,500–10,000** (Orthobullets) · **1 in 2,500**; long QT in **0.42% of elite athletes** (TO)

Who diagnosed **< 20 yr** · rarer in females (Orthobullets) · **SCD mostly in females < 40 yr** (TO)

Type	Gene / chromosome	Channel	Share	Key fact
LQT1	**KCNQ1, chr 11**	K^+ efflux (loss of function)	**45% – commonest**	**Adrenergic surge** → swimming, diving into cold water (TO)
LQT2	**KCNH2, chr 7**	K^+	25–40%	**Most dangerous – most SCDs (females)** (TO)
LQT3	SCN5A, chr 3	**Na^+**	5–10%	–

LQT1–3 (TO) **~95%** of cases; **12 types** (LQTS-1 to 12)

Acquired drugs antiarrhythmics (**sotalol, amiodarone**), **macrolides, fluoroquinolones**, antipsychotics (haloperidol, olanzapine), cisapride (TO)

Presentation (TO) palpitations, dizziness, **unheralded syncope** (polymorphic VT) · brief **myoclonic jerks → misdiagnosed as epilepsy** · most asymptomatic (Orthobullets)

Torsades de pointes polymorphic VT → VF → SCD · 50% asymptomatic · **Romano-Ward syndrome** (abnormal repolarization → torsades, VT, SCD)

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5 Management

Return-to-play criteria

Condition	Return to play
HCM	**Absolute contraindication to vigorous exercise + competitive sport** · mild–moderate activity allowed
Commotio cordis	**No restrictions** · precautions against future chest blows
Long QT	By genotype, presentation, pacemaker need; intermittent ECG; **no exercise restriction** (Orthobullets) · **avoid all sports with adrenergic surges** (TO) – read the options
Syncope	**Medical evaluation first**
Cardiac condition found	**Avoid vigorous exercise + sport** (OB)

Hypertrophic cardiomyopathy – treatment

Lifestyle asymptomatic, mild LVH, no obstruction → **observation**, repeat echo if symptoms change, **hydration**, **avoid vigorous exercise**

Medical symptomatic → **beta-blocker (metoprolol) first line** (non-vasodilating) · calcium-channel blockers · **symptom relief only**

Prognosis **1–2% annual mortality**

Coronary artery disease

Features older patients; risk: hypertension, family history; mostly **asymptomatic** (chest pain, palpitations, syncope)

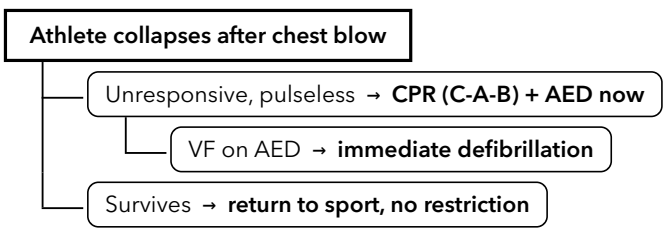
Treatment lifestyle · β-blocker, aspirin, ACE inhibitor, nitroglycerin, CCB · angioplasty + stenting, CABG

Commotio cordis – AED & defibrillation

Treatment **immediate cardiac defibrillation** · CPR alone inferior
✗ Wrong options: IV epinephrine · pericardial needle · rescue breathing · fluid bolus

Collapsed, apnoeic, pulseless pitcher **protect airway, start CPR, prepare to cardiovert**
✗ Wrong options: smelling salts · shade + reverse Trendelenburg · seizure precautions · slow move to spine board

Prognosis mortality **59%** (Orthobullets); **time to defibrillation predicts survival** ✓



Long QT syndrome – management (TO)

All patients **beta-blocker** unless contraindicated · avoid sports with **adrenergic surges**

ICD **resuscitated cardiac arrest**, **β-blocker-resistant symptoms**, β-blocker contraindicated · ± asymptomatic high-risk

On-field emergency – BLS, ACLS & collapsed athlete (TO)

Event sequence (TO) preparation (drills, AED location, ambulance route) → triage → **primary survey ABCDE** → resuscitate → **AMPLE** history → secondary survey → definitive care

Triage colour (TO)	Category
Red	**P1** — immediate, life-threatening but treatable
Yellow	P2 — urgent, can wait briefly
Green	P3 — minor
Blue	Expectant — poor prognosis
Black	Dead

High-quality CPR (TO)	Value
Rate	**100–120 / min**
Depth (adult)	**5–6 cm** · child **2 inches** (one hand) · infant **1.5 inches** (two-thumb)
Ratio	**30 : 2** · child with 2 rescuers **15 : 2**
Interruptions	**< 10 s** · full recoil · avoid over-ventilation
Rescue breaths	1 every **5–6 s** adult · 3–5 s child
Pulse check	Carotid **≤ 10 s**

AED (TO) pads right upper chest + left lateral chest · shock → **resume CPR 2 min** · survival **↓ 10% per minute** of delay · paediatric pads 1–8 yr (or adult pads antero-posterior) · **biphasic 120–200 J** / monophasic 360 J

Airway (TO) head tilt–chin lift · **jaw thrust if C-spine injury** · OPA sized corner of mouth → earlobe · NPA tip of nose → earlobe · LMA supraglottic · **cricothyroidotomy = last resort (CICO), avoid in children < 5–12 yr** · cuff pressure < 20 cm H_2O · **waveform capnography = gold standard tube confirmation** ($ETCO_2$ 0 = misplaced; ≥ 20 during CPR; **< 10 after 20 min → poor CPR**; abrupt ↑ to **≥ 40 = ROSC**)

ET-tube drugs (TO) **NAVEL**-type: naloxone, atropine, epinephrine (vasopressin), lidocaine at **2–2.5× IV dose** · **not amiodarone**

ACLS drug (TO)	Dose
Epinephrine (arrest)	**1 mg IV/IO every 3–5 min** (1:10,000)
Amiodarone (VF / pulseless VT)	**300 mg**, then **150 mg**
Lidocaine (no amiodarone)	1–1.5 mg/kg, then half
Adenosine (PSVT)	6 mg → 12 mg
Atropine	1 mg, max 3 mg
Magnesium	**1–2 g IV — torsades**

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Rhythms (TO)	shockable: **VF, pulseless VT** (150–250 bpm, QRS > 0.12 s) · non-shockable: **asystole, PEA** → CPR + epinephrine + reversible causes (**Hs & Ts**: hypoxia, hypovolaemia, H^+, hypo / hyperkalaemia, hypothermia, hypoglycaemia · tension pneumothorax, tamponade, thrombosis PE / MI, toxins)
Post-arrest (TO)	10 breaths/min, **SpO_2 92–98%**, $PaCO_2$ 35–45 · SBP > 90, **MAP > 65** · **TTM 32–36 °C**
Anaphylaxis (TO)	**IM adrenaline 1:1000**: adult **0.5 mg** · 6–12 yr 0.3 mg · < 6 yr 0.15 mg · + O_2, hydrocortisone 200 mg, chlorpheniramine, salbutamol
Choking (TO)	encourage cough → **5 back blows + 5 abdominal thrusts** → unconscious = BLS
Spinal / head injury on field (TO)	assume C-spine injury · collar (chin → suprasternal notch) · neutral head · **log roll ≥ 4 rescuers** · imaging by **NEXUS ("NSAID")**: Neuro deficit, Spinal tenderness, Altered alertness, Intoxication, Distracting injury · GCS **13–15 mild, 9–12 moderate, ≤ 8 severe**

6 Complications

HCM	SCD, arrhythmias, CHF, stroke

Anticoagulation-related wound complication

Toe lesions 9 days after aortic valve replacement, on anticoagulation	**emboli from cardiac surgery**; no gangrene / infection / sepsis → **observation ± late debridement** ✗ Wrong options: reverse anticoagulation + debridement / forefoot amputation · urgent arteriogram · 40 °C water bath



Embolic toe lesions after valve replacement

X vs Y

– – Active space – –

	Athletic heart syndrome	HCM
LVH	Physiological — **may mimic HCM**	Pathological, asymmetric septal
LV cavity	**Maintained**	**Diminished / nondilated**
Ventricle	**Left** (✗ Wrong options: right)	Left
SCD risk	**No extra risk**	**Most common cause < 35 yr**
Clinical	Asymptomatic, normal exam	Exertional symptoms, FHx SCD, systolic murmur
ECG	Sinus bradycardia, 1st-degree AV block, nonspecific ST-T	**Lateral T-wave inversion** = pathology; LVH + RA enlargement

Pathophysiology in options	Condition
Dynamic outflow obstruction by enlarged LV	**HCM**
Plaque blocking coronary arteries	CAD (angina / MI, unlikely in the young)
Direct impact during ventricular repolarization	**Commotio cordis**
Lengthening of QT interval	Long QT → VT, torsades
AV-node dysfunction	Heart block

How it's asked

"Most common cause of sudden death in athletes (TO)?"	→ **cardiomyopathy**
"Incorrect statement on SCD (TO)?"	→ **commoner in white American males** (✓ black athletes)
"Most important factor for SCD survival (TO)?"	→ **time from arrest to defibrillation**
"Most common SCD cause, 20–30 yr?"	→ **hypertrophic cardiomyopathy**
"High-school basketball death — autopsy?"	→ **HOCM**
"16-yr-old collapses on court — commonest cause?"	→ **HCM**
"Pre-participation ECG — all true EXCEPT?"	→ **ECG cannot detect causes of SCD**
"FHx of relatives dying in sport — best test?"	→ **echocardiogram**
"Most cost-effective HCM screen?"	→ **history of chest pain, syncope, FHx early cardiac death**
"Black freshman basketball player, no history?"	→ **routine history + physical**
"Seen in a well-conditioned athlete's heart?"	→ **decreased resting heart rate**
"Athletic heart syndrome vs HCM — true?"	→ **LVH may mimic HCM**
"Commotio cordis — treatment?"	→ **immediate cardiac defibrillation**
"AED is for which rhythm?"	→ **ventricular fibrillation**

– – Active space – –

"Pitcher hit by line drive, apnoeic, pulseless?"	→ airway, CPR, prepare to cardiovert
"MMA blow to precordium — true?"	→ time to defibrillation predicts survival
"Helmet to chest, AED shows VF — pathophysiology?"	→ direct impact during ventricular repolarization
"Highest SCD risk after blunt trauma?"	→ 22-yr-old African-American male football player
"Embolic toe lesions after valve replacement?"	→ observation, possible late debridement
"Long QT > 480 ms — most dangerous type (TO)?"	→ type 2
"Commotio cordis — return to sport?"	→ no restriction
"HCM diagnosed — sport?"	→ absolute contraindication to vigorous exercise
"Adult CPR compression rate + depth (TO)?"	→ 100–120/min, 5–6 cm
"Confirmation of ET-tube placement (TO)?"	→ waveform capnography
"Drug not given via ET tube (TO)?"	→ amiodarone
"IM adrenaline dose in adult anaphylaxis (TO)?"	→ 0.5 mg of 1:1000

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CONCUSSIONS (MILD TRAUMATIC BRAIN INJURY)

– Knee & Sports · 9 / 61 –

1 Basics

Definition & epidemiology

Definition	subset of **mild TBI → acute transient impairment of neurologic function** 2° to an **impulsive force transmitted to the head** (blow to head, face, neck or elsewhere)
Functional, not structural	**standard CT / MRI normal** in the vast majority ✗ Wrong options: MRI often diagnostic
LOC	**not required** for diagnosis · occurs in only **~10%**
Incidence	**1.6–3.8 million** sports concussions / yr (Orthobullets) · "nearly 4 million" (OB) · **5–9% of all sports injuries** · ~6% of NCAA football injuries · rising (awareness, youth)
Who / where	**TBI = leading cause of sport-related death** · **American football** = majority · **competition > practice**

Pathophysiology – neurometabolic cascade

Step	Clinical result
Abnormal **neuronal ionic flux**	Headache, **photophobia, phonophobia**
Altered neurotransmitter release (**glutamate**)	Impaired cognition, **amnesia**, slowed reaction time
Energy depletion (correcting imbalance)	**Vulnerable to second injury (second-hit)**

After concussion **re-injury risk significantly ↑** (biochemistry altered, neuroprotection disrupted)
✗ Wrong options: risk same as pre-injury

Risk factors & prevention (helmets, prior concussion)

Risk factor	Fact
Prior concussion	**Strongest** · **2–8×** (Orthobullets) · **2–6×** (OB) – read the options
Female	Higher risk than males (✗ Wrong options: male gender) – possibly oestrogen / cerebral blood flow
Age < 18 (younger)	Less "cognitive reserve" (✗ Wrong options: increasing age)
Player-to-player contact	**Football, wrestling, soccer, basketball** highest
Mood / learning disorders, migraine, anxiety / depression	**Complicate diagnosis + recovery** – not proven to predispose
Equipment quality	**No relationship established**

Helmets / headgear ↓ impact force · ↓ head + neck injuries in **youth hockey, alpine, equestrian, cycling, motor sports**

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Equestrian injuries	**helmets** = greatest opportunity ✗ Wrong options: knee pads, wrist guards, boots, quick-release stirrups
Risk compensation	protective gear may ↑ injury by enabling dangerous technique
Rule changes	work when one clear mechanism · **upper limb-to-head contact banned in American football (~50% of concussions)** · red cards for high elbows (soccer)
Preparticipation evaluation	number of prior concussions, symptoms, recovery length · **low-impact injury + disproportionate symptoms → particularly vulnerable athlete**

Genes & lab markers

Apolipoprotein E	genetic test under research (prevention)
CTE proteins	**tau**, **beta-amyloid**, **TDP-43** (after repetitive diffuse axonal injury)

Consensus guidelines (International Conference on Concussion in Sport)

Meets every 4 years	consensus statement · 1st 2001 · **grading removed 2004** · **all classification systems removed 2008** · latest focus = assessment tools + **timing of return to play**
SCAT origin (TO)	**2nd conference, Prague 2004**
Zurich 2008 (3rd)	**no same-day RTP even if symptoms resolve** · applies to **minimal- and noncontact** athletes · applies to **children ≥ 10 yr** (younger → age-appropriate checklists) · young elite athletes → more conservative · **CT contributes little** to early diagnosis (use only if structural lesion suspected) · multidisciplinary approach

3 Presentation

Symptoms & mandatory signs

Headache	**most common symptom — 70%** · myofascial tension (posterior neck, base of skull) vs post-traumatic (forehead / vertex pressure)
Somatic	dizziness, balance problems, **nausea / vomiting**, vision change, **diplopia**, **tinnitus**, photophobia, phonophobia
Cognitive	**"in a fog"** / slowed, poor concentration, forgetful
Emotional	**lability**, irritability, sadness · **sleep** — insomnia, drowsiness, more or less sleep
Mandatory signs	**LOC (~10%)**, **lying motionless > 5 s**, slow to get up, **confusion / disorientation**, clutching head, **amnesia**, **vacant look / blank stare**, motor incoordination, **ataxia / gait instability** → further evaluation ± removal (sport-specific)
Cervical muscle tenderness	**not a concussion sign** → C-spine evaluation, but no 5–10-day concussion protocol (✓ EXCEPT answer)
Handball player faints, plays on, progressively aggressive, can't recall last goal (TO)	concussion → **immediately remove from play area** + **Maddocks score** ✗ Wrong options: counsel 15 min · electrolyte drink · "plays best aggressive"

4 Diagnosis & Classification

Sideline evaluation

Primary survey	ABC · **altered mental status → assume C-spine injury** (collar + board) · **never leave player alone**
Secondary survey	**remove from field** + immediate **cognitive evaluation by licensed healthcare provider** · no provider → remove entirely + urgent physician referral
Diagnosis	**clinical judgement = gold standard** · no tool diagnoses concussion alone

Red flags – skull base fracture, diffuse axonal injury & emergent imaging

Red flag → ER for advanced imaging	Not a reason for transport
Deteriorating mental status, restless, agitated, combative	**Transient unilateral UE numbness** resolving (stinger)
Severe / worsening headache	Headache alone, normal cognition
Focal deficit – unequal pupils, abnormal EOM, motor / sensory	Mild headache + shoulder pain + **prior concussion**
Neck pain / tenderness (C-spine)	**Photophobia + dizziness < 15 min**, no focal / cognitive signs
Bloody otorrhea, **mastoid ecchymosis** (skull fracture), **blepharohematoma** (eyelid haematoma)	
GCS < 15, LOC, amnesia / confusion (OB list)	

LOC + bloody discharge from external auditory meatus	skull fracture → **transfer to ER for advanced imaging** ✗ Wrong options: SCAT5 · graduated RTP · practice tomorrow
Mastoid ecchymosis + eyelid haematoma, no LOC	**emergent transport** to ED
Diffuse axonal injury (severe closed head injury)	**no return to football / contact sports** ✗ Wrong options: return after 6 mo, 1 yr, 2 yr asymptomatic

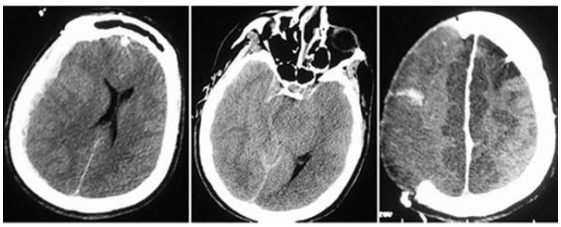
Key radiology findings

Modality	Fact
Imaging overall	**Usually unnecessary** · indicated for red flags above, or **subacute / chronic trauma with persistent symptoms**
CT head	**Most used acutely** · normal in vast majority · shows fracture, haemorrhage, contusion, mass effect, herniation
MRI brain	Superior detail · **DWI most sensitive to shear injury** · standard MRI normal
Functional MRI	↑ **cerebrovascular reactivity** in acute post-concussive period
Negative CT	**Does not clear RTP** · not diagnostic of concussion

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Most sensitive test for deficits after mTBI

neuropsychologic testing · its RTP role not fully defined
✗ Wrong options: head CT · MRI · skull radiographs · sideline assessment



Boxer sustains subdural hematoma
Left: right-sided subdural with midline shift
Middle: effacement of basal cisterns
Right: status-post craniectomy

Subdural haematoma in a boxer – midline shift, effaced cisterns, craniectomy

Assessment tools – SCAT

SCAT fact	Detail
Use	**Standardised tool, athletes ≥ 13 yr · Child SCAT for ≤ 12 yr (5–12)** (TO)
SCAT5 on-field	**Red flags, observable signs, Maddocks questions (memory), GCS, C-spine assessment**
SCAT5 off-field (private, quiet)	Background, symptom evaluation, cognitive screen (**SAC**), neuro screen, **mBESS**, delayed recall, decision
SCAT5 additions (TO)	On-field section, C-spine assessment, **10-word list** (≥ 5 min between immediate + delayed recall), neuro screen
SCAT5 time	**Minimum 10 min – can't be done correctly in less (TO, OB key)** · OB text: "< 10 min" / "on-field < 15 min" – read the options
SCAT history (TO)	SCAT 2005 → SCAT2 2009 → SCAT3 2013 → **SCAT5 2017** (5th conference)
SCAT6 (TO)	Ideally **within 72 h**, up to **7 days** · **> 7 days → SCOAT6** / Child SCOAT6
SCAT6 immediate (TO)	Observable signs, GCS, C-spine, **coordination + ocular / motor screen**, Maddocks
SCAT6 off-field (TO)	Background, symptoms, cognitive screen, coordination + balance, delayed recall, decision
Limits	**Not stand-alone** for diagnosis, recovery or RTP · rapid screen, not a comprehensive neuro exam
Cost	**Free**, no technology → widely used in **high school**

Current SCAT

SCAT6 (TO, recent key) · **SCAT5** (TO, FNB 2021 key; Orthobullets "most commonly used") – read the year / options

SCAT combines (TO)

GCS (consciousness) + **modified Maddocks** (orientation) + **PCSS** (21 symptoms, 4 categories – somatic, cognitive, emotional, sleep – 7-point Likert) + **SAC** (orientation, concentration, immediate + delayed memory) + neck, balance, coordination exam

Boxing doctor – recovery from prior concussion, fit for sport? (TO)

SCAT
✗ Wrong options: GCS · PCSS · SAC

Not a documentation tool (TO) — **SCAT** · documentation = **SOAP notes** (Subjective, Objective, Assessment, Plan of first aid), **daily sideline injury report**, **training room treatment log**, **Daily Red Cross List** (no / limited / full participation)

Assessment tools — ImPACT & neuropsychological tests

Test	Assesses
ImPACT	**Computer-based** · **verbal + visual memory, processing speed, reaction time, impulse control, symptoms** (attention, working memory) · vs **baseline** or historical controls
SAC	**Orientation** (month?), **immediate + delayed memory** (5 words × 3 trials + 1 at end), **concentration** (5 digits backwards, months in reverse) · pencil-and-paper · no reaction time, balance
BESS	Balance — **3 stances (bipedal, unipedal, tandem) × 10 s**
SOT	**Entire balance system** — vestibular, visual, somatosensory
King-Devick	**Saccadic eye movements**, language, concentration — read numbers L→R, **3 tests** · portable, **2 min**
GCS	**Depth + duration of impaired consciousness** · for **more severe** head injury
EyeBoxCNS	Eye movements · clinic computer — **not sideline**
SCORE	Smartphone app guiding parents + coaches
Helmet sensors	Measure head impact, **not brain → cannot diagnose SRC**

ImPACT logistics — preseason **baseline** · off-field, quiet · **fee-based, computer / tablet, ~30 min** → collegiate / pro · **not stand-alone**; return to baseline alone ≠ RTP

ImPACT baseline — stable over 2 yr in **college** (no annual retest) · **retest annually in high school / youth**

Low baseline neuropsych score — athlete's own reference → **not** a reason to withhold play (✓ EXCEPT answer)

Post-injury neuropsych testing — **not required for all**; no defined role in RTP criteria · RTP stays a **clinical decision**

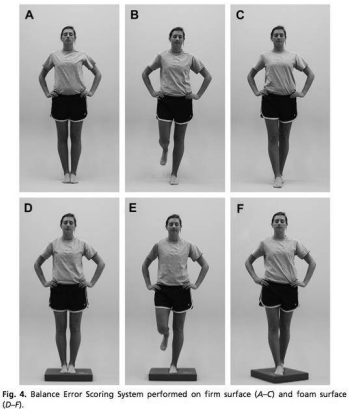
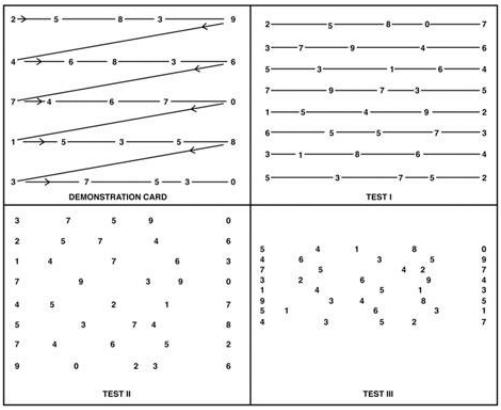


Fig. 4. Balance Error Scoring System performed on firm surface (*A–C*) and foam surface (*D–F*).

BESS — bipedal, unipedal, tandem on firm and foam surfaces

– – Active space – –



King-Devick test cards

5 Management

Head trauma in athlete

- Red flag (worsening GCS, focal deficit, otorrhea, Battle sign) → **ER + CT**
- Any sign / symptom of concussion → **remove, NO same-day RTP**
  - rest 24–48 h → **graduated RTP (≥ 24 h / stage, ≥ 1 wk)**
- Unilateral transient UE paresthesia, normal exam, no concussion signs → stinger → **same-day RTP**

Removal criteria — no same-day return to play

Remove from play (no same-day RTP)	Not a criterion
Any sign / symptom of concussion ("if concerned at all")	**Low baseline neuropsych testing**
No provider experienced in concussion present	Cervical muscle tenderness alone
History of prior concussion	Parental waiver / consent (doesn't permit RTP)
LOC · **amnesia**	Normal neuropsych testing or imaging
Positive exertional stress test (symptoms recur with exertion)	
Symptoms > 15 min	

Same-day RTP — **contraindicated in every diagnosed / suspected concussion**, even if symptoms resolve · test to pass for same-game return → **none**
✗ Wrong options: neuro exam · SCAT5 · ImPACT · K-D · SAC · SOT clears

Only same-day RTP — **stinger**: unilateral transient UE pain / paresthesia / deltoid weakness, **complete resolution**, normal exam, no concussion signs

Bilateral hand paresthesias / weakness after helmet-on-helmet hit — **spinal cord work-up**, no same-day RTP

NCAA — out ≥ 5–10 days + stepwise rehab — **momentary LOC**, **can't recall events before tackle**, **blank stare**, **gait instability**
✗ Wrong options: cervical muscle tenderness

Lacrosse ball to head, headache + photophobia, better after minutes	**hold out for rest of the day** ✗ Wrong options: RTP on resolution · CT + MRI · "≥ 2 wk rest"

Graduated return-to-play protocol & timing

Stage	Activity	Goal
1 Symptom-limited activity	Daily activities not provoking symptoms	Reintroduce work / school
2 Light aerobic	**Walking, swimming, stationary cycling**, HR **< 70% max** · no resistance	↑ HR
3 Sport-specific exercise	**Running** or skating drills · **no head impact**	Add movement
4 Non-contact training drills	Complex drills (passing) · **start progressive resistance / strength training**	Exercise, coordination, cognition
5 Full-contact practice	Normal training	Confidence, functional skills
6 Return to play	Normal game play	

Each stage	**≥ 24 h** (✗ Wrong options: 12 h · 48 h · 72 h · 96 h) → **earliest RTP = 1 week** (✗ Wrong options: 30 min · 24 h · 4 wk · 2 mo) · usually **7–10 days** (OB)
Number of steps	**6** (Orthobullets, 2012 consensus) · **5** (OB211942, counting asymptomatic day → full-contact practice) — same sequence
Symptoms recur at a stage	**24 h rest → drop back to previous asymptomatic stage**
Correct sequence	**stationary bike → running on a track → running + catching a football** ✗ Wrong options: squats / leg press before stage 4 · track before bike
14-yr-old, dizzy + can't focus after head clash	**RTP after ≥ 1-week graduated protocol** ✗ Wrong options: negative CT · ImPACT back to baseline · later in game · next day
Confirmed concussion with LOC	**RTP after graduated protocol while symptom-free** ✗ Wrong options: next day · after 7 days rest alone · neuropsychiatric clearance
1 wk later, headache + blurred vision at rest, cognition back to baseline	**withhold from all contact — symptoms at rest = absolute contraindication** ✗ Wrong options: graduated RTP · day-7 ImPACT · outpatient CT (symptoms may last up to 2 wk)
Friday concussion, still symptomatic → Monday	**sub-symptom threshold aerobic exercise after baseline symptom reassessment** ✗ Wrong options: strict rest until symptoms resolve · aerobic despite headache · accelerated drills · fixed 7 days regardless

Cognitive rest & predictors of prolonged recovery

Treatment	**cognitive + physical rest 24–48 h** → graduated RTP · **no prolonged strict rest (weeks)**
Cognitive rest	minimise concentration: reading, schoolwork, **video games**, texting, working online, chess → **limiting video games** ✓

– – Active space – –

Cognitive rest	**↓ symptom duration** ✓ · **↑ activity in post-concussion period → prolonged symptoms** (OB217826)
Early aerobic activity (days after injury)	**encouraged, speeds recovery** (OB217914, OB219980) – must stay **below symptom threshold** ✗ Wrong options: delay aerobic ≥ 1 wk
Prolonged recovery predictors	**subacute headache + depression symptoms** ✓ · **ADHD no longer** thought a significant risk factor
Natural history	cognitive deficits, balance + symptoms **improve within the first 2 weeks** · **small % persistent** ✓

6 Complications

Second impact syndrome

Definition	**second head trauma before first concussion's symptoms resolve** – second hit can be trivial
Mechanism	**catastrophic cerebral oedema** from **loss of cerebral autoregulation** ✗ Wrong options: venous / arterial microvascular bleeding
Course	**rapid** progression · **brain stem** often involved · CT shows oedema · **poor prognosis** · **mortality ~50%** ✗ Wrong options: gradual
Prevention	**restrict RTP until symptom-free** ✓ – the reason for no same-day RTP
Associations	**male**, **young age**, **American football** ✗ Wrong options: leading cause of CTE · short-term memory loss

Postconcussion syndrome

Definition	persistent symptoms (headache, confusion, lethargy) **> 10–14 days adults** · **> 4 weeks children** (younger → slower RTS)
Management	**RTP contraindicated** · formal **neuropsychiatric evaluation** · pharmacotherapy controversial · **self-limiting** (vs CTE)

Intracranial haemorrhage

Subdural haematoma	**most common**
Epidural haematoma	**lucid interval** before decline → **neurosurgical decompression + seizure prophylaxis**

Chronic traumatic encephalopathy

Definition	**progressive neurodegeneration** from **repetitive brain trauma** · **permanent** ✗ Wrong options: temporary neuronal derangement
Onset	**years–decades later**, **midlife** · **mean 42 yr** · boxing, football, hockey, wrestling ✗ Wrong options: by age 30
Early features	**behaviour / affect: apathy, irritability, suicidal ideation**, aggression, "short fuse", loss of impulse control · or cognitive: poor episodic memory, **executive dysfunction**, dementia
Late	depression, disinhibition + **parkinsonism** (bradykinesia, tremor, rigidity)

Pathology	tau, beta-amyloid, TDP-43 · **superficial cortical laminae II–III** (Alzheimer: layer III and deeper) · **gross changes present**: cerebral atrophy, enlarged ventricles, diffuse senile plaques ✗ Wrong options: no gross changes
Diagnosis	**post-mortem neuropathology only** · no consensus clinical criteria
45-yr-old ex-NFL running back, cognitive + mood + parkinsonism	**CTE** ✗ Wrong options: early Alzheimer (rare in 40s) · Parkinson's (usually > 60) · ALS (no cognitive change) · FTD
40-yr-old surgeon, ex-college football, worsening irritability + forgetting steps	**CTE** ✗ Wrong options: postconcussion syndrome

X vs Y

	Concussion	**Stinger**	**Cervical cord neurapraxia**
Symptoms	Cognitive, somatic, emotional	**Unilateral** UE pain, paresthesia, **C5-6 weakness**	**Bilateral** / ≥ 2 limbs
Same-day RTP	**Never**	**Yes** if complete resolution, full strength, painless neck ROM	**No**
Work-up	Clinical ± SCAT; CT if red flags	X-ray if recurrent	**MRI**

	Second impact syndrome	**Postconcussion syndrome**	**CTE**
Cause	2nd hit before recovery	Single concussion, slow recovery	**Repetitive** trauma
Timing	Minutes	> 10–14 d adult · > 4 wk child	Years–decades (mean 42)
Outcome	**~50% mortality**	**Self-limiting**	**Progressive**, post-mortem diagnosis

	ImPACT	**SCAT5**
Format	**Computer**, fee, ~30 min	Paper, **free**, ≥ 10 min
Setting	Off-field, baseline comparison	On-field + off-field
Users	College / professional	High school (widely)

How it's asked

"Handball player faints, aggressive, can't recall last goal (TO)?"	**→ immediately remove from play area**
"Remove from competition for each EXCEPT?"	**→ low baseline neuropsychological testing**
"Confused pro footballer, no LOC – same-day RTP?"	**→ return to play should not be allowed**

– – Active space – –

"Which athlete may return same day?"	→ **unilateral transient arm paresthesia / deltoid weakness that resolves (stinger)**
"Same-game return — hit to left side of neck, no LOC, normal cognitive exam, left arm paresthesias resolve after several minutes?"	→ **yes — transient unilateral brachial plexopathy (stinger)** ✗ Wrong options: LOC 10 s · amnesia · can't repeat words · bilateral paresthesias
"NCAA 5–10 day protocol for each EXCEPT?"	→ **cervical muscle tenderness**
"Zurich 2008 — symptoms resolved after injury?"	→ **may not return to play**
"Test to pass for same-game return?"	→ **none of the above**
"Headache + photophobia, better after minutes on sideline?"	→ **held out for the remainder of the day**
"Minimum time per RTP stage?"	→ **24 hours**
"Earliest full return if asymptomatic through protocol?"	→ **1 week**
"Correct activity sequence after symptoms resolve?"	→ **stationary bike → track running → running and catching**
"Safe return for 14-yr-old basketballer?"	→ **after a one-week minimum graduated protocol**
"Headache + blurred vision at rest 1 week later?"	→ **withhold from all contact activity**
"Confirmed concussion with LOC — when to return?"	→ **after graduated protocol while symptom-free**
"Still symptomatic on Monday after Friday concussion?"	→ **sub-symptom threshold aerobic exercise after reassessment**
"Which is not a documentation tool (TO)?"	→ **SCAT**
"Score to declare boxer fit after concussion (TO)?"	→ **SCAT**
"Which SCAT is current (TO)?"	→ **SCAT6 (recent key) · SCAT5 (FNB 2021 key)**
"Computer-based test of attention, memory, processing speed?"	→ **ImPACT**
"Most sensitive test for deficits after mTBI?"	→ **neuropsychologic testing**
"Tool correctly paired with description?"	→ **ImPACT — memory, processing speed, reaction time, impulse control**
"True about assessment tools?"	→ **SCAT5 must take at least 10 minutes**
"Describe ImPACT?"	→ **computer-based: verbal + visual memory, processing speed, reaction time, impulse control**
"Example of cognitive rest?"	→ **limiting video games**
"True after brief LOC in 15-yr-old?"	→ **cognitive rest decreases duration of symptoms**

"Correlates with prolonged recovery?"	→ subacute headache + depression symptoms
"Most accurate statement about SRC?"	→ improves within 2 weeks; small % persistent
"Second impact syndrome characterised by?"	→ preventable by restricting RTP until symptom-free
"Second impact syndrome — true?"	→ mortality approaching 50%
"CTE characterised by?"	→ apathy, irritability, suicidal ideation
"Retired running back, dementia + parkinsonism?"	→ chronic traumatic encephalopathy
"Ex-football surgeon, irritability + forgetfulness?"	→ chronic traumatic encephalopathy
"Diffuse axonal injury — return to football?"	→ no return to football
"Sign warranting emergent transport?"	→ mastoid ecchymosis + eyelid haematoma
"LOC + bloody discharge from ear canal?"	→ transfer to ER for advanced imaging
"Greatest equipment opportunity in equestrian injury?"	→ helmets
"Most increases concussion risk?"	→ history of previous head injury

– – Active space – –

DISCOID MENISCUS

– Knee & Sports · 10 / 61 –

1 Basics

Epidemiology

Definition	abnormal development → **hypertrophic, disc-shaped meniscus**
Incidence	**3–5%** of the population · highest in **Japanese + Korean** populations (TO)
Side	**lateral** » medial (✗ Wrong options: medial commonest) · **bilateral in 25%** (Orthobullets, OB) · 25–50% (OB) – read the options (✗ Wrong options: > 75%) · examine the other knee
Commonest cause	of **symptomatic clicking / clunking in a child's knee**
Discoid medial (TO)	much rarer · disc-shaped · linked with **trauma / tears**

Aetiology & pathology

Aetiology	**failure of apoptosis in utero** (Orthobullets, OB) · the normal meniscus develops from a cartilaginous anlage and **is never discoid** during development (OB)
Aetiology traps	✗ Wrong options: **COL5A1** (female ACL rupture) · landing biomechanics · quadriceps dominance (ACL) · softening of cartilage with intact surface (OCD)
Tissue	**disorganised type I collagen** – fewer fibres, heterogeneous course, lost continuity (Atay, Papadopoulos) · TO: ↓ collagen density, **mucinous change**, ↑ thickness, enlarged surface, **↓ peripheral vascularity**
Pathology traps	✗ Wrong options: incomplete ossification of ACL tibial insertion (tibial eminence fracture) · patellar ossification variant (bipartite patella) · proximal tibial dyschondrosis (Blount) · cartilage sleeve–ossific nucleus separation (patellar sleeve)
Why it tears	**↑ thickness**, **poor tissue quality**, **instability / lack of peripheral attachments**, ↓ vascularity
Tear site + pattern	most tears **posterior or middle** (Kramer) · commonest pattern **horizontal** (TO) · OB cases: longitudinal LM tear, radial tear

3 Presentation

Symptoms	pain, clicking, **mechanical locking** · often symptomatic in **adolescence** · most children **asymptomatic**
Snapping knee syndrome	commonest presentation of a symptomatic discoid · classic for the **unstable (Wrisberg) variant** · clunk going **flexion → full extension** · TO: "meniscal thud" at **terminal extension (10–20°)**
Age (TO)	**3–4 y**: often painless snapping · **8–10 y**: pain with activity, more acute
Exam	mechanical symptoms **most pronounced in extension** · **loss of extension** (lacks 10°, ROM 10–85°) · lateral fullness, lateral joint line tenderness

Painless clunk at 5–6 y, no trauma	**discoid lateral meniscus (abnormal meniscal morphology)** ✗ Wrong options: ACL agenesis (hypoplastic LFC, shallow notch, absent tibial spine) · medial plica (older athletes) · grade IV chondral flap (rare < 9 y) · PVNS · osteomyelitis (systemically ill)
12 y, snapping, lacks 10° extension, wide lateral joint	**discoid meniscus** ✗ Wrong options: OCD · medial plica (medial snapping) · Osgood–Schlatter (tibial tubercle) · Sinding-Larsen–Johansson (inferior pole patella)

4 Diagnosis & Classification

Key radiology findings

Study	Finding
Radiographs (AP + lateral)	**Normal in most children** · severe discoid: **widened lateral joint space (up to 11 mm)**, **squaring of lateral femoral condyle**, **cupping of lateral tibial plateau**, **hypoplastic lateral intercondylar spine**
MRI – study of choice, most sensitive	**≥ 3 consecutive 5-mm sagittal slices with meniscal continuity ("bow-tie" sign)** – pathognomonic
MRI sagittal / coronal	Thick, flat meniscus across the **whole lateral compartment** · ↑ lateral meniscus-to-tibia width ratio · tear in symptomatic cases
MRI pitfalls (TO)	**Vacuum phenomenon** mimics a tear in a discoid · **Wrisberg type mimics a floating meniscus**



Discoid lateral meniscus – widened lateral joint, squared lateral condyle

Most likely X-ray sign	**squaring of the lateral femoral condyle** (though radiographs are most often normal)

– – Active space – –

X-ray traps ✗ Wrong options: joint-space **narrowing** · cupping of the **condyle** · squaring of the **plateau** · **hyperplastic** spine · shallow notch (ACL) · Segond (ACL) · patella alta · chondrocalcinosis (CPPD) · tricompartmental OA

Imaging finding	Diagnosis
≥ 3 consecutive sagittal bow ties	**Discoid meniscus**
Double anterior horn · double PCL	**Bucket-handle tear** (double PCL usually medial)
T2-bright cyst within meniscus	**Meniscal cyst**
High T2 / PD signal in patellar cartilage	**Chondromalacia patellae**

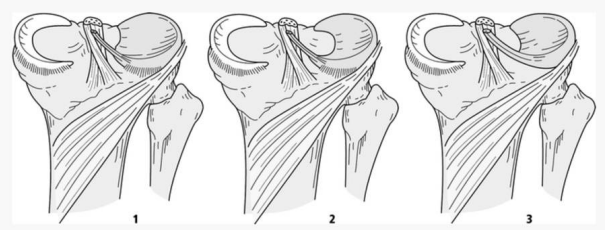
Diagnostic criteria

MRI criterion continuity of anterior + posterior horns on **≥ 3 consecutive 5-mm sagittal images**

Watanabe classification

Type	Shape / coverage	Posterior attachment	Key fact
I Complete	Disc / block – **entire** lateral plateau	Normal	**Most common** (OB) · instability commoner in complete + younger
II Incomplete	Semilunar – **≤ 80%** (OB) · < 80% / up to 80% (TO)	Normal	
III Wrisberg	Near-normal size + shape	**No meniscotibial (coronary) attachment – only Wrisberg MFL**	**Least common**, hypermobile → **snapping knee** · needs **stabilization**

Popliteomeniscal fascicles (TO, MRI) hypoplastic / torn posterior fascicles → gap between capsule and LM, rounded posterior edge, lost bow tie → **hypermobile lateral meniscus** (slips in and out, degenerates) · may coexist with discoid LM · **discoid medial meniscus = rare**



Watanabe classification

Frequency order complete > incomplete > Wrisberg (OB) · TO: complete + incomplete commoner, Wrisberg least common

Watanabe (1969) arthroscopic appearance · **plateau coverage + posterior attachment** (TO) ✗ Wrong options: "classified by Wrisberg" (TO)

Neuschwander (TO) Wrisberg type = "**lateral meniscal variant with absence of the posterior coronary ligament**" (not truly discoid)

Wrisberg type (TO) occurs **younger**, **no trauma**

Instability (Klingele)	**28% peripheral rim instability · anterior horn / anterior third commonest** > posterior horn > body · complete detachment least common · Good: tears + instability each **77%** ✗ Wrong options: instability uncommon
Tears at surgery (Klingele)	**69% (69.5%)**

5 Management

Algorithms

Discoid meniscus (OB / Orthobullets)

- Asymptomatic, no tear (incl. incidental at arthroscopy) → **observe / do nothing**
- Pain, mechanical symptoms, tear → **arthroscopic saucerization ± repair**
  - Unstable / detached rim (Wrisberg) → **saucerization + stabilization (repair)**
- Total meniscectomy (open or arthroscopic) → **avoid – early OA**

Discoid lateral meniscus (TO)

- Asymptomatic → **nonop, close follow-up**
- Child (usually complete), mechanical symptoms → **partial meniscectomy ± repair**
- Adult, complete, mechanical symptoms → **partial meniscectomy ± repair**
- Adult, incomplete
  - Pain ± mechanical symptoms → **partial meniscectomy ± repair**
  - Clicking only → **nonop, close follow-up**

Observation – asymptomatic discoid

Painless clicking, 8 y, incomplete, no tear	**observation only** ✗ Wrong options: diagnostic arthroscopy · saucerization · + microfracture · + stabilization
Direct blow, 8 y, asymptomatic discoid on MRI, normal X-ray	**observation** ✗ Wrong options: debridement + repair · ligament reconstruction · cast at 45° · immobiliser + NWB 6 wk
Incidental, untorn discoid at arthroscopy, 38 y with medial pain	**do nothing surgically** to the lateral meniscus · not treated prophylactically ✗ Wrong options: saucerization · meniscectomy · repair · microfracture

Operative criteria

Operate (saucerize ± repair)	Do not operate
Persistent pain, recurrent **locking**, swelling	**Asymptomatic** discoid without tear
Mechanical symptoms / loss of extension	Incidental finding at arthroscopy
Meniscal tear or **meniscal detachment**	Painless clicking only (TO: adult incomplete)

– – Active space – –

Saucerization ± repair / stabilization

Indication for saucerization	**symptomatic meniscal tear** ✗ Wrong options: asymptomatic abnormal shape · valgus instability · varus malalignment · hyperextension
11 y, 6 months snapping, 10° extension loss	**arthroscopic saucerization ± meniscal repair** ✗ Wrong options: PT strengthening · NWB 6 wk · **open lateral meniscectomy – historical** · microfracture
9 y, partial central tear of stable discoid	**arthroscopic saucerization** (partial meniscectomy) · repair only for peripheral instability ✗ Wrong options: total lateral meniscectomy (poor long-term results) · transplant · repair of central tear · cast 6 wk
13 y, torn discoid, effusion, failed 4 wk PT	**arthroscopic stabilization (repair vs partial meniscectomy) + saucerization** ✗ Wrong options: hinged brace · meniscectomy alone (premature OA) · open saucerization
Watanabe I with radial tear, locking, 14 y	**partial meniscectomy ± repair** ✗ Wrong options: observation · repair only (leaves discoid tissue to re-tear) · complete meniscectomy · transplant (viable tissue present)
Technique	debride to an **anatomic-looking meniscus** · **repair if detached** (Wrisberg) · saucerization = remove central disc, keep peripheral crescent
Rim	saucerize to a **6–8 mm peripheral rim** – keeps function, avoids re-tear of larger remnants
Saucerization alone vs with repair	**comparable** ROM, pain, complications · Carter: stabilization doesn't change outcomes or complications ✗ Wrong options: repair inferior
Small knees	**all-inside repair difficult** in young children (small knee, **deep posterior perforation** risk)

Outcomes

Saucerization	successful in **93% (28 / 30 knees)**; rest needed complete meniscectomy for complex tears (Good) · full flexion > 135°

Long-term degenerative change (OB)	Rate
Arthroscopic saucerization	~20%
Partial meniscectomy	~40%
Complete meniscectomy	~90%

Lee (< 15 y)	partial meniscectomy → better radiographic results / less OA than subtotal / total ✗ Wrong options: similar degenerative change

Total meniscectomy – TO view vs OB view

TO	**< 16 y: total meniscectomy may give the longest symptom-free period** · **Wrisberg type → total meniscectomy** (subtotal leaves an unstable rim) or saucerization + suture rim to capsule · stable complete / incomplete → defer unless grooving / chondromalacia · hypermobile medial segment, intact periphery → subtotal meniscectomy / saucerization

Conflict TO above vs OB / Orthobullets: **avoid complete meniscectomy** (~90% degenerative change); **repair / stabilize** a detached Wrisberg rim with saucerization — read the options

X vs Y

	Complete / incomplete	Wrisberg type
Shape	**Disc** / semilunar	**Near normal**
Posterior attachment	**Present**	**Absent** (Wrisberg MFL only)
Stability	Stable; symptomatic when torn	**Hypermobile**
Presentation	Asymptomatic until tear	**Snapping knee**, younger, no trauma
Treatment	Saucerize ± repair if symptomatic	Saucerize + **stabilize** (OB) · total meniscectomy (TO)

How it's asked

"6 y painless clunk, wide lateral joint space?"	**→ abnormal meniscal morphology (discoid)**
"True about discoid menisci?"	**→ squaring of affected condyle with cupping of tibial plateau**
"8 y trampoline, discoid on scope — imaging?"	**→ wide lateral joint, squared lateral condyle, cupped plateau, hypoplastic lateral tibial spine**
"Scope + 3 consecutive sagittal bow ties?"	**→ discoid lateral meniscus**
"5 y clicking, meniscal band filling lateral compartment?"	**→ discoid lateral meniscus**
"Lateral discoid — most likely X-ray finding?"	**→ squaring of the lateral femoral condyle**
"Flat LFC + cupped plateau, locked 10–85° — MRI?"	**→ longitudinal lateral meniscus tear**
"12 y snapping, lacks extension — diagnosis?"	**→ discoid meniscus**
"MRI horn continuity on 3 slices — which X-ray?"	**→ lateral joint widening + squared lateral condyle**
"Aetiology?"	**→ failure of apoptosis during in-utero development**
"Underlying pathology?"	**→ disorganised type 1 collagen**
"11 y snapping, 10° extension loss?"	**→ arthroscopic saucerization of the lateral meniscus and / or meniscal repair**
"Asymptomatic discoid after a direct blow?"	**→ observation**
"Painless clicking, incomplete discoid, no tear?"	**→ observation only**

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"Commonest indication for saucerization?"	→ **treatment of symptomatic meniscal tear**
"9 y discoid with central partial tear?"	→ **arthroscopic saucerization of the meniscus**
"Incidental untorn discoid at 38 y?"	→ **do nothing surgically to the lateral meniscus**
"True about saucerization?"	→ **a 6–8 mm peripheral rim is recommended**
"Torn discoid, failed PT?"	→ **arthroscopic meniscus stabilization and saucerization**
"Watanabe I with radial tear?"	→ **partial meniscectomy ± repair**
"Snapping knee – which type (TO)?"	→ **Wrisberg type**
"Deficient popliteomeniscal fascicles on MRI (TO)?"	→ **hypermobile lateral meniscus**

EAR, EYE, MOUTH INJURIES

– Knee & Sports · 11 / 61 –

1 Basics

Epidemiology

Nose	**nasal bones = most commonly injured facial bone** · nasal fractures **up to 50%** of sports facial fractures · basketball 26.4%, baseball 22.0% · elbow (basketball), ball (baseball), head (soccer)
Eye	baseball, basketball, racquetball, boxing, martial arts · **hyphema = most common intraocular sports eye injury (24%)**
Ear	**auricular haematoma**: wrestling, boxing, martial arts
RTP after facial trauma	~**3 weeks** (limited evidence) · full face shields

4 Diagnosis & Classification

Key radiology findings

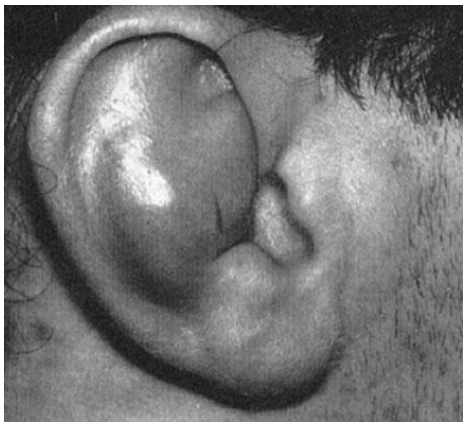
Test	Use
CT	**Facial fractures** (study of choice) · **emergent CT** for ↓ acuity / field cut → traumatic optic neuropathy, orbital + optic canal fracture, nerve sheath bleed, foreign body, brain injury
Fluorescein + UV light	**Corneal abrasion** / corneal foreign body only
Otoscopy	**First step** in auricular haematoma – **TM perforation**
Penlight	Septum – **septal haematoma**, CSF leak, cribriform injury

5 Management

Auricular haematoma

Mechanism	blunt **shear separates perichondrium from cartilage** → blood collects
First step in ED	**otoscopic evaluation** (TM rupture; rule out skull fracture) ✗ Wrong options: compressive dressing + PCP referral · formal surgical decompression · needle decompression **before** otoscopy · head MRI
Treatment	**aspiration (urgent needle decompression)** + **wrapping (compressive dressing)** to prevent re-accumulation · **formal surgical evacuation → recurrent or delayed** cases · prevent with **headgear**
Untreated	**cauliflower ear** (fibrous / abnormal cartilage), perichondritis, infection, necrosis

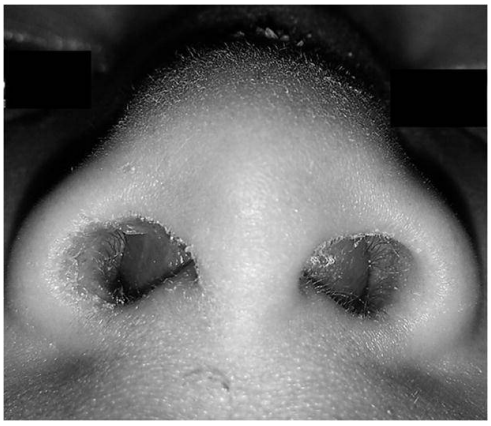
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Acute auricular haematoma

Nasal septal haematoma

What	sub-perichondrial / subperiosteal septal haematoma (submucosal vessel rupture) · uni- or bilateral (bilateral → severe trauma) · **any age** ✗ Wrong options: mostly children
Commonest symptom	**nasal congestion / obstruction (~95%)** · exam: **purple-red bulging septum** ✗ Wrong options: epistaxis
Treatment	**urgent decompression (I&D) under local anaesthesia** + anterior packing ✗ Wrong options: phenylephrine packing (that's for epistaxis)
I&D prevents	**septal cartilage (chondral) avascular necrosis → saddle nose** · **septal abscess within 72 h** (common, not unlikely) ✗ Wrong options: deviated septum · recurrent epistaxis · olfactory nerve damage · turbinate swelling



Septal haematoma — bilateral purple septal bulge

Ocular injury — emergent CT

Injury	Key finding	Treatment
Traumatic mydriasis	**Dilated pupil**, **contusion of iris sphincter**, transient	**Bed rest**
Corneal abrasion	**Fluorescein stain + UV light**	Topical antibiotics + NSAIDs, patch, **non-urgent** ophthalmology
Hyphema	**Blood in anterior chamber** (haze) — may mean vitreous / retinal injury	Patch + **ER / emergent ophthalmology**
Ruptured globe	**Sunken** eye, vitreous exudate	Emergent
Orbital wall fracture	**Extraocular muscle entrapment**	Emergent
Retinal detachment	**Bright flashes**, stabbing pain, **field cut**	Emergent
Traumatic optic neuropathy	↓ acuity, ↓ colour vision, field cut	**Emergent CT**

Red-flag symptoms	vision loss, ↓ acuity, floaters / flashers, diplopia
Boxer: hyphema + ↓ acuity + ↓ colour vision + field cut	**emergent CT** ✗ Wrong options: patch + ophthalmology in 2 days · fluorescein · high-dose steroids before CT · observation
Traumatic optic neuropathy	**blunt 78%** / penetrating 22% · sports + MVA · **optic canal fracture on CT = poor prognosis** · treatment controversial (steroids, optic canal decompression, sheath fenestration) · children: only 29–44% improve

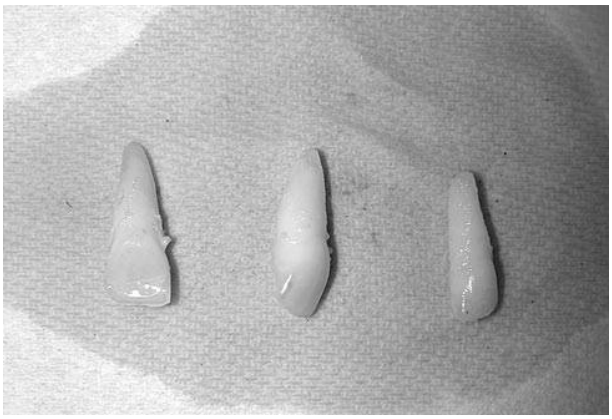
Mandible fracture — airway

Most serious acute complication	**airway obstruction** ✗ Wrong options: blood loss · CVA (carotid injury, rare) · periodontal disease (late) · hearing loss
Why	**mandible suspends the tongue anteriorly** → fracture + supine → **tongue falls back** · soft-tissue swelling · worst with **multiple mandible** or combined maxillary-mandibular-nasal fractures
Airway	intubation often impossible → **surgical airway**

Dental avulsion — milk storage & replantation

Avulsed tooth	**medical emergency** · survival depends on **time out of socket** + **periodontal ligament** damage
Handling	hold by **crown, never root** · **rinse** debris with water / saline · **never brush or scrub** ✗ Wrong options: toothbrush · hydrogen peroxide
Replant immediately	**immediate repositioning / replantation** → bite on towel → dentist / ER ✗ Wrong options: wait for dentist · emergency root canal
If not replanted	keep **moist**: **milk (whole)**, saline, **saliva** / buccal fold, buffered solution kits · **rinse with saline → milk** = usual answer ✗ Wrong options: ice · water · carbonated drink
Children	replanting or carrying in mouth → **aspiration risk**
Dentist	as soon as possible, **preferably within 1 hour**

– – Active space – –



Avulsed incisors

X vs Y

	Auricular haematoma	Septal haematoma
First step	**Otoscopy**	Penlight inspection
Treatment	**Aspiration + compressive wrap**	**Urgent I&D under LA + packing**
Complication	**Cauliflower ear**	**Saddle nose**, septal abscess

How it's asked

"Several front teeth knocked out – initial steps?"	**→ rinse in saline, place in milk**
"Emergent management of tooth luxation?"	**→ immediate repositioning or replantation**
"Three teeth out (figure) – next step besides calling dentist?"	**→ pour saline on teeth, then place in milk**
"Unstable jaw, malocclusion – most serious acute complication?"	**→ airway obstruction**
"Hyphema, ↓ acuity + colour vision, field cut?"	**→ emergent CT**
"Purple bulging septum – true?"	**→ septal avascular necrosis is a potential complication**
"I&D of septal haematoma prevents?"	**→ chondral necrosis**
"Auricular haematoma – best next step in ED?"	**→ otoscopic evaluation**

EXERCISE SCIENCE

– Knee & Sports · 12 / 61 –

1 Basics

Contraction types – isokinetic, isotonic, isometric & eccentric

Type	Definition	Example	Fact asked
Isotonic	**Constant muscle tension (force)** as length changes · improves motor performance	Free-weight biceps curls · leg extensions + curls	**Concentric + eccentric are isotonic** · force–velocity (force constant)
Isometric	**Constant length**, tension ∝ external load · causes hypertrophy (Orthobullets)	Push against an immovable object · plantarflex against a wall 15 s	**Length–tension relationship**
Concentric	Muscle **shortens**, tension ∝ load	Curl with elbow flexing · slow single-leg heel rise	–
Eccentric	Muscle **lengthens** under constant force	Curl with elbow extending · "negative" of a lift	**Most efficient strengthening · highest muscle forces**
Isokinetic	**Constant velocity, variable resistance** (✗ Wrong options: constant effort at variable speed)	**Cybex / dynamometer** · bike at constant pedal velocity, varied resistance	Objective strength testing in rehab · force–velocity (velocity constant)
Plyometric	**Rapid eccentric → concentric** (stretch-shortening)	**Box jumps**, bounding, hopping	Power sports · late rehab
Open chain	Distal end **moves freely**	Seated leg extensions + curls	–
Closed chain	Distal end **fixed**	Squat with planted foot, leg press	–

Eccentric = injury contraction — **muscle strain at myotendinous junction** · **DOMS** · **distal biceps rupture** (forced eccentric load, + hook test)
✗ Wrong options: isometric · concentric · isokinetic · plyometric

Achilles tendinitis rehab after rest + CAM boot — **eccentric strengthening**: rise on both legs on a 1-inch board, **lower slowly on the injured leg only** + gastrocnemius-soleus stretching · avoid steroid injection
✗ Wrong options: box jumps (plyometric) · wall push (isometric) · slow single-leg rise (concentric) · delay rehab

Overstretched muscle — sarcomeres generate little active tension → **< 30% maximal force**

Rule of 10 (TO, isometrics) — **10 s contraction × 10 reps × 10 times/day** · isometrics **contraindicated in hypertensive / cardiac patients** (BP spikes) · **kinetic-chain classification = Steindler 1955** · early rehab closed chain (safe) → late open chain (targeted)

Mobility vs flexibility (TO) — **mobility** = available ROM, set by **capsuloligamentous** structures → ROM / mobilisation · **flexibility** = beyond ROM, **musculotendinous** stretchability → static / PNF stretching · **ballistic stretching obsolete** (tears) · **PNF most effective** for flexibility

– – Active space – –

Physical activity vs exercise (TO)

activity = any movement spending energy · **exercise = planned, structured, repetitive** · proprioception = joint position sense; **kinesia** = sense of movement · hydrotherapy: buoyancy + hydrostatic pressure → early PWB ROM · biofeedback with isokinetic machines

Isokinetic in tennis (TO)

slow + fast speed isokinetic → maximal resistance at every point of ROM, uniform strength, concentric + eccentric · add **core strengthening (lower back + hip)** → kinetic-chain power transfer
✗ Wrong options: ballistic · plyometric (power, added once baseline strength exists) · fartlek (trains energy systems, not muscle)

Open vs closed kinetic chain exercise

Characteristic	Closed chain	Open chain
Distal segment	**Fixed** against considerable resistance	**Free** in space
Stress pattern (TO)	**Linear** (axial loading)	**Rotary**
Joint axes (TO)	Multiple	**One primary**
Joint segments (TO)	**Both move simultaneously**	One stationary, other mobile
Moving joints (TO)	**Multiple**	Isolated joint
Planes (TO)	Multiple (triplanar)	One
Muscles (TO)	**Significant co-contraction** → predictable movement	**Isolated** group, minimal co-contraction
Pattern (TO)	Functional	Often non-functional
Loads (OB)	**Compressive** → ↓ shear, translation, distraction	↑ shear
Use (OB)	**Early rehab**; more often **lower limb**	Later; maximal single-muscle strengthening
Examples	Squat, leg press, bicycle, push-up	**Seated leg extension**, hamstring curl, waving a hand, swing-phase foot

Closed-chain limits (OB)

individual muscles **not maximally strengthened**, not all joint motion normalised · joint compression ↑
✗ Wrong options: distal end free · mainly upper limb · ↓ compression · single joint

After ACL reconstruction

closed chain (quads + hamstrings co-contract, protects graft) · **open-chain leg extension → strong anterior tibial pull → graft stress** · one EMG study: **more hamstring co-activation with isokinetic** than closed chain

Push-up on a weight plate (TO)

closed chain dynamic (distal fixed + up-down movement; instability adds core work) · static push-up = plank (isometric)

– – Active space – –



Push-up on weight plate — closed chain dynamic

Shoulder rehab stage (OB)	Exercise
Early	**Closed chain** — axial load → **glenohumeral compression ↓ rotator cuff demand · co-contraction of periscapular + rotator cuff**
Early–mid	**Facilitated active motion** — proximal segment motion, diagonal patterns
Later	**Resistive active motion** — **open chain** with extrinsic loads (weights, tubing)
Late (return to sport)	**Plyometrics** — slow = stabilisation; fast = stretch-shortening

Muscle fibre types

	Type I	Type IIA	Type IIB
Name	**Slow oxidative** ("slow red ox")	Fast oxidative-glycolytic	**Fast glycolytic**
Metabolism	Aerobic (O2 needed)	Aerobic + anaerobic	**Primarily anaerobic**
Recruitment	**First** (smallest units)	Intermediate	**Last**
Speed / force	Slow, less forceful	Intermediate	**Fastest, most powerful**
Fatigue	**Resistant**	Intermediate	**Most fatigue-prone**
Training	↑ % with endurance (OB)	↑ % with endurance (Orthobullets) · ↑ with resistance (OB)	↓ with resistance training · **DOMS mainly here**

– – Active space – –

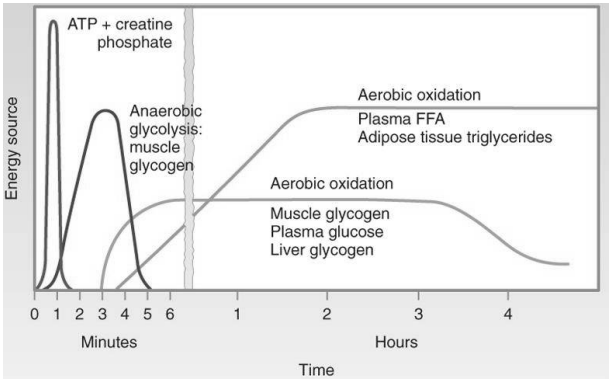
Energy systems & nutrition (ATP-CP, glycogen, fatty acids)

	ATP-CP (alactic)	Glycolytic (lactic acid)	Aerobic
Fuel	**ATP + creatine phosphate**	Muscle glycogen → lactate	**Glycogen + fatty acids** (Krebs, oxidative phosphorylation)
Duration	**< 10 s** high intensity	**2–3 min** (Orthobullets) · **1–4 min** glycogen + lactic acid (OB)	Endurance · **> 4 min** (OB) · predominates from **~60 s** (TO)
Fibre	Type II	Type II	**Type I**
O2	Not needed	Not needed	Required
ATP yield	Immediate, tiny stores	**Low**; lactic acidosis after several min	**High**

Energy continuum at maximal effort (TO) **20 s: PCr 40% · glycolysis 50% · aerobic 10% · 40 s: 5% · 80% · 15%**

Marathon **muscle glycogen early (< 90 min) → free fatty acids after ~90 min** (final hour = FFA) · liver glycogen early, secondary · lactate = intense exercise in low glucose

✗ Wrong options: creatine phosphate · muscle / liver glycogen late



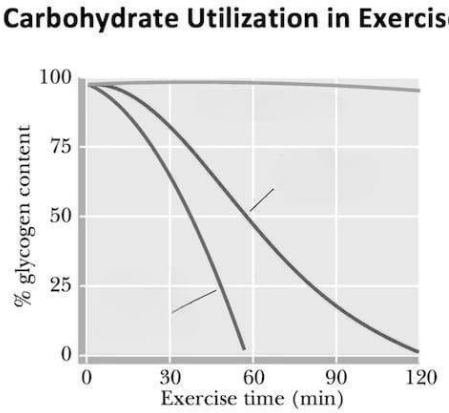
Energy substrates over time

Substrate curve labels (OB) A = ATP + CP · B = muscle glycogen (anaerobic) · **C = plasma FFA + adipose triglycerides · D = muscle glycogen, plasma glucose, liver glycogen**

Glycogen stores (TO) CHO = **1–2%** of energy stores · **80% muscle (300–400 g)**, 14% liver (70–100 g), 6% blood glucose

Glycogen depletion curve (TO)	Exercise	Fuel
A – flat	**Light** (walk, warm-up)	Plasma FFA oxidation, low steady glycogen use
B – gradual fall	**Moderate** (jogging)	↑ glycolysis supplementing aerobic
C – steep fall	**Heavy** (sprints, HIIT)	**Anaerobic glycolysis** + PCr

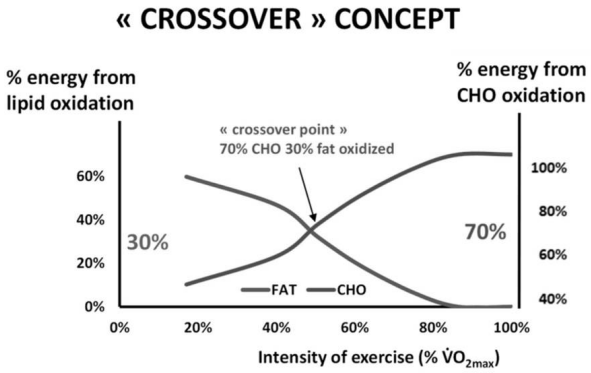
– – Active space – –



Glycogen depletion: light, moderate, heavy

RER = VCO2 / VO2 (TO, respiratory gas analyser)	Fuel
0.7–0.8	**Lipids**
1.0	**Carbohydrate**
> 1.0	Significant anaerobic contribution
↓ RER with	Exercise duration, dietary fat, age, VO2max, % type I fibres
↑ RER with	Intensity, dietary CHO, male sex

Crossover point (TO) fat → CHO predominance at **70% CHO / 30% fat** · **below it → mainly fat oxidation** · **training shifts it right** (more fat) · **sympathetic activity shifts it left**
✗ Wrong options: protein above it · same for everyone · fat ↑ beyond it



Crossover concept

– – Active space – –

Caffeine (TO) – methylxanthine	Fact
Ergogenic	**↑ FFA + glycerol**, **↑ catecholamines**, **↑ muscle fat oxidation (glycogen sparing)** · not **↑ carbohydrate oxidation**
Muscle	↑ SR Ca2+ release, ↑ troponin / myosin Ca2+ sensitivity, ↓ SR reuptake · ↑ Na+/K+-ATPase (K+ uptake, Na+ efflux)
WADA	**Removed January 2004** · old urine limit **12 µg/ml** ≈ 10 mg/kg (~700 mg / 70 kg)
Dose	Drinks 30–200 mg / serve · tablets 100–200 mg · **250–300 mg mild diuresis** · **~500 mg (7 mg/kg) → dehydration**

Carbohydrate loading ↑ CHO **3 days** before + ↓ activity → **maximises muscle glycogen** for events **> 90 min** · best technique = **maintain normal diet** (Orthobullets)
✗ Wrong options: liver glycogen / 60 min · fructose (makes liver, not muscle, glycogen) · diuresis

Fluids core temperature + HR rise **∝ water debt** at onset · replace to **maintain pre-practice weight** · **low-osmolarity (< 10%)** CHO-electrolyte fluid → ↑ gut absorption · **glucose polymers ↓ osmolarity**

Training adaptations – capillary density, fibre types, protein synthesis, periodisation

	Endurance (aerobic)	Strength (resistance)
Key adaptation	**↑ capillary density (most important)**	**↑ cross-sectional area (hypertrophy)**; hyperplasia unlikely
Mitochondria	↑ size, number, density · ↑ respiratory capacity → ↑ VO2max	↑ (Orthobullets weight training) · not aerobic only (OB)
Enzymes	↑ Krebs, fatty acid, respiratory chain · **no change in glycolytic enzymes**	–
Fuel	Fatty acids > glycogen over time	Immediate sources
Fibres	↑ oxidative capacity I, IIA, IIB	↑ IIA, ↓ IIB (OB)
Early gains	–	**↑ motor unit recruitment** ± synchronisation; **hypertrophy after 4–6 wk**
Force	–	Max force **∝ physiologic CSA**
Load	Dynamic exercise ↑ cardiac output via **↑ stroke volume**	High load, low reps

Weight training (Orthobullets) ↑ CSA, strength, mitochondria, capillary density, thickened connective tissue · **adult gains = hypertrophy**

Marathoner vs powerlifter (OB) **marathoner: ↑ capillary density**
✗ Wrong options: ↑ CSA in marathoner · ↑ mitochondria / fat use / type I in powerlifter

Myofibrillar protein synthesis (OB) **resistance exercise above 60% 1-repetition maximum (1-RM)** (anabolic ceiling; sigmoidal dose response)
✗ Wrong options: aerobic (↑ oxidative capacity only) · 30% 1-RM high reps · GH / testosterone (no acute MPS effect; driven by autocrine / paracrine + mechanotransduction)

Periodisation — **planned variation in intensity + duration over a predefined time** (Orthobullets, OB) · **planned manipulation of training variables (load, sets, reps) to maximise adaptation** + prevent overtraining (TO)
✗ Wrong options: rest interval · gradual increase / decline

Cycles (OB) — **macro = training year** · **meso = 3–6 months** · **micro = 3–6 weeks** · goal = peak performance via progressive overload

Models (TO) — **linear** (one attribute, e.g. 4 wk strength) vs **non-linear** (strength + power + endurance together)

Training term	Meaning
Progressive overload (OB)	Stress above what the body is used to → adaptation
Stacking (OB)	Multiple performance-enhancing drugs / supplements for synergy
Static progressive (OB)	Splinting for contracted joints, not exercise
Interval (TO)	Near-maximal work (~**90% max HR**) alternating with recovery; **work : rest ~1 : 1**
Fartlek (TO)	Continuous steady pace + sprints → aerobic + anaerobic; distance athletes; up to 45 min
Circuit (TO)	≥ **6 stations** in sequence

State (TO)	Performance	Recovery
Supercompensation	**Balanced training** → dip → rebound **above baseline**	–
Functional overreaching	Temporary decline → supercompensation	Short
Non-functional overreaching	Short-term decrement ± maladaptation	**Days–weeks**
Overtraining syndrome	**Long-term** decrement (unbalanced training / non-training stress / rest)	**Weeks–months**

Continuum recovery (TO, Meeusen 2006) — **FOR: days to ~2 weeks** (supercompensation; e.g. pre-season camp) · **NFOR: weeks–months**, full recovery with rest · **OTS: months or longer**, may end a career · no improvement after **2 weeks of rest** → think OTS

OTS hallmark (TO) — **inability to sustain intense exercise + ↓ sport-specific performance despite maintained / ↑ load** ("unexplained underperformance syndrome") · commonest in **high-volume endurance sports** (swimming, triathlon, cycling, rowing)

OTS type (TO)	Sport	Features
Parasympathetic	**Aerobic / endurance**	Fatigue, depression, **bradycardia**, ↑ HRV, low catecholamines
Sympathetic	**Anaerobic** (sprint, jumps, throws)	Restlessness, irritability, insomnia, **tachycardia, hypertension**

– – Active space – –

OTS diagnosis (TO) — **clinical diagnosis of exclusion** · **no diagnostic biomarker** (HRV, CK, glutamine, cortisol only supportive) · needs: ↓ performance for weeks–months **despite rest** + **mood disturbance** + other causes excluded (anaemia, thyroid, infection, asthma, diabetes) · cannot diagnose without a rest period · biggest risk = **load–recovery imbalance**, monotonous training

OTS hypothesis (TO)	Mechanism
Glycogen	Depletion → fatigue, catabolic state, ↓ IGF-1
Central fatigue	Low glycogen → ↑ FFA displace **tryptophan** from albumin → competes with **BCAA** → ↑ brain **5-HT**
Glutamine	Falls after > 2 h / repeated exercise → **immunosuppression, infections**
Oxidative stress	ROS → ↓ citrate synthase, mitochondrial dysfunction
HPA axis	**↓ testosterone : cortisol ratio** · blunted ACTH, GH, LH, FSH
Cytokine	IL-1β, IL-6, TNF systemic inflammation

OTS management (TO) — no proven drug · **rest ≥ 2 weeks** → reassess → very low volume (5–10 min/day), **↑ volume before intensity** · training diary (RPE Borg 6–20, resting HR, sleep, mood) · prevention: periodisation, tapering, **> 6 h between bouts**, **≥ 1 rest day / week** · **soreness > 72 h = DOMS, not OTS**

Strength training in children

Safety — age-appropriate, supervised **weight training + plyometrics are safe** · no ↑ injury, **no growth-plate damage**, no ↑ OCD

Mechanism — gains from **↑ neuromuscular activation + coordination** / efficiency of muscle action ("muscle memory") · **not hypertrophy** (insufficient testosterone) · gains **reversible** if stopped

Biomechanics of racing & ball sports (drag, force-time, force platform, cycling posture)

Drag (TO)	Cause	Reduce by
Surface (friction / skin)	Fluid friction at the boundary layer	Smooth suits, shaving, waxing skis / surfboards
Pressure (form)	High pressure in front, low-pressure wake behind	**↓ frontal area + streamlining**, **drafting**
Wave	Surface waves (open-water swimming)	Lane markers + gutters in pools

Flow (TO) — laminar at low velocity · **turbulent boundary layer → later separation → ↓ pressure drag** · "drag crisis" · **dimples / scuffing** → low drag at lower velocity

Not useful to ↓ drag (TO) — **increasing frontal area**
✗ Wrong options: streamlining · drafting · smooth suits

Drafting (TO) — **cycling**, car racing (also swimming) · leader uses more energy · running **1 m behind → ~7% ↓ metabolic cost**
✗ Wrong options: shooting · baseball · ice hockey

Force–time principle (TO) Newton's 2nd law → **impulse–momentum** · longer force application → safer deceleration · **catch with arm / glove extended (A)** · land with legs extended to flex through range
✗ Wrong options: catching close to body (B)

Cycling posture (TO) **anterior pelvic tilt** flattens lumbar spine → ↓ **frontal area** (wind = main retarding force) · **low stack-height cleat** (pedal spindle → cleat top) · 1st MT base over pedal axis · narrow hips → cleats outside, wide hips → inside · toe-out → rotate cleat in

Force platform (TO) measures **ground reaction force** (gait biomechanics)
✗ Wrong options: ball spin · jump force · shoe friction

Sports clothing, footwear & equipment (TO)

Tight body suits ↓ surface drag, ↓ **turbulence** (bonded seams follow contour), ↑ **buoyancy**, ↑ speed (✗ Wrong options: **decreased buoyancy**)

Children in Delhi summer **moisture-wicking garments** (capillary action; polyester, polypropylene, nylon, merino wool)
✗ Wrong options: cotton (absorbs **up to 7%** of own weight, damp, hinders cooling) · lightweight dry · dark heavy

Footwear (TO)	Fact
Pitch (heel drop)	**Heel height − forefoot height**; range 0–14 mm · lower = closer to barefoot forefoot strike
~6 mm	**Midfoot strike**
~12 mm	**Heel strike** (false: "pitch > 12 mm gives marathoners a midfoot strike")
Achilles problems	**Slightly ↑ pitch** (↓ stretch load) + flexibility + eccentric strength
Sole flexion point	**At 1st MTP** · too proximal → ↓ stability · too distal → ↓ sagittal 1st MTP motion

2 Mechanism

Muscle injury healing – TGF-β & early mobilisation

Injury	Key facts
Soreness (DOMS)	**Oedema + inflammation in connective tissue** → ↑ intramuscular pressure · **type IIB** · **eccentric** · starts at MTJ, spreads · cytoskeletal disruption · **peaks 24–72 h** (Orthobullets)
Strain	**Myotendinous junction**, **eccentric** contraction · inflammation → fibrosis
Contusion	Blunt → haematoma · ECM within 2 days, peaks 5–21 days · **myositis ossificans most apparent at 4 wk**
Laceration	Near MTJ, abnormal contour · **dense scar via myofibroblasts ← TGF-β** · regeneration + reinnervation unpredictable, incomplete

Scar cytokine **TGF-β** (myofibroblast differentiation + proliferation) · **regeneration from 7–10 days ← IGF-1, FGF, NGF**
✗ Wrong options: VEGF

– – Active space – –

Immobilisation after muscle injury — **≤ 3–5 days**, then progressive strengthening + stretching · **immediate motion → large dense scar**, poor fibre penetration · **2 wk → atrophy limiting outcome**, ↑ fatty infiltration, less scar, less regeneration, disorganised fibres

Immobilisation (Orthobullets) — atrophy, ↓ tension, **↑ fatigability**, fatty infiltration · faster in **single-joint** muscles · **non-linear** (most in first days; myofibril loss) · worse **without tension → quadriceps > hamstrings** when knee held in extension

4 Diagnosis & Classification

Genes & lab markers

Marker	Fact
CK after eccentric exercise	↑ **24–48 h**, peak **3–6 days** (OB)
CPK normal (TO)	**40–350 U/L**
CK > 5,000 U/L (TO)	Serious muscle disturbance
CPK > 10,000 U/L (TO)	Clinically significant exertional rhabdomyolysis threshold
Rhabdomyolysis urine	Dipstick + for blood but **no microhaematuria** (myoglobinuria)
TGF-β	Scar after laceration
IGF-1, FGF, NGF	Muscle regeneration

Diagnostic criteria

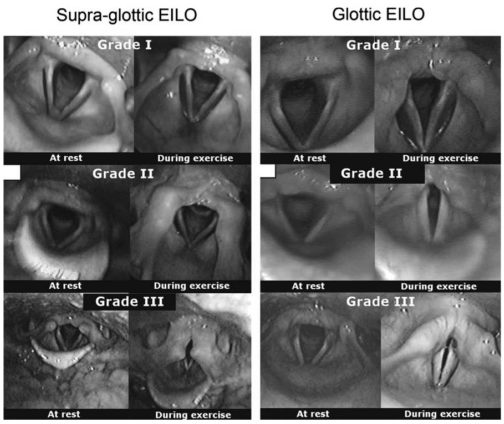
Condition	Criterion
Exertional rhabdomyolysis (TO)	**CPK > 10,000 U/L + signs of muscle damage**
Haematuria needing work-up	**Persists > 7 days** after stopping exercise or **age > 50** (Orthobullets) · **persists > 72 h** (OB)
EILO	**Negative BDR + negative bronchoprovocation, positive CLE**
High-altitude illness (TO)	Rapid ascent **> 8000 ft (2500 m)** · HAPE > 2500–3000 m

Medical conditions in athletes (laryngeal obstruction, haematuria)

EILO — mistaken for exercise-induced bronchoconstriction · **females** · dyspnoea + wheeze · includes exercise-induced laryngomalacia + vocal cord dysfunction · differentials: EIB, hyperventilation, cardiac

Test	EIB (exercise asthma)	EILO
Bronchodilator reversibility (β2-agonist)	**Positive**	**Negative**
Bronchoprovocation (methacholine, mannitol, EVH)	**Positive**	**Negative**
Continuous laryngoscopy during exercise (CLE)	–	**Positive – gold standard**

EILO level — **supraglottic** (arytenoids, closure above glottis) vs **glottic** (**vocal cord closure**) · optimal treatment under study



Supraglottic vs glottic EILO

Exercise-induced haematuria	microscopic **16.7–46.7%**, macroscopic **3.1–19.2%** · risk: intensity, posture, age, heat, altitude, renal disease · bladder / kidney contusion, renal vasospasm · asymptomatic · stop exercise, **repeat UA at 48–72 h**
Persistent haematuria differential	**sickle cell trait → papillary necrosis**, UTI / STI (pyuria, dysuria), kidney contusion (flank blow), stone (flank pain) ✗ Wrong options: pre-workout (only ↑ susceptibility) · mononucleosis · rhabdomyolysis (myoglobin, no RBCs)

Cardiorespiratory fitness testing (VO2 max, ventilatory threshold, Bruce protocol) (TO)

VO2 max	max O2 taken in + used at maximal exercise · **best indicator of cardiovascular fitness** · units **ml/kg/min** · rises with fat loss ✗ Wrong options: l/kg/min · ml/g/min · ml/kg/hr
Measurement	direct (treadmill, ergometer, step test; accurate, costly) vs indirect (Astrand charts, HR, RPE)

Threshold	Definition
Aerobic threshold (Orthobullets)	Effort where anaerobic pathways become significant
Anaerobic (lactate) threshold (Orthobullets)	**Lactate production > removal**
Lactate threshold (TO)	Substantial ↑ blood lactate · **predicts endurance better than VO2max**
Ventilatory threshold (TO)	**Ventilation rises disproportionately to O2 uptake** · **55–75%** of max aerobic capacity · H+ buffered by HCO3− → extra CO2 · synonyms: lactate threshold, OBLA, HR deflection point, maximal lactate steady state
Low intensity 30–50% (TO)	Ventilation coupled to O2 (depth, diaphragm) · ageing blunts maximal ventilation

Lactate balance (TO)	blood lactate = **appearance vs disappearance**, not production alone · produced **even at rest** (~20 mmol/kg/day) · **LDH: pyruvate + NADH → lactate + NAD⁺** (regenerates NAD^+ so glycolysis continues) · ~99% dissociated → lactate + H^+ · lactate = fuel, gluconeogenic precursor, signal · **fatigue multifactorial** (H^+, Pi, ↓ PCr), not lactate alone

– – Active space – –

Lactate shuttle (TO) **MCT4 exports** lactate from glycolytic muscle · **MCT1 imports** into oxidative muscle, **heart, brain** (→ pyruvate → TCA) and liver / kidney (**Cori cycle**) · proton co-transport 1:1 · active recovery speeds clearance

Lactate zone (TO)	Fibres	Blood lactate
Below LT1	**Type I**; fat; production = clearance	**0.5–2 mmol/L**
LT1 = aerobic threshold	**Type IIa** join	First systematic rise (~2–2.5) · = **VT1**
LT1–LT2	More IIa	Rises, stabilises; sustainable ~30–45 min
LT2 = anaerobic threshold / OBLA	**Type IIx**	Exponential rise, **~4 mmol/L** · = **VT2 (respiratory compensation point)**

	OBLA	MLSS
Definition	Workload at **fixed 4 mmol/L**	**Highest constant workload** with lactate steady
Found by	Incremental test	Several constant-load tests
Nature	Same for everyone ("4 mmol trap")	Individual, true steady state
Training	Definition unchanged	**Workload ↑** (↑ mitochondria + MCT1 uptake, not ↓ production)

![Ventilation and blood lactate vs VO2: LT, VT, OBLA, respiratory compensation](../../TO notes/attachments/TOn-notes-energy-systems-to-exercise-intensity-lactate-threshold-p04-f010.jpg)

Training effect (TO) lactate curve **shifts right** (lower lactate at same workload) · trained reach OBLA at higher % VO2max · graph stem "lactate stable → rises → steep" = **progressive fibre recruitment + glycolytic flux, set by production / transport / removal**
✗ Wrong options: abrupt aerobic → anaerobic switch · lactate only when O2 delivery stops · lactate = O2 debt

RER (TO) ~0.7 rest / fat · 1.0 CHO · **> 1.0 near maximal** (HCO_3^- buffering adds CO_2)

CPET normal (TO)	Value
Peak VO2	**> 84% predicted**
Ventilatory AT	**> 40%** of peak VO2 (40–80%)
Max HR	> 90% age-predicted · HR reserve < 15
BP	< 220/90
Ventilatory reserve	MVV − VEmax > 11 L or VEmax/MVV < 85%
VE/VCO2 at AT	**< 34**

Field tests (TO) VO2max: CPET, **1.5-mile Cooper test**, **Yo-Yo intermittent recovery** · lactate from earlobe / fingertip · training zones from LT2: Z1 < 55% (< 2 mmol) → Z2 55–75 → Z3 75–90 → **Z4 threshold 90–105%** → Z5 VO2max 105–120% (> 8 mmol)

Test (TO)	Fact
Bruce protocol	**Most common treadmill test** (Robert A Bruce) · speed + incline ↑ **every 3 min** · **7 stages** · stop at **85% max HR** or HR > 115 for two stages
Modified Bruce	Starts flat: **1.7 mph 0% → 1.7 mph 5%**, then stage 1 of standard · elderly / sedentary
Bruce VO2max	$14.76 - 1.379T + 0.451T^2 - 0.012T^3$ (T = minutes)
Astrand	Ergometer 6 min to HR > 120, then +25 W / min · **Borg RPE 6–20**

Body composition – bulking/cutting & bioimpedance (TO)

Body mass analyser **electrical impedance measurement**
✗ Wrong options: body fat analysis · dry mass · electron spectroscopy

BIA weak alternating current; Ohm's law R = V/I · current passes faster through **hydrated fat-free tissue** · **more muscle → more water → ↓ impedance** · phase angle read at 50 kHz

BIA current **800 µA at 50 kHz** (answer key; explanation text writes "800 mA") · source electrodes dorsal foot + wrist, detectors at wrist styloids + between malleoli · Siri equation · 5 mm girth or 1.5 cm height error → up to 2% change

Bulking vs cutting never simultaneous · bulking = high CHO + protein · cutting = calorie restriction preserving muscle · **ratio 3 : 1**, bulking 8–12 wk → **gain 12 wk before, cut 4 wk after competition**

5 Management

Choosing the exercise

- Protect healing graft / joint → **closed chain**
- Achilles tendinitis after rest → **eccentric + calf stretch**
- Shoulder: early → closed chain → resistive → **plyometrics**
- Tennis player muscle training (TO) → **slow + fast isokinetic**

Therapeutic modalities – iontophoresis & aquatic exercise

Goals ↓ inflammation, ↑ local blood flow, ↑ tissue compliance · cryo / heat, massage, ultrasound, electrical stimulation, iontophoresis

Iontophoresis **direct (galvanic) current drives charged medication through skin** · steroids, LA, salicylates, NSAIDs · tendinitis · deep delivery evidence conflicting · shoulder tendinitis **< 45 yr** responds to steroid iontophoresis
✗ Wrong options: phonophoresis = ultrasound · TENS · EMG · laser · magnetic fields

– – Active space – –

Contraindications **cardiac pacemaker** (current-sensitive), drug allergy

Iontophoresis indication	Drug
Hyperhidrosis	Tap water, glycopyrrolate
Muscle spasm	Magnesium sulphate, calcium chloride
Oedema	**Hyaluronidase**
Adhesive conditions	Iodine
Inflammation	Dexamethasone, hydrocortisone, prednisone, lidocaine, salicylates
Calcific tendinitis + myositis ossificans	**Acetic acid**
Open wounds	Zinc oxide, tolazoline

Aquatic exercise **buoyancy ↓ vertical ground reaction force** → ↓ joint stress · prevents secondary lower-limb injury · **less abrupt HR rise** (lower max HR) · **↑ O2 consumption** vs land
✗ Wrong options: ↑

FITT exercise prescription (TO)	Aerobic	Resistance
Frequency	**5 days/week**	**3 days/week**
Intensity	**55–70% HRmax** (HRmax = **220 − age**)	**50–75% RM** (beginner 50–60, intermediate 60–75, advanced > 75)
Time	**20–30 min/day**	2–5 sets × 2–5 reps / muscle
Type	Aerobic	Isometric / isotonic / isokinetic

Fuel by duration (TO, rehab lecture) < 1 min **creatine phosphate** · 1–20 min **glycolysis** · > 20 min **fats** (Krebs)

Rehab ladder after ACLR (TO, Mohindra)	Time	Focus
Immediate	0–2 days	Brace, pain control, gentle isometrics
Early	2–3 wk	Isometrics + passive / active-assisted ROM
Late	3–8 wk	Isotonic (closed → open chain), cardio, balance
Advanced	2–6 months	Plyometrics, PNF, agility, sport drills
RTS	**~6 months**	**≥ 85% strength vs contralateral on isokinetic testing**

Taping – rigid, McConnell, Mulligan, Low-Dye & kinesiology (TO)

– – Active space – –

Tape (TO)	Description	Use
Rigid / athletic	Non-elastic **zinc oxide**	Stabilisation (ankle, thumb, wrist) · ankle: anchors → stirrups (medial → lateral) → figure-of-eight → **heel lock** · ↓ inversion sprain recurrence **50–70%** · **loses effect after 20–30 min**
McConnell	Rigid brown tape over **Fixomull** underlay · **Jenny McConnell**	**PFPS / patellar maltracking** – corrects glide, tilt, rotation (pull medially) · max **18–24 h**
Mulligan	Rigid tape during pain-free glide (**mobilisation with movement**) · **Brian Mulligan**	Tennis elbow, PFPS, impingement, ankle instability
Low-Dye	Non-elastic arch support, strips lateral → medial	**Plantar fasciitis**, pes planus, MTSS · 24–48 h
Kinesiology (K-tape)	Elastic cotton, **140–180%** stretch · **Kenzo Kase 1970s**, popular Beijing 2008	Pain, swelling, proprioception

K-tape properties (TO) 100% cotton, ~0.3 mm, **heat-activated acrylic** glue (rub after applying), latex-free, wave pattern · lasts **3–5 days** · **colour has no functional effect** · apply **30 min before activity** · **no doping risk** · aids proprioception, **does not strengthen**

K-tape mechanism (TO) **lifts skin** → decompresses **nociceptors + lymphatics** → ↓ pain + swelling · ↓ dorsal-horn input → ↓ reflex hypertonus · anchor + tail **no tension**, only **therapeutic zone** stretched · **under-stretch preferred**

K-tape application (TO)	Direction	Stretch
Weak muscle → facilitation	**Origin → insertion**	**15–25%** (e.g. weak VMO)
Tight muscle → inhibition	**Insertion → origin**	**10–15%** (e.g. tight ITB)
Ligament / tendon support	Centre outward	**50–75%**
Lymphatic drainage	**Fan / web** cut	0–15%
Trigger point	Star / cross	~10%

Cuts (TO) I strip (linear) · **Y** two heads (deltoid) · **X** large muscle (hamstrings) · fan (oedema) · star (trigger points)

Contraindications (TO) open wounds, dermatitis / eczema, unhealed scars, acrylic allergy, **DVT / thrombophlebitis**, malignancy, active infection, fragile skin · patch test 24 h · remove along hair growth, re-tape after 24 h gap · K-tape fits **POLICE / PEACE & LOVE**, rigid fits PRICE

Stretching (PNF, gastrocnemius) (TO)

PNF = stretching · muscle **contracted maximally → relaxes → stretched** actively / passively by partner · Golgi tendon organ response relaxes the muscle · OB wording: muscle stretch used to augment contraction

✗ Wrong options: strength · resistance · injury-prevention technique

– – Active space – –

Gastrocnemius vs soleus (TO) gastrocnemius plantarflexes ankle + assists knee flexion → stretch with **knee straight** · soleus acts at ankle only → stretch with **knee bent**

Calf stretch (TO)	Position	Hold
Straight-leg gastrocnemius	Hands on wall, **back knee straight**, heel down, lean in	**20–30 s**
Achilles / soleus lunge	**Rear knee bent**, weight on rear leg	10–30 s × 3
Soleus	**Both knees slightly bent**, heels down	30–60 s × 2, 3 / day

Exertional rhabdomyolysis (TO)

Warning signs swelling + soreness beyond expected · screen with **urinalysis**: scanty, **dark brown "Coca-Cola"** urine, protein + blood · **definitive = CPK**

Treatment **fluids + maintain urine output** (renal protection) · fasciotomy only for severe compartment syndrome

Biceps swollen after workout, normal urine, CPK 8500, transaminases 1.5–2× **admit to ward for IV hydration**
✗ Wrong options: muscle biopsy · oral hydration + recheck 12–24 h · ICU fasciotomy

High-altitude illness (TO)

HAI acute mountain sickness, **HACE**, **HAPE**

HAPE non-cardiogenic oedema; **exaggerated hypoxic pulmonary vasoconstriction**, capillary "stress failure" · **2–5 days** after arrival · dry cough, exertional dyspnoea → rest dyspnoea, **pink frothy sputum** (advanced) · cyanosis, tachycardia, tachypnoea, fever ≤ 38.5 °C, rales mid-zones · patchy opacities, often **right middle lobe** · rare after **1 wk acclimatisation**

First step **oxygen** (TO answer) · **immediate descent = definitive** · remote, no care → **descent first priority** · portable hyperbaric chamber · **nifedipine 30 mg SR 12-hourly**
✗ Wrong options: hypoxia chamber · oral nifedipine first

Prevention nifedipine, tadalafil, dexamethasone, **acetazolamide** 500–1000 mg/day 8–12-hourly or 125 mg BD (start **24 h before ascent**)

Low back pain prevention – core strengthening (TO)

LBP prevention in athletes answer key not provided by source · options: core strengthening, hamstring stretching alone, foot posture correction, intermittent NSAIDs · module filed as **core muscle strengthening**

X vs Y

Constant	Isotonic	Isometric	Isokinetic
Held constant	**Tension / force**	**Length**	**Velocity**
Relationship shown	Force–velocity	**Length–tension**	Force–velocity

	Iontophoresis	Phonophoresis
Driver	**Electrical (direct) current**	**Ultrasound**

How it's asked

"Push-ups on a weight plate (TO)?"	→ **closed chain dynamic**
"Open chain after knee surgery?"	→ **seated leg extensions**
"Distal end fixed, axial load, co-contraction in ACL rehab?"	→ **closed chain exercise**
"Closed chain in shoulder rehab?"	→ **co-contraction of periscapular + rotator cuff muscles**
"Leg extensions + leg curls (figure)?"	→ **open chain**
"Closed chain differs by?"	→ **compressive nature of applied loads**
"Early rehab, safe scapular + cuff co-contraction?"	→ **closed kinetic chain**
"Closed vs open chain difference?"	→ **predictable movement from co-contraction**
"Strengthening at constant speed?"	→ **isokinetic**
"Isokinetic training example?"	→ **bicycling at constant pedal velocity with varied resistance**
"Most efficient strengthening (figures)?"	→ **eccentric (Figure B)**
"Constant during isotonic exercise?"	→ **muscle tension**
"EXCEPT — muscle biology definitions?"	→ **isokinetic: constant effort through a variable speed** (false)
"DOMS mainly from?"	→ **eccentric exercise**
"Distal biceps rupture contraction?"	→ **muscle lengthens during contraction**
"Length–tension at constant length?"	→ **isometric**
"Achilles tendinitis rehab movement?"	→ **both legs up, lower slowly on the affected leg (eccentric)**
"Caffeine — false (TO)?"	→ **increased carbohydrate oxidation**
"Glycogen depletion curves A–C (TO)?"	→ **A light, B moderate, C heavy**
"Below the crossover point (TO)?"	→ **mainly fat oxidation**
"Primary energy source, marathon runner (TO)?"	→ answer key not provided · OB: **muscle glycogen < 90 min, free fatty acids after**
"Carbohydrate loading mechanism?"	→ **maximises muscle glycogen for exercise > 90 minutes**
"Final hour of a 3-hour marathon?"	→ **free fatty acids**

– – Active space – –

"Fatty acids + liver glycogen on the curve?"	→ **C and D**
"Anaerobic activity < 10 s?"	→ **ATP and creatine phosphate**
"Periodisation in resistance training (TO)?"	→ **manipulation of training variables to maximise adaptations**
"Red line on performance-time graph (TO)?"	→ **supercompensation**
"Muscle training for a 24-year-old tennis player (TO)?"	→ **slow + fast speed isokinetic**
"Planned variation of intensity + duration over time?"	→ **periodization**
"Endurance training adaptation?"	→ **↑ capillary density of active muscles**
"Most important endurance adaptation?"	→ **↑ capillary density**
"Induces myofibrillar protein synthesis?"	→ **resistance exercise above 60% 1-RM**
"Type I fibres?"	→ **slow oxidative, first recruited**
"Marathoner vs powerlifter — true?"	→ **↑ capillary density in the marathoner**
"Weight training + plyometrics in an 11-year-old?"	→ **safe; gains from neuromuscular activation + coordination**
"Weight training in a prepubescent 10-year-old?"	→ **increases efficiency of muscle action**
"Cytokine for scar in lacerated muscle?"	→ **TGF-beta (TGF-β)**
"Partial laceration immobilised 2 weeks?"	→ **atrophy limiting final functional outcome**
"Galvanic current + medication?"	→ **iontophoresis**
"Iontophoresis defined?"	→ **transcutaneous medication delivery by electrical current**
"Aquatic exercise EXCEPT?"	→ **vertical ground reaction force increases** (false)
"Albuterol for wheeze, −ve BDR / BPT, cords close on CLE?"	→ **glottic level EILO**
"Persistent microscopic haematuria after 72 h rest?"	→ **sickle cell trait (papillary necrosis)**
"PNF is a type of (TO)?"	→ **stretching**
"Wall calf stretch, back knee straight (TO)?"	→ **straight-leg gastrocnemius stretch**
"Tight body suits — all EXCEPT (TO)?"	→ **decreased buoyancy**
"Clothes for children in a Delhi summer (TO)?"	→ **moisture-wicking garments**
"Footwear — false (TO)?"	→ **pitch above 12 mm gives marathoners a midfoot strike**
"Biceps swollen after workout, CPK 8500 (TO)?"	→ **admit to ward for IV hydration**
"Not useful to reduce drag (TO)?"	→ **increasing frontal area**
"Drafting reduces pressure drag in (TO)?"	→ **cycling**

"Force–time principle when catching (TO)?"	→ athlete A — arm extended to absorb over time
"Good cycling posture (TO)?"	→ position A, anterior pelvic tilt, low-stack-height cleats
"Force platform measures (TO)?"	→ ground reaction force
"BIA current setting (TO)?"	→ 800 µA at 50 kHz
"Body mass analyser principle (TO)?"	→ electrical impedance measurement
"Bulking + cutting timing (TO)?"	→ gain 12 weeks before, cut 4 weeks after
"Pink frothy sputum after rapid ascent — first step (TO)?"	→ oxygen
"LBP prevention in athletes (TO)?"	→ answer key not provided · module = core muscle strengthening
"Ventilation rises out of proportion to O2 intake (TO)?"	→ ventilatory threshold
"Commonest treadmill stress protocol (TO)?"	→ Bruce protocol
"VO2 max units (TO)?"	→ ml/kg/min
"Hallmark of overtraining syndrome (TO)?"	→ inability to sustain intense exercise, ↓ sport-specific performance despite maintained load
"Performance drop for weeks with mood change, recovers after rest (TO)?"	→ non-functional overreaching
"Fixed 4 mmol/L lactate workload (TO)?"	→ OBLA (vs MLSS = highest steady-state workload)
"Lactate transporter exporting lactate from muscle (TO)?"	→ MCT4 (MCT1 imports)
"Rule of 10 in isometrics (TO)?"	→ 10 s × 10 reps × 10 times/day
"Target HR for moderate aerobic exercise (TO)?"	→ 55–70% of (220 − age)
"K-tape direction to inhibit a tight muscle (TO)?"	→ insertion → origin, 10–15% stretch
"Taping for patellofemoral pain / maltracking (TO)?"	→ McConnell taping

– – Active space – –

EXERTIONAL COMPARTMENT SYNDROME

– Knee & Sports · 13 / 61 –

1 Basics

Epidemiology

Definition exercise-induced, **reversible ischaemia** of muscle within a compartment

Rank **2nd commonest exercise-induced leg syndrome** after **medial tibial stress syndrome**

Who **males > females**, **3rd decade**, **runners** · **long-distance runners + military recruits** (TO) · **paediatric CECS → more common in females** in running sports (OB)

Sites **leg** commonest, then thigh + forearm (OB) · forearm 2nd (another OB) · **volar forearm** with repetitive gripping

Leg compartment involved	Share
Anterior	**~70% – most common** (Orthobullets) · **anterior (TO)**
Anterior + lateral	10% (Orthobullets) · **95% of CECS in anterior + lateral** (TO)
Posterior	**Less predictable surgical outcome**; more often linked to **PAES** (accessory medial gastrocnemius head)

Pathophysiology

Mechanism local metabolism can't clear metabolic waste · **↑ muscle volume in a non-compliant closed compartment** (TO)

Muscle swelling (OB) fibres swell up to **20× resting size → 20% ↑ muscle volume and weight**

Aetiology (OB) **limited osseofascial expansion**, **↑ fascial hernias**, **arteriole dysregulation** · **↓ capillary density** vs asymptomatic

Fascial hernias **40% of ECS vs 5% asymptomatic** (Orthobullets) · **palpable in 40–60%** (TO) · near the **intermuscular septum of anterior + lateral compartments, where the SPN exits**

Risk factors – anabolic steroids (TO)

Risk factor (TO)	Why
Anabolic steroids	↑ muscle volume
Creatine	↑ muscle volume
Eccentric exercise	**↓ fascial compliance**
Rear-foot landing / over-pronation	Differential loading of individual muscle groups

Traps (TO)	✗ Wrong options: inversion + supination of foot on landing · "most recover with activity modification, eccentric exercise + stretching" · posterior + medial calf pain as commonest symptom

3 Presentation

Symptoms	**aching / burning** or cramping leg pain · **paraesthesias over dorsum of foot** · **reproduced by exercise, relieved by rest** · often **bilateral** (vs acute CS)
Timing	starts at a **predictable time / distance / intensity** · **~10 min into exercise**, resolves **~30–40 min** after stopping · patient can predict how long pain lasts
Exam at rest	**most likely normal** ✗ Wrong options: Tinel at fibular neck · firm compartments · 1st web numbness · weak dorsiflexion
Exam findings when present	**↓ sensation 1st web space**, **↓ active ankle dorsiflexion** (drop foot)
Exam after provoking exercise (TO)	tender muscles, **pain on passive stretch**, **firm compartments**, palpable fascial hernia

4 Diagnosis & Classification

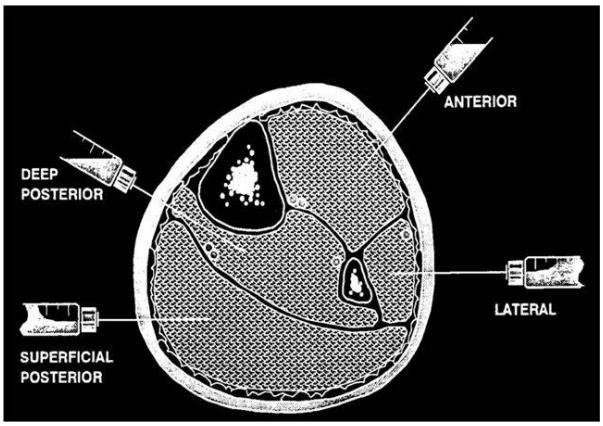
Key radiology findings

Study	Use
Radiographs	**Normal** – exclude other pathology
MRI	**Not diagnostic** – excludes others: **marrow oedema = tibial stress fracture**, **ring-enhancing collection = abscess**
Near-infrared spectroscopy	Muscle deoxygenation; **normal within 25 min** of stopping
Compartment pressures	**Required / gold standard**

Diagnosis – pressure measurement protocol & thresholds

Protocol	limb **relaxed + consistent position** · **rest, 1 min post-exercise, 5 min post-exercise** (± **15 min**, some authors) ✗ Wrong options: 30 min continuous · rest + 1 min only · 60 min continuous
Stryker catheter (TO)	commonest device · TO protocol = **pre-exercise + 5 min** after exercising to severe symptoms
Forearm (OB)	**simulated grip activity → forearm compartment pressures** ✗ Wrong options: CTA · MRA · EMG / NCS · angiography

– – Active space – –



Compartment pressure measurement sites

Diagnostic criteria

Time point	Orthobullets / OB	Pedowitz (OB, TO)	TO (FNB)
Rest (pre-exercise)	**> 15 mmHg**	**≥ 15**	**> 15** (✗ Wrong options: > 10)
1 min post-exercise	**> 30 mmHg**	**≥ 30**	**> 30** (✗ Wrong options: ≥ 20)
5 min post-exercise	**> 20 mmHg**	**≥ 20**	**> 20**
15 min post-exercise	**> 15 mmHg** (some authors)	–	–

How many criteria listed with "and" (Orthobullets) · **"one or more"** (Pedowitz, TO) · "or" / "and/or" (OB) – **any one value above threshold = consistent** in the MCQs

Shin-pain lecture criterion (TO) **> 25 mmHg post-exercise** or **↑ > 10 mmHg from baseline** · confirm with ICP or **NIRS**

Values in the options	Verdict
Rest 10 · 1 min 35 · 5 min 25	**Normal rest, ↑ 1 + 5 min** (OB)
17 · 32 · 21 · 16 (rest, 1, 5, 15 min)	**Confirms ECS** (OB)
5 min = 25	**Most consistent** (✗ Wrong options: rest 10 · 1 min 15 / 25 · 5 min 15) (OB)
Anterior 25 at 5 min	**Positive** (✗ Wrong options: superficial posterior 10 at 15 min) (OB)
Pre 18 · 1 min 58 · 5 min 33	**All point to CECS** (TO)

5 Management

Exertional compartment syndrome

- First line, ≥ 3 months → **activity modification, NSAIDs, rest / ice / stretching**
  (leg: rarely effective; works only if activity stopped completely)
- Forearm, no treatment yet → **NSAIDs + PT (stretching, endurance)**
- Refractory → **fasciotomy of involved compartments**

Nonop success	**forearm PT ~80%** vs **leg PT < 50%** (OB) · all CECS get a **nonop trial first**
Definitive (TO)	**fasciotomy** in non-responders · mainstay = **subcutaneous fasciotomy** (OB)
Surgery (TO, shin-pain lecture)	**fasciotomy / fasciectomy** of affected compartment only (no clear evidence favouring fasciectomy) · **periosteal stripping = revision only** · **best results: nerve entrapment** · **poor: deep posterior CECS** (incomplete release) · high pre- / post-op pressures → poorer prognosis
Post-fasciotomy rehab (TO)	limited WB **3–5 days** → swim / cycle after suture removal (~**2 wk**) → light jog **4–6 wk** → **full RTP 8–12 wk**
Anterior vs lateral (TO)	anterior: pain lateral to shin + **1st web paraesthesia (DPN)** · lateral: pain anterior to fibula + **dorsum of foot (SPN)** · conservative: dry needling, myofascial release, ↓ heel height, **midfoot / forefoot strike, shorter stride** · refractory: **Botox** or fasciotomy

Fasciotomy – recurrence & SPN risk

Incision	Releases	Landmark
Lateral	**Anterior + lateral**	**12–15 cm above lateral malleolus** · protect **SPN** · may see hernia
Medial	Posterior compartments (if pressures ↑)	Mid-tibia, **posterior border**
Endoscopic	Same	Smaller incisions, **similar complications**

Structure at risk (single anterolateral incision)	**superficial peroneal nerve** crossing near the anterior / lateral fascial border ✗ Wrong options: sural (posterolateral) · saphenous (medial) · peroneal artery (deep) · posterior tibial artery (posteromedial)
Success	**anterior > 80%** · **deep posterior ~60%** · not a "home run" (multifactorial) · no op vs nonop trials
Recurrence / failure	**up to 20%** at mean **2 yr** · **fibrosis / scar** · risk = **isolated (anterior + lateral) release** · **four-compartment fasciotomy ↓ reoperation**
Not risk factors (OB)	age < 23 (better success + satisfaction), higher pre-op pressures, bilateral involvement

Forearm exertional compartment syndrome

Sports	**motocross, rock climbing, gymnastics, rowing** (repetitive gripping) · **volar compartment** commonest
Features	pain, tightness, weakness · **numbness / paraesthesia rare** (vs leg)
Gymnast, 5-min pressures 25 + 32, untreated	**NSAIDs, activity modification, PT** ✗ Wrong options: endoscopic / open fasciotomy · fasciectomy · observation

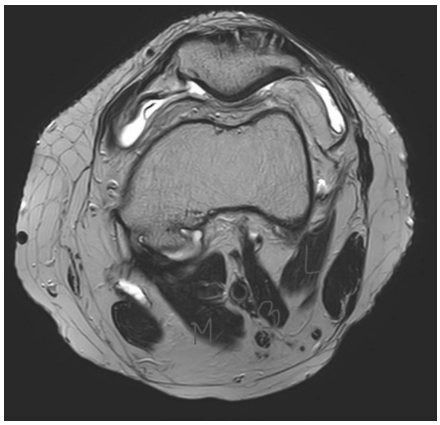
Differential – popliteal artery entrapment

Posterior leg pain, diminished pulses on resisted plantarflexion, sural paraesthesia	**PAES** (accessory slip of medial gastrocnemius on MRI)

– – Active space – –

Provocation **passive dorsiflexion + active resisted plantarflexion**

Traps ✗ Wrong options: intraseptal SPN entrapment (gives CECS-type symptoms) · flexor digitorum accessorius longus → **tarsal tunnel** · popliteal thromboembolism



Accessory medial gastrocnemius head (M, 3, L)

6 Complications

Complication	Fact
Nerve injury	**SPN – most common**
DVT	–
Recurrence	**Up to 20%**, mean 2 yr – **fibrosis**; isolated compartment release

X vs Y

	CECS	Acute CS
Side	Often **bilateral**	Unilateral
Course	**Resolves with rest**	Progressive
First line	Activity modification	**Emergent fasciotomy**

	CECS	Tibial stress fracture	PAES
Exam	**Normal at rest**	**Focal bony tenderness**	↓ pulses on PF / DF
Pain	Resolves with rest	Progresses to **pain at rest**	Claudication
Test	**Pressures**	**MRI marrow oedema**	Arteriogram
Compartment	**Anterior / lateral**	–	**Posterior**

How it's asked

"Resting pressure NOT a criterion (TO)?"	→ **> 10 mmHg** (criterion is > 15)
"Most common compartment (TO)?"	→ **anterior**
"Pre 18, 1 min 58, 5 min 33 (TO)?"	→ **all the above point to CECS**

"NOT a criterion (TO)?"	→ **1 minute postexercise pressure greater than or equal to 20 mm Hg** (criterion is ≥ 30)
"Bilateral leg pain, dorsal foot paraesthesia, normal X-ray — next?"	→ **post-exercise compartment pressure measurement**
"Pressure most consistent with CECS?"	→ **5 min post-exercise 25 mmHg**
"Optimal diagnostic protocol?"	→ **rest, immediately post-exercise, 5 min post-exercise**
"Normal rest, elevated 1 + 5 min?"	→ **rest 10, 1 min 35, 5 min 25**
"Set that confirms ECS?"	→ **17 / 32 / 21 / 16 mmHg**
"Exam in clinic?"	→ **normal examination**
"Positive confirmatory test, normal X-rays?"	→ **anterior compartment 25 mmHg at 5 min**
"Recurrence risk after fasciotomy?"	→ **isolated anterior + lateral release**
"Structure at risk, anterolateral incision?"	→ **superficial peroneal nerve**
"Forearm CECS, untreated, pressures 25 / 32?"	→ **NSAIDs, activity modification, PT**
"Rock climber, volar forearm pain after 20 m?"	→ **simulated grip then forearm compartment pressures**
"Posterior leg pain, ↓ pulses on resisted PF, MRI accessory head?"	→ **popliteal artery entrapment syndrome**
"Post-exercise CS — true (TO)?"	→ **anabolic steroids are a risk factor**

– – Active space – –

EXERTIONAL HEAT ILLNESSES

– Knee & Sports · 14 / 61 –

1 Basics

Epidemiology & risk factors

Cause	**heat gain > heat loss** (production > dissipation)
When / who	↑ in warmer months · highest in **American football**, running, cycling, adventure races
Risk factors	**poor heat acclimatisation**, **dehydration**, sleep deprivation, **prior heat illness** · conditioning, body composition, **protective equipment**, environment
Heat stroke	**3rd leading cause of death in athletes** (after cardiac disorders + cervical trauma)
Football preseason	**~74% of EHI = heat cramps** · peak in **first 14 days** (esp. first 7) · risk ↑ when **WBGT ≥ 82°F (27.8°C)**
Spectrum	mild (heat oedema, rash, **cramps**, **syncope**) → moderate (**exhaustion**) → severe (**heat stroke**) · heat cramps **don't** predispose to heat stroke

3 Presentation

Spectrum of exertional heat illness

Condition	Key features	Treatment
Heat syncope	**Transient LOC**, peripheral vasodilation, ↓ cardiac output, **normal temperature** · rapid recovery supine	**Supine + legs elevated**, fluids
Heat cramps / EAMC	Painful spasm of large muscles · ↓ hydration, **↓ Na + Cl** · random muscles, spasms **1–3 min**	**Stretching** + oral fluids with **NaCl**
Heat exhaustion	Can't continue · **core < 39°C (102.2°F)** · **no CNS dysfunction** · **profuse sweat**, pale moist skin, nausea, vomiting, headache, weakness	Cooling, **oral** rehydration, monitor · loosen clothes, ice axilla + groin, cool sprays
Hypernatraemic heat exhaustion	**Inadequate water replacement** · thirst, oliguria	Water
Heat stroke	**Hyperthermia + CNS dysfunction + loss of thermoregulation** ± multi-organ failure	**Immediate cold-water immersion**
Exertional rhabdomyolysis	Muscle pain, weakness, **↑ CK**, **myoglobinuria**	Cooling, **aggressive IV fluids**, ± dialysis

– – Active space – –

Condition	Key features	Treatment
Exercise-associated collapse (EAC)	**Conscious** collapse **after finishing** · **loss of venous return** on stopping · **normal mental status, temp, vitals** · **most common in marathon tent (59–85%)**	Rule out serious causes → **oral fluids + Trendelenburg** → discharge
Exercise-associated postural hypotension	**SBP falls ≥ 20 mmHg** standing vs supine	
Exercise-associated hyponatraemia	**Na < 135 + mental status change** · dilutional	Severe → **EMS + IV hypertonic saline**

Heat exhaustion core temp **< 39°C (102.2°F)** (Orthobullets, OB) · **98.6–104°F** (OB211458) – read the options

Heat stroke – definition, CNS dysfunction & cold-water immersion

Key discriminator from heat exhaustion **CNS dysfunction** ✓
✗ Wrong options: temp > 38°C · hypotension / dehydration · cramps + dizziness (dizziness isn't CNS dysfunction) · profuse sweating

CNS signs confusion, disorientation, **bizarre behaviour (often first sign)**, staggering / **ataxia**, seizures, delirium, coma · **CNS depression = most important marker**

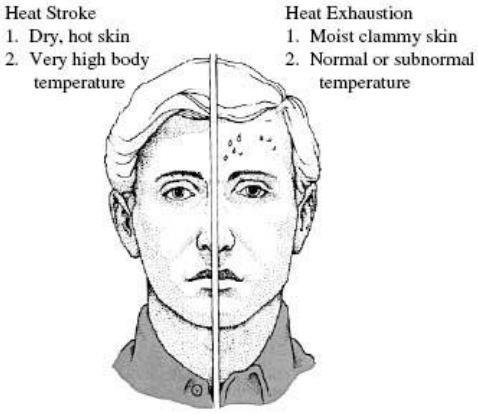
Heat stroke core temperature	Source
> 40°C (104°F)	Orthobullets (printed "> 102.2°F or 40°C"), ACSM / AFP (OB)
> 41°C (105.8°F)	OB1428, OB8705
> 105°F	OB211458
May reach **106–110°F (41.1–43.3°C)**	OB8735

Sweating classic teaching: **anhidrosis, hot dry skin** (OB "hallmark") · **exertional** heat stroke **often sweat-soaked, pale** (OB219812) · sweating is an **inconsequential criterion** (OB211458) · dry, flushed skin → **non-exertional / classic** heat stroke

Temperature **rectal = only reliable** · not oral, temporal, axillary, auricular (ACSM – don't use for diagnosis)

Trigger for immediate cooling **collapse + mental status change** · rectal > 40°C + any CNS change → cool
✗ Wrong options: temporal temp 39.4°C · anhidrosis · tachypnoea · hypotension (both conditions can show collapse, hypotension, tachypnoea)

– – Active space – –



Heat stroke (dry hot skin, very high temp) vs heat exhaustion (moist clammy skin)

5 Management

Athlete collapses in heat

- Mental status change (± rectal > 40°C) → heat stroke → **cold-water immersion ON SITE**
  - then IV fluids + EMS → **cool to 39°C first, transport second**
- Alert, sweating, temp < 39°C → heat exhaustion → **shade, cool, oral fluids**
- Conscious, normal temp + vitals after finishing → EAC → **oral fluids, Trendelenburg**
- Endurance event, heavy water intake, confused → hyponatraemia → **hypertonic saline**

Heat stroke treatment

First step	**rapid cooling: ice / cold-water immersion** = most effective whole-body method ✗ Wrong options: IV fluids first · acetaminophen (antipyretic COX block, no effect) · supine + leg elevation (heat syncope) · IV antibiotics · benzodiazepines / Valium (may worsen) · electrolyte or low-osmolarity drinks (cramps / syncope) · warm bath · reverse Trendelenburg
Cooling target	**0.2°C / min** to **39°C** · other: cooling blankets, evaporative (fans, sprays), ice to large muscle groups, internal cooling, cool IV fluids
Cool first, transport second	whole-body cooling **on site before evacuation** unless other medical emergency (Orthobullets, ACSM, OB219812) · OB8735 key = **transport to ED** with IV fluids + cooling **during transfer** – choose immersion if offered
Outcome	**cooling within 1 h of onset → usually full recovery** · on-site immersion safe (half-marathon: 68% treated on site, 59% discharged home)
Then	IV resuscitation + treat complications · EMS activation
RTP after heat stroke	**no evidence-based guideline** · **rest 7 days** → gradual increase over **next month** → cleared when **heat tolerance returns** (medically supervised)

Exercise-associated hyponatraemia & collapse

Ironman, 60°F, heavy bottled-water intake, collapses	**hyponatraemia** ✗ Wrong options: hypernatraemia · hypothermia · subendocardial MI · berry aneurysm
Cause	**excess free (hypotonic) water** not replacing **sweat Na losses** · usually **> 3 h** ingestion · endurance athletes (triathlon, ultra-marathon, marathon)

Na thresholds	**< 135** (Orthobullets prints mg/dl; OB mEq/L) · up to **10% of ultradistance athletes ≤ 135 mEq/L** (mostly asymptomatic) · symptomatic usually **~125 mEq/L**
Presentation	**mental status change**, peripheral paresthesias, **extremity swelling**
Correction	**severe acute → rapid correction** (avert **brainstem herniation**) · mild → **oral salty solutions** · **too-rapid correction → central pontine myelinolysis** (OB) – read the options
Prevention	fluid balance · **sports drinks (4–8% carbohydrate) for exercise > 1 h** · water best early; electrolyte drinks later prevent dilutional hyponatraemia
Soldier collapses after 2-mile run, conscious, rectal 37.9°C, BP drop 10 mmHg, recovers	**exercise-associated collapse** ✗ Wrong options: heat stroke (no CNS change) · cramps · postural hypotension (needs ≥ 20 mmHg) · hyponatraemia

Prevention – wet bulb globe temperature

Wet bulb temperature thermometer **wrapped in a wet cloth** → reflects **evaporative (sweat) cooling** capacity in heat + humidity

Wet bulb temperature	Action
82–86°F	**1:1 practice : rest**
> 86°F	**Postpone / cancel** participation (✗ Wrong options: > 75 · > 83 · > 90 · > 93°F)

Other prevention **heat acclimatisation** (gradual ↑ activity in heat), hydration, activity modification, **immersion tub on site at high-risk events**

X vs Y

	Heat exhaustion	Heat stroke
CNS	**Intact**	**Dysfunction** (confusion, ataxia, seizure)
Core temp	**< 39°C**	**> 40°C** (OB also > 41°C)
Skin	Profuse sweat, pale, moist	Classic dry hot · exertional often sweaty
Treatment	Cool, oral fluids	**Cold-water immersion on site**

	Heat syncope	EAC
Timing	During heat exposure	**After finishing**
Mechanism	Vasodilation, ↓ cardiac output	**Loss of venous return** on stopping
Treatment	Supine, legs up, fluids	Oral fluids, Trendelenburg

How it's asked

"Hyperthermia + CNS dysfunction + loss of thermoregulation?"	→ **heat stroke**
"41°C, dry skin, disoriented, convulsing – most important step?"	→ **rapid cooling with ice immersion**

– – Active space – –

"Best description of heat stroke?"	→ **hyperthermia, CNS dysfunction, loss of thermoregulatory function**
"Collapsed, confused, not sweating, 80°F + 80% humidity?"	→ **transportation to the emergency department**
"Ataxic, disoriented, 105°F, no sweat?"	→ **heat stroke**
"Lineman confused, 104.5°, no sweat — initial step?"	→ **rapid ice water immersion**
"Finding that mandates immediate cooling in collapsed soldier?"	→ **mental status changes**
"Feature differentiating heat stroke from exhaustion?"	→ **presence of CNS dysfunction**
"Ironman collapse after drinking lots of water?"	→ **hyponatraemia**
"Conscious collapse after 2-mile run, normal temp?"	→ **exercise-associated collapse**
"Wet bulb threshold to postpone sport?"	→ **greater than 86 degrees Fahrenheit**

FEMORAL NECK STRESS FRACTURES

– Knee & Sports · 15 / 61 –

1 Basics

Epidemiology

Incidence	**0.04%** of military personnel (**20 / 10,000 recruits / yr**) · **9%** of lower-limb stress fractures
Share of sport stress fractures	**3–5%** (Orthobullets) · **3%** (TO)
Who	age 16–56 · **female > male** (TO: higher in female athletes) · most often caucasian
Sports (TO)	**marathon / long-distance running**, basketball, gymnastics, ballet · Orthobullets: track + cross-country, military recruits

Risk factors

Group	Risk factors
Demographic	**Female**, caucasian, older age · **poor baseline fitness** (TO)
Activity	**Military**, track / cross-country, high training volume · **sudden ↑ training intensity** (TO)
Medical	**Low BMI**, **↓ BMD**, **energy deficiency** (expenditure > intake), tobacco · **acute weight loss** (independent, Armstrong)
Anatomical	**FAI**, **coxa vara**, abnormal running gait · **coxa profunda + acetabular retroversion** (pincer) (TO)
Triad	**Female athlete triad** (TO + Orthobullets)

FAI association	present in **50%** of FNSF · **cam 42%**, **pincer 78%**
Female athlete triad	**amenorrhoea + eating disorder + osteoporosis** · **2–4× risk** · consider in every female athlete with a stress fracture
Triad mechanism	HPG-axis dysregulation → **↓ oestrogen** (needed for osteoblast maturation) → **↑ osteoclast relative to osteoblast activity** · OCP ↑ BMD
Workup	endocrine / metabolic bone workup · training, diet + menstrual history

Genes & lab markers

3× risk if absent	**calcitonin receptor C allele** · **vitamin D receptor C-A haplotype**

Anatomy & biomechanics

Proximal femur	neck-shaft angle **130 ± 7°** · anteversion **10 ± 7°** · **calcar** = strongest part (dense posteromedial bone)
Head blood supply	**MFCA predominant** · LFCA · artery of ligamentum teres (~**10%**) · ↑ displacement → ↑ vascular disruption
Capsular ligaments	iliofemoral (Y of Bigelow, AIIS → IT line) resists **ER + extension** · ischiofemoral resists IR · pubofemoral resists abduction + ER in extension

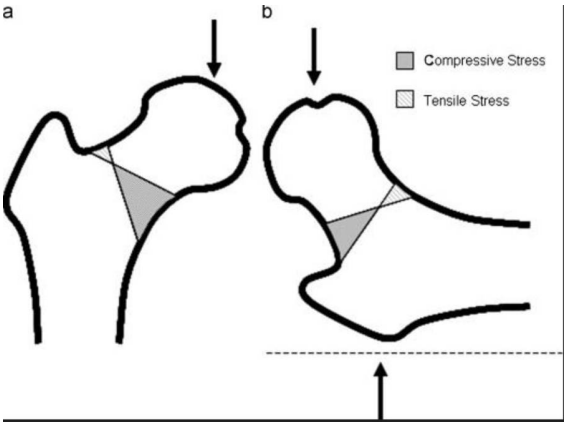
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Load **3–5× BW jogging · 8.4× BW running**

Mechanism repetitive loading > elastic limit → microfractures outpace osteoblastic repair · **fatigued hip muscles lose shock absorption** (Boden)

Muscle strain abductors → highest compressive strain **superolateral** neck · extensors / flexors → tensile vs compressive strain posterior / anterior cortex · rectus femoris (knee extensor) loads neck least

	Compression side	Tension side
Site	**Inferomedial** neck (calcar)	**Superolateral** neck
Propagation	Starts inferomedial → superiorly · **45° → stable oblique**	Starts superolateral → inferiorly · **90°** to cortex → **unstable transverse**
Cause	Compressive load	Bending; **abductors fatigue**
Risk	Low (if < 50%)	**High – displaces**
Rx	Nonop if < 50%	**Screw fixation**



Compression (inferomedial) vs tension (superolateral) stress zones

3 Presentation

History **overuse** running, **recent ↑ training**, high-impact activity

Symptoms **insidious thigh / groin pain** ± radiates to knee · ↑ with weight-bearing, ↓ with rest · pain while running → then walking · completion → crack / pop + can't bear weight

Exam groin tenderness (**62%**), pain at extremes of motion (**79%**), antalgic gait · pain with **SLR, log roll, axial load**, passive IR

No mechanical symptoms no impingement signs (vs FAI / labral tear)

4 Diagnosis & Classification

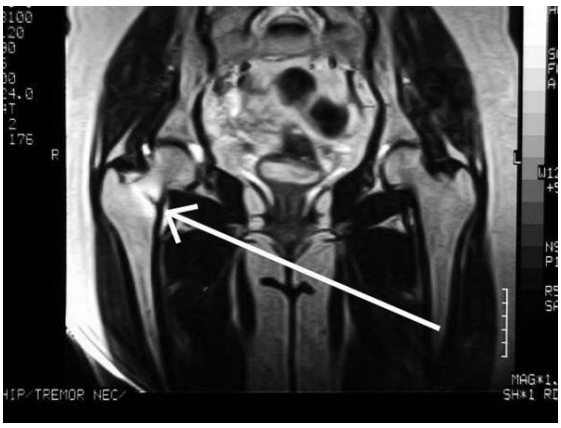
Key radiology findings

Study	Findings	Facts
Radiographs	AP pelvis, AP hip, cross-table lateral	**90% initial normal** · 50% normal at 4–6 wk repeat · changes take **6–8 wk**
X-ray late signs	**Grey cortex sign**, **linear lucency**, endosteal callus, **sclerotic line** (inferior neck = compression)	Visible line → MRI not needed, ↑ chance of displacement
MRI	Oedema on **STIR / fat-sat T2** · **low-signal line on T1** · hip effusion	**Study of choice** if X-ray negative · **sens 100%, spec 100%** · defines side + % width
MRI effusion	Hip effusion	**8× risk of propagation**
Bone scan	↑ uptake (remodelling)	Only if **MRI contraindicated** · sens 93–100%, spec 76–95%, **false + 32%**

Gradual groin pain in runner, normal X-rays
MRI hip
✗ Wrong options: IA steroid (labral tear) · oral steroids · EMG / NCS (radiculopathy) · CT for sports hernia

Slender runner, inferior neck sclerosis on X-ray
urgent MRI to define extent
✗ Wrong options: aspiration · immediate fixation before extent known · protected WB with 2-wk films · return to running

Normal X-ray, superolateral marrow oedema on MRI
tension-side fracture (alpha 45°, LCEA 30° = normal → no FAI surgery)



Compression-side stress fracture on MRI

Fullerton–Snowdy & Shin classifications

Fullerton–Snowdy (X-ray + bone scan)	Shin (X-ray + MRI)
Type I – tension (superolateral) (TO numbering)	Compression: **oedema, no fracture**
Type II – compression (inferomedial)	Compression: **line < 50%**
Type III – displaced	Compression: **line > 50%**
–	Tension-sided
–	Displaced

– – Active space – –

Fullerton–Snowdy (TO) — **most commonly used** classification · Orthobullets lists the three groups unnumbered

MRI compression subgroups (TO) — no line · line **< 50%** · line **≥ 50%** of neck width

Rohen-Quinquilla & Steele classifications

Rohen-Quinquilla (MRI)	Steele (X-ray + MRI)
Low grade I – endosteal oedema **≤ 6 mm**	Compression: oedema, no fracture
Low grade II – oedema **> 6 mm**, no fracture	Fracture **< 50%, no effusion**
High grade III – fracture **< 50%** neck width	Fracture **< 50% with effusion**
High grade IV – fracture **> 50%** neck width	Fracture **> 50% with effusion**
–	Tension-sided
–	Displaced

High- vs low-risk stress fracture sites

High risk (TO)	Low risk (TO)
Femoral neck – tension side	**Femoral shaft**
Patella – tension side	Medial tibia
Anterior tibial cortex	Ribs
Medial malleolus	Ulnar shaft
Talar neck	**1st–4th metatarsals**
Dorsal navicular cortex	–
5th MT proximal metaphysis	–
Great toe sesamoids	–

High risk (TO) — tension sites → progression to complete fracture, delayed / nonunion, refracture, worse if diagnosis delayed

Low risk (TO) — compressive side · favourable natural history · respond to activity modification · rarely recur or go to nonunion

Differential diagnosis

Condition	Clue
Early OA	Older, ↓ IR, joint-space narrowing + sclerosis
Labral tear	Snapping, young, FAI · MRA
Chondral defect	Overlaps labral / FAI / dysplasia · MRI
Rectus strain	Sudden onset, AIIS tenderness, pain on resisted flexion / extension
Osteonecrosis	Steroids, alcohol, sickle cell · crescent sign, head flattening
Osteoid osteoma	Night pain, relieved by NSAIDs, nidus
Lumbar disc	Positional pain, back pain, paraesthesia, positive SLR

5 Management

Femoral neck stress fracture on MRI

- Compression, line < 50%, no effusion → **NWB / protected WB + activity restriction**
- Compression, line > 50% or effusion or progression → **percutaneous cannulated screws**
- Tension side → **percutaneous screw fixation (urgent)**
  (MRI stress change without line → nonop per Orthobullets)
- Displaced → **urgent ORIF, anatomic reduction**

Operative criteria

Finding	Treatment
Compression, line < 50%	**Nonoperative** – NWB / protected WB, activity restriction
Tension-side stress change, **no line** (MRI)	Nonoperative (Orthobullets)
Tension-side fracture	**Cannulated screw fixation**
Compression, line > 50%	**Cannulated screws** (or strict NWB + weekly X-rays – Shin)
Compression + hip effusion	**Cannulated screws** (8× progression)
Progression of compression fracture	Cannulated screws
Displaced	**ORIF** – cannulated screws or sliding hip screw

TO treatment table	Incomplete (< 50% width)	Complete (> 50% width)
Compression	**Conservative** – unless significant pain or **can't straight-leg raise**	Surgical – **cannulated screws or DHS**
Tension	Surgical – **DHS**	Surgical – **DHS**
Displaced	–	**Immediate reduction + fixation** – DHS ± derotation screw
Atypical tension	Conservative	Surgical – DHS

Implant choice conflict — **3 cannulated screws, inverted triangle** (Orthobullets, OB) · **DHS** for tension-side (TO table) – read the options

Initial management – protected weight bearing

Compression < 50% – NWB vs protected WB — **NWB + activity modification** (TO; ✗ Wrong options: protected WB with crutches · screws · bed rest + calcium / vitamin D) · **crutches with protected WB** (OB; ✗ Wrong options: skeletal traction · calcium · ORIF · pulsed US) – read the options

Nonop protocol (Orthobullets) — **NWB 6 wk → ↑ 25% BW per week** to painless FWB → **unrestricted at 3–4 months** · **75–100%** heal and return

OB protocol — protected WB + close observation **6–8 wk** · cessation of running until healed

Nonop cons — refracture (case reports), progression

No role — corticosteroids, calcium, pulsed ultrasound, skeletal traction in acute FNSF

– – Active space – –

Low-risk stress fracture (TO)	relative rest + activity modification · not healed or symptoms > **4–6 wk** → immobilisation + NWB or surgery
High-risk incomplete fracture (TO)	**immobilisation + strict NWB** · worsening / progression → fixation · complete high-risk fracture → strongly consider surgery

Tension vs compression side – surgical fixation

Tension side	high risk of displacement → **percutaneous screws as soon as diagnosed** (Shin & Gillingham) → avoids displacement + harms of bedrest, allows early crutch WB ✗ Wrong options: cam / pincer arthroscopy · NSAIDs + ↓ mileage · PWB
Nonop tension side	rare reports: strict NWB with **3+ wk bed rest** then prolonged crutches
Compression > 50%	**percutaneous screw fixation** ✗ Wrong options: oncology referral · valgus intertrochanteric osteotomy · arthroscopic labral repair · I&D
Recruit, 10% weight loss, compression line 60%	**compression-sided; surgical fixation** ✗ Wrong options: complete fracture · crutches only · tension-sided
Deciding factor	**compression vs tension side + % width** · bone density **not** important
Displaced fatigue fracture	**urgent surgery**
Screw technique	**three 7.0 / 7.3 mm cannulated screws, inverted triangle** (> two): **inferior calcar** (highest load to failure), posterosuperior, anterosuperior · start **at or above lesser trochanter** (avoid stress riser) · parallel, maximal spread · threads in head, not across fracture · washer vs GT penetration · percutaneous or mini-open lateral
Screw outcomes	prevent displacement · outcomes ≈ nonop lower-risk FNSF · **more military separation**
Screw complications	cutout, **varus collapse**, implant failure, neck shortening, nonunion

ORIF of displaced FNSF

Principle **anatomic reduction** + **early surgery** ↓ AVN

Approach	Interval	Note
Smith-Petersen (anterior)	Superficial **TFL (SGN) / sartorius (femoral)** · deep **gluteus medius (SGN) / rectus femoris (femoral)** – internervous	Best view of whole neck · implants via separate lateral incision
Watson-Jones	**TFL / gluteus medius** (both SGN) – intermuscular	Same incision for reduction + implant · poor subcapital view

Reduction aids	K-wire / Schanz joysticks, pointed clamps, **medial buttress plate** · fix with cannulated screws or DHS · autograft vs nonunion
ORIF complications	infection, AVN, nonunion, cutout, malreduction, implant failure · lower return to activity than nondisplaced

6 Complications

Complication	Incidence	Risk factors	Treatment
Progression / completion	**14%**	**Hip effusion (8×)**, delayed diagnosis	Cannulated screws
AVN	Rare if incomplete · **5–42% displaced**	Delay to surgery, initial displacement, **varus malreduction**	Pre-collapse: core decompression / vascularised fibula · collapse: arthroplasty
Nonunion	Rare if incomplete · **9–44% displaced**	Delay, noncompliance	**Valgus intertrochanteric osteotomy**
Varus malunion	**5–33%**	–	Revision ORIF + graft, arthroplasty
Refracture	Case reports	After nonop	–

Fracture size and progression size **not** associated with progression (Orthobullets complications) · **> 50%** width → **high risk of progression** (OB, Robertson & Wood) – read the options

Prognosis

Outcome	Nonoperative	Operative
Return to military	**~75%**	**~40–45%**
Return to sport	**30–100%**	**48–56%**

Negative predictors displacement, high athletic demand (vs recreational), delayed treatment · natural history → progression + displacement

X vs Y

	FNSF	FAI / labral tear
Pain	Insidious, **weight-bearing / axial load**	Flexion, sitting, **FADIR**
Mechanical symptoms	**None**	Click, catch, lock
Imaging	**MRI oedema / line**	Alpha angle, crossover, MRA tear
Rx	Rest / screws	Arthroscopy

How it's asked

"Marathoner, groin pain, normal X-rays, pain on SLR + IR?"	→ **MRI of the hip**
"Slender XC runner, inferior neck sclerosis – next?"	→ **urgent MRI**
"Marathoner, compression fracture > 50% on MRI?"	→ **percutaneous screw fixation**
"Superolateral neck oedema, normal X-ray?"	→ **percutaneous screw placement**
"Marine recruit, compression line 60%?"	→ **compression-sided fracture; surgical fixation**

– – Active space – –

"Athlete, limp, compression line < 50% (TO)?"	**→ non-weight-bearing + activity modification**
"Same TO stem, repeated paper?"	**→ NWB + activity modification** ✗ Wrong options: protected WB with crutches
"Recruit, compression fracture ⅓ neck width on MRI?"	**→ crutches with protected weight-bearing**
"Low-risk stress fracture (TO)?"	**→ femoral shaft** ✗ Wrong options: tension patella · anterior tibial cortex · medial malleolus

– – Active space – –

FEMOROACETABULAR IMPINGEMENT (FAI)

– Knee & Sports · 16 / 61 –

1 Basics

Epidemiology & natural history

Definition	abnormal femur–acetabulum contact → **labral damage**, chondral injury, progressive hip pain
Who	deformity common + often asymptomatic · symptomatic with **extreme ROM** (ballet, gymnastics, martial arts) · **hockey goalies, gymnasts, dancers** → labral tears (repetitive FADIR)
Onset	active young adult, slow-onset groin pain, often after minor trauma
Elite athletes	prone to **cam** → linked to ↑ training intensity during skeletal development
Natural history	early hip dysfunction + arthritis · **FAI = most common cause of OA in nondysplastic hips** ✗ Wrong options: Perthes · traumatic labral tear · repetitive high-impact sport · dashboard injury
Paediatric FAI (TO)	diagnose only with **all 3: hip pain + restricted ROM + imaging morphology** · **cam = most common in adolescents**, develops during physeal maturation **11–14 yr** (young athletic males) · **mixed = most common overall** · causes: DDH sequelae, SCFE, Perthes, **AIIS avulsion malunion**, dysplasias, acetabular retroversion
Synonyms (TO)	**cervicoacetabular impingement** · **acetabular rim syndrome**
Deep-flexion activities (TO)	squatting, **Indian toilets**, cross-legged sitting, gymnastics · age **25–50** · also normal morphology with excessive ROM (dancers)

Anatomy (capsule, labrum, ligamentum teres)

Acetabulum	ilium + ischium + pubis · femoral head anteverted **15°** relative to condyles
Capsular ligaments	**iliofemoral (Y ligament of Bigelow)**, ischiofemoral, pubofemoral
Iliofemoral	**strongest** capsular ligament → restrains **extension + ER** → key restraint vs **microinstability** · **cut during capsulotomy**
Labrum	horseshoe fibrocartilage, joined inferiorly by **transverse acetabular ligament** · ↑ acetabular volume + **suction seal**
Labral seal	negative intra-articular pressure, retains synovial fluid · no labrum → ↑ cartilage loading + ↓ resistance to distraction
Ligamentum teres	cotyloid fossa → femoral head · **negligible** blood supply in adult
Ligament arrangement	iliofemoral, pubofemoral + ischiofemoral form a **helical spiral** around the proximal femur

– – Active space – –

Cam vs pincer pathomorphology & chondrolabral damage patterns

	Cam	Pincer
Side	**Femoral**	**Acetabular**
Typical patient	**Young athletic males**	**Active middle-aged women**
Site	Anterolateral neck / anterosuperior head-neck junction	Anterosuperior acetabular rim
Morphology	**↓ head-neck ratio**, **aspherical head**, **↓ femoral offset**, **femoral neck retroversion**	**AS rim overcoverage**, **acetabular retroversion**, **protrusio**, **coxa profunda**
Mechanism	Aspherical head (radius > acetabulum) jams in → **shear at chondrolabral junction**	Normal neck abuts abnormal rim → **crushes labrum**
Labrum	**Chondrolabral separation** – outside-in detachment, anterosuperior	**Intrasubstance tear** / contusion (coup)
Cartilage	**Delamination** anterosuperior – **precedes** labral tear	**Contrecoup posteroinferior** acetabulum (head levered back)
X-ray	**Pistol grip**, ↑ alpha angle, ↓ offset ratio	**Crossover sign**, posterior wall sign, ↑ LCEA

FAI (TO) **dynamic conflict**: early abutment of proximal femur on acetabulum · intra-articular impingement = cam or pincer

Cam (TO) aspherical **antero-superior head-neck junction** rotates in → compression + shear at chondrolabral junction → **chondrolabral separation**, labral degeneration, **cartilage detached from subchondral bone**

Pincer (TO) **excessive acetabular wall** → labrum compressed between femoral neck and rim

Mixed cam + pincer **most common** type (up to **80%**)

Only one lesion needed cam or pincer alone can cause symptoms

SCFE epiphysis displaced posteriorly → prominent anterior neck, ↓ IR → **cam / pistol grip** (variable patterns) · externally rotated limb

After PAO acetabulum may become **retroverted** → pincer

FAI labral tears **anterosuperior, 10–2 o'clock**

Pathologic anatomy of cam **morphologic abnormality of femoral head** · impingement in **flexion**
✗ Wrong options: acetabular retroversion (pincer) · posteroinferior labral tears · femoral anteversion · osteonecrosis

Cam pathology in a gymnast **↓ femoral head-neck offset**
✗ Wrong options: AS rim overcoverage (pincer) · neck anteversion (cam = retroversion) · ↑ head-neck ratio · spherical head

Horizontal growth plate sign pistol grip + **epiphyseal scar extending lateral to the best-fit circle** of the head → cam



Bilateral pistol grip deformity (cam)

Lesion pattern seen	Means
AS cartilage delamination → then labral tear at chondrolabral junction	**Cam** ✓
AS labral tear at labral-capsular junction → then delamination	Wrong order
Posteroinferior acetabular cartilage lesion	**Pincer contrecoup**
Posterosuperior acetabular eburnation · posterosuperior head fraying	OA
Non-spherical head + AS chondral-labral separation (scope)	**Cam** ✓
Anterior labral contusion + intrasubstance tear + posteroinferior chondral softening	**Pincer** ✓
Lateral acetabular cartilage softening + partial labral detachment	Cam
Posteriorly oriented acetabulum + AS labral tear	Retroversion (pincer)
Non-hemispherical acetabulum + **hypertrophic labrum**	**Dysplasia**
Tense iliopsoas + anterior labral tear · tear at **4 o'clock** + bursal inflammation	**Internal (iliopsoas) impingement**
AS femoral head cartilage flap + haemarthrosis	AVN
Gluteal enthesopathy + bursal scarring	GTPS (lateral pain, abductor weakness)
Ligamentum teres tear + patulous anterior capsule	Hip instability

Frequency (TO) **mixed > cam > pincer** · cam: **cartilage first → labrum later** · pincer: **labrum first → cartilage later**

3 Presentation

Symptoms

Pain activity-related **groin pain ↑ with hip flexion** · worse with **prolonged sitting / driving** · better standing / lying · deep groin pain on squatting

Mechanical clicking / popping (may be absent) · gluteal or trochanteric pain from aberrant gait

– – Active space – –

Byrd C sign (TO) — patient cups hip: **index finger over anterior hip + thumb over posterior trochanteric region → intra-articular hip pathology**
✗ Wrong options: knee · ankle · shoulder

Intra-articular injection relief — temporary relief confirms intra-articular source

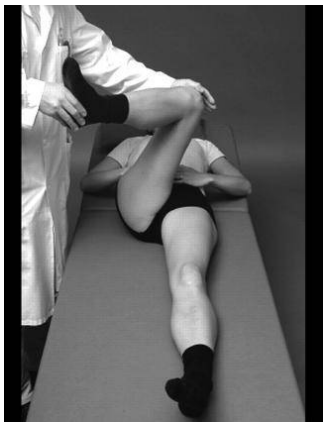
Clinical examination – FADIR & loss of internal rotation

Hallmark — **↓ flexion + ↓ internal rotation** (flexion < 90°, IR < 5°)
✗ Wrong options: ↑ ER · ↑ abduction · ↓ ER · ↓ adduction

Anterior impingement test (FADIR) — **supine, hip passively flexed 90°, adducted, internally rotated** → deep groin pain
✗ Wrong options: abduction / ER combinations

Posterior impingement test — **extension + ER** → occasionally positive

FADIR in FAI — most sensitive exam · positive in cam, pincer and AS chondrolabral tears



Anterior impingement test (flexion, adduction, internal rotation)

Exam finding	Points to (not FAI)
Stinchfield – groin pain on resisted hip flexion	Femoral neck fracture, **iliopsoas tendonitis**, abscess
FABER pain	Lumbar spine, iliopsoas, **sacroiliitis** (not routine for FAI)
Trendelenburg gait	Abductor weakness
Half sit-up pain + pubic ramus tenderness	**Athletic pubalgia**
Greater trochanter tenderness	Trochanteric bursitis
Pain with axial loading, female athlete triad	**Femoral neck stress fracture**
Symphysis tenderness, resisted adduction / abduction	Osteitis pubis
Pain on resisted hip flexion / sit-up	Adductor / hip flexor strain

Red-flag ROM (TO) — flexion **< 90–100°** · IR **< 10°** / absent · **obligatory ER on flexion**

Muscle tightness mimics FAI (TO) — **Ober** (ITB) · **Ely** (rectus femoris) · **90–90** (hamstrings) · Stinchfield = intra-articular

4 Diagnosis & Classification

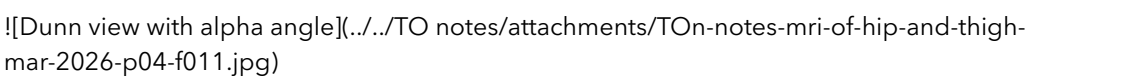
Key radiology findings

View / study	Technique	Use
AP pelvis + true lateral	Lateral with hip in **15° IR**	Recommended routine views (Orthobullets)
Frog-leg lateral	–	Alpha angle (Orthobullets method)
45° modified Dunn (TO)	Supine, **45° flexion, 20° abduction, neutral rotation**	**Best view for cam** (✗ Wrong options: frog-leg · cross-table lateral · 90° sitting view)
Cross-table lateral (TO)	Hip in **10° IR**	Anterior head-neck offset ratio
90° Dunn / seated (TO)	Seated, **90° flexion, 20° abduction, neutral rotation**	Beta angle
False profile (faux profil)	**Standing**, affected hip on cassette, ipsilateral foot parallel, **pelvis rotated 65°**	**Anterior coverage** (DDH + FAI) → ACEA (✗ Wrong options: AP · inlet · outlet · frog lateral)
CT ± 3D	–	Adjunct for structure · **femoral version** (anteversion / retroversion)
MRI / MR arthrogram	In-line with femoral neck	**Best for labrum + cartilage** · fraying, tears, chondral damage, subchondral cysts
Bone scan	–	**Not** for FAI (used for FNSF when MRI impossible)

MRA triad (Kassarjian)	**AS labral tear + AS cartilage lesion + ↑ alpha angle**
Cam on MRI	axial T2: **loss of sphericity of anterior head-neck** (alpha angle measurable, e.g. 80°) · fluid-sensitive images: **labrum separated from chondral surface** · coronal T1: no neck oedema ✗ Wrong options: stress fracture
Alpha angle method (other OB views)	best-fit circle on head + line along neck axis → angle where head exits the circle · AP, frog-leg lateral or Dunn
False profile extra	shows cam deformity + **os acetabuli** (small ossicles beside acetabular roof)
Cam lesion views (TO)	frog-leg, cross-table lateral, 45° modified Dunn · anterolateral cam best on lateral + Dunn · quantified by **alpha angle + head-neck offset ratio**
Ideal AP pelvis (TO)	symmetric obturator foramina · **coccyx 1–3 cm above symphysis** · teardrops visible · lesser trochanters minimally visible · pincer signs: crossover, posterior wall, **ischial spine sign**, coxa profunda · **labral cyst = chronic labral tear** · 3T MRI replacing MR arthrogram
First radiological sign (TO)	cam → **anterolateral migration of femoral head** · pincer → **posteroinferior joint-space narrowing** (counter-coup)
Coxa profunda (TO)	acetabular floor **touches Kohler's (ilioischial) line** · protrusio → head **medial** to Kohler's line
Posterior wall sign (TO)	**head centre lateral to posterior wall** → acetabular retroversion
AP pelvis for FAI (TO)	hip **internally rotated 15°**

– – Active space – –

Alpha angle – where to measure (TO, radiology)	**Dunn / modified Dunn 45°, false profile or MRI oblique axials along the neck** · **not on AP or orthogonal MRI axials** · abnormal **> 55°** (some use **60° or 65°**, activity-dependent) · **MRI coronals often miss FAI**
False profile pelvis rotation	**65°** (current chapter) · **about 60°** (TO radiology) – read the options
GRE axial pincer clue (TO)	posterior acetabular wall should lie **lateral** to a line from the anterior wall · **in line → think pincer**
FAI + sports groin (TO)	young athletes with FAI often also show pubic symphysis irregularity, sclerosis, instability



Radiographic measurements

Measurement	How	Normal	Abnormal
Alpha angle	Line head centre → neck centre + line head centre → point where head **loses sphericity** (anterolateral head-neck junction)	**< 42°** (Orthobullets) · **≈ 42°** (TO) · **< 55°** (one OB)	**> 42°** head-neck offset deformity · **> 50–55° cam** · **> 50° diagnostic** (one OB)
Head-neck offset ratio	(anterior head line − anterior neck line) ÷ head diameter	**> 0.17**	**< 0.17 cam** · **< 0.15 → 95% PPV for FAI** (TO, Beaulé)
Beta angle (TO)	Acetabular rim–head centre line vs head centre–loss of sphericity line (Wyss, open MRI)	–	**< 30° → impingement** (cam, pincer, mixed) (Brunner)
LCEA (Wiberg)	AP pelvis / hip	**Under 40°** (Orthobullets) · **25–45°** (one OB) · **25–39°** (another OB)	**> 35°** overcoverage → poor arthroscopy outcome · **< 25°** inadequate coverage · **< 20°** dysplasia
ACEA (Lequesne)	False profile	**Over 20°** (Orthobullets) · **25–50°** (OB)	**< 20°** structural instability / dysplasia
Tönnis (roof) angle	Acetabular index (inclination)	**Above 0°** (Orthobullets) · **0–10°** (OB)	**< 0° ↓ inclination → overcoverage (pincer)** · **> 10° ↑ inclination → dysplasia**
Head to ilioischial line	Medial head	**< 10 mm** lateral	**> 10 mm lateral → dysplasia** · medial to line → protrusio

Alpha angle (Nötzli, MRI)	**74°** symptomatic vs **42°** asymptomatic → ↓ head-neck concavity in impingement
Tönnis grading (TO)	**OA grade → cannot quantify FAI** ✗ Wrong options: alpha angle · beta angle · anterior head-neck offset ratio



Alpha angle on frog-leg lateral

Pincer/acetabular retroversion – crossover sign

Normal AP pelvis	anterior rim runs medially + distally, **diverging** from the more vertical posterior rim · walls meet superolaterally
Crossover sign	**anterior rim line crosses posterior rim line** ("figure 8") → crossing occurs more distally → **acetabular retroversion → pincer** · the two wall lines **should never cross** normally (two OB explanations word it as the **posterior wall crossing over the anterior**)
Seen on	**AP pelvis / AP hip** · hip AP normal head-neck + popping in gait extension → **AP pelvis shows crossover** ✗ Wrong options: normal film · ↑ femoral offset · SI sclerosis · coxa valga
Crossover = ?	**excessive acetabular retroversion** ✗ Wrong options: low acetabular index · deficient anterior column · labral detachment · pelvic discontinuity
Crossover in FAI	**retroversion in pincer** ✗ Wrong options: retroversion in cam · protrusio (head medial to ilioischial line) · femoral anteversion / retroversion (CT)
Other pincer signs	posterior wall sign, protrusio, coxa profunda, ↑ LCEA, negative Tönnis angle
Pincer AP pelvis	bilateral **acetabular over-coverage + cephalad retroversion** · bilateral cam + crossover = mixed



Crossover sign – anterior (red) and posterior (blue) rims

– – Active space – –

FAI types

Type	Defect	Classic sign
Cam	Femoral head-neck (aspherical)	Pistol grip, alpha > 50–55°
Pincer	Acetabular overcoverage / retroversion	Crossover sign
Mixed	Both — **most common (up to 80%)**	Both

Differential diagnosis

Condition	Discriminator
Femoral neck stress fracture	Female athlete triad / energy deficiency · **axial-load pain**, no impingement · **linear lucency / sclerosis of inferior or posteromedial neck** · T2 oedema
DDH	**Shallow acetabulum, ↓ anterolateral coverage**, low CEA, hypertrophic labrum
Os acetabuli	Unfused secondary ossification centre of **anterosuperior acetabulum**
Iliopsoas tendonitis / internal impingement	Resisted hip flexion pain · Rx tenotomy
Sacroiliitis	Pain with prolonged standing, low back, **FABER** · **SI joint sclerosis** + T2 signal
Osteitis pubis	Central pain, symphysis tenderness · **peri-symphyseal sclerosis + osteolysis** · NSAIDs, rest
AVN	Insidious, pain with impact / inclines, risk factors, not end-range
Displaced femoral neck fracture	ORIF (young) · arthroplasty (old)
Other	Ischiofemoral impingement, adductor strain, athletic pubalgia, lumbar radiculopathy, hip instability

Compression-sided FNSF (distractor) NWB if **< 50%** neck · percutaneous screws if **> 50%**

5 Management

Symptomatic FAI

- Minimal symptoms, no mechanical symptoms → **activity modification, PT, NSAIDs**
- Failed nonop, mechanical, non-arthritic → **arthroscopic osteoplasty + labral repair**
  (no viable labrum → labral reconstruction)
- Large femoral deformity (residual SCFE / Perthes) → **open surgical hip dislocation (Ganz)**
- Significant acetabular retroversion / dysplasia → **periacetabular osteotomy**
- Arthritic, end-stage → **THA (resurfacing vs THA controversial)**

Operative criteria

Procedure	Indication
Nonoperative	Minimal symptoms, **no mechanical symptoms** — always first
Arthroscopic osteoplasty	Symptomatic + **mechanical**, **failed nonop**, **non-arthritic**
Labral repair	Viable tissue, tear at **chondrolabral junction** (vascular acetabular margin)
Labral debridement	Intrasubstance tears
Labral reconstruction	No viable labral tissue
Open surgical dislocation	Previous gold standard · **preserved cartilage, correctable deformity, reasonable expectations** · **residual SCFE / Perthes**
PAO	**Acetabular deformity with significant retroversion** · dysplasia
THA	Arthritic, end-stage
Hip preservation contraindicated	**Tönnis grade 2–3** or **< 2 mm joint space** (Harris)

Treatment – nonoperative first

Initial Rx	**activity modification + NSAIDs** + PT (core / hip strengthening, kinetic chain) — even pro athlete with labral tear on MRI
PT to ↑ passive IR	avoid: worsens impingement + chondrolabral damage
Untreated cam FAI, pro footballer	**activity modification + NSAIDs** ✗ Wrong options: open dislocation (more morbidity) · isolated labral repair (leaves cam) · PAO (no acetabular deformity)
Hockey player, 3 wk pain, no mechanical symptoms	**activity modification, PT, NSAIDs** ✗ Wrong options: bone scan · any surgery yet
Nonop extras (TO)	avoid deep flexion (Indian toilets, squatting, cross-legged) · core strengthening · **short course oral steroids** for acute inflammatory phase (symptom relief only)
Paediatric (TO)	non-op first (activity modification, stretching, NSAIDs) → **up to 50% improve** · arthroscopy: 70° scope, interportal capsulotomy, **treat cause first (osteochondroplasty), then labral repair** · children: **osteochondroplasty, avoid physis** · **labral repair > debridement** (suction seal)

Arthroscopic osteoplasty with labral repair

Failed 12 wk PT + NSAIDs, cam + labral tear	**arthroscopic labral repair + femoral osteoplasty** ✗ Wrong options: more PT · surgical dislocation (slower return) · debridement · PAO (no dysplasia)
Arthroscopic over open	less soft-tissue trauma, **↓ AVN risk**, **quicker return** · **equivalent results** to open
Repair vs debridement	repair / reconstruction → **better PROMs (HHS, SF-12, VAS) + less pain** ✗ Wrong options: ↑ OA risk · no difference · ↑ pain · ↑ nerve injury
Always treat bone	labral debridement or repair **alone** → suboptimal; address cam / pincer as well · femoral osteotomy no role · minimal arthritis → arthroplasty not indicated

– – Active space – –

Pincer + AS chondrolabral tear, failed 6 mo PT	**pincer osteoplasty + labral repair** ✗ Wrong options: cam osteoplasty (no cam) · labral reconstruction · more PT + isolated repair
Cam + labral detachment, minimal arthritis	**femoroacetabular osteoplasty + labral repair** ✗ Wrong options: debridement · femoral osteotomy · hemi / THA
Mixed FAI, wrestler with Trendelenburg + IR 10°	**hip arthroscopy + osteoplasty** ✗ Wrong options: abduction orthosis · DEXA · bone scan · osteotomy
Hockey goalie, failed PT + iliopsoas bursa injection	**FADIR reproduces pain → femoroacetabular osteoplasty** ✗ Wrong options: PAO (dysplasia) · iliopsoas tenotomy (internal impingement / snapping) · FABER · loose body removal
Technique	**capsulotomy** to reach peripheral compartment (cam) · trim head-neck · rim trim → **labral refixation if destabilised** · debridement vs repair vs reconstruction
Evidence caveat	long-term halt of arthritis unproven (Gerdeman)

Open surgical hip dislocation (Ganz)

Approach	lateral decubitus, **Kocher-Langenbeck** incision, gluteus maximus split → **digastric trochanteric flip** (fragment mobilised anteriorly) → capsulotomy → **anterior dislocation**, ligamentum teres transected
Advantage	wide exposure, **preserves external rotators + MFCA → no ↑ AVN**
Cam + anterior labral tear (classic)	**anterior transtrochanteric dislocation + labral repair + femoral osteoplasty** ✗ Wrong options: THA · arthroscopic debridement alone · arthroscopic repair alone · posterior dislocation
Alternative	direct anterior approach → limited posterior acetabular view · combined arthroscopic + limited open → early results promising
Complication	**trochanteric hip pain**
Alternatives (TO)	**Smith-Petersen (anterior)** approach · pincer: **acetabular rim trimming / reorientation osteotomy** (retroversion correction) · cam osteoplasty / cheilectomy – **do not resect > 30% of neck width**

Return to sport

Adolescent, open treatment	**median 7 months** ✗ Wrong options: 4–6 wk · 3 mo · 12 mo · 16 mo
Pro athlete, no OA	return at same level **> 85–90%** (Orthobullets) · **> 85%** (OB) · ≈ **90%** after arthroscopy · **> 90%** (OB, athletes)
With concomitant microfracture	**92%** return (Byrd and Jones) · 5 transient neurapraxias, all resolved

Poor prognostic factors – arthritis & subchondral cysts

Poor outcome after FAI surgery	NOT a poor prognostic factor
Preop OA / Outerbridge III–IV cartilage	Age < 20 / adolescent (mean 17.6 yr – good results)
Tönnis ≥ 1 (Orthobullets) · **subchondral cyst = Tönnis ≥ 2** (OB)	**Professional athlete** · football player
Subchondral cysts / oedema on MRI	**Alpha angle > 60°** (treated by osteoplasty) · cam prominence
Age > 45	Posterior labral tear
BMI > 25–30 (one OB) · **BMI > 30** (another OB) – weaker than OA	Labral tear **repaired**
LCEA > 35° (overhang hard to modify arthroscopically)	LCEA 25° (normal)
Relative femoral retroversion (anteversion < 5°)	–
Labral **debridement** without repair	–

OA on AP — **lateral joint-line sclerosis + acetabular subchondral cyst** · MRI chondromalacia + **subchondral oedema**

OA + FAI — inferior results + **earlier conversion to THA** → counsel on limited benefit · symptomatic care → eventual THA

6 Complications

Complication	Fact
Femoral neck fracture	Femoroplasty · limit resection depth **< 30% of neck diameter** + multiple fluoroscopy views
Heterotopic ossification	–
Residual deformity	Commonest cause of revision · prevent with multiple fluoroscopy views
Capsular deficiency	No capsular closure → instability
Neurapraxia	Traction (see hip arthroscopy)

Failed arthroscopy – residual FAI & subspine impingement

Revision rate — **5–10%** · causes: **under-resection**, over-resection, capsular deficiency, cartilage injury, unaddressed dysplasia

Commonest cause of revision — **residual FAI** (persistent structural deformity)
✗ Wrong options: progressive OA · HO · iatrogenic soft-tissue injury · AVN

Persistent FADIR pain, cartilage intact — **alpha angle 60°** = residual cam
✗ Wrong options: Tönnis angle 8° · LCEA 25° · ACEA 27° · head 7 mm from ilioischial line (all normal)

Commonest arthroscopy complication — **iatrogenic chondral injury** · commonest neurological → **traction pudendal** palsy

Revision options — repeat arthroscopy, arthroscopy + limited open capsulorrhaphy, PAO, surgical dislocation, arthroplasty

– – Active space – –

Revision success	complete cam resection on **Dunn 45°** (alpha angle, anterior offset, head-neck ratio) → better outcome
Subspine (AIIS) impingement	large AIIS spur → **pain on passive hip flexion > 90° in neutral rotation** · overhang below AIIS must be addressed
Subspine imaging	**AIIS spur** on AP + frog-leg lateral (abuts femur in frog-leg position) · MRI: spur **dark, tendon-like on T1** + **head-neck oedema** on T2
Postop alpha 47° + periportal capsulotomy, still painful	**residual subspinous impingement** ✗ Wrong options: inadequate cam (47° < 50–55°) · OA beyond Tönnis 2 (needs large cysts + joint-space loss) · plication (none done) · microinstability (no interportal capsulotomy)
Periportal capsulotomy	small separate portal openings, not joined → no repair or plication needed

Postoperative capsular deficiency

Cause	iliofemoral ligament incised in capsulotomy → **close after interportal + T-capsulotomy**
Risk factors	no capsular closure, unaddressed laxity, trauma, overaggressive rehab, ignoring ROM limits
Imaging	Dunn view: no residual cam / pincer · **T2 FS coronal MRI: superior capsular defect + extracapsular fluid**
Presentation	pain **different from preop**, **pain / apprehension with hip extension + ER**, positive dial test, **loss of recoil on log roll**, gross instability
Treatment	**capsular repair** · **capsular reconstruction** (autograft / allograft) for large irreparable defects

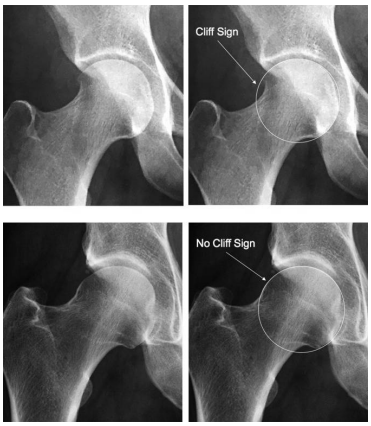
Postop exam finding	Diagnosis
Pain with extension + ER	**Capsular deficiency** ✓
Pain with FADIR	Residual impingement
Pain on resisted hip flexion at 90°	Iliopsoas tendonitis
Pain on passive flexion > 90° in neutral rotation	Subspine impingement
Loss of motion	Capsular adhesions

Microinstability – iliofemoral ligament & hyperlaxity

Strongest restraint	**iliofemoral ligament** ✗ Wrong options: ischiofemoral · ligamentum teres · pubofemoral · inguinal
Causes	bony / DDH, connective tissue disorder (Beighton), post-traumatic, microtrauma, iatrogenic, idiopathic
Pathology	abnormal head translation → microtrauma, capsular stretch, ↑ shear

Test	Sensitivity	Specificity
Abduction-hyperextension-ER	**80.6%**	89.4%
Prone instability	33.9%	**97.9%**
Hyperextension-ER	71%	85.1%

Imaging	**cliff sign** (loss of head sphericity; in **> 89%** of microinstability) · anterior recess widening, thin anterior capsule, capsular defect, labral tear (MRA)
Gymnast, Beighton 5, cliff sign after labral repair	**PT first** (core + hip stabilisers) → **> 70% resolve** ✗ Wrong options: steroid injection · PAO · capsular plication / repair (only after failed PT)



Cliff sign of hip microinstability

Ischiofemoral impingement

Mechanism	**lesser trochanter abuts lateral ischium** · **↓ native femoral offset predisposes** ✓
Presentation	**deep posterior hip pain**, worse after long runs · no relief from hip-joint injection
IFI test	**passive hip extension in contralateral decubitus** · pain on extension + ER ✗ Wrong options: FADIR (that is FAI)
MRI	**↓ ischiofemoral space** · **quadratus femoris** signal (nerve to quadratus femoris, not femoral nerve) ✗ Wrong options: iliofemoral
Treatment	stretching, steroid injection into ischiofemoral space → recalcitrant: **partial lesser trochanter resection** · MRI: ↑ signal around quadratus femoris tendon

X vs Y

Postop / hip pain	Provocation
FAI / residual cam	**FADIR**
Ischiofemoral impingement	**Extension (+ ER), contralateral decubitus**
Capsular deficiency / microinstability	Extension + ER, AB-HEER
Subspine impingement	Passive flexion > 90°, neutral rotation
Iliopsoas	Resisted hip flexion (Stinchfield)

	FAI	Dysplasia
Coverage	Normal / ↑ (LCEA > 35°, Tönnis angle negative)	↓ (LCEA < 20°, Tönnis angle > 10°)
Labrum	Tear, crushed / separated	**Hypertrophic**
Surgery	Osteoplasty ± labral repair	**PAO** ± labral repair

– – Active space – –

How it's asked

"Volleyball player / 40-yr-old, restricted FADIR, abnormal head-neck on AP?"	→ femoroacetabular impingement
"Commonest cause of OA without DDH?"	→ FAI
"Pathologic anatomy of cam?"	→ morphologic abnormality of femoral head
"Soccer player, insidious groin pain, FADIR pain, normal coverage?"	→ cam-type FAI
"Post-SCFE pinning, FADIR pain, normal resisted flexion?"	→ cam-type FAI
"Cartilage lesion with cam?"	→ AS chondrolabral delamination preceding labral tear
"Gymnast — underlying pathology?"	→ decreased femoral head-neck offset
"Pincer — wear pattern at arthroscopy?"	→ anterior labral contusion + intrasubstance tear + posteroinferior chondral softening
"Cam — arthroscopic findings?"	→ non-spherical head + AS chondral-labral separation
"Radiograph supporting pincer?"	→ crossover sign
"Crossover sign indicates?"	→ excessive acetabular retroversion
"Normal hip AP, popping in extension — pelvis AP shows?"	→ acetabular crossover sign
"AP of female with FADIR pain shows?"	→ acetabular retroversion
"Hockey goalie, crossover sign?"	→ acetabular retroversion in pincer impingement
"Pincer + AS chondrolabral tear, failed PT?"	→ pincer osteoplasty + labral repair
"Best view for cam (TO)?"	→ 45° modified Dunn
"Cannot quantify FAI on X-ray (TO)?"	→ Tönnis grading
"Standing view, pelvis 65° to film?"	→ false profile
"Hockey goalie, FADIR + squat pain, managed with PT — imaging?"	→ alpha angle 70° + AS labral tear
"Hurdler, flexion-IR pain — image?"	→ cam lesion
"Byrd C sign (TO)?"	→ intra-articular hip pathology
"Supine position provoking FAI pain?"	→ IR, passive flexion to 90°, adduction
"Commonest physical finding?"	→ ↓ flexion + ↓ internal rotation
"Exam finding with cam on films?"	→ pain with flexion + adduction + IR
"Collegiate athlete, bilateral pistol grip — exam?"	→ positive anterior impingement test

"Hockey goalie failed PT + injection — position + next step?"	→ **FADIR, femoroacetabular osteoplasty**
"Cam + anterior labral tear — classic open approach?"	→ **anterior transtrochanteric dislocation + labral repair + femoral osteoplasty**
"Labral detachment from cam, minimal arthritis?"	→ **femoroacetabular osteoplasty + labral repair**
"Wrestler, mixed FAI, degenerative labral tear?"	→ **hip arthroscopy + osteoplasty**
"Adolescent return to sport after open FAI surgery?"	→ **7 months**
"Pro footballer, untreated cam FAI — initial Rx?"	→ **activity modification + NSAIDs**
"Pro athlete, no OA — return to same level?"	→ **> 85%**
"Lacrosse, failed 12 wk PT, cam + labral tear?"	→ **arthroscopic labral repair + femoral osteoplasty**
"Hockey player, 3 wk pain, no mechanical symptoms?"	→ **activity modification, PT, NSAIDs**
"Labral repair vs debridement?"	→ **improved PROMs**
"Factor most linked to poor FAI surgery outcome?"	→ **degenerative arthritis**
"17-yr-old linebacker, BMI 32 — worst prognostic factor?"	→ **acetabular subchondral cyst**
"Poor prognostic indicator for arthroscopy?"	→ **subchondral bone cysts on MRI**
"Persistent pain after FAI arthroscopy — which measurement?"	→ **alpha angle 60°**
"Commonest reason for revision after FAI arthroscopy?"	→ **residual FAI**
"4 mo after labral repair, capsular defect on MRI — exam?"	→ **pain with hip extension + ER**
"Alpha 47°, periportal capsulotomy, AIIS spur?"	→ **residual subspinous impingement**
"Strongest ligament against microinstability?"	→ **iliofemoral ligament**
"Gymnast, Beighton 5, cliff sign?"	→ **physical therapy**
"Deep posterior hip pain, quadratus femoris oedema — true statement?"	→ **decreased native femoral offset predisposes**

– – Active space – –

HAMSTRING INJURIES

– Knee & Sports · 17 / 61 –

1 Basics

Epidemiology

Incidence	**30% of new lower-extremity injuries in athletes** · ↑ 4% / yr in soccer over ~15 yr
Sports	rapid acceleration: **soccer, track and field, football** · sprinters, gymnasts (TO)
Most common muscle injury in athletes (TO)	**biceps femoris** ✗ Wrong options: adductor longus · gastrocnemius · rectus femoris
Two-joint muscles (TO)	exposed to acceleration / deceleration forces → **hamstrings, quadriceps, gastrocnemius** most often injured
Which hamstring (TO)	**biceps femoris ~83%** > semimembranosus ~12% > semitendinosus ~5%
Site – adult	**myotendinous junction (MTJ)** = most common rupture site, sprinting
Site – skeletally immature	**ischial tuberosity avulsion** · also **water skiers** · **10% of pelvic avulsion fractures** in the skeletally immature (Orthobullets) · ischial tuberosity **~50%** of pelvic apophyseal avulsions (OB, Rossi – see AIIS / ASIS chapters) – read the options

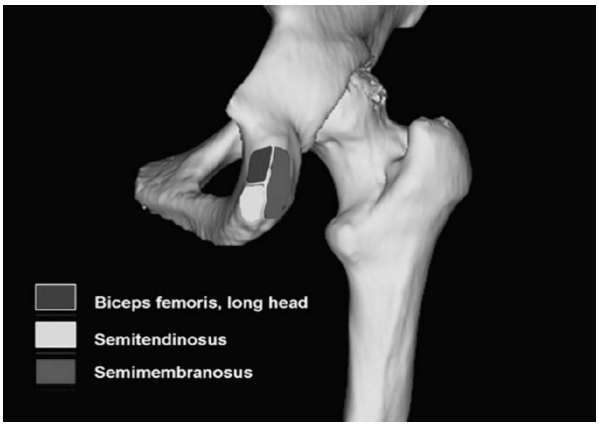
Risk factors

Previous hamstring injury	**×6 reinjury risk** (weak scar) → **greatest predictor** of proximal avulsion
Strength	**hamstring : quadriceps ratio < 0.6** · side-to-side difference **> 10–15%**
Other	inadequate warm-up · **↓ hip extension** (Orthobullets) · ↓ baseline hip flexion (OB) · leg-length difference (shorter leg = tighter hamstrings) · ↑ age, high training demand, ↑ BMI, **tight hip flexors** (OB)

Anatomy

4 muscles	**semimembranosus, semitendinosus, biceps femoris long head + short head** · **gracilis is NOT a hamstring** (pubic origin) (OB) · TO text lists "semitendinosus and gracilis medially, biceps femoris laterally" – read the options
Origin	**all from ischial tuberosity except short head** (**linea aspera**)
Footprint	**conjoint tendon (ST + BF long head) medial** (inferomedial / posteromedial), **BF more proximal than ST** · **SM = separate, most lateral** (superolateral / anterolateral), slightly proximal · **SM closest to sciatic nerve**

– – Active space – –



Ischial tuberosity footprint: BF long head proximal-medial, ST distal-medial, SM lateral

Muscle	Insertion	Nerve
Semimembranosus	Posterior **medial tibial condyle**	**Tibial** division of sciatic
Semitendinosus	Superomedial tibia – **pes anserine**	**Tibial** division
BF long head	**Fibular head**	**Tibial** division
BF short head	Fibular head, BF long head, lateral capsule	**Common peroneal** division

Blood supply **inferior gluteal + profunda femoris** arteries

Distances origin **~6 cm proximal** to inferior border of gluteus maximus · **sciatic nerve 1.2 cm lateral** to origin · **pudendal nerve 2–3 cm superomedial**

Biomechanics cross **hip + knee** (except short head – knee only) · **BF long head + SM** most active in **hip extension** · **ST + BF short head** most active in **knee flexion**

Muscle regeneration – satellite cells

Satellite cells dormant in muscle until injury → **regenerate skeletal muscle** + new satellite cells
✗ Wrong options: osteoblast signalling · periosteum · chemotherapy binding · **sonic hedgehog = limb bud**

2 Mechanism

Mechanism & origin anatomy

Hip flexion + knee extension **eccentric** contraction at **end of swing phase** (maximal fibre elongation) → **biceps femoris**
✗ Wrong options: sartorius · rectus femoris · adductor magnus · TFL

Intramuscular / MTJ injury **sudden take-off phase of running** · acute injuries usually **BF**

Chronic repetitive stretch hip flexion + knee extension → **proximal semimembranosus** (OB)

Older patients same hyperflexion in **slips and falls**, water-skiing

– – Active space – –

Adolescent hurdler ischial tuberosity avulsion = **"hurdler's fracture"** · same mechanism in adult → **MTJ rupture**
✗ Wrong options: muscle belly · tendinous insertion · ramus fracture · sports hernia

Hamstring strain (TO)	Type I	Type II
Mechanism	**High-speed running**	**Slow stretch** – hip flexed + knee extended (**high kicking, splits**)
Site	**BF long head, proximal MTJ**	**Semimembranosus near ischial tuberosity**
Acute loss of function	**More marked**	Less dramatic
Rehabilitation	**Shorter**	**Longer**

Tennis leg (TO) **medial head of gastrocnemius** torn at muscle belly–aponeurosis junction

3 Presentation

History sudden posterior thigh pain running / kicking / jumping ± **"pop"** · **pain with sitting** = proximal avulsion

Ecchymosis posterior thigh → proximal avulsion or high-grade MTJ tear · **latent ecchymosis** appears **~1 week** after injury (haematoma tracking)

Palpable mass middle ⅓ posterior thigh → **MTJ rupture**

Gait **"stiff-legged"** (avoids hip + knee flexion)

↑ Popliteal angle hip 90°, knee 90° → extend knee slowly → angle at pain vs other side

Strength prone knee flexion at 90° weak · **resisted knee flexion prone painful** (TO)

Neuro may have peroneal weakness (foot drop)

Provocative test (pain = +)	Method	Sens / spec
Puranen-Orava	Heel on raised surface, reach for toes	0.76 / 0.82
Bent-knee stretch	Supine, hip + knee max flexed, **slow** passive knee extension	0.84 / 0.87
Modified bent-knee stretch	Same, **rapid** full extension	**0.89 / 0.91 – best**

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
X-ray: **AP pelvis, AP + lateral femur**	**Bony avulsion of ischial tuberosity**
MRI – gold standard	**Number of tendons + retraction**, sciatic nerve position (chronic)
MRI T1	↑ signal = **tendinopathy**
MRI T2	↑ signal = **partial tear**
MRI partial avulsion	**"Sickle sign"** at hamstring origin
MRI complete rupture	**Ischial tuberosity denuded of tendon**, haematoma, retraction (e.g. **> 5 cm**) · e.g. BF long head + ST (conjoint) torn on axial
US + MRI (TO)	Support diagnosis · **MRI better** for oedema, haemorrhage, tear; BAMIC grading
CT	**No indication** (✗ Wrong options: urgent CT)



Bony avulsion of the ischial tuberosity

Hamstring tear MRI classification

Grade	MRI
1	T2 ↑ signal about tendon / muscle, **no fibre disruption**
2	T2 ↑ signal + fibre disruption **< ½ width**
3	Fibre disruption **> ½ width**

British Athletics Muscle Injury Classification (BAMIC) (TO)

Location (TO)	Site
a	**Myofascial** (surface / fascia)
b	**Muscle–tendon junction**
c	**Intratendinous**
d	**Tendon avulsion**

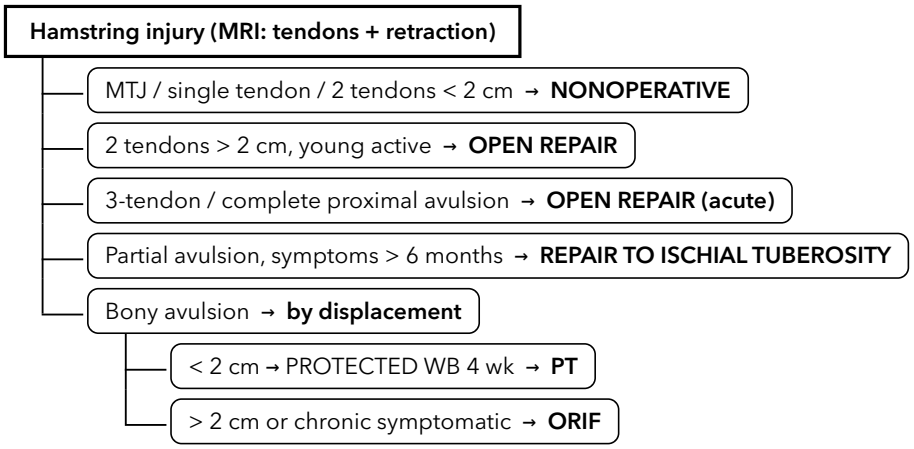
– – Active space – –

Grade (TO)	Injury	Severity
0	No structural damage, soreness	Mild
1	Small tear, minimal fibre disruption	Mild
2	Moderate tear, partial fibre disruption	Moderate
3	Extensive tear	Severe
4	**Full-thickness, complete rupture**	Very severe

Prognosis for return to play (TO)

Predictor (TO)	Effect
MRI-negative strain	**Early RTP**
More proximal (near ischial tuberosity)	**Prolonged RTP**
↑ **craniocaudal length** on MRI	Longer RTP
Pain-free walking > 1 day	**4× more likely RTP > 3 weeks**
Days to pain-free jogging	**1–2 d → < 2 wk · 3–5 d → > 2 wk · > 5 d → > 4 wk**
Intramuscular conjoint tendon injury	RTP delayed **3–4 months**
Pain on passive SLR, longer tender length	Longer RTP
Stopped within 5 min, ↑ max pain	Longer RTP

5 Management



Nonoperative care

Indications	**most injuries · all single-tendon tears · 2 tendons < 2 cm retraction · MTJ rupture** · less active / comorbid
Regimen	**rest, ice, NSAIDs, protected WB 4 wk** → stretching + strengthening · heals in **up to 6 wk** · OB: WBAT, gentle stretching, over 4–6 wk ✗ Wrong options: NWB 6 wk
Single tendon < 2 cm	ruptured tendon **scars to intact tendons**
Return	strength **≥ 90% of contralateral**
Modalities	massage, ultrasound, electrical stimulation

PRP acute strains, **high-level athletes** · **within 24–48 h**, US-guided · RTP **3–5 days earlier** (NFL, low-level data)

Operative criteria

Tendon repair	ORIF
Proximal avulsion ruptures	**Bony avulsion > 2 cm displaced** (fragment big enough for hardware)
Partial avulsion, failed nonop 6 months	**Chronic symptomatic bony avulsion**
2 tendons > 2 cm retraction, young (< 50 y) active	
3-tendon tears (all avulsed)	

Complete tears nonop → poorer outcomes → **repair** · **acute complete** > chronic partial for improvement after surgery

Arthroscopy no intra-articular pathology · endoscopic repair = case reports only

Proximal avulsion — surgical indications & repair

22-yr sprinter, complete proximal avulsion retracted > 5 cm
surgical repair first-line
✗ Wrong options: ATFL signal = low ankle sprain → rest, ice, early rehab · PCL avulsion → nonop, quadriceps strengthening · partial cuff tear → NSAIDs + PT · compression-sided femoral neck stress fracture < 50% width → activity restriction + NWB

Hurdler, 3 tendons avulsed (BF, SM, ST)
open surgical repair of all tendons to the ischial tuberosity
✗ Wrong options: NSAIDs + NWB 6 wk · stretching · CT · hip arthroscopy

42-yr, acute 3-tendon tear, massive retraction, no bony fragment
primary suture-anchor repair + 4–6 wk PWB ± brace · bracing optional
✗ Wrong options: ORIF · Achilles allograft + sciatic neurolysis · PRP + NWB · crutches 6 wk

Partial avulsion, ischial pain at 6 months
repair to ischial tuberosity
✗ Wrong options: sciatic decompression · labral repair · repair to intertrochanteric crest / femoral shaft (nonanatomic)

Approach **prone**, knee free to flex · **transverse incision in gluteal crease → "T" extension** distally for large retracted / chronic tears · fascia usually intact; vertical fascial incision → haematoma

Footprint scrape with **elevator / curette, avoid burr** (sciatic) · **4–6 suture anchors**, knee flexed

Chronic, unable to re-approximate **Achilles allograft** bridge (≈ acute results, small studies)

Post-op **PWB 4–6 wk, knee flexed 40°** ± knee / hip brace

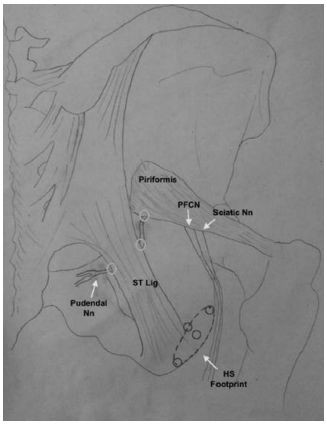
ORIF technique direct reduction, **partially / fully threaded screws + washers** ± suture anchors / interference screws

Outcomes **80% RTS at 6 months** · complications **up to 23%** (23.17%) · **chronic > acute (< 6 wk)** complications · 91% return of strength after operative repair (OB)

– – Active space – –

Surgical nerve risk (sciatic, pudendal)

Nerve	Position vs ischial tuberosity	Fact
Sciatic	**1.2 cm lateral**	**Most commonly injured** in repair · neurolysis just lateral to hamstring sheath · transient palsy (retraction)
Pudendal	**2–3 cm superomedial**	**Aggressive Hohmann superomedially** → sexual dysfunction, dyspareunia, impotence, pain sitting / defecating → use glenoid retractors
Posterior femoral cutaneous	Lateral, beside sciatic	With inferior gluteal artery deep to gluteus max · transient posterior thigh numbness
Inferior gluteal	Below piriformis, under gluteus max	Retracted with gluteus max · 5 cm cephalad to its inferior border
Superior gluteal	Far superolateral	Exits **above** piriformis (OB) · "inferior to piriformis, then superior" (other OB) – not in field
Nerve to obturator internus	Medial, with pudendal	Rarely encountered



Pudendal nerve superomedial and sciatic / PFCN lateral to the hamstring footprint

Rehabilitation & recurrence prevention (eccentric training)

Phase (TO)	Exercise
Acute (painful)	**Isometric** (no joint movement), ice, compression, elevation, NSAIDs as needed
Subacute (pain-free walking)	**Isotonic** = concentric + eccentric → **emphasis on eccentric**: **Askling L-protocol + Nordic hamstring curls**
Adjunct	**Lumbopelvic stability (balance board)**, adductor + glute strengthening

Recurrent hamstring injury, 15-yr footballer (TO) — **eccentric training**
✗ Wrong options: balance board · isotonic · isometric

Nordic hamstring exercise — kneel, heels held, lean forward to prone + return · **↓ injuries 50–70%**

– – Active space – –



Nordic hamstring exercise

Prevention	add abdominal, hip + quadriceps strengthening · **correct hamstring–quadriceps imbalance**
Recurrent sprinter returning – advise (TO)	work on **hamstring strength** · assess **hip position** while running · **lumbopelvic control** · false: **"avoid high-speed training"** – sprinter must train at high intensity
Terminal swing (TO)	hip max flexion **~30°** → controlled by **hamstrings + glutes**; hamstrings slow the extending knee before heel strike → **eccentric** load = injury phase · hip adductors switch eccentric → concentric
Anterior pelvic tilt (TO)	strains lumbar spine + hamstrings, impairs gluteal function → lumbopelvic control prevents recurrence

6 Complications

Complication	Fact
Recurrence	**Most common – 12–31%** · weakness, H:Q imbalance, premature return
Peroneal nerve injury	**Distal non-insertional** injuries · usually self-resolves
Sciatic nerve injury	**8% of surgical cases** · chronic scarring → **nerve exploration**
Hamstring syndrome	Buttock / ischial pain after **nonop avulsion**, worse sitting, stretching, sprinting, hurdling, kicking · hamstrings scarred to sciatic → **surgical release + sciatic decompression**
Ischial tuberosity nonunion	**Bony avulsion > 2 cm treated nonop → ORIF ± bone graft**

Prognosis	unpredictable · **84% regain strength, 89% endurance** · poor: **severely retracted**, **chronic scarred to sciatic nerve**

X vs Y

	Adolescent	Adult
Same hurdling mechanism	**Ischial tuberosity avulsion (hurdler's fracture)**	**MTJ rupture**
Surgery	**> 2 cm displacement**	**2 tendons > 2 cm (young) or 3 tendons**

– – Active space – –

	Sciatic nerve	Pudendal nerve
Position	**1.2 cm lateral**	**2–3 cm superomedial**
Injured by	Lateral retraction, burr, chronic scar	**Superomedial Hohmann**

How it's asked

"Most common muscle injury in athletes (TO)?"	**→ biceps femoris**
"Adolescent hurdler avulsion — adult equivalent?"	**→ hamstring myotendinous junction rupture**
"Hip flexion + knee extension injures?"	**→ biceps femoris**
"Ischial footprint labels 1-2-3 (figure)?"	**→ 1 = biceps femoris, 2 = semitendinosus, 3 = semimembranosus**
"Hamstring with a separate ischial attachment?"	**→ semimembranosus**
"Hurdler, MRI 3 tendons avulsed — next step?"	**→ open surgical repair**
"Partial avulsion, ischial pain at 6 months — surgery?"	**→ repair to the ischial tuberosity**
"Which patient gets surgery first-line?"	**→ 22-yr sprinter with complete proximal hamstring tear**
"Acute 3-tendon tear, massive retraction?"	**→ primary suture anchor repair, PWB 4–6 weeks ± brace**
"Retractor superomedial to ischial tuberosity — nerve?"	**→ pudendal nerve**
"Nerve most at risk lateral to ischial tuberosity?"	**→ sciatic nerve**
"Role of satellite cells?"	**→ regenerate skeletal muscle after injury**
"Recurrent hamstring injury — training (TO)?"	**→ eccentric**
"Recurrent sprinter — which advice is NOT given (TO)?"	**→ avoid training at higher speeds**

HIP ARTHROSCOPY

– Knee & Sports · 18 / 61 –

1 Basics

Overview & indications

Why hard deep joint, **high congruity** (vs knee / shoulder) → steep learning curve

Growth **250% ↑** in US 2007–2011 (Orthobullets) · **88-fold ↑** since late 1990s (OB)

Advantage lower morbidity than open arthrotomy, faster early recovery

Intra-articular indications	Extra-articular indications
FAI lesions, labral tears	**AIIS subspine impingement**
AVN – diagnosis + staging	Capsular tears / instability
Loose bodies, **chondral defects**	**Iliopsoas tendon** pathology
Ligamentum teres injury	**Gluteal tears**, trochanteric bursal debridement
Synovial disease, **septic arthritis**	**Ischiofemoral impingement**, **deep gluteal syndrome**
Impinging osteophytes, unexplained mechanical symptoms	**Piriformis release**, **proximal hamstring tears**

Contraindications (Orthobullets) **advanced DJD**, **hip ankylosis**, **acetabular dysplasia**, **joint contracture**, **severe osteoporotic bone**, significant protrusio acetabuli

Compartments

Compartment	Traction?	Contents
Central	**Yes**	Acetabular fossa, head articular surfaces, ligamentum teres, rim + labrum, AIIS + capsule
Peripheral (lateral to labrum)	No	**Head-neck junction**, **iliopsoas tendon**, **zona orbicularis**, medial synovial fold
Lateral (peritrochanteric)	No	IT band + bursa, gluteus medius / minimus tendons
Deep gluteal space	No	Piriformis + **sciatic nerve**, hamstring origin (ischial tuberosity)

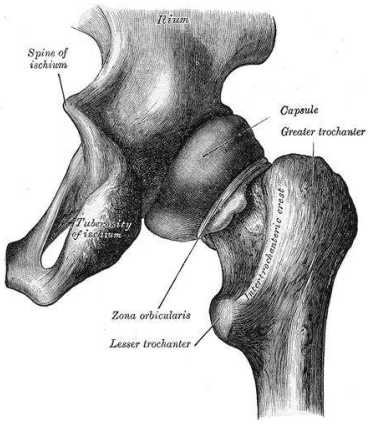
Arthroscopic landmarks – zona orbicularis

Zona orbicularis circular ring of capsule · **landmark for iliopsoas** tendon lying **immediately deep** to it
✗ Wrong options: pectineus · sartorius · adductor brevis · rectus femoris

Use pericapsular **iliopsoas release** for **internal snapping hip** (tendon over iliopectineal eminence or femoral head) · alternative: release at lesser trochanter

Stability proximal–middle capsule incl. zona orbicularis → greatest influence on stability during distraction (Ito)

– – Active space – –



Zona orbicularis

3 Presentation

Clinical sign of intra-articular pathology (C sign)

Byrd C sign (TO)	**index finger over anterior hip + thumb over posterior trochanteric region → intra-articular hip pathology** ✗ Wrong options: knee · ankle · shoulder

4 Diagnosis & Classification

Key radiology findings (pre-arthroscopy screen)

Measurement	Normal	Implication
Tönnis grade (OA)	0–1	**Grade 2–3 → contraindication** (failure, conversion to THA)
LCEA	**> 25°** (28° normal)	**< 20° → dysplasia** (e.g. 15°)
ACEA	–	**< 20° → dysplasia**
Tönnis (roof) angle	**0–10°** (5°, 8° normal)	**> 10° → dysplasia** (e.g. 15°)
Alpha angle	**≤ 42°** (40° normal)	**> 42° → cam** → treatable, not a contraindication
Head-neck offset ratio	**≥ 0.17** (0.2 normal)	**< 0.17 → cam** → treatable

5 Management

Setup, positioning & joint access

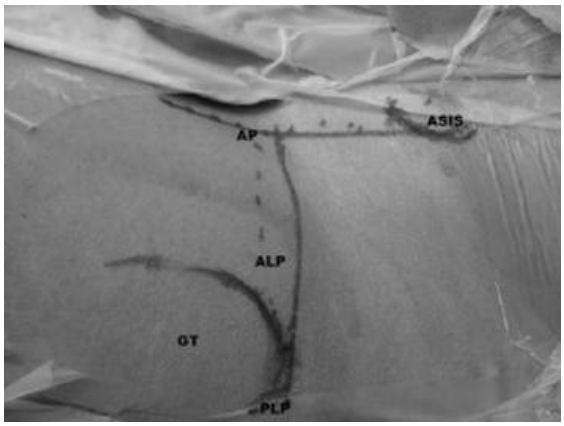
Kit	**70° arthroscope**, long cannulas + guides, flexible instruments · fluoroscopy + arthroscopy towers both visible
Position	**supine or lateral decubitus**, fracture / dedicated table · **well-padded perineal post**
Traction	**~50 lb** to distract · Trendelenburg ↓ force needed

Access spinal needle at **AL portal** into central compartment under **fluoroscopy** (avoid head + labrum) → saline distension or venting (air arthrogram) → guidewire, cannula, scope → later portals under direct vision

– – Active space – –

Portals & nerve injury (LFCN, sciatic)

Portal	Location	Use	At risk
Anterolateral (AL)	**2 cm anterior + 2 cm superior** to anterosuperior GT (Orthobullets) · **1 cm proximal + 1 cm anterior** to GT tip (TO)	**First** (fluoroscopy, only blind portal) · **primary viewing**	**Superior gluteal nerve** (> 4 cm proximal, TO) · **chondrolabral injury = biggest risk (TO)**
Anterior (AP)	Line down from **ASIS** × line across top of GT, moved lateral · ~**6 cm distal to ASIS** through sartorius + rectus femoris (OB)	Central compartment (**4:30–8:30 o'clock**)	**LFCN** + **lateral femoral circumflex artery** · femoral NV bundle if medial
Distal anterolateral (DALA)	**3–5 cm distal to AL** · traction off, neutral or **45° flexion**	Peripheral compartment (neck)	**Ascending branch LFCA**
Mid-anterior (MAP)	Between anterior and AL	As anterior	–
Posterolateral (PL)	**1 cm posterior + 1 cm proximal** to posterosuperior GT tip (Orthobullets, TO) · **2–3 cm posterior** to GT tip (OB) · **neutral rotation**	Posterior joint	**Sciatic nerve** (2.9 cm) · deep branch **MFCA** (> 1 cm) (TO)



Hip arthroscopy portals – AP, ALP, PLP relative to ASIS and greater trochanter

LFCN distance from anterior portal **within 3 mm** (TO) · **15.4 mm** average (OB, Robertson) – read the options

Portal roles (TO) **AL = viewing** · **PL = shuttling** · **mid-anterior (intersection of ASIS and GT lines) = working** · traction to **10–12 mm joint distraction** · nerves at risk: LFCN, superior gluteal, sciatic

LFCA distance (TO) femoral nerve + ascending LFCA ~**4 cm** · terminal LFCA branches **as close as 2 mm** · transverse LFCA branch **10.1 mm** (OB)

LFCN at portal level divided into **≥ 3 branches** · runs from inguinal ligament along sartorius, variable branching

Protect LFCN **"nick and spread"**: skin-only incision + blunt dissection · stay **lateral to vertical line from ASIS** (femoral artery / vein safe)

– – Active space – –

Medialised anterior portal	no better view · ↑ risk to **femoral vessels + LFCN** (Thorey) · too medial → **femoral nerve**
Anterolateral thigh numbness	**LFCN from anterior portal** · AL / distal lateral portals can also hurt LFCN but anterior = greatest risk ✗ Wrong options: AL portal → SGN · traction → pudendal · fluid extravasation · orthosis
Sciatic nerve + rotation	**ER brings sciatic closer** to PL portal → never ER during entry; IR moves it away (OB, Orthobullets) · ↑ risk with **IR or ER** (TO) – read the options
Standard central portals	AL, anterior, PL (Byrd) · reference for all others

Traction-related complications (pudendal nerve)

Complications mostly due to	**traction** ✗ Wrong options: lateral positioning · supine positioning · DVT · HO
Pudendal nerve	**commonest neurovascular complication** · compression by **perineal post → numb inner thigh + hemiscrotum / groin** ✗ Wrong options: obturator (no scrotal supply) · LFCN (anterolateral thigh) · pudendal from DALA portal
Pudendal rates	**1.8%** (62/3405), **4.3% with post** + longer traction · all resolve 6 wk–3 mo (Habib) · **2%** sensory neuralgia, resolve 3 wk–6 mo (Pailhe)
Prevent pudendal	limit traction force + time, padded post **> 8–10 cm** diameter, **post-less** surgery
Peroneal nerve	traction or ill-fitting boot → **dorsum of foot** · prevent: **intermittent traction release**, **total traction < 2 h**, **adequate muscle paralysis**
Overall rates	complications **3.3%**, nerve injury **0.9%** → **pudendal + LFCN** commonest (Nakano) · Byrd 10-yr series: only 2 complications
Obturator nerve	medial thigh sensation + adductors · deep + medial to AL portal → not at risk

Operative criteria – contraindications (arthritis, dysplasia, age)

Factor	Contraindication / poor prognosis	Not a contraindication
Arthritis	**Tönnis grade 2–3**	Acetabular chondrosis (treat at scope)
Dysplasia	**LCEA < 20°**, **ACEA < 20°**, **Tönnis angle > 10°**	LCEA 28°, Tönnis angle 5–8°
Cam	–	**Alpha 60°**, offset ratio 0.2 (treat it)
Age	> 60 → conversion to THA	**< 40 → positive** prognostic (27, 25 yr)
Smoking	↑ poor outcome	**Not** a contraindication
Tear site	–	**Anterosuperior** (commonest)

Dysplasia + isolated labral repair	fails (primary cause unaddressed) → **PAO ± arthroscopic labral repair**
Dysplasia outcomes	failure **29% vs 2.5%** FAI-only (Vahedi) · **32.2%** failure (Larson) → avoid even mild–moderate dysplasia
Failed arthroscopy (Bogunovic)	**37% OA, 43% under-treated FAI, 15% dysplasia** (one OB) · **residual FAI 68%, dysplasia 24%** (another OB) – all visible on preop imaging
ANCHOR / Clohisy	standardised radiographs, always read with history + exam

Conversion to THA after arthroscopy

Risk factor (Kester)	Odds ratio
Obesity (BMI > 30)	**5.6**
Age > 60	**3.4** – most predictive in a 64-yr-old (✗ Wrong options: BMI 27)
Hip OA	2.4
Surgeon **< 40 cases / yr**	1.9
Tobacco	1.9
Female	1.8

Kester conversion **5.9%** at 14.7 mo · ipsilateral knee OA not a risk factor

Degen revision **13.2%** at 1.7 yr · THA **9.6%** · OA HR 2.72, **age > 50** HR 2.09, loose body removal HR 1.61 · protective: non-white HR 0.74, surgeon **> 164 cases / yr** HR 0.42

Age threshold **> 50–60** depending on study · protective: male, non-white, high-volume surgeon

Rehabilitation

Protocol preop PT · foot-flat partial WB → full WB · **early ROM** · strengthening after full ROM · **iliopsoas tendinitis common**

6 Complications

Complication	Cause / risk	Prevention / Rx
Iatrogenic chondrolabral injury	**Commonest reported complication** · scope / instruments (AL portal blind)	Instruments under direct vision
Instability	Capsular insufficiency · imprecise capsulotomy, **no capsular closure**	Careful capsulotomy, **closure / plication**
Traction neurapraxia	**Pudendal** (post), **peroneal** (boot / traction)	Post-less, < 2 h, intermittent release
Portal nerve injury	SGN (AL), **LFCN (anterior)**, **sciatic (PL, ER)**, ascending LFCA (DALA)	Lateral portals, nick and spread
Fluid extravasation	Long surgery, high pump pressure · rare **abdominal compartment syndrome**	Low pressure · **iliopsoas tenotomy last**
Heterotopic ossification	–	**Routine postop NSAIDs**
Conversion to THA	Obesity, age > 60, female, tobacco, low-volume surgeon, OA	Patient selection

– – Active space – –

"Most common" (read the stem)	Answer
Reported complication overall	**Iatrogenic chondral injury** (Orthobullets, OB)
Complications related to	**Traction** (OB)
Neurovascular complication	**Pudendal nerve** (traction / post)
Nerve injuries (Nakano)	**Pudendal + LFCN**
Portal-related nerve	**LFCN** (anterior portal)
Postop dislocation	**Anterior**

Postoperative dislocation

Pattern — **anterior** dislocation commonest after arthroscopy · hip in **extension + ER** (Duplantier)

Management — **immediate closed reduction → restrict extension + ER**
✗ Wrong options: emergent ORIF (only irreducible / fracture) · restrict flexion + IR (posterior precautions) · 24 h traction first · vascular surgery consult

Sequel — progressive arthritis possible (Wylie)

X vs Y

Numb area after hip scope	Nerve	Cause
Anterolateral thigh	**LFCN**	**Anterior portal**
Groin / perineum / hemiscrotum / inner thigh	**Pudendal**	**Perineal post** compression
Dorsum of foot	Peroneal	Traction / boot
Abductor weakness	Superior gluteal	AL portal
Posterior thigh / foot	Sciatic	PL portal in ER
Medial thigh (no scrotum)	Obturator	Not portal-related

How it's asked

"Lateral upper-thigh numbness after hip scope — which portal (TO)?"	**→ anterior**
"Anterior portal — 2 structures at maximum risk (TO)?"	**→ LFCN + lateral femoral circumflex artery**
"Sciatic nerve most at risk with?"	**→ posterior peritrochanteric portal with limb in external rotation**
"Numbness in yellow zone after 3 portals?"	**→ LFCN from portal A (anterior)**
"Anterior portal — structure at greatest risk?"	**→ lateral femoral cutaneous nerve**
"Anterolateral thigh numbness — which part of surgery?"	**→ use of the AP (anterior) portal**
"Hip arthroscopy complications mostly related to?"	**→ use of traction**
"Numb inner thigh + hemiscrotum after scope with post?"	**→ pudendal nerve, direct compression**

"General contraindication to hip arthroscopy?"	→ **Tönnis grade 3**
"64-yr-old converted to THA — strongest preop predictor?"	→ **age > 60**
"Contraindication to isolated arthroscopic labral repair?"	→ **LCEA 15°**
"Dysplasia sign contraindicating isolated repair?"	→ **Tönnis angle 15°**
"Zona orbicularis is the landmark for?"	→ **iliopsoas**
"Dislocation in recovery, hip extended + ER?"	→ **immediate closed reduction, restrict extension + ER**
"Byrd C sign (TO)?"	→ **intra-articular hip pathology**

– – Active space – –

HIP LABRAL TEAR

– Knee & Sports · 19 / 61 –

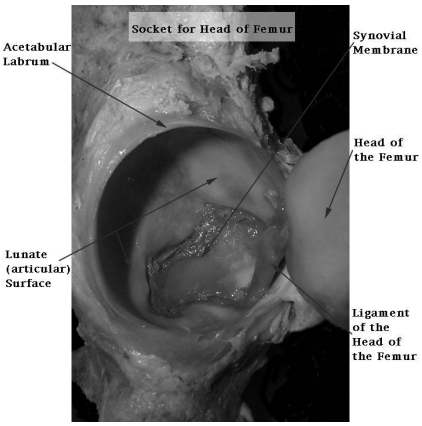
1 Basics

Epidemiology & etiology

Commonest association	**acetabular dysplasia = highest incidence** (Orthobullets, OB) · **FAI = most common cause** of labral pathology (another OB) – read the options
Who	all ages · commonly **active females**
Site	**anterosuperior** (chondrolabral junction) – low collagen content + fibres parallel to chondrolabral junction · FAI tears 10–2 o'clock
Asymptomatic	**69%** of asymptomatic adults have a labral tear on MRI · 73% some hip pathology (Register) → a tear alone ≠ indication for surgery
Causes	**FAI**, **dysplasia** (floppy / redundant / **enlarged** labrum), **trauma** (dislocation / subluxation), **capsular laxity** (↑ translation), **degeneration** (edge loading)
Effect	tear ↓ resistance to motion → **initially relative hip instability** ✓ · commonest associated lesion: **chondral injury**
Traps	tears **are** related to DJD (✗ Wrong options: unrelated) · occur **with or without** abnormal morphology (✗ Wrong options: only with) ✗ Wrong options: caused by ligamentum teres lesions · posteroinferior quadrant
Dysplasia vs FAI labrum (Leunig)	both tear **anterosuperior** · labrum **enlarged** in 12 dysplastic hips, none with FAI → anterosuperior rim = initial fatigue site
Consequence	↑ friction, cartilage consolidation + strain → accelerated degeneration (Safran)

Anatomy

Shape	horseshoe, **continuous with transverse acetabular ligament**
Two parts	articular = **fibrocartilage** · capsular = dense connective tissue
Vascularity	from capsule + synovium at acetabular margin → **only peripheral ⅓ vascularised**
Innervation	**highly innervated** (mechanoreceptors + nociceptors) · **nerve to quadratus femoris** + **obturator nerve**
Function	adds **22% surface area** + **33% volume** · **negative-pressure suction seal** → retains synovial fluid, frictionless motion



Acetabular labrum and transverse acetabular ligament

3 Presentation

Symptoms mechanical **pain + snapping** (internal snapping), **locking**, vague groin pain · pain with sitting (anterior pathology)

Provocative test	Manoeuvre
Anterior labral tear	**Flexed + ER + abducted → extension + IR + adduction** (painful catch)
Posterior labral tear	**Flexed + adducted + IR → abduction + ER + extension**
FADIR	Positive in **AS chondrolabral tear** and cam FAI

Intra-articular steroid relief diagnostic + therapeutic · confirms intra-articular source
✗ Wrong options: sacroiliitis

Distractors FNSF: no mechanical symptoms, axial-load pain, not linked to dysplasia · trochanteric bursitis: lateral GT tenderness · AVN: risk factors

4 Diagnosis & Classification

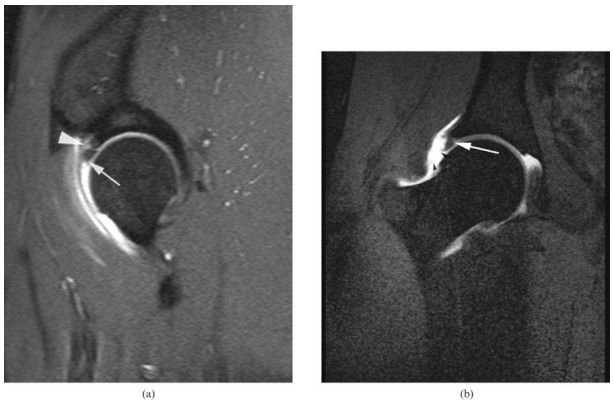
Key radiology findings

Study	Finding / use
Radiographs	Exclude other pathology · **dysplasia**, arthritis, acetabular cysts
ANCHOR views (Clohisy)	AP pelvis, cross-table lateral, **45° + 90° Dunn**, frog-leg, false profile
MR arthrogram	**Study of choice / gold standard** · **92% sensitive** (Orthobullets) · **> 90%** (OB)
MRA finding	**Dye extravasation into labrum** · fluid deep to anterior chondrolabral junction
MRA + injection	Intra-articular **lidocaine + steroid** → diagnostic + therapeutic

FAI triad on MRA ↓ head-neck concavity → **↑ alpha angle + chondrolabral tear** · impingement + no crossover → **abnormal alpha angle + chondrolabral tear**
✗ Wrong options: acetabular retroversion · HO · AS · coxa vara

Normal values in stems **alpha 40° normal** (> 42° cam) · **head-neck offset ratio 0.20 normal** (< 0.17 cam)

– – Active space – –

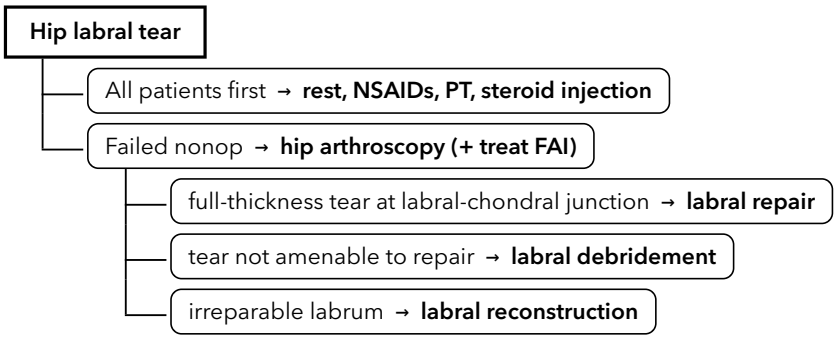


MR arthrogram — labral tear

Konan classification (acetabular chondral lesions)

Element	Grading
Cartilage grade	**0** normal → **4** full-thickness defect, exposed bone
Site A	**<** $\frac{1}{3}$ of rim-to-cotyloid-fossa distance
Site B	$\frac{1}{3}$–$\frac{2}{3}$
Site C	**>** $\frac{2}{3}$
Reliability	Intra-observer **0.81** · inter-observer **0.88**

5 Management



Operative criteria

Procedure	Indication	Outcome
Nonoperative	**Initial treatment for all**	No long-term data
Arthroscopic repair	Failed nonop + **full-thickness tear at labral-chondral junction**	Better healing + function than partial labrectomy
Arthroscopic debridement	Failed nonop + tear **not amenable to repair**	**70–85%** short-term relief
Labral reconstruction	Failed nonop + **irreparable** labrum	**Allograft ≈ autograft** PROMs

Treatment – arthroscopic repair

Failed rest, PT, NSAIDs + 1-day injection relief	**arthroscopic treatment of labral tear ± FAI** ✗ Wrong options: repeat injection · pinning (no fracture) · loose body removal · PAO (no dysplasia)
Address cause	both labrum and FAI bone must be treated for symptomatic gain · AP pelvis may show retroversion
Repair technique	**knotless suture anchors** preserve labral anatomy
Debridement	remove unstable labrum + synovitis · **limited WB × 4 wk** · **flexion + abduction limited 4–6 wk**
Approach (Matsuda)	open dislocation, mini-open and arthroscopy all improve pain + function · open → **highest complication rate** · mini-open → **LFCN injury** · all-arthroscopic ≥ others, fewest major complications

X vs Y

	Dysplasia labrum	FAI labrum
Labrum	**Hypertrophic / floppy / enlarged**	Normal size, crushed / separated
Site	Anterosuperior	Anterosuperior
Bony fix	**PAO**	Osteoplasty

How it's asked

"Avid runner, MRA dye in labrum?"	**→ labral tear**
"Impingement test +, no crossover – MRI shows?"	**→ abnormal alpha angle + chondrolabral tear**
"True statement about hip labral tears?"	**→ initially lead to relative hip instability**
"Dysplasia, snapping, pain flexed-ER-abducted → extended-IR-adducted?"	**→ labral tear**
"FADIR pain, no axial-load pain – which image?"	**→ MRA with anterior labral tear (fluid deep to chondrolabral junction)**
"Failed PT, NSAIDs, rest, brief injection relief?"	**→ arthroscopic treatment of labral tear and possible FAI**

– – Active space – –

HISTORY AND PHYSICAL EXAM OF THE KNEE

– Knee & Sports · 20 / 61 –

1 Basics

Fabella anatomy

Fabella (TO)	accessory sesamoid ossicle **almost always in lateral head of gastrocnemius**, rarely medial head (can be in both) · **~20% (10–30%)** of population · **bony or cartilaginous**
Articulation (TO)	**often articulates directly with lateral femoral condyle** → "no bony attachment" is the FALSE statement
Function (TO)	stabilisation of the medial femoral condyle (as stated in source) + **fabella complex**: plantaris, gastrocnemius, arcuate, **fabellofibular**, fabellopopliteal, oblique popliteal ligaments

Normal values

ROM	**10° recurvatum → 130° flexion** · rotation minimal in extension · **at 90°: 45° ER, 30° IR** · ab / adduction 0° in extension, few degrees at 30°
Quadriceps girth	measured at **10 cm (VMO)** + **15 cm (quadriceps)**
Tibial step-off	medial plateau **~1 cm anterior** to MFC at 90°

3 Presentation

History & exam by lesion

Lesion	History	Exam
ACL	Non-contact, awkward landing, **"pop"**, **immediate swelling**	**Lachman**, pivot shift, large haemarthrosis
PCL	**Dashboard**, fall on **plantarflexed** foot, posterior pain	**Sag**, **posterior drawer (90°)**, quad active
MCL	Blow to outside of knee, medial pain	Valgus instability
LCL	Varus injury, lateral pain	Varus instability
PLC	Lateral + posterior pain, usually combined	**Dial + at 30°**
Meniscus	**Mechanical** (catching, locking), joint line pain, **delayed swelling**	**Joint line tenderness**, McMurray
Patella	Fall on **dorsiflexed** foot, **2 "pops"**, anterior pain, stairs	Apprehension, **MPFL tenderness**, effusion, crepitus, ↑ Q angle

Inspection / palpation	scars, swelling, quadriceps atrophy, gait (antalgia, stride), standing alignment · joint line, pes bursa, patellar tendon, ITB · effusion: ballot, milking

Neurovascular exam

Nerve	Sensation	Motor
Obturator	Medial thigh	Thigh adduction
Femoral	Anterior thigh	Knee extension
Sciatic	Posterolateral calf	Knee flexion
Peroneal	**Dorsal foot**	**Toe extension**
Tibial	Plantar foot	Toe flexion

Vascular popliteal, DP, PT pulses · **ABI < 0.9 abnormal**

ACL tests

Lachman **most sensitive** test · **PCL tear → "false" Lachman** (posterior subluxation)

Lachman grade	Translation
1	**< 5 mm**
2 A / B	**5–10 mm**
3 A / B	**> 10 mm**
A / B	**A = firm endpoint**, **B = no endpoint**

Pivot shift **extension → flexion**, reduces at **20–30°** · needs full relaxation (easier under anaesthesia) · mimics giving way

KT-1000 quantifies anterior laxity · slight flexion, **10–30° ER**

Gait **quadriceps avoidance** (doesn't actively extend)

PCL tests

Posterior drawer (90°) **most accurate** for PCL

Posterior sag hips + knees 90°, ankles supported → tibia sags vs other side

Quadriceps active test quad contraction at 90° → **anterior reduction of tibia** = +

Collateral tests

Stress	30° only	0° + 30°
Valgus	**Isolated MCL**	**MCL + ACL and / or PCL**
Varus	**Isolated LCL**	**LCL + ACL and / or PCL**

Varus opening + ↑ ER at 30° **LCL + PLC**

MCL grade (opening) **I 0–5 mm · II 6–10 mm · III 11–15 mm** (this topic) · 1–4 / 5–9 / ≥ 10 mm (MCL topic exam) – read the options

Anterior drawer, tibia in ER positive → associated injury · **grade III MCL often with ACL + posteromedial corner** tears

– – Active space – –

Posterolateral corner – dial test

Test	Technique	Positive means
Dial test	ER asymmetry at 30° + 90°	**> 10° at 30° only → isolated PLC · 30° + 90° → PLC + PCL**
Varus stress	0° vs 30°	0° → LCL + cruciate · 30° → LCL
Posterolateral drawer	Hip 45°, knee 80°, foot **ER 15°**, posterior drawer + ER	↑ posterolateral translation
Reverse pivot shift	Knee **90°**, ER + valgus, **extend**	Tibia reduces with clunk at **~20°** (ITB flexor → extensor)
ER recurvatum	Suspend leg by toes, supine	Leg falls into ER + recurvatum
Gait	–	**Varus thrust** or hyperextension thrust
Peroneal nerve	–	↓ dorsal foot sensation, weak dorsiflexion

Dial threshold **> 10°** (Orthobullets) · 10–15° (OB) · 15° (TO) – see PCL / PLC chapters

Reverse pivot direction 90° → extension (Orthobullets) · extension → flexion (TO, PLC chapter) – read the options



External rotation recurvatum test

Isolated PLC (OB) blow from medial side, lateral tenderness, Lachman **1A**, normal posterior drawer, **20° ↑ ER at 30°**, symmetric at 90° → **PLC reconstruction** (repair or reconstruction in high-level athlete)
✗ Wrong options: BTB ACL · ACL + PLC · PCL · PCL + PLC

Veltri (OB) isolated PCL **< 10 mm** posterior laxity → **nonop** · **> 10 mm or combined** → reconstruct

Meniscus tests

Findings **joint line tenderness** · effusion

McMurray flex knee, hand on medial side, **externally rotate**, extend → palpable pop / click → **medial meniscus** tear

Patella tests

Haemarthrosis large · **absent swelling → ligamentous laxity / habitual** dislocation

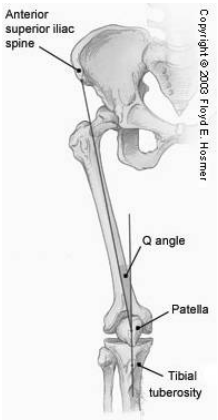
Tenderness **medial (MPFL)**

Passive translation quadrants (midline = 0) vs other side · normal **< 2 quadrants** · medial border to lateral trochlear edge = **2 = abnormal**

J sign excessive lateral translation in extension, "pops" into groove in early flexion → **patella alta**

– – Active space – –

Also apprehension · ↑ **Q angle**



Q angle

How it's asked

"ER 20° ↑ at 30°, symmetric at 90°, normal drawer and Lachman – surgery?"	**→ posterolateral corner reconstruction**
"NOT true about fabella (TO)?"	**→ no bony attachment (it articulates with lateral femoral condyle)**
"Fabella usual site (TO)?"	**→ lateral head of gastrocnemius (rarely medial, can be both)**
"Fabella composition (TO)?"	**→ bony or cartilaginous**

– – Active space – –

IDIOPATHIC CHONDROMALACIA PATELLAE

– Knee & Sports · 21 / 61 –

1 Basics

Epidemiology & pathophysiology

Who	**adolescents + young adults** · **F > M** · PF pain = **most common condition affecting adolescents** (OB) · mainly adolescent females
Pain generator	unclear, **multifactorial** · roughened / damaged **undersurface cartilage of patella** · biologic, mechanical + emotional causes (OB)
Proposed factors	limb malalignment, muscle weakness, chondral lesions, **maltracking** (narrow contact points → **↑ PF contact pressure → pain**) · **dynamic valgus** = excessive medial movement of limb in weight bearing (OB)
Pain receptors	**anterior fat pad + joint capsule = highest potential** · subchondral bone weak
Miserable malalignment	**femoral anteversion + genu valgum + external tibial torsion / pronated feet** → ↑ Q-angle · assess tibial + femoral rotation

Associations & aetiology (TO)

Associated with chondromalacia (TO)	**rheumatoid arthritis**, **alkaptonuria**, **recurrent haemarthrosis**
EXCEPT (TO)	**ankylosing spondylitis**

3 Presentation

Pain	**diffuse peripatellar / retropatellar**, insidious, vague · always weigh physical, mental + social elements
Aggravated by	**climbing or descending stairs** · **theatre sign** (prolonged sitting, knee bent – long car rides, back seat) · **squatting / kneeling** · rising from a deep seat
Exam	**pain on patellar compression** with ROM or **resisted (open-chain) knee extension** · crepitus · quads atrophy · full ROM, **no effusion**, stable ligaments
Maltracking signs	↑ femoral anteversion / tibial ER, lateral subluxation or **loss of medial mobility**, positive apprehension
Differential	quadriceps or patellar tendinitis, **saphenous neuroma**, post-operative neuroma

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
X-ray (AP, lateral, notch ± Merchant)	Often **normal** · chondrosis, shallow sulcus, patella alta / baja, **lateral tilt**
Merchant view	Knee **45° flexion**, beam **30°** · lateral tilt = possible contributor, **not specific** for PF pain
CT	PF alignment, fracture, trochlear geometry, **TT-TG**, limb torsion · TT-TG CT is for **recurrent dislocation**
MRI	**Best for articular cartilage** · **T2** best · abnormal cartilage = **high signal** · only if intra-articular / intra-osseous lesion suspected

Outerbridge MRI classification

Grade	Cartilage
0	Normal
I	Surface intact, heterogeneous, **high signal**
II	Fissures + fragmentation extending down to articular surface
III	**Partial-thickness defect**, focal ulceration
IV	**Exposed subchondral bone**



Chondromalacia of the patella on axial MRI

5 Management

Chondromalacia / PF pain

- All patients → **rest, NSAIDs, quads (VMO) + core + hip PT, ≥ 1 yr**
- Failed rehab + Outerbridge 2–3 → **arthroscopic debridement**
- Tight lateral retinaculum + lateral tilt → **lateral retinacular release**
  (severe, failed extensive PT → realignment osteotomy / MPFLR)

– – Active space – –

Physical therapy – quadriceps, core & hip strengthening

Mainstay	rest, rehab, NSAIDs for **minimum 1 year** · **NSAIDs > steroids** · activity modification
Rehab focus	**VMO**, **core**, **closed-chain short-arc quads**, **hip external rotators** · low-impact aerobic exercise
15-yr-old runner, 5 wk, normal X-rays	**PT focused on quadriceps and core strengthening** ✗ Wrong options: FAI hip series · CT for TT-TG · Graston / ASTYM / iontophoresis (chronic tendinopathy) · chest CT + oncology
Failed 4 wk closed-chain PT	**continued PT with addition of hip exercises** ✗ Wrong options: steroid injection · MRI · diagnostic arthroscopy · patellar taping (adjuvant only)
Exercise that ↑ pain (TO)	**leg extension** = open chain against resistance in functional range → ↑ contact pressure ✗ Wrong options: leg press · squat · lateral step
Best quads exercises (TO)	**closed kinetic chain** (mini-squat, lateral step, high-saddle cycling) in **0–30°** (minimal pressure) · **leg press tolerated** · mini-squat + **hip adduction (ball)** favours VMO
Least PF stress (TO)	**CKC restricted to 0–45° + OKC restricted to 90–45°** ✗ Wrong options: OKC full 90–0° · CKC 60–90° + OKC 30–0° · deep squats 0–90°

Operative options

Procedure	Indication
Arthroscopic debridement (mechanical, radiofrequency)	**Outerbridge grade 2–3** chondromalacia
Lateral retinacular release (open or arthroscopic)	**Tight lateral retinaculum, loose medial capsule, lateral tilt**
Maquet (anterior tubercle elevation)	Elevate **≤ 1 cm** → else **skin necrosis**
Fulkerson (anteromedialisation)	↑ Q-angle, instability (controversial) · not with **superior medial arthrosis** (scope first) or skeletal immaturity
Elmslie-Trillat, MPFL reconstruction	Severe symptoms after failed extensive PT

X vs Y

	Closed kinetic chain	Open kinetic chain
Examples	Leg press, mini-squat, lateral step	**Leg extension**
Safe PF arc	**0–45°** (best 0–30°)	**90–45°**
PF pain	Tolerated	**↑ pain** (full-range resisted extension)

How it's asked

"Exercise likely to increase PF pain (TO)?"	→ **leg extension**
"PFJ-sparing quads prescription (TO)?"	→ **CKC 0–45° + OKC 90–45°**
"Associated with chondromalacia EXCEPT (TO)?"	→ **ankylosing spondylitis**

"Adolescent, theatre sign, normal X-rays — next?"	→ **PT: quadriceps + core strengthening**
"Failed closed-chain PT, normal sunrise views?"	→ **continue PT, add hip exercises**

– – Active space – –

– – Active space – –

ILIAC CREST CONTUSION (HIP POINTER)

1 Basics

Hip pointer	**contusion / traumatic haematoma over the iliac wing** · direct blow or crush in **contact sports** (hockey player checked into the boards)
Haematoma	around iliac wing → may bleed into **hip abductor muscles**
Muscles from the crest (any can be involved)	sartorius, TFL, **gluteus medius**, abdominals (transversus, obliques)
Adolescent	**rule out iliac apophysis avulsion** (see ASIS chapter)

3 Presentation

Exam	**point tenderness over iliac crest** · **contusion + haematoma** · hip weakness · **↓ ROM**



Hip pointer: bruising and haematoma over the iliac crest

4 Diagnosis & Classification

Key radiology findings

X-ray + MRI	usually **unremarkable** · MRI may show a **large haematoma**

5 Management

Nonoperative (mainstay)	**ice, compression, rest** (extended, until symptoms settle), NSAIDs ± muscle relaxants
Therapy	stretch muscles about the crest · **leg on maximum stretch (extension)** ✗ Wrong options: flexion (that is quadriceps contusion)
Corticosteroid injection	near the crest, **some advocate**

Return to play **extra padding** after acute phase

How it's asked

"Hockey check, point tender iliac crest, bruise (photo)?"	→ **hip pointer** ✗ Wrong options: hip flexor strain · apophysitis · piriformis syndrome · trochanteric bursitis

– – Active space – –

ILIOTIBIAL BAND FRICTION SYNDROME

– Knee & Sports · 23 / 61 –

1 Basics

Epidemiology

Incidence	**2–15%** of knee overuse injuries · **most common cause of lateral knee pain in runners + cyclists** (TO)
Who	**runners, cyclists**, repetitive flexion–extension · first seen in **US Marine recruits (1975)**; skiers, hockey, basketball, soccer (TO) · **slightly more common in women** (TO)

Anatomy

Origin	continuation of **tensor fascia lata** (Orthobullets) · distal fascial continuation of **TFL + gluteus medius + gluteus maximus** (TO)
Insertion	**Gerdy tubercle** (lateral tibial plateau) + partly **lateral femoral supracondylar ridge** (TO) · runs **superficial to vastus lateralis**
Iliopatellar band (TO)	anterior extension to lateral patella → **prevents medial patellar translation**
Innervation	**superior gluteal nerve (L4, L5, S1)** · synergists = **hip abductors**
Function (TO)	**knee extensor < 30° flexion**, **flexor > 30°**; moves **posterior** to lateral epicondyle with flexion · anterolateral stabiliser (OB)



Iliotibial band over the lateral femoral condyle

Pathophysiology

Mechanism	ITB shifts forward + back over **lateral femoral condyle / epicondyle** → friction, tension, inflammation · also compression + bursal inflammation (OB)
Impingement zone	**30° of knee flexion** = knee angle at **foot strike** (TO)
Pathoanatomy	compression / irritation of tissue **deep to ITB** (TO: highly innervated **fat pad** under ITB) · cysts or **bursitis of lateral synovial recess** · condylar bone oedema · **ITB itself rarely abnormal**

Associated — **patellofemoral syndrome** (tight ITB) · **medial compartment OA** (↓ medial joint space → varus) · **greater trochanteric pain syndrome**

2 Mechanism

Presentation & running risk factors

Risk factors	Orthobullets	TO
Training	**Sudden ↑ intensity / mileage**, poor shoe support	Tilted surfaces, hills, abrupt training change, **downhill running**
Anatomy	**Genu varum / recurvatum**, LLD, **excessive foot pronation**, **weak hip abductors**, **tight ITB**	Tight ITB + weak hip abductors
Biomechanics	Quads–hamstring imbalance, ↑ landing force, **↑ knee flexion at heel strike**	**↑ hip adduction**, **↑ knee IR** + IR velocity, **↑ knee flexion at heel strike**, **↑ maximal foot inversion**

Foot conflict — **excessive pronation** (Orthobullets) vs **↑ maximal foot inversion** (TO) — read the options

Hockey player, lateral knee pain on running (TO) — **increased knee flexion at heel strike**
✗ Wrong options: ↑ foot eversion · ↑ knee external rotation · ↑ hip abduction

3 Presentation

Symptoms — **activity-related lateral knee pain** in endurance athletes (e.g. **10 min into a run**) · worse with terrain / mileage change, **downhill** · relieved by rest

Site — **lateral femoral condyle / epicondyle** · tenderness **proximal to lateral joint line** (TO) · can be **insertional at Gerdy tubercle** (OB)

26 yr woman, no trauma, pain after downhill running, normal exam (TO) — **ITB syndrome**
✗ Wrong options: lateral meniscus tear · LCL tear · lateral hamstring strain

Exam — swelling over ITB, **foot + knee malalignment**, crepitus, ↓ hip / knee motion, **weak hip abduction**, **pain on single-leg squat**

- - Active space - -

4 Diagnosis & Classification

Diagnosis — provocative tests

Test	Technique	Positive
Ober (Orthobullets)	**Side-lying, symptomatic side up, knee 90°**; hip from **flexion + abduction → extension + adduction**	**Pain, tightness or clicking** over ITB
Ober (OB)	Lying on unaffected side; leg **abducted + extended**	**Can't adduct** the leg
Ober (TO)	Lateral recumbent, knees 90°; examiner abducts + extends hip, then adducts	**Restriction + lateral knee pain**
Noble compression (TO)	Supine, **knee 90°**; press **lateral epicondyle** proximal to joint line; extend knee (+ hip) to 0°	**Pain at ~30° flexion** — **specific for ITBFS**



Ober test positioning

Key radiology findings

Study	Finding
X-ray (AP, lateral ± oblique / skyline)	**Usually normal**; may show **medial JSN**, patellar malalignment, fracture
MRI	Rule out **lateral meniscus tear, LCL injury**; signal in **lateral synovial recess**, ITB or periosteum · **↑ T2 signal deep to ITB at lateral epicondyle + ITB thickening** · **negative MRI does not exclude ITBS**
US / MRI (TO)	**ITB thickening**

Differential — other lateral knee tests

Finding	Means
Flexion/abduction → extension/adduction pain (Ober)	**ITB syndrome**
Pop on rotation + extension (McMurray)	Meniscal tear
Varus laxity at 30°	**Isolated LCL** (LCL inserts on **fibula**, not Gerdy)
ER asymmetry **> 10° at 30°** (dial)	**PLC** (LCL, popliteus, popliteofibular ligament)
"Clunk", IR + valgus, extension → flexion (pivot shift)	**ACL**; reduction at **20–30°**

5 Management

ITB friction syndrome

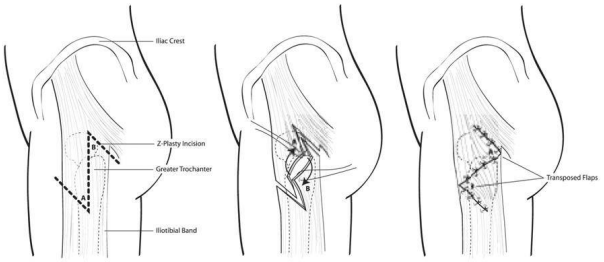
- Acute → **REST, ICE, NSAIDs (oral / topical), activity modification**
  - fails → **CORTICOSTEROID INJECTION (ITB bursa)**
- Then → **PT: ITB / gluteal stretching, HIP ABDUCTOR strengthening, foam roller**
- Refractory > 6 months → **SURGERY (release, excision, Z-plasty, bursectomy)**

Diagnosis (Ober test) & stretching

Initial treatment	**home stretching program + cross training** ✗ Wrong options: bursectomy · Z-lengthening · partial ITB excision
PT	stretch **ITB, lateral fascia, gluteals** · deep transverse friction massage · **hip abductor strengthening** · proprioception · **myofascial release / foam roller** (TO) · strengthening **after** inflammation settles
Training changes	**change shoes every 300–500 miles** · avoid sudden mileage ↑
Return to running (TO)	wk 1: alternate days, **flat surface** → daily fast-paced, **no downhill** → after **3–4 wk** ↑ distance / frequency → hills + cambered surfaces only if pain-free · relapse → rest + restart
Prognosis	**50–90% improve in 4–8 wk** nonop (Orthobullets) · most fully relieved, return in **6–8 wk** (TO)

Operative criteria

Procedure	Indication / detail
Surgery overall	**Refractory > 6 months** nonop (TO)
Excision of cyst, bursa or **lateral synovial recess**	Failed nonop + soft-tissue pathology **without ITB signal change**; arthroscopic or open; may cause **chronic effusion + pain**
Elliptical ITB excision	Chronic, failed nonop · open, lateral distal femur incision · **2 × 4 cm ellipse of posterior band** over lateral epicondyle
Resection of posterior half of ITB (TO)	Resistant cases
Z-plasty lengthening	**Refractory only**
Others (TO)	Percutaneous / open release (triangle over LFC), **bursectomy** (bursa + fat pad), arthroscopic debridement · modern focus: **resect lateral synovial recess** + innervated synovial fat



Z-plasty lengthening of the iliotibial band

– – Active space – –

X vs Y

	Ober test	Noble test
Position	**Side-lying**, knee 90°	**Supine**, knee 90°
Manoeuvre	Hip abduct + extend → **adduct**	**Press lateral epicondyle**, extend knee
Positive	Tight ITB, pain, can't adduct	**Pain at 30°**

How it's asked

"Runner, lateral epicondyle pain, Ober + — initial?"	**→ home stretching program and cross training**
"Cyclist, pain at Gerdy tubercle after ↑ training — exam finding?"	**→ pain with hip brought from flexion/abduction to extension/adduction while side-lying** ✗ Wrong options: McMurray · varus at 30° (LCL) · dial (PLC) · pivot shift (ACL)
"Hockey player, lateral knee pain — running risk factor (TO)?"	**→ increased knee flexion at heel strike**
"Lateral knee pain after downhill running, normal exam (TO)?"	**→ iliotibial band syndrome**

INFECTIOUS DISEASES IN ATHLETES

– Knee & Sports · 24 / 61 –

1 Basics

Overview

Skin infections **10–15% of time-loss injuries** in collegiate athletes (OB, Johnson) · NCAA + NFHS return-to-play rules · mandatory **team skin checks**

Infection	Organism	Spread	Typical sport
CA-MRSA	S. aureus (PVL +)	**Direct skin contact**; **abrasions ("turf burns")**; shared razors, towels, soap	Contact sports, wrestling
Herpes gladiatorum	**HSV-1** (80%, OB)	**Skin-to-skin**; blister fluid, saliva, mucosa	**Wrestling (2–7%)**, rugby
Tinea	Dermatophytes	Direct contact, broken skin; survives off host in warm, moist places	**Wrestling**
Impetigo	**S. pyogenes, S. aureus**	Highly contagious	Wrestling
Acne mechanica	Friction, heat, occlusion	– (not infectious)	**Padded sports** (hockey, football)
Pitted keratolysis	**Corynebacterium**	Moist feet	Runners
Mononucleosis	**EBV** (herpesvirus)	**Saliva** – kissing, shared cups / utensils, sneeze / cough	Any
HIV	CD4 (T-helper) cells	↑ haemophiliacs, IV drug users, MSM	Any

– – Active space – –

Genes & lab markers

Marker	Meaning
Panton-Valentine leukocidin (PVL)	**Cytotoxin that defines CA-MRSA** (not HA-MRSA) → lyses WBCs, tissue necrosis → rapid abscess, "spider bite"
mecA → PBP2a	Low β-lactam affinity → **methicillin resistance** (all MRSA)
SCCmec type IV	**CA-MRSA**; lacks extra elements → **less multidrug resistance** (HA-MRSA SCCmec carries them)
β-lactamase	In most S. aureus; methicillin resists it
D-zone test	Erythromycin-resistant isolate → clindamycin + erythromycin discs → **D-shaped zone = inducible clindamycin resistance**
Heterophile Ab (Monospot)	**87% sensitive, 91% specific**; may stay positive
VCA IgG + IgM	**97% sensitive, 94% specific**
CBC	Absolute + relative **lymphocytosis, > 10% atypical lymphocytes**
LFTs (TO)	Mildly ↑ in **~90%** of IM
KOH prep	**Hyphae → tinea**
Wood's lamp red fluorescence	**Corynebacterium** (pitted keratolysis; erythrasma too)
EBV (TO)	**Double-stranded DNA gamma herpesvirus**; isolated **1964 from Burkitt lymphoma**
AIDS	HIV + **CD4 < 200** or opportunistic infection

2 Mechanism

MRSA risk	**exposed skin from abrasions** ↑ risk significantly · sharing hygiene products · prevention: cover compromised skin + good hygiene
HSV	incubation **2–14 days** (Orthobullets) · up to 14 days (OB) · **5–10 days** (OB) · head, neck, shoulders · eye contact → **herpetic conjunctivitis**
EBV	incubation **30–50 days** · **transmissibility low → isolation unnecessary** · transmission to teammates possible in acute phase (OB) ✗ Wrong options: blood-borne spread more severe
IM epidemiology	annual incidence **1–3% of college freshmen / students** (OB)

3 Presentation

CA-MRSA – presentation

CA-MRSA	**community-acquired methicillin-resistant Staphylococcus aureus**; ↑ in sport
Lesion	boil / pimple · **"spider bite"** · **pustules on an erythematous base**
Wrestler, 6-cm elbow lesion "spider bite"	**CA-MRSA** ✗ Wrong options: psoriasis · tinea · HSV

Abscess → amoxicillin → recurs with fever + pain	**CA-MRSA** · up to **70%** of players hospitalised ✗ Wrong options: group A strep (amoxicillin covers it) · tinea · HSV
7-day ankle swelling after slide tackle + malaise, normal X-ray	**MRSA abscess**



CA-MRSA abscess

Herpes gladiatorum – presentation

Lesion	**clusters of fluid-filled vesicles on an erythematous base** · grouped **1–2 mm** clear vesicles → rupture → moist ulcers / crusted plaques
Primary infection	constitutional symptoms; burning, tingling, stinging · bacterial infections = larger pustules
DDx	impetigo, tinea corporis, eczema



Herpes gladiatorum vesicles on the neck

Other skin infections – impetigo, tinea, pitted keratolysis

Tinea types	pedis, corporis, capitis, cruris (by site) · **corporis** excludes feet, groin, scalp, bearded face, nails
Tinea corporis ("ringworm")	**scaly red circular patches** · well-demarcated plaque, **raised border, central hypopigmentation** · hypo- / hyperpigmented dry, dusty oval patches · highly contagious → minor epidemics

– – Active space – –



Tinea corporis

Impetigo	fluid-filled blisters → **crusting after a few days** · **golden-brown crusts around nose + mouth** (OB)
Acne mechanica / folliculitis	**red papules**, inflamed follicles under padding
Pitted keratolysis	**shallow crater-like pits on weight-bearing sole** (palms less often), thickened skin, **foul odour**, sweating · **red fluorescence on Wood's lamp** (Corynebacterium)

Infectious mononucleosis – presentation

Prodrome (3–5 days)	**malaise, myalgia, nausea, headache**
Hoagland's triad	**fever, pharyngitis, lymphadenopathy** (Orthobullets, OB) · TO "triad": fatigue + pharyngitis (chief complaint), **posterior cervical >> axillary / inguinal nodes**, athlete with only fatigue / ↓ performance
Pharyngitis	30% (Orthobullets, GAS) · **exudative** (white / grey pseudomembrane) in 50% · exudative pharyngitis + **tonsillar enlargement** classic (TO)
Lymphadenopathy	**posterior cervical chain**, lasts 2–3 wk
Rash	**after ampicillin / amoxicillin**; petechial / maculopapular / urticarial · **10–40%** (TO)
Other (TO)	**posterior palatine petechiae (⅓)**, **jaundice 10–15%**
Splenomegaly	palpable only **50%** (OB) · lymphocytic infiltration pushes spleen beyond rib cage · resolves **4–6 wk** (TO)

IM symptom (TO)	Frequency
Sore throat	**95%**
Cervical lymphadenopathy	80%
Fatigue	70%
Upper respiratory symptoms	65%
Headache · ↓ appetite	50% each
Fever	47%
Myalgia	45%

Course	symptoms resolve **4–8 wk** (Orthobullets) · acute phase **7 days to 2–3 wk**, recovery up to **2–3 months** (TO) · most findings ≤ 10 days; **fatigue + cervical nodes ≥ 3 wk** (TO)

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
IM imaging	**Generally unnecessary**
Ultrasound	**First choice if imaging** – noninvasive, reliable, no radiation · confirms **normal spleen size**; baseline variability limits RTP use (TO)
CT	**Exclude splenic rupture** (stable patient)
FAST	Free fluid + unstable → **laparotomy**



Splenic rupture after mononucleosis (CT)

Diagnostic criteria

Condition	Criteria
AIDS	**HIV positive + CD4 < 200** or **opportunistic infection**
Tinea	KOH scraping → **hyphae**
Pitted keratolysis	Crater-like pits + odour + **red Wood's fluorescence**
IM	Heterophile Ab (Monospot); VCA IgG / IgM; lymphocytosis > 10% atypical

– – Active space – –

5 Management

Return-to-play criteria

Infection	Return to play
CA-MRSA	After **72 h on antibiotics** (e.g. TMP-SMX) (OB) · **cover lesions** · NCAA bacterial: **no new lesions 48 h** + **72 h antibiotics** + **no exudative lesions** (OB)
Herpes gladiatorum	**No new lesions 72 h** + **≥ 5 days (120 h) antivirals** + **all lesions dry / crusted** + no systemic symptoms (NCAA)
Tinea corporis	**≥ 72 h topical antifungal** + lesions **covered** + **screened before competition**
Tinea capitis (scalp)	**14 days** antifungal (NCAA, OB)
Molluscum contagiosum (pox virus, painless papule)	Lesions **removed (curetted)** + **covered with Tegaderm** before play (NCAA, OB)
Impetigo	**All lesions clear of crusting**
Mononucleosis	**No contact sport 3–5 wk** (up to 3 months) **until splenomegaly resolves** – see sources below
HIV	**HIV alone does not bar competition** · universal precautions · out until **bleeding stops**

CA-MRSA – treatment & prevention

Lesion	Treatment
Small	**Topical mupirocin**
Larger	**I&D + oral TMP-SMX + rifampin**
Severe	**I&D + IV antibiotics**

Abscess
incise, drain, culture first; antibiotics = necessary adjunct · empiric anti-MRSA cover until sensitivities
✗ Wrong options: topical antibiotics + occlusive dressing · training-room debridement + oral antibiotics · ice only

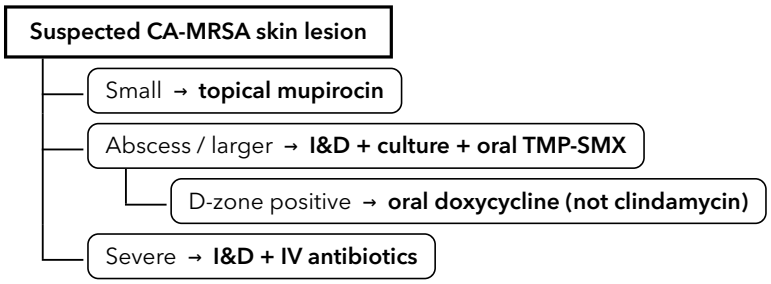
Superficial, no systemic illness
oral antibiotics
✗ Wrong options: IV

Erythromycin-resistant + D-zone positive
switch clindamycin → oral doxycycline
✗ Wrong options: continue clindamycin · IV vancomycin · IV cefazolin · **rifampin alone** (rapid resistance)

Mupirocin small + I&D / TMP-SMX large
= **MRSA boil**
✗ Wrong options: impetigo · molluscum contagiosum · ringworm · HSV

Most effective team prevention
cover lesions with occlusive dressing during sport
✗ Wrong options: separate locker room · prophylactic antibiotics for teammates · ban from team events

Hygiene
showering, hand washing, breathable clothing, shower sandals, **no shared towels / equipment**, daily skin surveillance, disinfect equipment, restrict contact



Herpes gladiatorum – antivirals & return to play

Treatment	**acyclovir, valacyclovir, famciclovir** for any active lesion → **inhibit viral DNA polymerase / inactivate viral DNA synthesis** ✓
Duration	**5 days regardless of agent** ✗ Wrong options: famciclovir 3 days
Traps	✗ Wrong options: TMP-SMX (MRSA) · unroof / debride · RTP 48 h after starting · streptococci on culture (impetigo) · wrestle if covered · out 2 weeks · keflex 48 h
Wrong-drug distractors (OB)	clotrimazole → **ringworm** · mupirocin → **impetigo** · erythromycin → **erysipelas**

Tinea – treatment

Topical antifungal	**tinea corporis, cruris, pedis** · **topical ketoconazole 14 days, RTP after 72 h** ✓ · imidazoles (clotrimazole, ketoconazole, miconazole) or allylamines (terbinafine, naftifine)
Systemic antifungal	**tinea capitis** (oral itraconazole; griseofulvin first line) + severe cases · **oral fluconazole** → widespread, failed topical, immunocompromised ✗ Wrong options: for an isolated lesion
✗ Wrong option: Bactrim	antibiotic, no role in tinea
No sport	while infection active

Impetigo, acne mechanica, pitted keratolysis – treatment

Impetigo	**erythromycin, topical Bactroban (mupirocin)** first line · no sport while active
Acne mechanica	**observation first** (resolves after season) · **tretinoin** (keratolytic) if severe · wash immediately after play, moisture-wicking clothing
Pitted keratolysis	keep feet dry, **topical clindamycin / erythromycin / mupirocin**, antiperspirants; oral antibiotics if severe · clindamycin 1%–benzoyl peroxide 5% gel effective (OB)

Infectious mononucleosis – splenomegaly, splenic rupture & return to play

Supportive	fluids, acetaminophen, rest · **no isolation** · **valacyclovir not needed**
Strep pharyngitis	**IM penicillin once** or **oral penicillin 10 days**; erythromycin if allergic · **NO amoxicillin** (rash)
Corticosteroids	shrink tonsils if dysphagia / dehydration (Orthobullets) · **no role for antivirals or corticosteroids** (OB) – read the options
Other	airway management if respiratory distress · **stool softener** (↓ Valsalva straining)

– – Active space – –

Rest (TO)	rest **3 weeks**, then light activity · no strenuous activity, weights, contact, roughhousing first **21 days** · spleen still palpable at 3 wk → light activity until **28 days**
RTP prerequisites (TO)	**afebrile, hydrated, asymptomatic, no palpable liver or spleen** · individualise (contact vs noncontact, in- vs off-season) · explain splenic risk — **full recovery may take months**
Not RTP markers	negative heterophile test (stays positive) · fatigue / sore throat better (may resolve before spleen) · lymphocytosis resolved · 48 h after antibiotics

Mono RTP rule	Source
3 weeks — contact sport after 3 wk of illness if no ongoing signs	**TO**
3 wk after symptoms resolve, no splenomegaly	OB
4 wk after onset (seen at 1 wk → 3–4 wk more)	OB
4 wk after onset AND normal-sized spleen	OB
3–5 wk off until splenomegaly resolves	OB, Orthobullets
No exertion 3 wk, then **light non-contact activity**	OB (consensus)

Diagnosed 2 wk ago, asymptomatic 1 wk	**RTP in 2 weeks if no splenomegaly** ✗ Wrong options: now · valacyclovir course

6 Complications

Splenic rupture

Rate	**0.1–0.5%** (Orthobullets, OB) · 0.1–0.2% (OB) · **< 1%** (TO) — most feared complication
Timing	**most in first 3 weeks** (21 days) (Orthobullets, OB, TO) · within a month (OB)
Mechanism	**sudden ↑ portal venous pressure** · **50% atraumatic Valsalva** (rowing, weightlifting) + 50% external trauma (Orthobullets) · **up to 86% atraumatic** (OB) — read the options
Normal-sized spleens	rupture reported in nonathletes + normal spleens too
Returned at 2 wk, tackled, severe left abdominal pain	**splenic rupture** ✗ Wrong options: liver contusion · sports hernia · iliac / ASIS avulsion
Signs	LUQ pain, **left shoulder pain (Kehr's sign)**, fatigue
Unstable + FAST positive	**exploratory laparotomy** ✗ Wrong options: CT · IV fluids alone · angiography · Monospot

Other complications of mononucleosis

Long-term performance (TO)	**chronic fatigue** (> 6 months) ✓ ✗ Wrong options: thrombocytopenia · myocarditis · hepatitis
Rare, ~1% (TO)	airway obstruction, strep pharyngitis, meningoencephalitis, haemolytic anaemia, thrombocytopenia
Orthobullets list	aplastic anaemia, Guillain-Barré, meningitis / encephalitis, neuritis, lymphoma, HUS, DIC
Malignancy link (TO)	**Burkitt, Hodgkin lymphoma, nasopharyngeal carcinoma**

HIV wound care **compressive dressing**; participation stops until all bleeding ceases

X vs Y

Appearance	Diagnosis	Treatment
Pustule / boil, "spider bite", abscess	**CA-MRSA**	Mupirocin · I&D + TMP-SMX
Grouped vesicles on erythematous base	**HSV**	Acyclovir / valacyclovir 5 days
Ring, raised scaly border, central clearing	**Tinea corporis**	Topical antifungal ≥ 72 h
Golden-brown crusts, nose / mouth	**Impetigo**	Erythromycin, mupirocin
Red papules under pads	Acne mechanica	Observe · tretinoin
Well-demarcated raised erythema, face	Erysipelas	Erythromycin
Sole pits + odour + red Wood's	**Pitted keratolysis**	Topical clindamycin
Reddish-brown patches (Corynebacterium)	Erythrasma	–
Thick red skin, silver-white flaky patches	Psoriasis	–
Keratin plugs in follicles, no odour	Keratosis pilaris	–
Scaling + itch, no pits, no fluorescence	Tinea pedis	Topical antifungal

	CA-MRSA	HA-MRSA
Defining element	**PVL cytotoxin**	–
SCCmec	**Type IV**	Multidrug elements
Drug resistance	**Less multidrug**	Multidrug resistant
Common in	Healthy athletes, close contact	Healthcare settings

How it's asked

"Safe return after mononucleosis (TO)?"	**→ 3 weeks**
"Sore throat + cervical nodes – complication hurting long-term performance (TO)?"	**→ chronic fatigue syndrome**
"Rash, tonsillar exudate, LUQ mass – true?"	**→ RTP 3 weeks after asymptomatic**
"Mono 1 wk, splenomegaly, wants to play tomorrow?"	**→ safe after 3–4 weeks**
"Lacrosse player with mono – RTP?"	**→ 4 weeks after onset AND normal-sized spleen**
"Mono 2 wk ago, asymptomatic 1 wk?"	**→ RTP in 2 weeks if no splenomegaly**
"Splenomegaly + lymphocytosis – RTP?"	**→ extended rest + resolution of splenomegaly**
"Mono, tackled at 2 wk, left abdominal pain?"	**→ splenic rupture**
"LUQ + left shoulder pain, unstable, FAST +?"	**→ exploratory laparotomy**
"MRSA, D-zone positive – definitive antibiotic?"	**→ oral doxycycline**
"Abscess recurs after amoxicillin?"	**→ CA-MRSA**

– – Active space – –

"Best prevention of MRSA spread?"	→ occlusive dressing over lesions during sport
"Fluctuant ankle lesion, malaise — treatment?"	→ incision and drainage
"Wrestler's 'spider bite'?"	→ CA-MRSA
"Genetic element defining CA-MRSA?"	→ Panton-Valentine leukocidin
"Vesicles on chest wall before NCAA final?"	→ out until scabbed + no new lesions 72 h
"HSV — true?"	→ treatment inactivates viral DNA synthesis
"Facial vesicle cluster — RTP?"	→ after 5 days on valacyclovir
"Neck vesicles — treatment?"	→ acyclovir, RTP after 5 days + crusting
"Mupirocin small, I&D + TMP-SMX large — figure?"	→ MRSA boil
"Ringworm on back — treatment + RTP?"	→ topical ketoconazole 14 days, RTP after 72 h
"Sole craters, odour, red Wood's lamp?"	→ pitted keratolysis
"MRSA return to play?"	→ after 72 h of antibiotics
"Impetigo return to play?"	→ all lesions clear of crusting
"HIV-positive athlete?"	→ may compete; universal precautions

KNEE IMAGING

– Knee & Sports · 25 / 61 –

4 Diagnosis & Classification

Key radiology findings – high-yield signs

Finding	Meaning
Segond sign	Small **lateral tibial** avulsion → **ACL tear**
Reverse Segond	Small **medial tibial** avulsion → **PCL tear**
Arcuate sign	**Fibular head** avulsion → **PLC injury**
Pellegrini-Stieda	**Medial femoral condyle** avulsion / calcification → **chronic MCL**
Patella alta	Patellofemoral pathology
Patella baja	**Arthrofibrosis**
Fairbanks changes	Post-meniscectomy DJD: **square condyle, peaked eminences, ridging, narrowing**
Lateral MFC lesion	**OCD**
Crescent-shaped MFC lesion	**SONK**
Square LFC + cupped lateral plateau	**Discoid meniscus** (3 consecutive sagittal MRI slices)
Bipartite patella	Must differentiate from fracture



Segond fracture (lateral tibial avulsion)

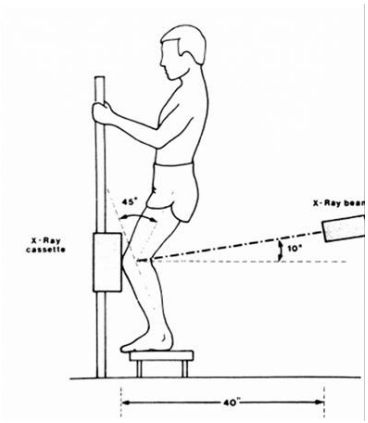
Weight-bearing flexed PA & mechanical axis views

Weightbearing films — preferred for joint space · **always** when patient can bear weight (shows mechanical loading)

Rosenberg view = weight-bearing PA in 45° flexion — **most sensitive for degenerative change** · best for **early tibiofemoral OA, posterior wear** · earliest cartilage loss in **30–60° flexion zone** (missed on extension films), esp. **lateral compartment**
✗ Wrong options: non-WB AP · WB AP · non-WB PA · Merchant

– – Active space – –

Dervin 45° PA > AP vs arthroscopy for lateral wear → **"screening radiograph of choice"** for knee OA



Method for obtaining the PA flexion knee radiograph

Merchant / sunrise patellofemoral space, tilt, alignment

Medial pain, varus, old arthroscopy, normal supine PCP films (OB) **next: mechanical axis + weight-bearing radiographs**
✗ Wrong options: reassurance + unloader brace · steroid injection · MRI · PT (no diagnosis yet)

Young medial OA option (OB)	When
Medial unloader brace, steroid injection	Non-surgical, isolated medial OA
Valgus HTO	Disrupted mechanical axis, excessive varus · medial overload after meniscectomy · osteochondral defects (Rossi) · also address slope
Meniscal transplant	Normal / corrected axis, stable, minimal arthritis (Amendola)
UKA	Some cases; arthroplasty usually avoided at this age

Brouwer valgus HTO ↑ pain + function in medial OA · no non-op control · no osteotomy proven better

Stress radiographs

Stress view	Use
Varus–valgus	**Physeal fractures in children** · MCL / LCL injuries
Posterior stress	**Isolated PCL: 10–12 mm** posterior displacement · **PCL + PLC: > 12 mm**

MRI

Plane	Evaluates
Sagittal	Menisci, cruciates, cartilage, extensor mechanism
Coronal	Menisci, cruciates, **collaterals**, cartilage
Axial	**Patellofemoral joint**, cruciates, popliteal fossa

Sequence	Signal	Use
T1	**Water dark, fat bright**	**Anatomy**, not pathology
T2	**Fat dark, water bright**	**Oedema + pathology**
STIR	Fat suppression	Better imaging **with prostheses**

Meniscus	**medial extrusion > 3 mm** → severe degeneration, large tear or **root tear** · radial tears ↑ after prior meniscal surgery (**32% vs 14%**) · **3.0 T MRI > 90%** accuracy / sensitivity / specificity
ACL	↑ signal, thickening, cysts in / around ACL = common + **clinically insignificant** (no instability)
Bone bruise	near **sulcus terminalis** · osteochondral injury
PLC	LCL + popliteus tendon routinely seen (others rarely) · **oedema posterior to popliteus tendon** → underlying PLC injury
Reading rules (TO)	diagnose on **sagittal + coronal**; **axials key for patellofemoral + popliteus tendon** · normal **ACL greyer than PCL** (fibrofatty; AM taut, PL fibrillar) — high signal between intact taut fibres ≠ sprain · normal **PCL uniformly dark, buckled in extension** · fluid in popliteal sulcus + magic-angle signal normal · **fabella** in lateral gastrocnemius = normal
Osteophyte vs enthesophyte (TO)	osteophytes at **articular margins** · spurs at **tendon insertions = enthesophytes**
Metal (TO)	paramagnetic artefact mimics oedema → request **MAVRIC** (metal artefact reduction) MRI
Partner injuries (TO)	**isolated popliteus tear does not occur** → look for FCL, ALL, cruciate · **every ACL tear → look for lateral meniscus posterior root tear** (near ACL footprint; MFLs can hide it) · PCL injury → check **popliteal artery** flow void

Ultrasound

Uses	**fluid collections**, bursae · **arthrofibrosis after TKA → synovial thickening + neovascularity**

How it's asked

"Most sensitive view for degenerative changes?"	**→ weight-bearing PA in 45° flexion (Rosenberg)**
"Medial pain, normal supine films from PCP — next step?"	**→ mechanical axis + weight-bearing radiographs**
"Lateral femoral condyle + inferomedial patella contusions (TO)?"	**→ transient lateral patellar dislocation**
"Loss of extension months after ACLR, graft intact (TO)?"	**→ cyclops lesion**

– – Active space – –

LATERAL PATELLAR COMPRESSION SYNDROME

– Knee & Sports · 26 / 61 –

1 Basics

Cause	**tight lateral retinaculum → excessive lateral tilt without excessive patellar mobility**
Miserable triad	**femoral anteversion + genu valgum + external tibial torsion / pronated feet** → ↑ Q-angle, worse PF dysplasia
Q-angle	normal **17° in females** (OB) · 21° = increased

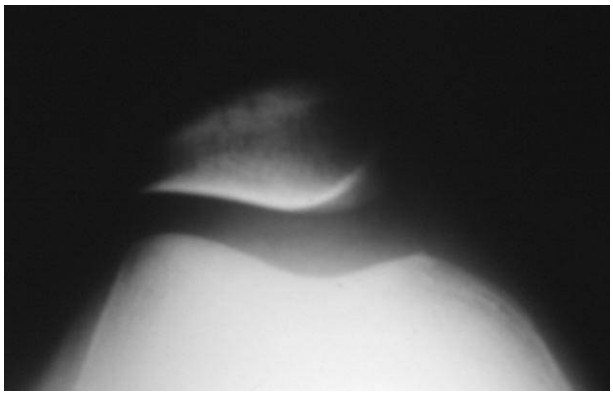
3 Presentation

Symptoms	pain with **stair climbing** · **theatre sign** (pain sitting for long periods)
Exam	**pain on patellar compression**, moderate **lateral facet tenderness**, **cannot evert lateral edge of patella** · stable knee, no effusion

4 Diagnosis & Classification

Diagnosis – patellofemoral pain syndrome

Merchant / sunrise — **lateral patellar tilt** (neutral or negative tilt)



Lateral patellar tilt on Merchant view

TT-TG (CT)	trochlear trough line ⊥ posterior condylar line, superimposed on tubercle slice · **> 20 mm → medialising TTO** needed · 14 mm = normal
13-yr-old gymnast, normal X-rays, negative tilt, pain on resisted open-chain extension	**patellofemoral syndrome** ✗ Wrong options: bipartite patella · distal femoral stress fracture · patella alta · baja
PF syndrome (OB)	no clear cause · quadriceps + **dynamic malalignment** contribute · X-rays + exam · MRI only if intra-articular / intra-osseous lesion suspected · **low-impact aerobic exercise**

5 Management

Lateral patellar compression

- Initial → **NSAIDs, activity modification, closed-chain quads + VMO rehab (prolonged)**
- Refractory + objective lateral tilt, no instability → **arthroscopic lateral release**
- Refractory + distal / lateral facet lesions, ↑ Q → **realignment osteotomy**

Initial management – closed-chain strengthening

Mainstay	nonop for an **extensive period** · NSAIDs, activity modification · **VMO strengthening** + **closed-chain short-arc quads**
First step (Q-angle 21°)	**closed chain exercises with focus on quadriceps and hamstring strengthening** + isometrics ✗ Wrong options: lateral release · tubercle elevation + medialisation · strict immobilisation / NWB · **open-chain seated leg extensions** (↑ PFJRF)

Operative criteria – lateral retinacular release indications

Lateral release indication	Detail
Objective lateral tilt	**Neutral or negative tilt** on Merchant
Refractory pain	After extensive rehab
Cannot evert lateral patellar edge	Tight lateral retinaculum
No instability	Ideal candidate
Medial glide < 1 quadrant	–
Lateral glide < 3 quadrants	–

9 months, Q 15°, < 1 quadrant medial glide, can't evert, TT-TG 14 mm	**lateral retinacular release** ✗ Wrong options: anterolateral TTO (↑ TT-TG) · **Maquet** (tightens lateral retinaculum, worsens negative tilt) · **Elmslie-Trillat** (TT-TG normal) · medial plica resection
Arthroscopic findings	superior portal: **medial facet does not articulate with trochlea at 40° flexion**
Technique	secure **haemostasis** · after release patella **passively tilts 80°**
Outcome	↑ Lysholm in < 40 and > 40 yr · fibrosis at release site rare
Complications	persistent / worse pain · **medial patellar instability**

Patellar realignment osteotomies

Procedure	Movement	Indication	Avoid
Maquet	**Anteriorisation**	**Distal pole lesions only**	Elevate **> 1 cm** → **skin necrosis**
Elmslie-Trillat	**Medialisation**	**Instability with lateral translation** (not isolated tilt)	**Medial facet arthrosis**
Fulkerson	**Anteromedialisation**	**Lateral + distal pole lesions**, ↑ Q-angle (controversial)	**Superior medial arthrosis** (scope first), **skeletal immaturity**

– – Active space – –

How it's asked

"Anterior knee pain, Q-angle 21° — first step?"	**→ closed chain quadriceps + hamstring strengthening**
"Refractory, can't evert patella, TT-TG 14 mm?"	**→ lateral retinacular release**
"13-yr-old, normal radiographs, pain on resisted extension?"	**→ patellofemoral syndrome**

– – Active space – –

LCL INJURY OF THE KNEE

– Knee & Sports · 27 / 61 –

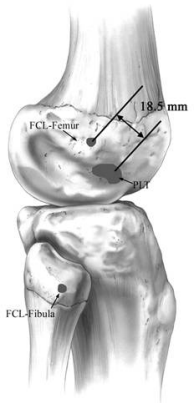
1 Basics

Epidemiology

Isolated	**extremely rare (< 2% of knee injuries)** · gymnasts, tennis players
Combined	**7–16%** of knee ligament injuries, mainly with **PLC** · also ACL, PCL

LCL anatomy – femoral insertion relative to popliteus & epicondyle

Shape	tubular, cord-like · 2–3 mm thick · 4–5 mm wide · **66 mm** long (anatomy) · **~70 mm** (graft-sizing text)
Femoral origin	**posterior (3.1 mm) + proximal (1.4 mm) to lateral epicondyle** · **posterior + proximal to popliteus** origin ✗ Wrong options: anterior · distal · directly on epicondyle
LCL ↔ popliteus distance	**18.5 mm** · popliteus always attaches **anterior** to LCL (proximal + anterior fifth of popliteal sulcus) (LaPrade, 10 cadavers)
Fibular insertion	**anterolateral fibular head**, covers 38% of width · **most anterior structure on proximal fibula** · order **LCL → PFL → biceps femoris**
Blood supply	anterior tibial recurrent + **inferolateral geniculate** arteries



FCL (LCL) femoral footprint 18.5 mm posterior-proximal to popliteus tendon (PLT)

Biomechanics

Varus restraint	**primary at 5° and 30°** · **55% at 5°, 69% at 30°** (LCL note) · 5° (55%) and **25°** (69%) (PLC note) – read the options
Also	secondary restraint to posterolateral rotation at **< 50°** flexion · resists varus in full extension with ACL + PCL
Position	**behind the axis of rotation → tight in extension, lax in flexion** · tensile strength 750 N

– – Active space – –

28 lateral stabilisers (OB) 3 primary: **LCL, popliteus tendon (PT), PFL** · unreconstructed PLC → ↑ force on ACL / PCL grafts → failure

Structure (James)	Restrains	Peak
LCL	**Varus** + rotation **in extension**	**0–30°**
Popliteus tendon + PFL	**Rotation in flexion** – most rotational stability	**60–90°**
Lateral meniscus	2° restraint to anterior translation	–
Lateral gastrocnemius	No significant rotational role	–

Medial stabilisers (LaPrade)	Role
sMCL	**Primary valgus** (proximal) + 2° rotational (distal) at all angles
Deep MCL	2° valgus + rotation
POL	**Primary rotational** restraint

2 Mechanism

Traumatic **direct blow / force to medial knee** · varus stress, external tibial rotation, hyperextension

3 Presentation

Symptoms instability near full extension · stairs · **cutting / pivoting** · lateral joint line pain + swelling

Exam ecchymosis · tender LCL insertion · **figure-of-4 position** → palpable cord-like LCL · **varus (lateral) thrust** or hyperextension gait · **common peroneal nerve** exam

Test	Finding	Meaning
Varus stress 30° only	Laxity	**Isolated LCL**
Varus stress 0° + 30°	Laxity	**LCL ± ACL / PCL** (cruciate)
Dial test	**> 10° ↑ ER at 30°**	**LCL + PLC**
Dial at 30° + 90°	↑ ER	PCL + PLC
Lachman 1A	**< 5 mm**, firm endpoint	ACL intact

Varus laxity at 30° + dial + at 30° only, Lachman 1A (OB) **LCL + PLC**
✗ Wrong options: isolated LCL (no dial) · ACL · PCL + PLC (+ at 90°)

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
Weightbearing AP, lateral	Asymmetric lateral joint widening
Varus stress X-ray	Quantifies lateral opening
MRI	**Modality of choice** · **most tears off fibular insertion** · medial compartment bone contusions (hyperextension-varus)
MRI sensitivity	**95%** vs EUA **58%**



Varus stress radiographs

LCL grading

Grade	Clinical (lateral opening vs other side)	MRI
1	**0–5 mm**	Fluid around ligament, intact
2	**6–10 mm**	Partial fibre tear
3	**> 10 mm, no firm endpoint**	Complete tear (midsubstance or insertion)

5 Management

LCL injury

- Isolated grade I / II (stable at 0°) → **limited immobilisation + ROM + rehab (RTS 6–8 wk)**
- Isolated grade III, acute (< 2 wk), avulsed from fibula → **repair**
- Grade III subacute / chronic or midsubstance with varus instability → **reconstruction**
- Rotatory / posterolateral instability → **LCL + PLC reconstruction**

Diagnosis & treatment of LCL / PLC injury

Nonop progressive ROM, quadriceps + hamstring strengthening · 6 wk casting heals but → **stiffness** · unrecognised PLC → progressive varus / hyperextension laxity

– – Active space – –

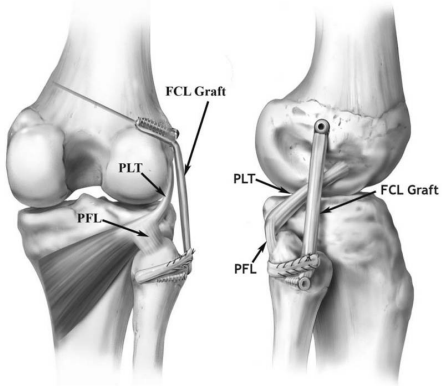
Repair	**acute (< 2 wk) grade III avulsion** · failure **up to 40%** · repair fails more in chronic **(> 3 wk)** (OB)
Reconstruction	better than repair · **6% failure at 3 yr** · **anatomic + semitendinosus autograft** best
Chronic isolated grade 3 + peroneal paraesthesias, athlete (OB)	**LCL reconstruction + peroneal neurolysis** · Moulton: semitendinosus graft, 43 pts, ↑ Lysholm + WOMAC ✗ Wrong options: PT + hinged brace (grade 1–2 only) · PRP · repair (chronic) · popliteus / PFL reconstruction (no PLC)
LCL heals worse than MCL	lower threshold to operate in grade 3 (Grawe)

Operative criteria

Procedure	Indication
Nonop	**Isolated grade I / II**, no instability at 0°
Repair	**Isolated acute (< 2 wk) grade III, avulsed from anatomic site (fibula)**
Reconstruction	Subacute / chronic (> 2 wk) grade III with varus instability · acute **midsubstance** grade III
LCL + PLC reconstruction	**Rotatory** or **posterolateral** instability (better if acute)

Techniques

Lateral approach	interval **ITB (superior gluteal n.) – biceps femoris (sciatic n.)** · second interval in ITB for femoral insertion · **peroneal neurolysis** · traction suture (knee extended) tests repairability · suture anchors
Graft	**semitendinosus autograft preferred** (closer to LCL length; patellar tendon ~50 mm; stronger than gracilis; less saphenous irritation) · alternatives patellar tendon / Achilles allograft
Tunnels	fibular: lateral fibular head → posteromedial styloid, distal to PFL · femoral: ~3 mm posterior to epicondyle, exit anteromedial
LCL + PLC	**Larson** fibular-based figure-of-eight (LCL + PFL) · transtibial double-bundle split Achilles · anatomic LCL + popliteus + PFL (bifid graft)



Anatomic LCL reconstruction

6 Complications

– – Active space – –

Complication	Fact
Persistent varus / hyperextension laxity	**Grade III treated nonop · missed PCL or PLC**
Peroneal nerve injury	**Up to 44%** of multiligament injuries with LCL / PLC
Stiffness	Prolonged immobilisation
Physeal arrest	Errant lateral condylar fixation in skeletally immature
Graft harvest	Saphenous neurapraxia, MCL injury

How it's asked

"LCL femoral attachment vs popliteus?"	**→ posterior and proximal**
"Two structures giving most rotational stability in flexion?"	**→ popliteus tendon + popliteofibular ligament (C + D)**
"LCL femoral insertion vs lateral epicondyle?"	**→ posterior and proximal**
"Varus laxity at 30°, dial + at 30° not 90°, Lachman 1A?"	**→ LCL + PLC**
"Chronic isolated grade 3 LCL + peroneal symptoms, athlete?"	**→ LCL reconstruction + peroneal neurolysis**

– – Active space – –

MCL KNEE INJURIES

– Knee & Sports · 28 / 61 –

1 Basics

Epidemiology – most commonly injured knee ligament

Most common (TO)	**MCL = most commonly injured knee ligament** · **ACL = most common needing surgery** ✗ Wrong options: ACL · MPFL · ATFL
Incidence	**40%** of knee ligament injuries (under-reported) · **8%** of athletic knee injuries · **males** > females · skiing, rugby, football, soccer, ice hockey
Associated ACL tear	**most common** associated injury · **~95%** of non-isolated MCL injuries · **ACL + MCL = most common multiligament injury** · hemarthrosis suggests it · ↑ with MCL grade (III > II > I)
Meniscus	**medial > lateral** · up to **5%** of isolated MCL injuries

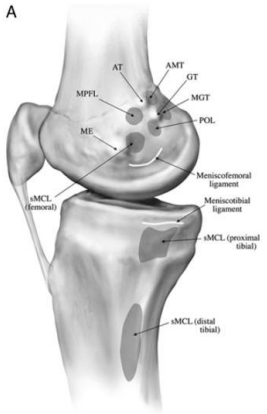
Anatomy

	Superficial MCL (sMCL)	Deep MCL (dMCL)
Layer	**II** (with PMC ligaments, MPFL)	**III** (with capsule)
Femur	Medial epicondyle · **3.2 mm proximal + 4.8 mm posterior** to ME · **1 cm anterior + distal to adductor tubercle**	Meniscofemoral ligament
Tibia	Proximal **11.2 mm** below joint line (soft tissue) · distal **6 cm below joint line**, on bone, **deep + posterior to pes**	Meniscotibial ligament
Function	**Primary valgus stabiliser at all angles** · max at **25° (78%)** · 2° to ER + AP translation	**2° valgus stabiliser**, max in **full extension**

Blood supply	superior + inferior medial geniculate arteries
Other medial stabilisers	**POL** (resists IR in full extension; 2° valgus; primary rotational restraint) · oblique popliteal ligament · posterior capsule · dynamic **semimembranosus complex** (5 attachments) · vastus medialis, medial retinaculum, pes (sartorius, semitendinosus, gracilis)

Anatomic sMCL reconstruction landmarks

Graft footprints (OB)	femur **3.2 mm proximal + 4.8 mm posterior to medial epicondyle** (between AT and ME) → tibia **6 cm distal to joint line** · location **B : 3** on figure
Distractors	tibial point 1 = **proximal** sMCL attachment · point 2 = **pes anserinus** ✗ Wrong options: distal
Wijdicks	anatomic augmented repair ≈ anatomic reconstruction (18 matched pairs); neither restores native kinematics · **LaPrade** – sMCL + POL anatomic reconstruction (28 pts) → ↓ valgus gapping, ↑ scores



LaPrade medial knee anatomy (sMCL, POL, adductor tubercle)

2 Mechanism

Mechanism of valgus injury

Commonest **valgus stress**, knee in slight **flexion + external rotation**

Contact (more common) **direct blow to lateral knee → high-grade / complete** · rupture usually at **femoral insertion**

Non-contact commoner in **skiing** · pivot / cut with valgus + ER → **low-grade**

MCL mechanism (TO)	Fact
Valgus direct blow to outer knee	**Most common**
Non-contact valgus + twist (foot planted)	Basketball, skiing, tennis · often **with ACL**
Hyperextension + valgus	Less common · gymnastics, skiing, football
Fall with knee forced into valgus	Catching ski edge, falling sideways

Level of the blow (TO) hit on lateral **distal leg**, knee slightly flexed, foot planted → **forced flexion, valgus → ACL tear** · force higher (thigh / knee), knee nearer extension → **MCL tear**
✗ Wrong options: forced valgus MCL · varus LCL · valgus ACL without flexion

ACL (TO)	Fact
Contact vs non-contact	**~30% vs ~70%**
Classic	**Forced flexion + valgus, foot planted**, femoral adduction, tibial ER, ankle eversion (pivot / cut)
Landing	Stiff (extended) or valgus collapse · volleyball, basketball, gymnastics
Deceleration	Sudden stop · soccer, skiing, football
Hyperextension	Tibia moves forward
Risk factors	Poor core stability · **stiff landing (knee flexion < 20°; source prints "<200")** · weak hip abduction · ↑ knee valgus · **heel-strike landing**

– – Active space – –

PCL (TO)	Fact
Dashboard	Posterior force on proximal tibia, knee flexed
Hyperflexion	Foot plantarflexed, fall on bent knee
Hyperextension	Often **ACL + PCL**
Rotational + varus / valgus	Rugby, football, wrestling
Fall on knee, foot plantarflexed	Wrestling, ice hockey, American football

3 Presentation

History / symptoms **"pop"** · medial joint line pain · difficulty walking

Exam medial tenderness, ecchymosis, effusion · **saphenous nerve** exam · check ACL, PCL, patellar dislocation, medial meniscus

Valgus stress	Meaning
30° flexion	**Isolates sMCL** · gap vs other knee = grade
0° (full extension)	**Posteromedial capsule or cruciate** injury

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
AP + lateral	**Usually normal**
Stress X-ray (skeletally immature)	Gapping through **physeal fracture**
X-ray (chronic)	**Pellegrini-Stieda** calcification at medial femoral insertion
MRI	**Modality of choice** · location, extent, other injuries

Pellegrini-Stieda lesion

Definition (TO) **ossification of MCL at / near proximal (femoral) insertion on MFC** · **chronic** MCL tear / deficiency
✗ Wrong options: fibular LCL · tibial sMCL · tibial dMCL

Pellegrini-Stieda disease / syndrome (TO) lesion **+ medial knee pain or restricted ROM**

– – Active space – –



Pellegrini-Stieda calcification at the medial femoral MCL origin

Look-alike (TO)	Site	Association
Pellegrini-Stieda	**Medial femoral condyle** (MCL origin)	**Chronic MCL**
Segond fracture	**Lateral proximal tibia** avulsion (= **ALL**)	**ACL**
Reverse Segond	**Medial tibial plateau** cortical avulsion	**PCL** mainly · MCL, MM, probably ACL
Stieda process	Elongated **lateral tubercle of posterior process of talus**	Anatomical variant

Classification

AMA (1966, laxity alone)	Opening
Grade I	**3–5 mm**
Grade II	**6–10 mm**
Grade III	**> 10 mm**

Hughston modification	Endpoint	Injury
I (mild)	**Firm, no laxity**	Stretch / few fibres
II (moderate)	**Firm ± mild laxity**	Partial tear
III (severe)	**No endpoint**	Complete · **1+ 3–5 mm, 2+ 6–10 mm, 3+ > 10 mm**

Gap at 30° → grade **1–4 mm I · 5–9 mm II · ≥ 10 mm III** (Orthobullets exam) · AMA 3–5 / 6–10 / > 10 mm · **0–5 / 6–10 / 11–15 mm** (Orthobullets knee exam topic) – read the options

5 Management

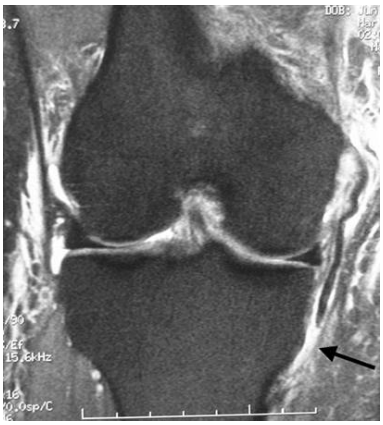
MCL injury

- Grade I → **NSAIDs, rest, therapy (RTP 5–7 days)**
- Grade II → **hinged brace + therapy (RTP 2–4 wk)**
- Grade III, stable in extension, no cruciate → **brace (RTP 4–8 wk)**
- Grade III + multiligament / Stener / entrapment → **acute repair**
  - Chronic or poor tissue → **reconstruction**

Therapy	immediate quad sets, SLR, hip adduction above knee · cycling + progressive resistance

Natural history & healing by tear location (nonoperative grade III)

Proximal (femoral) tears	**heal most reliably** (vascularity) · ↓ residual valgus laxity · more **calcification** (temporary pain + stiffness)
Distal (tibial) tears	heal worse, residual valgus laxity · **Stener-type lesion** = torn MCL **flipped over pes anserinus insertion** → can't heal
Shelbourne	grade III at **femoral origin / midsubstance → nonop** · **tibial origin → primary repair**
Combined injuries	heal less reliably than isolated
Grade III + ACL tear	**ACL does NOT mandate MCL repair** (EXCEPT answer) · grade III = > 10 mm valgus opening, can form Stener lesion



Distal grade III MCL rupture, coronal MRI

ACL reconstruction delay by MCL grade	OB (MCL topic)	TO (Knee Dislocation, Trauma)
Grade I	**3–4 wk**	**2 wk**, no motion restriction
Grade II	**4–6 wk**	**within 2 wk**; brace locked 30° × 7 days
Grade III	**6–8 wk**	**3 wk**; brace locked 30° × 10–14 days

Delay conflict	OB 3–4 / 4–6 / 6–8 wk vs TO 2 / ≤ 2 / 3 wk – read the options

Operative criteria

Option	Indication
Acute repair, grade III	**Multiligament** injury · **displaced distal avulsion (Stener-type)** · **entrapment of torn end in medial compartment**
Subacute repair, grade III	Instability despite nonop · **> 10 mm medial opening in full extension**
Reconstruction	**Chronic** injury · **inadequate tissue**
All surgical candidates	**Diagnostic arthroscopy** first

Repair technique avulsions → **suture anchors at 30° flexion** · interstitial → anterior advancement to origins · **internal brace** · medial approach

Reconstruction graft semitendinosus autograft / hamstring, tibialis anterior or Achilles allograft

Prophylactic knee bracing

Hinged prophylactic brace **trend to ↓ MCL tears only** · functional bracing may help **interior linemen**
✗ Wrong options: ACL · ankle (may ↑ ankle stress)

Evidence Albright: non-significant trend · **West Point RCT: 3.4 vs 1.5 / 1000 exposures** (unbraced vs braced), MCL in defensive players · Pietrosimone SR: inconsistent → can't advocate or discourage

6 Complications

Loss of motion · **saphenous nerve** injury · **laxity** (distal injuries)

How it's asked

"Grade III MCL – all true EXCEPT?"	**→ requires repair when ACL torn**
"MCL tear that heals most reliably?"	**→ proximal**
"Prophylactic hinged bracing ↓ which injuries?"	**→ MCL tears**
"sMCL anatomic reconstruction extents on figure?"	**→ B : 3 (3.2 mm prox / 4.8 mm post to ME → 6 cm below joint line)**
"Hit on lateral distal leg, foot planted (TO, FNB 2023)?"	**→ forced flexion, valgus and ACL tear**
"Same stem, FNB 2024 (TO)?"	**→ answer not provided by source – same stem → forced flexion, valgus, ACL tear**
"Pellegrini-Stieda lesion (TO)?"	**→ ossification at femoral attachment of MCL**
"Identify condition (TO)?"	**→ Pellegrini-Stieda lesion**
"Most commonly injured knee ligament (TO)?"	**→ MCL**

– – Active space – –

MENISCAL CYSTS

– Knee & Sports · 29 / 61 –

1 Basics

Definition — local collection of **synovial fluid** within or next to the meniscus, **most often from a meniscal tear**

Incidence — no general-population data · **1–4% of knee MRIs** · no age trend

Association — **meniscal tear** – almost always present, the most common associated lesion (TO) · also articular cartilage injury, ACL tear (Orthobullets) · TO: false: "common with ACL tear"

Type	Site	Usual tear
Perimeniscal	Small fluid collection **within** the meniscus · medial → **posterior horn** · lateral → **anterior horn / mid-portion**	**Radial or vertical**
Parameniscal (e.g. Baker)	**Extruded outside the meniscus – most common** · usually between **semimembranosus + medial head of gastrocnemius**	**Horizontal or complex**

Side — **medial slightly commoner, 2:1** (Orthobullets – literature conflicting) · **medial > lateral** on recent radiology reviews; lateral was the old teaching (TO)

Lateral meniscal cysts (OB) — **myxoid degeneration** spreading from the meniscal centre outward · **horizontal cleavage tears** · most often from a **middle-third LM tear**
✗ Wrong options: discoid meniscus · PLC injury · vertical tears · popliteus tear

2 Mechanism

One-way valve — the tear lets synovial fluid out, not back → fluid concentrates into gel-like material

3 Presentation

History — ± recent trauma · may be **asymptomatic**

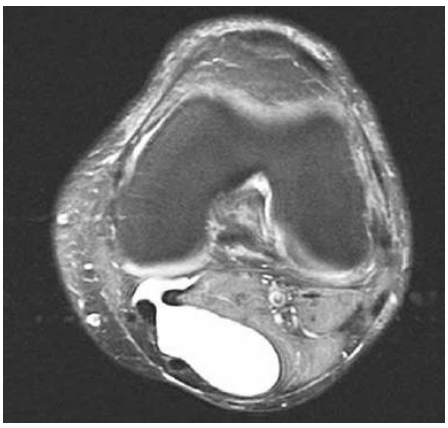
Symptoms — pain at the **medial / lateral joint line or back of knee** · **locking + clicking** · delayed / intermittent swelling · weakness or claudication (**neurovascular impingement**)

Exam — **popliteal mass, best seen in extension** · joint line tenderness · palpable mass · crepitus

4 Diagnosis & Classification

Key radiology findings

Study	Finding
Radiographs	**Normal** in young patients with acute meniscal injury or cyst
MRI	**Most sensitive test** for both the **cyst and the tear** · cyst = **bright T2**
MRI mimics	**Necrotic tissue, nerve sheath tissue, pus** look like cysts on T2 → **IV contrast** may be needed



Parameniscal (Baker) cyst between semimembranosus and medial gastrocnemius — axial T2 MRI

Parameniscal cyst on MRI — **implies a meniscal tear** (Meniscal Tears chapter)

5 Management

Meniscal cyst

- Small perimeniscal or parameniscal → **rest, NSAIDs, physiotherapy (first line)**
- Isolated Baker cyst, young patient → **aspiration + US-guided steroid injection**
- Tear communicates with joint, not repairable → **arthroscopic partial meniscectomy + decompression**
- Symptomatic parameniscal / tear not open to joint → **open posterior cyst excision**

Nonop — first line · trial to gauge pain response · may work with degenerative tears

Aspiration + steroid — for **isolated Baker cyst in the young** · **poor results in older patients with degenerative tears + cysts**

Arthroscopic debridement, decompression + partial meniscectomy — perimeniscal cyst with a **tear not amenable to repair** (complex, degenerative, radial) · decompress fully · **incomplete resection → recurrence** · repair the meniscus if the tear allows (Orthobullets summary)

Tear doesn't open into joint (OB) — arthroscopy **followed by open cystectomy**

Open posterior excision — **prone**, curved incision over popliteal fossa · interval **medial gastrocnemius ↔ semimembranosus** · sharp dissection of cyst down to capsule · incomplete resection → recurrence

– – Active space – –

How it's asked

"Lateral meniscal cyst — associated pathology?"	**→ middle-third lateral meniscal tear**
"True about meniscal cysts (TO)?"	**→ most common associated lesion is a meniscal tear**

MENISCAL TEARS

– Knee & Sports · 30 / 61 –

1 Basics

Epidemiology & pathophysiology

Incidence	very common · **most common indication for knee surgery** · ↑ risk in **ACL-deficient** knees
Medial > lateral	overall · **exception: acute ACL tear → lateral tears commoner** (alpine skiers: LM > 4× MM) · **chronic ACL → medial** (PH MM = secondary restraint)
MM tears (TO)	**5–6×** commoner than LM (relative fixation to capsule / MCL) · LM circular, more mobile
ACL + meniscal injury (TO)	meniscal tear in **> 50% of acute** and **> 80% of chronic** ACL tears
Acute tears	young athletes · **twisting on a loaded, flexed knee**
Degenerative tears	older (**> 40 y**, loss of elasticity) → **posterior horn medial meniscus**
Commonest pattern (TO)	**radial** in a normal meniscus · **horizontal** in a discoid meniscus

Meniscal structure & collagen (TO)

Feature (TO)	Fact
Arteries	**Medial, lateral + middle geniculate** (from popliteal) → **perimeniscal capillary network** in synovium / capsule
Vascular rim	**Peripheral 10–30% of MM**, **10–25% of LM** (Orthobullets: MM 20–30%)
Avascular part	Remaining **65–75%** fed by **synovial fluid diffusion / mechanical pumping** (joint motion)
Age	**Fully vascular prenatal → just after birth** · 10–30% at age 10 · **peripheral 10–30% at maturity = red-red**
Nerves	**Posterior articular branch of posterior tibial nerve** + terminal **obturator + femoral** · lateral capsule = **recurrent peroneal branch of CPN**
Nerve fibres	Concentrated in the **periphery + anterior / posterior horns** (✓ horns rich in mechanoreceptors)
Mechanoreceptors	**Ruffini endings, Pacinian corpuscles, Golgi tendon organs**
ECM	**Water 72%, collagen 22%** + cells
Collagen	**Type I predominates** (+ II, III, V, VI) · collagen = up to **75% of dry weight** · **collagen (type I) gives tensile strength** (✗ Wrong options: type II)
Fibres	Type I **circumferential in deep layers** (parallel to periphery) · **radial superficially**
Hoop stress	**Axial load → tensile stress along circumferential fibres** (✓ generated on axial loading)

– – Active space – –

Why peripheral tears heal	↑ **blood supply** (geniculate vessels at capsular attachment) ✗ Wrong options: elasticity · type I / II collagen · GAG (distribution similar, no effect on healing)

2 Mechanism

O'Donoghue unhappy triad (TO)

Triad (TO)	**ACL + MCL + medial meniscus** · **lateral force on knee with foot fixed** · O'Donoghue **1950** ✗ Wrong options: LCL · PCL · lateral meniscus
Later arthroscopic studies (TO)	**lateral meniscus tear** commoner with **acute ACL** tear → triad questioned
Bucket handle (TO)	twisting injury, **fixed leg with flexed knee** struck by another player
Root tear	MMPRT: **low-energy attritional** · **descending stairs / downhill**, rising from a seat, deep squat → **pop** · LPRT: **with ACL** (high energy)

3 Presentation

Symptoms	pain localised medial / lateral · **mechanical (locking, clicking), esp. squatting** · **delayed / intermittent swelling** · effusion
Joint line tenderness	**most sensitive physical examination finding**
Locked knee	**cannot actively or passively extend** (past 40°, past 30°, ROM 20–90°) → **displaced bucket handle in notch** · **MRI indicated** with mechanical symptoms or locked knee
Root tear	**pop** in loaded flexion (rising from seat, stairs) · **posteromedial** JLT · **less catching / locking** than flap tears (meniscus extrudes) · exam **lacks specificity** ✗ Wrong options: exam = gold standard

Provocative tests

Test	Technique	Positive
McMurray (Orthobullets / OB)	Flex knee, hand on joint line, rotate tibia, bring **flexion → extension**	Palpable **pop / click + pain** · **ER → medial**, **IR → lateral**
McMurray (TO)	Knee **> 90° flexion**, full **IR (LM)** or full **ER (MM)**; more flexion → **more posterior segments**	**Thud / click felt** (sometimes heard)
Apley	**Prone, 90° flexion, axial compression**	Pain with ER (MM) / IR (LM)
Thessaly	Standing on affected limb, twist IR / ER at **5° then 20°** flexion	Joint line pain, locking / catching
Ege	**Squat** with knees max ER or IR	Pain / click

McMurray stress – conflict	**valgus force** for medial meniscus (OB) · **no varus / valgus stress** (TO) – read the options
Deep flexion + ER pain (TO)	**posterior horn medial meniscus** tear ✗ Wrong options: body of MM · discoid LM · anterior horn LM
Acute ACL + joint line tenderness	**lateral meniscus tear** ✗ Wrong options: MM · popliteus · proximal tib-fib · pes bursitis

MM tenderness, intact ACL + PCL, dial 5°	**medial meniscal tear only** (dial negative → no PLC)
Exam traps	no valgus end-point = MCL · no varus end-point = LCL · apprehension = patellar instability · **posterior drawer + quadriceps active test** = PCL · **pivot shift** (clunk with flexion, IR, valgus from extension) = ACL · 3-quadrant lateral translation + **J-sign** = patellar dislocation · posterior sag = PCL
Most specific test (TO)	**Thessaly** (WB rotation at 20°) · **joint line tenderness = most reliable** (TO) · swelling **delayed > 24 h** (synovial irritation, not haemarthrosis)

Differential diagnosis

MCL sprain	pain with **valgus stress at 30°** (isolates superficial MCL), medial gapping
Plica syndrome	**medial parapatellar** pain, palpable cord
Osteochondral lesion	presents similarly → **MRI** differentiates

4 Diagnosis & Classification

Genes & lab markers

RF + anti-CCP positive, synovitis at arthroscopy	**active rheumatoid arthritis → absolute contraindication to meniscal allograft**

Key radiology findings

Study	Finding
Radiographs	**Normal** in young acute tear · **meniscal calcification** in crystal arthropathy (**CPPD**)
MRI	**Most sensitive** test · **high false-positive** rate
MRI tear criterion	**Grade III signal** = linear high signal **reaching the superior or inferior surface**
Parameniscal cyst	**Indicates a meniscal tear**
Extrusion	**Root tear** · < 3 mm extrusion not significant
Which compartment (sagittal)	**Lateral**: convex "anvil" tibia + **fibula** visible · **medial**: concave "champagne glass" tibia

– – Active space – –

MRI sign	Means
Double PCL	**Bucket-handle tear of MM** – low-signal fragment in notch **parallel, anteroinferior to PCL** · **specificity 100%** for flipped bucket handle
Absent bow tie (TO)	**Bucket handle** – < 2 bow ties on consecutive sagittal images (normal 9–12 mm → 2–3 images at 3 mm slice, 1 mm gap) · **more accurate than double PCL** · sensitivity 88.4%
Fragment in notch	Bucket handle · ≥ 1 displaced-fragment sign sensitivity 90.7% · commonest sign with absent bow tie + double PCL
Double anterior horn / double meniscus	**Flipped meniscal tear** (TO) · **flipped lateral meniscal tear** (OB210282) · **bucket handle**, fragment anterior to native anterior horn, often LM (Orthobullets, OB7436)
Empty / ghost meniscus (TO)	**Complete radial tear**
Ghost sign (OB / Orthobullets)	**Root tear** – no identifiable meniscus on sagittal images, or ↑ signal replacing dark meniscus · ± extrusion on coronal
Cleft sign (TO)	**Radial tear**
Floating meniscus (TO)	**Meniscotibial (coronary) ligament avulsion** – fluid under posterior horn
Vacuum phenomenon (TO)	Nitrogen gas (negative pressure, position) → **mimics tear in discoid meniscus**
Meniscus within meniscus (TO)	Wedge of high signal **not reaching surface → NOT a tear**
Ramp (TO)	Thin fluid line **PH MM ↔ posteromedial capsule** (most specific) ± posteromedial tibial oedema



Double PCL sign – bucket-handle fragment in the notch

Double PCL prerequisite (TO)	**intact ACL + PCL** · **double delta sign** = flipped meniscus (TO) · **tram-track** = PCL mucoid degeneration (not meniscal)
Double anterior horn – conflict	**flipped meniscus tear** (TO, with "bucket handle" as a wrong option; OB210282: flipped LM tear) · **bucket handle** (Orthobullets, OB7436) – read the options
Ghost – conflict	"ghost / empty meniscus" = **complete radial tear** (TO) · "ghost sign" = **root tear** (OB / Orthobullets) – a radial tear within 1 cm of the root is a root tear

Bucket-handle MRI distractors bone bruise **mid LFC + posterior lateral tibial plateau** = ACL · **medial patellar facet** bruise = patellar instability · fluid around lesion at **posterolateral MFC** = unstable OCD · T2 signal at **posterior tibial sulcus** = PCL

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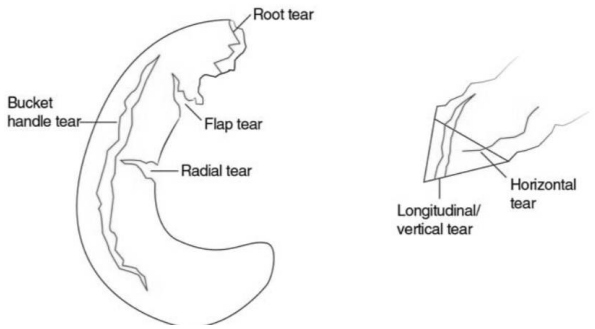
Tear classification

Zone	Distance from MCJ (TO)	Blood supply	Healing (TO)
Red-red (outer third)	**0–3 mm**	Rich	**> 90%** – repair all
Red-white (middle third)	**3–5 mm** (5 mm = classic)	Few vessels	**60–80%** – repair, often with augmentation
White-white (inner third)	**> 5 mm**	**Avascular**	**< 50%** – "almost never heals"

Rim width distance from tear to **meniscocapsular junction** · **smaller rim = better blood supply = heals** · repairable **< 5 mm** (OB) · best **< 4 mm** (Orthobullets Meniscus) · Henning: healing up to **5 mm** with synovial rasping – read the options

Other descriptors position (anterior, middle, posterior third, root) · size

Pattern	Key fact
Vertical / longitudinal	Common, **esp. with ACL**; in line with fibres (separated, not disrupted) → **repair when peripheral**
Bucket handle	Displaced vertical tear → **notch**; complete tears **> 1 cm** unstable → **block to extension**
Oblique / flap / parrot beak	**PH–body junction**; **mechanical locking**; repair rarely possible → **excise to stable rim**
Radial	Cuts **circumferential fibres**; **complete to MCJ = posterior root tear** biomechanically
Horizontal (cleavage)	**Older**, degenerative · **meniscal cysts**
Complex / degenerative	Not reparable
Root	Avulsion or **radial tear within 1 cm** of root → **= total meniscectomy**
Ramp	PH MM meniscocapsular / meniscotibial separation **with ACL**



Meniscal tear patterns

Root tears & meniscal avulsion

Root function (TO) converts **axial load → hoop stress** · loss → **extrusion**, altered kinematics, ↓ contact area, **↑ peak contact pressure** → early OA

– – Active space – –

Peak contact pressure ↑ **25%** (= total meniscectomy) · ↑ **most in flexion** · ↑ ER + lateral tibial translation · repair restores normal
✗ Wrong options: extension

Partial-thickness tears oblique / parrot beak **< 70% thickness** → no significant pressure change

Posterior > anterior roots esp. **medial (least mobile)** · anterior horn injury rare → **iatrogenic**: ACL tibial tunnel reaming (**posterolateral** tunnel) (TO), tibial nailing

	Medial posterior root (MMPRT)	Lateral posterior root (LPRT)
Patient	**Middle-aged / elderly women**, ↑ age, **obesity**, **varus**, sedentary	**Young, active**
Nature	**Degenerative**, attritional	**With ACL tear** (TO: not degenerative)
Linked with	**Chondral injury**, extrusion → **rapid medial OA**	Suspect with **grade 3 Lachman + 3+ pivot shift** (TO)
Treatment	Root repair if joint preserved	**Always repair with the ACL** (TO)



Ghost sign – absent meniscus at the root

Meniscal avulsion (TO) **meniscotibial (coronary) ligament** torn → **floating meniscus** · **repair whenever possible** (suture to anatomic position)
✗ Wrong options: "both meniscotibial + meniscofemoral torn"

Ligaments (TO) meniscofemoral → posterior femur · **meniscotibial = coronary**, short peripheral bands, part of **layer 3 (deepest)** of lateral capsule

Floating-meniscus mimic (TO) **Wrisberg-type discoid** (no posterior tibial attachment)

Ramp lesions (TO)

Definition (TO) longitudinal / oblique **peripheral tear of PH MM** → meniscocapsular or meniscotibial disruption · Staubli + Rauschning **1994** · lies in **red-red zone** → heals well

Anatomy (TO) posterior junction = **meniscocapsular ligament (superior) + meniscotibial ligament (inferior)** · with PH MM resist **anterior tibial translation + posteromedial rotation**

ACL link (TO) **9–40%** of ACL-injured knees · pivot shift subluxes medial plateau → avulses PH · ↑ with **chronic ACL / delayed reconstruction**

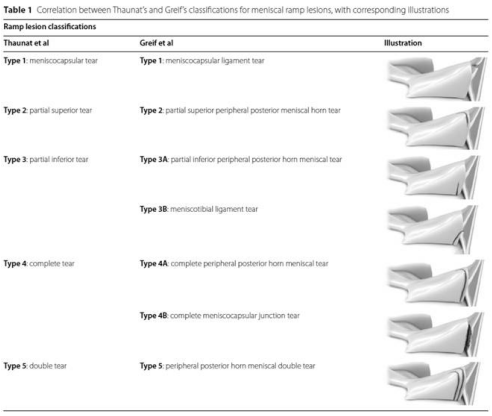
Risk factors (TO) posteromedial tibial **bone contusion**, chronic ACL, **steep medial tibial + meniscal slope**, gradual lateral tibial slope, **varus > 3°**, **male < 30**, concomitant **lateral meniscal tear**

Diagnosis (TO) exam non-specific · **MRI sensitivity 48–65%** · **gold standard = arthroscopy through a posteromedial portal** or **trans-notch (Gillquist) view**
✗ Wrong options: MRI · McMurray · standard anterior portals (blind spot)

Differential (TO) **zip lesion ("Wrisberg rip")**: tear at Wrisberg MFL–PH LM junction with ACL, **> 1.4 cm** mediolateral from lateral PCL edge (vs Wrisberg pseudotear) · flipped tear · root tear

Other classifications (TO) Seil (extent, capsular injury, **adherent → self-heal vs dehiscent → repair**) · Thaunat · Greif (extended Thaunat)

Thaunat (TO21733124)	Greif	Stability
1 Meniscocapsular tear (isolated, posterosuperior)	1 Meniscocapsular ligament tear	Stable
2 Partial superior tear	2 Partial superior peripheral PH tear	Stable
3 Partial inferior tear – "hidden"	3A partial inferior PH tear · **3B meniscotibial ligament tear**	Unstable
4 Complete tear	4A complete peripheral PH tear · 4B complete MCJ tear	Unstable
5 Double tear	5 Peripheral PH double tear	Unstable



Thaunat and Greif ramp classifications

Thaunat & Fayard 2016 (TO25178973)	Description
1	Partial meniscocapsular separation – mildest, stable on probing
2	**Complete meniscocapsular separation – classic, most common**
3	Meniscocapsular tear **+ partial longitudinal extension** into PH
4	Complete longitudinal tear + meniscocapsular separation – large unstable fragment
5	**Double longitudinal tear** – most severe, uncommon

Ramp classification – conflict the two TO explanations describe Thaunat types differently · **"hidden" = type 3** · types 1–2 stable, 3–5 unstable – read the options

Meniscal calcification (TO)

Causes (TO) **trauma, ochronosis, pseudogout (CPPD), degenerative arthritis / OA, hyperparathyroidism** · **NOT rheumatoid arthritis**, **NOT osteopetrosis**

– – Active space – –

Meniscal cysts (TO)

Association (TO) **almost always an underlying meniscal tear** (most common associated lesion) · synovial fluid **extrudes through the tear**

Side (TO) **medial > lateral** on recent radiology reviews (old teaching: lateral)
✗ Wrong options: lateral commoner · equal · "common with ACL tear"

Discoid meniscus (TO)

Discoid (TO) **most frequent congenital meniscal malformation** · **lateral > medial** · highest prevalence **Japanese + Korean** · classified by **Watanabe 1969** (arthroscopic)
✗ Wrong options: "classified by Wrisberg"

Watanabe (TO)	Shape / coverage	Posterior attachment	Fact
Complete (I)	Block / disc, **entire** lateral plateau	Normal	Commoner (with incomplete); usually asymptomatic
Incomplete (II)	Semilunar, **< 80%** / up to 80%	Normal	Usually asymptomatic; torn → lateral JLT, clicking, effusion
Wrisberg (III)	**Near-normal size + shape**	**No coronary / popliteomeniscal attachment – only Wrisberg MFL**	**Least common**, most unstable, **younger**, no trauma → **snapping knee**

Structure (TO) **↓ collagen density, disorganised network** · mucinous change (like degenerative meniscus) · ↑ thickness, enlarged surface, **↓ peripheral vascularity**

Commonest tear in discoid (TO) **horizontal** (repetitive microtrauma after mucoid degeneration)

Snapping knee syndrome (TO) **Wrisberg type** · also a torn stable discoid whose tear spreads to the posterior tibial attachment · audible / palpable **"meniscal thud" at terminal extension (10–20°)**, clunk at end of flexion · **3–4 y**: often painless snapping · **8–10 y**: pain with activity, more acute

Symptoms (TO) insidious popping / snapping without trauma, pain, giving way, effusion, quadriceps atrophy, ↓ ROM, locking

Neuschwander (TO) Wrisberg type = "**lateral meniscal variant with absence of the posterior coronary ligament**" (not truly discoid)

Discoid medial (TO) much rarer · **disc shaped** · linked with **trauma / tears**

5 Management

Algorithms

Meniscal tear

- Degenerative, older → **nonop first (rest, NSAIDs, PT)**
  ("noninferior" to arthroscopic partial meniscectomy)
- Young, locked knee / displaced bucket handle → **arthroscopic reduction + repair**
- Peripheral vertical / longitudinal, young → **repair**
- Root tear → **root repair (see below)**
- Complex, degenerative, flap, white-white → **partial meniscectomy**

Posterior root tear

- Minimal arthritis (KL ≤ 2), BMI < 35 → **transtibial pull-out root repair**
- With ACL tear → **ACL reconstruction + root repair**
- Established arthritis → **arthroplasty (UKA / TKA)**

Operative criteria — repair indications (TO)

Patient factors (TO)	Tear factors (TO)
Younger (< 40), active	**Red-red / red-white** — ideal, not mandatory
No significant comorbidities	**Simple pattern**: vertical, longitudinal, **bucket handle**, **root**, **ramp**
BMI < 30	**< 3 months old**
Willing to follow rehab	**Associated ACL reconstruction**
	Reducible without excess tension
	Lower threshold for complete radial tears

NOT a repair indication (TO) — **horizontal tear of 2 cm**
✗ Wrong options: associated ACL tear · 1-month-old bucket handle · 2-month-old root tear

Best indication (TO) — **capsulomeniscal junction tear** = most peripheral red-red, maximum healing · **best zone = red-red**
✗ Wrong options: red-white · white-white · fresh radial in young

Best candidate (Orthobullets) — **peripheral red-red**, **low rim width**, **vertical + longitudinal**, **1–4 cm**, **root tear**, **acute + ACL reconstruction**

Favouring repair (TO) — length **> 10 mm** · **unstable (> 3 mm displacement on probing)** · **acute (< 6 wk)** · young active · **concurrent ACL reconstruction**

– – Active space – –

Partial meniscectomy instead **not reparable (complex, degenerative, radial) · repair failed > 2 times**

Repair prognosis	Good	Poor
Pattern	**Vertical + longitudinal**	Radial, horizontal, complex, degenerative
Location	**Red-red, small rim width**	Red-white junction, **large rim width**, white-white
Knee	**ACL reconstructed**	**Unaddressed ACL insufficiency** (meniscus strained as secondary stabiliser)
Most important factor	**Width of the meniscal rim**	✗ Wrong options: days from injury · medial vs lateral · growth hormone · dominant leg

Stärke (OB) biology matters more than technique · ↓ healing in complex / degenerative, central, unstable-knee tears

Timing / length – conflict time from injury, side and length **not consistently shown to affect healing** (OB) · **acute < 6 wk** and **> 10 mm** favour repair (TO) · **1–4 cm** ideal (Orthobullets) – read the options

Age young → **repair** (better OA prevention + return to sport) · older → nonop or partial meniscectomy

Bucket-handle tears – locked knee & repair

Young athlete, locked knee, double PCL (TO) **arthroscopic repair**
✗ Wrong options: partial meniscectomy · RICE 2 wk then exercises or return to sport

Locked knee, LM bucket on MRI **arthroscopic lateral meniscectomy or repair** · large bucket handles usually **red-red → reparable**; badly frayed → partial meniscectomy
✗ Wrong options: medial side · transplant (not acute) · passive stretching (more damage) · steroid

Displaced LM bucket handle, 3-mm rim, 16 y **inside-out vertical mattress sutures** · standard for bucket handles = **inside-out**
✗ Wrong options: partial / subtotal meniscectomy · all-inside arrows / darts · horizontal mattress

12 days, lacks extension, LM bucket handle **arthroscopic repair + immediate mobilisation** (or short immobilisation) · **no prolonged immobilisation**
✗ Wrong options: rest + PT · loose-body removal · PCL reconstruction · immediate ACL reconstruction

Fixation inside-out (**gold standard**), outside-in, all-inside → **equivalent outcomes** in recent studies

Bucket handle vs radial repair **equivalent clinical outcomes + reoperation rates** (VAS, IKDC, Tegner)
✗ Wrong options: higher reoperation for bucket handle

Radial repair – conflict radial tears "generally not amenable to repair" (OB, Orthobullets meniscectomy indications) · **equivalent to bucket-handle repair** (newer OB, Orthobullets recent evidence) – read the options

Meniscal repair — healing & outcomes

Repair success	Orthobullets	OB	TO
Overall	**70–95%**	–	Inside-out 70–90%
With ACL reconstruction	**90% (highest)**	**> 90%** · Barber **92%**	**85–95%**
Intact ACL / stable knee	**60% (modest)**	60% · Barber **67%**	50–60% (no ACL reconstruction)
Untreated ACL deficiency	**30% (poor)**	–	–

Why ACL reconstruction helps — **haemarthrosis** from tunnel drilling → **growth factors** (+ marrow-derived stem cells, TO), restored stability · **2nd-generation all-inside: no difference** with / without ACL reconstruction (current literature, Orthobullets) · Barber: no change to accelerated ACL rehab needed

Treated at ACL reconstruction — heals best — **meniscal tear (> 90%)**
✗ Wrong options: chondral defect 60–70% · grade III MCL (19% reoperation for arthrofibrosis) · PLC (90% objective stability) · PCL (85–90%)

Longitudinal tear 5 mm from MCJ, 28 y (TO) — **red-white zone; concurrent ACL reconstruction improves healing (marrow-derived stem cells)** (keyed) · (✗ Wrong options: white-white → meniscectomy only · red-red, inside-out "to spare saphenous") · isolated red-white repair → **fibrin clot / PRP**

Biological augmentation (TO) — **fibrin clot** (venous blood; PDGF, TGF-β, VEGF, MSCs; **Arnoczky**) · **PRP** (mixed evidence; red-white + white-white) · **marrow stimulation** (trephination / microfracture) · **rasping** (Orthobullets)

Suture configuration — **vertical mattress = strongest** (captures circumferential fibres)

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Repair techniques

(TO + Orthobullets)	Inside-out	Outside-in	All-inside
Status	**Gold standard** (Henning 1983)	Warren 1985, Morgan 1991	**Most common**; modern standard for posterior horn
Best for	**Mid-body + posterior horn**, bucket handle, large complex, **revision**	**Anterior horn** (technique of choice) + body	**Posterior horn** of both menisci; with ACL reconstruction
Method	Long needles in → out via **zone-specific cannulae**; 2-0 braided non-absorbable, **vertical mattress 3–5 mm apart**	**18-G spinal needle** out → in; Morgan **mulberry knot**	Suture-anchor devices (FasT-Fix: 2 polymer anchors + pre-tied sliding knot)
Incision	**Posterior safety incision** 2–4 cm (TO) / 3–4 cm (TO) for posterior half	Small "poke" incisions	**None** – portals only
Strength	**Highest** – pull-out **80–120 N**	Suture-dependent; may be < inside-out	**60–80 N**; some give horizontal mattress
Advantage	Versatile, surgeon control, longest results	Simplest, **lower NV risk** (anterior entry)	**Fast (2–3 vs 15–20 min per suture)**, no incision, no assistant
Main risk	**Saphenous (medial) / peroneal (lateral) nerve**	**Chondral damage** (repeated passes); poor posterior reach	**Implant failure**, migration, **cyst**, chondral injury, breakage, cost $300–600 per device

1st-generation all-inside (TO) rigid absorbable **arrows / darts / staples** → migration, cartilage damage, foreign-body reaction, cysts → abandoned

Hybrid repair (TO) combined techniques → **bucket-handle tears** · suture **UHMWPE** (FiberWire / FiberTape) · vertical or horizontal mattress · all-inside usable anywhere but **costly**

Augmentation (TO, Mohindra) synovial abrasion / **rasp** · **vascular access channels** (refenestration) · PRP / fibrin clot · **concurrent ACLR = natural biological stimulus**

Inside-out indications (TO) posterior horn, middle third, peripheral capsular, bucket handle · **contraindicated**: **root tears**, severe degeneration, avascular-zone tears

Open repair uncommon; trauma, knee dislocation

Turman & Diduch gold standard = **inside-out vertical mattress** · all-inside for special cases (concurrent ACL reconstruction)

Failure inside-out ≈ all-inside · **synovitis** with absorbable sutures / implants

Inside-out approach intervals & nerves at risk

	Medial meniscus	Lateral meniscus
Incision	**Posterior to MCL**; incise **sartorius fascia**	Between **ITB and biceps**
Retract	**Pes + semimembranosus posteriorly**	Biceps posteriorly → **lateral gastrocnemius posteriorly** (+ medially – shields NV)
Deep interval	**Medial head gastrocnemius ↔ capsule**	**Lateral head gastrocnemius ↔ capsule**
Nerve at risk	**Saphenous nerve** (infrapatellar branch) + vein – needle or **suture entrapment**	**Common peroneal nerve** – foot drop, the more feared
Avoid	Medial incision **too posterior**	Splitting **below biceps** · keep incision **anterior to biceps + lateral gastrocnemius**
Also at risk	–	**Popliteal vessels**, **popliteus**, **lateral inferior genicular artery** (posterolateral capsule at joint line)



Lateral inside-out exposure – ITB / biceps interval, gastrocnemius retracted

Interval traps ✗ Wrong options: medial: semimembranosus–medial gastrocnemius (too posterior) · sartorius–gracilis (splits pes) · lateral: lateral gastrocnemius–biceps (too posterior, peroneal in field) · gastrocnemius retracted anteriorly · split gastrocnemius · LCL retracted

ITB–biceps split, gastrocnemius retracted posteriorly **inside-out lateral meniscus repair**
✗ Wrong options: two-incision ACL · tibial inlay PCL (direct posterior / posteromedial) · peroneal nerve exploration (below biceps)

Saphenous injury commonest with **inside-out medial** repair (> all-inside) · **peroneal** with lateral repairs

Maximum risk, inside-out LM repair (TO) **peroneal nerve** · lateral geniculate artery also at risk (TO keyed 1 and 2)
✗ Wrong options: popliteus · lateral inferior genicular artery · saphenous

Complex bucket handle, mid-body → PH MM (TO) **inside-out for the PH needs a safety incision to protect saphenous nerve + vessels**
✗ Wrong options: outside-in for PH · all-inside "only option" for mid-body · all-inside less chondral damage

Root repair

Indication **no / minimal arthritis (KL ≤ 2)**, adequate tissue, **BMI < 35** (TO) · young patients first-line · now also older patients

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Technique (TO)	**transtibial (transosseous) pull-out** · 2 × No. 2 FiberWire → tunnel from **anteromedial tibial cortex to root footprint** → tied over a **cortical button** in extension · **anatomic anchoring** ✗ Wrong options: vertical mattress
Healing	**> 90%** (OB) · **70–90% if repaired within 3–6 months** (TO) — read the options
Outcome	**better function than partial meniscectomy** · contact pressures restored · TKA at 5 y much lower (0% in some studies)
Natural history, nonop / partial meniscectomy	**~30% TKA at 5 y** · **up to 28% TKA at mean 3.2 y** (OB) · rapid medial OA
Which arthroscopic lesion to repair?	**root tear** ✗ Wrong options: degenerative tear · complex white-white tear · parrot beak → partial meniscectomy
Surgery first-line?	**18-year-old female, medial root tear on imaging, pain with deep flexion after landing awkwardly off a jump** ✗ Wrong options: 40 y stable MCL, 4 mm gapping · isolated PCL with negative dial · Osgood–Schlatter · Meyers–McKeever type 1 tibial eminence → NWB long-leg cast
ACL rupture + medial root tear, 22 y	**ACL reconstruction + root repair** ✗ Wrong options: meniscectomy · all-inside horizontal mattress · staged inside-out · delayed transplant (only if repair fails)
43 y medial root tear, wants to avoid TKA	**arthroscopic root repair** ✗ Wrong options: PT · steroid · partial meniscectomy (no hoop stress) · transplant (preserve native first)
60 y MMPRT with extrusion	**accelerated medial compartment OA** ✗ Wrong options: ACL-associated · excision better · "30% TKA regardless of treatment"

Ramp repair (TO)

Principle (TO)	peripheral vascular → **repair, not resection**
Stable (types 1–2) (TO)	small (< 1 cm) partial tear, stable on probing → may leave (ACL haemarthrosis aids healing) · most surgeons still repair
Unstable (types 3–5) (TO)	**posteromedial portal**, **all-inside** (FasT-Fix) or inside-out / suture anchor · rasp edges · 1–3 sutures · **with ACL reconstruction (essential)**
Missed ramp (TO)	↑ **anterior tibial translation + internal rotation** → **ACL graft failure**, persistent laxity, OA
Outcome (TO)	healing **85–95%** · better AP stability, fewer re-ruptures

Partial & total meniscectomy

Partial meniscectomy	**> 80% satisfactory** · **50% Fairbank changes** (osteophytes, flattening, joint-space narrowing) · **minimise resection** (OA ∝ amount removed) · **no thermal probes** · early AROM (10 wk immobilisation harmful — dog model)
Predictors of success	**age < 40**, normal alignment, minimal arthritis, **single tear**
Meniscectomy principles (TO)	excise **only the unstable fragment** · preserve **2–4 mm peripheral rim** · **cut first, then shave** (balance edges) · **bucket handle: reduce first, then resect** · indications: complex, degenerative, white-white tears

Total meniscectomy historical · **20% significant arthritis + 70% radiographic change at 3 y** · **100% arthrosis at 20 y** (Orthobullets) vs **50–70% radiographic OA by 20 y**, contact pressure **↑ 235%** (TO) — read the options

Rehabilitation after repair (TO)

Weight bearing (TO) **allowed in extension** (axial load compresses the tear) · nonop: PWB, ROM as tolerated (Orthobullets)

Flexion (TO) limit to **90° for 4–6 wk** (deep flexion loads the posterior horn) · Orthobullets: avoid > 90°

Return (TO) no deep squat / pivot for **4–6 months** · contact sport at **6–9 months**

Repair rehab — TO (Mohindra) table **0–6 wk brace, NWB, isometrics** → 6–12 wk PWB, ROM **0–90°**, avoid hamstring activation → 3–6 months full WB, closed chain → **> 6 months** sport · vs WB allowed in extension (above) — read the options

Meniscal allograft transplantation

Indication controversial salvage · **young (< 50)** after **total / near-total meniscectomy**, esp. **lateral**, with pain in that compartment

Criteria to proceed (OB)	Contraindications
Prior total meniscectomy	**Inflammatory arthritis (RA: + RF, + anti-CCP, synovitis) — absolute**
Age ≤ 50	**Diffuse / advanced arthritis**, **flattened femoral condyle** or tibial plateau
BMI < 30	**Outerbridge IV** (absolute, OB) · grade III + IV (Orthobullets)
Compartment pain, **recurrent effusions**	Untreated **tibiofemoral subluxation**, synovial disease
≥ 2 mm joint space on 45° flexion WB AP	**Previous joint infection**, **skeletal immaturity**, **marked obesity**
Stable ligaments (ACL reconstruction before / with)	**Uncorrected malalignment** (HTO before / with)
Normal alignment, no cartilage loss	**Ligament insufficiency / instability**

NOT contraindications **age > 30**, prior ACL reconstruction with allograft, **Outerbridge I–II**, BMI 28, healed HTO, intact ACL, 4 mm joint space · focal Outerbridge III if treated · malalignment not absolute (correct it)

Grade IV — conflict **absolute contraindication** (OB853, OB4892) · **relative** if not addressed by cartilage restoration (OB1429) — read the options · Rijk: grade IV → **up to 50% graft failure within 2 y**

Graft **fresh-frozen (−80 °C)** = standard, **no effect on outcome** (cell viability irrelevant — host repopulates) · **gamma irradiation degrades collagen** · cryopreservation / freeze-drying abandoned

Sizing on radiographs · mismatch **> 5–10% → poor** (15% = negative) · **oversized → extrusion** · **undersized → poor congruity, ↑ load**

Technique **bone plugs at each horn or a bone bridge** + peripheral vertical mattress sutures

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MAT technique (TO) **bone plug** (cylindrical blocks in tibial tunnels) vs **bone bridge** (single **keyhole** bone bridge → slot) – **bridge preferred** (TO) · indication: **symptomatic young post-meniscectomy knee**, no advanced OA / malalignment, intact ligaments

MAT graft preservation (TO)	Feature
Fresh	Best viability · **1-week shelf life**
Fresh-frozen (−80 °C)	**Commonest** · good balance
Lyophilised (freeze-dried)	Long storage · **weak biomechanics**
Cryopreserved (glycerol)	**Best cell viability** of stored grafts · costly

Outcomes Cole: **77% satisfied** at ≥ 2 y · return to sport **6–9 months** · 10 y: pain + function still better · re-tears / extrusion common

OA after MAT – conflict most show radiographic progression at 10 y (Orthobullets) · **protective against OA** (OB) – read the options

Healing – conflict graft **incorporates at 8–12 weeks** (OB) · **8–12 months to heal fully** (Orthobullets) – read the options

Varus + cartilage loss after medial meniscectomy, 34 y **isolated MAT would fail (alignment + cartilage loss)**
✗ Wrong options: literature doesn't support MAT · too old · viscosupplementation · medial closing-wedge HTO (varus-producing)

Discoid meniscus – treatment (TO)

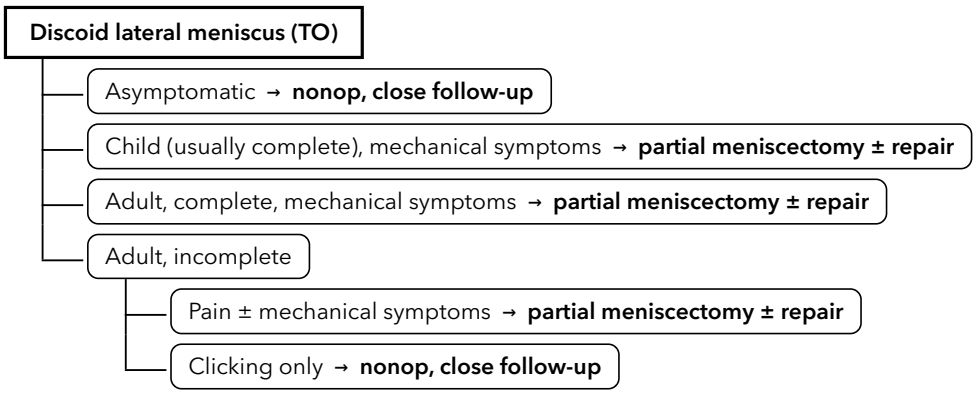
Stable complete / incomplete (TO) **defer** unless grooving, chondromalacia or other compartment damage

Torn complete / incomplete, hypermobile medial segment, intact periphery (TO) **subtotal meniscectomy / saucerization** of mobile fragment

Young child (TO) saucerized rim may remodel (vascular immature meniscus) · but **< 16 y: total meniscectomy may give the longest symptom-free period** (TO)

Wrisberg type (TO) **total meniscectomy** (subtotal leaves an unstable rim) · alternative: **saucerization + suture rim to capsule**

Total meniscectomy – conflict TO lines above vs OB / Orthobullets (Discoid Meniscus chapter): **avoid total meniscectomy** (~90% degenerative change); **saucerize to 6–8 mm rim**, **repair / stabilize** a detached Wrisberg rim – read the options



6 Complications

– – Active space – –

After meniscal surgery (Orthobullets)	Rate
Saphenous neuropathy	**7%** – commonest
Arthrofibrosis	**6%**
Sterile effusion	2%
Peroneal neuropathy	1%
Superficial infection	1%
Deep infection	1%

Allograft (MAT) complications **4–36%** · **graft tear from acellularity = most common late complication (> 50%)** · acute rejection < 1% (**late rejection never documented**) · loss of fixation ~1–2%, acute · chronic infection < 1%

All-inside devices breakage, iatrogenic chondral injury, migration, cyst

X vs Y

	Bucket handle	Medial root tear
Patient	Young athlete, high energy, ± cruciate	Middle-aged / older woman, obese, low energy
Symptom	**Locked knee** (no full extension)	Pop, posteromedial pain, little locking
MRI	**Double PCL**, absent bow tie, fragment in notch	**Ghost sign**, **extrusion**
Treatment	Reduce + **repair** (inside-out vertical mattress)	**Transtibial pull-out root repair**

	Medial meniscus tear	Lateral meniscus tear
Frequency	**Commoner overall**	Less common
ACL	**Chronic** ACL	**Acute** ACL
Degenerative	**Posterior horn**	–
Root tear link	Chondral injury, OA	ACL
McMurray	**ER**	**IR**
Repair nerve	**Saphenous**	**Common peroneal**

How it's asked

"Young cricketer, can't extend, double PCL (TO)?"	**→ arthroscopic repair**
"Double PCL sign?"	**→ bucket-handle meniscal tear**
"Displaced LM bucket handle, 3-mm rim – repair?"	**→ inside-out vertical mattress sutures**
"Locked knee, LM bucket handle on MRI?"	**→ arthroscopic lateral meniscectomy or repair**

– – Active space – –

"Wrestler – coronal, sagittal, axial MRI?"	→ bucket-handle medial meniscus tear
"Locked 20–90°, arthroscopic bucket handle – which MRI?"	→ double PCL sign
"Catching / locking, longitudinal MM tear – pre-op MRI?"	→ double PCL sign
"Double PCL on MRI – exam finding?"	→ can't actively or passively extend past 40°
"Best indication for meniscal repair (TO)?"	→ capsulomeniscal junction tear
"NOT an indication for MM repair (TO)?"	→ horizontal tear of 2 cm
"Best zone for repair (TO)?"	→ red-red
"Longitudinal tear 5 mm from MCJ (TO)?"	→ red-white; concurrent ACL reconstruction improves healing
"Why do peripheral tears heal better?"	→ increased blood supply
"Bucket handle 12 days, lacks extension?"	→ arthroscopic repair + immediate mobilisation
"Most important factor in repair healing?"	→ width of the meniscal rim
"Positive prognostic factor after repair?"	→ vertical and longitudinal tear
"Which arthroscopic lesion should be repaired?"	→ meniscal root tear
"Treated with ACL reconstruction – heals best?"	→ meniscal tear
"Surgery as first-line treatment?"	→ 18-year-old female with medial root tear (deep-flexion pain after a jump)
"Bucket-handle vs radial repair outcomes?"	→ equivalent clinical and reoperation rates
"Maximum risk, inside-out LM repair (TO)?"	→ peroneal nerve
"At risk in lateral meniscal repair (TO)?"	→ peroneal nerve (+ lateral geniculate artery)
"Complex bucket handle into PH MM (TO)?"	→ inside-out needs a safety incision (saphenous)
"ITB–biceps split, gastrocnemius retracted posteriorly?"	→ inside-out lateral meniscus repair
"Saphenous injury commonest with?"	→ inside-out medial repair
"Capsular exposure, inside-out LM?"	→ ITB–biceps interval, lateral gastrocnemius retracted posteriorly
"Inside-out vs all-inside medial – more likely?"	→ saphenous nerve injury
"Medial inside-out – interval + nerve?"	→ capsule ↔ medial gastrocnemius; saphenous nerve
"False about meniscal avulsion (TO)?"	→ both meniscotibial and meniscofemoral ligaments torn
"True about root tears (TO)?"	→ LM posterior root tear with ACL tear should always be repaired

"60 y, ghost sign, extrusion — true?"	→ accelerated medial compartment OA
"ACL rupture + medial root tear?"	→ ACL reconstruction + root repair
"57 y, pop on stairs, root tear — true?"	→ commonly associated with meniscal extrusion
"Root tear work-up — true?"	→ ghost sign on MRI
"Tear biomechanically equal to posterior root avulsion?"	→ complete radial tear
"64 y obese woman, pop rising from seat — MRI?"	→ medial meniscus posterior root tear
"Biggest contact-pressure change if unrepaired?"	→ radial tear to the meniscocapsular junction
"43 y root tear, wants to avoid TKA?"	→ arthroscopic meniscal root repair
"NOT a contraindication to medial MAT?"	→ age over thirty
"Negative effect on MAT EXCEPT?"	→ fresh-frozen graft
"Absolute contraindication to MAT?"	→ rheumatoid arthritis
"Late pain 3 y after MAT?"	→ graft tear due to acellularity
"MAT contraindication on work-up?"	→ flattening of the femoral condyles
"Why not isolated MAT (varus, cartilage loss)?"	→ alignment + cartilage loss → failure
"Loss of bow-tie sign (TO)?"	→ bucket-handle tear
"Double PCL sign (TO)?"	→ bucket-handle tear of medial meniscus
"Ghost meniscus sign (TO)?"	→ radial tear
"Empty meniscus sign (TO)?"	→ complete radial tear
"Double anterior horn sign (TO)?"	→ flipped meniscus tear
"Pain in deep flexion + ER (TO)?"	→ posterior horn medial meniscus tear
"Exam for MM tear on MRI?"	→ painful click, flexion → extension with rotation (McMurray)
"Arthroscopic video — tear type?"	→ medial parrot-beak tear
"MM tenderness, intact ACL / PCL, dial 5°?"	→ medial meniscal tear only
"Acute ACL + joint line tenderness?"	→ lateral meniscus tear
"Double meniscus sign, fibula on sagittal?"	→ lateral meniscus tear
"Discoid — all true EXCEPT (TO)?"	→ classified by Wrisberg
"True about Wrisberg-type discoid (TO)?"	→ normal in shape
"Commonest tear in discoid meniscus (TO)?"	→ horizontal
"Snapping knee syndrome — which type (TO)?"	→ Wrisberg type
"False about the meniscus (TO)?"	→ type 2 collagen gives tensile strength
"Hidden ramp lesion (TO)?"	→ type 3

– – Active space – –

"Gold standard for ramp lesion (TO)?"	→ **arthroscopy via posteromedial portal**
"Most specific clinical test for meniscal tear (TO)?"	→ **Thessaly test**
"Prerequisite for the double PCL sign (TO)?"	→ **intact ACL and PCL**
"Preferred MAT fixation technique (TO)?"	→ **bone bridge (keyhole slot)**
"NOT a cause of meniscal calcification (TO)?"	→ **rheumatoid arthritis**
"Meniscal calcification NOT a feature of (TO)?"	→ **osteopetrosis**
"True about meniscal cyst (TO)?"	→ **most common associated lesion is a meniscal tear**
"Unhappy triad (TO)?"	→ **ACL + MCL + medial meniscus**

MENISCUS

– Knee & Sports · 31 / 61 –

1 Basics

Function

Load transmission	carries **50% of load in extension, 85% in flexion** · ↑ congruency → ↑ contact area → ↓ point loading
Shock absorption	meniscus **more elastic than articular cartilage**
Hoop stress	vertical compressive load → **circumferential "hoop" tension** in the circumferential fibres → load spread, ↓ femorotibial contact stress
Stability	deepens the tibial surface · **secondary stabiliser** · **posterior horn MM = main secondary stabiliser to anterior translation** · menisci become **primary stabilisers in the ACL-deficient knee**
Lateral meniscus	less stabilising · **2× the excursion of the MM**
Load share (TO)	**LM bears ~70%** of lateral compartment load (more than MM) · LM circular, mobile; MM C-shaped, fixed to MCL · LM additionally stabilised by **meniscofemoral ligaments (Humphrey anterior, Wrisberg posterior)**
Thickness (TO)	**12–16 mm** (TO – verify: usual meniscal height ~3–5 mm at periphery; 9–12 mm is width)
Other roles (OB)	proprioception, lubrication, nutrition of the joint, cartilage protection / OA prevention

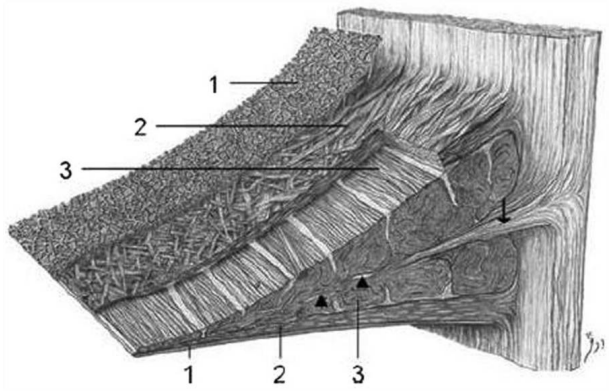
Composition & collagen

Tissue	**fibroelastic cartilage (fibrocartilage)**: collagen, proteoglycan, glycoproteins, cells
Water	**65–75%** (Orthobullets) · **72% water + 22% collagen** (OB, TO)
Collagen share	**type I = 90%** of meniscal collagen (Orthobullets, OB) · **type I = 75% of ECM dry weight** (TO) · + II, III, V, VI · collagen gives **tensile strength**
Predominant type (TO)	**type I** ✗ Wrong options: type II · "type II in red-red + type I elsewhere" · "type I in red-red + type II elsewhere"
Zonal collagen (TO)	**red-red: type I** (like ligament / tendon) · **white-white: more type II** (like hyaline cartilage) (false: "type II predominates in red-red")

Collagen	Tissues
Type I	**Meniscus (fibrocartilage)**, **ACL**, **annulus fibrosus**, **bone**, **tendon**
Type II	**Articular (hyaline) cartilage** (✗ Wrong options: same as meniscus)

Fibre layer (OB)	Orientation	Role
Superficial network + lamellar layer	**Radial**	Structural integrity; tie fibres together
Central main layer	**Circumferential (longitudinal)**, parallel to periphery	**Converts compression → hoop stress**; captured by **vertical mattress** sutures

– – Active space – –



Collagen layers of the meniscus — radial superficial, circumferential central

Horizontal cleavage tear (OB) usually excised (avascular inner zone, degeneration) · Ahn: all-inside **vertical mattress** + **notch drilling (marrow) → 91% healing**

Gross anatomy

	Medial meniscus	Lateral meniscus
Shape	**C-shaped**, semicircular, triangular cross-section	**Near circular** — horns close together, next to ACL
Length (TO)	**40–45 mm — longer / "larger"**; ~27 mm wide	**32–35 mm — shorter**
Width (Orthobullets)	9–10 mm	10–12 mm
Thickness	3–5 mm	4–5 mm
Plateau covered (TO)	**51–74% — less**	**75–93% — more**
Horns (TO)	**Posterior horn ≫ anterior**	Horns equal
Fixation	**Rigid peripheral (coronary) fixation → less mobile**	**Fewer capsular attachments → more mobile**
AP displacement (OB)	0.43 mm	0.66 mm
Artery (Orthobullets)	**Medial inferior genicular** → peripheral **20–30%**	**Lateral inferior genicular** → peripheral **10–25%**

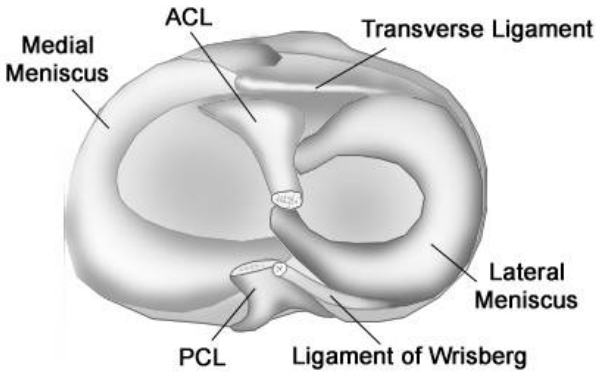
Kinematics both menisci move **anterior with extension, posterior with flexion** · **MM: less anterior translation in extension + less posterior movement in flexion**
✗ Wrong options: more translation · fewer capsular attachments

TO "not true" trap "MM larger than LM" ✓ (longer) · "LM covers more surface" ✓ · "PH MM larger than anterior horn" ✓ · **"type II predominates in red-red"** (false)

Attachments

Transverse (intermeniscal) ligament joins the **anterior horns** of MM + LM

Transverse ligament on MRI	its junction with the **anterior horn LM mimics a tear** on sagittal images → follow it over successive images ✗ Wrong options: LM / MM tear · anterior meniscofemoral ligament (a posterior structure) · ligamentum mucosum
Ligamentum mucosum (infrapatellar plica)	best on sagittal · **intercondylar notch → anterior fat pad**
Coronary (meniscotibial) ligaments	fix the meniscus **peripherally** to the tibia
Meniscofemoral ligaments	from the **posterior horn LM** into the **PCL / MFC near the PCL insertion** · **Humphrey = anterior** to PCL · **Wrisberg = posterior** · **both in 46%, at least one in 100%** (TO)



Meniscal anatomy – transverse ligament, ligament of Wrisberg

Blood supply & healing

Middle genicular artery	**posterior horns** (+ posterior capsule, ACL, PCL) → **primary supply to PH MM** ✗ Wrong options: medial inferior / superior genicular · lateral geniculates
Inferior geniculates	medial inferior → peripheral 20–30% MM · lateral inferior → peripheral 10–25% LM · anastomose anteriorly → fat pad, synovium, patellar tendon · **lateral superior + inferior → lateral retinaculum**
Perimeniscal capillary plexus (TO)	from superior + inferior medial / lateral geniculates · menisci lose vessels after birth → **only peripheral 10–25% perfused at maturity** (TO; meniscal-tears TO: 10–30%)
Central 75%	nourished by **diffusion** from synovial fluid · limited / no intrinsic healing
Vascular rim (TO, Mohindra)	outer **2–4 mm** via inferior geniculate branches · red-red vascular · red-white partial · white-white avascular
Peripheral 25% (red zone)	heals by **fibrocartilage scar** · **fibrochondrocyte** = healing cell · **tears < 4 mm** from periphery heal best
Innervation	peripheral two-thirds: **type I + II nerve endings** · **posterior horns: highest mechanoreceptor density**

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5 Management

Meniscal root tear – tibial fixation

Root tear	biomechanically **= total meniscectomy** (loss of hoop tension → ↑ peak contact pressure) → chronic pain, instability, **rapid OA** · an independent tear subtype
52 y, PH MM root tear, no arthritis on X-ray	**fixation to the tibia (root repair)** = gold standard ✗ Wrong options: NSAIDs + PT · partial meniscectomy · total meniscectomy (Fairbank changes) · UKA (only if arthritis)
Nonop (Krych)	poor outcomes, worsening OA, **> 33% arthroplasty at mean 32 months**
Partial meniscectomy vs repair (Chung)	repair better clinically + radiologically · **lower arthroplasty at 5 y**
Age (LaPrade)	transtibial double-tunnel pull-out improves **< 50 and > 50 y equally**, medial or lateral
Pre-existing / rapid arthritis	**UKA / TKA**
TKA after unrepaired root tear – cross-check	> 33% at 32 months (OB, here) · ~30% at 5 y · up to 28% at 3.2 y (OB, Meniscal Tears chapter) – read the options

X vs Y

	Red zone (peripheral 25%)	White zone (central 75%)
Blood supply	Geniculate capillary plexus	**Diffusion** only
Collagen (TO)	**Type I**	More **type II**
Healing	**Fibrocartilage scar** (fibrochondrocyte)	Little or none

How it's asked

"Predominant collagen in meniscus (TO)?"	**→ type I**
"NOT true of the menisci (TO)?"	**→ type 2 collagen predominates in red-red zone**
"Medial vs lateral meniscus – true?"	**→ MM: less anterior translation in extension, less posterior in flexion**
"Primary blood supply to PH MM?"	**→ middle genicular artery**
"Structure joining anterior horns on MRI?"	**→ transverse meniscal ligament**
"Torn meniscus – same collagen as?"	**→ ACL and annulus fibrosus (type I)**
"Root tear, no arthritis – definitive treatment?"	**→ fixation to the tibia**

NECK INJURIES IN ATHLETES

– Knee & Sports · 32 / 61 –

1 Basics

Spectrum & epidemiology

Spectrum	**ligament sprain → burner / stinger → spear tackler's spine → cervical fracture → transient quadriplegia → quadriplegia**
Who	contact sports, **football + rugby** · better protective gear → ↓ incidence · **9–15%** of ~12,000 new spinal cord injuries / yr from contact / collision sports
Catastrophic	**~6 quadriplegic events / yr** (US high school + college football; more at college level)
Highest rate of cord injury	**bilateral facet dislocation**

Predisposing conditions (↑ severity, may bar play)

Underlying conditions	**previous C-spine trauma** (fracture, ligament injury), **cervical stenosis**, **congenital odontoid hypoplasia**, **os odontoideum**, **Klippel-Feil** anomalies

2 Mechanism

Spear tackling & axial loading

Mechanism	**axial load (compression) on a flexed neck** · most injuries during **tackling**

Spear tackling (leading with crown of helmet) – **most common mechanism of neck injury in football** → gradual **cervical stenosis + loss of lordosis**

Catastrophic injury	**axial loading of the subaxial spine during spear tackling** → fracture, dislocation, ligament disruption ✗ Wrong options: arm-tackle traction root avulsion (neurotmesis, most severe "stinger") · lateral bending · flexion-distraction / whiplash cut block · face-mask rotation
Boden series (196 catastrophic injuries, 13 yr)	**subaxial 95**, C1-C2 9, combined 7, unknown 41 · **43 cord neurapraxia**, 1 root avulsion · **defensive backs + spear tackling** = greatest risk

#	Spear tackler's spine – all 4 criteria
1	**Developmental cervical canal stenosis**
2	**Persistent straightening / reversal of lordosis**
3	Post-traumatic radiographic abnormalities
4	**Documented spear-tackling technique**

Spear tackler's spine	**contraindication to contact sports**

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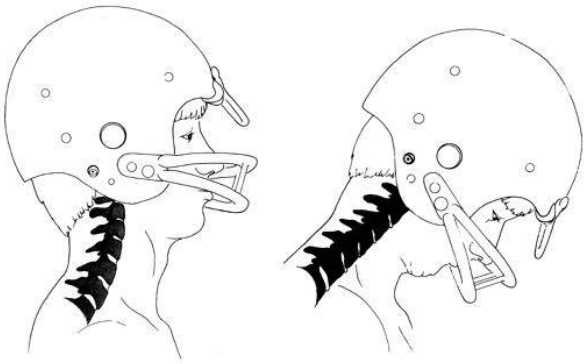


Figure 1

Spear tackling — straightened cervical spine takes the axial load

3 Presentation

On-field management of suspected cervical spine injury

Assume spinal injury	any athlete with **lost or altered consciousness**
Most important prehospital step	**maintain neutral cervical alignment** ✗ Wrong options: assess off field first · high-dose steroids (weak evidence, surgeon's discretion)

Prone player, can't move limbs

1. → **log-roll SUPINE with C-spine stabilised**
2. → **remove FACEMASK (airway)**
3. → **remove helmet + shoulder pads together (if safe)**
4. → **hard collar, controlled backboard (log-roll)**
   (equipment kept: towel rolls taped beside helmet)

(✗ Wrong options: **Traps** — **hard collar with helmet still on**) · keeping player **prone** · **helmet off before facemask**

Helmet + pads	**NATA now allows on-field removal of both** by trained personnel if neutral alignment kept (Orthobullets, Mills consensus) · older advice: **remove facemask, retain helmet** (Schroeder & Vaccaro) · cadaver C5-6: **helmet alone → kyphosis, pads alone → lordosis, both retained → neutral** — remove or retain **together**
Exam	inspect deformity · palpate spinous processes (**step-off**, pain) · motor, sensation + reflexes in **all 4 limbs**

Transient quadriplegia (cervical cord neurapraxia)

Features	neurapraxia of the cord → **bilateral upper + lower limb** pain, paresthesia, weakness → resolves **minutes–hours** (Orthobullets) · **10 min–48 h** (OB, stinger chapter)
Mechanism	hyperextension, hyperflexion or axial load · 3 of 16 NFL disc herniations presented with transient paresis

Key radiology findings

Study	Fact
Radiographs — indications	**Recurrent stingers** · neurologic symptoms / **transient quadriplegia** · C-spine trauma series
Canal diameter (SAC)	Normal **~17 mm** · **< 13 mm = stenosis** · **< 10 mm = absolute stenosis** (OB)
Torg-Pavlov ratio	**Canal ÷ vertebral body** (sagittal) · normal **1.0** · stenosis **< 0.8** (Orthobullets) · **< 0.82** (Pavlov + Torg original)
Torg ratio limits	**Technique dependent**, **low PPV / not predictive**, **inaccurate in large athletes** (big bodies skew ratio)
Torg ratio < 0.8 prevalence	**49% of asymptomatic pro football players**
MRI	**Bilateral neurologic symptoms** → stenosis, **loss of CSF around cord**, cord signal change
"Functional" stenosis	**Loss of CSF / cord deformation on MRI or CT myelogram** — bone ratios alone can't define it

Torg original data canal **14 mm** in symptomatic athletes vs **18.9 mm** controls

SAC at injury level vs deficit **10.5 mm complete** · 13.1 mm incomplete · 15.9 mm root · 16.7 mm none



Lateral C-spine — a = vertebral body, b = canal (SAC); Torg-Pavlov ratio = b ÷ a

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5 Management

Return-to-play contraindications (cord signal change, fusion, transient quadriparesis, Torg ratio)

✓ May return to play	Condition
Stinger	**Complete resolution**, normal strength + ROM
Transient quadriplegia	**Normal MRI**
Congenital stenosis (Torg < 0.8), no instability	✓ – counsel risks (✗ Wrong options: absolute contraindication)
Solid 1- or 2-level ACDF	Asymptomatic, **no T2 signal change**
Transient paralysis, **resolved T2 change, SAC 14 mm**	✓ RTP

Case-by-case	Condition
Solid **1-level ACDF, continued T2 change**	Individual decision
Transient paralysis, **never had T2 change** (e.g. SAC 9 mm)	Individual decision
Transient paralysis, **resolved T2 change, SAC ≥ 10 mm** (e.g. 11 mm)	Individual decision
Torg type 2 Klippel-Feil below C3, asymptomatic	**Relative** contraindication

No return to play	Condition
Cord signal change on MRI	**Absolute** · multiple transient quadriparesis episodes + MRI cord injury → avoid contact
Transient quadriplegia + **severe / functional stenosis**	No RTP
Spear tackler's spine	No RTP
Cord neurapraxia + **ligamentous instability**	No RTP
C0–C2 anomalies	**Odontoid hypoplasia, os odontoideum, atlanto-occipital fusion** – absolute
Torg type 1 Klippel-Feil (long fusion mass)	Absolute
Torg type 2 Klippel-Feil (1–2 segments)	With limited motion, degeneration or instability
3-level ACDF	Even if solid, asymptomatic, **no T2 change**
Continued T2 change after **2- or 3-level ACDF**	No RTP
Resolved T2 change + SAC < 10 mm after transient paralysis	No RTP
Cervical myelopathy history	No RTP

Bilateral UE numbness > 15 min, Torg 0.80 — **no RTP until advanced imaging (MRI)**
✗ Wrong options: Torg has high PPV · "low PPV → return now" · only useful in pros · congenital stenosis = absolute bar

Transient quadriplegia + residual paresthesia, Torg 0.7, C5-6 disc compressing cord, canal 9 mm, lost lordosis	**disqualify from further contact play** ✗ Wrong options: waiver · graduated RTP · PT + epidural · decompression + fusion then RTP
Disc herniation in athletes	commonest = **radiculopathy after one event** · disc abutting cord → nonop allowed but **no RTS until symptoms improve + repeat imaging shows no cord compression** · usual **6-week conservative trial**
Operative	as for any traumatic spinal injury

X vs Y

	Stinger	Transient quadriparesis
Lesion	Root / brachial plexus	**Spinal cord**
Limbs	**Unilateral** arm	**Bilateral**, arms ± legs
Imaging	Radiographs if recurrent	**MRI**
RTP	Same day if fully resolved	Only if **MRI normal**, no instability / severe stenosis

	Torg-Pavlov ratio	MRI "functional" stenosis
Measures	Bony canal ÷ body width	**CSF reserve around the cord**
Predicts injury	**Poorly** (low PPV)	Better — bars contact sport

How it's asked

"Sequence for prone player who can't move limbs?"	**→ log-roll supine, facemask off, helmet + pads off, hard collar, backboard**
"Most important prehospital consideration?"	**→ maintain neutral cervical spine alignment**
"Commonest mechanism of catastrophic neck injury in football?"	**→ axial loading of the subaxial spine with spear tackling**
"Absolute contraindication after transient arm paralysis?"	**→ cord signal change on MRI**
"Bilateral numbness > 15 min, Torg 0.80?"	**→ no RTP until advanced imaging obtained**
"Absolute contraindication to collision sports?"	**→ asymptomatic, no T2 change after solid 3-level ACDF**
"Transient quadriplegia, Torg 0.7, disc on cord, canal 9 mm?"	**→ disqualify from further play**

– – Active space – –

OSGOOD SCHLATTER'S DISEASE (TIBIAL TUBERCLE APOPHYSITIS)

– Knee & Sports · 33 / 61 –

1 Basics

Epidemiology & pathophysiology

Definition	**osteochondrosis / traction apophysitis of the tibial tubercle** → paediatric anterior knee pain
Who	**boys > girls** · **bilateral 20–30%** · **jumpers (basketball, volleyball)**, sprinters, soccer, gymnastics, track · non-athletes too
Age	boys **12–15 yr** (Orthobullets, OB) · boys **5–12 yr** (OB) · girls **8–12 yr** – read the options
Cause	repetitive pull of **extensor mechanism / patellar tendon** · **growth spurt**: bone lengthening outpaces muscle–tendon flexibility → ↑ tension on apophysis
Patellar tendon strain ratio	**↓ in OSD** (elastography, Enomoto: 18 vs 42 legs) → **less compliant tendon** → force reaches tibial tubercle **faster**

Anatomy – tibial tubercle ossification

Age	Tubercle
< 11 yr	**Cartilaginous**
11–14 yr	**Apophysis forms** (secondary ossification centre)
14–18 yr	Apophysis **fuses with tibial epiphysis**
> 18 yr	Epiphysis + apophysis **fused** to tibia

3 Presentation

Symptoms	dull **anterior knee pain**, worse with **kneeling**, jumping, squatting, stairs · often **bilateral**
Exam	**enlarged, tender tibial tubercle** · **pain on resisted knee extension** (most painful) · no effusion, no limp, stable knee

4 Diagnosis & Classification

Key radiology findings

Study	Finding
Lateral knee X-ray	**Irregularity + fragmentation of tibial tubercle** · early: apophysis **separation** · late: fragmentation + **intratendinous ossicles** · may be normal
X-ray role	Diagnosis is **clinical**; films rule out other causes
MRI	**Not essential** · soft-tissue swelling, **thickened oedematous inferior patellar tendon**, fragmented ossification centre
Mature patient	**Persistent ossicle** after tibial epiphysis fuses



Lateral radiograph — fragmentation of the tibial tubercle

Diagnosis & differential (Sinding-Larsen-Johansson, sleeve fracture)

Condition	Distinguishing fact
Osgood-Schlatter	**Tibial tubercle** tenderness, chronic, walks normally
Sinding-Larsen-Johansson	Chronic **apophysitis / minor avulsion of inferior patellar pole**; **10–14 yr**, esp. **cerebral palsy**
Patellar sleeve avulsion	**Inferior pole** tenderness, cortical elevation of inferior patella, **can't walk**
Tibial tubercle avulsion fracture	**Acute**, ambulation difficulty
Tibial spine avulsion	Acute, can't bear weight, big effusion, **Lachman +**
Bipartite patella	Normal variant (failed fusion of 2 ossification centres); **painless, incidental**
Others	Jumper's knee, **proximal tibial osteochondroma**, Lyme (isolated effusion), plica, fat-pad impingement, infection, malignancy

– – Active space – –



Sinding-Larsen-Johansson — inferior patellar pole

5 Management

Osgood-Schlatter

- First line → **NSAIDs, rest, ice, activity modification, strapping, quads stretching**
  - severe, not responding → **CAST 6 WEEKS**
- Skeletally mature + persistent symptoms (~10%) → **OSSICLE EXCISION ± TUBERCLEPLASTY**

Nonop outcome	**90% complete resolution** · **self-limiting** but **resolves only when growth stops** (physeal closure)
Persistent pain	**60.5%** still had knee pain at median **3.75 yr** (Guldhammer) → "self-limited" dogma questioned
Cast complication	**quadriceps wasting**

Persistent symptomatic ossicle — excision

Surgery indications	pain localised to tubercle, limits activity, **failed nonop**, **persists into skeletal maturity**
Procedure	**ossicle excision + tubercleplasty** (resect prominence) → **relieves symptoms, return to sport expected** ✗ Wrong options: never indicated · prednisolone injection · hamstring lengthening · RTS unlikely

Surgical outcome (Pihlajamäki)	Result
178 military recruits, 10 yr	**85% no pain** at operative site · **75% back to full activity**
107 patients (117 knees), median 10 yr	Good Kujala / VAS, no long-term harm · but **39% still pain on kneeling**
Mital (16 knees)	Only **1** did not return to sport → combine excision + tubercleplasty

X vs Y

	Osgood-Schlatter	Sinding-Larsen-Johansson	Tibial tubercle fracture
Site	**Tibial tubercle**	**Inferior patellar pole**	Tibial tubercle
Onset	Chronic, activity-related	Chronic	**Acute**
Walking	Normal	Normal	**Difficult**
X-ray	Fragmentation, ossicles	Inferior pole fragmentation	Displaced fracture

How it's asked

"13 yr, bilateral tibial tubercle pain on jumping / stairs?"	→ **Osgood-Schlatter disease** ✗ Wrong options: Lyme · plateau stress fracture · patellar tendinopathy · SLJ
"11 yr, growth spurt, pain on resisted extension — radiograph?"	→ **Figure C: tubercle irregularity / intratendinous ossification** ✗ Wrong options: A tubercle avulsion · B sleeve fracture · D bipartite patella · E tibial spine
"Decreased patellar tendon strain ratio — radiograph?"	→ **Figure B: fragmented tibial tubercle apophysis** ✗ Wrong options: normal · tubercle fracture · sleeve · loose body
"18 yr, OSD since 13, failed nonop — surgery?"	→ **ossicle excision and tubercleplasty will relieve symptoms and return to sport is expected**

– – Active space – –

OSTEITIS PUBIS

– Knee & Sports · 34 / 61 –

1 Basics

Definition	**inflammation of the pubic symphysis** from **repetitive microtrauma** · excessive strain on pubic bone
Sports	**repetitive kicking** or repetitive **hip adduction / abduction** · soccer, hockey, football, running · most common in **young football players** (OB)

Anatomy – pubic symphysis

Joint	anterior articulation of the hemipelves · cartilage-covered rami + **fibrocartilage disc** · **very stable**
Attachments	adductors (magnus, brevis, **longus**, gracilis), **rectus abdominis**, pectineus
Ligaments	superior, inferior, anterior, posterior pubic ligaments

3 Presentation

Pain	**vague, ill-defined anterior pelvic / groin pain**, dull ache · sharp stabbing pain running, kicking, changing direction · worse with hip adduction / abduction · ± adductor spasm · loss of groin flexibility
Exam	**localised tenderness directly over the symphysis** · pain on **resisted rectus abdominis testing**
Sports hernia contrast	tender in the **abdomen / inguinal region, not the pubis** (comparison table in Athletic Pubalgia chapter)
No fever	excludes osteomyelitis ✗ Wrong options: aspiration + antibiotics

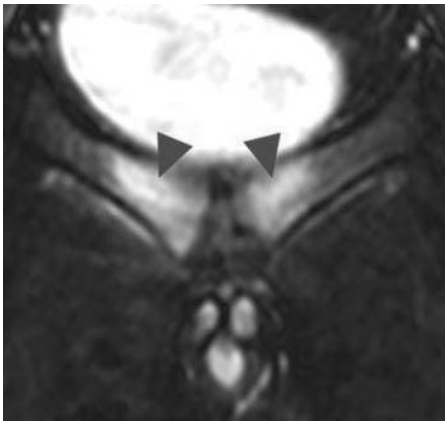
4 Diagnosis & Classification

Key radiology findings

Study	Finding
AP pelvis	**Bony erosions + irregularity**, osteolytic pubis, **symphysis widening (diastasis)**, degenerative change · may be **negative early**
MRI	**Bone marrow oedema** (early)
Bone scan	↑ uptake at symphysis



Osteitis pubis: erosions, irregularity and early widening of the symphysis



MRI: bone marrow oedema either side of the symphysis

Differential diagnosis

Image in the options	Diagnosis
Erosions + irregularity + early widening	**Osteitis pubis**
Widening, **no underlying bony abnormality**	Pubic symphysis diastasis
Femoral neck, compression side	Compression-type femoral neck stress fracture
Inferior pubic ramus	Pubic ramus stress fracture
Pelvic bony lesion	Pelvic osteochondroma

Differential athletic pubalgia, pubic ramus stress fracture, femoral neck stress fracture, inguinal hernia, oncologic disease (rare)

5 Management

Nonoperative (vast majority) **rest, NSAIDs, rehabilitation, activity modification, gradual return to play**

Soccer player, symphysis tenderness, negative X-rays **rest, NSAIDs, rehab, gradual RTP**
✗ Wrong options: aspiration + antibiotics · hernia referral · symphyseal plating (no instability) · MRI

Steroid injection into symphysis **controversial**

Course **self-limiting**, may take **several months**

– – Active space – –

How it's asked

"Which image is osteitis pubis?"	**→ Figure C — erosions, irregularity, early symphysis widening**
"Symphysis tenderness, pain on resisted rectus testing, no fever — next?"	**→ rest, NSAIDs, rehabilitation, gradual return to play**

OSTEOCHONDRITIS DISSECANS

– Knee & Sports · 35 / 61 –

1 Basics

Epidemiology & location (lateral aspect of MFC)

Definition acquired, potentially reversible idiopathic lesion of **subchondral bone** → delamination + sequestration ± articular cartilage involvement + instability

Juvenile form **open physes**, age **10–15** (median **13.1** Orthobullets · 13 OB) · **adult form** = skeletally mature

Typical patient **13-year-old male**, lesion on **lateral aspect of medial femoral condyle** · **M:F 5–6:1**

Age bands (TO) **juvenile < 12–14 yr** (open physis, heals) · **adult > 14 yr** (closed physis, poor healing, surgery) · OCD overall **13–21 yr** · sites also LFC, **lower pole patella** · **commonest cause of loose body in adolescent knee** (TO)

Sites **knee (most common)**, capitellum, talus · knee osteochondrosis on **0.002–0.003%** of knee radiographs

Knee location	Share
Posterolateral MFC (lateral aspect of MFC) – "classic"	**70%** (Orthobullets, OB) · **60–70%** (OB)
Inferior-central lateral femoral condyle	**15–20%** (~15%)
Femoral trochlea	**< 1%**
Patella, trochlea	Infrequent
Baseball catchers	**Posterior femoral condyle** (hyperflexion loading – OB, cartilage chapter)

Etiology & pathoanatomy

Causes **hereditary, traumatic, vascular** (multifactorial) · **adult form → vascular** · microtrauma · mechanical (**discoid meniscus**, **tibial spine impingement**) (OB)

Cascade softening of cartilage with **intact surface** → early cartilage separation → **partial detachment** → **osteochondral separation + loose bodies**

3 Presentation

Pain **vague, poorly localised, activity-related**

Mechanical symptoms catching, locking → **advanced disease** / unstable flap or loose body

Other **recurrent effusions**, localised tenderness, stiffness, swelling, antalgic gait

Wilson test **pain on internal rotation of tibia during extension between 90° and 30°, relieved by tibial external rotation**

– – Active space – –

Wilson (TO) seated, 90° flexion, IR, actively extend → pain at **~30°** (**tibial spine impinges lesion**)

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
Weight-bearing AP + lateral	Lesion (**incongruence of lateral aspect of MFC**), **physeal status**; may be **read as normal** → MRI if pain persists
Tunnel (notch) view	**Knee flexed 30–50°** – shows posterior condyle lesion
X-ray sign	Well-circumscribed **sclerotic subchondral bone + radiolucent halo**; **sclerosis = poor prognosis**
MRI	**Size**, subchondral bone + cartilage status, **signal around lesion**, **loose bodies → stability**
MRI unstable	**Fluid (high T2) behind / circumferential to fragment** (**rim sign**), fluid **breaching cartilage** / breaks in cartilage + oedema; cysts deep to lesion = subchondral involvement
Bone scan	**Open physes + activity behind lesion → 100% healed**; **closing physes → none** (Paletta)
Arthroscopy	**Gold standard for instability** – break in cartilage or **mobile flap on probing**



Tunnel (notch) view showing an OCD lesion

Diagnostic criteria – stable vs unstable lesion

	Stable	Unstable
MRI T2	**No ↑ T2 signal deep to fragment**	**High-signal fluid line behind / circumferential** (DeSmet → less likely to heal)
Subchondral plate	Intact	**Articular fracture through subchondral plate**
Cyst	None	**≥ 5 mm fluid-filled cyst deep to lesion**
Cartilage	**Intact cap**, no fragmentation	**Focal osteochondral defect**, fluid through cartilage
Arthroscopy	Firm on probing	**Hinged, ballotable**, mobile flap, detached

De Smet instability signs (TO, cartilage chapter) T2 fluid line **≥ 5 mm** · cysts **≥ 5 mm** · focal cartilage defect **≥ 5 mm** · fluid through cartilage into bone



MRI – fluid signal behind an OCD fragment

Clanton classification (Clanton & DeLee)

Type	Finding
I	**Depressed** osteochondral fracture
II	Fragment attached by **osseous bridge**
III	**Detached, non-displaced** fragment
IV	**Displaced** fragment

ICRS OCD **types II + III = unstable** → often surgery (OB)

OCD stage (TO, Mohindra)	Lesion	Procedure
I	Intact cartilage, stable	**Retrograde drilling** (fluoroscopy, from non-articular side)
II	Partial cartilage breach	**Fragment fixation** (Herbert screw / bioabsorbable pins)
III	Complete breach, fragment in crater	Debride crater → refix / OATS / ACI
IV	**Free loose body**	Loose-body excision + cartilage restoration (OATS / ACI / BMAC)

Fixation (TO) **Herbert screw = best compression** · alternatives bioabsorbable pins, PLLA plugs, fibrin glue

– – Active space – –

Prognosis – open physes & lesion stability

Good prognosis (nonop)	Poor prognosis
Open distal femoral physes – best predictor	**Closing / closed physes**, **adult form** (symptomatic → **DJD** if untreated)
Younger age	Older adolescent
Classic site: lateral aspect of MFC	**Lateral femoral condyle, patella**
Size < 2 cm (e.g. 1.5 cm)	> 2 cm
Intact surface, no fragmentation on MRI	**Synovial fluid behind lesion** on MRI
No sclerosis on X-ray	**Sclerosis** on plain radiographs

No proven effect — weight / BMI, gender, trauma history, ethnicity, family history, degree of sport, **concomitant tibial apophysitis**, cartilage thickness

Location conflict — classic MFC site = **good prognosis** (Orthobullets, OB) vs **typical location not predictive of healing** (OB, DeSmet / Paletta question) – read the options

Adult OCD — rarely heals spontaneously; **very low nonop healing rate**

Prognosis (TO) — good: juvenile, **< 1 cm**, medial condyle, stable · poor: adult, **> 2 cm**, patellar / lateral, unstable · **OA progression rare in OCD** (unlike SONK)

5 Management

OCD – first decision

- Open physes + stable MRI → **RESTRICTED WB / ACTIVITY MODIFICATION ± brace**
  - fails 3–6 mo / approaching physeal closure → **SUBCHONDRAL DRILLING**
- Unstable / hinged, salvageable → **ARTHROSCOPIC OR OPEN FIXATION**
- Unsalvageable, large / bone loss → **OCA or MACI (± bone graft)**
- Asymptomatic adult → **NONOP**

Arthroscopic finding (OB algorithm)

- Stable → **TRANSCHONDRAL DRILLING**
- Unstable, reducible → **FIXATION ± GRAFT**
- Unstable + fragmentation / osteolysis → **FIXATION + GRAFT**
- Unstable + chondral damage → **FIXATION + GRAFTS, CHONDROCYTE TRANSPLANT or OSTEOCHONDRAL GRAFT**

Operative criteria

Indication	Source
Impending physeal closure	Orthobullets, Cahill
Clinical signs of instability / detachment during treatment	Orthobullets, Cahill
Expanding lesion on plain films	Orthobullets
Failed nonop (persistent symptoms in compliant patient)	Orthobullets, Cahill
Persistently ↑ or worsening **bone-scan activity**	Cahill
Mechanical locking (loose body), displaced fragment	OB
Closed physes, unstable lesion	Orthobullets, OB

Cahill series — **204 JOCD**: **50% needed surgery** · fragment detachment **34%**, persistent symptoms / combined **26%** of failures

Procedure	Indication	Outcome
Subchondral drilling (K-wire / drill)	**Stable** lesion on arthroscopy; **transchondral (antegrade) or retrograde**	**Fibrocartilage**; better if immature; radiographic healing **~4.4 months**
Fixation	**Acute unstable** lesion on arthroscopy or MRI, **> 2 cm**	**85% healing** in JOCD
Chondral resurfacing	**> 2 × 2 cm**	–
Osteochondral grafting	**Arthroscopic < 3 cm** · **arthrotomy > 3 cm**	Allograft plugs, autograft OATS
Knee arthroplasty	**> 60 yr**	–

Stable juvenile lesions – activity modification & restricted weight bearing

Stable, open physes — **restricted WB + bracing / activity modification** · **50–75% heal without fragmentation** · also **asymptomatic adults**

Duration — **NWB 6–8 wk** (OB) · activity restriction **3–9 months** (OB) · cease sport 4–6 months (OB, osteonecrosis chapter)

Juvenile protocol (TO) — stage I–II → **immobilise + NWB 6 wk**, full WB by **3 months** · stage III–IV or **3 months failed** nonop → surgery · **adult: surgery if symptomatic or > 1 cm**

11 yr, intact cartilage, Wilson + — **non-weight bearing for 6–8 weeks**
✗ Wrong options: removal · ORIF · microfracture · full WB

11 yr catcher, 14 × 14 mm, no sclerosis / cysts — **stable, period of modified weightbearing**
✗ Wrong options: fixation · OCA (larger lesions)

10 yr, 1 × 1 cm LFC lesion, open physes — **activity modification**
✗ Wrong options: microfracture · antegrade drilling · fixation · excision

13 yr, stable LFC lesion, no prior treatment — **restricted weight bearing**
✗ Wrong options: **aggressive physical therapy** (worsens) · excision · drilling yet · OCA

Immature distal femur OCD — outcomes **good**, usually heal without surgery (Schenck & Goodnight)

– – Active space – –

Stable lesion failing nonoperative care — drilling

16 yr, stable 1.5 × 1 cm LFC, 6 months failed restriction	**arthroscopic drilling of the subchondral bone** · near maturity → unlikely to heal nonop; drilling **high success** ✗ Wrong options: 6 more months · aggressive PT · screw fixation · OATS
Drilling only works	**stable** lesions **without circumferential fluid** on MRI

Unstable lesions — fragment fixation or salvage

Principle	**salvage the fragment + restore native surface whenever possible** · stabilise **arthroscopically or open**
Fixation options	**cannulated screws, Herbert / headless compression screws**, bone pegs, K-wires, bioabsorbable screws / darts, metal or biocomposite · may need **hardware removal**
Unstable OCD, 15 yr	**arthroscopic or open reduction and internal fixation** ✗ Wrong options: loose-body removal · 9 months restriction · drilling
Most indicated for fixation	**16-year-old boy, closed physis, 2.5-cm hinging OCD lesion from the medial femoral condyle** ✗ Wrong options: intact cap lesions → drill if failed · 5.2 cm → OCA
Catcher, failed 8 wk NWB + 4-month RTP, MRI unchanged	**attempted salvage of the osteochondral fragment** (drilling vs screw fixation) · **immature → better salvage** ✗ Wrong options: play on NSAIDs · more PT / restriction · MACI first
20 yr, acute unstable 1.5 × 3 cm, no chronic bone change	**arthroscopic assisted internal fixation of lesion** ✗ Wrong options: PT + steroid · abrasion · MACI (needs supportive bone bed) · OATS (< 2 × 2 cm unsalvageable)
Chronic / poor tissue / no bone on fragment	**loose-body removal + debridement**, later cartilage restoration
16 yr lineman, ~2.5 × 2 cm unstable lateral-MFC lesion, circumferential fluid	**arthroscopic fixation of the lesion with headless compression screws** · microfracture **out of favour for large lesions** · same OB item calls him **skeletally mature** (explanation) and **skeletally immature** (figure legend) — fixation either way ✗ Wrong options: bracing · MACI stage 1 · OATS · retrograde drilling
Cartilage flap	lets **subchondral fluid accumulate** behind the fragment
Microfracture technique	awl to **1–1.5 cm** below surface · **NWB 4–6 wk + CPM**

Unsalvageable / failed lesions — osteochondral allograft

OCA after failed fixation + microfracture	**reliably high rate of patient satisfaction** · **survivorship > 90% at 10 yr** (✗ Wrong options: failure > 30% at 5 yr) · incorporation by **creeping substitution** (✗ Wrong options: cutting cone) · not replaced by fibrocartilage · **higher BMI → failure to return to sport** (✗ Wrong options: low BMI)

OCA over MACI — **subchondral bone loss > 1cm** · MACI needs minimal bone loss; add **bone graft if > 1 cm**
✗ Wrong options: age · bipolar (worse with any procedure) · location · > 2 cm² (fits both)

Defect size (OB)	Treatment
< 2 cm²	Chondroplasty, microfracture, **OATS**
> 2 cm²	**OCA or MACI**; **OCA** if bone loss / revision
Chronic > 2 up to 4–6 cm²	**OATS** (limited by harvest sites)
> 6 cm²	**OCA** (Treatment E) or MACI

31 yr, 2.25 × 2.75 cm untreated OCD, MPTA 87°, Mikulicz line 4 mm medial to lateral tibial spine — normal alignment → **OCA (Treatment E)**
✗ Wrong options: HTO · OATS · microfracture · screw

Figure D key — A **OATS** · B **microfracture** · C **HTO** · D **headless compression screw** · E **OCA**

X vs Y

	Juvenile OCD	Adult OCD
Physes	**Open** (10–15 yr)	**Closed**
Cause	Multifactorial	**Vascular**
Nonop healing	**Good** (50–75%)	**Very low**
First line	**Restricted WB** if stable	Surgery if symptomatic / unstable; nonop if asymptomatic
Prognosis	Better	**Worse → DJD**

	Drilling	Fixation	OCA
Lesion	**Stable**, intact cartilage	**Unstable / hinged**, salvageable	**Unsalvageable**, large, bone loss
Example	Failed nonop, near maturity	Acute, bone on fragment	Failed fixation, > 6 cm²

How it's asked

"Most likely location of JOCD?"	**→ lateral aspect of the medial femoral condyle**
"Most common presentation?"	**→ 13-year-old male with a lesion on the lateral aspect of medial femoral condyle**
"Most likely site on figure (15 yr, soccer)?"	**→ posterolateral medial femoral condyle (D)**
"Most important determinant of nonop success?"	**→ presence of open physes**

– – Active space – –

"Best predictor of nonop success?"	→ **open femoral physis** ✗ Wrong options: high signal behind lesion (unstable)
"Good prognosis EXCEPT?"	→ **sclerosis of the lesion on plain radiographs**
"Greatest impact on prognosis?"	→ **skeletal immaturity** ✗ Wrong options: tibial apophysitis · sport level · ethnicity · family history
"Warrants initial nonop management?"	→ **absence of increased T2 signal deep to lesion with open physes**
"11 yr, Wilson test +, intact lesion?"	→ **non-weight bearing for 6–8 weeks**
"Catcher, 14 × 14 mm lesion — describe + next step?"	→ **stable, period of modified weightbearing**
"10 yr, sclerotic lesion with lucent halo, open physes?"	→ **activity modification**
"13 yr, stable LFC lesion, untreated?"	→ **restricted weight bearing**
"16 yr, stable LFC lesion, 6 months failed?"	→ **arthroscopic drilling of the subchondral bone**
"15 yr, unstable lesion on MRI?"	→ **arthroscopic or open reduction and internal fixation**
"Which vignette most needs reduction + fixation?"	→ **16-year-old boy, closed physis, 2.5-cm hinging lesion of the MFC**
"Failed nonop, MRI unchanged — next?"	→ **attempted salvage of the osteochondral fragment**
"Acute unstable 1.5 × 3 cm lesion, 20 yr?"	→ **arthroscopic assisted internal fixation of lesion**
"Mature, unstable lesion on MRI?"	→ **arthroscopic fixation with headless compression screws**
"OCA for failed OCD — true?"	→ **results in a reliably high rate of patient satisfaction**
"OCA over MACI alone?"	→ **subchondral bone loss > 1 cm**
"Untreated OCD > 6 cm², normal alignment?"	→ **osteochondral allograft (Treatment E)**
"Commonest cause of loose body in an adolescent knee (TO)?"	→ **osteochondritis dissecans**
"Stage I adult OCD with intact cartilage (TO)?"	→ **retrograde drilling**

OSTEONECROSIS OF THE KNEE

– Knee & Sports · 36 / 61 –

1 Basics

Epidemiology

Who	**women : men 3 : 1** · **women < 55 yr with risk factors** · found **after knee arthroscopy in middle-aged women**
Distribution	usually **> 1 compartment** or even **metaphysis** · **80% bilateral** · **multifocal** common
Prognosis	**self-limiting** condition (Orthobullets)

Risk factors & pathophysiology

Group	Risk factors
Classic	**Alcoholism**, **steroids** (endogenous or exogenous), **sickle cell**, **hypercoagulable states**, **trauma**
Dysbaric	**Decompression sickness ("the bends")**
Marrow-replacing	**Gaucher disease**
Systemic	**SLE**, inflammatory bowel disease, transplant patient
Infective / drugs	Viruses (**CMV, hepatitis, HIV**, rubella, rubeola, varicella), **protease inhibitors** (HIV)

Mechanism	↓ blood flow to bone secondary to risk factors
Spontaneous osteonecrosis (SONK)	likely a **subchondral insufficiency fracture** · also linked to **meniscal root tear**

3 Presentation

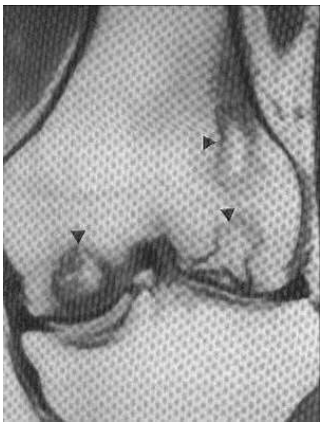
Pain	**with weight bearing**, especially **sitting → standing**

4 Diagnosis & Classification

Key radiology findings

Study	Finding
X-ray (first line)	AP, lateral, Merchant · **wedge-shaped lesion** · diagnostic only in **advanced** disease
MRI	**Most useful** · **highest sensitivity + specificity** · needed for **early / subclinical** disease
MRI signal	**T1 dark** · **T2 bright (marrow oedema)**

– – Active space – –



MRI of osteonecrosis of the distal femur

Differential diagnosis

Condition	Distinguishing fact
Osteochondritis dissecans	**Lateral aspect of MFC**, males **15–20 yr** (this note) · **10–15 yr**, median 13 (OCD chapter) – read the options
Transient osteoporosis	**Young–middle-aged men**; **multiple joints in 40%** (transient migratory osteoporosis)
Occult fracture / bone bruise	Trauma, weak bone, overuse
SONK	Subchondral insufficiency fracture; meniscal root tear

5 Management

Osteonecrosis of the knee

- All → **NONOP FIRST (NSAIDs, limited WB, quads, activity modification)**
- Small unstable fragments → **ARTHROSCOPIC REMOVAL**
- Extra-articular lesion → **CORE DECOMPRESSION**
- Large symptomatic lesion, young, failed nonop → **OSTEOCHONDRAL ALLOGRAFT**
- Large area / collapse / multiple compartments → **TKA**

Nonop outcome favourable, but **less so than nonop management of SONK**

Overall observation + **treat the underlying systemic condition** · surgery for **advanced disease with severe subchondral collapse**

Operative criteria

Procedure	Indication
Diagnostic arthroscopy	**Remove small, unstable fragments**
Core decompression	**Extra-articular lesions**
Osteochondral allograft	**Large symptomatic lesions, younger patients, failed nonop**
TKA	**Large area of involvement**, **collapse**, **multiple compartments**

Juvenile OCD — activity modification

12½ yr, stable small JOCD, untreated	**activity modification** + limited WB until pain resolves ✗ Wrong options: arthroscopic stability check · transarticular drilling with 0.045 K-wire · excision + microfracture · mosaicplasty
Sport cessation	**4–6 months** may allow healing
Surgery for JOCD	**unstable** lesions · no radiographic healing + **still symptomatic after 6 months** nonop · **approaching skeletal maturity** · loose body → fixation, drilling or regenerative procedure (depends on subchondral sclerosis)
Stable in situ lesion failing nonop	**transarticular or retroarticular drilling** → good results
Excision alone	**poor at 5 yr**; microfracture benefit unclear · symptomatic after excision + microfracture → **mosaicplasty**
JOCD vs adult	better healing potential, but **up to 50% fail to heal nonop** (Wall et al.) · 50–75% heal without fragmentation (Orthobullets, OCD chapter)

X vs Y

	Osteonecrosis (risk-factor)	SONK
Cause	**Risk factors** (steroids, alcohol, sickle cell, hypercoagulable)	**Subchondral insufficiency fracture** ± meniscal root tear
Extent	**Multiple compartments, metaphysis, 80% bilateral, multifocal**	–
Nonop result	Favourable	**Better**

How it's asked

"Stable small JOCD, 12½ yr, untreated — initial?"	→ **activity modification** ✗ Wrong options: drilling · excision · mosaicplasty
"Most sensitive / specific imaging?"	→ **MRI (T1 dark, T2 bright)**
"Extra-articular lesion — surgery?"	→ **core decompression**
"Collapse or multicompartment disease?"	→ **TKA**

– – Active space – –

PATELLAR INSTABILITY

– Knee & Sports · 37 / 61 –

1 Basics

Epidemiology

Incidence	**2–3% of all knee injuries** (Orthobullets; 2–3% of knee presentations, TO) · first-time dislocation **5.8 / 100,000 / yr** · recurrence risk 3.8 / 100,000
Who	**2nd–3rd decade** · usually 10–25 yr (OB) · **10–17 yr** high risk (TO) · young active females
Hemarthrosis	acute dislocation = **2nd most common cause of traumatic knee hemarthrosis**
Recurrence after 1st dislocation	**15–60%** (Orthobullets) · **17–49% → up to 70% after a 2nd dislocation** (TO) · **~1/3** (OB) · 15–50% at 2–5 yr (OB) – read the options

Risk factors for first & recurrent dislocation (age, prior episode, trochlear dysplasia, laxity)

Risk factor	Fact
Previous instability event	**Strongest predictor** of persistent instability (✗ Wrong options: younger age · ↑ Q-angle · male · lateral tilt)
Age < 20 at 1st dislocation	↑ recurrence · **age > 20 = protective**
Recurrence predictors (OB)	**Age, open physes, patella alta, trochlear dysplasia, ↑ TT-TG**
NOT recurrence predictors (OB)	**MPFL rupture**, **female sex**, CD 1.1 (normal), **TT-TG < 12 mm**
Contralateral knee	High + lateral patella, lateral hypermobility, dysplastic VMO → ↑ risk (congenital)
Miserable malalignment	**Femoral anteversion + genu valgum + external tibial torsion / pronated feet → ↑ Q-angle**
Osseous	Patella alta, trochlear dysplasia, lateral tilt (in extension), **LFC hypoplasia**
Muscle	**VMO weakness** · overpull of **ITB + vastus lateralis**
Systemic	Laxity – **Ehlers-Danlos**, **Marfan**, family history
Anatomic cut-offs (TO)	**CDI ≥ 1.2**, **patellar tilt ≥ 20°**, **TT-TG ≥ 20 mm**, trochlear dysplasia

Predisposing EXCEPT	**insufficiency of the vastus lateralis** · VMO insufficiency + tight lateral retinaculum = predisposing ✗ Wrong options: excess femoral IR · external tibial rotation · LFC hypoplasia · ↑ Q-angle
Beighton 6/9, hip IR 70°, genu valgum, patella alta	**patellar instability** ✗ Wrong options: ACL rupture · patellar tendon rupture · MFC osteochondral lesion (dislocation gives **lateral** condyle lesions) · lateral bucket-handle

Sex females ↑ **incidence** (↑ Q-angle) but female sex **not** a predictor of recurrence (OB)

MPFL anatomy & function (Schöttle point, restraint 0–30°)

Most important static stabiliser (TO) **MPFL** in a structurally normal knee
✗ Wrong options: VMO · Q-angle · TT:TG ratio

Stabiliser (TO)	Structures
Static	Trochlear shape (lateral edge elevated, contains patella after 30°), patellar shape / position, medial retinaculum, **MPFL**
Dynamic	**Vastus medialis (VMO)** – attaches to MPFL; stabilises in early flexion
Deeper flexion (> 30°)	**Bony** – trochlear morphology, patellar height, tracking

MPFL size 4.5–6.4 cm × 1.9 cm · **208 N** tensile strength

MPFL (TO, Mohindra) **60–70%** of restraint to lateral translation · lies in **layer II** of medial knee (Warren & Marshall; between medial retinaculum / layer I and capsule / layer III) · also patellomeniscal + patellotibial ligaments

Femoral origin **between medial epicondyle and adductor tubercle** ("saddle") · **proximal to sMCL** · proximal + posterior to medial epicondyle · **anterior + distal to adductor tubercle**
✗ Wrong options: on adductor tubercle = adductor magnus · distal to sMCL · lateral wall of MFC = PCL

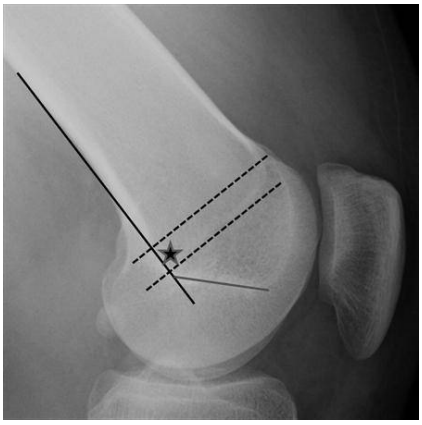
sMCL femoral attachment ~3.2 mm proximal, 4.8 mm posterior to medial epicondyle (OB)

Patellar insertion **junction of proximal + middle thirds of medial border** (Orthobullets) · **proximal half** of medial border (OB) · undersurface of VMO

Function **primary passive restraint to lateral patellar translation 0–30°** (Orthobullets, TO, OB) · **0–20°** (other OB, PF-joint note) – read the options · lax in deeper flexion

Function traps valgus at 30° = **MCL** · ER + posterior tibial translation = **POL** (AMRI) · knee flexion = **gastrocnemius** · hip adduction = **adductor magnus**

Schöttle point (lateral X-ray) **1 mm anterior to line extended from posterior cortex of shaft** · **2.5 mm distal** to **adductor tubercle** (Orthobullets) · to **posterior origin of MFC** (OB) · **proximal to posterior point of Blumensaat's line** · true lateral with **both posterior condyles superimposed** · = **isometric point** of the graft (OB)



Schöttle point – MPFL femoral attachment

– – Active space – –

Schöttle traps	ACL femoral = posterior + distal to Blumensaat (in notch) · PCL = lateral wall MFC, posterior to Blumensaat · LCL = lateral femoral condyle
Paediatric	femoral fixation **more anterior / distal** to Schöttle point (physis)

Patellofemoral mechanics – contact, sulcus angle, Q-angle (TO)

Maximum PF contact (TO)	**90°** · contact area ↑ 30° → 90° · **> 90°** quadriceps tendon contacts trochlea · **90–135° odd facet** contacts MFC · **45°** in the Orthobullets PF-joint note — read the options ✗ Wrong options: 30° · 60° · 120°
PFJRF (TO, this topic)	**level walking < stair ascent < stair descent < running < squat < deep squat < jumping** · **descent > ascent** · maximal in deep squat + jumping · the PF-joint topic's TO places **jumping just after walking** — read the options
Q-angle	**ASIS → centre of patella → tibial tubercle** · **14° M / 17° F** (TO) · 14° M / 18° F (OB) · 13° M / 18° F (Orthobullets PF joint) · **measure in slight flexion** (full extension = falsely normal, patella unengaged)
↑ Q-angle (TO)	**↑ femoral anteversion**, **external tibial torsion**, **lateral tibial tubercle**, **genu valgum** ✗ Wrong options: medialised tuberosity · internal tibial torsion · genu varum
Standing alignment (TO)	< 11° normal · **≥ 20° = risk factor needing correction**

Patellofemoral pain risk factors (TO)

Risk factors (TO)	flat / pronated feet, **gastrocnemius tightness**, **patellar instability without laxity**, excessive femoral anteversion + tibia vara
EXCEPT (TO)	**ligamentous laxity without patellar instability** (not necessarily painful)

2 Mechanism

Commonest	**noncontact twisting, knee extended, foot externally rotated** (external tibial rotation, foot fixed)
Reduction	reflex **quadriceps contraction** reduces patella → **osteochondral fracture as patella relocates**
Less common	**direct blow to medial knee** (knee-to-knee, helmet)
Impact	**medial patellar facet** strikes **lateral trochlear ridge / LFC** → bone bruises at both

3 Presentation

Symptoms	instability, anterior knee pain, painful **"pop" / "clunk"**, often **spontaneous reduction**
Acute signs	**large hemarthrosis** · **medial tenderness over MPFL** / medial facet / medial epicondyle · no swelling → laxity / habitual
Patellar translation	quadrants (midline = 0) vs other side · **normal < 2** · medial border to lateral trochlear edge = **2 quadrants = abnormal** · acute dislocation: **> 2 quadrants at 30°**

Apprehension	guarding with passive lateral translation · **moving patellar apprehension test 0–45°** most sensitive in chronic / lax knees (TO)
Medial apprehension (TO)	after **over-aggressive lateral release**, **over-tensioned MPFL**, **overcorrected tibial tubercle**
J sign	lateral in extension → "pops" into groove in early flexion (inverted J) · **associated with patella alta**
Glide / tilt (TO)	glide **> 3 quadrants** lateral = excessive · **tilt test** = cannot lift lateral border off femur (tight lateral retinaculum) · apprehension at **20–30°** · risk cut-offs: **TT-TG normal < 15 mm, pathological > 20 mm** · **Q angle > 20° (females)**
Tight lateral retinaculum (TO)	**cannot evert patella to neutral** (parallel to table) · repeat at **EUA** → decides **lateral lengthening**
Beighton (TO)	thumb to forearm, MCP hyperextension, elbow hyperextension, knee recurvatum (bilateral) + palms to floor → **/9**
Rotational exam (TO)	prone femoral anteversion (greater trochanter parallel to floor) + **thigh-foot axis**
Aspiration	tense effusion · **fat globules → fracture**

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
X-ray	Exclude fracture / loose body – **medial patellar facet (most common)**, LFC
AP (± scannogram, TO)	**Malalignment**, OA, angular deformity
Lateral	**Trochlear dysplasia** + **patellar height** · **Blumensaat line → inferior pole at 30°**
Sunrise / Merchant	**Lateral tilt**, lateral PF angle, congruence + sulcus angles
CT	**TT-TG**
MRI	Loose bodies, **bone bruise medial patella + LFC**, **MPFL tear**, TT-PCL, trochlear depth

Bone bruise – dislocation	**medial patellar facet + anterior / lateral LFC** ✗ Wrong options: patellar tendon / quadriceps rupture · ACL · PCL · LCL
Bone bruise – ACL	LFC at **sulcus terminalis (central)** + **posterior lateral tibial plateau**; patella spared
Trochlear dysplasia (TO)	present in **96%** with objective dislocation · trochlear groove normally **0.8 mm posterior** to anterior femoral cortex line

Lateral X-ray sign	Meaning
Crossing sign	Trochlear floor reaches anterior border of condyles → **flat groove** at that point; **96%** of true dislocations
Double contour	Anterior LFC border **anterior** to MFC border → convex groove / **hypoplastic medial facet**
Supratrochlear spur	Proximal (superolateral) trochlea → **more severe** dysplasia

– – Active space – –

Trochlear depth (TO) **< 3 mm** on axial MRI = dysplasia (sensitive + specific)

Patellar instability classification

Type	Features
Acute traumatic	M = F; may follow direct blow
Chronic patholaxity	Recurrent subluxation; **more in women**; malalignment
Habitual	**Painless**, every flexion; **proximal** cause (tight ITB, vastus lateralis); no effusion

Types (TO) **recurrent = commonest form** in practice · habitual = dislocates every flexion, relocates in extension · **congenital** = from birth, severe bony dysplasia · acute = first-time with haemarthrosis

Patellar height

Index	Measure	Normal	Alta	Baja
Blumensaat	Lateral at **30°**	Inferior pole at line	Above	Below
Insall-Salvati	**Patellar tendon length ÷ patellar length**	**0.8–1.2**	**> 1.2**	**< 0.8**
Blackburne-Peel	Plateau line → lower articular margin ÷ articular length	**0.5–1.0**	**> 1.0**	**< 0.5**
Caton-Deschamps	Lower articular margin → anterior tibial plateau ÷ articular length	**0.6–1.3** (~1)	**> 1.3** (TO) · **≥ 1.2** risk / **> 1.2** surgery (TO)	**< 0.6**
Plateau-patella angle	–	**20–30°**	–	–

Insall-Salvati (TO) **ratio of patellar tendon to patella** (✗ Wrong options: patella to tendon) · **> 1.2 = alta** (✗ Wrong options: baja) · tendon measured along its **posterior surface**, lower pole → tibial insertion

Best index (TO) **Caton-Deschamps "best"** · Blackburne-Peel **more accurate than IS** (lower articular margin easier to define)

Best index – conflict **Insall–Salvati = best indicator of patellar height** (TO, Mohindra notes) · Caton-Deschamps (TO, above) – read the options

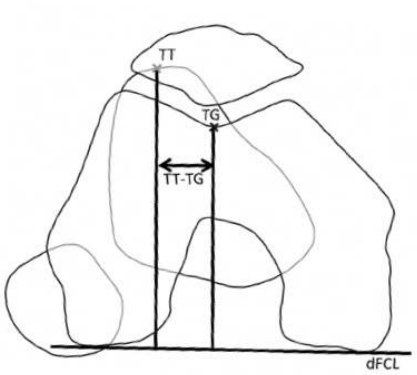


Blumensaat, Insall-Salvati, Blackburne-Peel, Caton-Deschamps

Axial & rotational measurements

Measure	Normal	Abnormal / meaning
Lateral PF angle	**> 11°** (Orthobullets) · **> 0°, opens laterally in 97%** (TO)	**< 11°** = excess tilt (OB) · subluxation: **parallel 60%**, opens medially 40%
Congruence angle	**−6°** (medial = negative)	**> +16°** (lateral) → subluxation
Sulcus angle	**138–142°** (TO) · Merchant **138° ± 6** · **130–145°** (TO answer) · **137° ± 8°** (OB)	**> 140°** (Orthobullets, OB) · **> 145°** (OB, Dejour A) · **> 150°** (TO) – read the options
Patellar tilt	**< 5°** (OB)	**> 20°** → tight lateral retinaculum, lateral release / lengthening
TT-TG (CT)	**9–13 mm** (Orthobullets) · **8–10 mm** (OB)	**> 20 mm** highly associated with instability · **< 12 mm** not a risk factor
TT-PCL (MRI, TO)	**11.9 ± 4.67 mm**	**> 20 mm** pathological · **> 24** in TO procedure table

TT-TG perpendiculars from posterior condylar line to **tibial tubercle** and **trochlear groove** · highly specific for recurrence (TO)



TT-TG distance

TT-PCL (TO) tubercle midpoint → **medial border of PCL** · tibial references only → **negates femoral rotation**

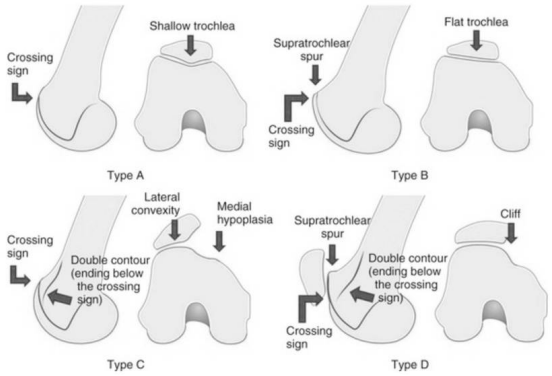
Dejour classification of trochlear dysplasia

Type	Lateral X-ray	Axial	Trochleoplasty
A	Crossing sign	Shallow, symmetric, concave · sulcus **> 145°**	**None** – not high-grade
B	Crossing + **supratrochlear spur**	**Flat or convex**	**Sulcus-deepening**
C	Crossing + **double contour**	**Convex lateral facet + hypoplastic medial**	no deepening (no spur) · lateral facet elevating (controversial)
D	Crossing + **spur + double contour**	**Cliff** (vertical asymmetric facets)	**Sulcus-deepening**

Severity **A + C mild** · **B + D severe** (spur = what is resected)

– – Active space – –

Dejour – TO lecture wording	A shallow trochlea · B flat + spur · C asymmetric facets, lateral convexity · D **cliff** + spur · **higher grade → higher recurrence** · Dejour C / D → trochleoplasty (TO) vs B / D (above) – read the options



Dejour classification

5 Management

Lateral patellar dislocation

- First-time, MRI: no fragment / loose body → **nonoperative: brace + PT**
- First-time + osteochondral fracture / loose body → **arthroscopic removal vs fixation ± MPFL repair**
  (fragment on X-ray → MRI before surgery)
- Recurrent, normal bony anatomy → **isolated MPFL reconstruction**
- Recurrent + bony risk factor → **MPFLR + TTO / distalisation / trochleoplasty**

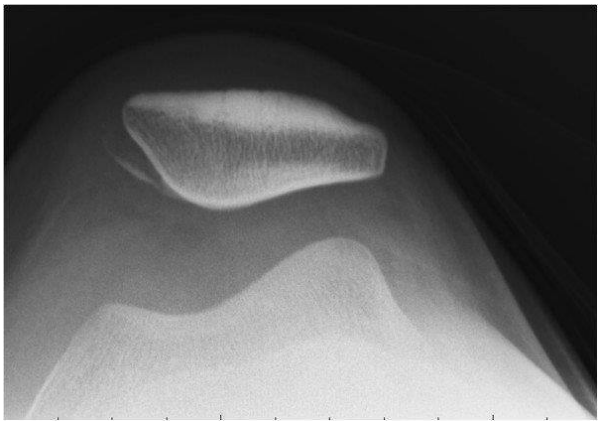
Nonoperative care

Indications	**first-time dislocator without loose body / intra-articular damage** · **habitual dislocator**
Protocol	short immobilisation → **6 wk controlled motion** (Orthobullets) · **2 wk brace locked at 0°** → ROM brace (TO)
First-time protocol (TO, Mohindra)	immobilise **in extension 3 weeks** → quads / VMO rehab · MRI for osteochondral fragment / MPFL · read the options vs 2 wk / 6 wk above
Rehab	**closed chain short arc quads**, core + hip (abductors, gluteals) · patellar sleeve / **"J" brace**
First dislocation (TO)	**operative only for osteochondral fracture** · recurrence = secondary indication ✗ Wrong options: MPFL reconstruction · always conservative · MPFL + retinaculum repair
Surgical indication after 1st dislocation (TO)	**X-ray shows osteochondral fracture** ✗ Wrong options: complete MPFL rupture on MRI · MPFL laxity · grade B dysplasia
13-yr-old female, no prior history of knee injury, MRI findings Figure A (MPFL rupture, no fracture / loose body)	**bracing + PT** ✗ Wrong options: recurrent + TT-TG 26 → distal realignment · Figure B: MPFL rupture + medial facet fracture + loose bodies → arthroscopy · recurrent despite PT → MPFLR

Acute dislocation – MRI findings, osteochondral fragments & initial management

– – Active space – –

Most consistent injury (TO)	**MPFL tear** ✗ Wrong options: patellotibial / patellomeniscal ligaments · LFC trochlea fracture · VMO disruption
MPFL tear site	**patellar insertion** commonest (Orthobullets) · **femoral attachment** common avulsion site (OB) · **soft-tissue avulsion** most common pattern; midsubstance + soft-tissue avulsion > **bony avulsion** (OB) – read the options
MRI after dislocation (TO)	contusion pair **LFC + inferomedial patella** (axials) · fragments may lie in **lateral suprapatellar or posterior recesses** · repeated episodes → MPFL / medial retinaculum **thick, separated from bone by fluid cleft** (insufficient) · **TT-TG on axials**: bicondylar line, perpendicular through deepest trochlear groove → most prominent tibial tuberosity (PT-TG uses patellar tendon midpoint)
Fragment origin	**medial patellar facet (most common)** > **LFC** / lateral trochlear ridge · **transverse patella fracture not associated** ✗ Wrong options: lateral facet · odd facet · medial / central trochlea



Medial patellar facet osteochondral fragment after dislocation

Radiograph after self-reduced dislocation, 3 quadrants	**Figure C = medial patellar facet fragment** ✗ Wrong options: A lateral facet fracture · B knee dislocation · D LFC fracture (less common) · E inferior pole / transverse fracture
Fragment on X-ray + large effusion	**next = magnetic resonance imaging (MRI)** ✗ Wrong options: NSAIDs + PT · aspiration + cortisone · ORIF · arthroscopy before MRI
Fragments missed on X-ray	**up to 40%** of first dislocations have large fragments → MRI
12-yr-old, 1st dislocation, large fragment in lateral recess + partial MPFL tear on MRI	**arthroscopic loose body removal versus osteochondral fragment fixation** ✗ Wrong options: aspirate + brace · PT · immediate isolated MPFLR · TTO + MPFLR ± trochleoplasty
Fixation	screws / pins if enough bone; else removal (arthroscopic or open)

– – Active space – –

Vertical intra-articular patella fracture after dislocation (TO) **internal fixation (ORIF)**
✗ Wrong options: arthroscopy · cylinder cast · patellectomy

Patella fracture (TO)	Rule
Commonest mechanism	**Direct impaction on flexed knee**; indirect = eccentric quads contraction
Nonop	Displacement **< 4 mm**, step-off **< 2 mm**, **active SLR intact** → brace ~10° flexion, neutral rotation (avoid baja)
Surgery	Displacement **> 4 mm** and / or step-off **> 2 mm**, extensor disruption
Other ORIF	Open fracture, articular gap **> 3 mm**, loose bodies, **osteochondral fractures**, paediatric **sleeve fracture**

MPFL repair vs reconstruction

MPFL repair **acute first-time dislocation with bony fragment** / avulsion with intact midsubstance · within first few days · no studies prove it beats nonop · not after many dislocations (attenuated tissue)

MPFL reconstruction **recurrent instability without significant malalignment** · autograft or allograft

Graft **gracilis or semitendinosus** (stronger than native), two-tailed · **open physes → medial quadriceps turndown on suture anchors** (no femoral tunnel, TO)

Graft choice **all equal rates of postoperative instability** (allograft = autograft = quadriceps = synthetic) · infection + PROs also equal

MPFLR technique (TO) **ipsilateral gracilis most preferred** · superficial quadriceps technique (TO: "Konkani") · femoral point between **adductor tubercle + medial epicondyle** · **fix at 30–45° flexion** (TO) vs 30° (OB) · graft tight only on lateral translation · **always diagnostic arthroscopy first** · **overtightening → medial facet overload**

Complications (TO) knee stiffness, **patella fracture**, continued instability · femoral graft position is critical

MPFL reconstruction – femoral tunnel malposition

Femoral tunnel	Result
Too proximal ("high and tight")	**Graft tight in flexion** → loss of flexion / failure, overconstrains medially
Too distal ("low and loose")	Graft **loose in flexion**
Too posterior	Not tighter in flexion than a proximal tunnel
Schöttle point	Correct – anteroproximal quadrant

Proximal tunnel traps ✗ Wrong options: tight in extension (minimal MPFL tension in extension) · lateral tilt · patella baja · cause = **cam-shaped distal femur**

Tension at **30° flexion** (OB)

Operative criteria

Finding	Procedure
Recurrent, no significant malalignment	**Isolated MPFLR**
TT-TG > 20 mm (CT) · TT-PCL > 24	**TTO** (AMZ / Elmslie-Trillat) ± MPFLR
Patella alta, CDI > 1.2	**Distalisation** ± MPFLR
Dejour B or D	**Trochleoplasty + MPFLR** ± TTO
Dejour A or C	MPFLR ± TTO
Femoral anteversion > 25°	**Femoral derotation osteotomy + MPFLR**
Lateral tilt > 20° / tight lateral retinaculum	Lateral release (TO) / **lengthening** (OB) + MPFLR
Genu valgum > 10°, ≥ 6 mo growth	**Guided growth** (8-plate, staples)

Surgical selection – isolated MPFL vs tibial tubercle osteotomy (TT-TG) vs trochleoplasty

Isolated MPFLR (TO)	**TT-TG ≤ 20 mm**, **normal or Dejour A trochlea**, **CDI < 1.2**, **tilt < 20°**
Isolated MPFLR (OB)	**IS < 1.2**, **TT-TG < 20 mm**, **sulcus < 140°**, **negative crossing sign** · e.g. **Insall-Salvati ratio 1.15, TT-TG distance 15 mm, sulcus angle 138°, negative crossing sign** ✗ Wrong options: sulcus 150° + crossing · TT-TG 21 · IS 1.3 · IS 1.25 + TT-TG 20 + sulcus 150°
Best isolated-MPFLR candidate	**recurrent patellar dislocation, 3 quadrants lateral translation at 30 degrees knee flexion, presence of J sign, TT-TG of 15mm** ✗ Wrong options: first-time (surgery controversial) · TT-TG 24 → consider TTO
Residual instability after isolated MPFLR	**severe trochlear dysplasia = most important predictor** · failure risks: dysplasia, ↑ TT-TG, **age < 20** · IS ratio + tibial slope **not** factors
14-yr-old recurrent, TT-TG 20, sulcus 150°	**presence of trochlear dysplasia places him at risk for recurrent instability** ✗ Wrong options: MPFLR alone poor (it does well) · IS 1.2 protective (normal) · age protective · slope 15°
Recurrent, TT-TG 12, CDI 1.2, no dysplasia, 7 dislocations	**MPFL reconstruction alone** ✗ Wrong options: MPFL repair (attenuated) · TTO medialisation / distalisation · trochleoplasty
3rd dislocation, mild valgus, CDI 1.2, TT-TG 14.5 mm, mild dysplasia, wants fast RTS	**MPFL reconstruction** ✗ Wrong options: MPFL repair · TTO alone · MPFLR + TTO (longer recovery, no proven benefit) · lateral release
TT-TG 20 exactly	isolated MPFLR still good · some add TTO only if **> 20** · TTO importance controversial
TTO threshold	usually **not** done if TT-TG < 15–20 mm (e.g. 13 mm → no TTO)

Tibial tubercle osteotomy (Fulkerson / AMZ)

Fulkerson	oblique osteotomy → tubercle **anterior + medial** (anteromedialisation, AMZ) · medialisation restores TT-TG, anteriorisation **unloads PF** · cut **long (> 5 cm) + thick (> 0.75 cm)** for **2 screws** (OB)

– – Active space – –

Indication (OB)	**TT-TG > 20 mm** (e.g. **25mm**) + recurrent instability ± cartilage restoration ✗ Wrong options: open apophysis · medial defect · sulcus 165° · tilt 25°
Indication (TO)	**grade 3 and 4 chondromalacia patella and recurrent patellar dislocation** (**lateral facet or distal pole** arthritis) + TT-TG > 20 mm ✗ Wrong options: habitual dislocation · grade 1–2 · Osgood-Schlatter
Contact pressures after AMZ	**↑ proximal & medial → proximal / medial facet arthritis = contraindication** · medial patellar defect = relative contraindication ✗ Wrong options: distal · lateral
Correction target	**TT-TG 10–15 mm, never < 10** (Orthobullets) · **~10 mm** (TO) · TO: medialisation = TT-TG − 20 mm
Other TTOs (TO)	**Elmslie-Trillat = direct medial** · **Maquet = direct anterior** · TTO when **TT-TG > 20**, ↑ TT-PCL, **CDI > 1.2**
Distal realignment (TO)	**medial shift → TT-TG > 20 mm** · **distal shift → patella alta** · **AMZ (Fulkerson)** → tracking + ↓ contact pressure · **Roux–Goldthwait** adjunct · proximal realignment = MPFLR (cornerstone), VMO advancement, lateral release
Distalisation (TO)	for **CDI > 1.2** → target **CDI 1** on C-arm · usually **~10 mm** bone removed
Open tibial apophysis	**absolute contraindication** → growth arrest → **recurvatum** · paediatric: **preserve physis**
TTO complications (TO)	**tuberosity nonunion**, painful hardware, **tibial shaft fracture** · longer recovery
Skin approach	midline incision → **infrapatellar branch of saphenous nerve** → inferolateral numbness

Trochleoplasty

Best suited (TO)	**Dejour type B and D** (spur) ✗ Wrong options: A + B · B + C · C + D
Indications (TO)	**high-grade dysplasia** + instability / maltracking, **no established OA** · revision with residual instability · rarely done in USA
Not for (TO)	**flatness alone** (spur must be present) · type A (no trochlear procedure) · type C (lateral facet elevating instead)
Contraindications (TO)	**open physes**, **PF arthrosis**
Complications (TO)	**arthrofibrosis / stiffness (most frequent)** · **trochlear shingle fracture** propagation (brittle subchondral bone in PF arthritis)
Techniques	arthroscopic or open sulcus deepening · open recession wedge
Recurrent + Dejour D (cliff + spur)	**MPFL reconstruction with trochleoplasty** ✗ Wrong options: MPFL repair ± VMO · TTO alone · MPFLR alone (residual instability)

Lateral release – lateral patellofemoral angle

Isolated lateral release	**not indicated for instability → iatrogenic medial instability**
Lateral lengthening vs release	lengthening: **better outcomes**, less quadriceps atrophy, **less medial instability** · only for **excess tilt / tightness after medialisation**

2nd dislocation, 3 quadrants, J sign, tight lateral retinaculum, TT-TG 13, CDI 1.15	**MPFL reconstruction + lateral retinacular lengthening** ✗ Wrong options: isolated release / lengthening · MPFLR + release · AMZ TTO
Main indication for isolated release	**lateral patellar (facet) compression syndrome**, no instability
Measurement for lateral release	**lateral patello-femoral angle** (**negative tilt**) on CT / sunrise ✗ Wrong options: congruence angle · Q-angle · sulcus angle · patellar height
Congruence + sulcus angles	assess **malalignment + instability**, not release

Other bony procedures

Guided growth	**genu valgum > 10°** + instability + **≥ 6 months growth** · tension band **8-plate** or **staples** (more rigid, faster)
Femoral osteotomy (TO)	severe valgus → distal femoral osteotomy · **anteversion > 25°** → supracondylar derotation + MPFLR · hemiepiphysiodesis in immature valgus

6 Complications

Complication	Fact
Recurrent dislocation	Nonop **15–60% at 2–5 yr** · highest if 1st dislocation **< 20 yr**
Medial dislocation + medial PF arthritis	**Almost exclusively iatrogenic** (prior stabilisation)
Inferolateral numbness	**Infrapatellar branch of saphenous nerve** (TTO midline incision)
MPFLR	Stiffness, patella fracture, recurrence, high-and-tight tunnel
TTO / trochleoplasty	Nonunion, tibial fracture / arthrofibrosis, shingle fracture

X vs Y

	Patellar dislocation	ACL tear
Bone bruise	**Medial patella + anterior / lateral LFC**	**LFC sulcus terminalis + posterolateral tibial plateau**
Patella involved	**Yes**	No

	High and tight	Low and loose
Tunnel	**Too proximal**	Too distal
Graft	**Tight in flexion**	Loose in flexion

– – Active space – –

	Dejour A / C	Dejour B / D
Spur	No	**Yes**
Severity	Mild	**Severe**
Trochleoplasty	No (C: lateral facet elevation)	**Sulcus deepening**

	Lateral release	Lateral lengthening
Instability	Not alone → medial instability	Adjunct to MPFLR
Outcome	Worse	**Better, less medial instability**

How it's asked

"Most important static stabiliser of patella (TO)?"	**→ MPFL**
"Most consistent injury after lateral dislocation (TO)?"	**→ MPFL tear**
"First dislocation – recommended treatment (TO)?"	**→ operative only in case of osteochondral fracture**
"First-time dislocation – surgical indication (TO)?"	**→ X-ray shows osteochondral fracture**
"Self-reduced dislocation, skyline shows vertical fracture (TO)?"	**→ internal fixation**
"Fulkerson osteotomy is done for (TO)?"	**→ grade 3–4 chondromalacia + recurrent dislocation**
"Sulcus-deepening trochleoplasty suits (TO)?"	**→ Dejour B and D**
"Maximum patella-trochlea contact (TO)?"	**→ 90°**
"Normal sulcus angle (TO)?"	**→ 130–145°**
"Q-angle increased by (TO)?"	**→ increased femoral anteversion**
"PF pain risk factors EXCEPT (TO)?"	**→ ligamentous laxity without patellar instability**
"Insall-Salvati – true statements (TO)?"	**→ > 1.2 = alta; ratio of patellar tendon to patella**
"Predisposing factor EXCEPT?"	**→ insufficiency of the vastus lateralis**
"Highest risk of persistent instability?"	**→ previous patellar instability event**
"Protective factor for recurrence?"	**→ age > 20**
"13-yr-old, 1st dislocation – recurrence risk factor?"	**→ age** ✗ Wrong options: MPFL rupture · sex
"Beighton 6/9, valgus, hip IR 70°, alta – at risk of?"	**→ patellar instability**
"Risk after MPFL reconstruction, sulcus 150°?"	**→ trochlear dysplasia → recurrent instability**
"Most likely MRI injury site?"	**→ soft-tissue avulsion of MPFL**
"Femoral MPFL landmark on lateral X-ray?"	**→ anterior to posterior cortical line, proximal to Blumensaat**

"Structure between medial epicondyle and adductor tubercle?"	→ **MPFL**
"Asterisk 1 mm anterior to posterior cortex — which reconstruction?"	→ **MPFL**
"Femoral attachment of injured structure (twisting, clunk)?"	→ **MFC, distal to adductor tubercle**
"Primary function of structure between AT and ME?"	→ **restraint to lateral patellar translation 0–30°**
"Graft tightens with flexion — position?"	→ **A (too proximal)**
"Where to drill the MPFL femoral tunnel?"	→ **A — Schöttle point**
"Femoral tunnel too proximal?"	→ **graft tight in flexion**
"MPFL graft with highest instability?"	→ **all equal**
"Loose fragment origin?"	→ **medial patellar facet**
"Axial MRI — medial patella + lateral LFC bruise?"	→ **lateral patellar dislocation**
"Medial facet + anterior LFC signal?"	→ **patellar dislocation**
"Which radiograph after self-reduced dislocation?"	→ **Figure C (medial facet fragment)**
"Nonop bracing + PT best for?"	→ **13-yr-old, first time, no fragment**
"Medial patellar avulsion fragment, effusion — next step?"	→ **MRI**
"12-yr-old, large fragment on MRI?"	→ **arthroscopic removal vs fixation**
"AMZ indicated by?"	→ **TT-TG 25 mm**
"Best isolated MPFLR candidate?"	→ **recurrent, 3 quadrants, J sign, TT-TG 15 mm**
"Imaging parameters for arthroscopy + isolated MPFLR?"	→ **IS 1.15, TT-TG 15, sulcus 138°, no crossing sign**
"Contact pressure after Fulkerson?"	→ **proximal + medial**
"Tight lateral retinaculum, TT-TG 13?"	→ **MPFLR + lateral retinacular lengthening**
"7 dislocations, TT-TG 12, CDI 1.2, no dysplasia?"	→ **MPFL reconstruction alone**
"3rd dislocation, TT-TG 14.5, wants fast return?"	→ **MPFL reconstruction**
"Cliff pattern + spur, 5 dislocations?"	→ **MPFL reconstruction + trochleoplasty**
"Measurement indicating lateral release?"	→ **lateral patello-femoral angle**
"17F, TT-TG 19 mm, shallow trochlea, recurrent dislocation (TO)?"	→ **trochleoplasty + proximal realignment (MPFLR)**
"MPFL lies in which layer of the medial knee (TO)?"	→ **layer II**

– – Active space – –

PATELLAR TENDINITIS

– Knee & Sports · 38 / 61 –

1 Basics

Risk factors & diagnosis

Incidence — up to **20% of jumping athletes** (Orthobullets) · prevalence **8.5%** in non-elite athletes (OB) – **volleyball highest (14.4%)**, soccer lowest (2.5%)

Who — **adolescents / young adults** · **M > F (2:1)** · **quadriceps** tendinopathy = older adults

Terminology (TO, IOC 2019) — preferred term **"tendinopathy"** · "tendinitis" outdated (minimal inflammatory cells) · tendinosis = degeneration without inflammation · cause = **load > adaptive capacity** → microfailure + dysrepair

Cook & Purdam stage (TO)	Features	Management (TO)
Reactive	Acute overload → **↑ proteoglycan** → thickened, swollen tendon	↓ load, taping, cryotherapy, **isometrics**
Dysrepair	Collagen disorganisation, **↑ cellularity, neovascularisation**	Eccentric loading, manual therapy, **shockwave**
Degenerative	**Irreversible**, hypocellular, apoptosis → **rupture risk**	Progressive strengthening, regenerative therapy
Reactive-on-degenerative	Reactive change over degenerate areas	Chronic overloaded athletes

Histopathology (TO) — **type I → type III collagen shift** (↓ tensile strength) · rounded tenocyte nuclei · **neovascularisation + nerve ingrowth → neurogenic pain** · clinical: start-up pain → better with warm-up → returns after activity

Risk factors — **poor quadriceps + hamstring flexibility** · **hamstring tightness** · male, **younger, taller, heavier**, type of sport (jumping) · patellar hypermobility · quadriceps atrophy · **hard surfaces** · **↑ training frequency**

Predisposing factor (OB) — **hamstring tightness**
✗ Wrong options: female · ↑ quadriceps strength · soft surfaces · low-frequency training

Risk factor EXCEPT — **older age**
✗ Wrong options: male · higher weight · quads / hamstring inflexibility · type of sport

Witvrouw vs Zwerver (OB) — **inflexibility = only significant factor**; height, weight, age, Q-angle, LLD, strength not (Witvrouw) · **younger, taller, heavier** = risk (Zwerver) – read the options

Malalignment — **not strongly associated** · malalignment → generalised PF syndrome, maltracking

Mechanism — **repetitive forceful eccentric contraction** of extensor mechanism · jumping, acceleration / deceleration, deep flexion

Histology	**degenerative, not inflammatory** · **micro-tears** at bone-tendon junction of **inferior pole** · angiofibroblastic + mucoid degeneration ✗ Wrong options: intra-substance inflammatory reaction
Theory (Rath, OB)	**impingement of deep tendon fibres on inferior pole** · flexion + quads contraction ↓ pain

Sinding-Larsen-Johansson & jumper's knee – site (TO)

Jumper's knee (TO)	tendinitis at **tendo-osseous junction of lower (inferior) pole of patella** · common in jumpers ✗ Wrong options: **superior pole**
Tooth sign (TO)	**periosteal reaction on anterior surface of patella** · seen in jumper's knee (TO)
Jumper's knee treatment (TO)	eccentric quads strengthening, avoid provoking activity, taping, **deep transverse friction massage**, cryotherapy
Sinding-Larsen-Johansson (TO)	**osteochondrosis of inferior pole of patella** · often **bilateral** · **overuse** traction by patellar tendon on partly cartilaginous lower pole · **active adolescents 10–14 yr**
SLJ clinical (TO)	pain + tenderness **at lower pole of patella** (↑ loading in flexion), subpatellar oedema · later tendon thickening, **fragmentation of lower pole**, bursitis
SLJ EXCEPT (TO)	**pain at lower end of patellar tendon** (false) – pain is at **proximal tendon / inferior pole**

3 Presentation

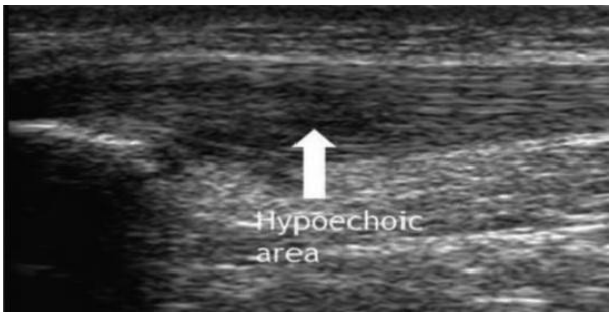
Symptoms	**insidious anterior knee pain at inferior border of patella** · early: **after** activity · late: **during** activity · **"movie theater sign"** (prolonged flexion)
Exam	swelling over tendon, tenderness inferior pole · local soft-tissue oedema but **no effusion**, no apprehension, no crepitus
Basset's sign	**tender at distal pole in full extension**, **not in full flexion → tenderness greater in extension than flexion** ✓

4 Diagnosis & Classification

Key radiology findings

Study	Finding
X-ray (AP, lateral, skyline)	**Usually normal** · chronic: **inferior traction spur (enthesophyte)**
Ultrasound	Tendon thickening, **hypoechoic areas**
MRI (chronic, surgical planning)	**Tendon thickening** (more diagnostic than oedema) · **↑ signal on T1 + T2** at proximal tendon (✗ Wrong options: decreased) · loss of posterior fat-pad border (chronic) · partial tear

– – Active space – –



Hypoechoic patellar tendon on ultrasound



Proximal patellar tendinopathy on MRI

Blazina classification

Stage	Orthobullets (3 phases)	OB illustration (4 stages)
I	Pain **after** activity only	Pain only after sports
II	Pain **during and after** activity	Pain at start, goes after warm-up, returns with fatigue
III	**Persistent pain with or without activity** (↓ performance)	**Constant pain at rest and with activity**
IV	–	**Complete rupture of patellar tendon**

Stage III pain **at rest** + with activity · **surgical indication** if refractory

5 Management

Patellar tendinopathy

- Early / Blazina I–II → **rest, activity modification, quads strengthening, eccentric PT**
- Blazina III, in season → **eccentric PT + rest from sport and jumping**
- Refractory / Blazina III / partial tear → **surgical excision + repair**

Nonoperative care – eccentric training & rest

First line ice, **rest, activity modification**, PT (**quads + hamstring stretching**, **eccentric** programme, **quadriceps strengthening**, isometric + short-arc work) · NSAIDs, moist heat, ultrasound, phonophoresis

Offloading taping or **Chopat's strap** ↓ tension across patellar tendon

Eccentric = isotonic with muscle lengthening ✓
✗ Wrong options: plyometric · isotonic shortening = concentric · isometric

As effective as surgery	**eccentric training** · also works for Achilles insertional tendinopathy ✗ Wrong options: electrotherapy · concentric · massage · taping (none proven)
Most recommended exercise (TO)	**decline squat** · **3 × 15 reps, twice daily**, single-leg eccentric squat, upright torso, on a decline board ✗ Wrong options: plyometric · treadmill protocol · concentric loading
Why eccentric (TO)	**promotes tendon remodelling** (collagen) · ↓ recurrence

Exercise (TO)	Dose / role
Eccentric (Alfredson)	**3 × 15 reps, twice daily, 12 weeks** – collagen alignment
Heavy slow resistance (HSR)	**3×/week**, slow tempo, progressive load
Isometric	**5 × 45 s holds** – **analgesic**, early (reactive) phase

Load management (TO)	keep pain **≤ 4/10** · **avoid complete rest** (matrix weakens) · **ESWT = best evidence for chronic Achilles + patellar tendinopathy** · PRP for chronic Achilles / lateral elbow · steroid weakens mid-substance tendon · prolotherapy (dextrose) mixed
Plyometric (TO)	rapid eccentric stretch → concentric of same muscle (jumping, hopping) · builds **power** · **late** rehab · early → can exacerbate (OB)
Corticosteroid injection	**contraindicated → patellar tendon rupture** · no benefit at 6 months (eccentric + heavy slow resistance keep effect)
PRP / sclerosing agents	**experimental**, inconsistent · PRP not first line
18-yr-old, 6 wk, plays through pain	**rest, activity modification, quadriceps strengthening** ✗ Wrong options: low-dose steroid · PRP · arthroscopic debridement · **no sport until pain-free** (complete rest → weakness + inflexibility)
Blazina III, in season, pain at rest	eccentric PT + **rest from sport and jumping** · playing through season → no improvement ✗ Wrong options: steroid · debridement without nonop trial · PRP · sclerosant

Refractory cases – surgical debridement

Indications	**Blazina stage III**, chronic pain / dysfunction refractory to nonop, **partial tears** · severe tendinopathy, extensor disruption
Surgery timing (TO)	after **6–9 months** of failed structured conservative care · debridement, longitudinal tenotomy, reconstruction ± graft
Technique	open or arthroscopic · **resect angiofibroblastic + mucoid degenerative tissue** · bone abrasion at insertion · suture repair / anchors
Rehab	**immobilisation in extension** → progressive ROM · WBAT
Outcome	**80–90% return** to sport · aching 4–6 months · 89% normal / near normal at 12 months (OB)
18 months pain, can't jump, MRI partial tear	**debridement and repair** ✗ Wrong options: cast · aggressive eccentric overload · ice + play on · steroid

– – Active space – –

Persistent pain with and without activity, failed PT (pole vaulter)	**surgical excision of the affected tissue** ✗ Wrong options: concentric stretching · intratendinous / intra-articular steroid · patellar chondroplasty

X vs Y

	Jumper's knee	Sinding-Larsen-Johansson
Lesion	Tendinopathy (degenerative)	**Osteochondrosis** of lower pole
Age	Adolescent / young adult athlete	**10–14 yr**
Site	Tendo-osseous junction, **inferior pole**	**Inferior pole** (proximal tendon)
Imaging	Thickening, ↑ T1 / T2 signal · tooth sign (TO)	Fragmentation of lower pole

	Patellar tendinopathy	Quadriceps tendinopathy
Age	**Adolescents / young adults**	**Older adults**

How it's asked

"Most recommended rehab exercise (TO)?"	**→ decline squat**
"Jumper's knee – all true EXCEPT (TO)?"	**→ involves superior pole of patella**
"Sinding-Larsen-Johansson – all true EXCEPT (TO)?"	**→ pain at lower end of patellar tendon**
"Predisposing factor for chronic patellar tendinitis?"	**→ hamstring tightness**
"Risk factor EXCEPT?"	**→ older age**
"Volleyball player, inferior pole pain – true?"	**→ tenderness greater in extension than flexion**
"Rehab as effective as surgery?"	**→ eccentric training**
"Best rehab protocol?"	**→ isotonic with muscle lengthening**
"Plays through pain, aching with prolonged sitting?"	**→ rest, activity modification, quadriceps strengthening**
"Blazina III in season – add to eccentric PT?"	**→ rest from sport and jumping**
"18 months pain, can't jump, partial tear on MRI?"	**→ debridement and repair**
"Persistent pain with and without activity, failed PT?"	**→ surgical excision of the affected tissue**
"Preferred term for chronic tendon disorders (IOC 2019, TO)?"	**→ tendinopathy**
"Exercise with immediate analgesic effect in reactive tendinopathy (TO)?"	**→ isometric holds (5 × 45 s)**

PATELLAR TENDON RUPTURE

– Knee & Sports · 39 / 61 –

1 Basics

Epidemiology & risk factors

Incidence	rare, **< 1 / 100,000 / yr**
Who	**3rd–4th decade** (Orthobullets) · **< 40 yr** (OB) · males in 30s–40s · **M > F**
Quadriceps vs patellar	**quadriceps tendon rupture more common (2:1)** · quadriceps **> 40 yr**, patellar **< 40 yr**
Systemic (→ bilateral)	**diabetes**, **SLE**, **RA**, **chronic renal disease** · also infectious + metabolic disorders
Local	**patellar degeneration (most common)** · previous injury · **patellar tendinopathy**
Drugs	**corticosteroid injection** · chronic steroids · **anabolic steroids**
Pathobiology	end-stage **chronic tendon degeneration** → eccentric load on a diseased tendon

Anatomy & biomechanics

Extensor mechanism	quadriceps tendon → patella → **patellar tendon** → tibial tubercle
Blood supply	**infrapatellar fat pad** + retinacular (**medial + lateral inferior geniculate arteries**)
Forces	**3× BW** ascending stairs · **17× BW** to rupture a normal tendon
After TKA	approach disrupts extensor blood supply → poorer healing · acute → primary repair · chronic → **allograft reconstruction**

2 Mechanism

Mechanism	**tensile overload**: **sudden quadriceps contraction with knee flexed** (jumping, missing a step) · eccentric load (basketball, football, soccer)
Flexion angle	**greatest tendon force at > 60° flexion** · PT : QT force ratio **> 1 at < 45°**, **< 1 at > 45°** (distal-pole contact gives quads advantage at low flexion)

Pattern	Fact
Avulsion ± bone from inferior pole of patella	**Most common** · strain at tendon-bone interface **3–4× midsubstance**
Midsubstance	Less common · reconstruction / augmentation if poor tissue
Distal avulsion from tibial tubercle	Least common

Associated	tibial tubercle avulsion, patella fracture, **TKA extensor mechanism rupture**

– – Active space – –

3 Presentation

History	**jump / sudden quads contraction**, **"pop"** · infrapatellar pain, **immediate swelling**, can't weight-bear · may mimic ACL (buckling, giving way)
Inspection	**high-riding patella**, **large hemarthrosis + ecchymosis**, **palpable gap below inferior pole**
Straight leg raise	**cannot actively SLR** or hold passively extended knee · retinaculum intact → active extension with **a few degrees of lag**

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
X-ray – AP + lateral at 30° flexion (± merchant)	**Patella alta** in complete rupture
MRI	**Most sensitive** · partial vs complete · site, degeneration, associated injury · **not needed** for complete tear
Ultrasound	Detects + localises · **operator-dependent**

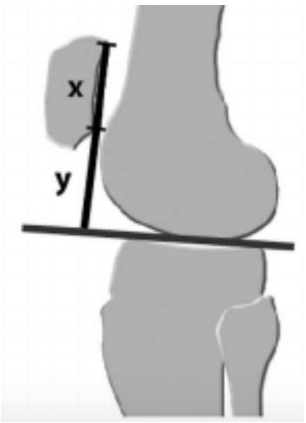


Patella alta after patellar tendon rupture

Diagnostic criteria – patella alta indices

Index	Calculation	Normal	Rupture (alta)	Baja
Insall-Salvati	**Tendon length (Y) ÷ patellar length (X)**	**0.8–1.2** (~1)	**> 1.2** (e.g. 1.3)	**< 0.8** → quadriceps rupture
Blackburne-Peel	Plateau line → lower articular margin (Y) ÷ articular length (X)	**0.5–1.0**	**> 1.0** (e.g. **Y/X 1.1**)	< 0.5
Caton-Deschamps	Anterior tibial plateau → lower articular margin (Y) ÷ articular length (X)	**0.6–1.3**	**> 1.3** (Orthobullets) · **> 1.2** (OB)	**< 0.6**

– – Active space – –



Blackburne-Peel ratio (Y / X)

Ratio traps	**X/Y** inverted ratios are meaningless · IS 0.95 and CD 1.1 = **normal**
Most reliable index	**Insall-Salvati** (highest intra- + inter-observer reliability on X-ray, CT, MRI; Verhulst) · **Blackburne-Peel** most useful – articular-surface based (Seil) – read the options
Diagnosis	complete tear = **exam + X-ray** · partial tear may need MRI

Classification

Type	Features	Treatment
Incomplete	**Extensor mechanism intact**	Nonop possible
Complete	**Patella alta + palpable defect**	**Surgical repair**

5 Management

Patellar tendon injury

- Partial, extensor mechanism intact → **brace locked in extension 6 wk, weight bearing**
- Complete, acute → **primary repair within 2 wk**
  (suture anchors: less gap formation)
- Chronic > 6–8 wk / degenerate / poor tissue → **tendon reconstruction**

Operative criteria

Indication	Procedure
Complete rupture, tendon can be approximated	**Primary repair** (end-to-end, transosseous, **suture anchors**)
Subacute **> 2 wk**	Repair + **quadplasty / scar release**
Chronic > 6–8 wk, severely disrupted / degenerate tendon	**Reconstruction**
Midsubstance tear, poor tissue	Reconstruction / **graft augmentation**
Revision or chronic	**Achilles allograft**

– – Active space – –

Treatment – primary repair (suture anchors vs transosseous)

Timing	**repair within 2 weeks** → prevents **extensor mechanism contracture** · **timing = most important prognostic factor**
End-to-end	needs a cuff of tissue on both sides · often **not feasible** (rupture at inferior pole, no proximal stump) · nonabsorbable locking sutures
Transosseous	**#5 nonabsorbable** sutures through **2 trans-patellar tunnels**, tied over top · ± **cerclage wire / tape** patella → tubercle
Suture anchors	**≥ 2 anchors** · **less gap formation** (0.9–4.1 mm vs 2.9–10.3 mm transpatellar) · **↑ pull-out strength / ultimate load to failure** (Orthobullets, OB) · load to failure **not different**, gap still less (Danaher, OB) – read the options
Anchor vs transosseous – no difference	**re-operation rate**, infection, PROMs, **return to sport**, release from care
Re-rupture	**~8% transosseous vs 1–2% anchors** (Orthobullets) · **7.5% vs 0%** (O'Dowd, OB)
Best for gap formation (footballer)	**suture anchor repair** ✗ Wrong options: end-to-end · rehab then autograft · Achilles allograft · transosseous
Anchor construct vs bone tunnels	**increased pull-out strength** ✗ Wrong options: ↓ re-operation · ↓ infection · better PROMs · quicker RTS
High load to failure + less gap	**primary repair with suture anchors** ✗ Wrong options: 2 or 3 transosseous tunnels · end-to-end · gracilis reconstruction
24-yr-old, patella alta after rebound	**primary surgical repair** ✗ Wrong options: PT then BPTB reconstruction · brace at 30° · medialising TTO + lateral release · arthroscopy
40-yr-old, alta on X-ray	**open surgical repair** ✗ Wrong options: MRI first · ice + observation · PT · arthroscopic repair
CKD + T2DM, 1 wk after fall	**primary repair of injured structure** ✗ Wrong options: + AMZ tubercle transfer (no instability) · immobilisation (partial tears only) · hamstring reconstruction first · CTA (pulses present)
Partial tear, extensor intact	**hinged brace locked in extension 6 wk**, weight bearing · progressive **active flexion / passive extension**

Tendon reconstruction

Grafts	ipsilateral **semitendinosus / gracilis**, **central quadriceps tendon-patellar bone** autograft, contralateral **BPTB** autograft or allograft, Achilles allograft
Technique	tendons through transosseous holes in patella + tibial tubercle → **circle graft** · retracted patella → quadplasty, scar release

Postoperative rehabilitation

Brace	**locked in extension, immediate weight bearing**, **6 wk** · brace-free by 6 wk
Early motion	from **7–10 days** · **passive extension + active flexion** ✗ Wrong options: any active extension

ROM exercises	**heel slides** (active closed-chain flexion) · **prone open-chain flexion** · both with passive extension
Active extension	**none for at least 4–6 wk** (OB) · permitted at **3 wk** (Matava, OB) – read the options
Later	open-chain strengthening (**leg extensions**) + weight-bearing resistance (**squats, lunges, leg press**)
Goals	**full extension + 120° flexion by 6 wk** · **full RTS at 6 months** (100% return at 6 months with cerclage repair, OB)
Chains	**closed** = distal segment fixed (squat, leg press, pull-up) · **open** = distal free (leg extension, hamstring curl) · open → more shear · open terminal knee extension = most anterior shear

6 Complications

Complication	Fact
Knee stiffness	**Most common** · risk: delayed surgery, delayed ROM · **MUA** if flexion < 120° at 6–12 wk · **lysis of adhesions** if < 120° after > 12 wk
Quadriceps atrophy	**2nd most common** · doesn't compromise strength return · PT
Re-tear	**~8% transosseous / 1–2% anchors** · acute + good tissue → repair · chronic / poor tissue → reconstruction
Infection	**~2%**
Delayed repair	Functioning extensor but **persistent quads atrophy + ↓ flexion**

X vs Y

	Patellar tendon rupture	Quadriceps tendon rupture
Frequency	Less common	**More common (2:1)**
Age	**< 40**	**> 40**
Patellar height	**Alta (IS > 1.2)**	**Baja (IS < 0.8)**
Common site	**Inferior pole of patella**	–

How it's asked

"Buckling, patella alta on X-ray + MRI?"	→ **patellar tendon rupture**
"Preoperative finding before tendon repair?"	→ **Insall-Salvati ratio of 1.3**
"Value suggesting rupture (Figures A–C)?"	→ **Y/X 1.1 in Blackburne-Peel**
"24-yr-old, alta after landing – management?"	→ **primary surgical repair**
"40-yr-old, pop, alta?"	→ **open surgical repair**
"CKD + diabetes, can't extend, 1 wk?"	→ **primary repair of injured structure**
"Best repair for gap formation?"	→ **suture anchor repair**
"Suture anchors vs bone tunnels?"	→ **increased pull-out strength**

– – Active space – –

"High load to failure, less gap?"	→ **primary repair with suture anchors**
"Rehab that limits stress on repair?"	→ **active flexion, passive extension**
"Immediate post-op active ROM exercise?"	→ **heel slides**

PATELLOFEMORAL JOINT

– Knee & Sports · 40 / 61 –

1 Basics

Anatomy & stabilisers

Articulation — **patella ↔ trochlear (intercondylar) groove · LFC diameter > MFC**

Bony constraint — **trochlear groove = primary restraint to lateral translation at > 30° flexion**

MPFL (this note) — **adductor tubercle → superomedial border of patella · primary restraint 0–20°** · instability note: between **medial epicondyle and adductor tubercle**, **0–30°** – read the options

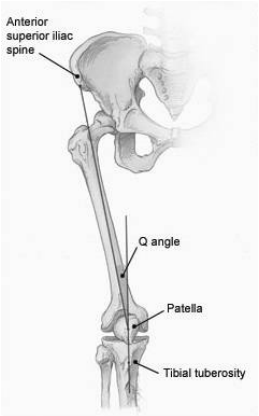
Passive restraint to lateral subluxation	Share of restraining force
MPFL (primary, at 20°)	**60%**
Medial patellomeniscal ligament	**13%**
Medial retinaculum	**10%**

Dynamic — **vastus medialis** resists **lateral** translation · vastus lateralis resists **medial** translation · quadriceps

Blood supply — superior medial + lateral geniculate · inferior medial + lateral geniculate · anterior geniculate · descending geniculate

Q-angle & patellofemoral contact pressures

Q-angle — **ASIS → centre of patella → tibial tuberosity** · **13° M / 18° F** (Orthobullets) · 14° M / 17° F (TO) · 14° M / 18° F (OB) – read the options



Q-angle

Effect of Q-angle — valgus axis → **laterally directed force → ↑ lateral facet pressure between 40–90° flexion** · 10° ↑ Q → ↑ peak pressure, lateral facet

Pressure vs flexion — ↑ from extension to **~90°** (maximum) · falls to **1/3 by 130°** · normally even medial / lateral

Maximum contact — **45°** (Orthobullets; OB: max bony contact ~45°) · **90°** (TO, instability topic) – read the options

– – Active space – –

Location deeper flexion → **higher pressure in proximal patella** (patella slides **caudally**)
✗ Wrong options: distal

Osteotomy medialisation and AMZ both **partially correct** high-Q lateral pressures, similarly

OB traps ✗ Wrong options: patella-trochlea articulation primary at 0–30° (MPFL is) · vastus medialis = primary restraint near extension · forces highest in extension

Biomechanics

Function transmits quads force to patellar tendon · **↑ extensor lever arm** · **patellectomy ↓ extension force 30%**

Activity (TO)	PFJRF (× body weight)
Level walking	**0.5–1**
Vertical jumping	**2.4–4.6**
Stair ascent / descent	**3.5–6** (descent > ascent) · 2–3 descending (Orthobullets)
Running	**4.5–7**
Squat	**Up to 7**
Deep squat	**Up to 20**

PFJRF order (TO, this topic) **walking < jumping < stair ascent < stair descent < running < squat < deep squat**
✗ Wrong options: jump after stair descent · ascent > descent · less force with more flexion

PFJRF order (TO, instability topic) walking < ascent < descent < running < squat < deep squat < **jumping** – read the options

Flexion **↑ flexion → ↑ PFJRF**

4 Diagnosis & Classification

Key radiology findings

View	Shows
AP	Alignment, fracture, arthritis
Lateral	**Patella alta / baja**, condylar dysplasia, arthritis, **transverse** patella fracture
Axial	**Malalignment**, trochlear depth, arthritis, **vertical** patella fracture
CT	Alignment, fracture, trochlear geometry, **TT-TG**
MRI	**Best for articular cartilage** · **T2** best sequence

Measurements patellar height (Insall-Salvati) · **lateral PF angle opens laterally** (normal) · **congruence angle −6°** (normal)

How it's asked

"PFJRF true statement (TO)?"	**→ walking < jumping < stair ascent < stair descent < running < deep squat**
"PF biomechanics – true?"	**→ higher Q-angle → ↑ lateral contact pressure 40–90°**

– – Active space – –

PCL INJURY

– Knee & Sports · 41 / 61 –

1 Basics

Epidemiology

Incidence	**5–20%** of all knee ligament injuries · **rare in isolation** → usually multiligament (PLC, ACL), knee dislocation
Sports (TO)	football, soccer, rugby, skiing → highest incidence of PCL tears

Anatomy

Origin (femur)	**anterolateral medial femoral condyle** · broad crescent-shaped footprint · arthroscopic view from **anterolateral portal** shows AL + PM bundle origins on MFC
Insertion (tibia)	**posterior tibial sulcus below (distal to) the articular surface** ✗ Wrong options: superior to articular margin
Size	38 mm long × 13 mm diameter · **30% larger than ACL** · strength 2500–3000 N · resists 95% of posterior tibial displacement
PCL (TO)	**38 mm** (Girgis) · **~1.5× stronger than ACL** · tibial insertion in PCL facet **~1–1.5 cm below posterior rim** · extrasynovial, own synovial sheath
Synovial status (TO)	**cruciates = intra-articular, extrasynovial** (own blood supply, not fed by synovial fluid) · collaterals = extra-articular, extrasynovial · menisci = intra-articular, intrasynovial (fed by synovial fluid)
Blood supply	**middle geniculate artery** + fat pad
Meniscofemoral ligaments	PCL lies between them · **Humphrey anterior, Wrisberg posterior** · arise posterior horn lateral meniscus → insert into PCL substance
MFL contribution (TO)	**~17%** of resistance to posterior drawer

Bundle	Tight in	Fact
Anterolateral (AL)	**Flexion**	Strongest · main posterior stabiliser at **90°** · **"PAL"** = PCL Antero-Lateral · reconstructed in single-bundle
Posteromedial (PM)	**Extension**	Reciprocal to AL
✗ Wrong option: anteromedial / posterolateral	–	**ACL** bundles (distractor)

Biomechanics

Function	**primary restraint to posterior tibial translation** · prevents hyperflexion / sliding · **isolated injury → greatest instability at 90° flexion**

Popliteus synergistic with PCL → controls ER, varus, posterior translation (see PLC)

Posterior translation (TO, Gollehon)	Amount
Isolated PCL	**8–10 mm**
Isolated PLC	**3–5 mm at 30°**
PCL + PLC	**> 15 mm** + ↑ ER

Secondary restraint (TO) PCL also resists **ER + varus**

2 Mechanism

Most common overall **direct blow to proximal (anterior) tibia, knee flexed (dashboard)** (TO, OB)

Most common in athletes **fall on flexed knee with foot plantarflexed** → posterior force on tibia
✗ Wrong options: foot dorsiflexed · blow to back of tibia · ER-valgus or IR-varus twist

Other (TO) non-contact **hyperflexion > hyperextension** (both less common) · hyperextension (stepping into hole) · rotational with varus / valgus · hyperextension often **ACL + PCL**

Clue in stem scars over **anterior proximal tibia + dorsum of ankle / forefoot** → foot plantarflexed + blow to tibia → **PCL** (can extend knee → not patella fracture, patellar tendon or tubercle avulsion; Lachman negative → not ACL)

3 Presentation

History high vs low energy · **posterior knee pain** · instability subtle / asymptomatic if isolated · ask about dislocation / nerve injury

Diagnosis mainly clinical (TO) **posterior drawer**; MRI best test for cartilage, ACL, meniscus **but not PCL** (TO) · MRI = confirmatory study (Orthobullets) – read the options

– – Active space – –

Clinical tests (posterior drawer, sag, quadriceps active, dial)

Test	Technique / positive	Meaning
Posterior drawer (90°)	Posterior force on proximal tibia · **most accurate** for PCL	Isolated: **> 10–12 mm neutral, 6–8 mm IR** · combined: **> 15 mm neutral, > 10 mm IR**
Posterior sag sign	Hips + knees 90°, ankles supported → tibia sags	Normal **medial plateau 10 mm anterior to MFC**; absent / posterior step-off = +
Quadriceps active test	Quad contraction at 90°	**Anterior reduction of tibia** = +
Reverse Lachman	Posterior translation near extension	PCL
False Lachman	Posterior subluxation mimics ↑ anterior travel	PCL (✗ Wrong options: ACL)
Varus / valgus at **0°**	Laxity	**Collateral + PCL** (cruciate)
Varus / valgus at **30° only**	Laxity	**Isolated MCL / LCL**
KT-1000 / KT-2000	Arthrometer	Less accurate than for ACL

Dial test (TO)

Dial result (ER asymmetry vs other side)	Lesion
> 10° at 30° only	**Isolated PLC**
> 10° at 30° AND 90°	**PLC + PCL**
ER 15° more at 30°, further ↑ at 90° (TO)	**PCL + PLC**
2° asymmetry / symmetric at 30° + 90°	No PLC → isolated PCL

Threshold **> 10°** (Orthobullets, OB) · **10–15°** (OB) · **15°** (TO supine description) – read the options

Prone dial (TO) **thigh–foot angle in prone position** (angle between sagittal plane of sole and body)
✗ Wrong options: foot–leg angle · tibial tuberosity angle supine · foot divergence

Supine dial (TO) leg over edge of bed, stabilise distal femur, hold heel → max ER at 30° → note **tibial tuberosity position** → repeat at 90°

Combined exam (OB) 2B reverse Lachman + sag + **varus at 0° / 30°** + negative dial + negative reverse pivot → **PCL + LCL**
✗ Wrong options: complete PLC (needs + dial / reverse pivot)



Posterior sag sign

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
AP + supine lateral X-ray	Avulsion fractures (acute) · **posterior tibiofemoral subluxation** · chronic → medial + PF arthrosis
Lateral stress view (knee 70°)	Asymmetric posterior displacement · **becoming gold standard** for diagnosing + quantifying · side-to-side **> 12 mm → PCL + PLC**
Kneeling stress X-ray	Quantifies posterior translation
Sagittal CT	Tibial PCL bony avulsion fragment
MRI	**Confirmatory** · essential to plan concomitant injuries in avulsions (TO)



Lateral stress radiographs in PCL injury

Imaging sign (TO)	Meaning
Normal PCL	**6 mm** low-signal curvilinear band
PCL tear	↑ signal, **thickened > 6 mm**, or discontinuity · partial = one bundle
Tram-track sign	**Mucoid degeneration** of PCL
Double PCL sign	**Bucket-handle meniscal tear**
Telos stress view	**Gold standard** (TO) · abnormal **> 11 mm** translation or **> 3 mm** side-to-side
Kneeling view	Most reliable (TO)
Arcuate sign	Fibular styloid avulsion → PLC

– – Active space – –

PCL grading

Grade (posterior tibial subluxation at 90°)	Injury	Translation
I	**Partial** tear	**1–5 mm**
II	**Complete isolated** tear	**6–10 mm** · anterior tibia **flush with femoral condyles**
III	Row blank in source table	–

Veltri (OB) isolated PCL **< 10 mm** posterior laxity → nonop · **> 10 mm or combined** → reconstruct

5 Management

PCL injury

- Isolated grade I / II → **protected WB + quadriceps rehab (RTS 2–4 wk)**
- Isolated grade III → immobilise in extension 4 wk → **quadriceps strengthening** *(surgery if bony avulsion or young athlete)*
- Bony tibial avulsion (displaced > 5 mm) → **ORIF (Burks–Schaffer posteromedial)**
- Combined (ACL / PLC / grade III MCL or LCL) → **PCL reconstruction + all lesions**
- Chronic, functionally unstable / varus → **reconstruction ± medial opening-wedge HTO**

Nonoperative management – quadriceps rehabilitation

Principle **quadriceps strengthening (knee extensors)** pulls tibia anteriorly → counteracts PCL deficit · ACL-deficient knee → hamstring rehab
✗ Wrong options: hamstrings · hip · gastroc-soleus

Grade II protocol (OB) **relative protection 10–14 days → ROM + gentle closed-chain quadriceps** strengthening · global strengthening at **4–6 wk** · RTS at full ROM + strength

Quadriceps-first, possibly definitive (OB) **isolated PCL rupture → closed-chain** quad strengthening (foot fixed) · no clear prognostic factors
✗ Wrong options: ACL rupture (hamstring rehab) · patellar tendon rupture · bucket-handle meniscal tear · meniscal root tear (surgery)

Prone ROM flexion done **prone** → limits posterior sag

Avoid **resisted hamstring** until healed (**4–6 wk**) · **plyometrics** early · prolonged immobilisation
✗ Wrong options: long leg cast 4 wk → stiffness · 12 wk in extension

Isolated grade II exam (OB) 2+ posterior drawer, 1+ stable Lachman, negative dial, no valgus laxity → **PT with quadriceps focus** · persistent anterior / medial pain or "loose" → reconstruct later
✗ Wrong options: delayed or acute reconstruction

Grade III isolated relative immobilisation **in extension 4 wk** + limited daily ROM → quad strengthening

Tibial avulsion – ORIF

Tibial-sided avulsion (TO) — **most common form of isolated bony PCL disruption** · mechanism: dashboard > sports hyperextension

Displaced > 5 mm (TO) — **surgical fixation (arthroscopic or open)** · MRI to plan concomitant injuries
✗ Wrong options: **long leg cast in full extension 4 wk**

Young active, displaced (OB) — **ORIF of fragment** · nonop historically → loss of motion · good results with repair of avulsions
✗ Wrong options: reconstruction · hinged brace + therapy · HTO

Exam — isolated PCL: posterior drawer **11 mm neutral / 7 mm IR**, stable varus / valgus, negative dial



Tibial PCL bony avulsion, lateral radiograph

Burks–Schaffer approach (TO) — **posteromedial approach for PCL avulsion** · **prone** · horizontal limb over popliteal crease + vertical limb over medial gastrocnemius
✗ Wrong options: ACL avulsion · TA repair

Interval (TO) — **medial gastrocnemius – semimembranosus** → blunt finger dissection to posterior capsule · **ligate middle geniculate artery** if needed · **preserve tibial nerve motor branch to medial head gastrocnemius** · retract medial gastrocnemius **laterally** (protects NV bundle) · vertical capsulotomy

Operative criteria

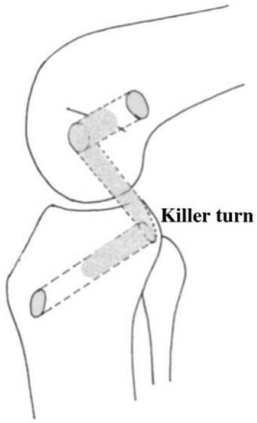
PCL repair / reconstruction indicated	Nonop
Combined: PCL + ACL or PLC	Isolated grade I, II
PCL + **grade III MCL or LCL**	Isolated grade III (unless avulsion / young athlete)
Isolated grade II / III with bony avulsion	
Isolated chronic, functionally unstable	
Chronic deficiency + varus → HTO	

Reconstruction technique – graft tensioning, tibial inlay, killer turn

Graft — usually **allograft** (Achilles, BPTB, hamstring, tibialis anterior) · no technique clearly superior · results < ACL reconstruction, residual posterior laxity common · **midsubstance primary repair fails** · success needs concomitant ligaments addressed

– – Active space – –

Single-bundle	reconstruct **anterolateral bundle** · **tension at 90° flexion** (ACL tensioned nearer extension; closest to isometric) ✗ Wrong options: 0° · 30°
Double-bundle	**AL at 90°, PM in extension** · biomechanical advantage; clinical advantage unproven · useful with PCL + PLC (rotational control; PLC grafts loosen)
Transtibial (arthroscopic)	drill anterior → posterior · **accessory posteromedial portal** 1 cm above joint line, posterior to MCL (**saphenous nerve** branches at risk) · fix graft at **90° with anterior drawer** · risk to popliteal vessels
Posteromedial corner view	**70° arthroscope** through notch (modified Gillquist) or posteromedial portal
Tibial inlay (open)	posteromedial incision between **medial gastrocnemius + semimembranosus** · graft bone block fixed directly to posterior tibia · screw **within 20 mm of popliteal artery** · ORIF of avulsions
Killer turn (TO)	sharp (~**90°**) bend of graft around **proximal posterior tibia** at tibial tunnel exit → abrasion, thinning, elongation → **main cause of residual laxity + graft failure** · Markolf: **thinning 40.6%, elongation 65.3% (9.8 / 15 mm)**



Killer turn at the posterior tibial tunnel exit

Reduce killer turn (TO)	Mechanism
Tibial inlay (Jakob, Berg)	No tibial tunnel · posterior arthrotomy · anatomic · bone-to-bone healing
Anterolateral tibial tunnel	↓ angulation, no ↑ translation vs anteromedial (theoretical: ↓ compressive force → ↑ translation)
Tunnel 15–20 mm deep from posterior tibial margin (jig)	↑ tunnel angle → smaller turn
Tunnel exit lower in footprint	One acute turn → two smoother turns
Aperture fixation (interference screw)	no suspensory (endobutton) "bungee cord" micromotion · stiffer construct
Smooth aperture	Punch / rasp
All-inside	Half tunnels (sockets) · bone stock · ↓ tunnel widening / convergence · ↓ "windshield effect"
✗ Wrong option: **tunnel exit superiorly (upper half of fossa)**	Does NOT avoid it
✗ Wrong option: **transtibial in place of inlay**	Does NOT help (transtibial causes it)

Inlay vs transtibial (OB) only advantage = **eliminates the 90° killer turn**
✗ Wrong options: less invasive · cosmesis · shorter time · faster rehab · better clinical results · normal kinematics

Rehabilitation after PCL reconstruction (OKC vs CKC)

PCL rehab (TO)	Time	Key steps
I	0–6 wk	Brace **0–90°**, prevent posterior tibial translation, **no hamstring activation**
II	6–12 wk	Gradual ROM, quads, closed chain
III	3–6 months	Proprioception, dynamic stability
IV	**> 6 months**	RTS: full strength, no lag, stable

Early immobilise in extension, **protect against gravity** · early motion **prone** · focus on quadriceps

Avoid **resisted hamstring curls (prone / OKC)** → posterior pull on tibia stresses graft

4 wk post-op, highest posterior shear (TO) **OKC resisted hamstring curls 0–90°** (esp. deep flexion **> 60°**)
✗ Wrong options: CKC leg press 0–60° · OKC quad extension 90→0° · CKC wall squat 0–45°

Allowed (OB, TO) quadriceps activation · closed-chain terminal extension · **non-WB resisted extension 60°→0°** · prone passive flexion + active terminal extension · ankle pumps · **CKC up to 60°**

PCL + PLC post-op (OB) **no active flexion, active extension 90→0°** · hinged brace + crutches, partial WB first 4 wk

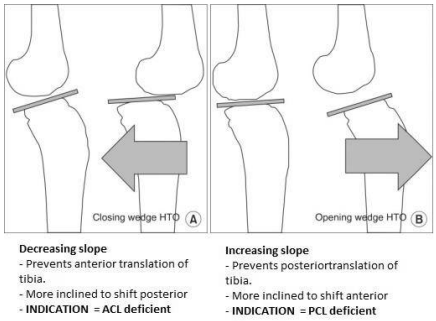
Kinetic chain (TO, Steindler 1955)	OKC	CKC
Distal segment	**Free** (leg extension, hamstring curl, SLR)	**Fixed** (squat, leg press, lunge, step-up)
Joints	Single (isolation)	Multi-joint, functional
Knee force	**Shear**	**Compressive** + quad–hamstring **co-contraction**
OKC extension	**Anterior tibial shear**, peaks **0–30°** → stresses ACL	–
ACL rehab	Restrict 0–45° first **12 wk** (or 90–45° only); full range at **12–16 wk**	**Early (2–4 wk)** – protects ACL
PCL rehab	**Extension safe** (anterior shear unloads PCL)	**Modify**: co-contraction → posterior shear, esp. **> 60°**
Patellofemoral	**90–45°** → lower PFJ force than deep CKC	**0–45°** → lower PFJ force than OKC
TKA	SLR, terminal extension (supplement)	Early sit-to-stand, partial squat
Fracture	Strength when WB restricted	Axial load → callus (**Wolff's law**)

Combined PCL / PLC / LCL injuries & slope-increasing osteotomy

Dial + at 30° and 90°, sag sign **PCL + PLC**
✗ Wrong options: ACL only · PCL only · PLC + ACL (dial + at 30° only + Lachman)

– – Active space – –

Chronic PCL + PLC, varus thrust, 10° varus	**HTO to INCREASE slope + correct varus, then PCL + PLC reconstruction** ✗ Wrong options: bracing · PCL alone · ACL + PCL (high failure if coronal alignment untreated)
HTO timing (TO)	chronic PCL / PLC varus knee → **HTO first** (correct varus, ↑ slope) → wait **6–8 months** → ligament reconstruction only if instability persists · grade II PCL **with PLC** injury → combined reconstruction
Sequence (TO)	always reconstruct **PLC with PCL** → prevents PCL graft overload · **missed PLC = most common cause of failed PCL reconstruction**
Medial opening-wedge HTO	**↑ tibial slope + tibia shifts anterior** · addresses varus + PCL deficiency ✗ Wrong options: decreased · posterior



Tibial slope: closing wedge (ACL deficient) vs opening wedge (PCL deficient)

Posterior compartment access (Gillquist portal / manoeuvre)

Gillquist portal (TO)	**central transpatellar portal** · midline **1 cm below inferior pole of patella** · used in ACL reconstruction after graft harvest ✗ Wrong options: anterolateral · anteromedial · supraclavicular
Gillquist manoeuvre (TO, 1979)	**visualise posterior compartment** · 5-mm 30° scope through patellar tendon → **between MFC and PCL** → swap to **70°** · sees posterior horn, meniscocapsular junction, **ramp lesion**, PCL, loose bodies ✗ Wrong options: PF tracking (superolateral portal)
Modified Gillquist (TO)	via **anterolateral or anteromedial** portal · McGinnis: inferior notchplasty of lateral MFC to pass scope (tight in intact cruciates / OA)

Knee portal (TO)	Site	Use
Anterolateral	1 cm above lateral joint line, 1 cm lateral to patellar tendon	**Main viewing, first portal** · misses anterior horn LM, periphery of posterior horn MM, PCL
Anteromedial	1 cm above medial joint line, 1 cm medial to tendon	**Main working** · views lateral compartment
Posteromedial	–	Posterior horn meniscal repair, **PCL**, posterior loose bodies
Superolateral	Lateral to quadriceps tendon, 2.5 cm above superolateral patella	**Patellofemoral articulation**
Central transpatellar (Gillquist)	Midline, 1 cm below inferior pole	ACL reconstruction · posterior compartment

Shoulder portal (TO)	Site	Use / at risk
Posterior (soft spot)	1–2 cm medial, 2 cm inferior to posterolateral acromion · aim coracoid	First, viewing · **axillary n.** inferior, **suprascapular a. + n.** medial
Anteroinferior (Wolf)	Lateral + inferior to coracoid	Anchors for capsulolabral repair · axillary + musculocutaneous n., axillary a., cephalic v.
Anterosuperior	Between coracoid + acromion, anterior to LHB	Outflow / retrieval in Bankart
Wilmington	1 cm lateral + 1 cm inferior to posterolateral acromion, through cuff	**Posterior SLAP**
5 o'clock	–	Inferior glenoid anchors at right angle (Bankart)
Neviaser (superior / supraclavicular)	Supraspinatus fossa: clavicle, scapular spine, medial acromion	SLAP anchors
Lateral subacromial	2–3 cm below lateral acromion, in line with posterior AC joint	Subacromial
Posterolateral subacromial	~2 cm below lateral acromion, in line with posterior edge	Scope for **cuff repair**
Anterolateral subacromial	2–3 cm below lateral acromion, in line with anterior edge	Subacromial

6 Complications

Loss of motion (TO) — **most common complication of PCL reconstruction** · **flexion loss > extension loss** (10–20° flexion loss) · graft malposition / inadequate rehab
✗ Wrong options: popliteal artery · residual instability · cartilage

Femoral tunnel (TO) — **more critical than tibial** · anterior + distal to isometric point → ↑ graft tension → **flexion loss** · extension loss / flexion contracture → prolonged immobilisation in flexion

Popliteal artery injury — lies **just posterior to tibial PCL insertion, separated only by posterior capsule** · risk ↑ with **transtibial drilling in knee extension** → drill in **flexion** · accessory posteromedial incision (finger retraction), oscillating drill, tapered bits
✗ Wrong options: saphenous nerve (that is posteromedial portal) · medial inferior geniculate artery

Residual posterior laxity — common; failure to reconstruct both bundles → rotational instability

PCL deficiency – medial & patellofemoral arthritis

Chronic PCL deficiency — ↑ contact pressure **patellofemoral + medial compartment** (varus) · late patellar + MFC chondrosis (controversial) · reconstruction may delay OA vs nonop
✗ Wrong options: isolated lateral · lateral without PF · isolated PF · medial only · normal (short term only)

– – Active space – –

Medial + PF degeneration, old dashboard injury → MRI	**PCL rupture** ✗ Wrong options: PHMM tear (medial only) · ACL · trochlear dysplasia (PF only) · MFC OCD
Young labourer, medial OA after old PCL injury	**medial opening-wedge HTO** (↑ slope helps PCL deficiency) ✗ Wrong options: UKA (needs cruciates) · TKA · isolated PCL reconstruction in arthritic malaligned knee · lateral closing wedge

X vs Y

	ACL deficient	**PCL deficient**
Rehab focus	Hamstrings · CKC	**Quadriceps** · OKC extension safe
HTO slope	**Decrease** (closing wedge)	**Increase** (opening wedge)
Graft tension angle	Nearer extension	**90° flexion**
Arthritis	–	**Medial + patellofemoral**

	Tibial inlay	**Transtibial**
Approach	Open posteromedial (med. gastroc–semimembranosus)	Arthroscopic + accessory posteromedial portal
Killer turn	**Eliminated**	**Present**
Proven clinical superiority	None	None

How it's asked

"MRI best investigation for all EXCEPT (TO)?"	**→ PCL tear (clinical diagnosis – posterior drawer)**
"PCL is (TO)?"	**→ intra-articular, extrasynovial**
"Dial test angle measured (TO)?"	**→ thigh–foot angle in prone position**
"Landed on flexed knee, foot plantarflexed (TO)?"	**→ PCL**
"Most common mechanism of PCL injury (TO)?"	**→ blow on front of upper tibia, knee flexed, foot plantarflexed**
"Isolated PCL tear in football player – mechanism?"	**→ fall on flexed knee, foot plantarflexed**
"Motocross scars over anterior tibia + dorsum of foot, negative Lachman?"	**→ PCL tear**
"True of injured structure on MRI after dashboard injury?"	**→ AL bundle tight in flexion, PM tight in extension**
"Untreated isolated PCL – long-term X-ray?"	**→ medial + patellofemoral degeneration**
"11 mm drawer neutral, 7 mm IR, ORIF planned – image?"	**→ tibial PCL bony avulsion**
"Muscle to strengthen for this footprint injury?"	**→ quadriceps**

"Grade II PCL nonop rehab?"	→ quadriceps strengthening + prone ROM as tolerated
"Initial rehab isolated grade 2 PCL?"	→ protection 10–14 days, then ROM + closed-chain quadriceps
"2+ posterior drawer, negative dial — treatment?"	→ PT with quadriceps focus
"Dial 2° asymmetry at 30° + 90° after dashboard injury?"	→ quadriceps strengthening
"Quadriceps strengthening as definitive treatment for which MRI?"	→ isolated PCL tear
"Grade 1 PCL at 18, medial OA at 35, labourer?"	→ medial opening-wedge HTO
"Compartments degenerate after PCL deficiency?"	→ medial + patellofemoral
"Medial + PF OA, old car accident — MRI shows?"	→ PCL rupture
"Killer turn avoided by all EXCEPT (TO)?"	→ tibial tunnel exiting superiorly (upper half of fossa)
"Most common complication of PCL reconstruction (TO)?"	→ loss of motion
"NOT effective for killer-turn failure (TO)?"	→ transtibial tunnel in place of inlay
"4 wk post-op, highest posterior shear (TO)?"	→ OKC resisted hamstring curls 0–90°
"Single-bundle PCL graft tension angle?"	→ 90 degrees flexion
"Avoid in early rehab after PCL reconstruction?"	→ prone hamstring curls
"Advantage of tibial inlay?"	→ eliminates the 90° killer turn
"Hyperextension injury, reconstruction planned — inlay advantage?"	→ avoidance of the "killer" 90-degree curve
"Transtibial drilling in full extension → risk?"	→ popliteal artery injury
"ER 15° more at 30°, more at 90° (TO)?"	→ PCL + PLC
"Chronic PCL + PLC, 10° varus, varus thrust?"	→ HTO increasing slope + correct varus + PCL & PLC reconstruction
"Sag sign + dial 20° at 30° and 90°?"	→ PCL + PLC
"Opening-wedge HTO for chronic PCL + PLC — slope / tibia?"	→ increased / anterior
"Reverse Lachman, sag, varus at 0° / 30°, dial negative?"	→ PCL + LCL
"Tibial PCL avulsion — NOT true (TO)?"	→ long leg cast in full extension for 4 weeks
"Burks–Schaffer approach (TO)?"	→ PCL avulsion fixation (posteromedial)
"Displaced PCL tibial avulsion, young infantryman?"	→ ORIF of avulsed fragment
"Gillquist portal (TO)?"	→ central portal through patellar tendon
"Gillquist manoeuvre purpose (TO)?"	→ visualise posterior compartment

– – Active space – –

"Most reliable clinical test for PCL (TO)?"	→ **posterior drawer test**
"Tram-track sign of the PCL (TO)?"	→ **mucoid degeneration**
"Most common cause of failed PCL reconstruction (TO)?"	→ **missed PLC injury**

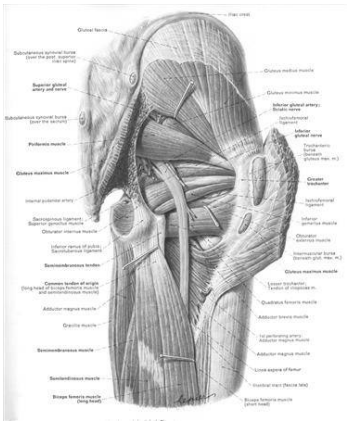
PIRIFORMIS MUSCLES SYNDROME

– Knee & Sports · 42 / 61 –

1 Basics

Sciatic nerve anatomy relative to piriformis

Normal exit	**inferior to (deep / anterior to) piriformis**, superior to superior gemellus · via greater sciatic foramen
Commonest pattern	**single nerve anterior to piriformis** · **79%** (OB, Pokorny) · **96%** (TO) – read the options
Atypical anatomy	**~21%** (Pokorny, 91 cadavers)
Commonest variant	**14%**: nerve splits above piriformis → **one branch anterior + one through** the muscle (OB) · usually **peroneal part pierces**, tibial part runs below / above (TO) · whole nerve may pass through the belly (TO)
Short external rotators (superior → inferior)	piriformis, superior gemellus, obturator internus, inferior gemellus, obturator externus, quadratus femoris



Sciatic nerve exiting deep to piriformis

Pathophysiology

Definition	sciatica-like pain from **extrapelvic sciatic nerve compression** at the hip · aka **back pocket / wallet sciatica** (TO)
Site	**anterior to piriformis** or posterior to obturator internus / gemelli at **ischial tuberosity level** (Orthobullets) · in piriformis at **greater sciatic notch** (TO)
Primary (TO)	anatomic: **split piriformis**, **split sciatic nerve**, anomalous nerve path · Orthobullets adds **bipartite piriformis**, tumours, **inferior gluteal artery aneurysm**
Secondary (TO)	trauma, **LLD**, cerebral palsy, narrowed sciatic foramen · macrotrauma to buttock (spasm) · microtrauma / overuse (long-distance walking, running, **dancing**) · **wallet in back pocket**
FAI link	↓ IR → short external rotator contracture → sciatic compression

– – Active space – –

3 Presentation

Pain	**posterior gluteal pain radiating down back of leg**, burning / aching
Pattern (TO)	worse **rising after sitting / squatting > 15–20 min** · pain on IR (cross-legged sitting) · **improves with ambulation, worse with rest**
Referred (TO)	spasm stresses **sacrotuberous ligament → pudendal** compression → groin / pelvic pain · **fibular (peroneal) branch** compression → paraesthesia sacrum → buttock → posterior thigh

Test (TO)	Positive when
FAIR (Orthobullets)	**Flexion, adduction, IR** stretches piriformis → reproduces symptoms
Pace sign	FAIR recreates **sciatic symptoms**
Piriformis sign	Supine, foot lies **externally rotated** · active return to midline → pain
Freiberg sign	Pain on **passive hip IR**
Beatty test	Side-lying on unaffected side, lifts top knee ~4 inches → sciatic pain
Lasègue	Hip + knee 90°, extend knee → posterior thigh pain



FAIR test

4 Diagnosis & Classification

Imaging	X-ray unremarkable · hip MRI usually unremarkable · **lumbar MRI** to exclude spinal cause · **EDS** document sciatic impairment

Differential – ischiofemoral impingement

Dancer, buttock pain → at rest, pain on hip extension + ER, quadratus femoris oedema (TO)	**ischiofemoral impingement** ✗ Wrong options: FAI · piriformis syndrome · quadriceps tendinopathy
IFI definition (TO)	chronic groin / buttock / hip pain + **narrowed space between lesser trochanter and ischial tuberosity**
Ischiofemoral distance on axial MRI (TO)	**normal > 2 cm · < 1.3 cm critical**

MRI (TO) **quadratus femoris ↑ T2 signal**, fibres intact · QF oedema can compress / displace **sciatic nerve** → radicular pain

Causes (TO) **iatrogenic after hip surgery**, hip trauma, **ischial exostosis**, primary in young athletes

IFI test (TO) pain on **extension + adduction + ER** (one TO explanation) · **extension + abduction + ER** (other TO explanation) · tenderness over ischiofemoral space

IFI treatment (TO) NSAIDs, guided steroid injection, PT → failure: **endoscopic resection of lesser trochanter**

Condition (TO)	Key features	Treatment
Ischiofemoral impingement	Buttock pain, extension + ER, **QF oedema**, distance < 1.3 cm	NSAIDs, injection → LT resection
Piriformis syndrome	Buttock → posterior leg, worse with sitting, **FAIR / Pace**	Stretch, NSAIDs → release + neurolysis
Snapping hip – internal	Anterior click · **iliopsoas over iliopectineal eminence**	Stretch → iliopsoas release / lengthening
Snapping hip – external	**More common** · lateral · **ITB over greater trochanter**	Stretch → ITB release / lengthening
Snapping hip – intra-articular	**Labral tear, loose body** · dynamic USG / MRI	**Arthroscopic loose body removal + labral repair**
Iliopsoas tendinopathy	Musculotendinous junction · pain on active flexion, stairs, getting out of car	Rest, stretch, NSAIDs, PT → release at LT / lengthening
Iliopsoas tendonitis after THR	Groin pain on active flexion · painful SLR, passive SLR pain-free	CT-guided steroid + stretch → **iliopsoas tenotomy**

Snapping hip (TO) coxa saltans / **dancer's hip** · ballet dancers, extreme flexion + abduction · **extra-articular more common** than intra-articular · resistant extra-articular → local steroid into sheath

Iliopsoas bursa (TO) **largest bursa in the body** · 3 causes of bursitis: **RA, acute trauma, overuse**

Snapping hip sign / extension test (TO) palpate inguinal crease, flex → abduct + ER → back to neutral → palpable click

Iliopsoas after THR – diagnosis (TO) **cross-table lateral** shows **cup prominent beyond anterior rim** (snapping over anterior lip) · relief after **USG / CT-guided local anaesthetic** into iliopsoas sheath confirms

5 Management

Piriformis syndrome
- First line → **rest, NSAIDs, muscle relaxants, ice, PT stretching, steroid injection**
- Refractory → **piriformis release + external sciatic neurolysis**

Nonoperative **stretch piriformis + short external rotators** · strengthen abductors + adductors (TO) · **corticosteroid injection near piriformis** · early conservative Rx most effective (TO)

Surgery only after failed conservative care · release piriformis tension + **explore sciatic notch** for fibrous bands / constrictions

– – Active space – –

How it's asked

"Most common sciatic nerve pattern at exit?"	→ **single nerve anterior to piriformis**
"Dancer, extension + ER pain, quadratus femoris oedema (TO)?"	→ **ischiofemoral impingement**
"MRI finding separating IFI from piriformis syndrome (TO)?"	→ **quadratus femoris oedema, ischiofemoral distance < 1.3 cm** ✗ Wrong options: piriformis syndrome (MRI usually normal)

PLICAE

– Knee & Sports · 43 / 61 –

1 Basics

Embryology & pathology

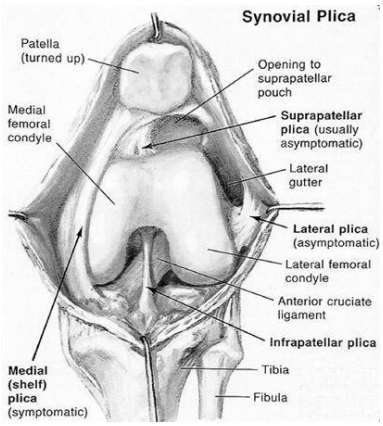
Origin (TO)	embryonic knee divided by **synovial folds** → absorbed by **3–4 months** → single cavity · **partial resorption** → persistent septations = **synovial plicae** (visible at arthroscopy)
Mediopatellar plica (TO)	not a septum remnant (Ogata & Uhthoff) · linked to **lateralised patella** = remnant of mesenchyme that was larger medially
Prevalence	medial patellar plica in **5–55%** of people; **few symptomatic** (OB) · mere presence ≠ symptoms (TO)
Plica syndrome	painful impairment from **thickened, inflamed synovial fold (usually medial)** · **50%** give history of **blunt anterior knee trauma** · repetitive sport, direct trauma, other knee pathology
Inflamed plica (TO)	**fibrotic, thickened, inelastic** → snaps over **femoral condyle** or under **patella** → pain, **synovitis of condylar margins**, **chondromalacia patellae** / cartilage wear

Plica anatomy, types & symptomatic plica

Plica	Course	Key fact
Infrapatellar = ligamentum mucosum	**Intercondylar notch → infrapatellar (Hoffa) fat pad**; narrow femoral origin (anterior notch), widens, **anterior to ACL**	**Most common / most frequently seen at arthroscopy**; **rarely symptomatic**
Suprapatellar	**Anterior femoral metaphysis → medial joint wall** (TO) · → posterior quadriceps tendon above patella (TO) · medial → lateral wall of suprapatellar space (Orthobullets); between joint + suprapatellar bursa	**Most persistent**; impinges **superomedial trochlea in flexion**; usually asymptomatic
Medial (mediopatellar, Lino band, shelf)	**Medial joint wall → synovium over infrapatellar fat pad** (from suprapatellar origin)	**Most common cause of plica syndrome**; **most significant / most symptomatic**; abrades **MFC**
Lateral	Lateral wall, **lateral patellar facet → infrapatellar region**; thin, 1–2 cm lateral to patella	**Least common**; existence controversial; asymptomatic

Frequency order (TO)	**infrapatellar > suprapatellar > mediopatellar** · Orthobullets: ligamentum mucosum most common
Conflict	**infrapatellar = most common plica** (TO, Orthobullets) vs **medial = most common type + most symptomatic** (TO FNB 2023) – read the options; **most symptomatic = medial** in all sources
Multiple plicae	not uncommon in one patient

– – Active space – –



Synovial plicae of the knee — medial (shelf) plica is the symptomatic one

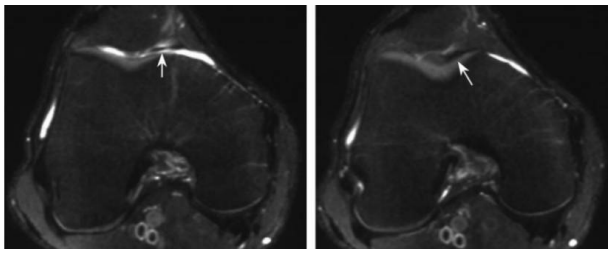
3 Presentation

Symptoms	**anterior / anteromedial knee pain**, **snapping, popping, clicking** with repetitive flexion–extension, **buckling**, **pain on sitting**, crepitus, swelling · worse with **poor quadriceps tone / muscle imbalance**
Cyclist (TO)	tenderness on **undersurface of medial patellar facet**; **deep flexion + full extension painful**
Exam	**medial parapatellar tenderness** · **painful palpable cord** rolled + popped under the finger · band painful with flexion–extension **with tibia internally rotated** · **McMurray negative**
Provocative test	knee held in **full extension**, examiner flexes against resistance, then **pushes patella medially** palpating medial border → **pain ± click = positive**

4 Diagnosis & Classification

Key radiology findings

Modality	Finding
Clinical	**Diagnosis is primarily clinical**
MRI	Detects plica but **low sensitivity**, non-specific · order only **after failed conservative** treatment
MRI symptomatic plica (TO)	**Thick ± synovitis** (normal plica thin)
MRI Sakakibara C (TO)	Axial FS PD **hypointense band covering anterior MFC**
MRI suprapatellar (TO)	**Sagittal**: band-like low-signal structure **posterior to patella**
Arthroscopy	**Gold standard for diagnosis + treatment**; arrow on PF view from inferolateral portal = **plica** (✗ Wrong options: loose body · meniscus · retinaculum · OCD)
Arthroscopy, suprapatellar (TO)	**Almost never diagnosed** — complete plica suggested only by a **short suprapatellar pouch** (or needle enters bursa)



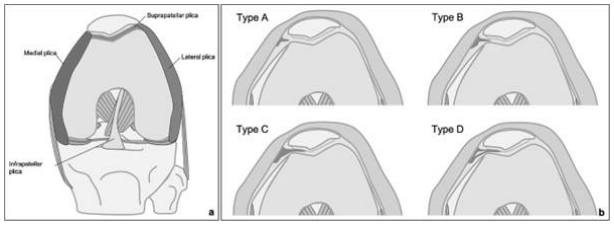
MRI – thickened medial plica (Sakakibara type C) covering the anterior MFC

Sakakibara classification of medial plica

Type (TO)	Arthroscopic appearance	Symptoms
A	**Cord-like elevation** in synovial wall	Unlikely
B	**Shelf-like**, does **not** cover anterior MFC	Unlikely
C	**Large shelf covering anterior surface of MFC**	**Trapped between MFC + patella** → thick, hard → **cartilage damage**
D	**Fenestrated** (central defect)	Trapped → cartilage damage

Types C + D (TO) impinge **medial patellar facet cartilage in flexion** + **MFC cartilage in extension → arthroscopic resection = treatment of choice**

Untreated medial plica syndrome (TO) can cause **Outerbridge grade IV** chondral lesions (subchondral bone exposed)



Plica locations and Sakakibara types A–D

Misfiled TO items (Brodie abscess, carpal tunnel tests)

Brodie abscess (TO)	Fact
Site / age	**Proximal tibia**, children + young adults
Course	**Indolent**; mild–moderate local tenderness; intermittent pain **worse at night**, > 2 wk; low-grade fever
Labs	WBC, ESR, CRP **normal or mildly ↑**
X-ray	**Cavity at metaphyseal–epiphyseal junction + rim of reactive new bone**
Diagnosis / organism	**Bone biopsy** confirms · **S. aureus** commonest
Treatment	**Curettage + bone grafting under an anti-staphylococcal antibiotic cover** (✗ Wrong options: curettage alone · observation · needle aspiration)

– – Active space – –

Carpal tunnel syndrome (TO)	Fact
Not a CTS test	**Murphy** (✗ Wrong options: Phalen · Durkan · Semmes-Weinstein)
Pathology	Carpal tunnel pressure **> 30 mmHg**
Commonest causes	**Idiopathic > hypothyroidism > pregnancy**
Risk factors	Female, obesity, pregnancy, hypothyroid, RA, trauma (distal radius fracture / malunion, carpal dislocation), repetitive motion, acromegaly, age, menopause, CRF, space-occupying lesion, OCP, CHF, diabetes, smoking, alcohol, mucopolysaccharidosis (children)
Symptoms	Pain, numbness, paraesthesia **first 3 digits + lateral half of 4th**; thenar atrophy
Riche-Cannieu anastomosis	Deep ulnar → recurrent motor median branch → **thenar strength preserved**
Sensory tests	2PD (fails at **5 mm**), **Semmes-Weinstein = most sensitive (early)**, vibrometry, texture, **hand diagram = most specific**
Phalen	Dorsal hands together **30–60 s**
Durkan	Thumb pressure over carpal tunnel just distal to wrist crease → symptoms **within 30 s**
Tinel	Tapping over nerve → tingling
CTS-6	Score **> 12 → 80%** probability
Imaging	X-ray limited · MRI if mass · **US CSA > 10 mm²** at pisiform / proximal TCL
NCV	**Distal sensory latency > 3.5 ms**, **motor latency > 4.5 ms**, CV **< 52 m/s** (less specific than latencies)
EMG	Traditional gold standard (AAOS: limited recommendation) · ↑ insertional activity, sharp waves, fibrillations, fasciculations, complex repetitive discharges

5 Management

Plica syndrome

- First line → **ACTIVITY MODIFICATION + PHYSICAL THERAPY (hamstring stretching), NSAIDs**
  - persistent → **INTRAPLICAL CORTICOSTEROID INJECTION (TO)**
- Failed conservative → **MRI (limited sensitivity)**
- Refractory / Sakakibara C-D → **ARTHROSCOPIC PLICA EXCISION**

Diagnosis & initial management

Most cases	**nonop**: activity restriction, NSAIDs, PT · avoid **repetitive jumping, squatting, lunging** · usually successful
PT content	**moist heat**, **hamstring stretching** · **resistive strengthening avoided early** (Orthobullets) · **quadriceps strengthening** part of rehab (OB) – read the options
Weightlifter, palpable painful anteromedial fold, McMurray –	**activity modification & physical therapy** ✗ Wrong options: diagnostic arthroscopy · MRI · PRP · patellar stabilising brace

Arthroscopic resection — **rare**: plica band syndrome **not responding to nonop · thickened Sakakibara C plica (TO) → arthroscopic excision**

– – Active space – –

X vs Y

	Infrapatellar (lig. mucosum)	Medial (shelf)	Suprapatellar	Lateral
Frequency	**Most common**	Common (most common type — one TO)	Most persistent	**Least common**
Symptoms	**Rare**	**Most symptomatic**	Usually none	None
Course	**Notch → fat pad**	**Medial wall → fat pad**	**Femoral metaphysis → medial wall**	Lateral facet → infrapatellar

	Sakakibara A / B	Sakakibara C / D
Covers anterior MFC	No	**C: yes**; D fenestrated
Cartilage damage	Unlikely	**Yes**
Treatment	Nonop	**Arthroscopic resection**

How it's asked

"Arrow on arthroscopic PF view (inferolateral portal)?"	→ **plica**
"Anteromedial fold, popping on squats — next step?"	→ **activity modification & physical therapy**
"Infrapatellar plica (TO)?"	→ **runs from intercondylar notch to fat pad** ✗ Wrong options: medial wall → fat pad = medial · metaphysis → medial wall = suprapatellar
"Most common symptomatic plica (TO)?"	→ **medial**
"Most significant plica causing knee pain (TO)?"	→ **medial**
"NOT true about synovial plica (TO)?"	→ **"Type C Sakakibara medial plica is not associated with articular cartilage damage"** (it is)
"Cyclist, hypointense band covering anterior MFC (TO)?"	→ **Sakakibara type C synovial plica — arthroscopic excision** ✗ Wrong options: type B ablation · MPFL cyst
"Brodie abscess — treatment (TO)?"	→ **curettage + bone grafting under an anti-staphylococcal antibiotic cover**
"Not a test for carpal tunnel syndrome (TO)?"	→ **Murphy**

– – Active space – –

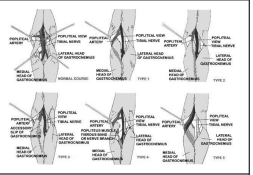
POPLITEAL ARTERY ENTRAPMENT SYNDROME

– Knee & Sports · 44 / 61 –

1 Basics

Definition	**constriction of the popliteal artery** by adjacent muscles, tendons or fibrous tissue · usually an **underlying anatomic abnormality**
Demographics	**male 4:1** · age **25–40 years**
Pathophysiology	↓ flow distal to the popliteal fossa → symptoms **mimic compartment syndrome**
Classic pathoanatomy (OB)	compression by the **medial head of gastrocnemius** · **accessory slip of medial gastrocnemius head** on MRI

Modified Whelan classification

Type	Anatomy	Image
I	**Normal medial gastrocnemius head**, popliteal artery takes an **aberrant course**	
II	**Medial gastrocnemius head located laterally**, artery not deviated	
III	**Abnormal muscle bundle from medial gastrocnemius head** surrounds + constricts artery	
IV	Artery entrapped by **popliteus**	
V	**Popliteal vein entrapped with the artery** in any of the above	

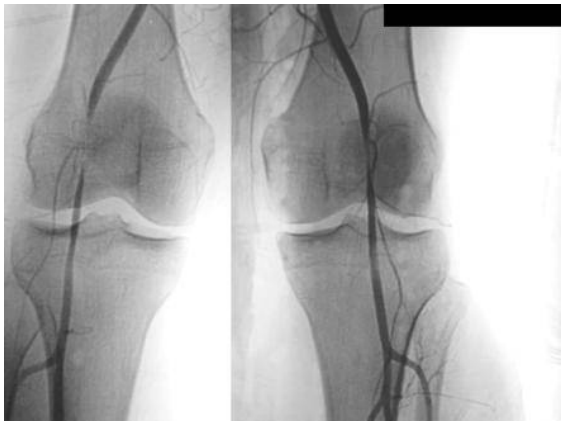
3 Presentation

Symptoms	**intermittent claudication** (OB) · **limb swelling often the primary complaint** · foot numbness, **toe tingling after vigorous exercise** · **calf cramping even after light exercise** as it worsens · may be asymptomatic
Exam	**diminished pulses with active plantarflexion or passive dorsiflexion** · cool posterior calf, paraesthesias · **easily confused with posterior compartment syndrome**
Sensory pattern (OB)	dysaesthesia in the **sural** distribution (posterolateral leg / foot)
Runner with pain, coolness, tingling (OB)	**pain with passive dorsiflexion + plantarflexion**, **compartment pressures normal → PAES** ✗ Wrong options: exertional compartment syndrome · tibial stress fracture · lumbar radiculopathy · piriformis

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
Radiographs	**Usually normal**
Doppler US	Less invasive; detects **pulse change during active plantarflexion / passive dorsiflexion**
Arteriogram	**Confirms diagnosis**, **~100% sensitivity → stenosis, obliteration, post-stenotic dilation**
MRA / CTA	Usefulness still under study · MRA can show **segmental popliteal occlusion** (OB)



Arteriogram — popliteal artery entrapment

Differential of exertional leg pain (OB)

Exertional compartment syndrome	needs **↑ compartment pressures**: rest > 15 · 1 min post > 30 · 5 min post > 20 · 15 min post > 15 mmHg (Orthobullets) → normal pressures exclude it
Lumbar radiculopathy / piriformis	neuropathy does not explain a **cool, crampy leg**
Tibial stress fracture	**not relieved by running**, no claudication signs
Other differentials	CECS, medial tibial stress syndrome, tibial / fibular stress fracture, fascial defects, nerve entrapment, **vascular claudication (atherosclerotic or PAES)**, lumbar disc herniation

5 Management

Popliteal artery entrapment

- Mild symptoms, rigorous exercise only → **activity modification + observation**
- Refractory / artery or vein damage → **vascular surgery**
  - release the entrapping structure (depends on type)
  - damaged artery → **saphenous vein bypass vs endarterectomy**

Natural history	**most patients eventually need surgery**
Approach	**posterior → better exposure · medial → used when bypass needed**

– – Active space – –

6 Complications

Complication	Rate
Surgical failure (entrapment returns)	**30%** → can need **amputation**
Wound infection	**< 5%**
DVT	**~10%**
Asymptomatic after surgery	**70–100%**

X vs Y

	PAES	Exertional compartment syndrome
Compartment	**Posterior** leg	**Anterior** (± lateral) commonest
Mechanism	Artery compressed (medial gastrocnemius)	↑ pressure in a closed compartment
Key sign	**↓ pulses with active plantarflexion / passive dorsiflexion**, cool leg	Normal exam at rest
Nerve symptoms	Sural distribution	**Dorsum of foot / 1st web**
Confirm	**Arteriogram** (Doppler US screen)	**Compartment pressures**

How it's asked

"Claudication, coolness, tingling; pain on passive DF + PF; normal pressures?"	**→ popliteal artery entrapment syndrome**
"Confirmatory test?"	**→ arteriogram (stenosis, obliteration, post-stenotic dilation)**

– – Active space – –

POSTEROLATERAL CORNER INJURY

– Knee & Sports · 45 / 61 –

1 Basics

Epidemiology

Incidence	**7–16%** of knee ligament injuries (Orthobullets) · **10–15%** (OB) – read the options
Isolated	**only 28%** isolated (Orthobullets) · **< 5%** isolated (OB) – read the options · usually combined with cruciate (**PCL > ACL**)
Missed PLC	**common cause of ACL (and PCL) reconstruction failure** · often missed (subtle exam + imaging)

Anatomy – LCL, popliteus, biceps femoris

3 major static stabilisers	**LCL (FCL)**, **popliteus tendon**, **popliteofibular ligament (PFL)**
Other static	lateral capsule thickening (meniscofemoral + meniscotibial ligaments) · arcuate ligament (variable) · fabellofibular ligament (variable)
Dynamic	**biceps femoris**, popliteus muscle, **ITB**, **lateral head of gastrocnemius**
Lateral quadruple complex (TO)	**ITB + LCL + popliteus tendon + biceps femoris** ✗ Wrong options: fabellofibular ligament · popliteofibular ligament
Arcuate complex	LCL + arcuate ligament + popliteus tendon
Full PLC (OB, Covey)	ITB, LCL, popliteus complex (dynamic MTU; static PFL, popliteotibial + popliteomeniscal fascicles), middle third lateral capsular ligament, fabellofibular, arcuate, posterior horn LM, lateral coronary ligament, posterolateral capsule

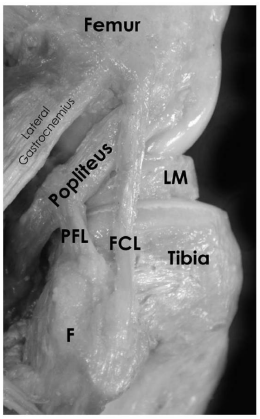
Structure	Femur	Fibula / tibia	Key fact
LCL	LFC, **proximal + posterior to popliteus**	**Most anterior** on fibular head · **anterolateral to PFL**	Runs **superficial to popliteus** · primary varus restraint
Popliteus tendon	**Distal + anterior to LCL** (✗ Wrong options: proximal) · through LM hiatus	Muscle → posteromedial tibia	**Dynamic internal rotator of tibia**
PFL	Popliteus musculotendinous junction	**Medial to LCL** on fibular head (✗ Wrong options: lateral)	Anterior + posterior divisions
Biceps femoris	–	**Posterior to LCL** on fibular head	Long head: direct arm posterolateral, anterior arm lateral to LCL · short head: superior fibular head, medial + posterior to LCL

Fibular insertion order (anterior → posterior)	**LCL → PFL → biceps femoris**

– – Active space – –

Biceps–capsulo-osseous ITB confluence injury correlates with **positive Lachman** (Terry & LaPrade)

PMC distractors **posterior oblique ligament** (from adductor tubercle region, posterior + proximal to MCL) + deep MCL (meniscofemoral + meniscotibial) → **medial**; no positive dial



Lateral knee: popliteus, PFL, FCL (LCL), fibula

Lateral layer	Structures
I	Iliotibial tract, biceps
–	**Common peroneal nerve between layers I and II**
II	Patellar retinaculum, patellofemoral ligament
III superficial	**LCL**, fabellofibular ligament
–	Lateral geniculate artery between superficial + deep III
III deep	Arcuate, coronary ligament, **popliteus tendon**, **PFL**, capsule

Function

Popliteus + PCL synergistic → control **external tibial rotation, varus, posterior tibial translation**

Popliteus primary action **internal tibial rotation** · popliteus complex reinforces posterior third of lateral capsule; restrains ER, posterior translation, varus
✗ Wrong options: ER · posterior / anterior translation

Popliteus + PFL maximal **in flexion** → resist ER

LCL primary varus restraint at **5° (55%)** and **25° (69%)** (PLC note) · **5° and 30°** (LCL note) – read the options

2 Mechanism

Mechanisms **blow to anteromedial knee** · **varus blow to flexed knee** · contact / non-contact **hyperextension** · **ER twisting** · knee dislocation · varus force + ER component (OB)

Associated **common peroneal nerve 15–29%** (Orthobullets etiology) · **~25%** (Orthobullets exam) · vascular injury

3 Presentation

Symptoms instability **in full extension** · difficulty reciprocating stairs, pivoting, cutting · lateral + posterior pain

Gait standing **varus**, **varus thrust** or hyperextension thrust

Peroneal nerve ↓ sensation dorsum of foot + weak ankle dorsiflexion

PLRI **lateral tibial plateau moves posterior to femoral condyle** on rotational stress → **posterolateral rotatory instability** · instability named by tibial position relative to femur
✗ Wrong options: AMRI · ALRI · PMRI

Diagnosis – dial test, reverse pivot shift & PLRI

Test	Technique	Positive means
Varus stress 30°	–	**LCL**
Varus stress 0°	–	**LCL + cruciate (ACL or PCL)**
Dial test	Prone or supine; 30° + 90°	**> 10° ER at 30° only → isolated PLC · 30° + 90° → PLC + PCL**
ER recurvatum	Supine, suspend legs by great toes	↑ recurvatum + varus + **anterior** tibial translation → **ACL + PLC**
Posterolateral drawer	Hip 45°, knee 80°, foot ER 15° · posterior drawer + ER	↑ posterolateral translation (lateral tibia ER on LFC)
Reverse pivot shift	Tibia ER + valgus	Clunk of reduction → **PLC** (PLRI)
Posterior drawer	90°	PCL (negative in isolated PLC)

Dial threshold **> 10°** (Orthobullets) · **10–15°** (OB) · **15°** (TO supine description) – read the options

Supine dial (TO) leg over bed edge, stabilise distal femur, hold heel, max ER at 30° → **tibial tuberosity position** vs other side → repeat at 90°

Prone dial (TO) both feet rotated at 30° then 90° · angle between **sagittal plane of sole and body** (thigh–foot angle)

Prone preferred (Jung) dial more clearly positive when **posterior sag is reduced**

Other PLC tests (TO) ER recurvatum · posterolateral drawer · reverse pivot shift

Dial ↑ at 30° + 90° – least likely torn? **posterior oblique ligament** (medial)
✗ Wrong options: FCL · popliteus · PCL · arcuate

+ Reverse pivot, negative posterior drawer PLC without PCL → expect **10° ↑ ER at 30°**
✗ Wrong options: at 90° · valgus opening · anterior drawer

– – Active space – –

LaPrade (OB) reverse pivot shift ↔ injury to **LCL, popliteal components, mid-third lateral capsular ligament** · LCL injury ↔ ↑ ER at 30° · **Gollehon: sectioning LCL → 20° ↑ ER at 30°**

Pivot tests (TO)	Movement	Instability
Pivot shift (Macintosh)	**Extension → flexion**, tibia **IR** + valgus · tibia relocates posteriorly	**Anterolateral** (ACL + lateral capsule / ALL)
Jerk test (Hughston & Losse)	**Flexion → extension** (reverse of pivot shift)	**Anterolateral**
Reverse pivot shift	Tibia **ER** + valgus; clunk	**Posterolateral** (PLC, ± PCL)

Reverse pivot direction start at **90°**, ER + valgus, **extend** → tibia reduces from posterior subluxation at **~20°** (ITB flexor → extensor) (Orthobullets) · knee moved **extension → flexion** to feel clunk (TO) – read the options



Reverse pivot shift: valgus and external rotation

ER recurvatum (TO) historically relative hyperextension + ER of tibial tubercle + varus · **LaPrade: sensitivity 7.5%** in 134 PLC patients · X-ray showed **anterior** tibial translation · **all positives = ACL + PLC** · doesn't identify which PL structures (multiple inferred)
✗ Wrong options: posterior

ER recurvatum yield identifies **~10%** of PLC injuries (Orthobullets) · **7.5% sensitivity** (TO) – read the options

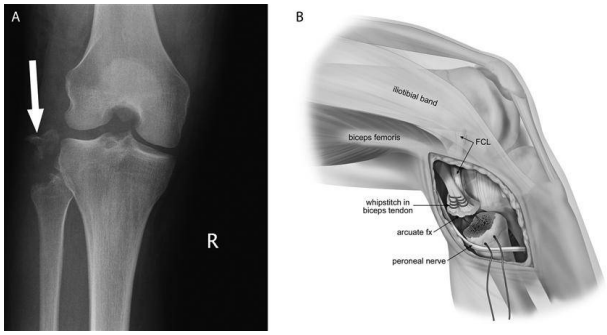
4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
AP X-ray	**Arcuate sign** = fibular head avulsion (LCL / biceps) · or femoral condyle avulsion
Varus stress X-ray (bilateral, **20° flexion**)	Side-to-side **2.7–4 mm → isolated LCL** · **> 4 mm → PLC**
Long-leg standing	**Chronic** PLC → **triple varus**, mechanical axis, need for **proximal tibial osteotomy**
MRI	LCL, popliteus, biceps · acute **bone bruise MFC + medial tibial plateau** · **coronal oblique thin slices through fibular head** best
MRI (sagittal T2)	Posterolateral tibia oedema + **oedema posterior to popliteus tendon**
MRI (coronal)	ITB, LCL, popliteus torn · coronary ligament torn from tibia → displaced LM

Arcuate sign (OB) **highest association with PLC**
✗ Wrong options: Segond (ALL + ACL) · Schatzker II lateral plateau (varus PLC injury → **medial** plateau injury commoner) · PCL avulsion · tibial spine avulsion (ACL)

Fragment size (Lee) arcuate ligament / PFL → **small styloid** avulsion · LCL / biceps → **larger** fragment · **Huang**: fibular head avulsion predicts **PLC + PCL** injury (13 pts; LCL > popliteus)



Arcuate fracture (fibular head avulsion)

Varus alignment (chronic)	Cause
Primary	Tibiofemoral malalignment
Secondary	LCL deficiency → ↑ lateral opening
Triple	Remaining PLC deficient → varus recurvatum



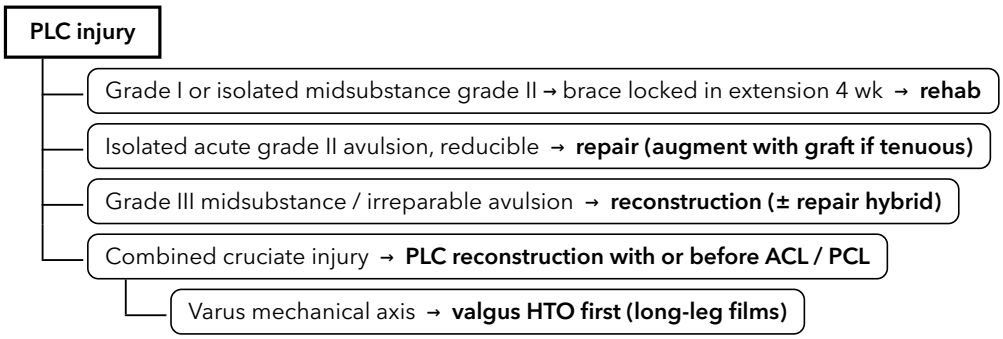
Bilateral varus stress radiographs

Modified Hughston classification

Grade	Examination	Findings
I	**0–5 mm** lateral opening on varus stress	Sprain, no tensile failure
II	**6–10 mm** opening · **6–10°** rotational instability on dial	Partial, moderate disruption
III	**> 10 mm**, **no endpoint**	Complete disruption

- - Active space - -

5 Management



Operative criteria

Option	Indication
Nonop (brace in extension 4 wk, quad rehab, RTS 8 wk)	**Grade I** · **isolated midsubstance grade II**
Repair	**Isolated acute grade II avulsion** · structures **anatomically reducible to attachment**
Fibular head avulsion	Screws or suture anchors
Reconstruction (hybrid)	**Grade III midsubstance** · avulsion not repairable / poor tissue
PLC + ACL / PCL reconstruction	Acute + chronic **combined** injuries
Valgus HTO	**Varus mechanical alignment** (else ACL + PLC reconstruction fail)

Treatment – reconstruction over repair

Repair → higher failure (OB)
midsubstance repair 40% failure (Orthobullets) · **Levy: 40% repair vs 6% (1 pt) reconstruction** · **Stannard: 37% vs 9%** (57 PLC injuries; both "near-normal" IKDC)
✗ Wrong options: quicker ROM · better stability · higher short-term scores · worse kinematics

After revision of failed repair
outcome scores ≈ primary reconstruction

ACL reconstruction + PLC repair vs reconstruction
33% vs 88% IKDC A / B

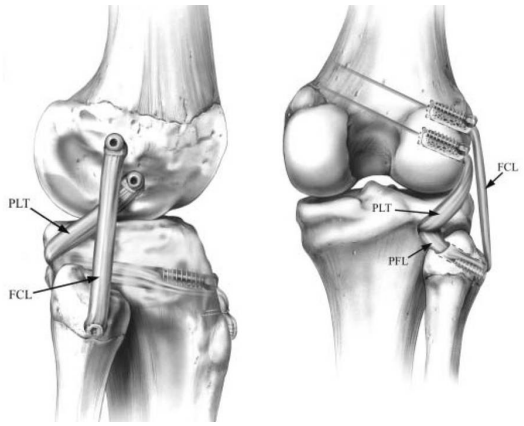
Outcomes
operative > nonop · **early treatment better** · anatomic reconstruction restores **rotatory but not all varus** stability

Fanelli–Larson PLC grade (TO)	Structures	Findings	Treatment (TO)
A	**Popliteus + PFL**	↑ ER at 30°, normal varus	**Augmented repair (modified Larson)**
B	Popliteus + PFL + **FCL**	↑ ER at 30°, **5–10 mm** varus opening	**Reconstruction (modified LaPrade)** + cruciate
C	Popliteus + PFL + FCL + **capsule** ± cruciate	↑ ER at **30° + 90°**, **> 10 mm** varus opening	Reconstruction (modified LaPrade) + cruciate

Larson rebuilds (TO) **LCL + popliteus** with single graft (TO) · LCL + PFL (table below) — read the options

– – Active space – –

Reconstruction	Graft / technique	Rebuilds
Fibular-based (Larson)	Soft-tissue graft (semitendinosus / Achilles) through fibular head tunnel → **figure-of-eight** → single femoral tunnel	**LCL + PFL**
Transtibial double-bundle	Split Achilles to femoral epicondyle isometric point · fibular limb + tibial limb	Fibular limb **LCL** · tibial limb (posterior tibia) **PFL**
LaPrade anatomic	**Two grafts** · graft 1 femoral LCL site → fibular head (lateral → medial) · graft 2 popliteus site · both dock in tibial tunnel posterior → anterior	**LCL + PFL + popliteus**



LaPrade anatomic PLC reconstruction

Alignment correction & combined ACL revision

Sequence	PLC reconstruction **same time or before** ACL / PCL → prevents early cruciate graft failure
Missed PLC + ACL graft (OB)	3+ Lachman, varus laxity at 30°, dial + at 30° dissipating at 90°, varus thrust → **revision ACL + PLC reconstruction** · unrecognised PCL → medial + PF OA without lateral laxity / thrust ✗ Wrong options: PCL · PMC · hamstring autograft alone
ACL + PLC + varus + MM tear (OB)	HTO + ACL + meniscus + PLC reconstruction; **NOT PCL reconstruction** (quad active negative, normal ER at 90°) · untreated varus thrust → ACL graft fails
Noyes (staged HTO + ACL, 41 pts, ~50% PLC)	4 yr: pain ↓ **71%**, giving way gone **85%**, light recreation **66%**
Grade 3 PLC laxity + varus after ACL + PCL reconstruction (OB)	**long-leg alignment films first** → primary varus → **medial opening-wedge proximal tibial osteotomy before PLC reconstruction** ✗ Wrong options: straight to fibular / tibial-fibular reconstruction · PT · bracing
Untreated PLC ↑ PCL graft force	Harner (↑ posterior translation, ER, in-situ graft force) · LaPrade (grade 3 PLC)

– – Active space – –

Rehabilitation

After PLC reconstruction (hybrid)	hinged brace, **NWB 6 wk** · ROM varies (passive 0–90° at once vs 2 wk immobilisation) · at 6 wk WB + closed-chain · **RTS 6–9 months**
After combined reconstruction	immobilise + **protected WB 4 wk** (long leg cast controls ER better than brace) · **passive ROM at 4 wk** (avoid arthrofibrosis) · **avoid active hamstring** · full active extension allowed
PCL reconstruction + PLC repair (OB)	**no active flexion, active extension 90→0°** · hinged brace, crutches, partial WB first 4 wk · quad sets + SLR ✗ Wrong options: patellar tendon repair · ACL ± MCL · ACL + PLC

6 Complications

Missed PLC	failure of ACL or PCL reconstruction
Peroneal nerve injury	**15–29%**
Arthrofibrosis	

X vs Y

	PLC	PMC
Key static	LCL, popliteus, PFL	**POL**, deep MCL, sMCL
Exam	**Dial**, varus, reverse pivot	Valgus, AMRI
Instability	**PLRI**	AMRI / PMRI

	Dial + at 30° only	Dial + at 30° and 90°
Lesion	**Isolated PLC**	**PLC + PCL**
Posterior drawer	Negative	Positive

	Repair	Reconstruction
Indication	Acute grade II **avulsion**, reducible	Grade III midsubstance, chronic, combined
Failure	**37–40%**	**6–9%**

How it's asked

"Lateral quadruple complex EXCEPT (TO)?"	**→ fabellofibular ligament**
"NOT part of lateral quadruple complex (TO)?"	**→ popliteofibular ligament**
"LCL relationship in PLC?"	**→ inserts anterolateral to PFL on fibula, superficial to popliteus**
"Normal anatomy of region injured (dial + at 30°)?"	**→ biceps femoris inserts posterior to LCL on fibular head**
"Function of structure at probe tip (popliteus)?"	**→ internal tibial rotation**
"Use of test in image (TO)?"	**→ detect PLC injury (dial test)**

"Positive reverse pivot shift (TO)?"	→ **PLC injury**
"Positive ER recurvatum test (TO)?"	→ **combined PLC + ACL injury**
"Lateral plateau moves posterior to femoral condyle on rotation?"	→ **posterolateral rotatory instability**
"↑ ER at 30° + 90° — least likely torn?"	→ **posterior oblique ligament**
"Finding indicating PLC + PCL?"	→ **dial + at 30° and 90°**
"+ Reverse pivot, negative posterior drawer — other finding?"	→ **10° ↑ ER at 30°**
"Radiograph most associated with PLC injury?"	→ **arcuate sign (fibular head avulsion)**
"Post-op: no active flexion, active extension 90→0°?"	→ **PCL reconstruction + PLC repair**
"ACL tear, dial + at 30° only, varus axis, MM tear — EXCEPT?"	→ **PCL reconstruction**
"Grade 3 PLC laxity + varus after ACL + PCL reconstruction?"	→ **long-leg alignment films**
"Failed ACL graft, varus laxity, dial + at 30° only?"	→ **revision ACL + PLC reconstruction**
"Acute PLC repair vs reconstruction?"	→ **higher rate of failure**
"Fanelli type B PLC injury (TO)?"	→ **popliteus + PFL + FCL, 5–10 mm varus opening**

– – Active space – –

– – Active space – –

PREPATELLAR BURSITIS (HOUSEMAID'S KNEE)

– Knee & Sports · 46 / 61 –

1 Basics

Frequency	**most common bursitis of the knee**
Anatomy	bursa **anterior to patella** · potential space → gliding of tissue over patella
Risk factors	**excessive kneeling** · **wrestlers** → **concern for septic bursitis**
Septic vs aseptic	**20% septic**
Organism	**S. aureus** = most common in septic prepatellar (and olecranon) bursitis

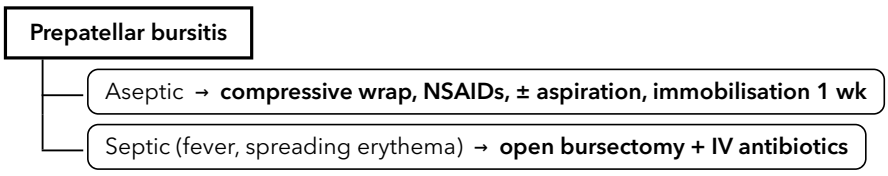
3 Presentation

Symptoms	pain, swelling over patella, history of kneeling · **warm**, especially if septic
Septic vs aseptic	hard to tell clinically · **extra-articular** → no septic-arthritis pain on knee motion

4 Diagnosis & Classification

Aspiration + Gram stain + culture	**rarely indicated** · use only to separate septic from aseptic

5 Management



Aseptic	compression, NSAIDs, padding, ice, ± aspiration, **1 wk immobilisation** · corticosteroid **controversial**
Septic	**complete bursectomy + systemic (IV) antibiotics** = standard · antibiotics + aspiration / percutaneous drain succeed in some
Recurrent after aspiration, fever 101.7°F, spreading erythema	**perform open bursectomy + start intravenous (IV) antibiotics** ✗ Wrong options: oral antibiotics 7 days · re-aspirate + steroid · padding + ice
Bursal resection	rarely needed · open or **arthroscopic**

How it's asked

"Febrile, erythematous prepatellar bursa after aspiration?"	→ **open bursectomy + IV antibiotics**

– – Active space – –

PROXIMAL TIB-FIB DISLOCATION

– Knee & Sports · 47 / 61 –

1 Basics

Incidence	**rare**, rarer isolated · **often missed** · 2nd–4th decades
Anatomy	fibular head articulates with facet on lateral tibial cortex (**separate from knee joint**) · anterior + posterior ligaments of fibular head · **common peroneal nerve** distal to joint, **posterolateral fibular neck**

2 Mechanism

Mechanism	**high energy** (horseback riding, parachuting) · fall on **flexed, adducted knee** · twisting of flexed knee in athletes (OB)
Associated	**posterior hip dislocation** (knee + hip flexed) · **open tibia–fibula fractures** · other knee / ankle fractures

3 Presentation

Symptoms	**lateral knee pain mimicking lateral meniscal tear** · instability · lateral prominence · popping / catching · dorsal foot numbness (peroneal)
Exam	fibular head tenderness · **compare both knees** (landmarks)

4 Diagnosis & Classification

Key radiology findings

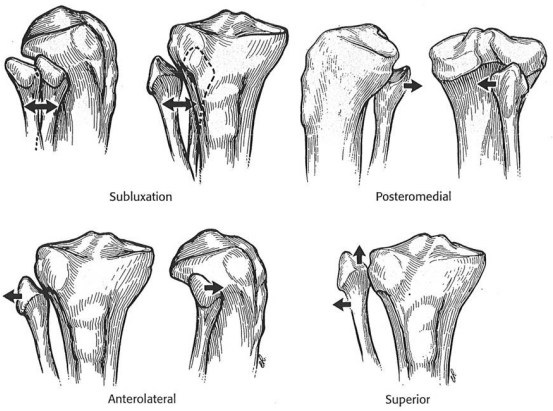
Study	Finding / use
AP + lateral of both knees	**Contralateral comparison views essential**
CT	Clearly confirms / excludes dislocation
MRI	Normal cruciates + menisci → points away from intra-articular cause

Ogden classification

Type	Fact
Subluxation	–
Anterolateral dislocation	**Most common**
Posteromedial dislocation	**Peroneal nerve injury usually with posterior** dislocations
Superior dislocation	–

4 types (Sekiya & Kuhn)	subluxation + 3 dislocations · diagnosis frequently missed

– – Active space – –



Ogden classification of proximal tibiofibular joint injuries

5 Management

Proximal tib-fib dislocation

- Acute → **closed reduction (knee flexed 80–110°)**
  (failed closed reduction → open reduction)
- Chronic pain / instability / missed / malreduced → **surgery**
  - soft-tissue (capsule) reconstruction, ORIF + pinning, arthrodesis, fibular head resection

Closed reduction	**flex knee 80–110°** (relaxes **LCL + biceps femoris**) → **pressure on fibular head opposite to dislocation** · usually successful
Post-reduction	immobilisation in extension vs early ROM → **controversial**
2 wk after rugby fall, popping + dorsal foot numbness, normal MRI (OB)	**closed reduction** ✗ Wrong options: meniscal arthroscopy · ACL reconstruction · OCD arthroscopy · EMG for anterior tarsal tunnel (deep peroneal entrapment)

6 Complications

Recurrence	
Common peroneal nerve injury	usually **posterior** dislocations · **> 35%** with concomitant proximal tibia fracture
Arthritis	rare, minimally symptomatic

How it's asked

"Lateral knee pop, catching, dorsal foot numbness, X-ray dislocation, normal MRI – next?" → **closed reduction of dislocation**

PROXIMAL TIBIOFIBULAR JOINT GANGLION CYSTS

– Knee & Sports · 48 / 61 –

1 Basics

Incidence	**< 1%** of knee MRIs for pain · age 20–76 · **median 67** (Orthobullets) · "middle-aged males" (OB) · **male 90%** · right leg ~70%
Location	**extraneural 80%** > intraneural 20%
Cause	repetitive microtrauma · **degeneration** · associated **OA**
Pathology	**mucin-filled, synovial-lined sac without true epithelial lining** · collagen walls · viscous (hyaluronic acid, mucopolysaccharides)
Anatomy	fibular head in groove behind lateral tibial ridge · capsule stabilisers: anterior + posterior tibiofibular ligaments, LCL, popliteus, biceps · **anterior tibial artery** just distal to PTFJ · **common peroneal nerve** around fibular neck → superficial + deep branches
Biceps femoris	long head (posterior on fibular head) → **tibial nerve** · short head (anterior) → **common peroneal nerve**
PTFJ function	anterolateral / posteromedial sliding → ↓ ankle torsion, disperses axial load

Classification	Site
Intraneural	Within epineurium
Extraneural	Outside epineurium

3 Presentation

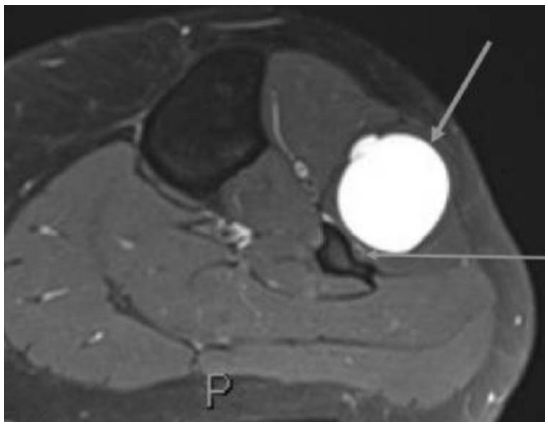
Symptoms	usually asymptomatic · pain, **mobile palpable mass distal to lateral joint line**, fullness, **paraesthesias dorsum of foot**
Neuro	**foot drop** from common peroneal compression

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
X-ray (AP + lateral knee / tibia)	Usually normal · ± sclerosis lateral proximal tibia
MRI	**Diagnostic tool of choice** · well-circumscribed homogeneous cyst · **T1 low, T2 high**
US	If MRI not possible · thin-walled fluid cyst
EMG / NCS	Extent of sensory + motor deficit
Histology	**Confirms** · no true epithelium, no atypia / mitoses

– – Active space – –



PTFJ ganglion, T2 MRI

Differential ABC, chondroblastoma, GCT, juxta-articular myxoma, **lumbar disc herniation**, PVNS, solid nerve tumours

5 Management

PTFJ ganglion

- Asymptomatic / first presentation → **observation, NSAIDs, aspiration**
- Neuro symptoms / failed nonop / progressive → **decompression + marginal excision**
- Recurrence after first excision → **PTFJ arthrodesis (or fibular head resection)**

Treatment	Recurrence
Aspiration alone	**80%** (Orthobullets) · **25–80%** (OB)
Decompression + excision (hockey-stick lateral approach, neurolysis, excise stalk)	**8–25%** · 1st excision success 7 / 8; repeat excision 3 / 3 recurred (Miskovsky)
PTFJ arthrodesis (6.5 mm partially threaded cancellous screw)	**< 5%** – lowest

Painful growing mass + dorsal foot paraesthesias (OB) **surgical decompression + cyst excision**
✗ Wrong options: NSAIDs / activity change · aspiration · arthrodesis / fibular head resection (reserved for recurrence)

Recurrent cyst Gulati: proximal fibular excision, good results in 9 · arthrodesis (Miskovsky: 2 fused, no recurrence)

6 Complications

Peroneal nerve dysfunction **up to 50%** · risk: **intraneural** · decompression; tendon transfer if complete palsy · prolonged neuro symptoms = poor prognosis

Recurrence **5–80%** · aspiration only highest

Other neurapraxia, perineural fibrosis · hardware irritation after arthrodesis

How it's asked

"PTFJ ganglion with peroneal paraesthesias — reduce recurrence?"	**→ surgical decompression and excision of cyst**

– – Active space – –

PULMONARY CONDITIONS IN ATHLETES

– Knee & Sports · 49 / 61 –

1 Basics

Airway disorders

Asthma	**transient lower airway obstruction** from hypersensitivity · **expiratory** · exercise trigger in an asthmatic = **exercise-induced asthma (EIA)**
EIB	**no underlying asthma** (usually; can coexist, OB) · transient **lower**-airway, **expiratory** obstruction after exertion
EIB risk factors	**winter / cold-weather sports** (drying + cooling of mucosa → oedema + constriction) · **endurance athletes** · high minute ventilation ✗ Wrong options: warm weather
EILO	transient **upper**-airway obstruction: paradoxical vocal-cord closure or supraglottic narrowing · **inspiratory** · prominent **psychological** component · types: **supraglottic, glottic (vocal cord dysfunction), mixed**
Exercise-induced anaphylaxis	rare, life-threatening **type I (IgE)** reaction to exertion · after a food allergen = **food-dependent EIA (FDEIA)**

Chest wall & vascular conditions

Pneumothorax causes	**spontaneous** (smoking, male, FHx, Marfan, Ehlers-Danlos) · **traumatic** (**contact athletes – ice hockey highest risk**) · **iatrogenic** (interscalene block)
Haemothorax	**most common result of major chest-wall trauma** · **30%** of rib fractures → pneumohaemothorax
PE in athletes	↑ risk: musculoskeletal injury ± immobilisation, repetitive motion (pitchers, tennis), long-distance travel

3 Presentation

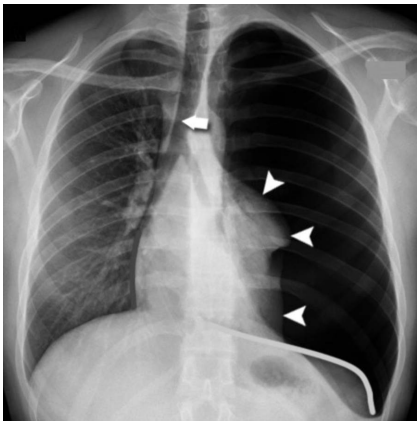
Asthma	wheeze, SOB, **chest tightness**; **expiratory wheeze**, accessory muscles
EIB	**triad: wheeze + SOB + cough** with exercise · "poor performance", fatigue, headache · exam often normal
EIB timing	onset **10–15 min after exertion**, resolves with **20 min rest** (Orthobullets) · 10–15 min after the **start** of exercise; resolves **15–20 min** after stopping (OB) – read the options
Pseudo-asthma syndrome	exercise-induced hyperventilation; no bronchoconstriction
EILO	**rapid onset**, **throat tightness / choking**, "breathing through a straw" · **inspiratory stridor** · **fails bronchodilators** (often misdiagnosed as EIA / EIB)
Exercise-induced anaphylaxis	onset **within 30 min** of exercise · pruritus, urticaria, GI cramps, SOB, throat tightness

Pneumothorax	sudden unilateral **pleuritic pain**, SOB · ↓ / absent breath sounds, **hyperresonant** · tension adds **tracheal deviation + JVD**, cardiopulmonary compromise
Tension pneumothorax on the field	dyspnoea, anxiety, **deviated trachea**, **hypertympanic** chest, ↓ breath sounds ✗ Wrong options: spontaneous (closed, no deviation) · haemothorax (dull) · EIB · HCM
Haemothorax	less pleuritic pain · **hypotension** if large · **dull to percussion**
PE	sudden pleuritic pain, SOB, syncope · tachycardia, **hypoxia**, hypotension if large

4 Diagnosis & Classification

Key radiology findings

Condition	Test	Finding
Asthma	PFT ± **methacholine** challenge	**↓ FEV1, ↓ FEV1/FVC**, reverses with bronchodilator
EIB	**Eucapnic voluntary hyperventilation (EVH)** = gold standard	**> 10% ↓ FEV1**; baseline PFT may be normal
EILO	**Continuous exercise laryngoscopy (CEL)** = gold standard	PFT: **blunted inspiratory loop**
Pneumothorax	**Ultrasound**	**Absent lung sliding**
Pneumothorax	**CXR**	**Pleural line**, absent lung markings · tension: **mediastinal shift + depressed hemidiaphragm**
Haemothorax	CXR	**Opacified hemithorax**
PE	**Helical CT** first line · **pulmonary angiography** gold standard	V/Q scan alternative
HAPE (TO)	CXR	Patchy opacities, often **right middle lobe**



Tension pneumothorax: mediastinal shift, depressed hemidiaphragm

– – Active space – –

Pneumothorax types

Type	Communication	Pleural pressure
Closed (e.g. bleb rupture)	None	**< atmospheric** → no tracheal deviation
Open	Air in on inspiration, out on expiration	**= atmospheric**, no tension
Tension	**One-way: in, cannot exit**	**↑ > atmospheric** → lung + mediastinal compression — **emergency**

5 Management

Return-to-play criteria

Condition	Return to play
Asthma / EIB	As tolerated · **active wheezing = contraindication**
Exercise-induced anaphylaxis	After **allergist review + comprehensive action plan** · always carry **EpiPen**, exercise with a partner
Pneumothorax	**Radiographic resolution, ~4 weeks** · pain limits RTP · **no flying 2–4 weeks**
PE	**~6 months** · on anticoagulation controversial · **contact athletes withheld** during anticoagulation

Exercise-induced bronchospasm

Acute	rest, controlled breathing, **inhaled β2-agonist (albuterol q15 min PRN)** · severe / refractory → **epinephrine + ER** · moderate–severe asthma maintenance → **inhaled glucocorticoids**
Prevention	**warm-up**, environmental control · **"refractory period"** (deliberate induction) works in **~50%** · **albuterol 2 puffs 15–20 min before** activity

EILO & anaphylaxis

EILO	nonoperative mainstay: **laryngeal control therapy** (lower-abdominal breathing, controlled exhalation through mouth), speech therapy, SSRIs, botulinum toxin · surgery (debulking) rare
Anaphylaxis prevention	no exercise **4–6 h after eating**, avoid temperature extremes, **avoid aspirin / NSAIDs** before exercise · acute: **epinephrine**, antihistamines, steroids

Tension pneumothorax – diagnosis & needle decompression

ATLS 10th, adults	**standard 14-gauge 4.5 cm angiocatheter (5 cm needle), 4th / 5th ICS mid-axillary line** ✓
Children	**2nd ICS mid-clavicular line** still recommended
Why changed	5 cm needle at 2nd ICS **fails in 22–50%** (chest-wall thickness) · risk to internal thoracic, subclavian, heart · no original evidence
Site evidence	5th ICS accurate **100%** vs just over **50%** at 2nd ICS (Inaba cadaver study) · either site reasonable, **2nd ICS slightly better in obese** (Azizi) — read the options

Site wording "2nd or 5th ICS" acceptable (Orthobullets, OB)

Failed decompression check catheter patency → **longer 8.25 cm needle → finger thoracostomy**
✗ Wrong options: any of these first

Traps ✗ Wrong options: chest tube first (hospital; supplies rarely on hand) · digital thoracostomy first · long needle first (visceral injury)

Pneumothorax

- Tension → needle decompression 5th ICS MAL → **chest tube in ED**
- > 20% (lung edge > 2 cm) → **chest tube + admit**
- < 20% (< 2 cm), stable → **O2, observe 4–6 h, repeat CXR**

Observation arm O2 ↑ pleural air absorption · repeat CXR before discharge + at **12–48 h**

Haemothorax & pulmonary embolism

Haemothorax **chest tube** first · repeat CXR at **4–6 h** and **24 h**

VATS / thoracotomy criterion	Threshold
Initial chest-tube output	**> 1500 cc**
Ongoing output	**> 200 cc/h for 3+ h**
Haemothorax size	Increasing
Despite **2 chest tubes**	Incomplete resolution

PE anticoagulation **3–6 months (minimum 3)** · IV heparin / Lovenox → oral

High-altitude pulmonary oedema (TO)

High-altitude illness (TO) **AMS, HACE, HAPE** · usual threshold **8000 ft (2500 m)** with rapid ascent

HAPE (TO) **noncardiogenic** oedema; lowlanders ascending rapidly to **> 2500–3000 m** · **exaggerated hypoxic pulmonary vasoconstriction**, ↑ PA pressure → capillary **"stress failure"** leak

Onset (TO) **2–5 days** after arrival · rare below 2500–3000 m or after **1 week** of acclimatisation

Symptoms (TO) dry cough, exertional dyspnoea, ↓ performance → rest dyspnoea, orthopnoea · **gurgling + pink frothy sputum = advanced** · cyanosis, tachycardia, temp ≤ **38.5 °C** · rales over middle lung fields

Prevention (TO) **nifedipine, tadalafil, dexamethasone, acetazolamide** (500–1000 mg/day 8–12-hourly or 125 mg BD, start **24 h before ascent**)

Treatment (TO) **supplemental oxygen = first line / first step** ✓ · **immediate descent = definitive** · portable hyperbaric chamber · nifedipine 30 mg SR 12-hourly
✗ Wrong options: descent by 1000 ft · hypoxia chamber · oral nifedipine first

Remote, no medical care (TO) **descent has first priority** — read the stem's setting

– – Active space – –

X vs Y

	EIA / asthma	EIB	EILO
Underlying asthma	**Yes**	**No**	No
Airway	Lower	**Lower**	**Upper (larynx)**
Phase	Expiratory	Expiratory	**Inspiratory**
Key symptom	Chest tightness, wheeze	Wheeze + SOB + cough	**Throat tightness, choking, stridor**
Spirometry	↓ FEV1, ↓ FEV1/FVC	**> 10% ↓ FEV1**	Blunted inspiratory loop
Bronchodilator	Improves	Improves	**No response**
Gold-standard test	PFT ± methacholine	**EVH**	**CEL**

	Simple pneumothorax	Tension pneumothorax	Haemothorax
Percussion	Hyperresonant	**Hypertympanic**	**Dull**
Trachea	Midline	**Deviated + JVD**	Midline
CXR	Pleural line	**Mediastinal shift, low hemidiaphragm**	Opacification
First step	O2 / observe or chest tube	**Needle decompression**	**Chest tube**

How it's asked

"Tension pneumothorax – ATLS next step?"	**→ standard 14-G 5 cm needle, 5th ICS mid-axillary line**
"Helmet to chest, deviated trachea, hypertympanic?"	**→ tension pneumothorax**
"Occasional β2-agonist for exertional wheeze – true?"	**→ FEV1 reduction on spirometry**
"Pink frothy sputum at altitude – first step (TO)?"	**→ oxygen**
"Throat tightness, stridor, inhaler fails?"	**→ EILO → continuous exercise laryngoscopy**
"Pneumothorax – return to play?"	**→ radiographic resolution, ~4 weeks**

– – Active space – –

QUADRICEPS CONTUSION

– Knee & Sports · 50 / 61 –

1 Basics

Epidemiology

Who **M:F 2:1** · **football, soccer, rugby** · competition > practice

Pathophysiology

Mechanism **direct blow** compresses anterior thigh muscles **against the femur**

Sequence small fibre tears → haemorrhage + swelling in **anterior compartment** → myonecrosis + **haematoma** → scar → muscle regeneration

Thigh compartment	Muscles
Anterior	Sartorius, rectus femoris, vastus lateralis / medialis / intermedius
Medial	Gracilis, adductor magnus / longus / brevis
Posterior	Biceps femoris, semitendinosus, semimembranosus

Muscle healing biology (immobilisation, losartan)

Prolonged immobilisation **↑ granulation tissue** (→ myositis ossificans) · scar contraction · poor organisation · **↓ tensile strength + ↓ tensile stiffness**
✗ Wrong options: ↑ stiffness · better tensile strength · ↓ MO

Brief immobilisation → mobilisation **parallel penetration of muscle fibres through (limited) scar**, in line with native muscle · ↑ myotubules, early nerve regeneration, ↓ inflammation
✗ Wrong options: lymphocyte / macrophage infiltrate · ↓ nerve fibrils · narrowed myotubes

Immobilisation alone **disorganised** fibre penetration through **dense** scar

Losartan (angiotensin II receptor blocker) **↑ muscle regeneration, ↓ fibrosis** (animal model) · ↓ apoptotic cascade · thought to act via **insulin-like growth factor** blockade (OB)
✗ Wrong options: aspirin · allopurinol · acetaminophen · bupropion

PRP + losartan ↑ angiogenesis + **follistatin**, **↓ phosphorylated Smad2/3**, ↓ fibrosis

3 Presentation

History collision / direct blow in competition

Pain **worsens over first 24–48 h** · worse with dynamic movement + **knee flexion**

Exam swelling, **ecchymosis, point tenderness** · compare firmness + circumference · palpable defect = partial / complete tear · ± knee effusion · **↓ knee flexion** · antalgic gait

Straight leg raise tests **extensor mechanism** · able (even with pain) → extensor mechanism intact

Distal NV exam screen for **thigh compartment syndrome**

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Use / finding
X-ray	Only for severe injury with **suspected fracture** · chronic → **myositis ossificans ("egg-shell" calcification)**
MRI	Concern for **extensor mechanism disruption** · best for soft tissue · **oedema / haematoma in anterior compartment**
CT for arterial extravasation	not part of initial work-up (no signs of active bleeding)

Jackson and Feagin classification

Grade (knee ROM at 24 h)	Knee flexion
Mild	**> 90°**
Moderate	**45–90°**
Severe	**< 45°**

5 Management

Quadriceps contusion

- All acute → 120° FLEXION 24–48 h + ICE + NSAIDs → **PT**
- Compartment syndrome → **THIGH FASCIOTOMY**
- Myositis ossificans → **OBSERVE**
  - painful, ↓ ROM / strength, MATURE (6–12 mo) → **RESECT**

Treatment – immobilisation in 120° flexion

Immobilise — **knee in 120° flexion** (ace wrap / hinged brace) **immediately**, **24–48 h** · cryotherapy, NSAIDs · start at once to **limit haematoma**



Immobilisation of the knee in 120° flexion

Then — stretching + active ROM, protected WB with crutches → sport-specific rehab once full pain-free ROM

17-yr, firm thigh, SLR with pain, NV intact — **cryotherapy, NSAIDs, hyperflexion splinting** (overnight → ↓ MO)
✗ Wrong options: CT for extravasation · emergent fasciotomy · **splint in extension** · MRI for extensor rupture

Knee immobilised in 120° flexion → injury? — **quadriceps contusion**
✗ Wrong options: iliac crest contusion (hip pointer leg in **extension**) · lesser trochanter avulsion · hamstring rupture · sports hernia

MRI next day — **haematoma in anterior compartment of thigh**
✗ Wrong options: meniscal, MCL, ACL, hamstring tears

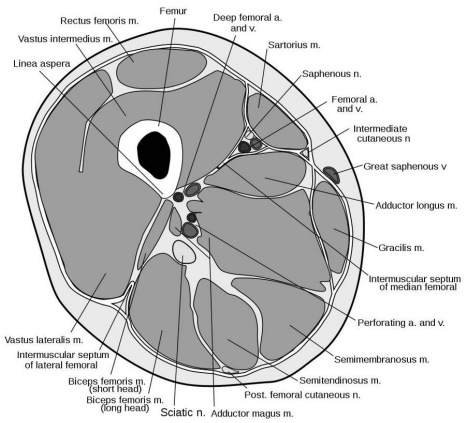
Study	Cohort	Result
Jackson	Midshipmen, **24 h hyperflexion**	RTP **3.5 days** · MO 1 / 23
Aronen	47 military, **120° for 24 h**	RTP **3.5 days**
Ryan	117 West Point cadets	Disability **mild 13 d, severe 21 d** · **MO 9%**

Disability time — **3–13 days** across studies · **13 days for mild** (Orthobullets) · time to return ∝ initial severity

Thigh fasciotomy

Indication — **compartment syndrome** (from rupture of **deep perforating branches of deep femoral vessels**)

Approach — **single anterolateral incision** → incise **fascia lata** (anterior) → retract **vastus lateralis medially** → incise **lateral intermuscular septum** (posterior) · ± **medial incision** for adductor compartment



Mid-thigh cross-section: anterior, medial and posterior compartments

6 Complications

Myositis ossificans

Incidence — **9–18%** (Orthobullets) · **5% and 9%** (OB study series) – read the options · more with **severe** contusions

Risk factors (Ryan) — **knee motion < 120°**, injury in football, **previous quadriceps injury**, **delay in treatment > 3 days**, **ipsilateral knee effusion**

Timing — develops **2–4 wk** after injury · serial X-rays → **"egg-shell" calcification**

Treatment — **observation** usually · **resect only mature lesions** (no growth, **6–12 months**) with pain + ↓ strength / ROM

– – Active space – –

Compartment syndrome	**thigh fasciotomy**
Prognosis	self-limited, but prolonged disability without treatment · early treatment speeds RTP

X vs Y

	Quadriceps contusion	Hip pointer
Site	Anterior thigh	Iliac crest
Position	**Knee flexed 120°**	**Leg in extension** (maximum stretch)
Chronic risk	**Myositis ossificans**	–

	Brief immobilisation → mobilisation	Prolonged immobilisation
Fibres	**Parallel** penetration, limited scar	**Disorganised**, dense scar
Tissue	↓ inflammation, early nerve regrowth	**↑ granulation tissue**
Mechanics	Better strength	**↓ tensile strength, ↓ stiffness**

How it's asked

"Immobilised in 120° knee flexion – diagnosis?"	**→ quadriceps contusion**
"Thigh contusion, SLR with pain, NV intact – next step?"	**→ cryotherapy, NSAIDs, hyperflexion splinting**
"MRI after flexion immobilisation shows?"	**→ haematoma in the anterior compartment of the thigh**
"Drug that improves muscle regeneration + ↓ fibrosis (animal)?"	**→ losartan**
"Prolonged immobilisation after contusion leads to?"	**→ increased granulation tissue production**
"Histology at 4 wk with early mobilisation?"	**→ parallel penetration of muscle fibre through scar tissue**

– – Active space – –

QUADRICEPS TENDON RUPTURE

– Knee & Sports · 51 / 61 –

1 Basics

Epidemiology & risk factors

Frequency	**more common than patellar tendon rupture** (2:1 in the patellar tendon rupture chapter)
Age	usually **> 40 yr** (Orthobullets, OB) · **patellar tendon rupture < 40 yr** (OB)
Sex + side	**M > F (up to 8:1)** · **nondominant limb > 2× as often**
Site	usually at the **insertion of the tendon on the patella**
Systemic risk factors	**renal failure**, **diabetes**, **rheumatoid arthritis**, hyperparathyroidism, connective tissue disorders
Local / drug risk factors	**steroid use** · **intra-articular injections (in 20–33%)** · previous knee surgery (OB)
Pathobiology (OB)	degeneration from **reduced blood supply** + **repetitive microtrauma** · **little evidence that inflammation is active** · high mechanical load + poor blood supply impede repair

Anatomy

Quadriceps tendon	**2 to 4 distinct layers** → matters for partial vs complete tear and for repair

2 Mechanism

Mechanism	**eccentric loading** of extensor mechanism · **foot planted, knee slightly bent**
Younger patients	usually **direct trauma**

3 Presentation

Diagnosis – patella baja & palpable defect

History	preceding pain from **underlying tendinopathy** · "pop" · large haematoma / effusion
Palpable defect	usually **within 2 cm of superior pole of patella**
Extensor function	**can't extend against resistance** · **no straight leg raise in complete rupture** · e.g. **extension lag 30°**
29-yr-old basketball, 30° lag, suprapatellar defect	**surgical repair of ruptured quadriceps tendon** ✗ Wrong options: MRI · diagnostic arthroscopy · immobiliser 6 wk then sport brace · limited WB 3 wk then PT

- - Active space - -

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
X-ray – AP + lateral knee	**Patella baja**
MRI	Only if **diagnosis uncertain** · **partial vs complete** tear



Patella baja after quadriceps tendon rupture (lateral radiograph)

Diagnostic criteria – Insall-Salvati index

ISI — **patellar tendon length ÷ patellar height** (superior → inferior pole) · tendon and patella ≈ equal, no variation > 20% (Insall + Salvati)

ISI	Meaning	Cause
< 0.8	**Patella baja**	**Quadriceps tendon rupture**
0.8–1.2	Normal	–
> 1.2	**Patella alta**	**Inferior pole patellar sleeve avulsion**, patellar tendon rupture

Worked example (OB) — patella **4 cm**, tendon **3 cm → ISI 0.75 = baja → quadriceps tendon rupture**

Distractors (OB) — ✗ Wrong options: OCD (no ISI change) · sleeve avulsion (alta, ISI > 1.2) · MPFL disruption (instability, abnormal **patellar tilt**) · nail-patella syndrome (**small or absent patellae**)

MRI ISI (Shabshin) — normal **0.79–1.52** · **females higher** ratios than males

Recurrent subluxation (Aglietti) — normal knees: ISI ~1, congruence angle −8, **Q angle 15°** · subluxating: **ISI 1.23**, congruence +16, **Q angle 20°**

– – Active space – –



Insall-Salvati index: tendon length ÷ patellar length

Classification

Type	Extensor mechanism	Treatment
Partial	**Intact**	Brace immobilisation
Complete	**Disrupted** (no SLR, baja)	**Primary repair to patella**

5 Management

Quadriceps tendon rupture

- Partial, extensor mechanism intact → **knee immobilisation in brace**
- Can't tolerate surgery → **knee immobilisation in brace**
- Complete, acute → **primary repair to patella (anchors or transosseous)**
- Chronic > 2 wk, retracted → **repair + V-Y lengthening (Codivilla) ± graft**

Operative criteria

Indication	Procedure
Complete rupture with loss of extensor mechanism	**Primary repair with reattachment to patella**
Chronic, retracted (**> 2 wk → up to 5 cm**)	Repair + **tendon lengthening (Codivilla V-Y)**
Tendon can't reach patella	**Auto- or allograft** tissue

Repair technique – transosseous vs suture anchors

Approach — **midline incision**

Sutures — **nonabsorbable**, **running locking** in tendon, free ends through **transpatellar drill holes** or tied to **suture anchors** · retinaculum → **heavy absorbable** sutures

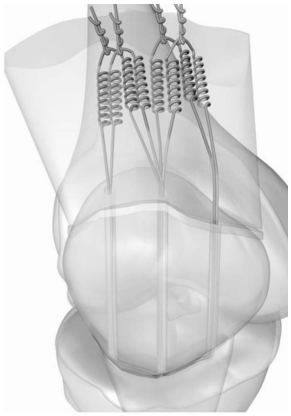
End-point — knee should **flex to 90° after repair**

Strength (OB) — ↑ strands across repair + ↑ suture calibre → ↑ strength but **wound complications**

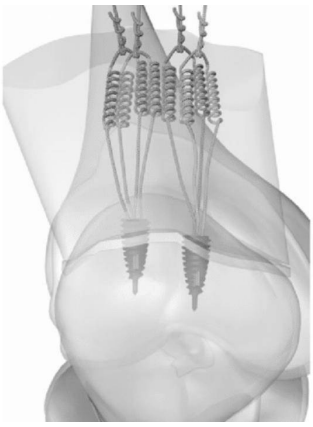
Biomechanics — **suture anchors → less gap formation + higher ultimate load to failure** (Orthobullets, OB)
✗ Wrong options: transosseous less gap · transosseous higher load

– – Active space – –

Petri (OB)	30 cadaver knees · **5.5-mm double-loaded anchors** vs transpatellar tunnels with No. 2 suture + **Krackow** → anchors less gap on cyclic loading, higher ultimate failure
Kindaya (OB)	**knotless suture-tape anchors** → less displacement, ↑ stiffness, ↑ load to failure vs transosseous **and** vs traditional anchors
Onggo (OB)	212 knees (114 quadriceps) → anchors smaller gap; higher load to failure in **2 of 5** studies
Conflict	anchors vs simple patellar holes **no apparent biomechanical advantage** (OB7683) vs anchors biomechanically superior (OB210177, OB219988, Orthobullets) — read the options
Clinical outcomes	**equivalent functional + clinical outcomes** ✓ (OB, Orthobullets) · possibly **better PROMs with anchors** (Orthobullets) vs **no PROM difference** (OB) — read the options
Mehta (OB)	transosseous: **higher ROM**, **lower complication rate** (not lower re-rupture) · **no Lysholm difference**
Transosseous vs anchors — true (OB)	**equivalent functional outcomes** ✗ Wrong options: less gap · higher load to failure · higher complication rate · worse PROMs



Transosseous patellar tunnel repair (A)



Suture anchor repair (B): less gap, higher load to failure

Postoperative care

Post-op	immobilisation in **brace, cast or splint** → progressive flexibility + strengthening

6 Complications

– – Active space – –

Complication	Fact
Strength deficit	**33–50%** of patients
Stiffness	Common
Functional impairment	**50% can't return to prior activity / sport**
Re-rupture	↑ with **pre-existing inflammatory arthropathy**

X vs Y

	Quadriceps tendon rupture	Patellar tendon rupture
Frequency	**More common**	Less common
Age	**> 40 yr**	**< 40 yr**
Patellar height	**Baja (ISI < 0.8)**	**Alta (ISI > 1.2)**
Defect	**Within 2 cm of superior pole**	Below inferior pole

	Suture anchors	Transosseous tunnels
Gap formation	**Less**	More
Ultimate load to failure	**Higher**	Lower
Complications (Mehta)	Higher	**Lower**
ROM (Mehta)	Lower	**Higher**
Functional outcome	**Equivalent**	**Equivalent**

How it's asked

"Patella 4 cm, tendon 3 cm – cause?"	→ **quadriceps tendon rupture** (ISI 0.75)
"Inferior pole sleeve avulsion – ISI?"	→ **> 1.2 (alta)**
"30° extension lag + suprapatellar defect – management?"	→ **surgical repair of ruptured quadriceps tendon**
"Anchors (B) vs transosseous (A)?"	→ **B: higher ultimate load to failure + less gap formation**
"Transosseous vs anchor repair – tell the patient?"	→ **equivalent functional outcomes**
"Which repair has lower complication rate (Mehta)?"	→ **transosseous**
"Chronic retracted rupture – added procedure?"	→ **Codivilla V-Y lengthening**
"Complication in 33–50%?"	→ **strength deficit**

– – Active space – –

QUADRICEPS TENDONITIS

– Knee & Sports · 52 / 61 –

1 Basics

Epidemiology

Who	**adult athletes** · **M:F 8:1** · often **recurrent**
Risk factor	**jumping sports**: basketball, volleyball, athletics (long jump, high jump)
Age contrast	quadriceps tendinopathy = **older adults** · patellar tendinopathy = adolescents / young adults (patellar tendinitis chapter)

Pathophysiology

Mechanism	**repetitive eccentric contractions** of extensor mechanism
Pathoanatomy	**microtears at bone-tendon interface** (superior pole of patella)
Quadriceps tendinosis	**chronic degeneration, no inflammation**
Jumper's knee	**patellar tendonitis** · usually **patellar tendon insertion at patella**, less often at tibial tubercle

Anatomy

Quadriceps tendon layer (trilaminar)	Muscle
Anterior (superficial)	Rectus femoris
Middle	Vastus medialis + vastus lateralis
Deep	Vastus intermedius

Blood supply	**medial, lateral + peripatellar arcades**
Nerve	**muscular branches of femoral nerve (L2, L3, L4)**

3 Presentation

History	overuse in a **jumping athlete** · **recent ↑ demand** / activity
Symptoms	pain at **superior border of patella** · worse with activity · swelling
Exam	**tender on deep palpation at quadriceps insertion** · **pain on resisted open-chain extension** · **can actively extend against gravity** ✓
Palpable gap	suggests **quadriceps tendon tear** (not tendonitis)
Also check	knee alignment, swelling, **patellar subluxation**

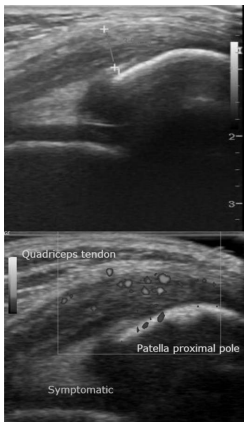
4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
X-ray — AP + lateral (± **sunrise / Merchant** for instability)	**Usually normal** · **tendon calcinosis** in chronic degeneration · varus / valgus alignment
X-ray — patellar height	Exclude rupture: **Blumensaat's line reaches inferior pole at 30° flexion** · **Insall-Salvati 0.8–1.2**
Ultrasound	Detects + localises disruption · **operator- and user-dependent** · tendinosis: **thicker, hypoechoic, power Doppler flow**
MRI	**Most sensitive** · **intrasubstance signal + tendon thickening**

Normal tendon width on US (OB) **2–3 cm** (figure shows 3.9 cm in tendinosis)

Imaging morphology (King, OB) localised thickening, **hypoechoic areas**, **↑ vascularity**



Quadriceps tendinosis on ultrasound: thick hypoechoic tendon with Doppler flow



Quadriceps tendinitis on sagittal MRI

– – Active space – –

5 Management

Quadriceps tendinopathy

- First line → rest, ice, NSAIDs, activity modification, PT (ROM → **eccentric)**
- Failed nonop → **PRP (leukocyte-rich) or polidocanol injection**
- Failed ≥ 3 months, severe → **arthroscopic / open debridement (very rare)**

Eccentric strengthening

Best rehab for overuse tendinopathy (TO)	**eccentric strengthening** (TO answer not given; eccentric = first line in OB) ✗ Wrong options: strict bed rest 1 month · plaster 1 month · steroid injection first line then strength training
Nonop	**mainstay** · rest until pain improves → PT **ROM first → eccentric exercises** · hamstring, quadriceps + core strengthening
Cortisone injection	**contraindicated → quadriceps tendon rupture**
CrossFit, 1 month pain, US tendinosis	**PT for eccentric strengthening** ✗ Wrong options: arthroscopic debridement · open debridement + repair (needs ≥ 3 months failed nonop) · leukocyte-poor PRP · jumper's knee strap
PRP	**leukocyte-rich** PRP effective for **patellar** tendinopathy · **not first line** ✗ Wrong options: leukocyte-poor
Dragoo RCT (OB)	23 patients, patellar tendinopathy · eccentric exercise + US-guided dry needling + **LR-PRP → faster recovery**, benefit **dissipates over time** · no failures at 26 wk
Injections after failed nonop (King, OB)	**PRP** + **sclerosing agents (polidocanol)** → good outcomes
Jumper's knee strap	**patellar tendon strap** → for **patellar** tendinopathy, not quadriceps tendinosis
Surgery	**quadriceps tendon debridement**, **very rarely** · only after **≥ 3 months** failed nonop (OB) · arthroscopic or open → good outcomes

Kinesiology taping

K-tape (TO)	"**kinesis**" = movement · **stretchy cotton / synthetic** tape mimicking **skin elasticity** + muscle extension capacity · invented by Japanese chiropractor **Kenzo Kase, 1970s**
vs classic rigid taping (TO)	classic = immobilisation / stability · K-tape → **joints stay mobile**
Stretch (TO)	**40–60% of original length** on average · up to **120–150%** by manufacturer
Mechanism (TO)	tape contracts → **lifts skin + subcutis off fascia** → space → ↑ blood + **lymph flow** (removes inflammatory factors) → ↓ pain + swelling · ↑ proprioception + muscle function · applied over **trigger points**
Wear (TO)	**3–5 days** · waterproof **acrylic adhesive** (avoid rubbing while bathing) · trim hair · **caution with anticoagulants** (skin haemorrhages)

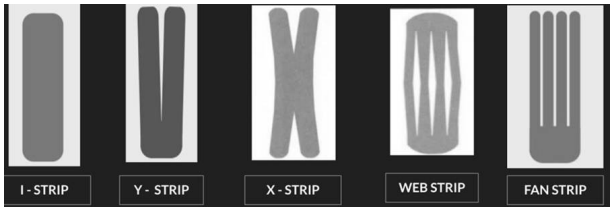
Pre-stretch (TO) manufactured on backing paper with **10% stretch** · still called "**unstretched**" · keep this pre-stretch on application

– – Active space – –

Method (TO)	Stretch applied to
Neutral technique ✓	**Tissue, not tape**
Stretch technique	**Tape, not tissue**

Application (TO)	Detail
Muscle	↑ or ↓ tone · ~**10%** stretch · **tonus decreasing = insertion → origin** · **tonus increasing = origin → insertion**
Ligament	Chronic tendinopathy, ligament overload, trigger points · **maximum stretch**, fixed to **both osseous insertions** → receptor stimulation
Corrective	Functional correction, e.g. **patellar maltracking** · **Y tapes**, maximum pre-stretch → shifts bone position
Lymphatic	**I tapes**, ~**25%** pre-stretch → skin lift → lymphatic drainage

Shapes (TO) **I, Y, X, web, fan** strips



Kinesiology tape shapes: I, Y, X, web, fan

Myths (TO) **"↑ performance" – false** (treatment only; **not banned** under anti-doping unless coated with banned substance; no active drug) · **colours differ – false** (cyan, magenta, beige, black – **same properties**; colour theory only: red stimulating, blue calming) · "can't heal without tape" – false

K-tape – true (TO) **neutral application stretches tissue, not tape** = best pick
✗ Wrong options: tonus decreasing origin → insertion · colours differ in stretch / hold · "40–60%" (only an average, varies up to 120–150%)

X vs Y

	Quadriceps tendinopathy	Patellar tendinopathy (jumper's knee)
Site	**Superior pole** of patella	**Inferior pole** of patella
Age	**Older adults**	Adolescents / young adults
Strap	No	**Patellar tendon (jumper's knee) strap**
Steroid injection	**Contraindicated** (rupture)	**Contraindicated** (rupture)

	Tendonitis / tendinosis	Quadriceps tendon rupture
Active extension against gravity	**Preserved**	Lost (no SLR)
Palpable gap	No	**Yes, within 2 cm of superior pole**
Patellar height	Normal	**Baja**

– – Active space – –

How it's asked

"Best rehabilitation for overuse tendinopathy (TO)?"	→ **eccentric strengthening**
"CrossFit athlete, US tendinosis, 1 month — treatment?"	→ **PT for eccentric strengthening exercises**
"Kinesiology taping — true statement (TO)?"	→ **neutral application stretches the tissue, not the tape**
"Tonus-decreasing K-tape direction (TO)?"	→ **insertion → origin**
"K-tape stretch capacity (TO)?"	→ **40–60% (up to 120–150%)**
"Do tape colours differ (TO)?"	→ **no — same properties**
"Most sensitive imaging?"	→ **MRI**
"Why avoid cortisone?"	→ **quadriceps tendon rupture**
"Deep layer of quadriceps tendon?"	→ **vastus intermedius**

RECTUS FEMORIS STRAIN

– Knee & Sports · 53 / 61 –

1 Basics

Who	**soccer + football** players
Anatomy	rectus femoris **crosses hip + knee → flexes hip, extends knee** · origin **AIIS (direct head)** → one of four quadriceps → **patellar tendon**
Site by chronicity	**acute → distal thigh** · **chronic → near origin**

Rectus femoris origin avulsion (AIIS)

Adolescents	proximal **bony avulsion of AIIS** at the **direct head** of rectus femoris (see AIIS avulsion chapter)
Track athlete, AIIS fragment above acetabulum	**rectus femoris** ✗ Wrong options: TFL · gluteus medius · gluteus minimus · biceps femoris



AIIS avulsion in an adolescent (arrow)

2 Mechanism

Mechanism	**sudden forceful eccentric contraction** · **sprinting from standing**, **kicking a ball hard**

3 Presentation

Pain site	Lesion
Anterior hip	Strain or **avulsion at AIIS**
Mid-thigh	Strain in **muscle fibres** (belly)
Distal (near knee)	**Most common finding**

Exam	tenderness over the injured zone · **palpable defect = full tear (unusual)**
Provocative test	**pain on resisted hip flexion or extension** (Orthobullets) · AIIS avulsion → **most pain on resisted knee extension** (OB) – read the options

– – Active space – –

11-yr-old hurdler, AIIS avulsion — most painful test (OB)	**resisted knee extension** (rectus contraction greatest) ✗ Wrong options: resisted adduction (adductors arise from **symphysis pubis**) · resisted abduction (**outer iliac crest**) · resisted hip extension (**posterior iliac crest**)

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
X-ray	**Adolescents with suspected AIIS avulsion** · otherwise **normal in most**
MRI	Strain extent · pseudotumour vs neoplasm
Ultrasound	Strain

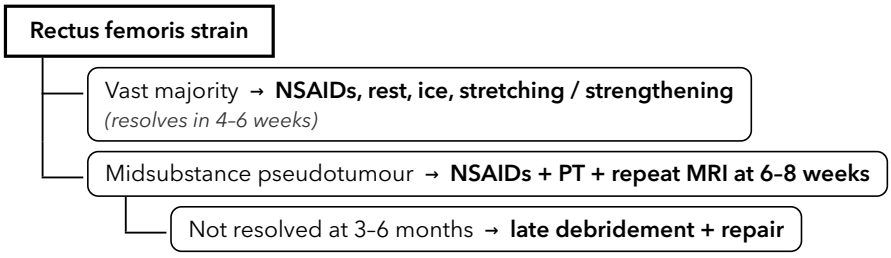


Rectus femoris strain on MRI (arrow)

Mid-substance strain / pseudotumour

14-yr-old, 3 wk, firm tender mid-rectus mass after soccer	**midsubstance partial rupture = "pseudotumour"**
MRI = injury, not tumour	**no mass effect on T1 · defined fluid signal on T2 · no significant gadolinium enhancement**

5 Management



Nonop	**definitive treatment for the vast majority · resolves within 4–6 wk**
Pseudotumour	**nonsteroidal anti-inflammatory drugs, physical therapy + repeat MRI scan in 6–8 weeks** to confirm resolution ✗ Wrong options: I&D of abscess · open biopsy · haematoma evacuation + musculotendinous repair · wide resection + radiotherapy

X vs Y

	Acute strain	Chronic strain	Adolescent AIIS avulsion
Site	**Distal thigh**	**Near origin**	**AIIS (direct head)**
Imaging	MRI / US	MRI / US	**X-ray**

How it's asked

"Teen track athlete, AIIS avulsion — muscle?"	→ **rectus femoris**
"Hurdler, AIIS avulsion — most painful test?"	→ **resisted knee extension**
"Firm mid-rectus mass, T2 fluid, no enhancement — management?"	→ **NSAIDs, PT, repeat MRI in 6–8 weeks**
"Most common pain site in rectus strain?"	→ **distal**

– – Active space – –

RIB STRESS FRACTURE

– Knee & Sports · 54 / 61 –

1 Basics

Incidence **uncommon** stress fracture site · elite > non-elite rowers · sweep (1 oar) = sculler (2 oars) (OB)

Site	Where	Who
First rib	**Anterolateral** · weak point = **subclavian artery groove**	**Baseball pitching**, basketball, weightlifting, ballet
Middle ribs 4–9	**Lateral + anterolateral**	**Competitive rowers**
Rowing (OB)	**Ribs 4–8 (or 5–9)**, anterolateral / lateral	Rowers
Middle ribs 4th–9th (TO)	–	**Rowers** – **serratus anterior** traction
Cough (OB)	**Anterior → posterior axillary line, ribs 5–9**	Chronic cough
Posteromedial ribs	–	**Novice golfers**

Risk factors **amenorrhoea**, osteopenia / osteoporosis, extreme overuse, **repetitive coughing** · OB: low BMD, **poor technique + ↑ training volume**, low calcium, low vitamin D, eating disorders, **low testosterone**, **depot medroxyprogesterone**

Rowing triggers (OB) **bigger blade** (↑ load per stroke) · ↑ volume incl. **long endurance rows at low stroke rate**

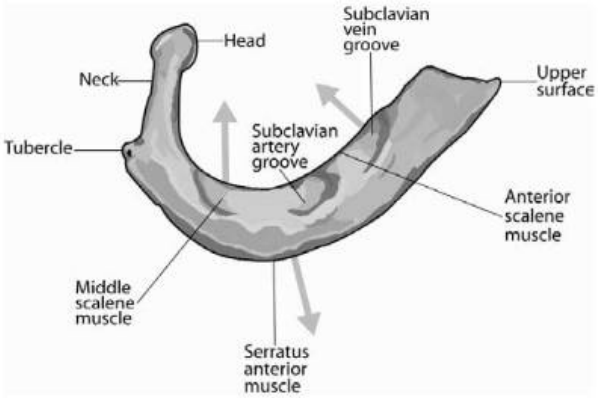
2 Mechanism

Repetitive contraction stress on rib from attached muscle · **muscles strengthen faster than bone** during training

Muscle fatigue ↓ ability to absorb + dissipate opposing forces

First rib **scalenes pull superiorly** vs **serratus anterior + intercostals pull inferiorly** → fails at subclavian groove

Rowers (OB, Karlson) deforming forces of **serratus anterior + external oblique** (attach ribs 5–9)



First rib stress fracture at subclavian groove

3 Presentation

– – Active space – –

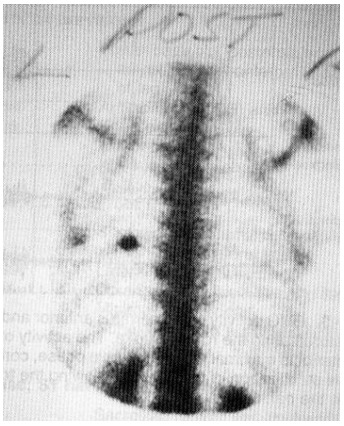
Symptoms	**insidious** pain · ↑ with **coughing, deep inspiration, overhead activity** · acute "**snap**" = completion of fatigued bone (throwing, batting, lifting)
Exam	**focal tenderness over the rib** · palpable **callus** when advanced

4 Diagnosis & Classification

Key radiology findings

Study	Use / finding
AP chest X-ray	**Negative in up to 60%** of rib fractures (any cause) · may show line or callus
Bone scan	X-ray negative + suspicion · ↑ activity · **rowing and cough fractures look identical**
CT	Suspected **pathologic** fracture · localises uptake at **costotransverse** region · clear fracture pattern
MRI	X-ray negative + suspicion · **no radiation** · **used more than bone scan in athletes** · marrow oedema ± line

Inspiration / expiration films for **pneumothorax**, which is **not associated**



Bone scan – rib stress fracture

5 Management

Nonop (majority, heal uneventfully)	**rest, analgesia, stop inciting activity**, correct training errors / mechanics
Duration conflict	**~4–6 weeks** (Orthobullets) · **1–2 weeks off rowing**, then **slow return** at low intensity, pain-free (OB) – read the options
Rower, cough-aggravated chest pain, hot rib on bone scan	**rest, analgesia, slow return to rowing** ✗ Wrong options: inspiration / expiration X-rays · operative stabilisation (only if recalcitrant) · scan differs from cough fracture · **more** protraction / retraction
Stroke modification (OB)	**less scapular protraction at catch + less retraction at finish** (less reach + pull-through) · proposed, **not clinically evaluated**

– – Active space – –

6 Complications

Non-union	the listed complication

How it's asked

"Olympic rower, bigger blade, rib uptake on bone scan – correct?"	**→ rest, analgesia, slow return to rowing**

SNAPPING HIP (COXA SALTANS)

– Knee & Sports · 55 / 61 –

1 Basics

Epidemiology

Aka (TO)	coxa saltans = **dancer's hip** · a **rare** disease (TO)
Who	athletes + dancers in **teens / twenties** · **ballet dancers** + athletes needing **repetitive extreme hip flexion + abduction** (TO)
Common in (TO)	**dancers** · applies to snapping hip generally **and** internal snapping ✗ Wrong options: gymnasts · long-distance runners · cyclists
Painless snapping	up to **10% of general population → normal** (OB)

Types & causes

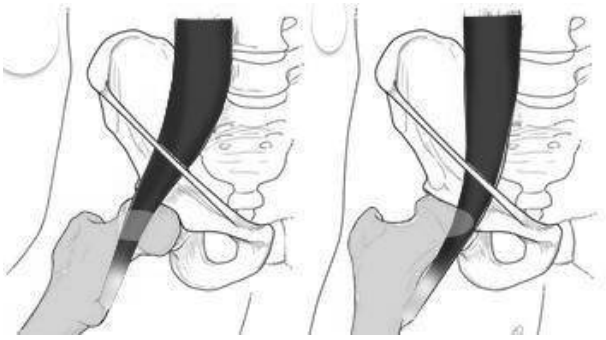
Type	Cause	Snap / pain
External – most common (Orthobullets, OB, TO)	**ITB over greater trochanter** · also **TFL or gluteus medius** over GT (OB)	**Lateral · visible**
Internal	**Iliopsoas** over **femoral head**, **iliopectineal ridge / eminence**, **lesser trochanter exostoses**, **iliopsoas bursa** · iliofemoral ligament (OB)	**Anterior · audible** click
Intra-articular	**Loose bodies** (traumatic, **synovial chondromatosis**), **labral tears**, osteochondral fragments (TO)	**Catching, locking**, clicking

Extra- vs intra-articular (TO)	**extra-articular more common** · divided into internal + external · **external more common than internal**
See vs hear	external snapping **seen** from across the room · internal snapping **heard** from across the room
Internal snapping cause (TO)	**thickened iliopsoas tendon snapping over iliopectineal eminence** ✗ Wrong options: labral tear · ITB over GT · FAI
Iliacus (OB)	may contribute as it overlies **anterior capsule + superior pubic rami**

Internal snapping (iliopsoas) – anatomy

Iliopsoas origin + insertion	**anterior surfaces + lower borders of L1–L5 transverse processes + iliac fossa → lesser trochanter** ✗ Wrong options: greater trochanter
Distractor structures (OB)	anterior → posterior acetabulum = **labrum** (tear → pain on **FADIR**) · **external lip of iliac crest → lateral tibial condyle (Gerdy's tubercle)** = **ITB** ✗ Wrong options: medial condyle
Snap mechanics	iliopsoas tendon **lateral to iliopectineal eminence in full flexion → snaps medially** as hip returns to neutral
Tendon bands (Philippon, OB)	single **28.3%**, **double 64.2%**, triple 7.5% · **psoas major most medial**, primary iliacus tendon lateral → **more than one tendon may need release**

– – Active space – –



Internal snapping: iliopsoas sliding over the femoral head

3 Presentation

Symptoms	**snapping** in / around hip · **painful or painless** · patient can **reproduce** it · worse with activity
Clicking / locking	suggests **intra-articular** pathology
Pain (TO)	tendon inflamed → snapping becomes painful · internal → **anterior** pain + audible click · external → **lateral** pain + click

External snapping (iliotibial band) – Ober test

Exam	palpate **greater trochanter** as hip **actively flexed** · **pressure stops the snapping** → confirms
Ober test	**limited hip adduction with hip extended** ✓ · lateral decubitus, **can't adduct across midline** ✗ Wrong options: adduction in flexion · abduction flexed / extended · bow-legged gait
Why	**TFL flexes + abducts** hip → contracture limits **extension + adduction**
Prone Ober (Gautam)	stabilises pelvis, holds extension → **quantifies** contracture



Ober test

Internal snapping – provocative manoeuvres

Source	Manoeuvre reproducing internal snap
Orthobullets	Passive **flexed + ER → extended + IR**
OB (Yen)	**Flexion-abduction-ER (FABER) → extension, adduction, IR**
OB (Frank, JAAOS 2013)	**Flexed + IR → extended + ER**
OB	Supine, **flex > 90° → extend to neutral**; accentuate with FABER in flexion, adduction + IR while extending
OB	**Flexion + extension**, or **flexion + ER + abduction**
TO	Rotate the **extended + adducted** hip IR / ER · **flex the extended hip** · **extend the flexed hip**

Manoeuvre conflict Orthobullets / Yen (**ER → IR** on extending) vs Frank (**IR → ER** on extending) – read the options; FABER → neutral is the commonest stem

Dancer, popping on FABER → neutral **internal snapping** · no pain on IR of flexed hip → **labral tear unlikely**

Released psoas tendon – pre-op finding **snapping with active hip flexion** ✓ · **FABER → neutral triggers popping** ✓

Finding (OB distractor)	Diagnosis
Positive Ober	**External** snapping (ITB / TFL)
Pain on IR + ER with hip + knee **extended** · IR only to neutral at 90° flexion, full ER	**Pincer FAI**
↓ IR, limited **FABER → FADIR**, positive impingement (flexion, adduction, IR)	**Cam FAI**
↓ active ROM, **catching, locking, grinding** on passive motion	**Intra-articular loose bodies**
Loose bodies on X-ray	**Synovial chondromatosis** → loose body removal
Labral tear on MRI	**Secondary to FAI** → repair vs debridement
Pain on half sit-up + tender pubic ramus	**Athletic pubalgia**

4 Diagnosis & Classification

Diagnosis **clinical** for external + internal (TO: mainly clinical in extra-articular) · **advanced imaging** usually needed for **intra-articular**

– – Active space – –

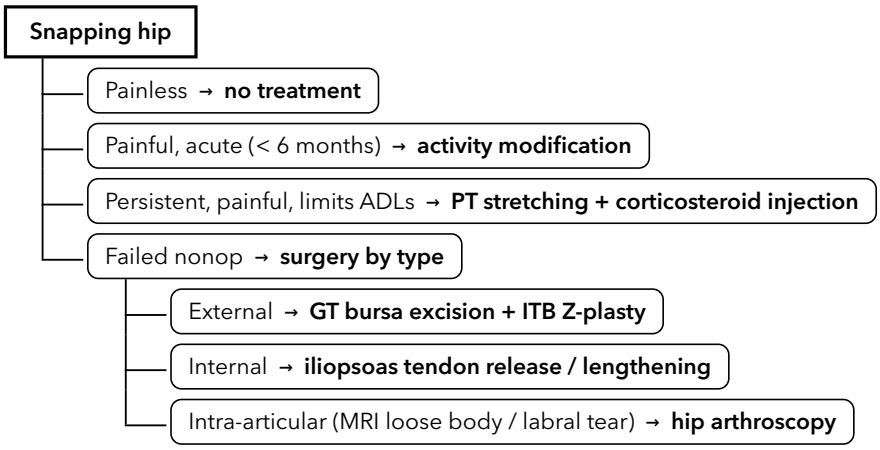
Key radiology findings

Study	Finding / use
X-ray – AP pelvis / hip	**Usually normal** · **rule out synovial chondromatosis** · occasional **calcification near lesser trochanter** (OB)
Dynamic ultrasound	**Best to confirm internal snapping** (shows tendon in motion) · shows snapping band (internal or external) · **guides diagnostic injection** (trochanteric bursa / iliopsoas sheath / joint)
MRI (often **arthrogram**)	**Rule out intra-articular** pathology + mimics · inflamed bursa · dynamic USG / MRI help **intra-articular** diagnosis (TO)
Iliopsoas bursography	Contrast into bursa under **fluoroscopy** → tendon seen · ± therapeutic injection · **radiation to pelvis**

Confirm internal snapping (OB ×2) — **ultrasound**
✗ Wrong options: conventional MRI · CT arthrogram · weight-bearing X-ray · fluoroscopy without contrast · bone scan · CT · plain radiographs

US anatomy (OB) — **hyperechoic oval iliopsoas tendon** anterior to **superior pubic ramus**, posterior to **hypoechoic iliacus**

5 Management



Nonop (TO) — **NSAIDs, ice, rest**, ultrasonic + interferential therapy, **stretching of iliopsoas + ITB → effective in majority** · **local anaesthetic + corticosteroid** into bursa / tendon sheath

Resistant (TO) — **surgical release + lengthening of iliopsoas tendon and ITB**

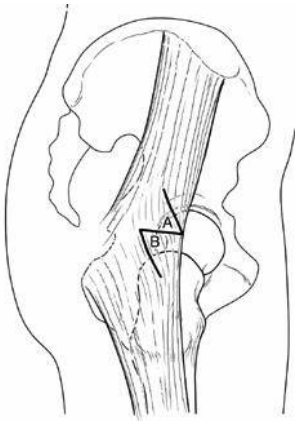
Intra-articular (TO) — **arthroscopic loose body removal + labral repair = treatment of choice**

Operative criteria

Indication	Procedure
Painful external snapping, failed nonop	**GT bursa excision + ITB Z-plasty**
Snapping after THR	**GT bursa excision + ITB Z-plasty**
Painful internal snapping, failed nonop	**Iliopsoas tendon release**
Intra-articular, failed nonop + **MRI loose bodies / labral tear**	**Hip arthroscopy** – loose body removal, labral debridement / repair

External snapping (iliotibial band) – Z-plasty

Mainstay	**stretching** (positive Ober = tight ITB) · often **PT alone** · occasional steroid injection for **profound GT bursitis**
Runner pattern (OB)	long symptom duration · **downward leg** (left leg running on left side of road)
Most reliable surgery	**Z-plasty of ITB** near **proximal origin** · may be **endoscopic** ✗ Wrong options: **release at Gerdy's tubercle — not described** · iliopsoas release · labral debridement · FAI debridement
Results (OB)	Z-plasty good in 10 patients (**Faraj**) · release good in 16 patients / 17 hips (**White**)



ITB Z-plasty lengthening

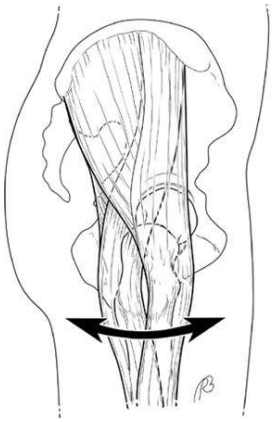
Internal snapping (iliopsoas) – release

Nonop success (Gruen, OB)	**63%** of 30 improve nonop · rest do well with **open lengthening**
Open approaches	anterior, medial, ilioinguinal, iliofemoral · **partial or complete** release · traditionally open → now **arthroscopic**
Arthroscopic routes	**endoscopic at lesser trochanter** · **transcapsular** at the hip joint: from **central compartment** (Orthobullets) · from **peripheral compartment** (OB) — read the options
Ilizaliturri RCT (OB)	**lesser trochanter (10) vs transcapsular (9)** → both ↑ WOMAC, **no difference** · both effective + reproducible
Regeneration (Marquez Arabia, OB)	27 patients · tendon **regenerates in all** by 23 months (**84%** of original circumference) · **all 5/5 hip flexion** · muscle fibres attach directly to proximal femoral shaft → no retraction
Post-op	**no hip flexion strengthening for 6 wk**

– – Active space – –

X vs Y

	External	Internal	Intra-articular
Structure	**ITB over GT**	**Iliopsoas over iliopectineal eminence / femoral head**	**Loose body, labral tear**
Frequency	**Most common**	Less	Least (extra > intra)
Site	**Lateral**, visible	**Anterior / groin**, audible	Deep, catching / locking
Key test	**Ober**, palpate GT during active flexion	**FABER → neutral (extension)**	FADIR (labrum), locking
Best imaging	Clinical	**Dynamic ultrasound**	**MRI arthrogram**
Surgery	**ITB Z-plasty** + GT bursa excision	**Iliopsoas release**	**Arthroscopy**



External snapping: ITB sliding over the greater trochanter

How it's asked

"Internal snapping hip common in (TO)?"	**→ dancers**
"Snapping hip syndrome common in (TO)?"	**→ dancers**
"Cause of internal snapping (TO)?"	**→ thickened iliopsoas tendon over iliopectineal eminence**
"Dancer, anterior popping FABER → neutral – confirm with?"	**→ ultrasound**
"Ballet dancer, painful popping on flexion + ER – best test?"	**→ ultrasonography**
"Pre-op finding before psoas tenotomy (MRI figure)?"	**→ FABER → neutral triggers popping**
"Psoas released arthroscopically – pre-op finding?"	**→ snapping with active hip flexion**
"Origin / insertion of structure in internal snapping?"	**→ L1–L5 transverse processes + iliac fossa → lesser trochanter**
"Positive Ober test?"	**→ limited hip adduction when extended**

"Runner, positive Ober, failed stretching – most reliable surgery?"	→ **Z-plasty of the iliotibial band**
"Painless lateral snapping, normal X-ray – structure?"	→ **iliotibial band**
"Most common type of snapping hip?"	→ **external (ITB)**
"Post-op restriction after iliopsoas release?"	→ **no hip flexion strengthening for 6 weeks**

– – Active space – –

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SPONTANEOUS OSTEONECROSIS OF THE KNEE (SONK)

– Knee & Sports · 56 / 61 –

1 Basics

Epidemiology

Who	**middle-aged + elderly females** · **women > 55 yr** (Orthobullets, OB) · **women > 60 yr** (OB5977) – read the options
Joints	**99% only one joint**
Site	**epiphysis of medial femoral condyle** (weight-bearing MFC) · also **lateral femoral condyle**, **patella** (OB) · **tibial plateau** (OB)
Pareek (OB)	223 patients with subchondral insufficiency fracture: **femur 69%**, tibia 55%, **medial compartment 89%** · > 5 yr: **47% surgery, 37% arthroplasty** (~$\frac{1}{3}$ **→ TKA**)
Ecker + Lotke (OB)	15 patients (16 knees), **medial tibial plateau** tenderness · first X-ray lucent in **9 / 16**, all 16 eventually · **TKA 9, UKA 3**, 2 refused, 2 resolved

Types of knee osteonecrosis

Type	Fact
Spontaneous (SONK)	**Most common** · idiopathic · MFC · elderly women
Secondary	Risk factors (**lupus, alcohol, sickle cell**) · more **diffuse** · see osteonecrosis chapter
Post-arthroscopic	**Rarest** · after **meniscectomy**, age **> 55 yr** (Di Caprio)

Pathophysiology

Core lesion	**subchondral insufficiency fracture** in poor-quality bone → **crescent-shaped** osteonecrosis
Cascade (OB)	**SIF of MFC → osteonecrosis + flattening → degenerative posterior horn medial meniscus tear**
Meniscus	**posterior horn medial meniscal tears**, may involve the **root** · **root injuries in up to 80%** of SONK (OB)
Root tear (Bhatia)	loss of **hoop stresses** → **extrusion** → ↑ tibiofemoral contact pressure ≈ **total meniscectomy**
Post-arthroscopic	mostly after **partial meniscectomy** → altered biomechanics + **loss of shock absorption** → insufficiency fracture → crescent osteonecrosis
Weight bearing	exacerbates · **treatment based on size + signal** (Zaremski)
Not AVN (TO)	previously thought vascular; now a **non-vascular subchondral insufficiency fracture** in osteopenic bone + **meniscal root tear** · **medial meniscus extrusion = hallmark** (> 50% on MRI)

3 Presentation

– – Active space – –

Onset — **sudden severe knee pain** · **worse at night** + with weight bearing

Exam — **effusion** · ROM limited by pain · **tenderness over MFC** / medial joint line · slight synovitis

Tibial plateau lesion (OB) — medial pain + tenderness, ROM normal, no deformity · **misdiagnosed as degenerative meniscus or OA**

Post-arthroscopic pattern — well at 2 wk after partial medial meniscectomy → **↑ medial pain at 6 wk** → MRI: **osteonecrosis**
✗ Wrong options: osteomyelitis (no infection signs) · advanced OA · inflammatory arthritis · osteochondral lesion

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
X-ray – **standing AP + lateral of hip, knee + ankle**	**Diagnostic in advanced disease** · **radiolucent subchondral lesion** · flattened MFC · **osteopenic area** in medial tibial plateau on lateral · early films may be normal
MRI	**Most useful study** · confirms + **sizes** lesion → guides treatment
MRI – lesion	**Crescent-shaped** · **subchondral hypointense crescent line**, cortical irregularity
MRI – signal	**T1 marrow hypointensity** + **T2 hyperintensity (oedema)** · **serpiginous border** · localised low-signal area + surrounding oedema

MRI timing (TO) — **subchondral fracture line + extensive marrow oedema** · diagnosis possible **within 48 h** of pain onset

Aglietti stage (TO)	X-ray
I	Normal
II	Subtle **flattening** of condyle
III	Subchondral **lucency**, no collapse
IV	**Subchondral collapse**

Condylar ratio (TO) — lesion width ÷ condylar width × 100 · **> 40% → poor prognosis**, rapid OA

– – Active space – –



SONK of the medial femoral condyle (radiograph)



Subchondral crescent line + marrow oedema of the MFC (MRI)

Diagnosis – MRI (osteonecrosis vs bone bruise / insufficiency fracture)

74-yr-old woman, acute medial pain, no trauma	**osteonecrosis** ✗ Wrong options: OA · RA · medial meniscal tear · transient osteoporosis
52-yr-old man, sudden atraumatic pain, worse at night, tibial plateau lesion	**osteonecrosis** ✗ Wrong options: CRPS (**MRI normal**) · OA (both MFC + MTP involved) · osteosarcoma (bone-forming) · inflammatory arthritis (whole knee)
28-yr-old skier, MRI after injury	**bone bruise** · **↑ T2 = haemorrhage from trabecular microfractures** · common after **ACL + MCL** injury ✗ Wrong options: osteosarcoma · bucket-handle tear · LCL tear · tibial spine avulsion

Concurrent injury with SONK (OB)	Likely?
Horizontal tear of posterior horn medial meniscus	✓ **Most likely**
Complex lateral meniscus tear	No (medial predilection)
PCL tear	No
Patellofemoral arthritis	No
LCL avulsion (lateral epicondyle)	No

Differential diagnosis

Condition	Who / key fact
SONK	**Elderly women**, **MFC (weight-bearing)**, 99% one joint
Osteochondritis dissecans	**Adolescent / young males**, **lateral aspect of MFC**
Transient osteoporosis	**Young–middle-aged men**
Bone bruise / occult fracture	**Trauma**, bone fragility, overuse
Secondary osteonecrosis	Risk factors, multifocal

Transient osteoporosis / BMES (TO) middle-aged men or **3rd-trimester pregnancy** · **hip > knee** · **diffuse marrow oedema, no fracture line** · resolves **2–3 months** · RSD variant? · Rx **NWB 6–8 wk + bisphosphonates ± calcitonin** · **pregnancy → rest only (bisphosphonates contraindicated)**

5 Management

SONK

- Most / pre-collapse / small (< 3.5 cm) → **NSAIDs, protected WB, quads PT, unloader brace**
- Failed nonop, unicompartmental → **UKA**
- Failed nonop, large (> 5 cm) / collapse / diffuse → **TKA**
- Younger with angular malalignment → **HTO**

Nonop **mainstay** (most resolve, **self-limiting**) · **NSAIDs, narcotics, protected WB** · **PT for quadriceps strengthening** · **unloader brace**, sometimes **bisphosphonates** (OB) · good results

Post-arthroscopic most nonop · **protected WB 4–8 wk** (Di Caprio)

SONK (TO, Mohindra)	Treatment
Stage I–II / ratio < 40%	**Restricted WB 6–12 wk + bisphosphonates**
Stage III–IV / ratio > 40%	**Subchondroplasty**, OATS, HTO or UKA
Stage IV, isolated medial	**UKA** – opposite compartment intact, **no flexion–varus deformity**, secondary causes excluded · **bisphosphonates 3 months pre-op**

Subchondroplasty (TO) image-guided **calcium phosphate (AccuFill)** injection → internal bone-graft substitute + scaffold

Timing (OB) most patients seen **post-collapse → arthroplasty** treatment of choice · pre-collapse treatment **controversial**

– – Active space – –

Operative criteria

Indication	Procedure
Failed nonop, **small lesion**, one compartment	**UKA**
Failed nonop, **large lesion / bone collapse**, diffuse / multicompartment	**TKA**
Angular malalignment (younger) · offload an osteochondral allograft	**HTO**
Younger / active (OB)	Microfracture, **osteochondral allograft**, HTO
Lesion size (OB)	**< 3.5 cm → nonop · > 5 cm + failed nonop → arthroplasty**

59-yr-old librarian, MFC lesion, failed nonop
UKA
✗ Wrong options: open biopsy (no matrix / periosteal reaction) · chest CT + bone scan (subchondral condyle not a metastatic site) · OATS (defect size + age) · valgus HTO (59 yr, no varus)

58-yr-old, antiphospholipid, crescentic collapse, failed PT / NSAIDs / PRP
arthroplasty
✗ Wrong options: core decompression (can't treat full-thickness collapse) · HTO (younger + angular deformity) · medial root repair (doesn't fix cartilage loss) · stem cells (not FDA approved)

Treatment – unicompartmental / total arthroplasty & outcomes

UKA for SONK – true
reliable long-term outcomes · **treatment of choice for severe localised disease** · TKA **rarely needed** as primary

UKA traps
✗ Wrong options: only for spontaneous (works for **secondary** localised too) · higher failure than OA (**comparable**) · tibial stress fracture commonest failure · TKA needed because UKA fails

UKA series (OB)	Result
Myers (systematic review)	TKA / UKA for SONK ≈ for OA (correct indications)
Parratte	Pain relief **96%** (≥ 3 yr) · 12-yr survivorship **96.7%** · spontaneous + secondary
Heyse (52)	**97.3%** satisfied · survival **93.1% at 10 yr**, **90.6% at 15 yr**
Bruni (84)	10-yr survival **89%** · revisions: **tibial subsidence 4**, aseptic loosening 3 · **no lateral progression**
Kaneko (61)	10-yr implant survival **90%**
Chalmers (46)	**Secondary ON less durable** than SONK · failure from **arthritis progression**

Commonest reason for revision after UKA
progression of arthritis (OB213949) · **aseptic loosening** most often (OB213790) · loosening + progression both (OB213790 answers) · subsidence + loosening, no progression (Bruni) – read the options
✗ Wrong options: aseptic loosening · bearing dislocation · implant failure · periprosthetic fracture

UKA contraindications (OB)
inflammatory arthropathy, **fixed deformity > 5–10°**, **motion < 90°**, **previous meniscectomy**, symptomatic other compartment

Bone marrow oedema adjuncts

Persistent BME (Akhavan)	prognostic for pain, dysfunction, cartilage damage · surgery: **core decompression**, bone-graft injections, arthroplasty
Zoledronic acid (Flores-Robles)	primary BMES (ankle > hip) · **75% complete response** at 12 months
Subchondroplasty (Cohen)	fluoroscopic **calcium phosphate** injection into bone marrow lesions · ↑ VAS + IKDC to 2 yr

X vs Y

	SONK	Secondary osteonecrosis
Who	**Women > 55 yr**	Risk factors (steroids, lupus, alcohol, sickle cell)
Extent	**One joint (99%)**, MFC epiphysis	Multiple compartments, **80% bilateral** (osteonecrosis chapter)
Cause	**Subchondral insufficiency fracture** ± root tear	↓ blood flow
UKA durability	**Better**	Less durable (Chalmers)

	SONK	OCD	Bone bruise
Patient	**Elderly woman**	**Adolescent male**	Young, trauma
Site	**Weight-bearing MFC**	**Lateral aspect of MFC**	After ACL / MCL injury
MRI	**Crescent**, T1 dark + T2 bright, serpiginous border	Osteochondral fragment	**↑ T2**, trabecular microfractures

How it's asked

"74-yr-old woman, acute medial pain – MRI diagnosis?"	**→ osteonecrosis**
"Skier, MRI after injury?"	**→ bone bruise**
"52-yr-old, sudden atraumatic pain, worse at night – MRI?"	**→ osteonecrosis**
"Medial pain 6 wk after partial meniscectomy, crescent ↑ T2?"	**→ osteonecrosis (post-arthroscopic)**
"Lupus, SONK on MRI – concurrent injury?"	**→ posterior horn medial meniscus tear (Injury C)**
"59-yr-old, MFC lesion, failed nonop – best treatment?"	**→ unicompartmental knee arthroplasty**
"UKA for SONK – true?"	**→ reliable long-term outcomes**
"Commonest reason for revision after UKA for SONK?"	**→ progression of arthritis**
"Crescentic collapse, failed PRP / NSAIDs – treatment?"	**→ arthroplasty**
"OCD vs SONK – site?"	**→ lateral aspect of MFC (OCD) vs weight-bearing MFC (SONK)**

– – Active space – –

"Hallmark MRI finding with SONK (TO)?"	**→ medial meniscus extrusion**
"SONK condylar ratio predicting poor prognosis (TO)?"	**→ > 40%**

– – Active space – –

STEROIDS & STIMULANTS

– Knee & Sports · 57 / 61 –

1 Basics

Anabolic steroids – mechanism & adverse effects (lipids, aggression, alopecia, surgical risk)

What they are	**synthetic derivatives of testosterone** (anabolic-androgenic steroids, AAS) · oral or injection · mimic testosterone + dihydrotestosterone → ↑ protein synthesis
Primary mechanism of ↑ muscle	**↑ production of messenger RNA** ✗ Wrong options: ↑ cortisol production · VEGF · ↓ HDL (HDL falls but doesn't affect muscle)
Other actions	**stimulate GH release** · **reverse / attenuate cortisol (catabolic hormones)** → ↓ catabolism · more intense + sustained workouts · ↑ strength, ↓ recovery time · used to **gain muscle + lose fat** · ↑ erythropoiesis ✗ Wrong options: ↑ fat

– – Active space – –

Examples **methandrostenolone (Dianabol)**: first made 1960, most used + abused steroid worldwide · **stanozolol**: Winstrol (oral) / Winstrol Depot (IM), derived from **dihydrotestosterone** · **metandienone**, **DHMCT** (TO doping cases)

AAS adverse effect (cardiometabolic)	Fact
HDL	**↓ HDL (20–70%)** + ↓ **apolipoprotein A1** (main HDL particle protein) ✓ (✗ Wrong options: ↑ HDL)
LDL	**↑ LDL > 20%** (OB) · **↑ or no effect** on LDL (OB) · ↑ LDL, hypercholesterolaemia (Orthobullets)
Coronary risk	Lipid changes → **CAD risk 3–6×** · premature CV events
MI in young athlete	**↑ LDL** (+ ↓ HDL) ✓ · ↑ O2 demand at peak exercise + accelerated atherosclerosis (✗ Wrong options: direct endothelial damage · ↓ BP · conduction change)
Blood pressure	**↑ systolic + diastolic** (hypertension) (✗ Wrong options: ↓ BP)
Heart	**LVH**, impaired diastolic filling, thrombosis
Ventricular arrhythmia	Cause of **sudden death** (✗ Wrong options: cause of CAD) · fatal cases in small case series
Coagulation	**↑ factor VII, IX + thromboxane** → hypercoagulable → **↓ bleeding time** (✗ Wrong options: ↑ bleeding time)
Liver	**Liver tumours** (Orthobullets) · **↑ liver enzymes** (OB) · oral AAS **induce hepatic enzymes** → less sensitive to anaesthetics (✗ Wrong options: impaired liver function)
Fluid	**Mineralocorticoid effect** + **diuretics used to mask** it → **fluid + electrolyte imbalance** ✓
Ventilation	Large muscle mass + high calories → ↑ O2 use + CO2 → **↑ ventilatory requirement** (✗ Wrong options: lowered O2 requirement)
Spleen	**No effect** (✗ Wrong options: splenomegaly)
Immune	Immunosuppression (OB)

– – Active space – –

AAS adverse effect (endocrine, skin, MSK, psych)	Fact
Gonadal	**Testicular atrophy (40–51%)** · ↓ gonadotropins + sex hormones (oestrogen, testosterone) → **↓ BMD**, **oligo- / azoospermia (↓ sperm count)** · uterine atrophy
Gynaecomastia	**10–34%**
Skin	**Acne (40–54%)** · cutaneous striae (34%) · injection-site pain (36%) · **alopecia / male-pattern baldness**
Female	**Irreversible deepening of voice** (Orthobullets)
Child	**Growth retardation**
MSK	**Tendon + muscle injury / rupture** (e.g. **pectoralis major**) · osteonecrosis of hip (OB) · ↑ body weight
Behaviour	**Aggression, hostility, violence = most common** ✓ · psychosis, depression, mania **rare** · suicidal behaviour · **no drug dependence** (✗ Wrong options: like narcotics)
Not linked	**Decreased vision** ✓ (methandrostenolone EXCEPT question)

Reversibility
alopecia = only nonreversible effect (OB) · Orthobullets also lists **irreversible female voice deepening** – read the options · on stopping: hypertrophy + training gains lost quickly, acne + personality effects reversible

Lipids after stopping
HDL / LDL ratios return to pre-existing levels (OB) · **changes persist after withdrawal** (Hartgens, OB) – read the options

Surgery + anaesthesia in AAS user
fluid and electrolyte imbalance ✓
✗ Wrong options: ↑ bleeding time (actually ↓) · impaired liver function · lowered O2 requirement · splenomegaly



Gynaecomastia in anabolic steroid user

– – Active space – –

Other hormones – HGH, androstenedione, DHEA, EPO, somatotropin

Agent	Source / mechanism	Effect	Adverse
Human growth hormone (HGH)	**Most abundant pituitary product** · direct anabolic (amino acids → protein) · **popular AAS substitute** · IGF-1 similar	**↑ muscle size, not strength**	**Myopathy, carpal tunnel syndrome, insulin resistance** (Orthobullets) · **acromegaly**, heart disease, cardiac failure, glucose intolerance (OB)
Androstenedione	Androgen from **adrenal glands + gonads**	Potent anabolic · **liver → testosterone**	As testosterone
DHEA	Natural, **adrenal cortex**	→ androstenedione → testosterone	**= testosterone** effects
Erythropoietin (EPO)	Stimulates Hb / RBC production · doping agent	**↑ O2-carrying capacity**	**↑ blood viscosity → stroke / MI** · ischaemic + thromboembolic events, HTN, flu-like symptoms, **hyperkalaemia** (OB)
Somatotropin	Growth hormone	**Type 1 hypertrophy**, type 2 atrophy	–
Glucagon	Fasting hormone	**Catabolic** to skeletal muscle	–

Glucagon

mobilises + uses stored nutrients to maintain fasting glucose → **catabolic** ✓ · acts **mainly on liver** (insulin: liver, adipose, muscle)
✗ Wrong options: anabolic · no effect · nootropic (cognition / memory) · ergogenic (any external influence enhancing performance)

Hormonal & metabolic effects (glucagon, endurance adaptations)

Endurance training adaptations

↑ storage + utilisation of intramuscular lipids ✓ → ↓ CHO contribution, **preserves muscle glycogen** · **type I** hypertrophy (✗ Wrong options: type II) · **↑ Krebs cycle enzymes** (✗ Wrong options: ↓) · **↓ rate of glycogen depletion** (depletion = fatigue) (✗ Wrong options: ↑) · **blunted catecholamine response** → ↓ HR at same intensity (✗ Wrong options: ↑ sympathetic activity)

Creatine – side effects

What it is

from **glycine, arginine + methionine** · **supplement, not an anabolic steroid** · → **phosphocreatine → ATP source** (phosphate storage substrate) · off-season weightlifting

Performance

mixed results (Orthobullets) · ↑ work in **anaerobic** trials but **cannot ↑ peak force** (Orthobullets) · ↑ strength accrual + **↑ force / power in short maximal bouts** (Terjung, OB) – read the options · mild ↑ body composition + strength, limited proof of performance / recovery (OB) · greatest effect in **loading cycle**; not running / swimming (Branch, OB) · does not reduce muscle-damage markers (Rawson, OB)

For endurance (TO)

not primarily endurance · benefits **short bursts of power, high-intensity short-duration**

Mechanism of dehydration

intracellular creatine **pulls water from intravascular → intracellular space** → **dehydration** ✓

Side effects	**dehydration**, **muscle cramping** (✓ more common than with testosterone), water retention, nausea · strains, stiffness (Terjung) · ↑ muscle injury · **renal insufficiency (rare)** ✗ Wrong options: tendon tears · ↓ HDL · hair loss, testicular atrophy, acne (testosterone)
High-school football (McGuine, OB)	**30%** use creatine · perceived benefit = faster recovery · biggest risk = **dehydration**

Stimulants – caffeine, ephedra, amphetamines

Class	**caffeine**, **ephedra**, **ephedrine**, **"ma huang"** (energy drinks), **amphetamines**, methylphenidate (TO)
Stimulant risks	**dehydration**, **impaired heat management**, **high blood pressure**, nervous system impairment
Caffeine dose	**2–3 mg/kg** improves performance · ↓ fatigue + ↑ alertness · stimulates CNS, **↓ perceived exertion** · ↑ endurance, reaction time, alertness · **↑ fat oxidation** (TO) ✗ Wrong options: **"no proven effect"** (TO)
Caffeine status	previously banned by IOC, **now allowed up to 12 µg/ml urine** (Orthobullets) · removed from WADA list **January 2004** (Exercise Science chapter) · **2018 WADA Monitoring Program, not prohibited** (TO) – read the options
Caffeine adverse	**↑ blood pressure** (OB)
Caffeine ban history (TO, PED lecture)	"IOC banned 1962 → removed 1972" (TO – verify: caffeine was a restricted IOC substance 1984–2003, removed from WADA list **Jan 2004**) – read the options
Ephedra	**cardiac arrhythmias (can be fatal)** · psychiatric, autonomic, GI symptoms, palpitations (OB) · "apparently effective" + weight-loss supplement (TO)
Other drugs (OB distractors)	**furosemide = most commonly used banned diuretic** · **α-ketoglutarate** = supplement claimed to **remove excess ammonia** · **propranolol** → ↓ anxiety → steadier hands, even HR, focus

Sports nutrition & supplements (protein, amino acids, carnosine, hydration) (TO)

Protein requirement (TO)	**g/kg/day**
RDA, healthy individual	**0.8** · resting 0.8–1
Endurance athlete	**1** (range 1.2–1.7)
High-performance / top elite	**1.6 = maximum required** ✓
Healthy athletes, heavy workers	1.2–1.6 · nitrogen-balance assay avoids excess
Physically active (strength or endurance)	1.2–2.0 (0.5–0.9 g/lb)

Leucine (TO)	**BCAA oxidised in endurance exercise** ✓ · enzyme **branched-chain oxo-acid dehydrogenase (BCOAD)**: ~4–7% active at rest → **~25%** at exercise onset by **dephosphorylation** · amino acid oxidation = only **1–6%** of endurance energy cost ✗ Wrong options: arginine, lysine, isoleucine
Amino acid metabolism (TO)	**liver** (> kidney) principal site · excess can't be stored / excreted → oxidised or → CHO + lipid · α-amino group removed → carbon skeleton → **pyruvate, acetyl-CoA, TCA intermediates**

– – Active space – –

Vitamin K (TO)

bone mineralisation; deficiency → stress fracture ✓ · with 25(OH)D + calcium regulates bone resorption / activation / distribution · **carboxylates osteocalcin** (mature bone cells, regulated by 25(OH)D) → binds Ca → mineralisation, ↓ bone loss, possibly ↓ fractures · deficiency → **non-carboxylated (inactive) osteocalcin** → Ca release from bone + **soft-tissue / arterial calcification** · 25(OH)D toxicity possible without vitamin K
✗ Wrong options: B, C, A

Carnosine (TO)

dipeptide of **beta-alanine + L-histidine** ✓ · discovered **1900, Gulevich (Kharkov)** · **vitamin B6** essential precursor · high in brain, skeletal + cardiac muscle, GI · **antioxidant**, anti-ageing, glycaemic control · **animal origin** → vegan deficiency · **nutritional supplement, not doping** · ↑ fatigue recovery · may prevent diabetes, atherosclerosis, cognitive impairment · loading benefits **high-intensity exercise** · side effects + mechanism need research
✗ Wrong options: glutamine / glutathione / alanine · arginine · methionine + cysteine

Dietary supplement (TO)

taken **by mouth**, contains "dietary ingredient" (vitamins, minerals, herbs / botanicals, amino acids, enzymes, organ tissues, glandulars, metabolites, extracts) · tablets, capsules, softgels, liquids, powders, bars · **legal – regulated as food, not drugs** · safe at recommended doses; excess may harm

Effectiveness (TO; classification debated)	Supplements
Apparently effective	**Creatine, ephedra, sodium bicarbonate, beta-alanine**
Possibly effective	**BCAAs**, green tea extract, calcium, hydroxymethyl butyrate (HMB)
Apparently ineffective	**Glutamine, carnitine**

Purpose (TO)	Supplements
Muscle building	Creatine, essential amino acids, HMB, glutamine
Weight loss	Green tea extract, carnitine, ephedra, caffeine
Performance	Creatine, sodium bicarbonate, caffeine, beta-alanine, glutamine
Recovery	Antioxidants

– – Active space – –

Supplement (TO)	Function	Performance effect	Source
Creatine	↑ ATP for short bursts	**High-intensity, short-duration**	Meat, fish
Caffeine	CNS stimulant, ↓ perceived exertion, ↑ **fat oxidation**	**Endurance**, reaction time, alertness	Coffee, tea, energy drinks
Sodium bicarbonate	**Buffers lactic acid** (✗ Wrong options: acidifier)	Anaerobic high-intensity · delays fatigue	Baking soda
BCAAs	**Essential amino acids** · muscle protein synthesis ✓	↓ soreness, **delays central fatigue** (compete with **tryptophan** uptake into brain)	Meat, dairy
Glutamine	Immune function, muscle repair	Recovery; small direct effect	Meat, dairy, eggs
Carnitine	Fat oxidation	Limited / inconsistent benefit	Red meat, dairy, avocado
Antioxidants (vit C, E, polyphenols)	↓ oxidative stress	Recovery · **excess / pre-workout impairs adaptation** (blunts mitochondrial biogenesis)	Fruit, vegetables, nuts

Endurance strategy (TO) **caffeine pre-workout + sodium bicarbonate pre-workout + BCAAs intra-workout** ✓
✗ Wrong options: creatine (power only) · carnitine (inconsistent) · pre-workout antioxidants

Sports drink (TO)	Carbohydrate	Purpose	Best for
Hypotonic	**< 6%**	Rapid hydration, minimal energy	**< 60 min**
Isotonic	**6–8%** (6–8 g / 100 ml)	Hydration + energy	**Endurance + team sports**
Hypertonic	**> 8%**	Energy, **slower absorption**	**Post-exercise recovery**

Isotonic drink (TO) Na+, K+, Mg2+, Cl− · osmolarity **~300 mOsm/kg (≈ plasma)** → fast absorption · Gatorade, Powerade, Lucozade Sport · homemade: **500 ml water + 30–40 g sugar + ¼ tsp salt + 200 ml orange juice** (K+) · for **> 60 min**, marathon, cycling, soccer, basketball, **hot + humid** · ↓ cramps

Soccer 90 min, hot + humid (TO) **isotonic sports drink, 6–8% CHO + electrolytes** ✓ · Exercise Science chapter: **low-osmolarity (< 10%)** CHO-electrolyte fluid
✗ Wrong options: water only · hypertonic · skip until after

Ergogenic aids & supplement pharmacology (TO)

Ergogenic aid (TO) any substance, treatment or technique that ↑ work performance · 5 types: **psychological** (hypnosis, imagery, placebo), **mechanical** (swimsuits, clap skates, compression), **pharmacological**, **physiological** (blood / gene doping), **nutritional** · placebo (saline "steroids") ↑ lean mass

– – Active space – –

Supplement regulation (TO)

DSHEA 1994: no proof of safety / efficacy needed before marketing; FDA acts only after adverse events · choose **USP seal** / **NSF Certified for Sport** · best reason to take supplements = **aid recovery** · prohibited: androstenedione, Chinese ephedra

Supplement (TO, PED lectures)	Key facts
Caffeine	**Adenosine receptor antagonist** → ↑ catecholamines, FFA mobilisation (glycogen sparing), SR Ca^{2+} · ergogenic ~**330 mg** (≈ 3–6 mg/kg), 2–3 h before · old IOC limit **12 µg/ml** (≈ 700 mg / 70 kg) · **> 7 mg/kg impairs hydration in heat** · > 9 mg/kg "penalisable" (TO) · TO MCQ: endurance via **↑ plasma K^+**
Creatine	Glycine + arginine + methionine (AGAT, GAMT) · **CREAT1** transporter · PCr + ADP + H^+ → ATP (CK) · **load 20 g/day (4 × 5 g) 5–7 days → 3–5 g/day ≥ 12 wk** · gains after **~12 wk with resistance training** · check urine + renal function first · **NCAA forbids teams providing it** (not WADA-banned)
Sodium bicarbonate	Buffers H^+ · **0.3 g/kg 1–2 h pre-event** · main side effect **GI upset / diarrhoea**
Beta-alanine	**Carnosine precursor** (rate-limiting) · **1.6–6.4 g/day**, sustained release · **paraesthesia > 800 mg per dose** · carnosine itself useless orally (hydrolysed) · vegetarians low carnosine
HMB	**Leucine metabolite** · 1.5–3 g/day · **mTORC1 activation + ubiquitin–proteasome inhibition** · strength + endurance · not on WADA list
Arginine	↑ GH (↓ somatostatin), creatine synthesis, **NO precursor** · 12 g/day × 8 wk → ↑ strength, **no change in VO2max / body composition**
Whey	Fast MPS via mTOR · ↑ glutathione (↓ oxidative stress) · 20 g/day · leucine 2–3 g per serve · CHO : protein **3 : 1**
L-carnitine	Long-chain FA into mitochondria · 1–2 g/day · ↓ CK leak, ↑ O2 delivery
Ephedrine (ma huang)	Sympathomimetic, thermogenic · **FDA banned 2004** · urine > 10 µg/ml = AAF
Others	Chromium picolinate (insulin action) · inosine (↑ 2,3-DPG) · octacosanol (grip) · ginseng · bee pollen

PED fact (TO)	Detail
Alcohol	Low BAC 0.02 may ↑ accuracy · banned in precision sports (**archery, shooting, modern pentathlon**)
Tibolone	Synthetic steroid · **↑ hand-grip strength in postmenopausal women** · WADA "other anabolic agent"
GH	191 amino acids · natural **22 kDa (main) + 20 kDa** · recombinant = 22 kDa only → **↑ 22:20 ratio = doping** · doping 10–25 IU 3–4×/wk → acromegaly, DM, carpal tunnel
Clenbuterol	Illegal β2-agonist → **type II hypertrophy**, ↓ fat
AAS effects MCQ	**Alopecia + gynaecomastia + oligospermia + altered thyroid function → all of the above**
CHO loading	↑ endurance **2–3%** · EPO ↑ VO2max **6–7.7%**
Use rates	High-school alcohol 55–90% · PED ~5%

2 Mechanism

Beta blockers – pharmacology (TO)

Action — **competitive antagonists** of adrenaline + noradrenaline at β-receptors · **β1: heart, kidney** · **β2: lungs, GI tract, vascular smooth muscle, skeletal muscle** · **β3: fat cells**

Haemodynamics — **↓ HR + ↓ cardiac output, no effect on stroke volume** · ↓ myocardial O2 demand

In endurance sport — ↑ perceived exertion, ↓ cardiac output + **VO2max**, **earlier fatigue + lactate threshold**, may worsen **EIB / asthma → avoid in asthmatics + endurance athletes**

Why banned in precision sports — **↓ HR, ↓ hand tremor, anxiolysis** → steadiness (**archery, shooting, billiards**)

Gene doping (TO)

Definition — non-therapeutic gene therapy · DNA inserted via vector to **enhance performance** (✗ Wrong options: restore missing gene function) · **banned by WADA in 2003** ✓ (✗ Wrong options: 2001, 2006, 2008)

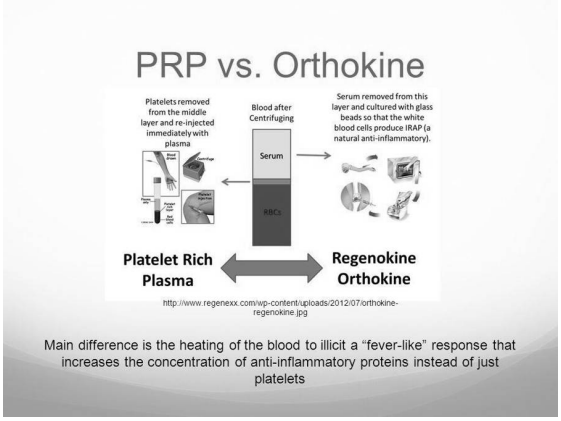
Gene doping (TO)	Detail
Viral vector	Efficient · **immune response may ↓ effect** · = **transduction**
Non-viral (bacterial plasmid)	**No immune response** · = **transfection**
Strength genes	**IGF-1, myostatin**
Endurance genes	**EPO**, **PPARδ** (fibre-type switch; TO text prints "fast type 1 → slow type 2"), PEP carboxylase, **VEGF**
Tissue repair genes	**BMPs**, endorphins
Detection – direct	**Doping vector**
Detection – indirect	Consequences: **transcriptomics (mRNA)**, **proteomics** (protein set), **metabolomics** (non-protein low-MW metabolites)
Risks	**Immune response to viral vector (SIRS)**, autoimmunity to recombinant protein, no expression, **insertional mutagenesis**

Blood spinning – Orthokine vs PRP (TO)

Orthokine (Kobe Bryant, knee) — **variation of blood spinning**, **prohibited by WADA** (TO) · **20 cc blood** + tube with "factors" · **heated + cultured with glass beads** ("fever-like" response) → spun → injected · platelets + lymphocytes aid healing · WBCs make **IRAP** (IL-1 receptor antagonist)

Orthokine vs PRP — both autologous blood · Orthokine **cultured at higher temperature** · **↑ growth factors** per procedure · **blocks IL-1 → anti-inflammatory** · FDA: too much manipulation → **unapproved drug**
✗ Wrong options: **"stored at 5 °C"** (EXCEPT answer)

– – Active space – –



PRP vs Orthokine preparation

Blood doping (TO) artificially recreates **high-altitude training** → banned · **EPO + CERA** (continuous erythropoietin receptor activator) give same effect **without blood removal** → banned in cycling + endurance sports

Pregnancy & performance (TO)

Pregnancy change (TO)	Timing / size
Blood volume	↑ throughout · **peak 28–30 wk**
Plasma volume	Starts **~8 wk**, plateaus 28–30 wk · **↑ 50–60%**
Red cell volume	**Starts at 18 wk** → term (38–40 wk) · **↑ 30%**
Physiological "anaemia"	Dilution · **Hb ↓ ~2 g/dl by 28 wk**
hCG	**Peaks 7–10 wk** (25,700–288,000 mIU/ml) then declines

Performance benefit from pregnancy **after 18 weeks** ✓ (RBC rise) · same TO text also says "**first 3 months** – natural RBC surplus" – read the options
✗ Wrong options: before 18 wk · before / after 10 wk

hCG men: **dose-dependent ↑ testosterone** → ergogenic · non-pregnant women: ovary → progesterone + oestradiol, **not testosterone**; 10,000 IU → T ↑ only ~1.1 ng/ml (≈ half daily diurnal swing) → no measurable muscle effect · normal female T **0.52–2.43 nmol/l** · pregnancy: hCG → **oestrogens + progesterone – not ergogenic**

Hormones oestrogen: uterus + breast enlargement, ↑ uterine / tubal contractility · progesterone: ↓ uterine contractility, sustains pregnancy

Pregnancy / abortion doping misuse of pregnancy ± abortion to ↑ RBC mass + O2 delivery · pregnant **2–3 months** then abortion **days before event** → up to **10% enhancement** ("MC method", artificial insemination) · "pregnancy dopes your blood" · sanctions extend to anyone who assists or covers up

3 Presentation

AAS user on exam **↑ body weight**, well-developed muscles, **testicular atrophy**, **gynaecomastia**, **acne**, alopecia · women: deep voice

17-year-old wrestler, aggression + hostility **stanozolol** (AAS) ✓
✗ Wrong options: furosemide · EPO · α-ketoglutarate · propranolol

Young male, pectoralis major tear, HDL 25 mg/dL, LDL 85, Hb 17.0

Dianabol (AAS) ✓
✗ Wrong options: creatine · GH · EPO · ephedra (none cause tendon tears or ↓ HDL)

4 Diagnosis & Classification

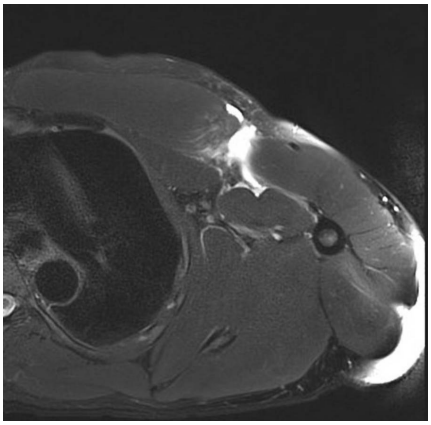
Genes & lab markers

Marker	Finding
Low HDL	**Should alert physician to steroid use** (Orthobullets, OB)
Testosterone : epitestosterone	**> 6:1** (Orthobullets – "blood testing")
LDL / apo A1	↑ LDL · ↓ apolipoprotein A1
Hb high-normal	↑ erythropoiesis (AAS); ↑↑ with EPO
Factors VII, IX, thromboxane	↑ (AAS)
Gonadotropins, oestrogen, testosterone	↓ (AAS)
Liver enzymes	↑ (AAS)
Potassium	↑ (EPO)
Glucose tolerance	↓ / insulin resistance (HGH)
Melatonin (TO)	Night **80–120 pg/ml** (~80% made at night) · day 10–20 pg/ml
Gene-doping targets (TO)	IGF-1, myostatin, EPO, PPARδ, PEP carboxylase, VEGF, BMPs, endorphins

Key radiology findings

Pectoralis major tear

T2 axial MRI: tendon tear with **retraction** · clinical: **loss of axillary fold** + arm bruising · think **AAS (Dianabol)** with ↓ HDL



Pectoralis major tendon tear (T2 axial MRI)

– – Active space – –

Diagnostic criteria

Adverse analytical finding (AAF) (TO)	Urine threshold
Salbutamol	**> 1000 ng/ml** (24-h urine)
Formoterol	**> 40 ng/ml**
Cathine	**> 5 µg/ml**
Ephedrine, methylephedrine	**> 10 µg/ml**
Pseudoephedrine	**> 150 µg/ml**
Caffeine	Old limit **12 µg/ml** (Orthobullets) · now monitoring only (TO)
Threshold drug **+ diuretic / masking agent**	**AAF at any quantity** unless TUE for **both**
Glucocorticoid	Above lab reporting level in **in-competition** urine

Urine sample suitability (TO)	Specific gravity
90 to < 150 ml	**≥ 1.005 refractometer** or **≥ 1.010 lab sticks**
≥ 150 ml (dilute urine)	**≥ 1.003, refractometer only**
Unsuitable sample	Doesn't invalidate analysis · **lab decides** with testing authority

Dysthymia (TO) DSM-5 **removed "dysthymia"** → merged with chronic MDD = **persistent depressive disorder (PDD)** · chronic, early onset, symptoms fluctuate but never fully resolve

PDD criteria (DSM-5-TR) (TO)	Requirement
A	**Depressed mood** most of the day, more days than not, **≥ 2 yr** adults · children / adolescents: irritable allowed, **≥ 1 yr**
B — **≥ 2 of**	Poor appetite / overeating · insomnia / hypersomnia · low energy / fatigue · low self-esteem · **poor concentration** / indecision · hopelessness
C	Never symptom-free **> 2 months** in the 2 yr (1 yr children)
D	MDD criteria may be continuously present for 2 yr
E	**Never manic / hypomanic**
F–H	Not psychotic disorder · not substance / medical · clinically significant distress
Not a feature	**Lack of interest** ✓ (✗ Wrong options: poor concentration · appetite / eating change · low mood)

WADA prohibited list – structure (TO)

Prohibited (TO, 2018 list)	Substances	Methods
At all times (in + out of competition)	Non-approved (e.g. veterinary) · **anabolic agents** · **peptide hormones, growth factors** + mimetics · **β2-agonists** · **hormone + metabolic modulators** · **diuretics + masking agents**	**Manipulation of blood**, **gene doping**, **chemical + physical manipulation**
In-competition only	**Stimulants**, **narcotics**, **cannabinoids**, **glucocorticoids**	–
Particular sports	**Beta-blockers**	–

WADA code (TO)	Class	Specified?
S1	Anabolic agents (DHMCT, metandienone)	**Non-specified**
S3	β2-agonists (terbutaline) · all times	**Specified**
S5	Diuretics + masking agents (HCTZ) · all times	**Specified**
M2	Chemical + physical manipulation (**IV infusions, since 2005**)	–
P1	Beta-blockers (incl. eye drops)	–

At-all-times class (TO)	Examples
Other anabolic agents	**Clenbuterol**, SARMs (andarine, LGD-4033, ostarine, RAD140), **tibolone**, zeranol, zilpaterol
EPO + erythropoiesis agents	EPO, darbepoetin (dEPO), EPO-Fc, **CERA**, EPO-mimetics (CNTO-530, peginesatide) · HIF activators (**argon, cobalt, xenon**, molidustat, roxadustat) · GATA inhibitor K-11706 · TGF-β inhibitors (luspatercept, sotatercept) · asialo EPO, carbamylated EPO
Peptide hormones	**hCG + LH** + releasing factors (buserelin, deslorelin, gonadorelin, goserelin, leuprorelin, nafarelin, triptorelin) · corticotrophins (corticorelin) · **GH**, GH fragments (AOD-9604, hGH 176-191), GHRH (CJC-1293 / 1295, sermorelin, tesamorelin), GHS (ghrelin, anamorelin, ipamorelin, tabimorelin), GHRPs (alexamorelin, GHRP-1 to 6, pralmorelin, hexarelin)
Growth factors	FGFs, HGF, **IGF-1**, MGFs, **PDGF**, thymosin-β4 (TB-500), **VEGF** + any affecting muscle / tendon / ligament synthesis, vascularisation, regeneration, fibre-type switching
Hormone + metabolic modulators	**Aromatase inhibitors** (anastrozole, letrozole, exemestane, formestane, aminoglutethimide, testolactone, 6-oxo, androstatrienedione, arimistane) · **SERMs** (raloxifene, **tamoxifen**, toremifene) · anti-oestrogens (**clomifene**, cyclofenil, fulvestrant) · **myostatin inhibitors** · AMPK activators (AICAR, SR9009), PPARδ agonists (GW1516), **insulins**, **meldonium**, trimetazidine
Diuretics + masking agents	**Desmopressin**, probenecid, **plasma expanders** (IV albumin, dextran, **hydroxyethyl starch**, mannitol) · **acetazolamide**, amiloride, bumetanide, canrenone, chlorthalidone, ethacrynic acid, **furosemide**, indapamide, metolazone, **spironolactone**, **thiazides (HCTZ)**, triamterene, vaptans (tolvaptan)
Diuretic exceptions	**Drospirenone**, pamabrom, **topical ophthalmic CAIs (dorzolamide, brinzolamide)**, felypressin in dental anaesthesia

– – Active space – –

In-competition class (TO)	Examples / rule
Stimulants	Cathine, ephedrine, methylephedrine, pseudoephedrine (thresholds above) · **cocaine** · **epinephrine** allowed only locally (nasal, ophthalmic, with LA)
Monitoring (not prohibited, 2018)	**Bupropion, caffeine, nicotine, phenylephrine**, phenylpropanolamine, pipradrol, synephrine
Narcotics	Buprenorphine, dextromoramide, **diamorphine (heroin)**, **fentanyl** + derivatives, hydromorphone, methadone, **morphine**, nicomorphine, oxycodone, oxymorphone, pentazocine, pethidine
Cannabinoids	Natural (cannabis, hashish, marijuana) + synthetic (Δ9-THC, cannabimimetics) · **except cannabidiol**
Glucocorticoids	**Oral, IV, IM, rectal** (2018) → **all injectable, oral, rectal** (2022) · betamethasone, budesonide, cortisone, deflazacort, dexamethasone, fluticasone, hydrocortisone, methylprednisolone, prednisolone, prednisone, triamcinolone
Beta-blockers (particular sports)	Acebutolol, alprenolol, **atenolol**, betaxolol, bisoprolol, bunolol, carteolol, carvedilol, celiprolol, esmolol, labetalol, levobunolol, metipranolol, **metoprolol**, nadolol, oxprenolol, pindolol, **propranolol**, sotalol, **timolol**

Not on the TO list **acetaminophen** (tibolone, cannabinoids, EPO all banned) – answer key not provided

Beta-blocker ban by sport (TO)	All times (in + out)	In-competition only
2018 WADA list	**Shooting** (ISSF, IPC)	Archery, automobile, billiards, darts, golf, ski jumping / freestyle aerials / halfpipe / big air, **underwater** (apnoea, spearfishing, target shooting)
Eye-drug explanation	**Archery, shooting**	Aeronautics, automobiles, billiards, boules, bridge, darts, golf, ninepin + tenpin bowling, powerboating, skiing / snowboarding
Angina case explanation	**Archery, shooting, underwater sports**	Billiards, golf, darts, automobile sports

Beta-blocker ban sport lists differ between TO explanations (above) – read the options · **hockey not on any list**

β2-agonist (TO)	Allowed inhaled (no TUE)	Urine AAF
Salbutamol	**≤ 1600 µg / 24 h** · divided: **≤ 800 µg / 12 h** (2018 list) · **≤ 600 µg / 8 h** (newer list) · **≤ 600 µg / 12 h, "earlier 800"** (FNB 2022 key)	**> 1000 ng/ml**
Formoterol	**≤ 54 µg / 24 h** (delivered)	**> 40 ng/ml**
Salmeterol	**≤ 200 µg / 24 h**	–
Vilanterol	**≤ 25 µg / 24 h** (newer list)	–
Terbutaline	**No threshold → TUE required** · prohibited at all times, even therapeutic	Any
All others	Fenoterol, higenamine, indacaterol, olodaterol, procaterol, reproterol, tulobuterol, arformoterol, levosalbutamol, tretoquinol (trimetoquinol) – prohibited	–

β2 rule — all selective + non-selective β2-agonists incl. optical isomers prohibited · urine above threshold = AAF unless **controlled pharmacokinetic study** proves therapeutic inhaled dose · salbutamol = albuterol (EIB prevention dose: Pulmonary Conditions chapter)

Salbutamol urine "permissible level" (TO) — **800 ng/ml** ✓ (safely below 1000 ng/ml; 1000 = borderline)
✗ Wrong options: 1600 · 1200

Doping control – sample collection (TO)

Step (TO)	Done by
1 Notification	**Chaperone**
2 Report to doping control station	**Athlete** – **ASAP, within 60 min** · delay allowed for **press conference, training session** · chaperoned throughout
3 Select collection vessel	**Athlete** – 1 individually sealed vessel, checks intact, keeps control
4 Provide sample	Observed by **DCO / DCA of same gender** – 2 people in washroom · 3 if **minor / disabled** (representative doesn't view)
5 Select doping control kit	**Athlete**
6 Divide sample	**Athlete** (representative if disabled) · **B > 30 ml**, rest in **A > 60 ml**
7 Seal A + B	**Athlete**
8 **Measure specific gravity**	**DCO** ✓ (✗ Wrong options: athlete)
9 Documentation	**DCO**
10 Analysis	**WADA-accredited laboratory**



A and B urine sample bottles

Urine volume — **minimum 90 ml** ✓ · **150 ml if dilute** → tell athlete **not to over-hydrate**
✗ Wrong options: 70, 120, 150

Partial sample (< 90 ml) — fill into **vial B**, use **partial kit** with **white lid** · "**Berlinger**" vial locks after **5–6 rotations** · later: collect interim sealing device, check number, pour extra urine into initial sample + mix

– – Active space – –



Sealing of a partial urine sample

Sample processing	athlete chooses kit, checks undamaged + **all code numbers match** · divides B then A, seals · **DCO doesn't handle equipment** unless asked · athlete controls sample until sealed
DCO duties	supplies: **3 A&B kits + 3 urine vessels per sample** · sets up DCS · ensures notification + chaperoning · ensures collection, identification, sealing, storage, dispatch to WADA lab · **overall responsible** even when blood taken
Chaperone	notifies athlete · accompanies + observes to and in DCS · may witness urine provision if trained
Blood control officer (BCO)	prepares athlete, **venepuncture**, aftercare / first aid
Blood volume	**20 ml once, divided into two samples** ✓ (10 ml per tube, range 10–15 ml) · "> 5 ml per tube, ~15–17 ml total" (other TO key) – read the options ✗ Wrong options: 10 ml twice · 15 ml twice · 10 ml once
Blood collection	confirm ID · **seated, feet on floor (not lying), ≥ 10 min** · **no exercise past 2 h** · athlete picks **BEREG kit (1 of 3)**, checks numbers · DCO / BCO don't touch materials · athlete sticks code label **longitudinally** on vacutainer · **non-dominant arm** usually · release tourniquet once blood flows · declare medication / supplements of **last 7 days**
Blood tubes	**purple = athlete biological passport (ABP)** · **yellow = GH or its markers** + other tests · kit: 1 disinfection pad, 2 cotton swabs, 1 band-aid, gloves, 3 accessory bags (tubes, butterfly needle)

Notable doping cases (TO)

Athlete (TO)	Substance	Detail
Prithvi Shaw (cricket)	**Terbutaline in cough syrup**	**Inadvertent**, for respiratory infection · BCCI ban **8 months**
Yusuf Pathan (cricket)	**Terbutaline** (cough syrup)	2018 · backdated **5-month** ban
Navjeet Dhillon (discus, 2018 CWG medallist)	**DHMCT** (S1)	Supplement, unknowingly · **3-year** AIU ban
Dhanalakshmi Sekar (sprint)	**Metandienone** (S1)	Failed dope test
Anish Kumar (para discus)	**Hydrochlorothiazide (HCTZ)** (S5, specified)	BP medicine
Kobe Bryant (basketball)	**Orthokine**	Knee therapy

Terbutaline — **SABA** (salbutamol + terbutaline = SABAs; salmeterol + formoterol = LABAs) · for bronchospasm in asthma, bronchitis, emphysema · ↑ time to exhaustion + muscle power output in cyclists · no help to a cricketer · asthmatic may use SABA / LABA with a **TUE certificate**

Anti-doping system, detection & sanctions (TO)

Bodies (TO) — **WADA 1999** (Lausanne → **Montreal**) · **NADA** (India) · federations test · **Anti-Doping Disciplinary Panel** (1st instance) → **Anti-Doping Appeal Panel** (appeal **within 14 days**) → **CAS = final, binding** · database **ADAMS**

Definitions (TO) — **in-competition = 12 h before competition → end of sample collection** · **strict liability** (athlete responsible regardless of intent) · ban covers **all sports** · prohibited association with banned coach · **3 whereabouts failures in 12 months = ADRV (Article 2.4)** · injured athlete still files whereabouts

Sanctions – conflict — 1st offence **2–4 years** (TO, Doping Part 1) · **4 years** (TO, medico-legal) – read the options · 2nd = **lifetime** · reduced (0–4 yr) if no intent / negligence or **substantial assistance** · support staff also sanctioned

Detection (TO)	Sample	Detects
GC-MS	Urine	**Anabolic steroids, stimulants**
LC-MS	Blood	Peptide hormones
Isoelectric focusing	Blood / urine	**EPO** (detectable ~3 days)
CIR / IRMS	Urine	**Endogenous vs exogenous** steroid (testosterone confirmation)
Athlete biological passport	Blood	Haematological + steroid profiles over time
GH	Serum	**22 : 20 kDa isoform ratio**
CO rebreathing	–	**Total Hb mass** (blood doping)

Marker thresholds (TO) — **T : E > 4 : 1** suspicious (TO) · > 6 : 1 (Orthobullets, above) – read the options · **nandrolone metabolite 19-norandrosterone > 2 µg/L**; detectable up to **18 months** · **metandienone up to 12 wk** · stanozolol ~10 days · **meldonium > 15 µg/ml** · codeine → morphine (ratio test)

Sample handling – TO doping lecture — urine **90–110 ml** · athlete may drink up to **1.5 L** if dilute · undress **nipple to mid-thigh** · A **30 ml** / B **60 ml** (TO lecture) vs B > 30 / A > 60 (table above) – read the options · BCO: athlete rests 10–20 min · to WADA lab **within 24 h**, room temperature (not frozen) · A positive = **AAF** → NADA checks TUE → B positive → provisional suspension · advance-notice testing only for minors, disabled, high-security / bio-bubble

TUE (TO)	Rule
Criteria (all 4)	Needed for a **diagnosed condition** · **no significant performance enhancement** · **no reasonable permitted alternative** · not due to prior prohibited use
When	Apply **≥ 30 days** before event (NADA / IF TUE committee)
Retroactive	Emergency / life-saving treatment (dehydration, anaphylaxis)

– – Active space – –

Prohibited method / substance (TO, M-class)	Status
M1 blood doping	Any autologous / homologous / heterologous transfusion · **perfluorochemicals, efaproxiral (RSR13)**, Hb-based O2 carriers · intravascular manipulation
Total Hb mass	Predicts aerobic capacity (not [Hb]) · plasma expanders alone don't help
EPO family	Epoetin α/β · **darbepoetin** (long t½) · **CERA (Mircera)** – slow, hard to detect · 60–350 U/kg SC × 4–6 wk
Cryopreserved RBCs	**–40 °C, 1 year** (vs RBCs 4–6 °C 35–42 days, platelets 20–24 °C 5 days)
Plasmapheresis – donor	**Prohibited (M1.1)**
Haemodialysis, **IV laser therapy** (M1.3)	**Banned**
Mannitol IV	**Banned (masking)** · **inhaled** allowed for asthma testing
Cobalt salts	**Prohibited** (erythropoiesis) · vitamin B12 fine
Trimetazidine	Metabolic modulator, prohibited in + out (**Sun Yang**)
Meldonium	Banned **1 Jan 2016** (**Sharapova**)
Modafinil	Stimulant S6, banned **2004** (BALCO era)
Hypoxic chambers, inhaled O2	Allowed
Colostrum	Not banned, discouraged (IGF-1)
PRP	Banned **2009–2011 → now permitted**; isolated growth factors (IGF-1, PDGF) prohibited

PRP biology (TO) platelets **~1,000,000 /µL** in ~5 ml · **α granules: PDGF, VEGF, TGF-β, IGF** · dense: ADP, serotonin · **95% GF released within 1 h**, synthesis 5–7 days · **double spin** (soft → hard) or buffy coat · **20–24 °C, never freeze** · citrate / ACD-A anticoagulant, **avoid EDTA** · activate with $CaCl_2$ / thrombin · **VEGF ↑ for 4 wk** after injection · **DEPA** classification = **D**ose, **E**fficiency, **P**urity, **A**ctivation · leukocyte-rich vs pure PRP / PRF

– – Active space – –

Case (TO)	Substance
Thomas Hicks 1904	**Strychnine + brandy** (marathon) – first known doping
Hans-Gunnar Liljenwall 1968 (TO prints 1967)	**Alcohol** (pentathlon shooting) – first Olympic disqualification
Ben Johnson 1988	**Stanozolol** → Dubin inquiry
BALCO (Marion Jones, Dwain Chambers, Barry Bonds)	**THG** (designer steroid), EPO, GH, modafinil
Lance Armstrong	EPO, transfusions, testosterone, **Actovegin** (calf-blood extract)
Maradona 1994	**Ephedrine**
Shoaib Akhtar 2006	**Nandrolone**
Narsingh Yadav	Metandienone
Robert Farah	**Boldenone** (contaminated beef)
Riccardo Riccò	**CERA (Mircera)**
Mohammad Asif	Nandrolone via **Keratyl eye drops**
Russia pre-Rio "Duchess cocktail"	Oxandrolone + metenolone + trenbolone

5 Management

Antihypertensives & other medications in sport (beta blockers, diuretics) (TO)

Hypertensive athlete – choose drug

- Needs prior permission → **DIURETICS (S5, banned at all times)**
- Shooter / archer / billiards → **NO BETA-BLOCKER**
- On propranolol → **HOCKEY ALLOWED**
- Training-neutral choice → **ACE INHIBITOR / ARB / CCB**

Diuretics **prohibited at all times; all specified substances** · masking agents
✗ Wrong options: nitrates · CCB · ACE inhibitors

Propranolol sportsman **hockey** ✓
✗ Wrong options: billiards · darts / archery · shooting

50 y, angina 3 yr ago, HTN on beta-blocker, half marathon in 8 months **change antihypertensive (ACEI / ARB / CCB) then full cardiac assessment** + BP monitoring ✓
✗ Wrong options: treadmill stress test on beta-blocker (HR blunted) · refuse clearance · **TUE – not needed** (banned only in particular sports, not tested)

– – Active space – –

Therapeutic use exemptions – glucocorticoids & IV fluids (TO)

Recurrent shoulder pain after intra-articular steroid	**give another dose + retroactive TUE** ✓ ✗ Wrong options: permission first · decline · "IA steroids banned for any reason"
Glucocorticoids (2022 list)	prohibited **in-competition** by **all injectable** (IV, IM, **intra-articular**, periarticular, peritendinous, intratendinous, epidural, intrathecal, intrabursal, intralesional, intradermal, SC), **oral** (oromucosal, buccal, gingival, sublingual) + **rectal** routes
GC permitted at all times, no TUE	inhalation, intranasal, eye drops, perianal, dermal, dental intracanal, topical
Out-of-competition GC	**not prohibited by any route** · but in-competition urine may still show it → AAF · clinical justification → **retroactive TUE** · no TUE → sanction
IV infusions (M2.2, since 2005)	**> 100 ml per 12 h prohibited** in + out of competition · **except** during **hospital treatment, surgical procedure, clinical diagnostic investigation** · **≤ 100 ml / 12 h permitted** unless substance prohibited
Why IV is banned	↑ **plasma volume** · **mask** prohibited substance · distort **athlete biological passport**
TUE needed for > 100 ml / 12 h outside hospital	doctor's office, hotel, home, tent, vehicle · event medical facility, first-aid station, start–finish line · IV clinics outside hospital
Prohibited substance given IV	**TUE always**, whatever volume or setting · non-prohibited substance > 100 ml outside the 3 exceptions → TUE for the **method**
Emergency	haemodynamically unstable → **never withhold IV fluids** · "how much in dehydrated state?" → **life-saving emergency, any amount** ✓
Team doctor, gastroenteritis, Ringer lactate in training camp	**100 ml / 12 h** ✓ (not hospital → TUE first, else sanction) ✗ Wrong options: 200 ml / 12 h · 200 ml / 24 h · excused because life-saving

IV infusion TUE criteria (TO)	Requirement
1	**Well-described diagnosis** / condition
2	**Permitted alternative not medically reasonable** (e.g. why not oral rehydration)
3	Given by **qualified medical professional**
4	**Adequate medical records** (substance, rate, history, exam, co-morbidities)

Respiratory & eye drugs under WADA (TO)

EIB at Olympics	**vilanterol** ✓ (**ultra-LABA, 24 h** vs LABA 12 h → fewer inhalations) · first-line asthma = **reliever + inhaled corticosteroid** · salbutamol, formoterol, salmeterol allowed inhaled at therapeutic doses (2018) ✗ Wrong options: terbutaline (TUE) · levosalbutamol · trimetoquinol

Glaucoma drop, hockey player, not doping

dorzolamide ✓ (topical CAI — **no diuretic effect** topically)
✗ Wrong options: timolol (β-blocker) · nandrolone · prednisolone

Eye preparation (TO)	WADA status
Dorzolamide, brinzolamide (topical CAI)	**Permitted**
Acetazolamide (systemic CAI)	**Prohibited in + out** — masking agent
β-blocker drops (timolol)	**Prohibited in particular sports (P1)** — systemic levels ≈ oral
Prostaglandin analogues, prostamides, apraclonidine, brimonidine	Permitted
Pilocarpine (topical)	Permitted (specifically exempted)
Antibacterial, antiviral, povidone-iodine	Permitted all routes
Antifungals	None prohibited · azoles exempt topically (imidazoles)
Topical + intracameral glucocorticoids	Permitted
NSAIDs, antihistamines, mast-cell stabilisers	Permitted all routes
Atropine, cyclopentolate, homatropine, tropicamide	Permitted
Phenylephrine	Monitoring list (2012) · not prohibited
Lidocaine, bupivacaine, proxymetacaine, tetracaine	Permitted all routes
Cocaine	**Prohibited in-competition**
Adrenaline	Local use only; else prohibited in-competition
Artificial tears, lubricants, fluorescein, verteporfin, pegaptanib, ranibizumab	Permitted

ADHD (TO)

Drug of choice

methylphenidate ✓ (stimulant; most commonly used; adults, teenagers, **children > 5 y**) · licensed: methylphenidate, lisdexamfetamine, dexamfetamine, atomoxetine, guanfacine
✗ Wrong options: propranolol · methylprednisolone · guanfacine

Sports psychology (self-efficacy, anxiety, mood disorders) (TO)

Individual zone of optimal functioning (IZOF)

each athlete has own preferred anxiety level ✓ · performance best when pre-competition anxiety within / near own zone
✗ Wrong options: drive theory · inverted-U · Troutwine athletic profile

– – Active space – –

Athletic self-efficacy — **strong belief in ability to perform successfully** ✓ · feeling competent to cope · covers ability, fitness, leadership, coaching · affects performance, success, **stress coping**
✗ Wrong options: vicarious experience · motivational behaviour · emotional arousal

Personality (TO)	Traits
Type A	Competitive, ambitious, perfectionist · impatient, time urgency, **stress-prone**
Type B	Relaxed, patient, creative
Type C	Analytical, detail-oriented, introverted, wants order / logic · emotional stress
Type D	**Anxious, worried, negative outlook**, introverted, suppresses emotions ✓ (injured athlete fearing never returning)

Immediate reduction of performance anxiety — cognitive restructuring / mindfulness / self-talk / goal setting – answer key not provided

Key psychological factor for return to sport — pain tolerance / psychological resilience / performance / physical strength – answer key not provided

Team physician – travel, jet lag, media & medical bag (TO)

Circadian system — master clock = **suprachiasmatic nucleus (SCN)** + peripheral clocks · synchronised by **zeitgebers**: **light (most critical)** via retinohypothalamic tract, sleep–wake, activity, social cues, meals · light timing, intensity, duration, spectrum matter

Jet lag — circadian desynchronisation after rapid travel across **≥ 3 time zones** · poor sleep onset / maintenance, fatigue, ↓ motivation + concentration, ↓ appetite, altered bowel · phase markers: **core body temperature, melatonin**

Jet lag intervention (TO)	Examples
Melatonin	**0.5–5 mg** · hypnotic + **chronobiotic**
Melatonin analogues	**Tasimelteon, ramelteon**
Sedatives	Benzodiazepines (**temazepam**)
Stimulants	Amodafinil, **caffeine** – ✓ **used, not depressants**
Non-drug	**Timed exercise**, sleep, light, meal timing / composition
Caveats	Drugs only **up to 4 days after arrival** · consider **DVT risk**

Jet lag odd one out — **depressants** ✓ (other options: glucocorticoids, melatonin analogues, sedatives)

Melatonin timing — **1–2 h before sleep**, up to **4 nights** after arrival ✓ · dose **1–5 mg** (child 1–2 mg) · after food, don't crush / chew · avoid with **caffeine, alcohol** · avoid in galactose intolerance, congenital lactase deficiency, glucose–galactose malabsorption · melatonin dose 0.5–5 mg (jet-lag table) vs 1–5 mg (dosing note) – read the options
✗ Wrong options: 30 min · 2–4 h · 3–6 h

Melatonin physiology — **N-acetyl-5-methoxytryptamine from serotonin, pineal** · retina → SCN → pineal · rises after sundown, **peaks 2–4 am** · sleep = ↑ melatonin + ↓ body temperature · extrapineal: retina, marrow, platelets, skin, lymphocytes, cerebellum, **GI enterochromaffin cells** (follows tryptophan intake) · child side effects: agitation, bed-wetting, dizziness, drowsiness, headache · patches, tablets, liquids, gummies, spray

Melatonin timing effect — **afternoon / evening → phase advance** · **early morning → phase delay**

↑ free tryptophan — protein with more Trp than LNAA · **CHO** (insulin drives BCAA into muscle → ↑ free Trp : BCAA) · high-fat meal (↑ FFA) · strenuous exercise

Media — team doctor may give only **own name + country / practice** ✓
✗ Wrong options: player name + injury · drugs / treatment · photo with athlete

Team physician ethics (TO)	Point
Conflict of interest	Personal physician = athlete's health · **team physician = team's health** + fitness to compete
Confidentiality	Contract may require sharing with management · **never with media**
Decision-making	Explain all alternatives, risks, benefits
Autonomy	Hierarchical team culture → players may not feel free to opt out
Informed consent	**Competent · risks + benefits explained · understood · voluntary**
Return to play	**Main ethical conflict** · **sports medicine physician decides** on athlete's short + long-term health, not result or coach
Advertising	Once illegal in USA · still illegal in **Italy, Germany**

Medical bags (TO)	Contents
Off-field (pre / post game)	Physio supplies bag (strapping), doctor supplies bag (suture box, otoscope, HIA tablets in rugby), **medications case (locked)**, recovery devices, medical folder (allergies, medications)
On-field	Physio pitch bag (tape, blood management), doctor pitch bag (player-specific drugs, inhalers, contact lenses)
Pitch-side	Splints bag, **emergency bag (defibrillator, analgesia, O2)**, vacuum mattress, spinal board + blocks, collar
Odd one out	**"Athlete medical bag"** ✓ – athletes only clear **sports kits** through customs

Bag rules — list agreed at **start of season** · check governing-body standards · **check customs rules for each country** + carry documentation · consistent packing · **checklist**; post-game list of items to replace

Medications case — oral analgesics, anti-inflammatories, antihistamines, antiemetics, Fucidin, chloramphenicol ointment, loperamide, Buscopan, GTN, aspirin, OCP, tranexamic + mefenamic acid

On-field tips — gloves · gauze + nose plugs under glove on **dorsal thenar of non-dominant hand** · saline pods · labelled contact lenses + pocket mirror · energy gels, hair bands · pocket mask, **methoxyflurane**, oropharyngeal airways in outer pockets

– – Active space – –

Medico-legal aspects of doping (TO)

Doctor's liability (TO) **team physician** answerable to NADA, WADA **and** Medical Council · private practitioner → Medical Council only · athlete consent does **not** absolve ethical liability · coach asks for pre-match steroid injection → **refuse** · OTC supplements not always safe · only **WADA-accredited labs** analyse samples

IPC section (TO)	Offence
336	Act endangering life – 3 months / ₹250
337	Hurt by negligence – 6 months / ₹500
338	Grievous hurt by negligence – 2 years
304A	**Death by negligence** – up to 2 years, non-bailable (TO)
284 / 276 / 351	Poisonous substance negligence / drug adulteration / assault

NDPS Act 1985 (TO) small quantity ≤ 1 yr or ₹10,000 · intermediate ≤ 10 yr + ₹1 lakh · commercial **10–20 yr** + ₹1–2 lakh · applies to doctors + athletes

Misuse of Drugs Act (UK) class (TO)	Examples
A	Cocaine, heroin, LSD, methamphetamine
B	Amphetamine, barbiturates, cannabis, codeine
C	Benzodiazepines, **anabolic steroids**, ketamine

Ethics (TO) anti-doping compromises **autonomy** (compelled samples), **privacy** (observed collection), **confidentiality** (public results) · avoid cough syrups with codeine, ephedrine, salbutamol · cheating: whereabouts abuse, urine substitution, masking diuretics, **micro-dosing**

X vs Y

Drug (OB)	Distinguishing adverse effect
AAS (Dianabol)	**↓ HDL, tendon tears**, aggression, ↑ Hb
Creatine	**Dehydration**, cramps, water retention, nausea
Human growth hormone	**Acromegaly**, cardiac failure, glucose intolerance, myopathy
EPO	**Hyperviscosity → thromboembolism**, HTN, hyperkalaemia
Ephedra	**Arrhythmias (fatal)**, palpitations, psychiatric
Caffeine	**↑ blood pressure**

	SABA	LABA	Ultra-LABA
Drugs (TO)	**Salbutamol, terbutaline**	**Salmeterol, formoterol**	**Vilanterol**
Duration	Short	**12 h**	**24 h**
WADA	Salbutamol threshold · **terbutaline needs TUE**	Thresholds	25 µg / 24 h

– – Active space – –

	PRP	Orthokine
Layer used	Platelets, re-injected at once	Serum **cultured with glass beads, heated**
Product	Platelets / growth factors	**IL-1 antagonist (IRAP)** + more growth factors

	Travel fatigue	Jet lag
Who	**All travellers**, any direction / time zones	**≥ 3 time zones**, east–west (transmeridian)
Cause	Cramped seats, mild hypoxia, climate change (trans-latitudinal), inactivity	**Circadian desynchrony**
Course	Acute or **chronic (cumulative)**	Episodic, **more severe + prolonged**

How it's asked

"Methandrostenolone – NOT linked?"	**→ decreased vision**
"Cyclist MI from steroids – mechanism?"	**→ ↑ LDL**
"Side effect of anabolic steroids?"	**→ decrease in high-density lipoprotein**
"Wrestler, aggression + hostility?"	**→ stanozolol**
"Primary mechanism of ↑ muscle?"	**→ ↑ messenger RNA production**
"Only nonreversible AAS effect?"	**→ alopecia**
"AAS user, concern for surgery + anaesthesia?"	**→ fluid and electrolyte imbalance**
"Most common behavioural effect?"	**→ increased aggression**
"Pec tear, HDL 25, Hb 17?"	**→ use of Dianabol**
"More common with creatine than testosterone?"	**→ muscle cramping**
"Substance that ↑ cellular ATP stores – side effect?"	**→ dehydration**
"Glucagon on skeletal muscle?"	**→ catabolic**
"Endurance training adaptation?"	**→ ↑ storage + utilisation of intramuscular lipids**
"Antihypertensive needing prior permission (TO)?"	**→ diuretics**
"Antihypertensive not for a shooter (TO)?"	**→ beta blocker**
"On propranolol – sport allowed (TO)?"	**→ hockey**
"Para discus thrower, BP medicine positive (TO)?"	**→ hydrochlorothiazide**
"Glaucoma eye drop not doping (TO)?"	**→ dorzolamide**
"50 y, angina, on beta-blocker, half marathon (TO)?"	**→ change antihypertensive medication, then full cardiac assessment**
"Gene doping banned by WADA (TO)?"	**→ 2003**
"Why Prithvi Shaw banned 8 months (TO)?"	**→ cough syrup containing terbutaline**

– – Active space – –

"Orthokine EXCEPT (TO)?"	→ **stored at 5 degrees**
"Navjeet Dhillon's substance (TO)?"	→ **DHMCT (dehydrochloromethyltestosterone)**
"Dhanalakshmi Sekar's substance (TO)?"	→ **metandienone**
"Yusuf Pathan's substance (TO)?"	→ **terbutaline**
"Recurrent pain after IA steroid (TO)?"	→ **give another dose, retrograde (retroactive) TUE**
"Permissible salbutamol in 24-h urine (TO)?"	→ **800 ng/ml** (AAF > 1000 ng/ml)
"EIB drug allowed at Olympics (TO)?"	→ **vilanterol**
"NOT banned by WADA (TO)?"	→ answer key not provided · **acetaminophen** by elimination
"IV fluids in dehydrated state (TO)?"	→ **life-saving emergency, so any amount**
"Gastroenteritis, Ringer lactate, not in hospital (TO)?"	→ **100 ml / 12 hours**
"Doping control step NOT done by athlete (TO)?"	→ **measuring specific gravity of urine**
"Minimum urine volume (TO)?"	→ **90 ml**
"Process shown — white lid (TO)?"	→ **sealing of partial urine sample**
"Blood for dope testing (TO)?"	→ **20 ml once, divided into two samples**
"Protein, high-performance athlete (TO)?"	→ **1.6 g/kg body weight**
"Amino acid oxidised in endurance exercise (TO)?"	→ **leucine**
"Vitamin for mineralisation, deficiency → stress fracture (TO)?"	→ **vitamin K**
"Carnosine composed of (TO)?"	→ **beta alanine + L-histidine**
"Correct supplement statement (TO)?"	→ **BCAAs — essential amino acids, muscle protein synthesis, ↓ fatigue**
"Evidence-based endurance supplements (TO)?"	→ **caffeine pre-workout, sodium bicarbonate pre-workout, BCAAs intra-workout**
"Soccer 90 min, hot + humid — hydration (TO)?"	→ **isotonic sports drink, 6–8% carbohydrates + electrolytes**
"Drug of choice for ADHD (TO)?"	→ **methylphenidate**
"Own preferred anxiety level theory (TO)?"	→ **zone of optimal functioning**
"Strong belief in own ability (TO)?"	→ **athletic self-efficacy**
"Injured athlete, negative, worried (TO)?"	→ **type D personality**
"Not a feature of dysthymia (TO)?"	→ **lack of interest**
"Immediate reduction of performance anxiety (TO)?"	→ answer key not provided
"Key psychological factor for return to sport (TO)?"	→ answer key not provided

"Jet lag – odd one out (TO)?"	→ depressants
"Melatonin timing for jet lag (TO)?"	→ 1–2 hours before sleep
"Only information team doctor gives media (TO)?"	→ own name + country of practice
"Medical bag – odd one out (TO)?"	→ athlete medical bag
"Performance benefit from pregnancy (TO)?"	→ after 18 weeks
"In-competition period (TO)?"	→ 12 h before competition until end of sample collection
"Whereabouts failures constituting an ADRV (TO)?"	→ 3 in 12 months
"Time limit for appeal after ADDP decision (TO)?"	→ 14 days
"Doping test distinguishing endogenous from exogenous testosterone (TO)?"	→ carbon isotope ratio (IRMS)
"GH doping detection (TO)?"	→ 22 : 20 kDa isoform ratio
"Beta-alanine dose-limiting side effect (TO)?"	→ paraesthesia (> 800 mg per dose)
"HMB mechanism (TO)?"	→ mTORC1 activation, ↓ ubiquitin–proteasome breakdown
"WADA status of PRP (TO)?"	→ permitted (banned 2009–2011); isolated growth factors prohibited
"First known doping case (TO)?"	→ Thomas Hicks 1904, strychnine + brandy

– – Active space – –

THE FEMALE ATHLETE

– Knee & Sports · 58 / 61 –

1 Basics

Physiologic sex differences

Women have	**↑ body fat %** · **↓ maximal O_2 consumption + haemoglobin** · **↓ cardiac output** · **↓ muscle mass + strength**
↑ incidence in women	**patellofemoral disorders**, **stress fractures**, **ACL injuries**

Triad definition & components (low energy availability, amenorrhoea, BMD; thyroid not included)

Component	Detail
Low energy availability	**± disordered eating** (with or without an eating disorder) · insufficient caloric intake for demand
Menstrual dysfunction	**Amenorrhoea** / oligomenorrhoea · energy imbalance, low body fat, hypothalamic–pituitary axis changes
↓ Bone mineral density	**Osteoporosis** (often presents as a **stress fracture**)

NOT components	**stress fracture (TO)** – a consequence, not a component · **thyroid dysfunction** (OB) · ligamentous laxity, ↑ ACL tears, hirsutism (OB)
Dysmenorrhoea in the options (TO)	counted under menstrual dysfunction → not the answer; **stress fracture** is the odd one out
Eating disorder	both "LEA **with** an eating disorder" and "LEA **without** an eating disorder" = components (OB)
Names	**female athlete triad**, coined **1992 by ACSM** (adolescent + young female athletes) · **anorexia athletica** = synonym (Orthobullets, OB) · **RED-S** (relative energy deficiency in sport) – **IOC 2014**, broader, **includes males**
Anorexia athletica criteria (OB)	**GI complaints**, **restriction of caloric intake**, **weight loss**, **menstrual dysfunction** · **requires ABSENCE of an affective disorder** (e.g. depression) · true osteoporosis relatively **uncommon** in female athletes → traditional triad criteria rarely met ✗ Wrong options: presence of affective disorder
Thyroid (OB)	**not a core component**; **may occur secondarily** to the metabolic + hormonal disruption ✗ Wrong options: core component · unrelated · primary cause · always present / needed for diagnosis
Associated (OB)	amenorrhoea, osteoporosis, **↓ immune function**, insufficient caloric intake · not **low LDL** → actually **↑ LDL** (hypoestrogenism)
Energy availability (TO)	**EA = (energy intake − exercise energy expenditure) ÷ fat-free mass (kg)** · **≥ 45 kcal/kg FFM/day optimal** · 30–44 subclinical · **< 30 = low EA → triad / RED-S** · RMR **< 90% predicted** = LEA marker

Continuum (TO) ACSM 1992 → **revised 2007 as a continuum** · **even one component = diagnostic** · menses: eumenorrhoea → luteal suppression → anovulation → **oligomenorrhoea (cycle > 35 days)** → **amenorrhoea (> 3 months)** · bone: normal → osteopenia → osteoporosis → stress fracture

RED-S (TO) IOC **2014** (update 2018) · **both sexes** · multisystem: reproductive, bone, endocrine, metabolic, **haematological (anaemia)**, CVS (bradycardia, hypotension), immune, GI, psychological, performance · males: **↓ testosterone** (restore EA first, **avoid early testosterone**)

Axis in LEA (TO)	Change
HPO	↓ GnRH → ↓ LH / FSH → ↓ oestrogen
HPT	**↓ T3, normal TSH** (euthyroid sick)
HPA	**↑ cortisol**
GH–IGF-1	**↓ IGF-1**
Leptin–ghrelin	**↓ leptin, ↑ ghrelin**

Screening tools (TO) **LEAF-Q** (females) **score ≥ 8 = high risk** · **IOC RED-S CAT**: **green** full participation · **yellow** modified training · **red** no participation until cleared

Differential of amenorrhoea (TO) PCOS (↑ LH:FSH, hirsutism) · pregnancy (β-hCG) · thyroid · **hyperprolactinaemia** → exclude before functional hypothalamic amenorrhoea · **peak bone mass by 25 yr**; adolescent BMD loss irreversible

Cardiovascular (OB) **endothelial dysfunction** (nitric oxide imbalance between vasodilators + vasoconstrictors) → atherosclerosis, ↑ future CVD · RED-S: **dyslipidaemia + ↑ visceral adiposity → precocious coronary atherosclerosis**

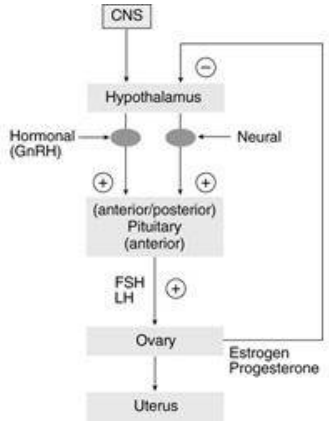
Pathophysiology – energy deficit & hormones

Root cause **low energy availability (LEA)** = cornerstone of RED-S + triad · drives the hormonal + bone deficits

Axis LEA disrupts **hypothalamic–pituitary–ovarian axis** (OB) · "**hypothalamic–pituitary–adrenal** axis disruption" (OB, RED-S) → reproductive hormonal dysfunction

Bone **↓ BMD from ↑ bone turnover** · loss of **oestrogen / progesterone** protective effect → osteoporosis → stress fractures

Undernutrition (TO) **↓ IGF-1 + ↓ leptin** → ↓ appetite stimulation, menstrual problems, impaired bone remodelling → **osteopenia + stress fractures**



Hypothalamic–pituitary–ovarian feedback loop

– – Active space – –

Genes & lab markers

Marker	Direction / value	Why
Cortisol	**↑**	Maintains blood glucose (OB)
Free T3	**↓**	Lowers basal metabolic rate (OB)
LH	**↓**	Diverts energy from reproduction (OB)
IGF-1, leptin	↓	Undernutrition (TO)
Oestrogen	↓	Hypoestrogenism from menstrual dysfunction
LDL	**↑**	Hypoestrogenism (OB) (✗ Wrong options: low LDL)
Serum calcium	**Normal**	Not a useful next test (OB)
25(OH)D	Normal **≥ 30 ng/mL** · **< 75.8 nmol/L = stress fracture risk**	TO
Iron status	**Ferritin / transferrin ratio**	TO – see Calcium & iron
COL5A1	**CC genotype under-represented** in females with ACL rupture	Orthobullets

Hormone trap (OB) **↑ cortisol, ↓ free T3, ↓ LH**
✗ Wrong options: ↑ T3 · ↑ LH · ↓ cortisol

Epidemiology – amenorrhoea & sports

Amenorrhoea prevalence	Source
Elite runners – nearly 50%	Orthobullets, OB
Vigorous exercisers **10–20%** · elite runners + professional ballet dancers **40–66%**	OB
Exercise-induced **5–25%** (by level + sport)	OB

Sports **running + gymnastics** · aesthetic / endurance (**gymnastics, cross-country**) · higher stress fracture risk: **gymnastics, cross-country, lacrosse** · multiple sports (not a risk factor)

Pregnancy & Title IX

Pregnancy **↑ estradiol → ligamentous laxity** · ↑ body weight → **16–32% ↑ O_2 consumption**

Title IX no person in the US excluded, denied benefits or discriminated against **on the basis of sex** in any educational programme receiving **federal financial assistance**

2 Mechanism

Sex differences in ACL injury

Risk — **2–10× males** (Orthobullets, this topic), esp. **pivoting sports** · ACL Tear chapter: **4.5 : 1** (Orthobullets), **2–9×** (OB), **~8×** (TO) – read the options

Factor (Orthobullets)	Detail
Landing biomechanics + neuromuscular control	**Conditioning + strength = biggest role** · land with knee in **more extension + valgus** (from **hip IR**)
Notch	**Smaller notch**
Ligament	**Smaller ACL**
Cyclic hormones	↑ risk in **first half (preovulatory phase)** of cycle
Alignment	Leg alignment
Genetics	**CC genotype of COL5A1** under-represented in females with ACL rupture

Menstrual phase conflict — **preovulatory** (Orthobullets, this topic) · **no consensus** (ACL Tear chapter) – read the options

Most significant contributor (OB) — **neuromuscular coordination + training – modifiable**
✗ Wrong options: ACL size · oestrogen levels · notch size · valgus alignment

Prevention — **neuromuscular training (jump training)**, plyometric conditioning ↓ ACL rate · NM training **↑ medial hamstring EMG → ↓ dynamic valgus** on sidecutting (Zebis)



Valgus medial knee collapse on landing

3 Presentation

Stress fracture as presentation – menstrual & nutritional work-up

First contact — often presents to the **orthopaedic surgeon / team physician with a stress (osteoporotic) fracture**

Stress fracture risk — **amenorrhoeic athletes 2–4× eumenorrhoeic** (TO) · **most frequent site = tibia** (TO) · menstrual irregularity → **more and multiple** stress fractures (Warren)

– – Active space – –

Sites (OB)	**weight-bearing cortical bone: tibia, metatarsals, femur** · not usually trabecular bone
NOT ↑ in amenorrhoeic athletes (OB)	femoral head AVN, ACL rupture, labral tear, FAI → answer = **stress fracture**
Amenorrhoea + Beighton 4 (volleyball)	at risk of **tension-side femoral neck stress fracture** (Figure A) · Beighton **> 5/9** = hypermobility (OB)



Tension-side femoral neck stress fracture (arrow)

Runner, tibial stress fracture, BMI 16, no menses 7 mo	**poor caloric intake** = **most common cause of amenorrhoea** in female athletes ✗ Wrong options: NF1 (tibial bowing, MPNST) · osteofibrous dysplasia (anterior eccentric tibial lesion in child) · EWS-FLI1 (Ewing) · focal fibrocartilaginous dysplasia (abrupt varus at metaphyseal–diaphyseal junction)
4th MT stress fracture + contralateral 1 yr ago	**amenorrhoea 12 months** ✗ Wrong options: daily OCP (helps regulate menses) · multiple sports · T-score −0.5 (normal) · BMI 26 (normal)
Thresholds (OB)	**BMI < 21** predictive of stress fracture (some) · **T-score < −1.0 osteopenia**, **< −2.5 osteoporosis**
Marine recruits (OB)	**~5%** of female recruits fracture in **13 wk** basic training · **~40% femoral / pelvic** (more severe) · biggest risk = **no menses in the previous year** (vs 10–12 menses) · lesser irregularity → trend only ✗ Wrong options: prior mileage · running frequency · self-rated fitness · race
Peak bone mass	hypoestrogenic athletes **may never reach peak bone mass** → possibly **irreversible** deficit · **neither calcium nor ↑ activity compensates** · lost BMD may never be regained

Runner's nipples (TO)

Cause (TO)	**irritation from rubbing against clothing** · **cold-weather sports** (cold → hard, prominent nipples) · common in **cyclists** (cold + perspiration) ✗ Wrong options: warm weather precipitates
Treatment (TO)	**tape nipples** to restrict movement · petroleum jelly, moisturisers, vitamin E soaks · **sports bra**
Sports bra (TO)	firm but elastic (room for padding), **non-abrasive**, absorptive · **crossed Y-shaped back, no hooks / fasteners** · **no seams over nipple** · personally fitted

4 Diagnosis & Classification

Diagnostic criteria

Term	Definition	Source
Secondary amenorrhoea	**Cessation of menses for 6 months after ≥ 1 normal cycle**	Orthobullets, OB
Secondary amenorrhoea	Absence of bleeding **6 months** or of **3–6 consecutive cycles** after normal menses (no pregnancy)	OB
Amenorrhoea (RED-S)	No menses for **≥ 3 months**	OB
Osteopenia / osteoporosis	T-score **< −1.0** / **< −2.5**	OB
Hypermobility	Beighton **> 5/9**	OB

Amenorrhoea cut-off conflict — **6 months** (Orthobullets, OB) · **3 months** (OB, RED-S) — read the options
✗ Wrong options: ↑ volume 3 mo · pain 2 mo · missed month after peak training · ↓ volume 2 mo

Pre-participation screening (TO) — triad domains = **eating patterns, menstrual cycles, stress fracture history** · source gives **no key**; the item **outside the triad** = **supplement / drug-abuse history** (inferred)

Work-up of a female athlete with a stress fracture

Any female athlete with a stress fracture — work up for triad: **menstrual history + nutritional consultation + bone density** (OB)

DEXA — female athlete with **amenorrhoea + stress fractures** (Orthobullets)

Runner, femoral neck stress fracture on MRI — evaluate **osteoporosis + anorexia + secondary amenorrhoea** (all of the above)

Prior 5th MT stress fracture, now tension-side femoral neck — fix + **obtain menstrual history**
✗ Wrong options: never compete again (can return when counselled) · serum calcium (normal) · psychiatry (only if eating disorder diagnosed) · ↓ training (not next step)

Femoral neck rule (OB) — **tension side → internal fixation** · compression side → rest, nonop (line > 50% or effusion → screws: Femoral Neck Stress Fractures chapter)

Key radiology findings

Study	Finding / use
Radiographs	**May be normal or equivocal** · screening films may miss stress reaction
X-ray femur	**Smooth cortical thickening, distal medial femoral diaphysis** = femoral stress injury
X-ray tibia (lateral)	**Anterior cortical thickening** = stress reaction · **"dreaded black line"** = poor prognosis → surgery
MRI	**Preferred** · rule out fracture in **high-risk areas (femur, femoral neck)** · **before return to sport** · stress reaction vs impending / complete fracture · STIR oedema (periosteum, cortex, marrow)
Bone scan	**Fallen out of favour** — radioisotope + relatively high ionising radiation
DEXA	BMD — amenorrhoea + stress fractures

– – Active space – –

Hip flexor tightness – modified Thomas test (TO)

Technique (TO) supine, **knees bent over table edge** · hold one knee to chest · test knee should stay at **90°**, hip + posterior thigh flat on table · **pass** = knee stays at 90° · **fail** = knee extends to < 90°

Finding (TO)	Interpretation	Muscles short
Thigh **touches** table, knee **stays flexed**	**Negative**	–
Thigh **off** table, knee **extends**	**Both one- + two-joint** flexors short	All below
Thigh touches table, knee extends	**Two-joint hip flexors short**	**Rectus femoris, TFL, sartorius**
Thigh **off** table, knee **flexes > 80°**	**One-joint** hip flexors short	**Iliopsoas, pectineus, adductor longus / brevis / magnus**
Thigh abducts as hip extends, **lateral patellar deviation**, thigh IR + leg ER	**TFL** short	TFL
Hip abduction + flexion + ER and knee flexion (**≥ 3** = tight)	**Sartorius** short	Sartorius

Muscle (TO)	Joints crossed	Actions
Iliopsoas	**1** (hip)	Hip flexion, ER
Pectineus	**1**	Hip flexion · abduction (as listed)
Adductor longus / brevis	**1**	Adduction, flexion, ER
Adductor magnus	**1**	Adductor part: flexion, adduction, ER · hamstring part: extension, IR
Rectus femoris	**2** (hip + knee)	Hip flexion, **knee extension**
TFL	**2**	Hip abduction, flexion, IR · knee extension
Sartorius	**2**	Hip flexion, abduction, ER · **knee flexion**

5 Management

Female athlete triad with stress fracture

- First / most critical step → **multidisciplinary nutrition + psychological counselling**
- Thigh pain at rest, equivocal X-ray → **stop training + MRI**
- Tension-side femoral neck → **fixation + menstrual history**
- Tibial stress reaction, no fracture line → **nonop + review diet and training**

Treatment – nutrition rather than bisphosphonates

– – Active space – –

Arm	Content
Psychological counselling	**Core** of treatment (Sabatini) · eating behaviours, self-esteem
Dietary management	**Establish energy balance** · ↑ weight / food intake, ↓ exercise · ↑ **calcium + vitamin D**
Training	**↓ intensity + cross-training** (↓ stress fracture risk) · **limit training until menses resume**
Supplements	**Calcium + vitamin D** for osteoporosis · consider **cyclic oestrogens / progesterones** for osteoporosis
OCPs	**May help amenorrhoea** (Orthobullets, OB) · **use OCPs to normalise menses** (TO)
Team	Physician, athletic trainer, **sports dietician**, mental-health provider, family, coaches

Least appropriate (OB)
bisphosphonates
✗ Wrong options: calcium + vitamin D · ↑ food intake · ↓ training · counselling

Advice NOT suitable (TO)
"avoid OCPs" · TO Rx = OCPs to normalise menstruation + **↑ body fat** via diet + prompt stress fracture care
✗ Wrong options: improve nutrition · increase body fat

Hormone conflict
OCPs beneficial (Orthobullets, OB, TO) · **hormone replacement does not fix the energy deficit; no longer primary Rx for functional hypothalamic amenorrhoea** (OB) — read the options

Most critical initial step (OB)
multidisciplinary nutritional + psychological counselling to correct energy deficit
✗ Wrong options: NSAIDs + crutches (NSAIDs impair bone healing) · oral oestrogen / progesterone · 6 wk complete rest (deficit persists → recurrence) · DEXA first

Tibial stress reaction, 3 mo amenorrhoea (15 yr)
discuss eating habits + training regimen · stop the precipitating activity, protected WB rehab
✗ Wrong options: keep running · CAM walker / casting (stiffness) · IM nail (no clear fracture line → nonop first)

Hurdler, vegan, BMI 18.1, femoral cortical thickening, pain walking
immediate cessation of training + MRI
✗ Wrong options: ↓ intensity + PT · dietician only · bone scan + oncology · bone stimulator / US (evidence limited for femoral)

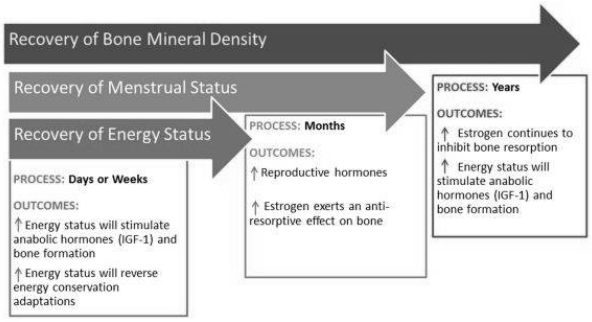
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Natural course appropriate diet + moderated exercise → **natural return of menses**

TO management detail	Content
Energy	**+300–600 kcal/day** initially · target EA ≥ 45 · **↑ intake before ↓ training**
Macronutrients	CHO 50–60%, protein 15–20%, fat 25–30% · **20–25 g protein post-exercise**
Ca + vit D	**1500 mg Ca + 1500–2000 IU vit D / day** (TO, triad lecture) · vs 2000 mg + 800 IU (TO, below) – read the options
Training	**↓ load 10–20%**, no double sessions until menses resume
Hormones (amenorrhoea > 6 months)	**Transdermal oestrogen + cyclic oral progesterone preferred** (bone, no hepatic effects) · **combined OCP symptomatic only – masks LEA** (TO) · read the options vs OCP rows above
Bisphosphonates	Severe osteoporosis only, after skeletal maturity, cautiously
Psychology	**CBT** preferred
Monitoring	DEXA every 6–12 months · vit D 6-monthly · RMR 3-monthly · menses monthly

Return to play (TO) normal menses (or testosterone) · **EA ≥ 45** · stable weight **≥ 90% ideal body weight** · **BMD (T / Z) ≥ −1.0** · MDT clearance

Recovery rates **energy status: days–weeks** (↑ IGF-1, reverses energy conservation) → **menses: months** (oestrogen anti-resorptive) → **BMD: years**



Triad components recover at different rates

Return to sport when properly counselled, may **return to high-endurance sport** (OB)

Calcium & iron requirements (TO)

Calcium for stress-fracture prophylaxis (TO) **2000 mg/day** + **vitamin D 800 IU/day** (= **200% + 150%** of recommended intake)
✗ Wrong options: 500 · 1000 · 1500 mg

Vitamin D (TO) intake up to **2000 IU/day** · normal **≥ 30 ng/mL** · **25(OH)D < 75.8 nmol/L** = stress fracture risk · supplementation ↓ stress fracture prevalence · high 25(OH)D → ↓ body fat, ↑ muscle mass

Vitamin D biology (TO) fat-soluble, first found in **cod liver oil** · from diet (fortified dairy, fish oils) or **UV irradiation of 7-dehydrocholesterol** in skin · active **1,25-dihydroxyvitamin D** regulates **> 900 gene variants**

Stress fractures in athletes (TO) prevalence **6.5–9.7%** · **metatarsals + tibia** commonest · risk factors: **female sex, age > 40 or < 18**, high arches, LLD, flat feet

Best iron test in female athlete (TO) **ferritin / transferrin ratio**
✗ Wrong options: ferritin · transferrin saturation · hepcidin

Why not ferritin (TO)	best marker in non-athletes, but an **acute-phase reactant** → ↑ with training inflammation even when stores are low
Female iron depletion (TO)	**menstrual loss**, **exercise-induced haemolysis**, **sweat loss**

X vs Y

	Female athlete triad	**RED-S**
Origin	**ACSM 1992**	**IOC 2014**
Who	Female athletes	**Includes males**
Core	LEA ± eating disorder, amenorrhoea, ↓ BMD / osteoporosis	LEA, reproductive dysfunction, low BMD (↑ turnover)
Scope	Narrow	**Broader spectrum** (performance, CV health)

Part of triad	**NOT part of triad**
Low energy availability ± eating disorder	**Stress fracture (TO)**
Menstrual dysfunction (amenorrhoea; dysmenorrhoea option – TO)	**Thyroid dysfunction**
↓ BMD / osteoporosis	Ligamentous laxity, ↑ ACL tear, hirsutism, **low LDL**

How it's asked

"Triad includes all EXCEPT (TO)?"	→ **stress fracture** ✗ Wrong options: dysmenorrhoea · LEA · low bone density
"Same TO stem, repeated papers (2015, 2016, 2019)?"	→ **stress fracture**
"Triad = amenorrhoea + osteoporosis + ?"	→ **low energy availability (± disordered eating)**
"Associated with triad EXCEPT?"	→ **low LDL levels** (LDL is raised)
"Not part of the triad?"	→ **thyroid dysfunction**
"Thyroid and the triad?"	→ **not a core component; may occur secondarily**
"Anorexia athletica criteria EXCEPT?"	→ **presence of an affective disorder**
"Secondary amenorrhoea definition?"	→ **absence of menstrual bleeding for 6 months**
"Hormones in low energy availability?"	→ **↑ cortisol, ↓ free T3, ↓ LH**
"Screening item NOT routinely assessed (TO)?"	→ no key given · **supplement / drug-abuse history** (inferred)
"Advice NOT suitable for triad (TO)?"	→ **avoid OCPs**
"Least appropriate treatment?"	→ **bisphosphonates**

– – Active space – –

"Most critical initial step, BMI 17, skips meals?"	→ multidisciplinary nutritional + psychological counselling
"Runner, femoral neck stress fracture — work-up for?"	→ osteoporosis, anorexia, secondary amenorrhoea (all)
"Tibial stress fracture, BMI 16, menses 7 mo ago — aetiology?"	→ poor caloric intake
"Amenorrhoea, Beighton 4 — at greater risk of?"	→ femoral neck stress fracture (Figure A)
"15 yr runner, tibial stress reaction, 3 mo amenorrhoea?"	→ discuss eating habits + training regimens
"Tension-side femoral neck after prior 5th MT fracture — next?"	→ obtain a menstrual history
"Marine recruit — biggest femoral / pelvic stress fracture risk?"	→ no menses in the year before training
"Recurrent MT stress fractures — likely finding?"	→ amenorrhoea for 12 months
"Hurdler, femoral cortical thickening, pain walking?"	→ stop training + MRI
"Biggest contributor to female ACL tears?"	→ neuromuscular coordination + training
"Daily calcium for stress-fracture prophylaxis (TO)?"	→ 2000 mg/day
"Best iron-status test in a female athlete (TO)?"	→ ferritin / transferrin ratio
"Runner's nipples — EXCEPT (TO)?"	→ warm weather precipitates it (cold does)
"Modified Thomas: thigh on table, knee extends (TO)?"	→ two-joint hip flexors short (positive)
"Energy availability threshold defining LEA (TO)?"	→ < 30 kcal/kg FFM/day
"LEAF-Q score indicating high risk (TO)?"	→ ≥ 8
"Preferred hormone therapy for amenorrhoea > 6 months (TO)?"	→ transdermal oestrogen + cyclic progesterone

– – Active space – –

TIBIAL SHAFT STRESS FRACTURES

– Knee & Sports · 59 / 61 –

1 Basics

Epidemiology – tibia most common site

Most common stress fracture bone	**tibia** in athletes + military training (OB) ✗ Wrong options: femur · fibula · navicular · 5th MT
Most common site (TO)	**posteromedial tibia, junction of middle + distal third** → answer "**distal third of tibia**" ✗ Wrong options: 2nd MT · fibula · femoral neck
Series (Ohta-Fukushima, OB)	370 athletes: **tibia 49.1%** > tarsals 25.3% > metatarsals 8.8% · evaluation **> 3 wk** after pain onset → prolonged recovery
2nd commonest	**2nd / 3rd MT shaft (March fracture)** (TO) · **tarsals** (OB series) – read the options
Who	**runners + military recruits** · after a **change in training routine** · incidence in runners **up to 20%** (OB)
Risk factors (OB)	**sudden ↑ training intensity**, **low BMD**, **menstrual irregularities**, **low energy availability** · always evaluate **female athlete triad**

Stress fracture sites (TO)

Site	Detail (TO)
Tibia	**Posteromedial**, **middle–distal third junction** – **most common**
2nd / 3rd MT shaft	**March fracture** – **2nd most common**
Other metatarsals	–
Medial malleolus	Jumpers + runners
Distal fibula	**Runner's fracture**
Femoral neck	**Inferomedial (compression) commoner** than superior (tension) · **tension more dangerous** → completes without fixation
Femoral shaft	Posteromedial or medial cortex
Navicular	**Central third** – runners / sprinters, ballet dancers, football players
Calcaneus	**Tuberosity** – military recruits, runners
Ribs	**Middle ribs (4th–9th)** – **rowers**, **serratus anterior** traction
Pubic ramus	**Female long-distance runners**

– – Active space – –

Site	Sport association (TO, stress-injury lecture)
Metatarsal	Running, ballet, **marching**
Patella	**Hurdling**, running
Navicular	Sprinting, jumping, football
Fibula	Running, **racewalking**, aerobics
Ulna	**Racquet sports**, gymnastics, volleyball
1st rib	**Throwing / pitching**
Ribs 2–10	**Rowing, kayaking**
Pars interarticularis	Gymnastics, **fast bowling**, ballet, volleyball
Scapula	Running with hand weights
Coracoid	**Trap shooting**
Talus	**Pole vaulting**

Commonest overall – conflict **metatarsals** most common overall; **pelvis + MT commoner in females** (TO, stress-injury lecture) · tibia (OB, TO above) – read the options

Risk factors (TO) LLD, pes planus / cavus, **hip ER deficit** · rapid ↑ in FIT (frequency, intensity, time) · fatigue, poor shock absorption · hard / downhill surfaces · worn shoes, **anti-pronation / motion-control shoes** · low vit D / Ca, RED-S



Bilateral distal-third tibial stress fractures (arrows)

Pathophysiology

Mechanism **repetitive submaximal loads** → resorption > formation · abrupt ↑ duration / intensity / frequency without rest → **↑ osteoclast activity** (OB)

Bone type conflict **linear microfractures in trabecular bone** (Orthobullets) · **weight-bearing cortical bone, not usually trabecular** (OB, Female Athlete chapter) – read the options

Healing **callus → woven bone → endochondral bone formation**

Continuum (TO) repetitive submaximal force → microdamage → remodelling imbalance → **stress reaction** (diffuse tenderness) → **stress fracture** (focal tenderness) → complete fracture · pain: diffuse after activity → focal, later at rest / night, **no relief with warm-up**

Bone-loading test (TO)	Positive
Hop test	Pain reproduced
Fulcrum test	Sharp pain → **femoral shaft** stress fracture
One-leg standing hyperextension	Pain → **pars** fracture

Why the anterior cortex fails **tension side** from **posterior muscle force** (Orthobullets) · **tensile forces + relative paucity of blood supply** in anterior midshaft (OB)

High- vs low-risk sites – tension vs compression

	Anterior cortex (tension)	Posteromedial cortex (compression)
Frequency	**Uncommon**	**Typical / most common**
Risk	**High** – delayed union, **nonunion**, **propagation to complete fracture**	**Low** – usually heals
X-ray	**"Dreaded black line"** (transverse anterior lucency)	Periosteal reaction, cortical thickening
Rx	Often **IM nail**	**Protected WB**

Greatest risk of delayed healing / nonunion (OB) **anterior cortex of tibia** · 2nd + 5th MT can delay / nonunite (OB)
✗ Wrong options: posteromedial tibia · distal fibula · inferior femoral neck (compression side) · **3rd MT – heals well**

High-risk sites (TO, Femoral Neck Stress Fractures chapter) tension femoral neck, tension patella, **anterior tibial cortex**, medial malleolus, talar neck, dorsal navicular cortex, 5th MT proximal metaphysis, great-toe sesamoids

Low-risk sites (TO, same chapter) femoral shaft, **medial tibia**, ribs, ulnar shaft, 1st–4th MT · **2nd MT** low risk (TO) vs can delay / nonunite (OB) – read the options

3 Presentation

History **change in exercise routine** · **insidious onset** · pain with running first → then **daily activities / walking**

Exam **pain directly over the fracture** · **point tenderness** (TO)

Army recruit, 6 wk training, posteromedial pain, point tenderness at distal–middle third junction (TO) **stress fracture**
✗ Wrong options: posteromedial tibial stress syndrome · gastrocnemius strain · bone contusion

MTSS clue (TO) **dull pain** ↑ with activity, relieved by rest · **diffuse tenderness along posteromedial tibial border** (vs **point** tenderness in fracture)

Differential (OB) medial tibial stress syndrome (shin splints), periostitis, infection, bursitis, neoplasm, **exertional compartment syndrome**, nerve entrapment

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Findings	Facts
Radiographs (AP + lateral)	**Lateral: "dreaded black line" anteriorly** (tension) · endosteal thickening · periosteal reaction + cortical thickening	Often diagnostic (Orthobullets) · **usually normal early** → later periosteal reaction, new bone, fracture line (OB)
MRI	**Earliest: T2 periosteal high signal** · marrow oedema · **T1 linear low-signal zone**	**Most sensitive** · **replacing bone scan** · normal X-ray + high suspicion → MRI
Tc-99m bone scan	**Focal uptake** in cortical and / or trabecular region	**Highly sensitive, low specificity** (OB)

Investigation of choice (TO)
MRI – **best combined sensitivity + specificity even early**; recommended for the whole bone-stress spectrum
✗ Wrong options: CT · clinical exam · bone scan

Modality (TO, stress-injury lecture)	Key facts
X-ray	Positive after **2–8 wk** (shin-pain lecture: 2–3 wk) · sensitivity **12–56%**, specificity **88–96%**
Scintigraphy	Positive **24–36 h** · sensitive, non-specific
CT	**Best for cortical fracture line**
MRI	**Gold standard**, early marrow oedema · sensitivity **68–99%**
Ultrasound	Periosteal thickening, callus · Doppler hypervascularity

MRI grading systems (TO)
Fredrickson–Kijowski (1 periosteal oedema → **4b** cortical line) · **Arendt** (1 STIR oedema → 4 line) · **Nattiv** (1 mild T2 oedema → 4 severe + line)

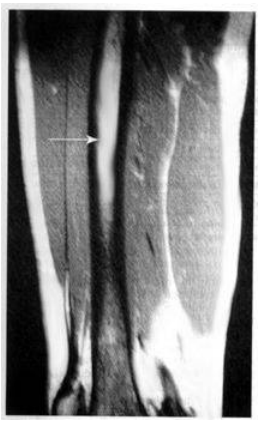
X-ray-negative suspected tibial stress fracture (TO)
MRI – **most sensitive + specific** · beats bone scan + CT for soft-tissue / oedema, **earlier** than bone scan · sensitivity ≈ bone scan but **far more precise for location + extent**

MRI vs bone scan (OB)
MRI better (Ishibashi)

Normal films, high suspicion (OB)
trial of rest + **serial radiographs** also appropriate



Lateral tibia – "dreaded black line" of the anterior cortex



MRI – tibial marrow oedema

MRI grading of bone stress injury (TO)

Grade (TO)	MRI signs
1	Mild–moderate **periosteal oedema on STIR**, **no marrow change**
2	Moderate–severe periosteal oedema on STIR **+ marrow change on T2WI**
3	Grade 2 **+ marrow change on T1WI**
4	**Fracture line visible**

5 Management

Tibial shaft stress fracture

- Posteromedial / X-ray normal, MRI oedema → **protected WB 4–6 wk, stop running**
- Anterior cortex, dreaded black line → **reamed IM nail**
  (esp. if the line violates the anterior cortex)
- Failed nonop, complete, or fixed deadline → **IM nail**

Operative criteria

Indication for IM nail (OB, Orthobullets)
High-risk fracture – anterior tibial cortex ("dreaded black line", esp. violating the cortex)
Failure of nonoperative management (no improvement with rest – Boden)
Complete fracture
Need to return on a specific timeline (athletes, military)

Posteromedial compression-side fractures – protected weight bearing

Standard (most cases)	**activity restriction + protected weight-bearing** · **avoid NSAIDs** (slow bone healing) · consider **bone stimulator** · correct nutritional deficiencies
Duration (OB)	**4–6 wk limited WB** → progress to full WB
Runner, normal X-ray, MRI oedema, positive bone scan	**protected WB 4–6 wk** ✗ Wrong options: immediate return · IM nail without nonop trial · long-leg cast · biopsy

– – Active space – –

Distance runner, proximal tibial pain now with walking, normal X-ray, MRI +	**protected WB with crutches** ✗ Wrong options: switch to elliptical · prescription NSAIDs · arthroscopy · ORIF
Pneumatic bracing + modified rest (OB, Batt)	effective, but return to full activity **~12 months**
Pulsed ultrasound (OB, Rue)	**no** effect on healing or return to duty (26 servicemen, 43 fractures)
Medial (posteromedial) tibial SF (TO)	cavus or over-pronated foot · rest, crutches, **Aircast 2–6 wk**, cross-training (swim, cycle) · return when **pain-free + no tenderness**

Site-specific treatment (TO)	Management
Femoral neck	Bed rest 1 wk → gradual WB · displaced → surgery
Anterior tibial cortex	**NWB 6–8 wk or IM fixation**
Medial malleolus	Cast 6 wk or fixation
Talus	NWB cast 6 wk or fragment excision
Navicular	NWB cast 6–8 wk or surgery
2nd MT base	NWB 2 wk → PWB 2 wk
5th MT base	Cast 6 wk or percutaneous screw
Sesamoid	NWB 4 wk

Adjuncts (TO)	**LIPUS** · **ESWT (avoid if pain > 5/10)** · TECAR (deep heat) · **teriparatide (PTH)** · short-term NSAIDs only · footwear: **rearfoot strikers → cushioned, mid / forefoot → stiff insoles**
Return-to-running (TO)	pain-free walking **≥ 5 days** → walk + jog 50% pace alternate days → 60–100% speed → consecutive days (2 run : 1 rest) → full · **anti-gravity treadmill**: after 1 wk pain-free walking, 3 × 5 min at 50–70% BW, +5–10% every 2 wk → ground at 90% BW
Most important factor in an athlete with a stress fracture (TO)	source gives **no key**; best fit = **correction of biomechanical + training errors** (inferred) (vs immobilisation 8 wk, gradual return, analgesics)

Anterior cortex tension-side fractures – nonunion risk & intramedullary nailing

Footballer, 6 mo pain, failed nonop, anterior linear lucency	**tibial IM nailing** – fastest WB + return ✗ Wrong options: posterior plate + graft · graft alone · cast · protected WB
Special-forces soldier, dreaded black line, deploys in 4 mo	**tibial IM nail** ✗ Wrong options: activity modification + vitamin D (high-risk site, no deficiency stated) · pulsed US · pneumatic brace NWB (up to 12 mo) · observe + repeat imaging (risk of completion)
Outcome (Varner, OB)	11 athletes, **reamed IM nail** for chronic anterior stress fracture → **return to sport ~4 months**
Plating	**anterior tension-band plating** described · **not posterior**
Anterior cortex nonop (TO)	offload (pneumatic Aircast / KT tape) · **stop NSAIDs + smoking** · **LIPUS** · IM nail ± graft if nonunion

IM nail caveat (OB) allows return to sport but **does not guarantee healing**

– – Active space – –

6 Complications

Anterior cortex **delayed union, nonunion, propagation to complete fracture · highest risk of all tibial sites**

Delayed diagnosis evaluation **> 3 wk** after pain onset → longer recovery

X vs Y

	Stress fracture	MTSS (shin splints)	CECS
Tenderness	**Point / focal**	**Diffuse posteromedial border**	Normal exam at rest
Pain	With running → **walking / rest**	Dull, worse at start, **↓ during training**	Starts ~10 min in, resolves ~30 min after
X-ray	**Dreaded black line** (anterior)	Normal · periosteal exostoses (late)	Normal
Bone scan	**Focal, intense, all 3 phases**	**Diffuse, longitudinal, delayed phase only**	–
Test	**MRI** (oedema / line)	Clinical · bone scan / MRI	**Compartment pressures**
NSAIDs	**Avoid** (slow healing)	Used	Used

How it's asked

"Most common site of stress fracture (TO)?"	→ **distal third of tibia**
"Same TO stem, 2014 paper?"	→ **distal third of tibia**
"Bone most often involved in athletes?"	→ **tibia**
"Greatest likelihood of delayed healing / nonunion?"	→ **anterior cortex of tibia**
"Recruit, posteromedial pain, point tenderness (TO)?"	→ **stress fracture** ✗ Wrong options: posteromedial tibial stress syndrome
"Investigation of choice for stress fracture (TO)?"	→ **MRI**
"X-ray-negative suspected tibial stress fracture (TO)?"	→ **MRI**
"Normal X-ray, MRI oedema, hot bone scan?"	→ **protected weight-bearing for 4–6 weeks**
"Runner, proximal tibial pain with walking, MRI +?"	→ **protected weight-bearing with crutches**
"Failed nonop, anterior linear lucency, quick return?"	→ **tibial intramedullary nailing**
"Soldier, dreaded black line, deploys in 4 months?"	→ **tibial intramedullary nail**
"Most important factor in athlete with stress fracture (TO)?"	→ no key given · **correct biomechanical + training errors** (inferred)
"Imaging best for a cortical stress fracture line (TO)?"	→ **CT**

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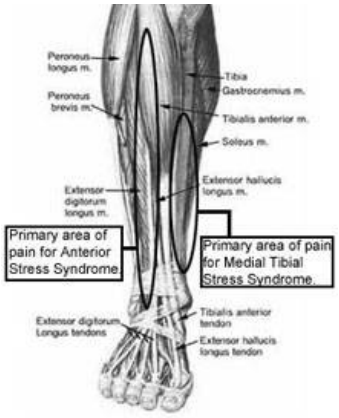
"Fulcrum test positive (TO)?"	→ **femoral shaft stress fracture**

TIBIAL STRESS SYNDROME (SHIN SPLINTS)

– Knee & Sports · 60 / 61 –

1 Basics

Definition	overuse / repetitive-load injury of the shin → **persistent dull anterior leg pain**
Incidence	**10–15% of running injuries** · **60% of leg pain syndromes**
Location	**medial (posteromedial) tibial stress syndrome = most common** · anterior (anterolateral) tibial stress syndrome · distal + posteromedial tibia
Risk factors	**poor shock absorption** (cement / uneven surfaces, improper shoes) · training errors (sudden ↑ intensity + duration) · **> 20 miles / week** · early-season hill training · previous lower-limb injury · **over-pronation / ↑ internal tibial rotation**
Pathophysiology	**traction periostitis** · anterolateral → **tibialis anterior** on tibia + interosseous membrane · posteromedial → **tibialis posterior + soleus**
Associated	**female athlete triad** (critical to diagnose + treat) · **tibial stress fracture** – **females 1.5–3.5× risk of progression**
Main culprit (TO)	**soleus** traction (+ tibialis posterior) · tenderness diffuse at **upper ⅔–lower ⅓ junction**, posteromedial · **pain improves after warm-up** (MTSS / tendinopathy) · extra risk factors (TO): **weak hip abductors**, calf tightness, low vit D / Ca, **RED-S**, prior stress fracture, female



Pain areas – anterior vs medial tibial stress syndrome

3 Presentation

Symptoms	**vague, diffuse pain along middle–distal tibia** that **↓ with running** (early) · later: pain earlier with more frequent training
Exam	**tenderness along posteromedial tibial border**, from **4 cm above medial malleolus** extending **up to 12 cm** proximally · **pes planus**, **tight Achilles**, **weak core**
Provocative test	**pain on resisted plantarflexion**

– – Active space – –

4 Diagnosis & Classification

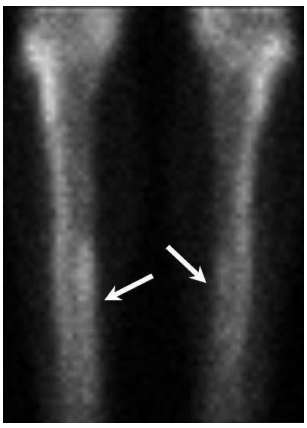
Key radiology findings

Study	Shin splints	vs stress fracture
Radiographs	**Normal first 2–3 wk** · late **periosteal exostoses**	**"Dreaded black line"**
3-phase bone scan	**Diffuse, longitudinal** uptake along **posteromedial (posterior) tibial cortex**, **delayed phase (3) only** · phases 1 (flow) + 2 (blood pool) normal	**Focal, intense** hyperperfusion + hyperaemia in phases 1 + 2 · **focal, fusiform** uptake in phase 3
MRI	**Periosteal oedema** → progressive marrow involvement · shows other soft-tissue injury	Marrow oedema ± line

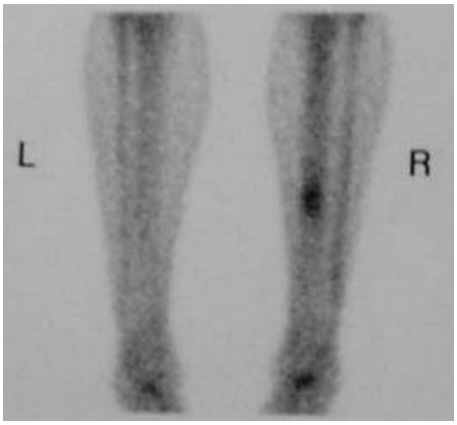
Distinguishing finding (OB)

bone scan: diffuse, longitudinal uptake of posterior tibial cortex = shin splints

✗ Wrong options: shin splints scan more intense · more diffuse tenderness in stress fracture · all 3 phases positive in shin splints · pain persists longer after activity in stress fracture



Bone scan – diffuse longitudinal uptake (shin splints)



Bone scan – focal uptake (stress fracture)

Differential diagnosis – exertional leg pain

Condition	Characteristics	Tissue
Anterior tibial stress syndrome	Diffuse anterolateral pain, worse at start, **↓ during training**	Periosteum
Medial tibial stress syndrome	Diffuse middle–distal pain, worse at start, **↓ during training**	Periosteum
Tibial / fibular stress fracture	Pain with running, **point tenderness**, **dreaded black line**	Bone
Exertional compartment syndrome	Starts **10 min** into exercise, **resolves 30 min** after · sensory / motor loss · ↑ anterior compartment pressure	Muscle + fascia
Leg tendinopathy	Achilles, peroneal, tibialis posterior	Tendon
Sural / SPN entrapment	Dermatomal symptoms	Nerve
Lumbar radiculopathy	Worse sitting (lumbar tension position)	Nerve
Popliteal artery entrapment	Vascular studies	Vessel

5 Management

Nonop – first line, works in vast majority — **NSAIDs**, rest, activity modification

Activity modification — **↓ running distance, frequency + intensity by 50%** · low-impact cross-training · stretching + strengthening · synthetic track · avoid hills, uneven / hard surfaces

Shoes — change every **250–500 miles** (lose shock absorption) · **orthotics** for pes planus

Therapy — strengthen **invertors + evertors of the calf** · **local phonophoresis with corticosteroids** may help

TO treatment — **soleus-focused stretching** + foam rolling · Ca + vit D3 + Mg · medial arch insoles / taping · **avoid anti-pronation (motion-control) shoes** (TO) · severe: Aircast or fascial release

Operative (failed nonop) — **deep posterior compartment fasciotomy + release of painful periosteum** · variable results, complete relief unlikely

6 Complications

Recurrence — common after resuming heavy activity

X vs Y

	Shin splints (MTSS)	Tibial stress fracture	CECS
Pain with running	**↓ during training** (early)	↑ → walking	**↑**, starts ~10 min in
Tenderness	**Diffuse posteromedial**	**Point**	Normal at rest
Bone scan	**Diffuse longitudinal, phase 3 only**	**Focal, all phases**	–
Cause	Traction periostitis	Bone fatigue	↑ compartment pressure

– – Active space – –

How it's asked

"Bone scan finding separating shin splints from stress fracture?"	**→ diffuse, longitudinal uptake of posterior tibial cortex in shin splints**
"MTSS vs tibial stress fracture on palpation (TO)?"	**→ diffuse vs focal tenderness**
"Key muscle culprit in MTSS (TO)?"	**→ soleus**

VISCERAL BLUNT TRAUMA

– Knee & Sports · 61 / 61 –

1 Basics

Overview

Injuries **abdominal wall**, **spleen**, **liver**, **kidney**, **testicular + penile**

Risk factors **contact sports**, **direct blows** · **deceleration** → spleen, liver, kidney

Commonest sport injuries (Rifat, OB) **abdominal wall muscle, liver, spleen** · **spleen = most frequent intra-abdominal organ** in sport

Organ	Rank / fact	Clue
Spleen	**Most common** abdominal organ in blunt trauma · **most common cause of death** from abdominal / visceral injury	**LUQ pain**, **Kehr's sign** (left shoulder), N/V · **mononucleosis**
Liver	**2nd most common**	**RUQ pain → right shoulder**
Kidney	Contact athletes, **blow to flank**	**Haematuria** (may be absent), flank ecchymosis
Abdominal wall	Commonest sport injury	**Localised pain**, pain on **trunk flexion / rotation**

3 Presentation

Warning pattern **localised pain → diffuse pain** = visceral injury

Spleen **nausea + vomiting** · **sharp LUQ pain → dull + diffuse** · shoulder pain (**Kehr's sign**) **before hypotension + collapse**

Signs (Juyia, OB) **referred shoulder pain**, **umbilical or flank ecchymosis**, **guarding / rebound** · high index of suspicion with any abdominal pain

Diaphragmatic spasm

"Wind knocked out" **direct blow to epigastrium** → diaphragmatic spasm → **observe**, resolves, **may continue play**

Abdominal wall injury / rectus sheath haematoma

Cause **epigastric or intramuscular vessel** injury

Clinical **acute localised pain** · **mimics surgical abdomen** (N/V) · **pain on active trunk flexion or rotation**

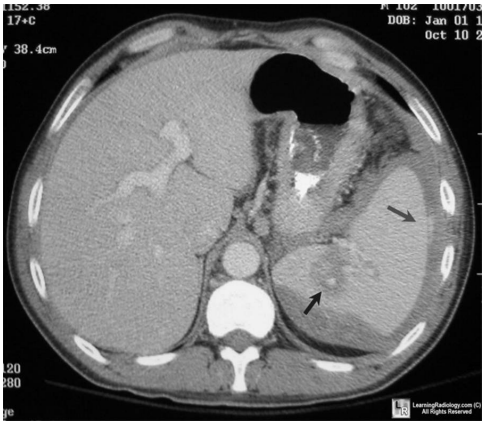
Treatment **ice + rest** (most) · **surgical decompression** if progressive + persistent pain

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Use
CT abdomen (± pelvis)	**Diagnosis for all organs** – wall haematoma, spleen, liver, kidney, bowel · grades severity, guides treatment
Scrotal ultrasound (urology)	**Testicular rupture / haematocoele**
Pelvic X-ray	Fracture → posterior urethral injury



Splenic injury on CT

5 Management

Hypotension after abdominal trauma – work-up

Athlete with abdominal blow

- Haemodynamically unstable → fluid resuscitation + transfer to ER → **abdominal CT**
- Stable, abdominal pain / haematuria → **ER + CT abdomen / pelvis**
- Epigastric blow, transient breathlessness → **observe, may play**

Footballer, BP 70/48, vomiting ×2, resolving arm numbness	**fluid resuscitation + transfer to ER for abdominal CT** ✗ Wrong options: antiemetics + sideline rest · non-emergent C-spine MRI (**stinger** resolved) · RTP when arm settles · head CT (no concussion signs)
RTP (Juyia, OB)	varies with organ + grade · **up to 6 months** after severe spleen / liver injury

Splenic rupture – mononucleosis risk

Mono	**acute splenomegaly** → ↑ rupture risk with trauma
Spread	**oral mucosa**: **sharing water bottles**, kissing ✓ ✗ Wrong options: creatine · spider bite (skin necrosis) · alcohol · NSAIDs on empty stomach (gastric irritation)
Timing	**4–21 days after infection** (OB, this topic) · **first 3 weeks** / within a month (infectious diseases chapter)

Farley (OB)	**> 8,000** IM patients → **9 splenic ruptures** (~0.1%; 0.1–0.5% in infectious diseases chapter)
17-yr-old, mono 1 month ago, tackled, LUQ pain, tachycardia	**splenic rupture** ✗ Wrong options: fractured kidney · ruptured appendix · ruptured gallbladder · pneumothorax
RTP after mono	**no contact sport until splenomegaly resolved** · radiographic surveillance **debated**
Spleen treatment	**observation** in most · **splenectomy only for severe** injury · range: observation → **laparotomy + packing ± splenectomy**

Liver injury

Treatment	**conservative: monitoring + rest**

Renal contusion – haematuria

Pattern	**males**, **single blow to the flank** · flank / abdominal pain, **ecchymosis**, **haematuria**
Lacrosse, right flank blow, pink urine > 5 RBC/hpf, stable	**renal laceration** ✗ Wrong options: splenic laceration (left side, intra-abdominal bleed not haematuria) · haemothorax (chest pain + dyspnoea) · liver laceration (no haematuria) · posterior urethral injury (**displaced pelvic fracture**)
Work-up	**CT abdomen + pelvis ASAP** (risk of instability with continued bleeding)
Treatment	**stable → nonop** · small subset → exploration, partial / complete nephrectomy · **urgent surgery: extensive bleeding with renal fracture or vascular pedicle injury**
Patel (OB)	grade III / IV: **contact sports + skiing / snowboarding** commonest · sports injuries more **isolated**, ↓ instability + mortality
Dangle (OB)	paediatric: **low-grade injuries in males** commonest · few need surgery
McAleer (OB)	**< 5%** of testicular / renal injuries from team sports, no organ loss → **solitary kidney / testis may not need restriction**



Right renal laceration with subcapsular haematoma (CT)

Penile & testicular injuries

Pudendal nerve neuropraxia	**cyclists** · **penile shaft numbness** · **seat modification**

– – Active space – –

Testicular rupture — usually a **kick → tunica albuginea rupture → haematocoele · painful firm scrotal mass, doesn't transilluminate · urology ultrasound** · observation

X vs Y

	Splenic injury	Liver injury	Renal injury
Side / blow	**Left**, LUQ	**Right**, RUQ	**Flank**
Referred pain	**Left shoulder (Kehr)**	**Right shoulder**	–
Urine	Normal	Normal	**Haematuria**
Bleeding	Intra-abdominal	Intra-abdominal	**Haematuria**
Treatment	Observe · **splenectomy if severe**	**Conservative** (monitor + rest)	Nonop if stable · surgery for renal fracture / **pedicle injury**

How it's asked

"Splenic rupture risk factor in hockey player?"	**→ sharing water bottles (mono)**
"Mono 1 month ago, tackled, LUQ pain + tachycardia?"	**→ splenic rupture**
"Peak window for splenic rupture after mono?"	**→ 4–21 days**
"BP 70/48 + vomiting after hit — next step?"	**→ fluid resuscitation + ER transfer for abdominal CT**
"Flank blow, pink urine, > 5 RBC/hpf — CT shows?"	**→ renal laceration**
"Most common cause of death from abdominal injury?"	**→ splenic injury**
"Cyclist with penile numbness?"	**→ pudendal neuropraxia — seat modification**
"Kicked, firm non-transilluminating scrotal mass?"	**→ haematocoele (tunica albuginea rupture)**